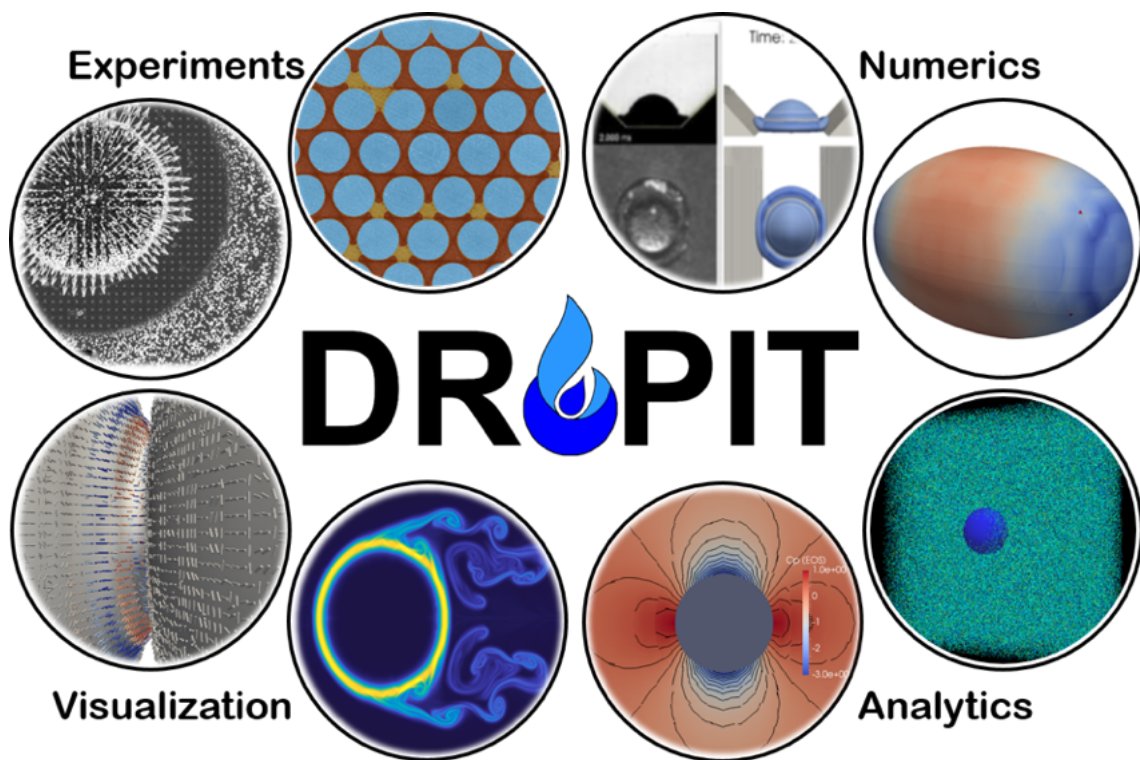


Final Report of the
International Research Training Group
entitled

GRK 2160/2:
“Droplet Interaction Technologies”
(DROPIT)



Universität Stuttgart
Germany



**UNIVERSITÀ
DEGLI STUDI
DI BERGAMO**

Table of Contents

1	GENERAL INFORMATION.....	1
1.1	GENERAL INFORMATION.....	1
1.2	PARTICIPATING RESEARCHERS.....	1
1.3	NUMBER OF DOCTORAL AND POSTDOCTORAL RESEARCHERS, QUALIFYING FELLOWS, AND STUDENT ASSISTANTS	3
2	SUMMARY	3
3	SCIENTIFIC PROGRESS REPORT	6
4	QUALIFICATION, SUPERVISION AND PARTNERSHIPS	9
5	PROJECT RESULTS	12
5.1	PROJECT RESULTS ACHIEVED BY THE DOCTORAL AND POSTDOCTORAL RESEARCHERS	12
5.1.1	PROJECT RESULTS ACHIEVED BY THE DOCTORAL AND POSTDOCTORAL RESEARCHERS (FINANCED FROM IRTG FUNDS)	14
5.1.2	PROJECT RESULTS ACHIEVED BY THE DOCTORAL AND POSTDOCTORAL RESEARCHERS (FINANCED FROM OTHER SOURCES)	38
5.2	PROJECT RESULTS GENERATED BY THE RESEARCH TRAINING GROUP	45
6	FURTHER INFORMATION.....	47
6.1	INFORMATION ON HOW THE PROJECT PROGRESSED, ITS STRUCTURAL IMPACT AND FUTURE PROSPECTS	47
6.2	INFORMATION ON QUALIFICATIONS AND STRUCTURAL IMPACT	66
6.2.1	LIST OF ALL PERSONS FUNDED BY THE RESEARCH TRAINING GROUP TO DATE	82
6.2.2	DURATION OF EMPLOYMENT AND FELLOWSHIP CONTRACTS.....	95
6.2.3	PROMOTION OF EQUALITY: FEMALE RTG PARTICIPANTS BY QUALIFICATION LEVEL	96
6.2.4	RESEARCH STAYS.....	100
6.2.5	CONFERENCE TRAVEL UNDERTAKEN BY (POST)DOCTORAL RESEARCHERS.....	102
6.2.6	STAYS BY GERMAN (POST-)DOCTORAL RESEARCHERS AT FOREIGN PARTNER UNIVERSITIES	113
6.2.7	RESEARCH STAYS BY (POST)DOCTORAL RESEARCHERS FROM ABROAD IN GERMANY	121
6.2.8	INDIVIDUAL REPORTS BY DOCTORAL AND POSTDOCTORAL RESEARCHERS AND, WHERE APPLICABLE, QUALIFYING FELLOW	124
	APPENDIX 1: LIST OF WORKSHOPS / SUMMER SCHOOLS WITHIN DROPIT.....	321
	APPENDIX 2: VISITING SCIENTIST PROGRAM.....	330
	APPENDIX 3: LIST OF COURSES.....	335
3.1:	SPECIALISED COURSES	335
3.2:	CORE COURSES	338
3.3:	TRAINING COURSES ON KEY SKILLS	340
	APPENDIX 4: SUPERVISION AGREEMENT OF DOCTORAL STUDIES (“PROMOTIONSVEREINBARUNG”)....	342

Final Report

International Research Training Group

GRK 2160/2: Droplet Interaction Technologies (DROPIT)

Universität Stuttgart

Institut für Thermodynamik der Luft- und Raumfahrt

Prof. Dr.-Ing. Bernhard Weigand

Università degli Studi di Bergamo

Dipartimento di Ingegneria e Scienze Applicate

Prof. Dr.-Ing. Gianpietro Elvio Cossali

First funding period: 01.10.2016 to 31.03.2021

Second funding period: 01.04.2021 to 30.09.2025

Period covered by this report: 01.10.2016 to 31.12.2025

25.03.2026

1 General Information

1.1 General information

DFG reference number: GRK 2160/2

Project number: 270852890

Title of the Research Training Group:

German: Technologien für Tropfeninteraktionen (DROPIT)

English: Droplet Interaction Technologies (DROPIT)

Name of the applicant universities:

Host university in Germany:

Universität Stuttgart

Keplerstraße 7

70174 Stuttgart

Host university in Italy:

Università degli Studi di Bergamo

Viale Marconi 5

24044 Dalmine

Foreign partner institutions:

Università degli Studi di Trento

Via Mesiano 77

38123 Trento

Spokesperson: Prof. Dr.-Ing. Bernhard Weigand

Official address: Universität Stuttgart, Institut für Thermodynamik der Luft- und Raumfahrt

Pfaffenwaldring 31, 70569 Stuttgart

Reporting period (entire funding period):

First funding period: 01.10.2016 to 31.03.2021

Second funding period: 01.04.2021 to 30.09.2025

1.2 Participating researchers

List of Principal Investigators (PI)

Last name, first name, academic title	Period of involvement	Research area
Germany: University of Stuttgart (US)		
Weigand, Bernhard Prof. Dr.-Ing. Spokesperson University of Stuttgart	01.10.2016 to 30.09.2025	Thermodynamics, Droplet dynamics, Heat transfer

Beck, Andrea Prof. Dr.-Ing.	01.10.2016 to 30.09.2025	Turbulence simulation & modelling, Discontinuous Galerkin Methods, High performance computing
Bringedal, Carina Jun. Prof., Ph.D.	01.04.2021 to 01.10.2022	Porous media, Homogenization, Phase field methods
Ertl, Thomas Prof. Dr. rer. nat.	01.10.2016 to 01.10.2023	Visualization, Computer graphics
Fasoulas, Stefanos Prof. Dr.-Ing.	01.10.2016 to 30.09.2025	Numerical methods, Astronautics
Helmig, Rainer Prof. Dr.-Ing.	01.10.2016 to 31.09.2025	Hydromechanics, Porous media
Lamanna, Grazia PD Dr.-Ing.	01.10.2016 to 30.09.2025	Thermodynamics, Droplet dynamics, Phase transition, Experimental methods
Munz, Claus-Dieter Prof. Dr. rer. nat.	01.10.2016 to 01.10.2022	Fluid mechanics, Mathematics, Numerical methods
Rohde, Christian Prof. Dr. rer. nat.	01.10.2016 to 30.09.2025	Mathematics, Numerical methods
Roth, Norbert Dr.-Ing.	01.10.2016 to 01.03.2019	Thermodynamics, Droplet dynamics, Experimental methods
Schulte, Kathrin Dr.-Ing.	01.04.2021 to 30.09.2025	Thermodynamics, Droplet dynamics, Direct numerical Simulation
Weiskopf, Daniel Prof. Dr. rer. nat.	01.04.2021 to 30.09.2025	Visualization, Computer graphics
Italy: Università degli Studi di Bergamo (UniBG)		
Cossali, Gianpietro Elvio Prof. Dr.-Ing. Spokesperson University of Bergamo	01.10.2016 to 30.09.2025	Theoretical and analytical modelling of two-phase flows
Colombo, Alessandro Prof. Dr.	01.10.2023 to 30.09.2025	Fluid mechanics, Numerical methods
Bassi, Francesco Prof. Dr.-Ing.	01.10.2016 to 30.03.2020	Fluid mechanics, Numerical methods
Botti, Lorenzo Alessio Prof. Dr.	01.04.2020 to 30.09.2025	Fluid mechanics, Numerical methods
Santini, Maurizio Prof. Dr.-Ing.	01.10.2016 to 30.09.2025	Exp. investigation of two-phase flow, Droplet dynamics, X-ray microcomputer-tomography
Tonini, Simona Prof. Dr.-Ing.	01.10.2016 to 30.09.2025	Droplet evaporation, Heat transfer modelling
Italy: Università degli Studi di Trento (UniTN)		

Dumbser, Michael Prof. Dr.-Ing.	01.10.2016 to 30.09.2025	Numerical methods, Fluid mechanics
------------------------------------	-----------------------------	------------------------------------

1.3 Number of doctoral and postdoctoral researchers, qualifying fellows, and student assistants

University of Stuttgart:

	Number
Individuals funded through the Research Training Group	
Doctoral researchers (full time, TV-L E13)	27
Postdoctoral researchers	4
Qualifying fellows	0
Student assistants	92
Individuals funded from other sources	
Doctoral researchers	8
Postdoctoral researchers	0

University of Bergamo:

Please note that all participating doctoral researchers and postdocs from the University of Bergamo are financed by the University of Bergamo. So, there is no complementary funding from any research foundation in Italy for the project.

	Number
Participating doctoral and postdoctoral researchers	
Doctoral researchers (full time)	8
Postdoctoral researchers	2

University of Trento:

Please note that the participating doctoral researcher from the participating institute at the University of Trento is funded from own financial sources.

	Number
Participating doctoral and postdoctoral researchers	
Doctoral researchers (full time)	4
Postdoctoral researchers	0

2 Summary

GRK 2160: Droplet Interaction Technologies (DROPIT)

This project focused on “Droplet Interaction Technologies”, which have applications in a variety of industrial processes. Recent technological advances and the need to develop adequate design tools

require the inclusion of interface kinetics and/or a detailed analysis of surface microstructures. Normally, complicated technical processes cover the underlying basic mechanisms. Thus, progress in the overall process modelling can hardly be gained.

Based on this starting point the project was launched in 2016 as an International Research Training Group (IRTG) between the University of Stuttgart and the University of Bergamo together with the University of Trento as a foreign partner institution. The clear focus of the IRTG was on the understanding of the underlying basic processes, which can influence dramatically the overall process. This has been done by investigating different spatial and/or temporal scales of the problem and by linking them through a multi-scale approach. The research of DROPIT had been organised in three thematic areas (TA): TA-A: Drop-Gas Interaction, TA-B: Drop-Wall Interaction, TA-C: Drop-Liquid Interaction. The research programme first focused on single droplets and fundamentals. This was extended in the second funding period to include multiple droplet interactions and extended physical phenomena, e.g. like heat transfer. Theoretical and numerical studies have been carried out which have been supported by selected experimental work and by the development of related new measurement techniques leading in some cases to patent applications. The three areas listed above are strongly interlinked with each other, which stimulated cooperation between the doctoral researchers.

The qualification programme of this IRTG included both specialised seminars and workshops, designed in such a way as to provide the doctoral researchers with the knowledge and skills to conduct independent research. The existing educational structures for doctoral researchers on both the Italian (ISA, DICAM) and the German side (GRADUS) was used for this, minimising logistic and administrative work, while promoting synergies and interactions among the existing educational and research programmes at the participating universities. Additional qualification measures were included in the programme like a yearly international workshop and a three-yearly international summer school. These initiatives had to be organised by the doctoral and the postdoctoral researchers, which provided the doctoral fellows a platform to put all management, organisational, interdisciplinary and scientific competencies acquired through the course and seminar programme into practice. Furthermore, the doctoral researchers were actively involved in proposing visiting researchers and new specialised courses. Thus, DROPIT was able to make a significant contribution to the topic under consideration in the course of its term, as evidenced by a large number of well-cited publications in highly ranked journals and at international conferences, among other things. The qualification programme led to the development of scientifically independent and excellently educated researchers who were able to find good positions in industry and academia.

GRK 2160: Technologien für Tropfeninteraktionen (DROPIT)

Dieses Projekt konzentrierte sich auf Technologien für Tropfeninteraktionen, die in einer Vielzahl industrieller Prozesse Anwendung finden. Die jüngsten technologischen Fortschritte und die Not-

wendigkeit, geeignete Auslegungswerkzeuge zu entwickeln, erfordern die Einbeziehung der Grenzflächenkinetik und/oder eine detaillierte Analyse der Oberflächen-Mikrostrukturen. In der Regel verdecken komplizierte technische Prozesse die grundlegenden Mechanismen. Daher sind Fortschritte bei der Gesamtprozessmodellierung kaum zu erzielen.

Davon ausgehend wurde das Projekt 2016 als Internationales Graduiertenkolleg (IRTG) zwischen der Universität Stuttgart und der Universität Bergamo zusammen mit der Universität Trento als ausländischen Partnerinstitution ins Leben gerufen. Der klare Schwerpunkt von DROPIT lag auf den zugrunde liegenden Detailprozessen, die den Gesamtprozess drastisch beeinflussen können. Dies geschah durch die Untersuchung verschiedener räumlicher und/oder zeitlicher Skalen des Problems und deren Verknüpfung durch einen Multiskalenansatz. Die Forschung von DROPIT wurde in drei Themenbereiche (TA) untergliedert: TA-A: Tropfen-Gas-Wechselwirkung, TA-B: Tropfen-Wand-Wechselwirkung, TA-C: Tropfen-Flüssigkeits-Wechselwirkung. Das Forschungsprogramm konzentrierte sich zunächst auf einzelne Tropfen und Grundlagen. In der zweiten Förderperiode wurde es dann auf die Wechselwirkungen mehrerer Tropfen und weiteren physikalische Phänomene, wie z. B. Wärmeübertragung, ausgeweitet. Es wurden theoretische und numerische Studien durchgeführt, die durch ausgewählte experimentelle Untersuchungen und die Entwicklung entsprechender neuer Messtechniken unterstützt wurden (von denen einige zu Patentanmeldungen führten). Die drei oben genannten Bereiche sind eng miteinander verknüpft, was die Zusammenarbeit zwischen den Doktoranden förderte.

Das Qualifizierungsprogramm des IRTG umfasste sowohl Fachseminare als auch Workshops, die so konzipiert waren, dass sie den Promovierenden das Wissen und die Fähigkeiten für die Durchführung unabhängiger Forschung vermittelten. Dazu wurden die bestehenden Strukturen für Doktorandenschulen sowohl auf italienischer (ISA, DICAM) als auch auf deutscher Seite (GRADUS) genutzt, wodurch der logistische und administrative Aufwand minimiert und gleichzeitig Synergien und Interaktionen zwischen den bestehenden Ausbildungs- und Forschungsprogrammen gefördert wurden. Zusätzliche Qualifizierungsmaßnahmen wie ein jährlicher internationaler Workshop und eine alle drei Jahre stattfindende internationale Sommerschule waren ebenfalls Teil des Programms. Diese Initiativen wurden von den Promovierenden und der Postdoktorandin bzw. den Postdoktoranden organisiert, wodurch sie alle im Rahmen des Kurs- und Seminarprogramms erworbenen Management-, Organisationskompetenzen sowie interdisziplinäre und wissenschaftliche Kompetenzen in die Praxis umzusetzen konnten. Darüber hinaus wurden die Promovierenden aktiv an der Auswahl von Gastwissenschaftlern und der Entwicklung neuer Fachkurse und Workshops eingebunden. Das Qualifizierungsprogramm führte somit zur Entwicklung wissenschaftlich unabhängiger und exzellent ausgebildeter Forscherinnen und Forschern, die gute Positionen in Industrie und Wissenschaft finden konnten. DROPIT konnte über seiner Laufzeit hinweg einen erheblichen Beitrag zum behandelten Themengebiet leisten, was unter anderem durch eine große Anzahl viel zitierter Publikationen in renommierten Fachzeitschriften und auf internationalen Konferenzen belegt wird.

3 Scientific Progress Report

Droplet interaction technologies find application in a large number of technological and industrial processes. These include processes like spray cooling, spray drying, droplet collisions for the generation of powders and encapsulated material, droplet evaporation and droplet-wall interaction in internal combustion engines. In all these applications, small scale thermo-fluid dynamics may have a very large impact on large scale flow pattern, leading to drag reduction or enhancement, heat transfer enhancement or depression, phase transition kinetics (e.g. drop condensation or nucleate boiling), acoustic impedance and optical reflection. The consequences of the presence of different length scales on macroscopic behavior and the associated enhancement of surface transport is still largely unexplored and mostly limited to the formulation of empirical correlations based solely on macroscopic observations. Here, it was the aim of the GRK 2160 to provide a systematic study of the interdependencies between small-scale and large-scale dynamics in the field of droplet interaction technologies. Due to the complexity of the problem, the analysis of such micro/macro interactions is not limited to one single aspect. Rather, an integrated approach was chosen that evolves along three parallel pathways, namely a numerical, experimental and theoretical approach. During the nine years of this project the research objectives were pursued continuously. This was possible because of the initially given research structure of the project. This structure consisted of three main thematic areas, TA-A to TA-C investigating different droplet interactions:

- **Thematic Area A: Drop-Gas Interaction**, with 6 subprojects
- **Thematic Area B: Drop-Wall Interaction**, with 6 subprojects
- **Thematic Area C: Drop-Liquid Interaction**, with 5 subprojects

The individual subprojects have been designed in a way that there were strong interactions between them. Each subproject was supervised by two to three principal investigators from Germany and Italy. This provided a basis for continuous and intense exchange among researchers from the different universities. Because of the initial selection of the international partners a strong added scientific value could be achieved in these bilateral collaborations, owing to the complementary expertise of the partners involved. For example, SP-B1 investigated droplet collisions with solid superhydrophilic surfaces and related phenomena, supervised by B. Weigand (University of Stuttgart) and G.E. Cosali (University of Bergamo). The project was characterized by close cooperation with other subprojects, e.g., SP-B5 and SP-A11, which analyzed the experimentally observed phenomena by Direct Numerical Simulations (DNS) [1]. The supervisors' complementary expertise facilitated a deep physical understanding of these phenomena, resulting in key publications that explain the mechanisms behind non-circular spreading (Fig. 1).

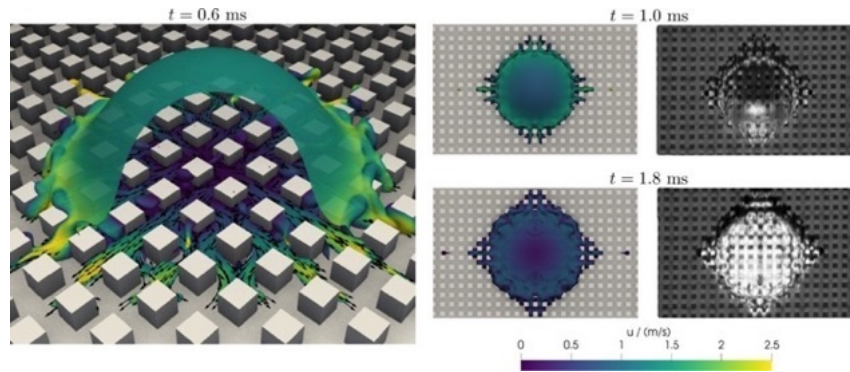


Fig. 1: Wetting behavior of droplet impacts onto micro-structured surfaces showing non-circular spreading behavior (left: Direct Numerical Simulation, right: experimental results) [1].

The research programme first focused on single droplets and fundamentals. It was extended in the second funding period to include multiple droplet interactions and extended physical phenomena, e.g., heat transfer.

Over its nine-year duration, this project has made significant progress and achieved excellent results in droplet interaction technology. The following highlight these achievements: In TA-A (droplet-gas interaction), new models were developed for oscillating and evaporating drops, which, for the first time, went beyond the quasi-steady assumptions previously used for these processes [2]. Simulations of small drops were carried out using Direct Simulation Monte Carlo simulations, leading to a deeper understanding of microscopic interfacial processes [3]. Highly resolved numerical methods were developed, resulting in significant enhancements to the open-source programme FLEXI (www.flexi-project.org) [4]. In TA-B (droplet-wall interaction), the new phenomenon of thermocapillary lamella recess was observed and subsequently explained through DNS results using the programme FS3D [5]. Highly resolved X-ray micro computed tomography was successfully employed to investigate drying phenomena and drop-surface interactions [6, 7]. Furthermore, new models for drop-wall interactions with a porous surface were established [7], and a new phase field approach was developed for compressible drop impingement [8]. In TA-C (droplet-liquid interaction), detailed micro-PIV experiments led to improved models for drop-film interactions [9]. In FP2, the shadow-imaging defocused particle tracing velocimetry (SD μ PTV) was developed to enable time and height-resolved measurements in the spreading lamella. For oblique drop-film impingement, the morphology was experimentally investigated in detail for the first time and explained using the enhanced DNS program FS3D [10]. Finally, advanced visualization techniques for droplet processes contributed significantly to the understanding of the complex phenomena investigated here [11]. Overviews about the project and its continuous progress have been presented at international conferences every year, with one such example provided in [12]. Furthermore, the progress was summarized in a first book [13].

To ensure the validity and verifiability of the research findings, several measures and approaches were implemented throughout the project. First, each subproject employed standard methods to an-

analyze the uncertainties inherent in their results. Furthermore, specialized workshops were established, focusing on experimental and numerical methods or on analytical and numerical methods. In these workshops, doctoral students from DROPIT showcased their specific methods. In addition, international experts were invited to share their expertise in experimental, numerical, and analytical methods. Consequently, the doctoral researchers gained a deeper understanding of the uncertainties associated with other subprojects, facilitating more robust comparisons of their findings and significantly enhancing the quality of the published results.

In addition, these workshops provided doctoral researchers with essential training in proper data management. Strict coordination is crucial when developing large computer code, necessitating the use of robust versioning systems. To address this, research groups working on numerical programs such as **FLEXI** or **FS3D** held annual two-to-three-day retreats. These meetings allowed researchers to collaborate on code modifications and version control strategies. Such diligence is critical for long-term reproducibility, ensuring that even years later, datasets can be accurately mapped to specific software versions. Additionally, these retreats fostered knowledge exchange among researchers working on the same code across different projects, e.g. SFB-TRR 75 or SFB 1313.

In the framework of DROPIT, unique experimental and numerical data have been produced. These research data were handled in accordance with the DFG guidelines for safeguarding good scientific practice. The IRTG used RAID-systems, where research data could be stored and easily exchanged between subprojects. Selected data have been archived in **DaRUS**, the central data repository of the University of Stuttgart. This secure platform assigns a DOI to each dataset for permanent identification. Consequently, the results are findable, accessible, interoperable and reusable, in accordance with the FAIR principles. There are no embargo periods for publication or use. A representative dataset is provided in [14].

Because of the importance to communicate the results of large research projects also to the general public, DROPIT researchers were also here very engaged. Key activities included public lectures for children, involvement in the “Day of Science” at the University of Stuttgart, and a cooperation with a Kindergarten. Additionally, project findings were featured through the contribution “Trickreiche Tropfen” [15] in the popular journal “Bild der Wissenschaft”.

References:

- **Palmeshofer, P., Wurst, J., Geppert, A.,** Schulte, K., Cossali, G.E., Weigand, B., 2025. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science* 682, 413–422. <https://doi.org/10.1016/j.jcis.2024.11.154>
- **Hu, Y.,** Tonini, S., Reutzs, J., Frey, M., Cossali, G.E., Weigand, B., 2024. Direct numerical simulation of an oscillating droplet under evaporating conditions. *Int. J. Heat Mass Transf.* 232, 125891. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125891>
- **Reschke, W.,** Pfeiffer, M., Fasoulas, S., 2020. Enabling Simulations of Droplets with the Direct Simulation Monte Carlo Method, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 57–68. https://doi.org/10.1007/978-3-030-33338-6_5

- **Mossier, P., Appel, D.**, Beck, A.D., Munz, C.-D., 2023. An Efficient hp-Adaptive Strategy for a Level-Set Ghost-Fluid Method. *J Sci Comput* 97, 50. <https://doi.org/10.1007/s10915-023-02363-7>
- **Palmetshofer, P., Geppert, A.**, Steigerwald, J., Arcos Marz, T., Weigand, B., 2024. Thermocapillary central lamella recess during droplet impacts onto a heated wall. *Sci Rep* 14, 1102. <https://doi.org/10.1038/s41598-024-51382-3>
- **Haide, R., Fest-Santini, S.**, Santini, M., 2022. Use of X-ray micro-computed tomography for the investigation of drying processes in porous media: A review. *Dry. Technol.* 40, 1731–1744. <https://doi.org/10.1080/07373937.2021.1876723>
- **Ackermann, S., Fest-Santini, S., Veyskarami, M.**, Helmig, R., Santini, M., 2023. Experimental validation of a coupling concept for drop formation and growth onto porous materials by high-resolution X-ray imaging technique. *Int J Multiphas Flow* 160, 104371. <https://doi.org/10.1016/j.ijmultiphaseflow.2022.104371>
- **Massa, F., Ostrowski, L.**, Bassi, F., Rohde, C., 2022. An artificial Equation of State based Riemann solver for a discontinuous Galerkin discretization of the incompressible Navier–Stokes equations. *J. Comput. Phys.* 448, 110705. <https://doi.org/10.1016/j.jcp.2021.110705>
- **Schubert, S.**, Steigerwald, J., **Geppert, A.**, Weigand, B., Lamanna, G., 2024. Micro-PIV study on the influence of viscosity on the dynamics of droplet impact onto a thin film. *Exp Fluids* 65, 69. <https://doi.org/10.1007/s00348-024-03800-5>
- **Stober, J.**, Santini, M., Schulte, K., 2025. Characterization of splashing and regime thresholds for oblique droplet impact on thin wall films. *Exp. Therm. Fluid Sci.* 167, 111493. <https://doi.org/10.1016/j.expthermflusci.2025.111493>
- **Straub, A.**, Karch, G., Steigerwald, J., Sadlo, F., Weigand, B., Ertl, T., 2023. Visual analysis of interface deformation in multiphase flow. *J Vis* 26, 1321–1338. <https://doi.org/10.1007/s12650-023-00939-x>
- Weigand, B., Lamanna, G., **Kumar, M.**, Tonini, S., Cossali, G.E., 2025. Project Overview and Selected Results of the International Research Training Group GRK 2160. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.
- Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), 2020. *Droplet Interactions and Spray Processes, Fluid Mechanics and Its Applications*. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-33338-6>
- **Palmetshofer, P., Wurst, J.**, 2022. Replication Data for: Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. <https://doi.org/10.18419/DARUS-4178>
- Breuer, R., 2023. *Trickreiche Tropfen, Bild der Wissenschaft*, 22-27.

4 Qualification, Supervision and Partnerships

The qualification programme of DROPIT included both specialised seminars and workshops, designed in such a way as to provide the doctoral researchers with the knowledge and skills to conduct independent research. The existing educational structures for doctoral researchers on both the Italian (ISA, DICAM) and the German side (GRADUS) was used for this, minimising logistic and administrative work, while promoting synergies and interactions among the existing educational and research programmes at the participating universities.

For the direct guidance and assessment of the doctoral researcher's development a joint Italian-German supervision was implemented. They were selected in order to fit best to the topic of the individual subproject. Within the qualification programme an individually conceived stay abroad was included for working scientifically in a different country.

The scientific independence of the doctoral researchers has been supported in several ways. In DROPIT, doctoral researchers had a prominent role in determining the evolution and outcome of their research project. Of course, all doctoral researchers had regular progress meetings with their supervisors focusing on details in their scientific work. Research outline and progress were presented regularly in yearly doctoral seminars and also in our Multiphase Flow Seminar. The doctoral researchers together with the postdoc organised summer schools. Here, they could gain a good insight in project management and logistic issues. Furthermore, the doctoral researchers had a very active role in defining the visiting researcher programme by proposing and inviting scientific guests with whom they wished to cooperate. They were also involved in the set-up of the agenda for the visits and could plan their time for individual discussions with the visitors.

In the yearly DIPSI (Droplet Impact Phenomena and Spray Investigations) workshop in Bergamo, the doctoral researchers were setting up a poster session with contributions from their ongoing work. Their posters have then been discussed with the participants of the workshop.

In general, doctoral and postdoctoral researchers were strongly encouraged to publish their work. This has been done on at least two conferences (e.g., ILASS, ICLASS) and in international well-known journals. At least one scientific contribution in a peer-reviewed journal was mandatory.

All this helped to support their scientific independence. The presentations at the ILASS and ICLASS conference were accompanied by an overview talk about DROPIT, see e.g. [12], in order to point out that several individual presentations of the DROPIT members are also present at the conference. With this, the whole IRTG could achieve a high visibility within the international community.

Furthermore, the doctoral researchers were actively involved in proposing new specialised courses and workshops. One of these new workshops was “DROPIT meet industry”. Here, representatives of industry met with the doctoral researchers to exchange and discuss industry related research topics. In additions, the doctoral researchers could establish contacts and partnerships for their future careers. DROPIT organized several workshops together and invited also visitors together with other research initiative like SFB-TRR 75 and SFB1313. By doing so, a valuable exchange between the different organization could be established. Figure 2 shows the elements of the qualification programme which strongly supported the progress of the doctoral researchers.

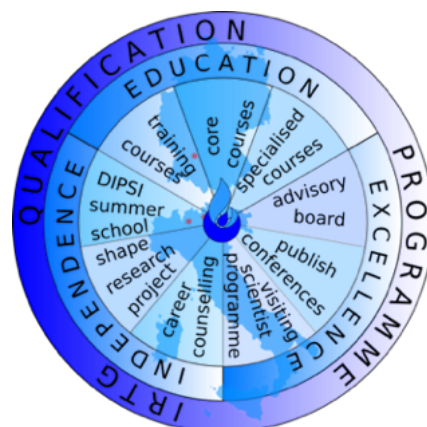


Fig. 2: Qualification program within GRK 2160.

The table below shows the record of the doctoral researchers financed by the IRTG. It can be seen that the average doctorate was around four years. This timeframe is considerably shorter than the average in engineering, despite the significant disruptions and challenges posed by the COVID-19 pandemic throughout the project's duration for the doctoral researchers.

Doctoral record of the doctoral researchers financed by the Research Training Group

Total number of funding recipients	Theses that have only been submitted to date	Completed doctorates	Average duration of doctorate in months	Interruptions in doctoral studies due to illness, maternity leave, parental leave etc. (>3 month)	Promotion dropouts
27	0	14	53	0	2

The University of Stuttgart offers an elaborate infrastructure that is tailored to implement gender equality and more specifically to attract, keep and promote the career chances of talented women within the university. The university has chosen a holistic approach to increase the participation of women in science, which targets female students from high-school to post-doctoral and professional levels (please see <https://www.uni-stuttgart.de/universitaet/profil/diversity-und-gleichstellung>).

In the framework of this IRTG, the mentoring activities have been expanded by setting up an international mentoring programme. The inclusion of foreign mentors enabled to improve the students' intercultural competence. This encompasses the enhancement of intercultural understanding and tolerance, the capability of communicating effectively and gaining more confidence in interacting with professionals from cultural backgrounds different to their own. In addition, researchers and PIs of DROPIT acted as mentors or role models. Here, e.g., one of the postdocs of DROPIT (A. Geppert) participated in TryScience, StartScience and had a mentee in the framework of the CyberMentor-programme (<https://www.cybermentor.de/index.php>). Today, Anne Geppert is department manager at Leibniz Institute for Materials Engineering (IWT) in Bremen. Moreover, two doctoral researchers (Martina Baggio and Karin Schlottke) received the very prestigious Amelia Earhart Fellowship.

5 Project Results

5.1 Project results achieved by the doctoral and postdoctoral researchers

Completed Doctoral Thesis from all doctoral researchers

J. Stober: Oblique Droplet Impact onto a Thin Wall Film, Doctoral thesis, University of Stuttgart, 2026.

P. Palmetshofer: Droplet impact with dry and wet structured surfaces of varying size and wettability, Doctoral thesis, University of Stuttgart, 2025.

W. Ren: Numerical Simulation of Drop Impact onto Textured Surfaces, Doctoral thesis, University of Stuttgart, 2025.

Y. Liu: Numerical investigation on high velocity liquid dynamics, Doctoral thesis, University of Stuttgart, 2024.

A. Straub: Visualization of multiphase flow at interfaces, Doctoral thesis, University of Stuttgart, 2024.

P. Mossier: An hp-Adaptive Strategy for Compressible Droplet Dynamics with Phase Transition, Doctoral thesis, University of Stuttgart, 2024.

D. Ferrari: Numerical methods for multiphase flows: from a simplified incompressible Baer-Nunziato model to a unified theory of compressible multiphase fluid and solid mechanics, Doctoral thesis, University of Trento, 2024.

M. Lipp: Capturing local details in fluid-flow simulations: options, challenges and applications using marker-and-cell schemes, Doctoral thesis, University of Stuttgart, 2024.

M. Gritti: Indagine sperimentale sull'evaporazione di gocce in flusso incrociato, Doctoral thesis, University of Bergamo, 2023.

M. Veyskarami: Coupled free-flow–porous media flow processes including drop formation, Doctoral thesis, University of Stuttgart, 2023.

L. Verzeroli: Nonconforming finite elements methods for nonlinear elasticity problems: Featuring Finite Deformations and Frictionless Contact Constraints, Doctoral thesis, University of Bergamo, 2023.

P. Foltyn: Droplet collisions with dry solid surfaces with variable wettability and topography, Droplet collisions with dry solid surfaces with variable wettability and topography, Doctoral thesis, University of Stuttgart, 2022.

S. Chiocchetti: High order numerical methods for a unified theory of fluid and solid mechanics, Doctoral thesis, University of Trento, 2022.

L. Ostrowski: Compressible multi-component and multi-phase flows: interfaces and asymptotic regimes, Doctoral thesis, University of Stuttgart, 2021.

C. Müller: Multiscale modeling of the evaporation process, Doctoral thesis, University of Stuttgart, 2021.

R. Haide: Characterization of Multi-Phase Fluid Displacement in Porous Media during Drying by means of X-Ray Micro Computed Tomography, Doctoral thesis, University of Bergamo, 2021.

S. Ackermann: A multi-scale approach for drop/porous-medium interaction, Doctoral thesis, University of Stuttgart, 2020.

M. Baggio: Volume of Fluid Numerical Simulation of Drop Impact on Superhydrophobic Complex Solid Surfaces Embedded in a Staggered Cartesian Grid, Doctoral thesis, University of Stuttgart, 2020.

W. Reschke: Particle-based modelling and simulation of heterogeneous reactions, Doctoral thesis, University of Stuttgart, 2020.

R. Bernard: Macro and micro dynamics of droplet impact onto wall-films made of different liquids, Doctoral thesis, University of Stuttgart, 2020.

J. Zeifang: A discontinuous Galerkin method for droplet dynamics in weakly compressible flows, Doctoral thesis, University of Stuttgart, 2020.

A. Geppert: Experimental investigation of droplet wall-film interaction of binary systems, Doctoral thesis, University of Stuttgart, 2019.

F. Massa: Implicit Discontinuous Galerkin methods with efficient time integration for incompressible variable density and compressible turbulent flows, Doctoral thesis, University of Bergamo, 2017.

N. Nguyen: Numerical methods for compressible multi-phase flows with surface tension, Doctoral thesis, University of Trento, 2017.

Completed Doctoral thesis that have not yet been published

Y. Hu: Modelling the evaporation of an oscillating droplet, Doctoral thesis, University of Bergamo, 2025 (submitted/under review).

5.1.1 Project results achieved by the doctoral and postdoctoral researchers (financed from IRTG funds)

1. Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications *(Joint publications with the foreign partners are marked in blue)*

- **Regener Roig, D.**, Crivellini, A., Colombo, A., 2026. Efficient entropy-conserving/stable discontinuous Galerkin solution of the multicomponent compressible Euler equations. J. Comput. Phys. 556, 114808. <https://doi.org/10.1016/j.jcp.2026.114808>
- **Mossier, P.**, Oestlinger, P., Jöns, S., Keim, J., Mavriplis, C., Beck, A.D., Munz, C.-D., 2026. Tackling compressible turbulent multi-component flows with dynamic hp-adaptation. Computers & Fluids 306, 106928. <https://doi.org/10.1016/j.compfluid.2025.106928>
- **Appel, D.**, **Mossier, P.**, Keim, J., Jöns, S., Beck, A., Munz, C.-D., 2026. A High-Order Framework for Compressible Droplet Dynamics, in: Ludwig, T., Bastian, P., Resch, M.M. (Eds.), High Performance Computing in Science and Engineering '23. Springer Nature Switzerland, Cham, pp. 379–393. https://doi.org/10.1007/978-3-031-91312-9_26
- **Hu, Y.**, Tonini, S., Cossali, G.E., 2026. An analytical/numerical evaporation model for an oscillating spheroidal droplet in a potential flow. International Journal of Heat and Mass Transfer 256, 127925. <https://doi.org/10.1016/j.ijheatmasstransfer.2025.127925>
- **Stober, J.**, Potyka, J., Ibach, M., Weigand, B., Schulte, K., 2026. DNS of the Early Phase of Oblique Droplet Impact on Thin Films with FS3D, in: Ludwig, T., Bastian, P., Resch, M.M. (Eds.), High Performance Computing in Science and Engineering '23: Transactions of the High Performance Computing Center, Stuttgart (HLRS) 2023. Springer Nature Switzerland, Cham, pp. 223–238. https://doi.org/10.1007/978-3-031-91312-9_16
- **Ferrari, D.**, Peshkov, I., Romenski, E., Dumbser, M., 2025. A unified HTC multiphase model of continuum mechanics. J. Comput. Phys. 521, 113553. <https://doi.org/10.1016/j.jcp.2024.113553>
- Krake, T., **Klötzl, D.**, Hägele, D., Weiskopf, D., 2025. Uncertainty-Aware Seasonal-Trend Decomposition Based on Loess. IEEE Transactions on Visualization and Computer Graphics 31, 1496–1512. <https://doi.org/10.1109/TVCG.2024.3364388>
- **Marianowski, C.**, Traub, C., Pfeiffer, M., Beyer, J., Fasoulas, S., 2025. Satellite design optimization for differential lift and drag applications. CEAS Space J 17, 11–29. <https://doi.org/10.1007/s12567-024-00550-2>

- **Mossier, P.**, Jöns, S., **Chiocchetti, S.**, Beck, A.D., Munz, C.-D., 2025. Numerical simulation of phase transition with the hyperbolic Godunov-Peshkov-Romenski model. *J. Comput. Phys.* 520, 113514. <https://doi.org/10.1016/j.jcp.2024.113514>
- **Palmetshofer, P.**, Hengsbach, S., Guttman, M., Worgull, M., Weigand, B., 2025. Azimuthal Variation of Apparent Contact Angles on Structured Surfaces Featuring Micrometric Ramps, Pyramids and Staggered Cubes at Two Different Inherent Wettabilities. *Langmuir* 41, 7312–7322. <https://doi.org/10.1021/acs.langmuir.4c04554>
- **Palmetshofer, P.**, **Saha, R.**, Bryce, J., **Geppert, A.**, Cossali, G.E., Weigand, B., 2025. Droplet impact onto sub-millimeter microstructures fully and barely submerged in a liquid film. *Phys. Fluids* 37, 082130. <https://doi.org/10.1063/5.0278343>
- **Palmetshofer, P.**, Weigand, B., 2025. Experimental study on splashing, spreading, and air entrapment dynamics in droplet impacts onto micrometric ramps, pyramids, and staggered cubes. *Phys. Fluids* 37, 022112. <https://doi.org/10.1063/5.0251394>
- **Palmetshofer, P.**, **Wurst, J.**, **Geppert, A.**, Schulte, K., Cossali, G.E., Weigand, B., 2025. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science* 682, 413–422. <https://doi.org/10.1016/j.jcis.2024.11.154>
- **Stober, J.**, Schulte, K., 2025. What governs droplet spreading on a wetted wall. *Phys. Fluids* 37, 032124. <https://doi.org/10.1063/5.0258742>
- **Stober, J.**, Heinemann, M., Santini, M., Schulte, K., 2025. Quantitative Analysis of the Asymmetry of Inner Wall Film Flow in Oblique Droplet Impacts. *Atomization Spray* 35. <https://doi.org/10.1615/AtomizSpr.2026062531>
- **Stober, J.**, Santini, M., Schulte, K., 2025. Characterization of splashing and regime thresholds for oblique droplet impact on thin wall films. *Exp. Therm. Fluid Sci.* 167, 111493. <https://doi.org/10.1016/j.expthermflusci.2025.111493>
- **Ferrari, D.**, Dumbser, M., 2024. A Semi-implicit Finite Volume Scheme for Incompressible Two-Phase Flows. *Commun. Appl. Math. Comput.* 6, 2295–2330. <https://doi.org/10.1007/s42967-024-00367-0>
- Gao, M., **Mossier, P.**, Munz, C.-D., 2024. Shock capturing for a high-order ALE discontinuous Galerkin method with applications to fluid flows in time-dependent domains. *Comput. Fluids* 269, 106124. <https://doi.org/10.1016/j.compfluid.2023.106124>
- **Geppert, A.**, **Stober, J.**, Steigerwald, J., Schulte, K., Tonini, S., Lamanna, G., 2024. Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. *Phys. Fluids* 36, 012005. <https://doi.org/10.1063/5.0179716>

- **Geppert, A.**, Tonini, S., Santini, M., **Fest-Santini, S.**, Lamanna, G., Cossali, G.E., Weigand, B., 2024. Recent results of the International Research Training Group GRK2160/2, in: Proceedings of 16th International Conference on Liquid Atomization and Spray Systems (ICLASS). Shanghai, China.
- **Hu, Y.**, Tonini, S., Reuttsch, J., Frey, M., Cossali, G.E., Weigand, B., 2024. Direct numerical simulation of an oscillating droplet under evaporating conditions. *Int. J. Heat Mass Transf.* 232, 125891. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125891>
- **Hu, Y.**, Tonini, S., Weigand, B., 2024. A numerical approach to the study of the evaporation of an oscillating droplet under unsteady conditions, in: Proceedings of 16th International Conference on Liquid Atomization and Spray Systems (ICLASS). Shanghai, China.
- **Klötzl, D.**, Krake, T., Heyen, F., Becher, M., Koch, M., Weiskopf, D., Kurzhals, K., 2024. NMF-Based Analysis of Mobile Eye-Tracking Data, in: Proceedings of the 2024 Symposium on Eye Tracking Research and Applications, ETRA '24. Association for Computing Machinery, New York, NY, USA, pp. 1–9. <https://doi.org/10.1145/3649902.3653518>
- **Palmetshofer, P.**, **Geppert, A.**, Steigerwald, J., Arcos Marz, T., Weigand, B., 2024. Thermocapillary central lamella recess during droplet impacts onto a heated wall. *Sci Rep* 14, 1102. <https://doi.org/10.1038/s41598-024-51382-3>
- Potyka, J., **Stober, J.**, **Wurst, J.**, Ibach, M., Steigerwald, J., Weigand, B., Schulte, K., 2024. Towards DNS of Droplet-Jet Collisions of Immiscible Liquids with FS3D, in: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.), High Performance Computing in Science and Engineering '22. Springer Nature Switzerland, Cham, pp. 197–212. https://doi.org/10.1007/978-3-031-46870-4_14
- **Saha, R.**, Weigand, B., 2024. Numerical simulation of droplet impact onto heated surfaces below the boiling point. *J. Phys.: Conf. Ser.* 2766, 012045. <https://doi.org/10.1088/1742-6596/2766/1/012045>
- **Schubert, S.**, Steigerwald, J., **Geppert, A.**, Weigand, B., Lamanna, G., 2024. Micro-PIV study on the influence of viscosity on the dynamics of droplet impact onto a thin film. *Exp Fluids* 65, 69. <https://doi.org/10.1007/s00348-024-03800-5>
- **Schubert, S.**, Stein, M., **Geppert, A.**, Lamanna, G., 2024. PIV measurements in the inner crown wall resulting from droplet impact on a thin liquid film, in: Proceedings of 16th International Conference on Liquid Atomization and Spray Systems (ICLASS). Shanghai, China.
- Steigerwald, J., Ibach, M., **Geppert, A.**, Weigand, B., 2024. Numerical investigation of drop–film interactions with a thixotropic liquid. *Journal of Non-Newtonian Fluid Mechanics* 329, 105259. <https://doi.org/10.1016/j.jnnfm.2024.105259>

- **Tietz, R.**, Fasoulas, S., Pfeiffer, M., 2024. Symmetric simulations of droplets with a particle based Vlasov-Enskog-Solver. Presented at the ICASF, 2nd International Conference on advanced earth science and foundation engineering, Mohali, India, p. 120002. <https://doi.org/10.1063/5.0187426>
- **Veyskarami, M.**, Bringedal, C., Helmig, R., 2024. Modeling and Analysis of Droplet Evaporation at the Interface of a Coupled Free-Flow–Porous Medium System. *Transp Porous Med* 151, 2745–2775. <https://doi.org/10.1007/s11242-024-02123-7>
- Wu, H., **Veyskarami, M.**, Schneider, M., Helmig, R., 2024. A New Fully Implicit Two-Phase Pore-Network Model by Utilizing Regularization Strategies. *Transp Porous Med* 151, 1–26. <https://doi.org/10.1007/s11242-023-02031-2>
- Heine, B., Veith, B., **Geppert, A.**, Selzer, T., Denis, T., Kaufmann, K., Maier, S., Schlun, M., Seemann, S., Weigand, B., 2024. From liquid to aerosol: Exploring droplet formation in vibrating mesh nebulization. *Respiratory Drug Delivery* 1, 111-122.
- **Ackermann, S., Fest-Santini, S., Veyskarami, M., Helmig, R., Santini, M., 2023.** Experimental validation of a coupling concept for drop formation and growth onto porous materials by high-resolution X-ray imaging technique. *Int J Multiphase Flow* 160, 104371. <https://doi.org/10.1016/j.ijmultiphaseflow.2022.104371>
- **Appel, D.**, Jöns, S., Keim, J., **Müller, C.**, Zeifang, J., Munz, C.-D., 2023. A narrow band-based dynamic load balancing scheme for the level-set ghost-fluid method, in: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.), *High Performance Computing in Science and Engineering '21*. Springer International Publishing, Cham, pp. 305–320. https://doi.org/10.1007/978-3-031-17937-2_18
- Gao, M., **Appel, D.**, Beck, A., Munz, C.-D., 2023. A high-order fluid–structure interaction framework with application to shock-wave/turbulent boundary-layer interaction over an elastic panel. *J. Fluids Struct.* 121, 103950. <https://doi.org/10.1016/j.jfluidstructs.2023.103950>
- Krake, T., **Klötzl, D.**, Eberhardt, B., Weiskopf, D., 2023. Constrained Dynamic Mode Decomposition. *IEEE Transactions on Visualization and Computer Graphics* 29, 182–192. <https://doi.org/10.1109/TVCG.2022.3209437>
- **Mossier, P.**, **Appel, D.**, Beck, A.D., Munz, C.-D., 2023. An Efficient hp-Adaptive Strategy for a Level-Set Ghost-Fluid Method. *J Sci Comput* 97, 50. <https://doi.org/10.1007/s10915-023-02363-7>
- **Müller, C.**, **Mossier, P.**, Munz, C.-D., 2023. A sharp interface framework based on the inviscid Godunov-Peshkov-Romenski equations: Simulation of evaporating fluids. *J. Comput. Phys.* 473, 111737. <https://doi.org/10.1016/j.jcp.2022.111737>

- **Stober, J.**, Santini, M., Schulte, K., 2023. Influence of Weber Number on Crown Morphology during an Oblique Droplet Impact on a Thin Wall Film. *Fluids* 8, 301. <https://doi.org/10.3390/fluids8110301>
- **Straub, A.**, Karch, G., Steigerwald, J., Sadlo, F., Weigand, B., Ertl, T., 2023. Visual analysis of interface deformation in multiphase flow. *J Vis* 26, 1321–1338. <https://doi.org/10.1007/s12650-023-00939-x>
- **Verzeroli, L.**, **Massa, F.C.**, Botti, L., 2023. A HDG formulation for nonlinear elasticity problems featuring finite deformations and frictionless contact constraints. *Finite Elements in Analysis and Design* 215, 103887. <https://doi.org/10.1016/j.finel.2022.103887>
- **Veyskarami, M.**, Michalkowski, C., Bringedal, C., Helmig, R., 2023. Droplet Formation, Growth and Detachment at the Interface of a Coupled Free-Flow–Porous Medium System: A New Model Development and Comparison. *Transp Porous Med* 149, 389–419. <https://doi.org/10.1007/s11242-023-01944-2>
- Boscheri, W., **Chiocchetti, S.**, Peshkov, I., 2022. A cell-centered implicit-explicit Lagrangian scheme for a unified model of nonlinear continuum mechanics on unstructured meshes. *J. Comput. Phys.* 451, 110852. <https://doi.org/10.1016/j.jcp.2021.110852>
- Botti, L., **Verzeroli, L.**, 2022. BR2 discontinuous Galerkin methods for finite hyperelastic deformations. *J. Comput. Phys.* 463, 111303. <https://doi.org/10.1016/j.jcp.2022.111303>
- **Foltyn, P.**, Ribeiro, D., Silva, A., Lamanna, G., Weigand, B., 2022. Influence of wetting behavior on the morphology of droplet impacts onto dry-patterned micro-structured surfaces. *Phys. Fluids* 34, 123322. <https://doi.org/10.1063/5.0124692>
- **Haide, R.**, **Fest-Santini, S.**, Santini, M., 2022. Use of X-ray micro-computed tomography for the investigation of drying processes in porous media: A review. *Dry. Technol.* 40, 1731–1744. <https://doi.org/10.1080/07373937.2021.1876723>
- **Klötzl, D.**, Krake, T., Zhou, Y., Hotz, I., Wang, B., Weiskopf, D., 2022. Local bilinear computation of Jacobi sets. *Vis Comput* 38, 3435–3448. <https://doi.org/10.1007/s00371-022-02557-4>
- **Klötzl, D.**, Krake, T., Zhou, Y., **Stober, J.**, Schulte, K., Hotz, I., Wang, B., Weiskopf, D., 2022. Reduced Connectivity for Local Bilinear Jacobi Sets, in: *2022 Topological Data Analysis and Visualization (TopoInVis)*. pp. 39–48. <https://doi.org/10.1109/TopoInVis57755.2022.00011>
- Lamanna, G., **Geppert, A.**, **Bernard, R.**, Weigand, B., 2022. Drop impact onto wetted walls: an unsteady analytical solution for modelling crown spreading. *J. Fluid Mech.* 938, A34. <https://doi.org/10.1017/jfm.2022.69>

- **Massa, F., Ostrowski, L., Bassi, F., Rohde, C., 2022.** An artificial Equation of State based Riemann solver for a discontinuous Galerkin discretization of the incompressible Navier–Stokes equations. *J. Comput. Phys.* 448, 110705. <https://doi.org/10.1016/j.jcp.2021.110705>
- Michalkowski, C., **Veyskarami, M.**, Bringedal, C., Helmig, R., Schleper, V., 2022. Two-phase Flow Dynamics at the Interface Between GDL and Gas Distributor Channel Using a Pore-Network Model. *Transp Porous Med* 144, 429–458. <https://doi.org/10.1007/s11242-022-01813-4>
- **Mossier, P.**, Beck, A., Munz, C.-D., 2022. A p-Adaptive Discontinuous Galerkin Method with hp-Shock Capturing. *J Sci Comput* 91, 4. <https://doi.org/10.1007/s10915-022-01770-6>
- **Straub, A.**, Boblest, S., Karch, G., Sadlo, F., Ertl, T., 2022. Droplet-Local Line Integration for Multiphase Flow, in: 2022 IEEE Visualization and Visual Analytics (VIS). IEEE, Oklahoma City, OK, USA, pp. 135–139. <https://doi.org/10.1109/VIS54862.2022.00036>
- **Chiocchetti, S.**, Peshkov, I., Gavriluk, S., Dumbser, M., 2021. High order ADER schemes and GLM curl cleaning for a first order hyperbolic formulation of compressible flow with surface tension. *J. Comput. Phys.* 426, 109898. <https://doi.org/10.1016/j.jcp.2020.109898>
- **Ferrari, D.**, Dumbser, M., 2021. A mass and momentum-conservative semi-implicit finite volume scheme for complex non-hydrostatic free surface flows. *Int. J. Numer. Method Fluids* 93, 2946–2967. <https://doi.org/10.1002/flid.5017>
- **Foltyn, P.**, Restle, F., Wissmann, M., Hengsbach, S., Weigand, B., 2021. The Effect of Patterned Micro-Structure on the Apparent Contact Angle and Three-Dimensional Contact Line. *Fluids* 6, 92. <https://doi.org/10.3390/fluids6020092>
- **Foltyn, P.**, Ribeiro, D., Silva, A., Lamanna, G., Weigand, B., 2021. Influence of wetting behavior on the morphology of droplet impacts onto dry smooth surfaces. *Phys. Fluids* 33, 063305. <https://doi.org/10.1063/5.0053539>
- **Foltyn, P.**, Rihm, L.K., Ribeiro, D., Silva, A., Weigand, B., 2021. Measurement of the lamella thickness during droplet impact onto differently wettable smooth surfaces using an extension of the Laser Pattern Shift Method with naturally occurring patterns. *Rev. Sci. Instrum.* 92, 105111. <https://doi.org/10.1063/5.0060407>
- **Foltyn, P.**, Ribeiro, D., Silva, A., Lamanna, G., Weigand, B., 2021. Effect of Surface Wettability on the Droplet Impact Morphologies on Dry Smooth Polycarbonate Surfaces, in: Proceedings of 15th International Conference on Liquid Atomization and Spray Systems (ICLASS). <https://doi.org/10.2218/iclass.2021.5907>

- Gabriel, A.-A., Li, D., **Chiocchetti, S.**, Tavelli, M., Peshkov, I., Romenski, E., Dumbser, M., 2021. A unified first-order hyperbolic model for nonlinear dynamic rupture processes in diffuse fracture zones. *Phil. Trans. R. Soc. A.* 379, 20200130. <https://doi.org/10.1098/rsta.2020.0130>
- Heinemann, M., Frey, S., Tkachev, G., **Straub, A.**, Sadlo, F., Ertl, T., 2021. Visual analysis of droplet dynamics in large-scale multiphase spray simulations. *J Vis* 24, 943–961. <https://doi.org/10.1007/s12650-021-00750-6>
- Koch, T., Gläser, D., Weishaupt, K., **Ackermann, S.**, Beck, M., Becker, B., Burbulla, S., Class, H., Coltman, E., Emmert, S., Fetzner, T., Grüninger, C., Heck, K., Hommel, J., Kurz, T., **Lipp, M.**, Mohammadi, F., Scherrer, S., Schneider, M., Seitz, G., Stadler, L., Utz, M., Weinhardt, F., Flemisch, B., 2021. DuMux 3 – an open-source simulator for solving flow and transport problems in porous media with a focus on model coupling. *Computers & Mathematics with Applications* 81, 423–443. <https://doi.org/10.1016/j.camwa.2020.02.012>
- **Liu, Y., Geppert, A.**, Chu, X., Heine, B., Weigand, B., 2021. Simulation of an Annular Liquid Jet with a Coaxial Supersonic Gas Jet in a Medical Inhaler. *Atomization Spray* 31, 95–116. <https://doi.org/10.1615/AtomizSpr.2021037223>
- Peshkov, I., Dumbser, M., Boscheri, W., Romenski, E., **Chiocchetti, S.**, Ioriatti, M., 2021. Simulation of non-Newtonian viscoplastic flows with a unified first order hyperbolic model and a structure-preserving semi-implicit scheme. *Comput. Fluids* 224, 104963. <https://doi.org/10.1016/j.compfluid.2021.104963>
- **Ren, W., Foltyn, P., Geppert, A.**, Weigand, B., 2021. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. *Sci Rep* 11, 18018. <https://doi.org/10.1038/s41598-021-97376-3>
- Ribeiro, D., **Foltyn, P.**, Silva, A., Lamanna, G., Weigand, B., 2021. The influence of Wettability on the Droplet Impact onto Micro-Structured Surfaces:, in: *Proceedings of 15th International Conference on Liquid Atomization and Spray Systems (ICLASS)*. <https://doi.org/10.2218/iclass.2021.5908>
- Steigerwald, J., **Geppert, A.**, Weigand, B., 2021. Numerical study of drop shape effects in binary drop film interactions for different density ratios, in: *Proceedings of 15th International Conference on Liquid Atomization and Spray Systems (ICLASS)*. <https://doi.org/10.2218/iclass.2021.6038>
- Steigerwald, J., Reuttsch, J., Ibach, M., **Baggio, M.**, Seck, A., Haus, B.K., Weigand, B., 2021. Direct Numerical Simulation of a Wind-Generated Water Wave, in: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.), *High Performance Computing in Science and Engineering*

- '19. Springer International Publishing, Cham, pp. 325–341. https://doi.org/10.1007/978-3-030-66792-4_22
- **Straub, A.**, Karch, G., Sadlo, F., Ertl, T., 2021. Implicit Visualization of 2D Vector Field Topology for Periodic Orbit Detection, in: Hotz, I., Bin Masood, T., Sadlo, F., Tierny, J. (Eds.), Topological Methods in Data Analysis and Visualization VI, Mathematics and Visualization. Springer International Publishing, Cham, pp. 159–180. https://doi.org/10.1007/978-3-030-83500-2_9
 - **Ackermann, S.**, Helmig, R., **Fest-Santini, S.**, 2020. Upscaling of Coupled Free-Flow and Porous-Medium-Flow Processes, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 139–148. https://doi.org/10.1007/978-3-030-33338-6_11
 - **Baggio, M.**, Weigand, B., 2020. Numerical Simulation for Drop Impact on Textured Surfaces, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 127–138. https://doi.org/10.1007/978-3-030-33338-6_10
 - Baumgartner, D., **Bernard, R.**, Weigand, B., Lamanna, G., Brenn, G., Planchette, C., 2020. Influence of liquid miscibility and wettability on the structures produced by drop–jet collisions. J. Fluid Mech. 885, A23. <https://doi.org/10.1017/jfm.2019.967>
 - **Bernard, R.**, **Vaikuntanathan, V.**, Lamanna, G., Weigand, B., 2020. A New Perspective for the Characterization of Crown Rim Kinematics, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 163–175. https://doi.org/10.1007/978-3-030-33338-6_13
 - Busto, S., **Chiocchetti, S.**, Dumbser, M., Gaburro, E., Peshkov, I., 2020. High Order ADER Schemes for Continuum Mechanics. Front. Phys. 8, 32. <https://doi.org/10.3389/fphy.2020.00032>
 - **Chiocchetti, S.**, **Müller, C.**, 2020. A Solver for Stiff Finite-Rate Relaxation in Baer–Nunziato Two-Phase Flow Models, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 31–44. https://doi.org/10.1007/978-3-030-33338-6_3
 - Dumbser, M., **Chiocchetti, S.**, Peshkov, I., 2020. On Numerical Methods for Hyperbolic PDE with Curl Involutions, in: Demidenko, G.V., Romenski, E., Toro, E., Dumbser, Michael (Eds.), Continuum Mechanics, Applied Mathematics and Scientific Computing: Godunov’s Legacy. Springer International Publishing, Cham, pp. 125–134. https://doi.org/10.1007/978-3-030-38870-6_17

- **Foltyn, P.**, Guttman, M., Schneider, M., **Fest-Santini, S.**, Wildenschild, D., Weigand, B., 2020. Fabrication and Evaluation Methods of Micro-structured Surfaces for Droplet Impact Experiments, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 71–86. https://doi.org/10.1007/978-3-030-33338-6_6
- Gaburro, E., Boscheri, W., **Chiocchetti, S.**, Klingenberg, C., Springel, V., Dumbser, M., 2020. High order direct Arbitrary-Lagrangian-Eulerian schemes on moving Voronoi meshes with topology changes. *J. Comput. Phys.* 407, 109167. <https://doi.org/10.1016/j.jcp.2019.109167>
- **Geppert, A.**, **Bernard, R.**, Weigand, B., Lamanna, G., 2020. Analytical Model for Crown Spreading During Drop Impact onto Wetted Walls: Effect of Liquids Viscosity on Momentum Transfer, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 177–190. https://doi.org/10.1007/978-3-030-33338-6_14
- **Haide, R.**, Santini, M., 2020. Use of X-ray Micro Computed Tomography for the Investigation of Drying and Salt Precipitation in a Regular Glass Bead Structure, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 87–100. https://doi.org/10.1007/978-3-030-33338-6_7
- **Müller, C.**, Hitz, T., Jöns, S., **Zeifang, J.**, **Chiocchetti, S.**, Munz, C.-D., 2020. Improvement of the Level-Set Ghost-Fluid Method for the Compressible Euler Equations, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 17–29. https://doi.org/10.1007/978-3-030-33338-6_2
- **Ostrowski, L.**, **Massa, F.**, Rohde, C., 2020. A Phase Field Approach to Compressible Droplet Impingement, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 113–126. https://doi.org/10.1007/978-3-030-33338-6_9
- **Ostrowski, L.**, Rohde, C., 2020. Compressible multicomponent flow in porous media with Maxwell-Stefan diffusion. *Math Meth Appl Sci* 43, 4200–4221. <https://doi.org/10.1002/mma.6185>
- **Ren, W.**, Reuttsch, J., Weigand, B., 2020. Direct Numerical Simulation of Water Droplets in Turbulent Flow. *Fluids* 5, 158. <https://doi.org/10.3390/fluids5030158>
- **Reschke, W.**, Pfeiffer, M., Fasoulas, S., 2020. Enabling Simulations of Droplets with the Direct Simulation Monte Carlo Method, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand,

- B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 57–68. https://doi.org/10.1007/978-3-030-33338-6_5
- **Straub, A.**, Ertl, T., 2020. Visualization Techniques for Droplet Interfaces and Multiphase Flow, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 203–214. https://doi.org/10.1007/978-3-030-33338-6_16
 - Tavelli, M., **Chiocchetti, S.**, Romenski, E., Gabriel, A.-A., Dumbser, M., 2020. Space-time adaptive ADER discontinuous Galerkin schemes for nonlinear hyperelasticity with material failure. J. Comput. Phys. 422, 109758. <https://doi.org/10.1016/j.jcp.2020.109758>
 - **Vaikuntanathan, V.**, **Bernard, R.**, Lamanna, G., Cossali, G.E., Weigand, B., 2020. On the Measurement of Velocity Field Within Wall-Film During Droplet Impact on It Using High-Speed Micro-PIV, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 215–223. https://doi.org/10.1007/978-3-030-33338-6_17
 - **Varma Raja Kochanattu, G.**, Cossali, G.E., Tonini, S., 2020. An Analytical Approach to Model the Effect of Evaporation on Oscillation Amplitude of Liquid Drops in Gaseous Environment, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 3–16. https://doi.org/10.1007/978-3-030-33338-6_1
 - **Zeifang, J.**, Kaiser, K., Schütz, J., **Massa, F.**, Beck, A., 2020. An Investigation of Different Splitting Techniques for the Isentropic Euler Equations, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 45–55. https://doi.org/10.1007/978-3-030-33338-6_4
 - **Zeifang, J.**, Schütz, J., Kaiser, K., Beck, A., Lukáčová-Medvidová, M., Noelle, S., 2020. A Novel Full-Euler Low Mach Number IMEX Splitting. CiCP 27, 292–320. <https://doi.org/10.4208/cicp.OA-2018-0270>
 - Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), 2020. Droplet Interactions and Spray Processes, Fluid Mechanics and Its Applications. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-33338-6>
 - **Baggio, M.**, Weigand, B., 2019. Numerical simulation of a drop impact on a superhydrophobic surface with a wire. Phys. Fluids 31, 112107. <https://doi.org/10.1063/1.5123593>
 - Beck, A., Bolemann, T., Flad, D., Krais, N., **Zeifang, J.**, Munz, C.-D., 2019. Application and Development of the High Order Discontinuous Galerkin Spectral Element Method for Com-

pressible Multiscale Flows, in: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.), High Performance Computing in Science and Engineering ' 18. Springer International Publishing, Cham, pp. 291–307. https://doi.org/10.1007/978-3-030-13325-2_18

- Fasoulas, S., Munz, C.-D., Pfeiffer, M., Beyer, J., Binder, T., Copplestone, S., Mirza, A., Nizenkov, P., Ortwein, P., **Reschke, W.**, 2019. Combining particle-in-cell and direct simulation Monte Carlo for the simulation of reactive plasma flows. *Phys. Fluids* 31, 072006. <https://doi.org/10.1063/1.5097638>
- Föll, F., Hitz, T., **Müller, C.**, Munz, C.-D., Dumbser, M., 2019. On the use of tabulated equations of state for multi-phase simulations in the homogeneous equilibrium limit. *Shock Waves* 29, 769–793. <https://doi.org/10.1007/s00193-019-00896-1>
- Ortwein, P., Copplestone, S.M., Munz, C.-D., Binder, T., **Reschke, W.**, Fasoulas, S., 2019. A particle localization algorithm on unstructured curvilinear polynomial meshes. *Comput. Phys. Commun.* 235, 63–74. <https://doi.org/10.1016/j.cpc.2018.09.024>
- **Reschke, W.**, Massuti-Ballester, B., Pfeiffer, M., Herdrich, G., Fasoulas, S., 2019. Validation of DSMC and CFD based catalysis modelling using plasma wind tunnel flows. Presented at the 31st International Symposium on rarefied gas dynamics: RGD31, Glasgow, UK, p. 070020. <https://doi.org/10.1063/1.5119574>
- Reuttsch, J., Ertl, M., **Baggio, M.**, Seck, A., Weigand, B., 2019. Towards a Direct Numerical Simulation of Primary Jet Breakup with Evaporation, in: Nagel, W.E., Kröner, D.H., Resch, M.M. (Eds.), High Performance Computing in Science and Engineering ' 18. Springer International Publishing, Cham, pp. 243–257. https://doi.org/10.1007/978-3-030-13325-2_15
- Schatz, K., **Müller, C.**, Gralka, P., Heinemann, M., **Straub, A.**, Schulz, C., Braun, M., Rau, T., Becher, M., Diehl, P., Marcello, D., Frank, J., Müller, T., Frey, S., Reina, G., Weiskopf, D., Ertl, T., 2019. Visual Analysis of Structure Formation in Cosmic Evolution, in: 2019 IEEE Scientific Visualization Conference (SciVis). IEEE, Vancouver, BC, Canada, pp. 33–41. <https://doi.org/10.1109/SciVis47405.2019.8968855>
- Terzis, A., Kirsch, M., **Vaikuntanathan, V.**, **Geppert, A.**, Lamanna, G., Weigand, B., 2019. Splashing characteristics of diesel exhaust fluid (AdBlue) droplets impacting on urea-water solution films. *Exp. Therm. Fluid Sci.* 102, 152–162. <https://doi.org/10.1016/j.expthermflusci.2018.11.002>
- Yang, G., **Vaikuntanathan, V.**, Terzis, A., Cheng, X., Weigand, B., Helmig, R., 2019. Impact of a Linear Array of Hydrophilic and Superhydrophobic Spheres on a Deep Water Pool. *Colloids and Interfaces* 3, 29. <https://doi.org/10.3390/colloids3010029>

- **Baggio, M.**, Weigand, B., 2018. A numerical method for handling arbitrary solid boundaries within the DNS multiphase code Free Surface 3D (FS3D), in: Proceedings of 14th International Conference on Liquid Atomization and Spray Systems (ICLASS). Chicago, USA.
- **Bernard, R., Geppert, A., Vaikuntanathan, V.**, Lamanna, G., Weigand, B., 2018. On the Scaling of Crown Rim Diameter during Droplet Impact on Thin Wall-Films, in: Proceedings of 14th International Conference on Liquid Atomization and Spray Systems (ICLASS). Chicago, USA.
- **Ostrowski, L.**, Rohde, C., 2018. Phase field modelling for compressible droplet impingement. Hyperbolic Problems: Theory, Numerics, Applications 586.
- Steigerwald, J., **Geppert, A.**, Ertl, M., **Bernard, R., Vaikuntanathan, V.**, Lamanna, G., Weigand, B., 2018. Direct Numerical Simulation of One- and Two-component Droplet Wall-Film Interactions within the Crown-type Splashing Regime, in: Proceedings of 14th International Conference on Liquid Atomization and Spray Systems (ICLASS). Chicago, USA.
- **Zeifang, J.**, Kaiser, K., Beck, A., Schütz, J., Munz, C.-D., 2018. Efficient high-order discontinuous Galerkin computations of low Mach number flows. Commun. Appl. Math. Comput. Sci. 13, 243–270. <https://doi.org/10.2140/camcos.2018.13.243>
- **Geppert, A.**, Terzis, A., Lamanna, G., Marengo, M., Weigand, B., 2017. A benchmark study for the crown-type splashing dynamics of one- and two-component droplet wall–film interactions. Exp Fluids 58, 172. <https://doi.org/10.1007/s00348-017-2447-2>
- **Geppert, A.**, Chatzianagnostou, D., Meister, C., Gomaa, H., Lamanna, G., Weigand, B., 2016. Classification of Impact Morphology and Splashing/Deposition Limit for N-Hexadecane. Atomization Spray 26, 983–1007. <https://doi.org/10.1615/AtomizSpr.2015013352>
- **Santini, M., Fest-Santini, S., Foltyn, P.**, 2016. On the local mass transfer rates around arbitrary shaped particles calculated by X-ray computed microtomography: Prospective for a novel experimental technique. Int. Comm. Heat & Mass Transfer 79, 135–139. <https://doi.org/10.1016/j.icheatmasstransfer.2016.11.001>

2. Category B – Any other form of published results

I. Proceedings (*Joint publications with the foreign partners are marked in blue*)

- Keim, J., **Konan, H.**, Rohde, C., 2025. A note on hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. Presented at the Sixth Workshop on Compressible Multiphase Flows: Derivation, Closure laws, Thermodynamics, 188–212, ESAIM Proc. Surveys, 78, EDP Sci., Les Ulis.

- Fuchs, T., **Molina Vogelsang, P.**, Lamanna, G., 2025. Shadowgraph Defocusing μ PTV Measurement of Drop Impact on a Wetted Surface. Presented at the 6th International Conference on Experimental Fluids Mechanics, Niagara-on-the-Lake.
- **Hu, Y.**, Tonini, S., Cossali, G.E., 2025. A Semi-Analytical and Numerical Model for the Evaporation of an Oscillating Droplet in a Potential Flow. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.
- Weigand, B., Lamanna, G., **Kumar, M.**, Tonini, S., Cossali, G.E., 2025. Project Overview and Selected Results of the International Research Training Group GRK 2160. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.
- **Kostelecky, A.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H., Helmig, R., 2025. Local thermal non-equilibrium processes in porous media: Comparison of different models from the pore- to the REV-scale. Presented at the InterPore 2025, Albuquerque, New Mexico, USA.
- **Molina Vogelsang, P.**, Fuchs, T., Lamanna, G., 2025. Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film. Presented at the ILASS Europe 2025, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.
- **Palmetshofer, P.**, Wirth, M., Steinhausen, C., Trtik, P., Lehmann, E., **Geppert, A.**, 2025. Neutron imaging investigation of liquids mixing during droplet impact onto liquid films. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization and Spray Systems, Lund, Sweden.
- **Saha, R.**, Inal, C., **Kumar, M.**, Weigand, B., 2025. DNS of droplet impact onto smooth and micro-structured surfaces. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.
- **Dhar, R.**, Schulte, K., 2024. Investigation of droplet-pool phenomena and droplet-film interaction for immiscible fluids, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- Ibach, M., Roth, N., Tonini, S., Cossali, G.E., Weigand, B., 2024. Analytical Study of Large Amplitude Shape Oscillations of Viscous Droplets, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- **Klötzl, D.**, Weiskopf, D., 2024. NMF-Based Analysis of Droplet-Wall-Film Interactions, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>

- **Kostelecky, A.**, Helmig, R., Stefansson, I., Berre, I., 2024. Local thermal non-equilibrium processes influenced by interfaces within porous media systems. Presented at the InterPore 2024, Qingdao, China.
- **Palmetshofer, P., Geppert, A.**, Weigand, B., 2024. Splashing Modes in Droplet Impacts onto Sub-Millimeter Microstructures, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2024_11
- **Saha, R.**, Weigand, B., 2024. DNS of dropelt impact onto heated surfaces, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- **Stober, J.**, Schulte, K., 2024. Investigation of the Asymmetry of the Flow Field at the Wall during an Oblique Droplet Impact onto a Thin Wall Film, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2024_12
- **Tietz, R.**, Fasoulas, S., Pfeiffer, M., 2024. Enskog-Vlasov Simulations in High Pressure Environments, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- **Geppert, A.**, Weigand, B., 2023. Ground spreading dynamics during single droplet impact onto a free-standing cubic surface element, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_1
- **Geppert, A.**, Lamanna, G., 2023. Towards spatial and time-resolved film thickness measurements during droplet impact onto thin liquid films. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- Weigand, B., **Geppert, A.**, Lamanna, G., Tonini, S., Cossali, G.E., 2023. Selected Results of the International Research Training Group GRK 2160 on "Droplet Interaction Technologies" (DROPIT). Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- **Geppert, A.**, Weigand, B., 2023. Spreading dynamics during single droplet impact onto a free-standing cubic surface element. Presented at the ICMF, 11th International Conference on Multiphase Flow, Kobe, Japan.
- **Hu, Y.**, Tonini, S., Cossali, G., Weigand, B., Reuttsch, J., Frey, M., 2023. Evaporation of an oscillating drop, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_4

- **Klötzl, D.**, Ibach, M., Weiskopf, D., Weigand, B., 2023. Relation between shear rate and structural parameter in non-Newtonian droplet oscillations via Jacobi sets and STL decomposition: a visual analysis, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_5
- **Marianowski, C.**, S. Fasoulas, M. Pfeiffer, 2023. Non-Equilibrium Particle Schemes to Simulate Liquid-Vapor Interfaces, European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- **Molina Vogelsang, P.**, **Schubert, S.**, Lamanna, G., 2023. An approach to investigate the 3D height evolution in the crown crater during early crown formation with the General Defocusing Particle Tracking (GDPT) method, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_6
- Munz, C.-D., Jöns, S., **Mossier, P.**, **Travnicek, A.**, Beck, A., 2023. The Use of Local Solutions to Model Phase Transitions in the Compressible Flow Regime, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_10
- **Palmetshofer, P.**, Heckel, J., **Geppert, A.**, Weigand, B., 2023. Experimental investigation of the height and angle of an uprising sheet occurring in near-simultaneous double droplet impacts onto smooth surfaces of varying wettability. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- **Palmetshofer, P.**, **Ren, W.**, Steigerwald, J., **Geppert, A.**, Weigand, B., 2023. Experimental and Numerical Investigation of Droplet Impacts onto Cubic Pillars with a Wetted Base. Presented at the ICMF, 11th International Conference on Multiphase Flow, Kobe, Japan.
- **Palmetshofer, P.**, Arcos Marz, T., Weigand, B., 2023. 360° Contact Angles of Water Droplets on Micrometric Ramps, Pyramids and Staggered Cuboid Surfaces at varying Wettability, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_7
- **Saha, R.**, Weigand, B., 2023. Numerical simulation of droplet impact onto structured surfaces, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_8
- **Stober, J.**, Santini, M., **Geppert, A.**, Schulte, K., 2023. Experimental Investigation of an Oblique Droplet Impact onto a Quiescent Wall Film with Two-Perspective High-Speed Imaging. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.

- **Stober, J., Schulte, K., Santini, M., 2023.** Experimental Observation of the Collision of a Crown with a thin Wall Film during Early Oblique Droplet Impact, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_9
- **Stober, J., Schulte, K., 2023.** Numerical Investigation of the Mechanism of Early Crown Formation during an Oblique Droplet Impact onto a Wall Film. Presented at the ICMF, 11th International Conference on Multiphase Flow, Kobe, Japan.
- **Appel, D., Zeifang, J., Beck, A., Munz, C.-D., 2022.** Narrow band-based dynamic load balancing for multiphase flow simulations, in: Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.6092/DIPSI2022>
- **Geppert, A., Foltyn, P., Weigand, B., 2022.** Effect of edge length and surface wettability on droplet impact onto a stand-alone cubic pillar, in: Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.6092/DIPSI2022>
- **Geppert, A., Foltyn, P., Weigand, B., 2022.** Effect of edge length and wettability on droplet impact onto a stand-alone cubic pillar. Presented at the ILASS Europe, 31st Conference on Liquid Atomization and Spray Systems, virtual.
- **Geppert, A., Santini, M., Lamanna, G., Cossali, G.E., Weigand, B., 2022.** Scientific Highlights of the International Research Training Group (GRK 2160/2) „Droplet Interaction Technologies" (DROPIT). Presented at the ILASS Europe, 31st Conference on Liquid Atomization and Spray Systems, virtual.
- **Palmetshofer, P., Foltyn, P., Geppert, A., Weigand, B., 2022.** Dynamic contact angle measurements of droplets spreading on surfaces of varying wettability, in: Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.6092/DIPSI2022>
- **Ren, W., Hu, Y., Weigand, B., 2022.** Splashing During Simultaneous Two Drops Impact onto a Micropillared Superhydrophilic Surface. Presented at the ILASS Europe, 31st Conference on Liquid Atomization & Spray Systems, virtual.
- **Stober, J., Schulte, K., 2022.** Numerical investigation of the mechanisms of early crown formation at an oblique drop impact onto a wall film, in: Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.6092/DIPSI2022>

- **Baggio, M.**, B. Weigand, 2019. Direct numerical simulations for drop impact on superhydrophobic patterned surfaces using the VoF method. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Paris, France.
- **Baggio, M.**, Weigand, B., 2019. On the implementation of Structured Surfaces to FS3D, in: Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2019_pp06-09
- **Bernard, R., Vaikuntanathan, V.**, Lamanna, G., Weigand, B., 2019. Crown rim kinematics for drop impact on wall-film. Presented at the 8th Conference on Bubble & Drop, Sofia, Bulgaria.
- **Foltyn, P.**, Roth, N., Weigand, B., 2019. Development and calibration of the laser pattern shift method for measuring the lamella topology during drop impact on walls. Presented at the ILASS Europe, 29th Conference on Liquid Atomization & Spray Systems, Paris, France.
- **Foltyn, P.**, Restle, F., Weigand, B., 2019. 360° Evaluation of Projected Contact Angles of Static Droplets on Structured Surfaces, in: Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2019_pp14-17
- **Ostrowski, L., Massa, F.**, 2019. An incompressible compressible approach for droplet impact, in: Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2019_pp18-21
- Reuttsch, J., **Varma Raja Kochanattu, G.**, Ibach, M., Kieffer-Roth, C., Tonini, S., Cossali, G.E., Weigand, B., 2019. Direct Numerical Simulations of Oscillating Liquid Droplets: a Method to Extract Shape Characteristics. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Paris, France.
- **Straub, A.**, Heinemann, M., Ertl, T., 2019. Visualization and Visual Analysis for Multiphase Flow, in: Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2019_pp25-27
- **Straub, A.**, Karch, G., Sadlo, F., Ertl, T., 2019. Implicit Visualization of 2D Vector Field Topology for Periodic Orbit Detection, in: Proceedings of TopoInVis, Nyköping, Sweden.
- **Zeifang, J.**, 2019. High order IMEX splitting methods for low Mach number flows, in: Proceedings of the 8th GACM Colloquium on Computational Mechanics for Young Scientists from Academia and Industry, Kassel, Germany.
- **Baggio, M.**, Weigand, B., 2018. On the implementation of Structured Surfaces to FS3D, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp37-39

- **Haide, R.**, Santini, M., **Fest-Santini, S.**, 2018. Experimental Investigations on Multi-Phase Systems with X-Ray Micro-CT, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp40
- Lamanna, G., Steinhausen, C., **Palmetshofer, P.**, Weigand, B., 2018. Influence of solubility effects and diffusion coefficient models on the drop vaporization rate ad high pressure, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp05-08
- **Reschke, W.**, Fasoulas, S., 2018. Gas-Kinetic Simulation of Microdroplet-Gas Interaction, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp48-51
- **Straub, A.**, Karch, G., Boblest, S., Kaufmann, J., Sadlo, F., Weigand, B., Ertl, T., 2018. Visual Analysis of Interface Deformation in Multiphase Flow, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp45-47
- **Varma Raja Kochanattu, G.**, Cossali, G., Tonini, S., 2018. Modelling of heat and mass transfer from spheroidal drops with general non-uniform Dirichlet boundary conditions, in: Proceedings of the DIPSI Workshop 2018: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2018_pp56-59
- Yang, G., **Vaikuntanathan, V.**, Terzis, A., Cheng, X., Weigand, B., Helmig, R., 2018. Impact of a linear array of hydrophilic and superhydrophobic spheres on a deep water pool. Presented at the 71st Annual Meeting of the APS Division of Fluid Dynamics Annual Meeting, Atlanta, USA.
- **Bernard, R.**, **Foltyn, P.**, **Geppert, A.**, Lamanna, G., Weigand, B., 2017. Generalized analysis of the deposition/splashing limit for one- and two-component droplet impacts upon thin films. Presented at the ILASS Europe, 28th Conference on Liquid Atomization & Spray Systems, Valencia, Spain.
- Karch, G., **Straub, A.**, Sadlo, F., Ertl, T., 2017. Implicit Visualization of Vector Field Topology. Presented at the TopoInVis, Tokyo, Japan.
- **Geppert, A.**, Terzis, A., Lamanna, G., Marengo, M., Weigand, B., 2016. On the Formation of Secondary Droplets from Crown Bottom Breakdown during Drop Impact on Very Thin Films. Presented at the ILASS Europe, 27th Conference on Liquid Atomization and Spray Systems, Brighton, UK.

- **Reschke, W.**, Binder, T., Kleinert, J., Mirza, A., Nizenkov, P., Pfeiffer, M., Fasoulas, S., Copplestone, S., Ortwein, P., Munz, C.D., 2016. Recent developments of DSMC within the reactive plasma flow solver PICLas. Presented at the 30TH International Symposium on rarefied gas dynamics: RGD 30, Victoria, BC, Canada, p. 130003. <https://doi.org/10.1063/1.4967629>

II. Technical and scientific reports

- Binder, T., Copplestone, S., Mirza, A., Nizenkov, P., Ortwein, P., Pfeiffer, M., **Reschke, W.**, Munz, C.-D., Fasoulas, S., 2018. Load balancing strategies for the DSMC simulation of hypersonic flows using HPC. <https://doi.org/10.48550/arXiv.1811.04742>
- Koch, T., Gläser, D., Weishaupt, K., **Ackermann, S.**, Beck, M., Becker, B., Burbulla, S., Class, H., Coltman, E., Fetzer, T., Flemisch, B., Grüninger, C., Heck, K., Hommel, J., Kurz, T., **Lipp, M.**, Mohammadi, F., Schneider, M., Seitz, G., Scholz, S., Weinhardt, F., 2018. Du-MuX 3.0.0. <https://doi.org/10.5281/ZENODO.2479595>
- **Foltyn, P.**, Roth, N., Weigand, B., inventor; University of Stuttgart, assignee. Verfahren zur Messung der Schichtdicke einer optisch transparenten Schicht, insbesondere einer Flüssigkeitsschicht. European Patent application 19 192 074.3. 2019 Aug 16.
- **Foltyn P.**, Weigand B., and Roth N., Measuring device and method of determining a depth of field of an optical setup. US Patent 12,078,473, 2024, EP 4105594 B1, 2025

III. Posters

- **Konan, H.**, Rohde, C., 2025. Hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. Presented at the NumHyp25, Darmstadt, Germany.
- **Regener Roig, D.**, 2025. Towards discontinuous Galerkin solution of multiphase flows. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Rusconi, D.**, 2025. Weak supervision: A solution for image Segmentation in Challenging Datasets. Presented at the IMAGING25 Conference, Varenna, Italy.
- **Saha, R.**, Inal, C., Weigand, B., 2025. DNS of ethanol droplet impact onto a smooth surface. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.

- **Dhar, R.**, Schulte, K., 2024. Investigation of Droplet-Pool Phenomena and Droplet-Film Interaction for Immiscible Fluids. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Klötzl, D.**, Weiskopf, D., 2024. NMF-Based Analysis of Droplet Wall-Film Interactions. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Konan, H.**, Rohde, C., 2024. Approximation of the Navier-Stokes-Cahn-Hilliard system for two-phase flows. Presented at the 6th Workshop on Compressible Multiphase Flow, Strasbourg, France.
- **Saha, R.**, Weigand, B., 2024. DNS of droplet impact onto heated surfaces. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Wurst, J.**, Schulte, K., 2024. Simulation framework for capillary driven two-phase flow using PLIC-based contact line modelling. Presented at the Coupled Free Flow-Porous Medium Systems Workshop, Stuttgart, Germany.
- **Wurst, J.**, Schulte, K., 2024. Simulation framework for drop impacts on structured surfaces using PLIC-based contact line modeling. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ferrari, D.**, Dumbser, M., 2023. An explicit finite volume scheme for a unified hyperbolic model for multi-phase continuum mechanics. Presented at the NumHyp23, Bordeaux, France.
- **Geppert, A.**, Weigand, B., 2023. Ground spreading dynamics during single droplet impact onto a free-standing cubic surface element. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Hu, Y.**, Cossali, G.E., Tonini, S., Weigand, B., Reuttsch, J., Frey, M., 2023. Evaporation of an oscillating droplet. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Klötzl, D.**, Ibach, M., Weiskopf, D., 2023. Relation between shear rate and structural parameter in non-Newtonian droplet oscillations via Jacobi sets and STL decomposition: a visual analysis. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Molina Vogelsang, P.**, **Schubert, S.**, Lamanna, G., 2023. An approach to investigate the 3D height evolution in the crown crater during early crown formation with the general defocusing particle tracking (GDPT) method. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.

- **Palmetshofer, P.**, Marz, T., Weigand, B., 2023. 360° contact angles of water droplets on micrometric ramps, pyramids and staggered cuboid surfaces at varying wettability. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Saha, R.**, Weigand, B., 2023. Numerical simulation of droplet impact onto structured surfaces. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Stober, J.**, Santini, M., Schulte, K., 2023. Experimental Observation of the Collision of a Crown with a thin Wall Film during Oblique Droplet Impact. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Palmetshofer, P.**, Foltyn, P., Geppert, A., Weigand, B., 2022. Dynamic contact angle measurements of droplets spreading on surfaces of varying wettability. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Stober, J.**, Schulte, K., 2022. Numerical Investigation of the Mechanisms of early Crown Formation at an oblique Drop Impact onto a Wall Film. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ackermann, S.**, Helmig, R., 2019. Upscaling of coupled free-flow and porous-media-flow processes. Presented at the InterPore Conference, Valencia, Spain.
- **Ackermann, S.**, Helmig, R., 2019. Upscaling of coupled free-flow and porous-media-flow processes. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ackermann, S.**, Ostrowski, L., 2019. A multi-scale approach for drop detachment in a coupled free-flow/porous-medium-flow system. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Baggio, M.**, Weigand, B., 2019. On the implementation of structured surfaces in FS3D. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Foltyn, P.**, Restle, F., Weigand, B., 2019. 360° Evaluation of Projected Contact Angles of Static Droplets on Structured Surfaces. Presented at the DIPSI Workshop Droplet Impact Phenomena and Spray Investigation, Bergamo, Italy.
- **Geppert, A.**, Bernard, R., Weigand, B., Lamanna, G., 2019. Drop impact on wetted walls: An analytical solution for modelling the crown spreading based on stagnation-point flow. Presented at the Fundamental Fluid Dynamics – Challenges in Inkjet Printing Workshop, Leiden, Netherlands.

- **Ostrowski, L.**, 2019. Compressible Droplet Impingement. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ostrowski, L., Massa, F.**, 2019. An incompressible-compressible approach for droplet impact, Proceedings of the DIPSI Workshop 2019. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- Schegk, G., **Foltyn, P.**, Li, W.-J., Terzis, A., Weigand, B., 2019. Oxygen and Hydrogen Plasma Activation Effects on Capillary Thermodynamics in Fibrous Materials. Presented at the 9th International Colloids Conference, Sitges, Spain.
- **Straub, A.**, Heinemann, M., Ertl, T., 2019. Visualization and Visual Analysis for Multiphase Flow. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ackermann, S.**, Helmig, R., 2018. Upscaling of coupled free-flow and porous-media-flow processes. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Baggio, M.**, Kieffer-Roth, C., Weigand, B., 2018. Implementation of structured Surfaces in FS3D. Presented at the 2nd International Conference on Simulation Technology, Stuttgart, Germany.
- **Baggio, M.**, Weigand, B., 2018. On the implementation of Structured Surfaces to FS3D. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Chiocchetti, S.**, Dumbser, M., 2018. High order schemes for a first order hyperbolic formulation of compressible flow with surface tension. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Haide, R.**, Santini, M., **Fest-Santini, S.**, 2018. Characterization of Porous Media by X-Ray Micro Computed Tomography. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ostrowski, L.**, 2018. Compressible Droplet Impingement. Presented at the Fall School on Hyperbolic Conservation Laws and Mathematical Fluid Dynamics, Würzburg, Germany.
- **Ostrowski, L.**, 2018. Numerical Upscaling for Compressible Phase Field Flow. Presented at the 2nd International Conference on Simulation Technology, Stuttgart, Germany.
- **Reschke, W.**, 2018. Gas-Kinetic Simulation of Microdroplet-Gas Interaction. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.

- **Straub, A.**, Boblest, S., Karch, G., Ertl, T., 2018. Visual Analysis of Interface Deformation in Multiphase Flow. Presented at the 2nd International Conference on Simulation Technology, Stuttgart, Germany.
- **Straub, A.**, Karch, G., Boblest, S., Kaufmann, J., Sadlo, F., Weigand, B., Ertl, T., 2018. Visual Analysis of Interface Deformation in Multiphase Flow. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Varma Raja Kochanattu, G.**, Tonini, S., Cossali, G.E., 2018. Modelling of heat and mass transfer from spheroidal drops with general non-uniform Dirichlet boundary conditions. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ackermann, S.**, Helmig, R., 2017. Upscaling of coupled free-flow and porous-media-flow processes. Presented at the InterPore Conference, Rotterdam, Netherlands.
- **Ackermann, S.**, Helmig, R., 2017. Upscaling of coupled free-flow and porous-media-flow processes. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Baggio, M.**, Weigand, B., 2017. Numerical computation for drop impact on textured surfaces. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Bernard, R.**, Lamanna, G., Roth, N., Tonini, S., 2017. Micro and Macro Drop Impact Dynamics with Miscible Liquids. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Bernard, R.**, Vaikuntanathan, V., Lamanna, G., Weigand, B., 2017. On the Crown Rim Expansion Kinematics during Droplet Impact on Wall-Films. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Müller, C.**, Munz, C.-D., 2017. Sharp interface methods for droplet simulation. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ostrowski, L.**, 2017. Compressible effects in droplet interaction with textured walls. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Ostrowski, L.**, 2017. Numerical Upscaling for Droplet Impact on rough Walls. Presented at the Multiscale Methods Summer School, Hasselt, Belgium.
- **Reschke, W.**, 2017. Gas-Kinetic Simulation of Micro-droplet-Gas Interaction. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.

- **Straub, A.**, Ertl, T., 2017. Visualization of Droplet Internal Flow. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Zeifang, J.**, Beck, A., 2017. Mathematical and numerical modelling of droplet dynamics in weakly compressible multi-component flows. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.

IV. Preprints

- **Ostrowski, L.**, Rohde, C. 2025. An energy-stable Discontinuous Galerkin method for the compressible Navier-Stokes-Allen-Cahn System. <https://arxiv.org/abs/2510.23289>
- **Reschke, W.**, Pfeiffer, M., Fasoulas, S., 2020. Modelling and Simulation of Heterogeneous Reactions with Statistical Particle Methods. <https://doi.org/10.48550/arXiv.2007.06339>
- **Appel, D.**, Keim, J., **Mossier, P.**, Oestinger, P., **Travnicek, A.**, Beck, A., n.d. Recent Contributions to the High-Order Numerical Simulation of Multiphase and Multispecies Flows, To Appear in: High Performance Computing in Science and Engineering '25: Transactions of the High Performance Computing Center, Stuttgart (HLRS). Springer Nature Switzerland, Cham.
- **Dhar, R.**, Gösele, D., Saumet, P., Weigand, B., Schulte, K., n.d. The Role of Interfacial Tension in Direct Numerical Simulations of Drop-Film Interaction for Immiscible Fluids, To Appear in: High Performance Computing in Science and Engineering '25: Transactions of the High Performance Computing Center, Stuttgart (HLRS). Springer Nature Switzerland, Cham.
- **Kumar, M.**, **Saha, R.**, Potyka, J., Schulte, K., Weigand, B., n.d. Direct Numerical Simulations of Droplet Impact onto Heated Surfaces using the Program Free Surface 3D (FS3D), To Appear in: High Performance Computing in Science and Engineering '24: Transactions of the High Performance Computing Center, Stuttgart (HLRS). Springer Nature Switzerland, Cham.

V. Science communication

- Breuer, R., 2023. Trickreiche Tropfen, Bild der Wissenschaft, 22-27.

5.1.2 Project results achieved by the doctoral and postdoctoral researchers (financed from other sources)

1. Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

- Botti, L., Di Pietro, D.A., **Massa, F.**, 2026. Hybrid High-order formulations with turbulence modelling capabilities for incompressible flow problems. *Comput. Fluids* 305, 106915. <https://doi.org/10.1016/j.compfluid.2025.106915>
- Botti, L., Botti, M., Di Pietro, D., **Massa, F.**, 2025. Stability, convergence, and pressure-robustness of numerical schemes for incompressible flows with hybrid velocity and pressure. *Math. Comp.* 95, 1–28. <https://doi.org/10.1090/mcom/4049>
- Ghidoni, A., **Massa, F.**, Noventa, G., 2025. Equal-loudness-adaptive time integration for the high-order solution of direct aeroacoustics simulations. *Comput. Fluids* 301, 106776. <https://doi.org/10.1016/j.compfluid.2025.106776>
- Colombo, A., Crivellini, A., Ghidoni, A., **Massa, F.**, Noventa, G., 2024. p -adaptive discontinuous Galerkin solution of transonic viscous flows with variable time step-size. *Comput. Fluids* 282, 106392. <https://doi.org/10.1016/j.compfluid.2024.106392>
- **Kunberger, V.**, Weigand, B., 2024. Binary Collisions of Supercooled Water Droplets. *Atomization Spray* 34. <https://doi.org/10.1615/AtomizSpr.2024052339>
- **Liu, Y.**, Chu, X., Yang, G., Weigand, B., 2024. Simulation and analytical modeling of high-speed droplet impact onto a surface. *Phys. Fluids* 36, 012137. <https://doi.org/10.1063/5.0186883>
- Baumann, A., **Hoch, D.**, Niessner, J., 2023. Fates of Emitted Particles Depending on Mask Wearing Using an Approach Validated Across Spatial Scales. *Global Challenges* 7, 2300008. <https://doi.org/10.1002/gch2.202300008>
- Berger, S., Azimian, M., Eichheimer, P., **Hoch, D.**, Cheng, L., Niessner, J., 2023. Computational Fluid Dynamics-Based Optimization of Face Mask Filter Media Using Micro-Computed Tomography Scans. *Chem. Eng. Technol.* 46, 1039–1046. <https://doi.org/10.1002/ceat.202200460>
- **Hoch, D.**, Weber, A., Niessner, J., 2023. Modeling and Simulation of Coalescence in the Context of Oil Mist Filtration. *Transp Porous Med* 148, 47–66. <https://doi.org/10.1007/s11242-023-01908-6>

- **Liu, Y.**, Chu, X., Wang, W., Weigand, B., 2023. Large-eddy simulation, convective instability, and modal causality of coaxial supersonic air–water jets considering a swirl effect. *Phys. Fluids* 35, 063316. <https://doi.org/10.1063/5.0149856>
- Bassi, F., Botti, L.A., Colombo, A., **Massa, F.**, 2022. Assessment of an Implicit Discontinuous Galerkin Solver for Incompressible Flow Problems with Variable Density. *Applied Sciences* 12, 11229. <https://doi.org/10.3390/app122111229>
- Botti, L., **Massa, F.**, 2022. HHO Methods for the Incompressible Navier-Stokes and the Incompressible Euler Equations. *J Sci Comput* 92, 28. <https://doi.org/10.1007/s10915-022-01864-1>
- Ghidoni, A., **Massa, F.**, Noventa, G., Rebay, S., 2022. Assessment of an adaptive time integration strategy for a high-order discretization of the unsteady RANS equations. *Int. J. Numer. Method Fluids* 94, 1923–1963. <https://doi.org/10.1002/fld.5131>
- **Hoch, D.**, Weber, A., Niessner, J., 2022. Modeling and simulation of coalescence in the context of oil mist filtration using a distance map. Presented at the FILTECH 2022 Conference, Cologne, Germany.
- Bonalumi, F., Crua, C., Savina, I.N., Davies, N., Habsteson, A., Santini, M., **Fest-Santini, S.**, Sandeman, S., 2021. Bioengineering a cryogel-derived bioartificial liver using particle image velocimetry defined fluid dynamics. *Mater Sci Eng C Mater Biol Appl* 123, 111983. <https://doi.org/10.1016/j.msec.2021.111983>
- **Fest-Santini, S.**, Steigerwald, J., Santini, M., Cossali, G.E., Weigand, B., 2021. Multiple drops impact onto a liquid film: Direct numerical simulation and experimental validation. *Comput. Fluids* 214, 104761. <https://doi.org/10.1016/j.compfluid.2020.104761>
- Baumann, A., **Hoch, D.**, Behringer, J., Niessner, J., 2020. Macro-scale modeling and simulation of two-phase flow in fibrous liquid aerosol filters. *Engineering Applications of Computational Fluid Mechanics* 14, 1325–1336. <https://doi.org/10.1080/19942060.2020.1828174>
- **Fest-Santini, S.**, 2020. Image Processing of Two-Phase Data for Drop-Surface Interaction Obtained by X-Ray Microtomography, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), *Droplet Interactions and Spray Processes*. Springer International Publishing, Cham, pp. 101–111. https://doi.org/10.1007/978-3-030-33338-6_8
- **Hoch, D.**, Azimian, M., Baumann, A., Behringer, J., Niessner, J., 2020. Comparison of Voxel-Based and Mesh-Based CFD Models for Aerosol Deposition on Complex Fibrous Filters. *Chem. Eng. Technol.* 43, 2538–2547. <https://doi.org/10.1002/ceat.202000318>
- **Lipp, M.**, Helmig, R., 2020. A Locally-Refined Locally-Conservative Quadtree Finite-Volume Staggered-Grid Scheme, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.),

Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 149–159. https://doi.org/10.1007/978-3-030-33338-6_12

- **Massa, F.**, Bassi, F., Botti, L., Colombo, A., 2020. An Implicit High-Order Discontinuous Galerkin Approach for Variable Density Incompressible Flows, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 191–202. https://doi.org/10.1007/978-3-030-33338-6_15
- Noventa, G., **Massa, F.**, Rebay, S., Bassi, F., Ghidoni, A., 2020. Robustness and efficiency of an implicit time-adaptive discontinuous Galerkin solver for unsteady flows. Comput. Fluids 204, 104529. <https://doi.org/10.1016/j.compfluid.2020.104529>
- **Schlottke, K.**, Reuttsch, J., Kieffer-Roth, C., Weigand, B., 2020. Direct Numerical Simulations of Evaporating Droplets at Higher Temperatures: Application of a Consistent Numerical Approach, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 287–299. https://doi.org/10.1007/978-3-030-33338-6_23
- Bassi, F., Botti, L., Colombo, A., Ghidoni, A., **Massa, F.**, Noventa, G., 2019. On the Development of an Implicit High-Order Discontinuous Galerkin Solver for a Hybrid RANS-LES Model, in: Salvetti, M.V., Armenio, V., Fröhlich, J., Geurts, B.J., Kuerten, H. (Eds.), Direct and Large-Eddy Simulation XI. Springer International Publishing, Cham, pp. 75–82. https://doi.org/10.1007/978-3-030-04915-7_11
- Guilizzoni, M., Santini, M., **Fest-Santini, S.**, 2019. Synchronized Multiple Drop Impacts into a Deep Pool. Fluids 4, 141. <https://doi.org/10.3390/fluids4030141>
- Marzorati, S., Goglio, A., **Fest-Santini, S.**, Mombelli, D., Villa, F., Cristiani, P., Schievano, A., 2019. Air-breathing bio-cathodes based on electro-active biochar from pyrolysis of Giant Cane stalks. Int. J. Hydrogen Energy 44, 4496–4507. <https://doi.org/10.1016/j.ijhydene.2018.07.167>
- Bassi, F., **Massa, F.**, Botti, L., Colombo, A., 2018. Artificial compressibility Godunov fluxes for variable density incompressible flows. Comput. Fluids 169, 186–200. <https://doi.org/10.1016/j.compfluid.2017.09.010>
- **Massa, F.**, Noventa, G., Lorini, M., Bassi, F., Ghidoni, A., 2018. High-order linearly implicit two-step peer schemes for the discontinuous Galerkin solution of the incompressible Navier–Stokes equations. Comput. Fluids 162, 55–71. <https://doi.org/10.1016/j.compfluid.2017.12.003>

- Fraundorf, P., Osborn, D., **Lipp, M.**, 2017. Some novel uses for three-dimensional data from SPM and stereo SEM. *Microsc Microanal* 23, 1178–1179. <https://doi.org/10.1017/S1431927617006559>
- **Lipp, M.**, Savage, T.J., Osborn, D., Fraundorf, P., 2017. Laboratory evidence of slow-cooling for carbon droplets in red-giant atmospheres. *Microsc Microanal* 23, 2192–2193. <https://doi.org/10.1017/S143192761701162X>
- Santini, M., **Fest-Santini, S.**, Cossali, G.E., 2017. Experimental study of vortices and cavities from single and double drop impacts onto deep pools. *Eur. J. Mech. B Fluids* 62, 21–31. <https://doi.org/10.1016/j.euromechflu.2016.11.009>
- Santini, M., Guilizzoni, M., **Fest-Santini, S.**, Lorenzi, M., 2017. Characterization of highly hydrophobic textiles by means of X-ray microtomography, wettability analysis and drop impact. *J. Phys.: Conf. Ser.* 923, 012013. <https://doi.org/10.1088/1742-6596/923/1/012013>
- Santini, M., Marzorati, S., **Fest-Santini, S.**, Trasatti, S., Cristiani, P., 2017. Carbonate scale deactivating the biocathode in a microbial fuel cell. *Journal of Power Sources* 356, 400–407. <https://doi.org/10.1016/j.jpowsour.2017.02.088>
- Bassi, F., Botti, L., Colombo, A., Crivellini, A., Ghidoni, A., **Massa, F.**, 2016. On the development of an implicit high-order Discontinuous Galerkin method for DNS and implicit LES of turbulent flows. *Eur. J. Mech. B Fluids* 55, 367–379. <https://doi.org/10.1016/j.euromechflu.2015.08.010>
- Fraundorf, P., **Lipp, M.**, Savage, T., 2016. Analogs for Unlayered-Graphene Droplet-Formation in Stellar Atmospheres. *Microsc Microanal* 22, 1816–1817. <https://doi.org/10.1017/S1431927616009922>
- Noventa, G., **Massa, F.**, Bassi, F., Colombo, A., Franchina, N., Ghidoni, A., 2016. A high-order Discontinuous Galerkin solver for unsteady incompressible turbulent flows. *Comput. Fluids* 139, 248–260. <https://doi.org/10.1016/j.compfluid.2016.03.007>
- Bassi, F., Botti, L., Colombo, A., Ghidoni, A., **Massa, F.**, 2015. Linearly implicit Rosenbrock-type Runge–Kutta schemes applied to the Discontinuous Galerkin solution of compressible and incompressible unsteady flows. *Comput. Fluids* 118, 305–320. <https://doi.org/10.1016/j.compfluid.2015.06.007>

2. Category B – Any other form of published results

I. **Proceedings** *(Joint publications are marked in blue)*

- Gösele, D., **Wurst, J.**, Fricke, M., Schulte, K., 2024. Geometrical Realization of Contact Angle Hysteresis for Multiphase DNS using PLIC. Presented at the 5th International Conference on Numerical Methods in Multiphase Flows (ICNMMF-5), Reykjavík, Iceland.
- **Kunberger, V.**, Weigand, B., 2024. The influence of heat transfer on the outcome of droplet collisions of supercooled water droplet, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- **Wurst, J.**, Dreisbach, M., Stroh, A., Kriegseis, J., Schulte, K., 2024. Simulation framework for drop impact on micro-grooves using PLIC-based contact line modelling. Presented at the 5th International Conference on Numerical Methods in Multiphase Flows (ICNMMF-5), Reykjavík, Iceland.
- **Wurst, J.**, Schulte, K., 2024. Simulation framework for drop impacts on structured surfaces using PLIC-based contact line modelling, in: Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. <https://doi.org/10.13122/DIPSI2024>
- Bittel, L., Jüngling, N., **Hoch, D.**, Niessner, J., 2023. Generation of datasets using POD for artificial intelligence to predict and control an energy-efficient HVAC system. Presented at the Proceedings of the FILTECH Conference, Cologne, Germany.
- **Kunberger, V.**, Weigand, B., 2023. Binary Droplet Collisions of Supercooled Water Droplets. Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- **Kunberger, V.**, Weigand, B., 2023. Phase Change and Interaction Behaviour of Supercooled Water Droplets, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_11
- Baumann, A., **Hoch, D.**, Niessner, J., 2022. Investigation of the near-field and far-field protection effect of face masks using micro-scale and macro-scale CFD simulations. Presented at the FILTECH 2022 Conference, Cologne, Germany.
- **Hoch, D.**, Azimian, M., Baumann, A., Behringer, J., Niessner, J., 2022. Comparing mesh-based and voxel-based CFD models for aerosol deposition in complex fibrous filters. Presented at the FILTECH 2022 Conference, Cologne, Germany.

- **Kunberger, V.**, Weigand, B., 2022. Interactions of Supercooled Water Droplets on a Surface. Presented at the ILASS Europe, 31st Conference on Liquid Atomization & Spray Systems, virtual.
- Baumann, A., **Hoch, D.**, Niessner, J., 2019. Macro-Scale Simulation of Fibrous Liquid Aerosol Filters. Presented at the FILTECH 2019 Conference, Cologne, Germany.
- **Hoch, D.**, Baumann, A., Niessner, J., 2019. Micro-Scale Simulation of Fibrous Liquid Aerosol Filters. Presented at the FILTECH 2019 Conference, Cologne, Germany.
- **Hoch, D.**, Baumann, A., Niessner, J., 2019. Modeling and simulation of liquid aerosol deposition in textile filters. Presented at the Sixth bwHPC Symposium, September, Karlsruhe, Germany.
- **Schlottke, K.**, Weigand, B., 2019. The influence of the near droplet neighbourhood on droplet evaporation, in: Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.6092/DIPSI2019_pp22-24
- **Schlottke, K.**, Meister, C., Karch, G., **Geppert, A.**, Sadlo, F., Ertl, T., Weigand, B., 2016. Direct Numerical Simulation of Multi-Component Droplet Wall Film Interaction. Presented at the 9th International Conference on Multiphase Flow, Florence, Italy.

II. Technical and scientific reports

- Fraundorf, P., **Lipp, M.**, 2016. Molar standards & information units in the 'new-SI.' <https://api.semanticscholar.org/CorpusID:104094372>

III. Posters

- **Wendt, F.**, 2025. Effective equations for a compressible liquid-vapor flow with highly oscillating initial density. Presented at the 7th Workshop on Compressible Multiphase Flow, Strasbourg, France.
- **Kunberger, V.**, Weigand, B., 2024. The influence of heat transfer on the outcome of droplet collisions of supercooled water droplets. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Wendt, F.**, 2024. Derivation of a Baer-Nunziato mixture model for fluids with internal capillarity in 1D. Presented at the Analysis of Dissipation in Inviscid and Compressible Fluid Dynamics, Konstanz, Germany.

- **Wendt, F.**, 2024. Derivation of a Baer-Nunziato mixture model for fluids with internal capillarity in 1D. Presented at the 6th Workshop on Compressible Multiphase Flow, Strasbourg, France.
- **Wurst, J.**, Schulte, K., 2024. Simulation framework for capillary driven two-phase flow using PLIC-based contact line modelling. Presented at the Coupled Free Flow-Porous Medium Systems Workshop, Stuttgart, Germany.
- **Wurst, J.**, Schulte, K., 2024. Simulation framework for drop impacts on structured surfaces using PLIC-based contact line modeling. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Kunberger, V.**, Weigand, B., 2023. Phase Change and Interaction Behaviour of Supercooled Water Droplets. Presented at the DIPSI Workshop Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Hoch, D.**, 2019. Micro-scale simulation of fibrous liquid aerosol filters. Presented at the InterPore Conference, Valencia, Spain.
- **Lipp, M.**, 2019. A locally-refined quadtree finite-volume staggered-grid scheme. Presented at the International Congress on Industrial and Applied Mathematics (ICIAM), Valencia, Spain.
- **Schlottke, K.**, Weigand, B., 2019. Investigating the near droplet neighborhood using FS3D. Presented at the DIPSI Workshop, Droplet Impact Phenomena & Spray Investigations, Bergamo, Italy.
- **Lipp, M.**, 2018. Adaptive Staggered 2D Grids for DuMuX - Plans/Ideas. Presented at the Summer School on Domain Decomposition Methods, Nice, France.
- **Lipp, M.**, 2018. Adaptive Staggered 2D Grids for DuMuX - Plans/Ideas. Presented at the 2nd International Conference on Simulation Technology, Stuttgart, Germany.
- **Schlottke, K.**, Reuttsch, J., Weigand, B., 2018. DNS of Evaporating Droplets under Various Conditions. Presented at the 2nd International Conference on Simulation Technology, Stuttgart, Germany.

IV. Preprints

- Rohde, C., **Wendt, F.**, 2025. Mathematical Justification of a Baer-Nunziato Model for a Compressible Viscous Fluid with Phase Transition. <https://doi.org/10.48550/arXiv.2504.10161>

5.2 Project results generated by the Research Training Group

Publications with scientific quality assurance (joint publications are marked blue)

- Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), 2020. Droplet Interactions and Spray Processes, Fluid Mechanics and Its Applications. Springer International Publishing, Cham. <https://doi.org/10.1007/978-3-030-33338-6>
- **Palmetshofer, P., Saha, R., Bryce, J., Geppert, A., Cossali, G.E., Weigand, B., 2025.** Droplet impact onto sub-millimeter microstructures fully and barely submerged in a liquid film. Phys. Fluids 37, 082130. <https://doi.org/10.1063/5.0278343>
- **Stober, J., Santini, M., Schulte, K., 2025.** Characterization of splashing and regime thresholds for oblique droplet impact on thin wall films. Exp. Therm. Fluid Sci. 167, 111493. <https://doi.org/10.1016/j.expthermflusci.2025.111493>
- **Mossier, P., Jöns, S., Chiocchetti, S., Beck, A.D., Munz, C.-D., 2025.** Numerical simulation of phase transition with the hyperbolic Godunov-Peshkov-Romenski model. J. Comput. Phys. 520, 113514. <https://doi.org/10.1016/j.jcp.2024.113514>
- **Hu, Y., Tonini, S., Reuttsch, J., Frey, M., Cossali, G.E., Weigand, B., 2024.** Direct numerical simulation of an oscillating droplet under evaporating conditions. Int. J. Heat Mass Transf. 232, 125891. <https://doi.org/10.1016/j.ijheatmasstransfer.2024.125891>
- **Tietz, R., Fasoulas, S., Pfeiffer, M., 2024.** Symmetric simulations of droplets with a particle based Vlasov-Enskog-Solver. Presented at the ICASF, 2nd International Conference on advanced earth science and foundation engineering, Mohali, India, p. 120002. <https://doi.org/10.1063/5.0187426>
- **Geppert, A., Tonini, S., Santini, M., Fest-Santini, S., Lamanna, G., Cossali, G.E., Weigand, B., 2024.** Recent results of the International Research Training Group GRK2160/2, in: Proceedings of 16th International Conference on Liquid Atomization and Spray Systems (ICLASS). Shanghai, China.
- **Veyskarami, M., Bringedal, C., Helmig, R., 2024.** Modeling and Analysis of Droplet Evaporation at the Interface of a Coupled Free-Flow–Porous Medium System. Transp Porous Med 151, 2745–2775. <https://doi.org/10.1007/s11242-024-02123-7>
- **Geppert, A., Stober, J., Steigerwald, J., Schulte, K., Tonini, S., Lamanna, G., 2024.** Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. Phys. Fluids 36, 012005. <https://doi.org/10.1063/5.0179716>
- **Mossier, P., Appel, D., Beck, A.D., Munz, C.-D., 2023.** An Efficient hp-Adaptive Strategy for a Level-Set Ghost-Fluid Method. J Sci Comput 97, 50. <https://doi.org/10.1007/s10915-023-02363-7>

- **Ackermann, S., Fest-Santini, S., Veyskarami, M.,** Helmig, R., Santini, M., 2023. Experimental validation of a coupling concept for drop formation and growth onto porous materials by high-resolution X-ray imaging technique. *Int J Multiphas Flow* 160, 104371. <https://doi.org/10.1016/j.ijmultiphaseflow.2022.104371>
- **Verzeroli, L., Massa, F.,** Botti, L., 2023. A HDG formulation for nonlinear elasticity problems featuring finite deformations and frictionless contact constraints. *Finite Elements in Analysis and Design* 215, 103887. <https://doi.org/10.1016/j.finel.2022.103887>
- **Massa, F., Ostrowski, L.,** Bassi, F., Rohde, C., 2022. An artificial Equation of State based Riemann solver for a discontinuous Galerkin discretization of the incompressible Navier–Stokes equations. *J. Comput. Phys.* 448, 110705. <https://doi.org/10.1016/j.jcp.2021.110705>
- **Klötzl, D.,** Krake, T., Zhou, Y., Hotz, I., Wang, B., Weiskopf, D., 2022. Local bilinear computation of Jacobi sets. *Vis Computing* 38, 3435–3448. <https://doi.org/10.1007/s00371-022-02557-4>
- **Straub, A.,** Boblest, S., Karch, G., Sadlo, F., Ertl, T., 2022. Droplet-Local Line Integration for Multiphase Flow, in: 2022 IEEE Visualization and Visual Analytics (VIS). IEEE, Oklahoma City, OK, USA, pp. 135–139. <https://doi.org/10.1109/VIS54862.2022.00036>
- **Chiocchetti, S.,** Peshkov, I., Gavrilyuk, S., Dumbser, M., 2021. High order ADER schemes and GLM curl cleaning for a first order hyperbolic formulation of compressible flow with surface tension. *J. Comput. Phys.* 426, 109898. <https://doi.org/10.1016/j.jcp.2020.109898>
- **Ren, W., Foltyn, P., Geppert, A.,** Weigand, B., 2021. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. *Sci Rep* 11, 18018. <https://doi.org/10.1038/s41598-021-97376-3>
- **Ostrowski, L.,** Rohde, C., 2020. Compressible multicomponent flow in porous media with Maxwell-Stefan diffusion. *Math Meth Appl Sci* 43, 4200–4221. <https://doi.org/10.1002/mma.6185>
- **Föll, F., Hitz, T., Müller, C.,** Munz, C.-D., Dumbser, M., 2019. On the use of tabulated equations of state for multi-phase simulations in the homogeneous equilibrium limit. *Shock Waves* 29, 769–793. <https://doi.org/10.1007/s00193-019-00896-1>
- Bassi, F., **Massa, F.,** Botti, L., Colombo, A., 2018. Artificial compressibility Godunov fluxes for variable density incompressible flows. *Comput. Fluids* 169, 186–200. <https://doi.org/10.1016/j.compfluid.2017.09.010>

6 Further Information

6.1 Information on how the project progressed, its structural impact and future prospects

Notes and recommendations arising from the review of the renewal proposal

At the review of the renewal proposal on 20.07.2020 and 21.07.2020, which took place online because of the Corona Pandemic, the following notes and recommendations have been given by the reviewers. Because of the fact that the results of the review of the renewal proposal are given in German, in the following these notes are first listed in German and then have been translated into English by using deepl (<https://www.deepl.com/de/translator>):

Comment 1 (German):

„Die gemeinsamen Publikationen, an denen Wissenschaftlerinnen und Wissenschaftler aus unterschiedlichen Gruppen sowohl aus Deutschland als auch Italien beteiligt sind, sind Beweis für die gelungene Kooperation. Weitere Veröffentlichungen, worin vor allem die Ergebnisse numerischer Simulationen und Experimente zusammengebracht und publiziert werden, werden mit Freude erwartet.“

Comment 1 (English, translated):

The joint publications, involving scientists from different groups in both Germany and Italy, are proof of the successful cooperation. Further publications, primarily bringing together and publishing the results of numerical simulations and experiments, are eagerly awaited.”

Answer to Comment 1:

The project has produced numerous publications across projects, universities and methodologies, including joint publications of experimental and numerical results. By using high fidelity numerical simulations, it was possible to explain experimental results and to achieve large progress in the explanation of unsolved phenomena, e.g., for the non-circular spreading or for splashing phenomena of droplets impacting under an angle onto a film. More details are given in the section on Research Progress, where three examples have been selected in such a way as to reflect the benefit in using simulations and experiments together. Furthermore, we also published results showing the benefit in using together numerical simulations and analytical work. This is also shown in the section on the Research Progress. Furthermore, the reader is referred to **Chapter 5**, showing the complete list of publications.

Comment 2 (German):

„Vielversprechende Nachwuchswissenschaftlerinnen und -wissenschaftler, können sich zur Unterstützung ihrer Karriere nach Abschluss ihrer Promotion um eine Anschubförderung bewerben. Die Entscheidung über die Vergabe trifft das advisory board des internationalen Graduiertenkollegs. Es wird angeregt, diese Mittel in Zukunft noch transparenter und nachvollziehbarer in einem kompetitiven Bewerbungsprozess zu vergeben.“

Comments 2 (English, translated):

Promising young scientists can apply for start-up funding to support their careers after completing their doctorates. The decision on the award is made by the advisory board of the international research training group. It is suggested that these funds be allocated in an even more transparent and comprehensible manner in a competitive application process in the future.

Answer to Comment 2:

Within FP2 one start-up grant was given in 2022. The position was announced in our International Scientific Network. The applications have been evaluated by the Speaker of DROPIT and by the PI where the applicant would fit best. The final decision on the grant was then made by the Advisory Board, where the position and the applications have been presented. We found an excellent candidate (Simone Chiochetti) to whom the start-up grant was given. He is now employed at the University of Cologne as a Marie Curie Postdoctoral Fellow.

Comment 3 (German):

„Nichtsdestoweniger könnten die Promovierenden interkulturell von längeren Austauschen nur profitieren. Daher regen die Gutachtenden an, die Aufenthalte an den Partnerinstitutionen für insgesamt mindestens sechs Monate verpflichtend zu machen. Eine Verlängerung der Aufenthalte beim Partner ist auch für das angestrebte Ziel der Antragstellenden, einen gemeinsamen Abschluss (cotutelle) der letzten Kohorte zu erreichen, unabdingbar.“

Comment 3 (English, translated):

"Nevertheless, doctoral students could only benefit interculturally from longer exchanges. The reviewers therefore suggest making stays at partner institutions mandatory for a total of at least six months. Extending the stays at the partner institution is also essential for achieving the applicants' goal of obtaining a joint degree (cotutelle) for the last cohort."

Answer to Comment 3:

This comment addressed the third generation of doctoral researchers. The second generation was within the time of the Coronavirus Pandemic (11.3.2020 – end of 2022) and no longer research stays were possible.

For the third cohort of doctoral researchers, we have implemented the recommendation as follows: During the interview process, all doctoral researchers were informed that they would be required to spend several months at the partner university, and the associated opportunities for personal development were highlighted. This was repeatedly brought up during every annual doctoral seminar as well as in one-to-one meetings with their supervisors.

This approach resulted in longer research stays. Rishav Saha (B05), Patrick Palmetshofer (B01) and Jonathan Stober (C02) each spent a few months in Bergamo. Anna Mareike Kosteletzky (B06) went for a research stay of roughly 2,5 months to the NTNU Trondheim, because of their outstanding expertise in porous media thermodynamics. Unfortunately, one of the doctoral researchers of the third generation dropped out (Claudia Marianowski (A03)) before she could take a longer research stay. She moved to industry, because she had a very good offer. Furthermore, we could not fill one position (for subproject C04 - Visualization) because here the possible applicants preferred to go to industry because of the huge difference in salary in Informatics. Rishi Dhar (A02) and Kim-Sophie Ellenberger (A03) started late in DROPIT (on 01.02.2024 and 01.11.2024), so that they could not go for a longer stay abroad so far. The doctoral researcher Hasel Cicek Konan (B04) did her Master-thesis in Italy before joining DROPIT. Therefore, she decided to go only for shorter visits to Bergamo. However, it is planned that she goes to Bergamo for a longer research stay either at the end of 2026 or at the beginning of 2027. This will be financed by our own funds. Unfortunately, two of the doctoral students (Pablo Molina Vogelsang (C01), and Amalia Travnicek (A06)) were only able to attend for a few short stays. This was understandable due to personal/family reasons.

Because the PIs of DROPIT still have the opinion that it would be important to bring at least some of the doctoral researchers of the third generation to a joint degree (cotutelle), we will send Rishav Saha (B05) this year from May to August for further three months to Bergamo. The required document for his joint degree (cotutelle) is already set up. This will be financed by our own funds.

Finally, we would like to note that the PIs of DROPIT have made a real effort in this regard, without however being entirely successful. However, we have all noticed a massive change in the attitude of doctoral researchers since the beginning of the Coronavirus Pandemic. There has been very intensive collaboration, with this collaboration shifting heavily (since the beginning of the Coronavirus Pandemic) towards online meetings.

Research Progress

Droplet interaction technologies find applications in a large number of technological and industrial processes. These include spray cooling in the food and chemical industry, spray drying absorption for waste and pollutants treatment in process engineering, droplet collisions for the generation of powders and encapsulated materials in the food and pharmaceutical industry, drop evaporation and

droplet-wall interaction in propulsion systems of airplanes and rockets as well as in medical equipment like inhalers. All these applications have the following two features in common:

- The non-linear coupling between flow dynamics and transport processes
- The presence of multiple length scales, whereby small-scale fluid dynamics has a huge impact on the large-scale flow pattern

At the start of the first funding period (FP1) the consequences of the presence of different length scales on macroscopic properties and the associated amplification of surface transport were largely unexplored, being limited mainly to the formulation of empirical correlations based solely on macroscopic observations. Here DROPIT proposed for the first time a **systematic study for investigating the interdependencies between small-scale and large-scale dynamics** in the field of droplet interaction technologies. Due to the complexity of the problem, the analysis of such micro/macro interactions is not limited to one single aspect. Thus, an integrated approach had been chosen that evolves along three parallel pathways, namely a numerical, experimental and theoretical approach. **Numerically**, different **computational schemes** have been developed and compared, in order to assess their **capabilities in resolving accurately small-scale flow dynamics and transport**. The focus to the class of Discontinuous Galerkin schemes, which allow a high flexibility in local resolution, led to the development of novel numerical tools in FP1. The use of different approaches of interface tracking, sharp and diffuse interfaces, with incompressible and compressible bulk flow and their mutual comparison and validation stands at the forefront of computational multi-fluid dynamics. **Experimentally**, flow patterns and surface characterisation did include both large-scale and small-scale features in FP1. While standard optical techniques can be employed to visualize large-scale motion, **novel approaches were required for the characterisation of the small-scale patterns**. Here X-ray micro computed tomography (micro-CT; Univ. Bergamo) and scanning electron microscopy (SEM; Univ. Stuttgart, University of Karlsruhe (KIT)) were used for the characterization of different surfaces. Micro-PIV (which has been further developed in FP1 into Microscale Particle Tracking Velocimetry, micro-PTV) and a completely new developed measurement technique for film thicknesses and the evolution of thin films, **LASER Pattern Shift Method (LPSM)**, have been employed in FP1. At a **theoretical level**, the focus of the research activities was on the **coupling of length scales**, i.e. on identifying the mechanisms coupling small-scales at the interface with large scales in the flow. Depending upon the specific problem addressed, different tools have been employed (e.g. analytical methods, Monte Carlo methods, Direct Numerical Simulations, etc.). This **synergetic integration of different investigation methods** was aimed at achieving the following **scientific objectives**:

- to understand the physical mechanisms through which small-scale features lead to an amplification or quenching of bulk-transport properties (e.g. drag reduction, heat transfer enhancement or quenched vaporisation rate)

- to translate this understanding into sound physical models, while keeping the complexity of these models as low as possible to make them applicable for technical applications
- to develop more realistic computational models and boundary conditions for two-phase simulations in porous media and on micro-structured surfaces.

The above-mentioned integral approach for investigating the phenomena along the three parallel pathways, namely an experimental, theoretical and numerical line, which had already been defined in the first funding period (FP1) of the IRTG (01.10.2016-31.03.2021), was very successful. It led not only to new and very interesting scientific findings but also to the development of innovative numerical and experimental methods.

Since the start of the IRTG, the research programme of DROPIT is subdivided into three Thematic Areas (TA), depending upon the type of interface under investigation. These are:

- TA-A: Drop-Gas Interaction
- TA-B: Drop-Wall Interaction
- TA-C: Drop-Liquid Interaction

This structure enabled us to tailor directly the development of theoretical, numerical and experimental methods to the specificity of the different interfaces. Furthermore, due to the large variety of technical applications, the research activities in each thematic area have been restricted to a number of selected applications. As described in our initial research proposal, this was a necessary step, in order to comply with the overall goal of developing accurate predictive tools suited for engineering applications. The selected applications are:

- 1. TA-A: Optimisation of evaporation models for sprays:** The objective here was to improve the accuracy of current evaporation models by including the effect of small-scale processes on the global, drop-vaporisation rate. This entails non-equilibrium, thermodynamic effects at the drop interface, local curvature effects and finally multi-component effects.
- 2. TA-B: Analysis of multiple, surface length-scales and their impact on transport efficiency:** In TA-B the focus was to understand and model how interfacial-driven transport phenomena affect bulk-transport (e.g. momentum or energy transfer) on the macroscopic scale. This is of great importance e.g. for drop impacts on structured surfaces, which are used in a large variety of technical applications.
- 3. TA-C: Enhanced break-up in miscible binary mixtures:** There is a strong interest in the utilisation of bio-fuels and synthetic fuels in aviation and automotive applications for obtaining lower emissions. This led to an increased interest in drop collisions (on films or drops) with mixture droplets. The objective here was to understand how miscibility-gaps and thin-films fluid dynamics

influence the large-scale impact dynamics and resulting drop size distribution in liquid propulsion systems.

In all thematic areas, the research activities concentrated on the study and coupling of physical processes at different length scales both at numerical and experimental level, as schematically illustrated in Fig. 3. Furthermore, they are characterised by a multidisciplinary framework that foresees a strong link between the different areas like **applied mathematics, physics, numerical methods, experimental techniques** and **visualization**.

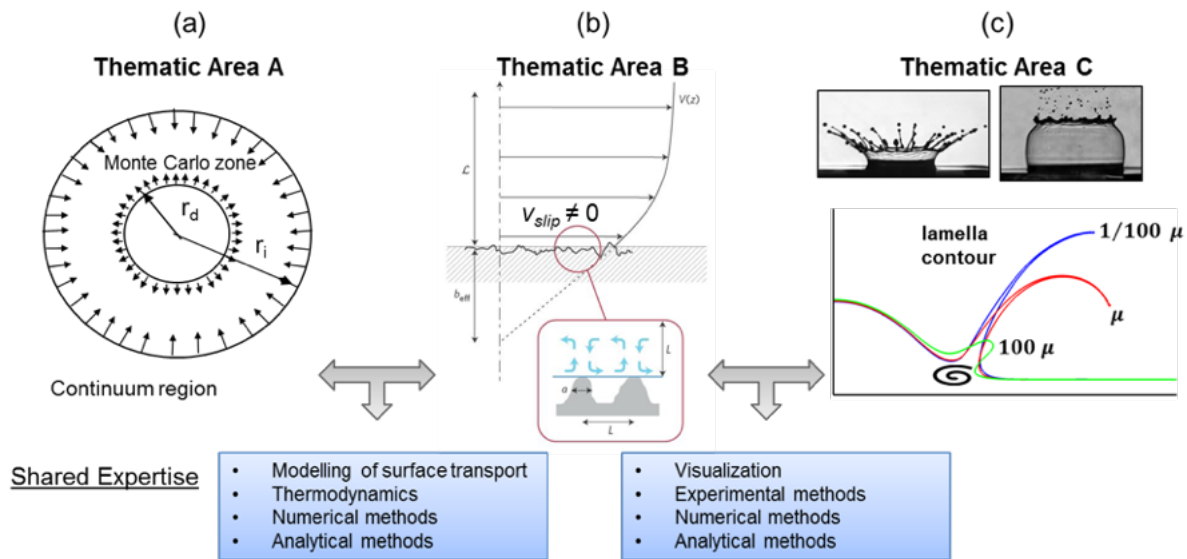


Fig. 3: Shared competences among the different thematic areas and schematic representation of multiple length scales coupling. (a) Gas-kinetic effects at the drop interface affect the mass and energy transport at macroscopic level (continuum); (b) Micro-scale flow patterns result in a significant increase of the macroscopic slip velocity (i.e. reduced momentum transport at macroscopic level, hence drag reduction); (c) micro-scale effects determine the shape of the splashing crown.

The very successful work of the **DROPIT** team within FP1 was summarized in 2020 in a book in the Scopus-listed book series “Fluid Mechanics and its Application”. It summarizes selected scientific findings: **G. Lamanna, S. Tonini, G.E. Cossali, B. Weigand: Droplet Interactions and Spray Processes. In Fluid Mechanics and its Applications 121, Springer, Heidelberg, Berlin, ISBN 978-3-030-33337-9, 2020.**

In the second funding period (FP2) the project followed the same structure as in FP1, enlarging the scientific scope to extended physical phenomena (e.g., heat transfer) and to multiple droplets.

In the following we will show some selected scientific research findings from FP2 in order to demonstrate progress and cooperation in the subprojects. The reader is referred to the individual reports of the second funding period (see **Chapter 6.2.8**) for a detailed description of the results achieved in

FP2. Furthermore, the progress of the project has continuously been reported on the important conferences for this subject (ILASS and ICLASS). Here, besides numerous presentations of the results of the subprojects, overview/progress presentations and posters were given:

- Lamanna, G., Tonini, S., Weigand, B., Cossali, G.E., 2028. Selected Results of the International Research Training Group (GRK 2160/1) "Droplet Interaction Technologies" (DROPIT), in: Proceedings of 14th International Conference on Liquid Atomization and Spray Systems (ICLASS), Chicago, IL, USA.
- Cossali, G.E., Weigand, B., Lamanna, G., Tonini, S., 2019. The International Research Training Group "Droplet Interaction Technologies" (DROPIT): Selected Results. Presented at the ILASS Europe, 29th Conference on Liquid Atomization & Spray Systems, Paris, France.
- Weigand, B., Lamanna, G., Tonini, S., Cossali, G.E., 2021. "Droplet Interaction Technologies" (DROPIT): Selected Results of the International Research Training Group GRK 2160, in: Proceedings of 15th International Conference on Liquid Atomization and Spray Systems (ICLASS), Edinburgh.
- **Geppert, A.**, Santini, M., Lamanna, G., Cossali, G.E., Weigand, B., 2022. Scientific Highlights of the International Research Training Group (GRK 2160/2) „Droplet Interaction Technologies" (DROPIT). Presented at the ILASS Europe, 31st Conference on Liquid Atomization and Spray Systems, virtual.
- Weigand, B., **Geppert, A.**, Lamanna, G., Tonini, S., Cossali, G.E., 2023. Selected Results of the International Research Training Group GRK 2160 on "Droplet Interaction Technologies" (DROPIT). Presented at the ILASS Europe, 32nd Conference on Liquid Atomization & Spray Systems, Naples, Italy.
- **Geppert, A.**, Tonini, S., Santini, M., **Fest-Santini, S.**, Lamanna, G., Cossali, G.E., Weigand, B., 2024. Recent results of the International Research Training Group GRK2160/2, in: Proceedings of 16th International Conference on Liquid Atomization and Spray Systems (ICLASS). Shanghai, China.
- Weigand, B., Lamanna, G., **Kumar, M.**, Tonini, S., Cossali, G.E., 2025. Project Overview and Selected Results of the International Research Training Group GRK 2160. Presented at the ILASS Europe, 33rd Conference on Liquid Atomization & Spray Systems, Lund, Sweden.

Selected Research Results of the three Thematic Areas

In the following, some selected research results of the three thematic areas are exemplified by discussing research results of one subproject for each thematic area in more detail.

TA-A: Drop-Gas Interaction

In this thematic area there were six subprojects planned and described in the proposal for FP2. One additional subproject was added to TA-A when Prof. A. Colombo (University of Bergamo) joined DROPIT on 1.10.2023 as a principal investigator (before he was as associated member). Thus, the new subproject: **SP-AX (Colombo, Beck): Entropy stable, high-order numerical simulation of multi-phase flows** was added into TA-A: As stated above, the research activities focused in TA-A mainly on the optimisation of droplet evaporation. The objective was to improve the accuracy of current models by including effects of small-scale processes on the global evaporation rate. This entails non-equilibrium, thermodynamic effects at the droplet interface (SP-A3), local curvature and oscillation effects (SP-A1, SP-A5) and multi-component and clustering effects (SP-A1, SP-A6). In addition, subprojects (SP-A2, SP-A4, SP-A6, SP-AX) focused on the development of numerical models for droplet dynamics and evaporation, by taking into account multi-scale and/or compressibility effects. In the following, selected results from one subproject (SP-A1) in TA-A are exemplified.

SP-A1: Modelling of deformed, multi-component time-dependent liquid drop evaporation, Doctoral Researcher: Yueqiao Hu, University of Bergamo (DROPIT member 01.10.2022 – 30.09.2025), Supervisors: G.E. Cossali, S. Tonini, B. Weigand

In this subproject, the research focused on an oscillating liquid droplet which evaporates in air. A hybrid analytical/numerical model for the evaporation of this oscillating droplet was proposed, by solving the transient species conservation equations in the gaseous phase under the assumption of potential gas flow. The moving boundary problem was solved in a generalized spheroidal coordinate system that describes the droplet oscillation by a continuously varying parameter.

The focus of this project was on overcoming the oversimplified assumption of quasi-steady mass and heat diffusion across the interface, as well as its influence on droplet evaporation. Thus, this subproject is one of the very few in literature to investigate this important effect.

Irrotational flow in the gas phase was supposed so that the gas velocity field is derived by solving the Laplace equation of the velocity potential. The gas was considered to be a mixture of air and vapor. The droplet shape was assumed elliptical and it oscillated between a prolate and an oblate shape. The results enlighten the transient effects on the evaporation characteristics in comparison to the predictions under the quasi-steady assumption: the vapour fluxes are no longer proportional to the Gaussian curvature, as found under the quasi-steady assumption; a phase shift appears between the oscillating evaporation rate and the droplet shape and its magnitude depends on the initial drop deformation. The droplet mass loss within one period may significantly differ from the quasi-steady case, with a possible effect on the oscillation dynamics. The model is compared with detailed

numerical simulations obtained by the FS3D (Free Surface 3D) code, to analyse the effect of the assumption of irrotational flow under the selected operating conditions. More details on the model and the results can be found in the individual report in **Chapter 6.2.8**.

The subproject in FP2 thus extended the phenomenological based work in FP1 by the former doctoral researcher Gautham Varma Raja Kochanattu (Phenomenological modelling of an oscillating drop under evaporating conditions) by including unsteady effects and, thus, not taking a quasi-steady assumption as it is normally done in literature.

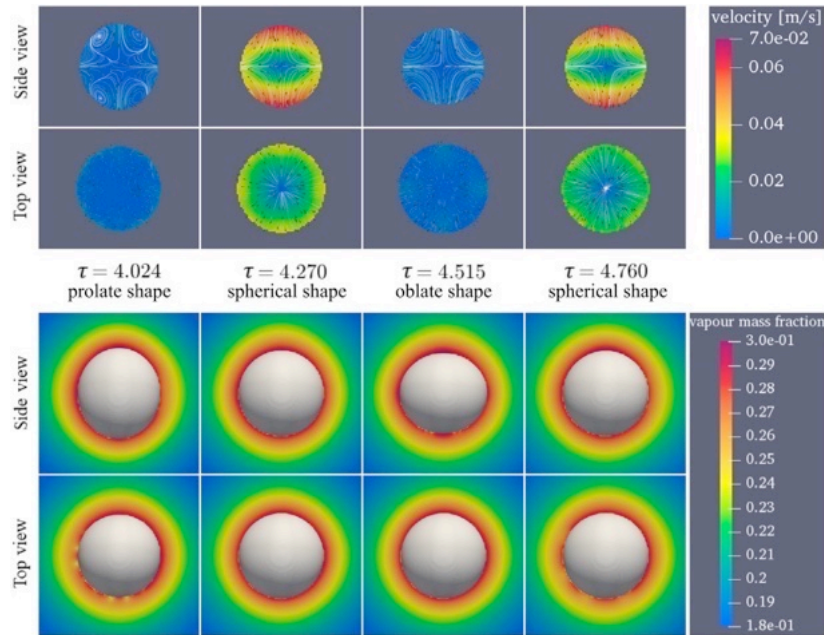


Fig. 4: Velocity field (top) and vapor mass fraction around the droplet (bottom) in a cross-section from the side and top view.

This subproject profited strongly of the international setup of DROPIT and the complementary knowledge of the different supervisors, e.g., by the comparison between detailed DNS calculations with sophisticated analytical modeling. The subproject resulted in several publications, from which two are mentioned below:

- **Hu, Y., Tonini, S., Cossali, G., Weigand, B., Reuttsch, J., Frey, M., 2023. Evaporation of an oscillating drop, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_4**
- Y. Hu, S. Tonini, G.E. Cossali, “An Analytical/Numerical Evaporation Model for an Oscillating Spheroidal Droplet in a Potential Flow.” *International Journal of Heat and Mass Transfer*, 256 (2026): 127925. <https://doi.org/10.1016/j.ijheatmasstransfer.2025.127925>

The first paper (highlighted in blue color) is a joint publication between researchers from Bergamo and Stuttgart and **is a result of the Summer School in 2022**, while the second paper is a paper of

the doctoral researcher together with his Italian supervisors. Both papers show very clearly how analytical and numerical work has been combined to solve complicated problems.

TA-B: Drop-Wall Interaction

TA-B consisted of six subprojects. The research activities are dedicated to the understanding and modelling of drop impact onto solid surfaces. This included textured surfaces (SP-B1, SP-B4, SP-B5), pretreated surfaces with hydrophobic or hydrophilic coatings (SP-B1, SP-B2) or even porous structures (SP-B3, SP-B6). The objective was to gain detailed understanding on how micro-scale structures can affect macroscopic detectable quantities and transport processes. Numerical simulations (SP-B4, SP-B5, SP-B6), are combined with detailed structure investigation by micro-CT (SP-B3) and experimental methods (SP-B1). In the following, selected results from one subproject (SP-B1) are exemplified.

SP-B1: Droplet collisions with solid superhydrophilic surfaces,

Doctoral Researcher: Patrick Palmetshofer, University of Stuttgart (DROPIT member 01.10.2022 – 30.09.2025), **Supervisors:** B. Weigand, G.E. Cossali

The work in this subproject built on the work in FP1 done by the previous doctoral researcher Patrick Foltyn, but extended it in several aspects: As already stated shortly in the public part in Chapter 3, droplet impacts onto micro-structured surfaces can lead to a very special spreading behaviour which might be non-circular (see Fig. 1, see **Chapter 3**). It was possible by the combination of very detailed experimental work in SP-B1 and DNS calculations in SP-A11 and SP-B5 to explain this different spreading behaviour. More details on the model can be found in the individual report in **Chapter 6.2.8**. Furthermore, the work in this subproject included studies on heated surfaces with droplet impacts. **Here, experimentally a completely new phenomenon was observed, the formation of a toroidal region of lower film thickness in the centre of the lamella formed during high Weber number water droplet impacts onto smooth heated walls.** This region formed around an air bubble, which was entrapped during the initial impact phase at the impact centre.

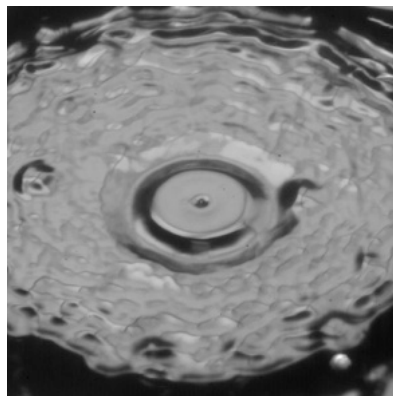


Fig. 5: Bottom view of a droplet impact onto a smooth heated sapphire glass surface, showing a thermocapillary lamella recess at the centre of the impact zone.

Different droplet sizes, impact velocities, surface wettabilities and temperatures were investigated experimentally. It was shown how this phenomenon can be explained by considering a two-step process involving thermocapillary convection in two separate regions: The temperature gradient along the surface of the entrapped air bubble caused heat conduction and induced flow that pumps warmer liquid to the lamella-ambient interface due to Marangoni effects. The resulting non-uniform temperature distribution then caused fluid acceleration in the radial direction, depleting the fluid volume around the bubble in a self-amplifying manner. With the help of direct numerical simulations of a stagnant liquid film with an enclosed bubble at the wall this explanation was confirmed.

In addition, in this subproject also simultaneous droplet impacts onto smooth surfaces have been investigated and have been studied by DNS in SP-B5.

This subproject profited strongly from the combination of detailed experimental investigations and DNS calculations using FS3D. It resulted in several publications, from which only two are mentioned below:

- **Palmetshofer, P., Wurst, J., Geppert, A., Schulte, K., Cossali, G.E., Weigand, B., 2025. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. Journal of Colloid and Interface Science 682, 413–422. <https://doi.org/10.1016/j.jcis.2024.11.154>**
- **Palmetshofer, P., Geppert, A., Steigerwald, J., Arcos Marz, T., Weigand, B., 2024. Thermocapillary central lamella recess during droplet impacts onto a heated wall. Sci Rep 14, 1102. <https://doi.org/10.1038/s41598-024-51382-3>**

The first paper (highlighted in blue color) is a common publication between researchers from Germany and Stuttgart and combines authors from different subprojects and associated projects. The second paper is on the thermocapillary central lamella recess. Both papers show very clearly how the combination of experimental and numerical work led to the derivation of a solution for this unknown phenomenon.

TA-C: Drop-Liquid Interaction

TA-C consisted of five subprojects. In SP-C1 and SP-C5, the main objective was to understand in great detail the drop impact onto a liquid film. This also included the feeding of the crown area, which needed the development of novel measurement techniques for micro-fluid dynamics through a collaboration among SP-C1 and SP-C5. Specifically, the micro-PIV approach was extended to enable velocity measurements in the crown lamella. Moreover, the particle tracking velocimetry (PTV) was extended to enable height-resolved velocity measurements in the crown crater by means of shadow-imaging defocused particle tracking (SD μ PTV). Results are published at the ILASS Europe 2025 in Lund and received the best paper award. An extended version of this contribution was invited by the editor-in-chief of the Atomization & Spray Journal with Open Gold Access and is currently under

revision. SP-C3 developed high-order numerical methods for multi-component incompressible flows. The subproject SP-C2, newly added in FP2, investigated an oblique drop impact onto a liquid film. Visualization methods developed in SP-C4, supported a better understanding of numerical and experimental results. An example of selected research findings in this area is also provided below.

SP-C2: Oblique droplet impact onto a wetted surface,

Doctoral Researcher: Jonathan Stober, University of Stuttgart (DROPIT member 15.04.2021 – 14.04.2025), **Supervisors:** K. Schulte, M. Santini

Droplet impacts onto thin films are fundamental to numerous technical and environmental processes, such as spray cooling, medical inhalers or spray coating. In these applications, droplets impact wetted surfaces at various velocities and angles. When the impact energy is sufficiently large, a crown forms from which secondary droplets detach — a phenomenon known as splashing.

To improve the efficiency of these applications, it is crucial to understand the impact dynamics and splashing behavior during oblique droplet impacts. While existing literature primarily focused on vertical impacts, this subproject investigated oblique impacts in detail, reflecting their prevalence in practical technical applications. To understand these dynamics, the project followed a dual approach: First, through detailed experimental investigations, a comprehensive regime map of all splashing outcomes was developed, as illustrated in Fig. 6.

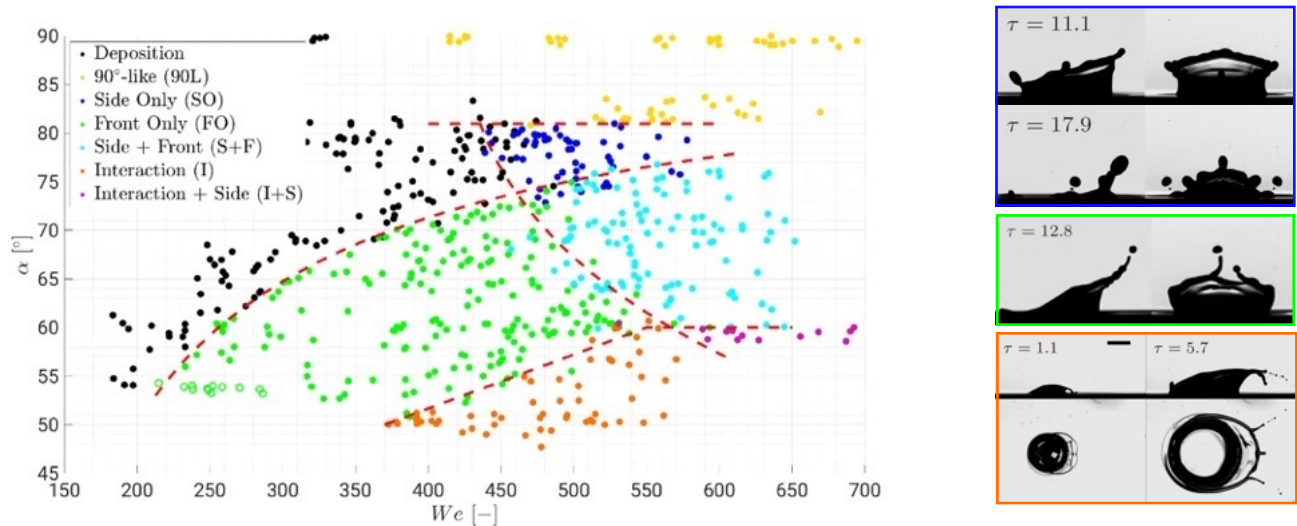


Fig. 6: Overview of different observed splashing types and threshold functions, $\delta = 0.22$, $Oh = 0.013$.

Second, Direct Numerical Simulations (DNS) were employed using FS3D to provide a deeper understanding of the underlying physical mechanisms. **These simulations made it possible for the first time to explain why the different splashing types observed in the experiments occur.** Figure 7 shows for example numerical results for the time development of the oblique mechanism for crown formation.

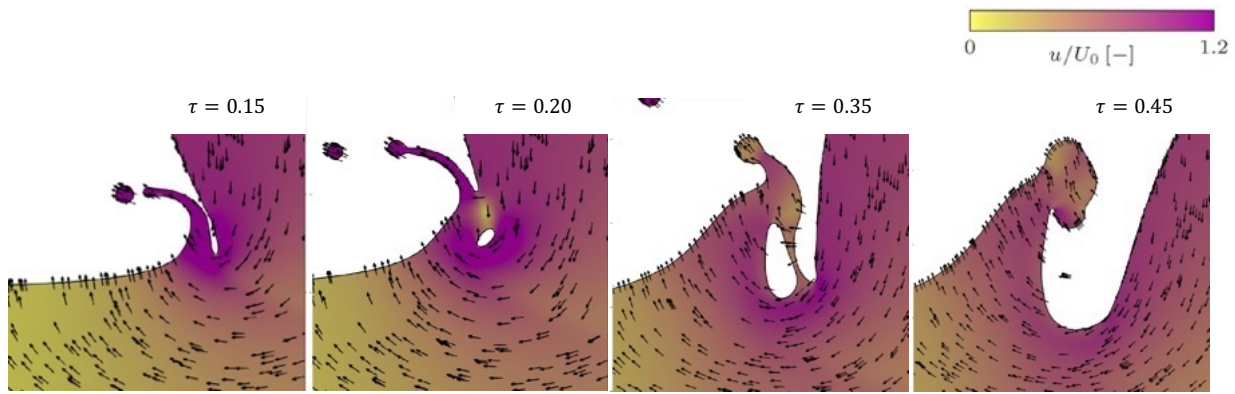


Fig. 7: Oblique mechanism of crown formation and velocity field on the back side ($\phi=180^\circ$).

Based on the classification of splashing types, regime thresholds have been derived and described using physically meaningful mathematical functions. These thresholds define the conditions under which specific splashing types occur, depending on the impact angle and Weber number.

This research benefited significantly from the expertise of our Italian partner, M. Santini, particularly regarding the generation of impacts at oblique angles. The project resulted in several publications that demonstrate the successful synergy between experimental and numerical methods in decoding and explaining complex splashing dynamics. Two joint publications are listed below as representative examples:

- **Stober, J., Santini, M., Schulte, K., 2025.** Characterization of splashing and regime thresholds for oblique droplet impact on thin wall films. *Exp. Therm. Fluid Sci.* 167, 111493. <https://doi.org/10.1016/j.expthermflusci.2025.111493>
- **Stober, J., Santini, M., Schulte, K., 2023.** Influence of Weber Number on Crown Morphology during an Oblique Droplet Impact on a Thin Wall Film. *Fluids* 8, 301. <https://doi.org/10.3390/fluids8110301>

Summarizing the progress in the research programme within **DROPIT**, it can be stated that excellent results have been achieved in the subprojects. As an international group with strong interdisciplinarity, we could make a large contribution to the research area under investigation. Advanced numerical methods (like DG methods for incompressible and compressible multiphase flow phenomena) have been further developed, high-level theoretical analyses (like for evaporating and oscillating droplets) have been established, and sophisticated new experimental methods (such as micro-PIV, LPSM, SDuPTV,...) have been further developed. Thus, the scientific goals set at the start of the project have been more than fulfilled. This success would not have been possible without the very good and fruitful cooperation between all partners of the University of Bergamo, the University of Trento and the University of Stuttgart. Thus, the added scientific value, which resulted from the

international cooperation lies in the very good cooperation and the complementary expertise between all partners. The research programme developed smoothly throughout both funding periods along the lines proposed in the original and the renewal proposal of this international research training group.

Personnel Developments, Visibility and Structural Impacts

During the nine year of research DROPIT reached a high visibility and had a lot of structural impacts on the involved universities. The PIs and doctoral researchers involved developed strongly during the duration of the programme. The doctoral and postdoctoral researchers have made impressive progress, including their personal development, and the careers of many principal investigators have also progressed during the course of the DROPIT project. In the following several examples are given first for the personnel development (in alphabetic order):

- **Martina Baggio** (doctoral researcher, Stuttgart) and **Karin Schlottke** (associated doctoral researcher, Stuttgart) received the prestigious „Amelia Earhart Fellowship Award“, Zonta International Foundation.
- **Andrea Beck** (PI, Stuttgart) was appointed as a Full Professor for Computational Fluid Dynamics at the University of Stuttgart in 2022.
- **Lorenzo Botti** (PI, Bergamo) was promoted as a Full Professor in 2024 at the University of Bergamo.
- **Carina Bringedal** (PI, Stuttgart) got a Full professorship at the Western University of Applied Sciences in Bergen, Norway. Therefore, she left the GRK in 2022.
- **Michael Dumbser** (PI, Trento) was promoted in 2018 as a Full Professor at the University of Trento.
- **Anne Geppert** (Postdoc from 2019 – 2024) moved in 2024 to the Leibniz Institute for materials-oriented technologies (IWT) at Bremen, where she is now leading the department on metal atomization and spray compacting.
- **Grazia Lamanna** (PI, Stuttgart) finished her habilitation in 2022 and is now Privatdozent. Since 2023 she was promoted to be an Academic Director at ITLR. Furthermore, since 2021 she is the Equal Officer for the whole University of Stuttgart.
- **Yanchao Liu** (associated doctoral researcher, Stuttgart) was selected for the „National Talent Program (1000-Talent-Plan)“ in China after he returned back to China.
- **Francescarlo Massa** (associated Postdoc, Bergamo) received a Research Fellow Tenure Track position in 2025.

- **Maurizio Santini** (PI, Bergamo) was promoted as an Associated Professor in 2017 and as a Full Professor in 2025 at the University of Bergamo.
- **Kathrin Schulte** was awarded the Margarete von Wrangell Scholarship by the Ministry of Science, Research, and Arts in Baden-Württemberg in 2018 and received a permanent position at the University of Stuttgart in 2023.
- **Simona Tonini** (PI, Bergamo) was promoted as an Associated Professor in 2018 at the University of Bergamo.
- **Stephanie Fest-Santini** (Associated Postdoc) received a Research Fellow A position in 2017, a Research Fellow B positions in 2021 and was promoted as an Associate Professor in 2024 at the University of Bergamo.
- **Vishak Vaikunthanatan** (Postdoc from 2026-2019) got a position as professor at the department of Mechanical Engineering, Shiv Nadar University, India.

Apart from these examples it can be stated that all doctoral and postdoctoral researchers, who worked within DROPIT obtained very attractive positions after they left the IRTG, e.g. at Zeiss, the German Patent and Trade Mark Office, at Kärcher, Tesat, and other companies and institutions. The reader is referred to the table in **Chapter 6.2.1** for more information. This clearly shows the importance of the research field investigated and the need for well-educated researchers in this area.

A High Visibility of the Project has been Reached Through the Following Activities:

There had been a large number of talks/lectures from international guests (see **Appendix 2**) and a large number of invited talks given by DROPIT members, which helped to make the IRTG visible in our research community. Furthermore, the already mentioned overview presentations at the international ILASS and ICLASS conferences helped to stimulate discussions on the programme and served as a basis to offer cooperation with international partners. Here just one example is given: Prof. Koji Hasegawa from the department of Mechanical Engineering at the Kogakuin University of Technology and Engineering in Japan heard about DROPIT at a conference and was invited from the doctoral researchers to Stuttgart for a scientific presentation about his work. Here he also saw the experimental and numerical work which had been done in DROPIT and was so fascinated that he spent his one-year sabbatical at Stuttgart (from 01.04.2024- 31.3.2025), strongly connected to the work in the IRTG.

Of course, the published book on results of DROPIT from 2020 and the large number of publications (see **Chapter 5**) in highly ranked international journals helped to achieve a high visibility of DROPIT. The yearly organized DIPSI workshop in Bergamo, where the doctoral researchers also presented their work in poster sessions and/or talks further stimulated international cooperation. In the two

Summer Schools of DROPIT, where several international groups worked on selected topics (see **Chapter 6.2 and Appendix 1**) a lot of new ideas were generated, from which several publications could be derived later on. High visibility in the general public could also be reached, e.g., by presentations on the “Day of Science” at Stuttgart University, lectures especially for kids on the topic of droplets, e.g., at the kids university in Heilbronn in 2020 and public lectures like “The wonderworld of droplets” in 2021, which was organized as a cooperation between the Heidelberg Academy of Sciences and Humanities and the Württemberg State Library or also in reports on DROPIT in the local newspapers of Bergamo...

In the very popular Journal “Bild der Wissenschaft“ with a circulation of more than 50,000 copies, a contribution about the droplet research was made in 2023 including also the DROPIT work. Below some impressions of the article by Breuer (R. Breuer, Trickreiche Tropfen, pp. 22-27, Das Geheimnis des Wassers, Bild der Wissenschaft, Januar 2023) which appeared in “Bild der Wissenschaft” are shown.



Fig. 8: Some impressions of the article by Breuer published in Bild der Wissenschaft in 2023.

Structural Impact, Knowledge Transfer and Influence on Future Projects

The project had also a strong structural impact at the involved Universities. First of all, the personnel developments mentioned above shall be noted. Then, the project helped to deepen and strongly enhance the cooperation between the University of Stuttgart and the University of Bergamo. In the preface of the book published about the DROPIT results in 2020 (G. Lamanna, S. Tonini, G.E. Cossali, B. Weigand: Droplet Interactions and Spray Processes. In Fluid Mechanics and its Applications 121, Springer, Heidelberg, Berlin, ISBN 978-3-030-33337-9, 2020) both rectors of Stuttgart/Bergamo stated:

“The International Research Training Group (IRTG) for Droplet Interaction Technologies (DROPIT), which has been established in October 2016 by the German Research Foundation (DFG), is an impressive example of the strategic partnership between the two Universities. The IRTG brings researchers from Bergamo and Stuttgart together to work on droplet dynamics. DROPIT is a structured doctoral program....”

This statement was confirmed and extended during their talks at the final colloquium of DROPIT held in Stuttgart from 31.07.2025 to 01.08.2025. The reader is referred to the **Appendix 1** for more details concerning the Final Colloquium.

There was also a joint research project with the Paul Scherrer Institute (CH) initiated on neutron imaging investigation of liquids mixing during droplet impact onto liquid films. The project had Dr. Anne Geppert as lead scientist and included several DROPIT doctoral and associated members. The result of this collaboration, partly financed by DROPIT, were presented at the ILASS-Europe 2025 conference in Lund, which represents one of the most successful measures in promoting the research and visibility of female, early career researchers.

There had also been intense collaboration with other initiatives at Stuttgart, e.g. with the **SFB-TRR 75**, the **SFB 1313** and also the **EXC 2075 – SimTech**. Especially, scientific guests have been invited together and workshops have been set up together with the above-mentioned initiatives. The reader is referred to the **Appendix** for a list of all workshops.

In fact, the presence of DROPIT as an IRTG in place stimulated discussion between the University of Stuttgart and the University of Bergamo which led to a new international master programme. This is explained in more detail in **Chapter 6.2**.

Transferring Knowledge and Influence on Future Projects

By introducing the workshop “**Future Perspectives**” already in FP1 and later the workshop “**DROPIT meets Industry**” in FP2 of the project, the doctoral researchers were able to show their work to selected companies and the companies gave presentations on their research. Thus, there was a good interaction for transferring knowledge to industry during the project. Furthermore, one associated project (**AS04, Yanchao Liu**) was added in Stuttgart to investigate droplet interactions at high velocities. This project investigated effects, which are present in medical inhalers and stimulated cooperation with the company PARI with whom we worked together on droplet interactions in inhalers. This included cooperation on numerical work, but also cooperation on experimental investigations. Two publications on this cooperation are listed below:

- **Liu, Y., Geppert, A.**, Chu, X., Heine, B., Weigand, B., 2021. Simulation of an annular liquid jet with a coaxial supersonic gas jet in a medical inhaler. *Atomization Spray* 31, 95–116. <https://doi.org/10.1615/AtomizSpr.2021037223>
- Heine, B., Veith, B., **Geppert, A.**, Selzer, T., Denis, T., Kaufmann, K., Maier, S., Schlun, M., Seemann, S., Weigand, B., 2024. From liquid to aerosol: Exploring droplet formation in vibrating mesh nebulization. *Respiratory Drug Delivery* 1, 111-122.

Also, in Bergamo transfer of knowledge was achieved during the time of this project. There the associated project **AS07 of Luca Vezzeroli** on micro-scale modelling and simulation of two-phase flow

processes in oil mist filtration, which was carried out in cooperation between the University of Bergamo and the company Serioplast in Italy, led to a knowledge transfer on the highly sophisticated used numerical methods and codes developed in Bergamo.

Within the research project inside DROPIT also two patent applications have been made:

- **Foltyn P.**, Weigand B., and Roth N., Measuring device and method of determining a depth of field of an optical setup. US Patent 12,078,473, 2024, EP 4105594 B1, 2025.
- **Foltyn, P.**, Roth, N., Weigand, B., inventor; University of Stuttgart, assignee. Verfahren zur Messung der Schichtdicke einer optisch transparenten Schicht, insbesondere einer Flüssigkeitsschicht. European Patent application 19 192 074.3. 2019 Aug 16.

For the invention from 2021 a European patent has been granted in 2025 and there is currently the discussion on how to bring this product into the market. One possibility would be to found a small start-up company for this. The first invention from 2019 was not followed further.

Furthermore, having the numerical computer programme **FLEXI** (www.flexi-project.org) as an open source code led to several possibilities to transfer knowledge to other researchers worldwide. In this programme new numerical solution schemes were implemented, which had been developed within the IRTG by different doctoral researchers.

In the framework of DROPIT, **unique experimental and numerical data have been produced**. The research data generated in this project have been treated according to the DFG guidelines for safeguarding good scientific practice and the corresponding guidelines. The IRTG used RAID-systems, where research data could be stored and easily exchanged between subprojects. The central depository of the University of Stuttgart **DaRUS** has been used for archiving selected experimental and numerical data. This secure database assigned a DOI for identification to the data. This makes the results findable, accessible, interoperable and reusable, in accordance with the FAIR principles. There are no embargo periods for publication or use. Some examples for datasets in DaRUS are shown below:

- **Saha, R.**, Inal, C., Kumar, M., Weigand, B., 2025. Numerical Data for: DNS of water droplet impact onto smooth surfaces at various static contact angles.
<https://doi.org/10.18419/DARUS-5365>
- **Wurst, J.**, 2025, "Replication Data for: PLIC-based contact line modeling for simulations of droplet impact onto smooth and structured surfaces", <https://doi.org/10.18419/DARUS-5496>, DaRUS, V1.

- **Stober, J.**, 2025. Numerical Data of the Asymmetric Inner Film Flow during Oblique Droplet Impact onto a Wall Film. <https://doi.org/10.18419/DARUS-5451>
- Steigerwald, J., **Schubert, S.**, 2024. Experimental and Numerical Data for: “Micro-PIV study on the influence of viscosity on the dynamics of droplet impact onto a thin film.” <https://doi.org/10.18419/DARUS-3887>
- **Palmetshofer, P., Wurst, J.**, 2022. Replication Data for: Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. <https://doi.org/10.18419/DARUS-4178>
- **Stober, J.**, 2022. Simulation Results of a Droplet Impact onto a Liquid Film. <https://doi.org/10.18419/DARUS-3102>

The Knowledge Developed in DROPIT by Several PIs also Helped them by Setting up New Projects.

For the **SPP 2410 - Hyperbolic Balance Laws in Fluid Mechanics: Complexity, Scales, Randomness (CoScaRa)** the studies in DROPIT and in SFB-TRR 75 (especially the strong interdisciplinary approach) served as a foundation for the organization of the Priority Programme. The **DROPIT PI Christian Rohde** is the coordinator of this project.

In Bergamo, the studies using X-ray micro-computed tomography (microCT) in porous media in DROPIT served as a foundation of using the technique also for biological materials. Within the large Italian project **ANTHEM: AdvANced Technology for Human centERed Medicine** the DROPIT (<https://fondazioneanthem.it/en/spoke-1-en/>) the **DROPIT PI Maurizio Santini** could establish a project within Spoke 1: Data and technology driven diagnosis and therapies, where he is responsible for Pilot 1.8: MicroCT-based 3D digital histology.

Within **DROPIT**, the first edition of the hybrid workshop **AI meets Scientific Computing: Bridging Theory and Practice** was held on 11 June 2025 in Bergamo, bringing together experts from academia and research centres to explore AI's transformative role in computer simulations. Because of the strong interest in this workshop it is planned to continue this workshop in the next years.

In general, the very good cooperation between the Universities involved led to long lasting research cooperation. This can be seen, e.g., in the fact that researchers from Stuttgart went to Bergamo (e.g. B. Weigand several times for two weeks as an invited professor – funded by the University of Bergamo or K. Schulte who will have a research stay in April and May 2026 in Bergamo – funded by the University of Stuttgart, or S. Tonini, who had a research stay for roughly two weeks in April 2024 at

the University of Stuttgart – funded by the University of Bergamo within the Outgoing Visiting Professors/ Researchers Programme 2023). All these visits will guarantee a sustainable intense cooperation going on, also after DROPIT has finished.

Comments Regarding the Proposal process and Programme Design

The participants of the GRK2160 - DROPIT can look back on extremely successful nine years of joint research. During this time, we have succeeded in gaining a much better understanding in the detailed processes of droplet interaction technologies, which also improved strongly the prediction capabilities of these processes.

The participants in this project believe that the results, data, and publications created here have already and will continue to have a major impact on scientific development in this field.

First of all, we would like to thank all the reviewers who, during the review of the preproposal, the implementation of the project and in the review of the renewal proposal helped us to further sharpen the profile of DROPIT, achieve all our goals, and to obtain these excellent results with their critical questions and very good comments. We consider the on-site reviews to be a very important tool, as they allow for detailed discussions with the reviewers, e.g., at the posters. Unfortunately, this was not possible for the review of the renewal proposal because of the Coronavirus Pandemic.

We would also like to express our sincere gratitude to all DFG staff members for their consistently friendly explanations and clarifications and for always being available to answer our questions. The smooth approval of the extensions for some doctoral researchers for some months during the Coronavirus Pandemic is very much appreciated. Nevertheless, this pandemic time posed many challenges to us.

All applicants and project staff of DROPIT would like to express their sincere thanks to the German Research Foundation (DFG) for its long-term funding of our work within the framework of GRK2160. In our opinion, only a funding instrument with this long-term perspective enables fundamental and interdisciplinary progress in such a broad field of knowledge as Droplet Interaction Technologies.

6.2 Information on qualifications and structural impact

Droplet interaction technologies find application in numerous fields of economic interest. Thus, potential areas of employment range from automotive and aerospace industries to food and pharmaceutical companies as well as in process engineering for the development of environmental-friendly treatment of waste and pollutants. The increased interdependence of economies and technologies together with the need of reducing the ecological footprint of industrial processes requires excellently

qualified researchers and engineers with an interdisciplinary background and international experience. Our qualification programme was especially designed to meet these requirements and its distinctive features are briefly summarized below.

Here, it should be pointed out that nowadays most universities have established well-structured doctoral schools. For the University of Stuttgart, we mention the Graduate Academy **GRADUS** (<https://www.gradus.uni-stuttgart.de>); for the University of Bergamo the Ph.D. programme in engineering and applied sciences **ISA** (now called **SUSTAIN**) and for the University of Trento, the doctoral school in Civil, Environmental and Mechanical Engineering **DICAM**. The main objectives of these Graduate Academies are

- to offer junior scientists high-quality interdisciplinary training by promoting collaborations among the faculties and the research institutions of the universities
- to ensure and develop standards of excellence within the university's doctoral research programmes
- to foster the development of each doctoral researcher into an independent researcher
- to train the doctoral researcher to effectively communicate the results and methods to an audience of international experts, both orally and in writing
- to promote the development of the skills necessary to formulate research projects and manage them in national and international contexts.

Furthermore, all information concerning admission, regulations, supervision, schedule of courses, scholarships and doctoral programmes are available on the online platform at the respective institutions. As pointed out in the original proposal for DROFIT, this provided a timely, transparent and efficient infrastructure that the doctoral researchers could use to plan and organise most effectively their individual doctoral curricula. Due to the above-mentioned characteristics, we decided already in the initial proposal to exploit the existing educational structures for the doctoral researchers by requiring that all graduates participating in the IRTG are associated to one of the available graduate academies: GRADUS, ISA or DICAM. This requirement, as described in the proposal for FP1, yielded a number of advantages:

- it ensured an optimal usage of resources by preventing duplication of infrastructure and associated costs;
- it ensured equal treatment and opportunities to all doctoral researchers;
- it allowed us to set up additional qualification measures for the IRTG doctoral researchers. The latter had been grouped into three main blocks: "Scientific and professional skills", "Supervision and advisory measures", "Internationalisation and self-management skills". They are shortly summarized hereafter.

Additional measures in support of scientific and professional skills

- Highly advanced courses on topics of relevance for the IRTG
- Training courses on key skills, ranging from the art of communication over ethics in science to leadership in academia.
- Inclusion of a patent officer in workshops, lecturing on practical patent drafting and application. This also can promote patent applications and the start-up of their own companies by the doctoral researchers after successfully finishing their studies.
- A biweekly “graduate school seminar” for learning to present and communicate research results

Additional measures in support of doctoral supervision

- A double or triple (Italian-German) supervisor system for each doctoral researcher
- An advisory board, composed of members from academia and industry

Additional measures in support of internationalisation and self-management skills

- A yearly joint doctoral meeting to discuss the scientific progress of the IRTG with the participation of members from the advisory board
- An exchange programme allowing the doctoral researchers to spend a long research visit in the partner country.
- The annual organization of an international workshop
- A three-yearly international summer school, organized directly by the doctoral researchers to foster their self-management skill.

With these measures, which have been described in our initial proposal, we were able to offer within DROPIT the doctoral researchers a unique environment, where they could learn to interconnect knowledge from different disciplines, get acquainted with cutting edge numerical and experimental methods and benefit from the educational programme. As shown below, this programme was especially tailored to the scientific goals of DROPIT.

Qualification Programme of DROPIT

The course and seminar programme aimed to providing the doctoral researchers with the necessary theoretical and experimental foundations, required for our strongly interdisciplinary approach. This also included management skills to ensure an efficient and effective deployment of resources, time planning, oversight of project’s goals, scientific achievement and their effective dissemination. Project and resource management, in fact, represent key qualifications for the professional development of the doctoral fellows.

The courses, which were given, are listed in the table in **Appendix 3**. It is distinguished between **core**, **specialised** and **training courses**. Some external specialised courses have been offered in the qualification programme in order to best train the doctoral researchers. As described in our initial proposal **core courses** are mainly part of the local universities’ general educational programmes.

They are typically held by the Principal Investigators (PIs) at the corresponding Italian or German university. They focus on theory and methods to ensure a common knowledge foundation among all young researchers, in order to facilitate the interaction within the IRTG. The courses have been given in a flexible manner. This means that, e.g., the course on “Dimensional Analysis” (normally a core course at the University of Stuttgart) has been given at the University of Bergamo as a compact course, because of the interest of several doctoral researchers at this place in this topic.

In order to make the courses taken by the doctoral researchers fully visible, they have been included into the individual reports of the doctoral researchers. Thus, the reader is referred to the individual reports given in **Chapter 6.2.8** for more details on this subject.

Specific, specialised courses for this IRTG

Appendix 3 provides a list of all given specialised courses, including information on lecturer, course duration, frequency and location, where the course took place. These specialised courses are at a more advanced level. Most courses were given by DROPIT principle investigators, and some by visiting scientists. Attendance was not limited to DROPIT doctoral researchers, but open to all doctoral researchers of the doctoral schools GRADUS, ISA and DICAM of the universities of Stuttgart, Bergamo and Trento, respectively.

These specialised courses were complemented by several international courses. These courses (like the CISM Course on Drops, Jets and Films or the Short Course on Atomization and Sprays) were found to be very good to provide to the doctoral researchers a detailed overview about the topics of droplets, sprays and interfaces. A list of these courses is given in **Appendix 3**.

Core courses

A list of the core courses is also provided in **Appendix 3**. The core courses are part of the local universities' general educational programmes. In addition, some courses are given at the Supercomputing Center in Stuttgart (HLRS). For the German side, doctoral researchers had to achieve 9 to 12 ETCS during their affiliation to DROPIT.

Training courses on key skills

A list of the training courses held on key skills is provided in **Appendix 3**. The training courses are not compulsory. Most of them are organised together with (or by) the respective Graduate Academy. Some training courses, like Scientific Writing, which are tailored especially to the need of DROPIT, have been organised by DROPIT alone. The training programme also includes courses that deal with the principles of good scientific practice, e.g.:

- Philosophy of science and ethics of technology
- Critical thinking for scientists
- Ethics of Science

- Ethics of Technology

Furthermore, in the **Supervision Agreement on Doctoral Studies** (“Promotionsvereinbarung”), please see **Appendix 4** for an example of this agreement, the working according to the rules of good scientific practice is further outlined and signed by the doctoral researcher and by the supervisor. Thus, each supervisor talks with her/his doctoral researcher about this topic. Furthermore, during several doctoral seminars this topic was highlighted and brought to the mind of all doctoral researchers.

Training on the collection, storage, editing and long-term provision of research data and on the use of computer programs and their versioning

The handling of data is very important and needs to be trained. In order to help the doctoral researchers to better understand what the needs of researcher working experimentally and numerically are, a workshop had been set up, where experimentally working doctoral researchers explained how they work, what data they produce and what problems evolve. The same was done by doctoral researchers working mainly numerically. The “**Workshop on Experimental and Numerical Methods**” took place in Bergamo (Italy) on 17.05.2018. This topic was also addressed in further workshops, e.g. “**Workshop on Mathematical and Numerical Modelling of Multiphase Flow**” in Stuttgart from 01.04.2019.-02.04.2019 and in the Workshop on “**Experimental and Numerical Methods in Science & Ethics in Science**”. Furthermore, a training on data storage was offered within the courses on key-skills, e.g.: “Research data management”.

In addition to data storage and handling, it is of great importance to coordinate the work on a larger computer code very strictly. This means also that the code needs a versioning system. Therefore, the groups working numerically on larger computer programs like **FLEXI** or **FS3D** take every year a 2-3 days’ retreat, where all involved doctoral researchers come together, discuss code changes and the versioning of the new code parts. This is extremely important, because after several years, it must still be clear which data were created with which program version. This promoted also exchange between the doctoral researchers who worked on the same code, but worked in different projects like EXC 2075, SFB-TRR 75 or SFB 1313.

Methods used to integrate student assistants into the qualification programme

Student assistants had been in close contact with the doctoral researchers, because they have been directly involved in their individual projects by supporting measurements, visualization and/or computations. Student assistants have also been integrated into the qualification programme in several ways. First of all, the specialised courses listed above were open for all students. In addition, interested student assistants could also attend the doctoral seminars and the Multiphase Flow Seminar,

where DROPIT doctoral researchers presented their work. Furthermore, the student assistants (and also other interested students) could attend the project presentations and talks of the summer school. This process resulted in a large number of students who started their Bachelor- and Master-thesis on topics of DROPIT after finishing their student assistant work. In addition, several students moved from Stuttgart to Bergamo or from Bergamo to Stuttgart for their thesis.

Visiting Scientists Programme

The visiting scientists programme represented an important cornerstone of our qualification measures for the doctoral researchers. Its goal was to promote scholarly exchange on numerical, theoretical and experimental methods for two-phase flows and, if feasible, to establish fruitful collaborations with international leading scientists. **The visiting scientists had been integrated in DROPIT in several ways. First, they presented an invited talk during their stay. Second, an intensive exchange between the visitor and interested doctoral researchers took place.** Thus, the doctoral researchers could benefit, already at an early stage, from an external and qualified feedback on the status of their own research. Furthermore, comments and suggestions from experts helped in their own revision and conceptualization of their research work, thus providing a unique learning experience. **In order to reach this goal, the doctoral researchers had a very active role in outlining the visiting researcher programme. They have been encouraged and motivated to propose names of visiting researchers.**

Thanks to the close geographical vicinity of Stuttgart and Bergamo some visitors could be hosted at both partner universities. All visitors of DROPIT are summarized in **Appendix 2** together with the time span of their visit, the purpose of the visit and the institution they are coming from.

Additional Qualification Measure and Research Visits to the Partner Institution

One of the educational objectives of this IRTG was that doctoral researchers must develop the ability to conduct research independently and effectively communicate the results and methods used to an audience of international experts, both orally and in writing. In order to develop their skills in this respect, the following measures had been included in **DROPIT**:

- to present and discuss their work in progress on the multiphase flow seminar
- to attend the yearly doctoral seminars and present the work progress
- to attend two scientific conferences and two DIPSI workshops during their doctoral studies
- to write at least one paper in a peer-reviewed journal and two conference papers

Furthermore, two additional qualification measures had been implemented in DROPIT: a yearly international workshop and a three-yearly international summer school. These initiatives build upon the previously described qualification measures and provided the doctoral fellows a platform to put

into practice all management, organisational, interdisciplinary and scientific competences acquired through the course and seminar programme. In the following, the qualification measures are explained in more detail:

Doctoral Seminars

The yearly **doctoral seminars** have been held in Bergamo and in Stuttgart:

- DROPIT Doctoral Seminar 2017, Bergamo (Italy), 24.05.2017
- DROPIT Doctoral Seminar 2018, Stuttgart (Germany), 11.04.-12.04.2018
- DROPIT Doctoral Seminar 2019, Bergamo (Italy), 15.05.-16.05.2019
- DROPIT Doctoral Seminar 2020, Stuttgart (Germany), 02.04.-03.04.2020

=> **Because of the spreading of the Corona virus this Doctoral Seminar had to be postponed to 25.06.-26.06.2020**

- DROPIT Virtual Doctoral Seminar 2021, 08.03.-09.03.2021
 - DROPIT Doctoral Seminar 2021, 08.09.-09.09.2021 (Bergamo) in presence and virtual
- => **This was the first Workshop which was (under very strict limitations) allowed in Bergamo**

- DROPIT Hybrid Doctoral Seminar 2023, Stuttgart (Germany), 01.03.-02.03.2023
- DROPIT Hybrid Doctoral Seminar 2024, Bergamo (Italy), 08.02.-09.02.2024
- DROPIT Doctoral Seminar 2025, Stuttgart (Germany), 28.01.-29.01.2025

During the doctoral seminars, the doctoral researchers presented their yearly progress in their research project.



Fig. 9: Kick-off and doctoral seminar on 08.09.2021 – 09.09.2021 in Bergamo (from left to right: G.E. Cosali, Ingrid Jung (General Consul of Germany in Milan), Remo Morzenti Pellegrini (Rector of University of Bergamo), Wolfram Ressel (Rector of University of Stuttgart), B. Weigand).

Multiphase Flow Seminar

In the weekly “Multiphase Flow Seminar” the doctoral researchers presented their work during the semester. In this seminar normally also students participated and thus, a good discussion of the work of the doctoral researchers could be achieved.

International Workshop on Droplet Dynamics (DIPSI) and DROPIT Workshops

The International Workshop builds upon the DIPSI Workshop on “*Droplet Impact Phenomena and Spray Investigation*”. This yearly workshop has a very long tradition and provides an opportunity to share recent knowledge on droplets and sprays in a variety of research fields and industrial applications. The workshop is always held in Bergamo every year. Here the doctoral researchers have actively been involved, e.g., by special poster presentations during the DIPSI workshop. This gave the doctoral researchers the chance to actively discuss their research results with international experts and to start scientific collaborations. The poster session was always introduced by a short talk about the international training group DROPIT, so that everyone was familiar with the concept of an IRTG. The doctoral researchers had to attend minimum two DIPSI workshops.



Fig. 10: DIPSI workshop 2019 in Bergamo



Fig. 11: DIPSI workshop 2024 in Bergamo

In the week of the DIPSI workshop normally also additional workshops, lectures and/or the doctoral seminar were held. Thus, in 2018 a day before DIPSI the **“Workshop on Experimental and Numerical Methods”** took place in Bergamo (Italy) on 17.05.2018. In 2019 the workshop **“Future Perspectives”**, took place from 10.10.2019-11.10.2019 in Stuttgart. This workshop was proposed by the doctoral researchers. It brought together participants from industry (Pari, BOSCH, Tplus, Boysen GmbH) and DROPIT members. Both groups presented their work and there has been a vital discussion about cooperation and job possibilities. In addition, a patent attorney also presented about patent application. This workshop was later renamed to **“DROPIT meets Industry”** and was given in Bergamo on 15.06.2023. In 2024 the workshop **“Experimental and Numerical Methods in Science & Ethics in Science”** was given in Bergamo. In 2025, there was a new workshop established: **“Workshop on AI meets Scientific Computing”** (11.06.2025) followed by the **“Workshop on Analytical and Numerical Methods in Science”** (12.06.2025). This gave the chance to talk here also about Ethics and Science and the relation between Ethics and AI and how they influence each other. There have been also workshops, which have been organised together with other research initiatives like SFB-TRR 75 or SFB1313. The reader is referred to **Appendix 1** for a complete list of all workshops. All these workshops had a structural impact on the involved universities. First of all, the DIPSI workshop will continue every year (the 18th edition will be held in 2026). Also, the new workshop “AI meets Scientific Computing” is planned to be held at least every second year.

International Summer Schools and Final Colloquium

During the nine years of the project two summer schools and a final colloquium have been organized. The summer schools were fully organised by the doctoral and postdoctoral researchers. Thus, they were responsible for the planning, budgeting, and logistic aspects of the summer schools. The summer schools were multinational and interdisciplinary in character. They were structured as intensive academic programmes. The main objective of the summer school was to endorse doctoral researchers to achieve scientific independence at an early point in their research career. Furthermore, it offered a natural platform to support teamwork and the ability to work in a multi-disciplinary and multinational team. Below a short report on both summer schools is given. For more details, the reader is referred to **Appendix 1**.

1st International summer school (04.09.2018 - 21.08.2018)

The first summer school of the IRTG was held in 2018. The aim of this summer school was to bring multi-disciplinary scientists together to work on research topics related to droplet interaction processes using a combination of experimental, numerical, and analytical tools. As a first step to achieve this, short summer school project proposals had been submitted by different teams, and were reviewed and approved. This resulted in five approved summer school projects. In addition, there were invited lectures and keynote speeches by experts in the field of droplet dynamics and related topics during the period of the summer school.

Projects

Diffuse Interface Models of Shear-Horizontal Type
 Christian Müller (SP-A4)
 Wolfgang Retsche (SP-A3)
 Shrawan Choudhary (SP-A4)
 Jonas Zillig (SP-A4)
 Francesco Sassi (University of Bergamo)
 Francisco Carlos Macos (University of Bergamo)
 Monica Haeffl (University of Bergamo)
 Andrea Nanni (University of Stuttgart)
 Stefania Fossati (University of Stuttgart)
 Fabian Hill (University of Stuttgart)
 Thomas Hitz (University of Stuttgart)
 Claus-Christoph Müller (University of Stuttgart)
 Michael Dumbser (University of Tübingen)
 Markus Kriebel (University of Tübingen)

Experimental and Numerical Investigation of Wetting Phenomena on Selected Structured Surfaces and Porous Media
 Patrick Polyn (SP-B1)
 Robert Hilde (SP-B2)
 Matthias Engel (SP-B2)
 Alexander Bräuer (SP-C4)
 Jan Stiller (University of Aachen)
 Stephanie Hest-Garten (University of Bergamo)
 Heinrich Götter (University of Bergamo)
 Doreen Wilmanns (University of Stuttgart)
 Thomas Goll (University of Stuttgart)
 Norbert Hoff (University of Stuttgart)

Coupling Porous Media and Free Flow with Help of an Upscaled Phase Field Model
 Lukas Odenwiler (SP-B4)
 Shoaib Mahmood (SP-B2)
 Boris Top (University of Stuttgart)
 Andre Mink (University of Stuttgart)
 Rainer Hering (University of Stuttgart)
 Jan Magnus (University of Stuttgart)
 Christian Nanni (University of Stuttgart)
 Lars von Wolf (University of Stuttgart)

Investigation on the Mixing Behavior for Droplet Impact onto Miscible and Immiscible Liquids
 Roman Barmann (SP-C1)
 Carina Penschke (TU Darmstadt)
 Gracia Lomana (University of Stuttgart)

Oscillation of an Evaporating Spheroidal Drop: Effect on Frequency and Amplitude
 Gauthier Verhaeghe (SP-A1)
 Christiano Silva Cassal (University of Bergamo)
 Shrawan Choudhary (SP-A4)
 Oliver Reiss (TU Darmstadt)
 Matthias Bach (University of Stuttgart)
 Jonathan Haderich (University of Stuttgart)
 Ralf Ischertke (University of Stuttgart)
 Bernhard Weigand (University of Stuttgart)

Farewell dinner

Our farewell dinner takes place in one of Stuttgart's most prestigious buildings, the television tower. There, you will also have a perfect view over the whole city center.

Together, we will get there using public transportation. Guided travel will be organized from conference center IBZ and the metro station on campus.

Date: September 21st
Departure: 16:00 at IBZ
 16:30 at S-Bahn Universität (on campus)
Route: S1/S2/S3 to 'Hauptbahnhof', U7/U15 to 'Ruhbank'

TV tower Stuttgart Fernsehturm

Contact

Dr. Visakh Vaikuntanathan
 visakh.vaikuntanathan@irg.uni-stuttgart.de
 Institute of Aerospace Thermodynamics
 University of Stuttgart
 Pfaffenwaldring 31
 70569 Stuttgart
 Tel.: +49 711 685 - 62017
 Fax: +49 711 685 - 62317

Further information

For more information, visit the DROPLIT website on project.uni-stuttgart.de/droplit/qualification/summer_school

DROPLIT Research Training Group 2160/1
 Droplet Interaction Technologies
 Prof. Bernhard Weigand
 Prof. Elvio Cossali

Summer School 2018

DROPLIT Summer School aims to bring together multidisciplinary scientists to work on droplet interaction related topics using a combination of experimental, numerical and analytical tools.

September 4th – 21st
 Stuttgart
 Germany

University of Stuttgart
 Germany

unibg
 University of Bergamo

UNIVERSITY OF TRENTO - Italy

DFG Deutsche Forschungsgemeinschaft

Fig. 12: Flyer of the first summer school in 2018.

The summer school began with an official kick-off meeting on 04.09.2018 in Stuttgart where the approved summer school project proposals were presented by the DROPLIT doctoral researchers. Subsequently, the project groups worked independently within themselves, collaborating with the

invited experts of each group over a period of two and a half weeks at various locations specified in the proposals. After this period of intensive work, all the participants met again for the final meeting on 21.09.2018 in Stuttgart, where they presented the work accomplished and prospects of collaborative work. This was followed by a farewell evening, taking place on the TV-tower in Stuttgart. During the summer school project, the teams worked in Stuttgart and in Bergamo.

2nd International summer school (10.06.2022 - 30.06.2022)

The second summer school in 2022 was held in the same way as the first one in 2018. The short summer school project proposals were submitted by different teams, and were reviewed and



Fig. 13: Participants of the DROPIT Summer School 2022 in Bergamo, Italy.

approved. This resulted in seven approved summer school projects. In addition, there were invited lectures and keynote speeches by experts in the field of droplet dynamics and related topics during the period of the summer school. The summer school began with an official kick-off meeting on 10.06.2022 in Stuttgart where the approved summer school project proposals were presented by the DROPIT doctoral researchers. Subsequently, the project groups spent three weeks working together with the invited experts of each group at their respective locations, as specified in the proposals. After this period of intensive work, all the participants met again for the final meeting on 30.06.2022 in Bergamo, where they presented the work accomplished and prospects of collaborative work. During the summer school project, the teams worked in Stuttgart and in Bergamo.

The work carried out in the individual groups at the two summer schools continued in several groups even after the summer schools had ended, leading to collaboration with various international groups. Furthermore, the work begun at both summer schools resulted in several publications, e.g.,

- **Foltyn, P.**, Guttman, M., Schneider, M., **Fest-Santini, S.**, Wildenschild, D., Weigand, B., 2020. Fabrication and Evaluation Methods of Micro-structured Surfaces for Droplet Impact Experiments, in: Lamanna, G., Tonini, S., Cossali, G.E., Weigand, B. (Eds.), Droplet Interactions and Spray Processes. Springer International Publishing, Cham, pp. 71–86. https://doi.org/10.1007/978-3-030-33338-6_6
- **Hu, Y.**, Tonini, S., Cossali, G., Weigand, B., Reuttsch, J., Frey, M., 2023. Evaporation of an oscillating drop, in: Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigations. Bergamo, Italy. https://doi.org/10.13122/DIPSI2023_4

Final Colloquium (31.07.2025 – 01.08.2025)

The Final Colloquium of DROPIT took place on 31.7.2025 – 01.08.2025. The photograph below shows the participants of the Final Colloquium.



Fig. 14. Participants of the Final Colloquium in Stuttgart in 2025.

The colloquium was structured in such a way to give all the participants an overview of the large progress, which had been achieved in the programme and in the individual subprojects. In addition, key note talks had been given by the two rectors of the involved Universities (Peter Middendorf for the University of Stuttgart and Sergio Cavaliere for the University of Bergamo) addressing the importance of this cooperation for both Universities.

We were really happy that also a large number of doctoral researchers of the first and second generation joined the Final Colloquium and gave a common talk with the last generation of doctoral researchers (third generation) about their subprojects.

In addition, a poster session was included, where detailed discussions on the individual subprojects took place.

Besides the talks of the doctoral researchers, also invited guests contributed to the programme. The invited guests were:

- **Remi Abgrall** (University of Zürich): “Uniformly high-order bound-preserving OEDG schemes for two-phase flows”. A joint work with Kailiang Wu and Ruifang Yan (Sustech, China)
- **Anne Geppert** (University of Bremen): “Melt atomization and Spray Forming” by Anne Geppert
- **Sergei Sazhin** (University of Brighton): “A combined analytical/numerical approach to modelling heat/mass transfer processes in droplets”.

Also, the final colloquium was organized by the doctoral researchers and the postdoc. The organisational setup, thus, was similar to the summer schools.

Because of the real positive feedback from several participants of the final colloquium, we are currently discussing to summarize the scientific progress of this project within a second book on this topic within the next two years.

Promotion of Equality: Female RTG Participants by Qualification Level

(see also Chapter 6.2.3)

The University of Stuttgart offers an elaborate infrastructure that is tailored to implement gender equality and more specifically to attract, keep and promote the career chances of talented women within the university. The university has chosen a holistic approach to increase the participation of women in science, which targets female students from high-school to postdoctoral and professional levels. The entire portfolio encompasses the following initiatives, e.g.: **Girl’s Day; TryScience; StartScience; Praktikums-Info-Börse; a gender- and diversity-sensitive Mentoring Programme; Femtec.Network; Service Uni & Familie and Service Gender Consulting**. Thanks to this elaborate gender concept, the University of Stuttgart is certified as “family-friendly higher education institution”. In this context, the present IRTG could benefit from a number of already established measures and programmes:

- To promote work/family balance
- To support women’s career chances in industry, science and academia
- To endorse women’s understanding and access to key technologies, which is also a key-aspect for fostering gender equality.

Concerning work/family balance, the university pre-school institutions and kids-vacation programme provide a first basis for child-care solutions. Supporting the career opportunities for women in the scientific and technical sector can take place within the framework of the Femtec.Network.

In the framework of this IRTG, the mentoring activities have been expanded by setting up an international mentoring programme as explained before. Each doctoral researcher had been assigned

an international mentor. For example, during the annual doctoral seminar there was the possibility to discuss career possibilities and career development strategies also with the Italian/German supervisors. In addition, during visits of the doctoral researchers at the partner university, this could be done. The inclusion of foreign mentors enabled to improve the researcher's intercultural competence. This encompasses the enhancement of intercultural understanding and tolerance, the capability of communicating effectively and gaining more confidence in interacting with professionals from cultural backgrounds different to their own.

In addition, researchers and PIs of DROPIT acted as mentors or role models. Here, e.g. the DROPIT postdoc Anne Geppert participated in **TryScience, StartScience** and had a Mentee in the framework of the CyberMentor-programme (<https://www.cybermentor.de/index.php>).

DROPIT also supported gender equality through pooling funds. The pooling fund is centrally located in the administration. With these funds, comprehensive gender equality measures have been implemented, which supported different DFG research projects and could be shared, e.g., equipment of a parent-child room on the Vaihingen campus or participation in the basic fee for flexible childcare with two providers.

In collaboration with **uniquUS**, career support measures were offered to female postdocs (e.g., A. Geppert), covering different aspects from visibility measures to the development of didactic, communication and management skills to foster their independence. Finally, a strong support was given also on promoting the visibility of female scientists and their work. This occurred in three parallel ways. First, in collaboration with **uniquUS** within the Pooling initiatives, the **Junior Professor Carina Bringedal** (former PI in DROPIT) and **Dr. Anne Geppert** (PostDoc) were selected for a scientific portrait in the programme "**Women in STEM make a difference**" to make their research visible to the general public and as successful role models for early career researchers (ECR) (<https://www.youtube.com/playlist?list=PLGuevu4esbwEy-0zrvy7GeeFD0Lra6cQa>).

Second, for the general public, the research topic of Anne Geppert was also proposed to **SWR** to be broadcasted in the TV show "**Das Wissen**" (<https://www.swr.de/swrkultur/wissen/tropfenforschung-eine-wissenschaft-fuer-alle-lebenslagen-100.html>).

This visibility measure opened also the opportunity for a joint research project with the Paul Scherrer Institute (CH) on neutron imaging investigation of liquids mixing during droplet impact onto liquid films. The project had Dr. Anne Geppert as lead scientist and included several DROPIT doctoral and associated members. The result of this collaboration, partly financed by DROPIT, were presented at the ILASS-Europe 2025 conference in Lund (S), which represents one of the most successful measures in promoting the research and visibility of female, early career researchers. Third, DROPIT also actively supported the participation of female ECRs in the **Fellowship-Programme Zia – Visible Women in Science**. During the one-year fellowship program, female ECRs gain a unique opportunity to support further their professional and personal development by interacting with female

role models, institutional representatives from academia, research, business, and the public sector. This enables also the creation of a wide, professional network while simultaneously improving their communication skills. In DROPIT, the doctoral researcher **Claudia Marianowski** could benefit from the ZIA Fellowship. Today she is working as mechanical and thermal engineer at TESAT.

As a big achievement, it can be noted that two doctoral researchers of DROPIT (**Martina Baggio** and **Karin Schlottke**) have won the very prestigious **Amelia Earhart Fellowship in 2019**. In an effort to carry out its mission that women have access to all resources and are represented in decision-making positions on an equal basis with men, **Zonta International offers the Amelia Earhart Fellowship**. The Amelia Earhart Fellowship was established in 1938 in honour of famed pilot and Zontian, Amelia Earhart. **The US\$10,000 Fellowship is awarded annually to up to 30 women worldwide pursuing Ph.D. degrees in aerospace engineering and space sciences**. See also: <https://foundation.zonta.org/Our-Programs/Educational-Programs/Amelia-Earhart-Fellowship>.

Finally, it should be stated that we believe that already in school females have to be educated in a way to interest them in engineering sciences. Therefore, DROPIT has started in FP1 to take additional measures to promote the topic of research of this IRTG to children. Thus, a lecture has been given in the “**kids university**” in Heilbronn on 15.01.2020 by K. Schulte and B. Weigand entitled: “**The wonder world of drops**”. Here roughly 250 kids participated. Furthermore, DROPIT participates in **TryScience** at the University of Stuttgart. In this workshop the children receive information about the courses of study and research, they carry out small experiments themselves, receive mini-lectures and visit laboratories and experimental halls. The CampusTour provides an insight into the everyday life of the students.

Impact of the Qualification Programme on Structure and on Future Projects

The qualification programme had a lot of impacts on the involved universities.

The “Multiphase Flow Seminar” at the University of Stuttgart will continue to serve as a fine example of how the work of doctoral researchers and invited guests can be presented to students, for instance, and how students’ interest in the doctoral researchers’ research areas can be sparked. This seminar continues to be led by A. Beck, C. Rohde and B. Weigand and attracts participants from the fields of aerospace engineering and mathematics.

The **visiting scientists programme** resulted in a strong visibility of DROPIT and the involved universities. A large number of cooperation with the visiting scientists are still continued after the end of the project. Several of them have been involved e.g., as reviewers on doctoral thesis.

At the University of Bergamo, the **DIPSI workshop** is continued after the end of the project. In 2026, the 18th edition of the workshop will take place on 19.06.2026 in Bergamo with the participation of several DROPIT members.

Also, the workshop “**AI meets Scientific Computing**”, which was held for the first time in 2025 in Bergamo shall be continued because of its strong success of it in 2025. This workshop involves scientists from Bergamo and from Stuttgart beside invited guests.

Furthermore, the course “**Dimensional Analysis**”, which was one of the core courses in the DROPIT project and which was also held in Bergamo has now been implemented by S. Tonini into the curriculum in Bergamo.

Because of the very good cooperation between the University of Stuttgart and the University of Bergamo several students decided for a stay at the other university for writing there a Bachelor- or a Master thesis. There had been several students using this opportunity during the past nine years. Also, after the end of the project this exchange process is continued, e.g. in the moment one student (Jonathan Behnk) has started his Bachelor thesis in Bergamo.

The intense cooperation between University of Bergamo and University of Stuttgart fostered new initiatives, such as the Double Master's Degree Program Mechatronics University of Stuttgart (USTUTT), Mechatronics and Smart Technology Engineering (UniBG).

As explained at the beginning of **Chapter 6.1** by answering the notes and recommendations arising from the review of the renewal proposal, we have not been entirely successful in getting all doctoral researchers to spend a longer time in Bergamo. However, we will now send doctoral researchers of the third generation for some remaining time to Bergamo in order to have a joint degree (cotutelle). One doctoral researcher, where already everything is planned (research stay from May-August 2026 planned, the document for cotutelle is set up) will be Rishav Saha. End of this year or at the beginning of next year Hasel Konan will also go for a longer research stay to Bergamo.

As a final remark, we would like to state that this project included the full time of the Coronavirus Pandemic, which had been especially in Bergamo extremely dramatically. We had to fight with a lot of challenges and we had to notice that the attitude of the doctoral researchers changed very much. Nevertheless, all participants enjoyed this project very much and a group, starting as cooperative scientists ended up as friend.

6.2.1 List of all persons funded by the Research Training Group to date

Doctoral record of IRTG doctoral researchers (financed by the DFG through RTG funds)

	Name	Supervisors	Field of study; university and degree date	Start of doctoral training	Funded mem- ber of IRTG (from / to)	For on- going doctoral studies: time to date (in months)	If appro- priate: interrup- tion in fund- ing (in months)	After submis- sion of thesis: time un- til sub- mission (in months)	Time of oral examination	Grade	Occupation(s) fol- lowing doc- torate	Cf. in- divid- ual re- port on page
1	Sina Ackermann	R. Helmig, M. Santini, B. Weigand	M.Sc. in Simula- tion Technology, University of Stuttgart; 03. 2016	01.10.2016	01.10.2016 / 30.09.2019	--	--	--	19.06.2020	Summa cum laude	Postdoc at University of Stuttgart, now with BOSCH	Report FP1
2	Martina Baggio	B. Weigand, M. Dumbser	M.Sc. in Aero- space Engineer- ing, University of Stuttgart, 12.2015	01.10.2016	01.10.2016 / 30.09.2019	--	--	--	16.07.2020	Summa cum laude	Diehl Defence	Report FP1
3	Lukas Ostrowski	C. Rohde, F. Bassi	M.Sc. in Mathe- matics, Univer- sity of Stuttgart, 10.2015	01.10.2016	01.10.2016 / 30.09.2019	--	--	--	10.02.2021	Summa cum laude	ZEISS SMT	Report FP1
4	Ronan Bernard	G. Lamanna, S. Tonini, N. Roth	M.Sc. in Me- chanical Engi- neering, KIT (Karlsruhe) & ENSAM (France), 04.2016	15.11.2016	15.11.2016 / 14.11.2019	--	--	--	27.07.2020	Summa cum laude	Recipient of DROPIT StartUp- Funding; now with Spraying Systems	Report FP1
5	Alexander Straub	T. Ertl, C.-D. Munz, M. Santini	M.Sc. in Com- puter Science, University of Stuttgart, 11.2016	01.12.2016	01.12.2016 / 30.11.2019	--	--	--	06.05.2024	Magna cum laude	Vector In- formatik GmbH	Report FP1
6	Wladimir Reschke	S. Fasoulas, F. Bassi	M.Sc. in Aero- space Engineer- ing, University of	01.01.2017	01.01.2017 / 31.12.2019	--	--	--	31.07.2020	Magna cum laude	Deutsches Patent- und Markenamt	Report FP1

			Stuttgart, 10.2015									
7	Christoph Müller	C.-D. Munz, C. Rohde, M. Dumbser	M.Sc. in Simula- tion Technology, University of Stuttgart, 02.2017	01.03.2017	01.03.2017 / 29.02.2020	--	--	--	12.04.2021	Summa cum laude	ZEISS	Report FP1
8	Jonas Zeifang	A. Beck, F. Bassi	M.Sc. in Aero- space Engineer- ing, University of Stuttgart, 02.2017	01.03.2017	01.03.2017 / 29.02.2020	--	--	--	30.06.2020	Summa cum laude	Postdoc at Hasselt Uni- versity, now at ZEISS	Report FP1
9	Patrick Foltyn	B. Weigand, G.E. Cossali, N. Roth	M.Sc. in Aero- space Engineer- ing, University of Stuttgart, 03.2017	01.05.2017	01.05.2017 / 31.03.2021*	--	--	--	30.08.2022	Summa cum laude	Alfred Kärcher Ver- triebs-GmbH	B1-1 p. 126
10	Weibo Ren	B. Weigand, M. Dumbser	M.Sc. in Aero- space Engineer- ing, University of Stuttgart, 05.2019	01.10.2019	01.10.2019 / 31.03.2023*	--	--	--	16.01.2026	Magna cum laude		B5-2 p. 144
11	Maziar Veyskarami	R. Helmig, M. Santini, C. Bringedal	M.Sc. in Reser- voir Engineering, Sharif University of Technology, 01.2017	01.10.2019	01.10.2019 / 31.03.2023*				12.07.2023	Summa cum laude	Postdoc at University of Stuttgart till 10.2023	B6-2 p. 149
12	Raphael Tietz	S. Fasoulas, L. Botti	M.Sc. in Aero- space Engineer- ing, University of Stuttgart, 04.2019	01.01.2020	01.01.2020 / 30.06.2023*	--	--	--	Still open		ZEISS	A3-1 p. 155
13	Stefan Schubert	G. Lamanna, S. Tonini	M.Sc. in Me- chanical Engi- neering, RWTH Aachen, 10.2019	15.01.2020	15.01.2020 / 14.07.2023*	--	--	--	Still open		Hensoldt	C1-2 p. 160
14	Daniel Appel	A. Beck, L. Botti	M.Sc. Mechan- ical Engineering, Technical Uni- versity of Mu- nich, 04.2019	01.03.2020	01.03.2020 / 31.08.2023*	--	--	--	Still open		University of Stuttgart	A6-2 p. 166

* Extension of the standard duration of doctoral positions in the IRTG due to the coronavirus pandemic

** Aborting of the doctoral project

15	Pascal Mossier	C.-D. Munz, C. Rohde, M. Dumbser	M.Sc. in Aerospace Engineering, University of Stuttgart, 01.2020	01.03.2020	01.03.2020 / 31.08.2023*	--	--	--	13.05.2024	Summa cum laude	Postdoc at University of Stuttgart	A2-2 p. 170
16	Andrej Baric	C. Rohde, L. Botti	M.Sc. in Mathematics, University of Stuttgart, 2020	01.03.2020	01.03.2020 / 31.03.2022**	--	--	--	--	--	--	B4-2 p. 175
17	Daniel Klötzl	T. Ertl, D. Weiskopf, M. Santini	M.Sc. in Mathematics, University of Tübingen, 2020	01.01.2021	01.01.2021 / 31.03.2024	--	--	--	Still Open		--	C4-2 p. 177
18	Jonathan Stober	K. Schulte, M. Santini, C. Bringedal	M.Sc. in Aerospace Engineering, University of Stuttgart, 03.2021	15.04.2021	15.04.2021 / 14.04.2025	--	--	--	04.02.2026	Summa cum laude	Tplus Engineering GmbH	C2 p. 183
19	Patrick Alois Palmetshofer	B. Weigand, G.E. Cossali	M.Sc. in Aerospace Engineering, University of Stuttgart, 2021	01.05.2021	01.05.2021 / 31.03.2025	--	--	--	04.03.2025	Summa cum laude	Boston Consulting Group	B1-2 p. 189
20	Anna Mareike Kostecky	R. Helmig, M. Santini, H. Dahle, S. Kjelstrup	M.Sc. in Simulation Technology, University of Stuttgart, 08.2022	01.10.2022	01.10.2022 / 30.09.2025	--	--	--	Still open			B6-3 p. 196
21	Hasel Cicek Konan	C. Rohde, L. Botti	M.Sc. in Applied Mathematics, University of Siena, 2022	08.12.2022	08.12.2022 / 07.12.2025	--	--	--	Still open			B4-3 p. 207
22	Claudia Marianowski	S. Fasoulas, L. Botti	M.Sc. in Aerospace Engineering, University of Stuttgart, 2022	01.01.2023	01.01.2023 / 30.09.2024**	--	--	--	--	--	TESAT	A3-3 p. 216
23	Pablo Molina Vogelsang	G. Lamanna, S. Tonini	M.Sc. in Aerospace Engineering, University of Stuttgart, 2021	16.01.2023	16.01.2023 / 31.12.2025	--	--	--	Still open			C1-3 p. 220
24	Rishav Saha	B. Weigand, M. Dumbser	M.Sc. in Chemical and Energy Engineering, Otto-von-Guericke-University Magdeburg, 11.2022	16.01.2023	16.01.2023 / 31.12.2025	--	--	--	Still open			B5-3 p. 226

25	Amalia Travnicek	A. Beck, L. Botti	M.Sc. in Aerospace Engineering, University of Stuttgart, 2022	01.04.2023	01.04.2023 / 31.12.2025	--	--	--	Still open			A6-3 p. 233
26	Rishi Dhar	K. Schulte, S. Tonini	M.Sc., RWTH Aachen, 01/2024	01.02.2024	01.02.2024 / 31.12.2025	--	--	--	Still open			A2-3 p. 243
27	Kim-Sophie Ellenberger	S. Fasoulas, L. Botti		01.11.2024	01.11.2024 / 30.09.2025	--	--	--	Still open			A3-4 p. 254

Doctoral record of RTG doctoral researchers (in Germany, funded by other sources)

	Name	Supervisors	Field of study; university and degree date	Start of doctoral training	Member of IRTG (from / to)	For on-going doctoral studies: time to date (in months)	If appropriate: interruption in funding (in months)	After submission of thesis: time until submission (in months)	Time of oral examination	Grade	Occupation(s) following doctorate	Cf. individual report on page
1	Anne Katharina Geppert	B. Weigand, G. Lamanna	Dipl.-Ing. in Aerospace Engineering; University of Stuttgart, 11.2011	01.01.2012	01.10.2016 / 17.04.2019	--	--	--	17.04.2019	Summa cum laude	Postdoctoral fellowship in DROPIT, Leibniz-Inst. für Werkstofforientierte Technologien, Bremen	Report FP1, AS01
2	Karin Schlottke	B. Weigand	Dipl.-Ing. in Aerospace Engineering; University of Stuttgart, 12.2013	17.03.2014	01.10.2016 / 30.03.2021	--	--	--	--	--	Research Scientist at HLRS in Stuttgart	Report FP1, AS02
3	Melanie Lipp	R. Helmig, C-D. Munz, R. Masson	M.Sc. in Physics, University of Stuttgart, 04.2017	28.06.2017	01.03.2019 / 14.06.2024	--	--	--	14.06.2024		Württembergische Versicherung	AS05 p. 264
4	Yanchao Liu	B. Weigand	M.Sc. in Aerospace Engineering, University of Stuttgart, 07.2017	15.09.2018	01.03.2019 / 30.03.2021	--	--	--	19.03.2024	Magna cum laude	Company in China	AS04 p. 259

5	Dennis Hoch	R. Helmig, J. Niessner, B. Weigand	Mech. Engineering, Hochschule Heil- bronn 09.2019	01.03.2019	01.03.2019 / to date	--	--	--	--		--	AS06 p. 269
6	Verena Kunberger	B. Weigand	M.Sc. in Aerospace Engineering, Uni- versity of Stuttgart, 2019	01.06.2019	01.02.2023 / 30.06.2024	--	--	--	Still Open		Tesat-Space- com GmbH & Co. KG	AS09 p. 280
7	Jonathan Wurst	K. Schulte	M.Sc. in Process Engineering, Uni- versity of Stuttgart, 09.2020	01.03.2021	01.01.2024 / 30.09.2025	--	--	--	Still Open			AS11 p. 290
8	Florian Wendt	C. Rohde	M. Sc. in Mathe- matics, University of Tübingen, 10.2023	01.01.2024	01.01.2024 / 30.09.2025	--	--	--	Still Open			AS10 p. 284

Doctoral record of doctoral researchers at the foreign partner institution

	Name	Supervisors	Field of study; university and degree date	Start of doctoral training	Member of IRTG (from / to)	For on- going doctoral studies: time to date (in months)	If appropri- ate: inter- ruption in funding (in months)2	After sub- mission of thesis: time until sub-mis- sion (in months)	Time of oral examination	Grade	Occupation(s) following doc- torate	Cf. in- divid- ual re- port on page
1	Francesco Carlo Massa (BG)	F. Bassi	M. Sc. Mechanical Engineering, University of Brescia, 07.2013	01.01.2014	01.10.2016 / 10.05.2017	--	--	--	10.05.2017	--	Research at University of Bergamo	Report FP1
2	Nguyen Tri Nguyen (T)	M. Dumbser, R. Klein, A. Hidalgo, M. Semplice	M. Sc. Mechanical Engineering, Vietnam	01.04.2014	01.10.2016 / 27.04.2017	--	--	--	27.04.2017	--	--	--
3	Robert Haide (BG)	M. Santini, T. Ertl	M. Sc. in Aero- space Engineer- ing, University of Stuttgart, 02.2017	01.10.2017	01.10.2017 / 12.07.2021	--	--	--	12.07.2021	--	Deutsches Patent- und Markenamt	Report FP1

4	Gautham Varma Raja Kochanattu (BG)	G.E. Cossali, S. Tonini, B. Weigand	M.Sc. in Advanced computational methods in Material science, Gdansk University of Technology, 10.2016	01.10.2017	01.10.2017 / 30.03.2021	--	--	--	--	--	--	--
5	Simone Chiocchetti (T)	M. Dumbser, C.-D. Munz	M. Sc. in Environmental Engineering, University of Trento, 03.2018	05.11.2018	01.05.2018 / 10.06.2022	--	--	--	10.06.2022	--	Start-up grant, Marie Skłodowska-Curie Post-doctoral Fellowship at University of Cologne	A4-1 p. 131
6	Manuel Giovanni Gritti (BG)	M. Santini, B. Weigand	M. Sc. in Civil Engineering, University of Bergamo, 03.2018	01.10.2018	01.10.2018 / 31.12.2021	--	--	--	03 / 2023	--	--	B2-2 p. 137
7	Luca Verzeroli (BG)	L. Botti	M.Sc. in Mechanical Engineering, University of Bergamo, 12.2017	01.10.2018	01.10.2018 / 30.09.2021	--	--	--	05.04.2022		Lovato Electric	AS07 p. 275
8	Yueqiao Hu (BG)	G.E. Cossali, S. Tonini, B. Weigand	M. Sc. in Aerospace Engineering, University of Stuttgart, 03.2022	01.10.2022	01.10.2022 / 30.09.2025	--	--	--	Thesis submitted		--	A1 p. 202
9	Davide Ferrari (T)	M. Dumbser, C.D. Munz, A. Beck	M.Sc. in Civil, Environmental and Mechanical Engineering, University of Trento	01.10.2023	01.10.2023 / 17.04.2024	--	--	--	17.04.2024		--	A4-2 p. 212
10	Daniel Regener Roig (BG)	A. Colombo	M.Sc. in Aeronautical Engineering, Universitat Politècnica de Catalunya, 11.2023	01.11.2023	01.11.2023 / 31.12.2025	--	--	--	Still Open		--	AX p. 238

11	Daniele Rusconi (BG)	M. Santini, S. Fest-Santini, D. Weißkopf	M.Sc. in Mechanical Engineering, University of Bergamo, 09.2024	01.11.2024	01.11.2024 / 31.12.2025	--	--	--	Still Open		--	B3 p. 249
12	Sara Rinaldi (T)	M. Dumbser, O. Zanotti	M.Sc. in Applied Mathematics, University of Rome, 07.2024	01.01.2025	01.01.2025 / 31.12.2025	--	--	--	Still Open		--	A4 p. 252

IRTG postdoctoral researchers (financed by the DFG through RTG funds)

Name	Project/topic	Doctorate (subject, date and place obtained)	Member of DROPIT (from / to)	Individual tasks relating to DROPIT	Occupation(s) after leaving DROPIT	Cf. individual report on page
Visakh Vaikuntanathan	Development of novel optical techniques for micro-fluid dynamics	Aerospace Engineering, Indian Institute of Science, 03.2016	01.05.2017 / 30.04.2019	Development of experimental methods for micro-fluid dynamic research, with focus of droplet impact on wetted walls. Mentoring, Supervision, Organization	Post-doctoral researcher in a DFG-funded project at ITLR.	Report FP1
Anne K. Geppert	Development of novel optical techniques for micro-fluid dynamics	Aerospace Engineering, University of Stuttgart, 04.2019	01.06.2019 / 30.09.2024	Development of an analytical modelling approach for crown propagation after drop impact. Coordination of all experimental methods used within DROPIT. Mentoring, Supervision, Organization	Head of Department at Leibniz-IWT Bremen	C5-2 p. 297
Manish Kumar	Direct numerical simulations of droplet evaporation		01.10.2024 / 31.12.2025 (50%)	Direct numerical simulations of droplet evaporation	Postdoc at the University of Stuttgart	C5-3 p. 305
Patrick Palmetshofer	Droplet impact on structured surfaces	Aerospace Engineering, University of Stuttgart, 03.2025	01.04.2025 / 15.06.2025 (50%)	Neutron Imaging investigations at Paul Scherrer Institute	Consultant at Boston Consulting Group	C5-4 p. 309

RTG postdoctoral researchers (funded by other sources)

Name	Project/topic	Doctorate (subject, date and place obtained)	Member of DROPIT (from / to)	Individual tasks relating to DROPIT	Occupation(s) after leaving DROPIT	Cf. individual report on page
Stephanie Fest-Santini (BG)	Usage of Micro-CT for detailed surface description	Heat transfer during cooling processes with intermittent sprays above the Leidenfrost point, Otto-von-Guericke University Magdeburg, 22-10-2009	01.10.2016 / 30.04.2020	Support function regarding data acquisition, reconstruction, segmentation and surface triangulation to guaranty high-quality microCT data for the thematic area B	Associate Professor at University of Bergamo	Report FP1
Francesco Carlo Massaro (BG)	High-order numerical methods for multi-component incompressible flows	Engineering and Applied Sciences, University of Bergamo, 10-5-2017	01.10.2016 / 31.12.2025	Development of a high-order implicit discontinuous Galerkin approach for the simulation of variable density incompressible flows with focus on the definition of the interface Godunov fluxes, the treatment of spurious oscillations, the positivity of the density	Researcher at University of Bergamo	AS02 p. 312

IRTG researchers who received start-up funding after their doctorates

Name	Aim of funding (e.g. individual project proposal on topic XY)	Date of doctorate	Funding period	Type of funding (position/fellowship, direct project costs, etc.)	Occupation(s) after start-up funding	Cf. individual report on page
Ronan Bernard	droplet-on-demand methods for additive manufacturing	27.07.2020	01.08.2020 / 31.03.2021	Position for writing a proposal and funds for experiments	Spraying Systems Co.	Report FP1
Simone Chiochetti	alternative subgrid generation strategies, involving nonconforming quadrilaterals and nested multi-level Voronoi cells	10.06.2022	16.05.2023 / 31.12.2023	Position for writing a proposal and travel funds	Marie Skłodowska-Curie Postdoctoral Fellowship at University of Cologne	Report p. 318

Qualifying fellows

Name	Supervisors	Academic qualification (type of qualification, subject, date and university)	Member of RTG (from / to)	Date of admission to doctoral programme	Occupation(s) after completion of qualifying phase	Cf. individual report on page
--	--	--	--	--	--	--

Student assistants (support staff)

Name	Supervisors	Course/field of study	Member of DROPIT (from / to)	Tasks in IRTG
Patrick Foltyn	G. Lamanna	Aerospace Engineering	01.12.2016 / 31.01.2017	FP1
Rupalee Baldota	S. Ackermann	Environmental Protection Technology	01.01.2017 / 31.03.2017	FP1
Tim Jupe	S. Ackermann	Environmental Protection Technology	01.01.2017 / 30.06.2017	FP1
Miriam Feil	S. Ackermann	Environmental Protection Technology	01.01.2017 / 31.03.2017	FP1
Florian Schwarz	S. Ackermann	Environmental Protection Technology	01.01.2017 / 31.03.2017	FP1
Marcel Hörmann	S. Ackermann	Environmental Protection Technology	01.01.2017 / 31.03.2017	FP1
Marcel Hörmann	S. Ackermann	Environmental Protection Technology	16.04.2017 / 30.06.2017	FP1
Florian Schwarz	S. Ackermann	Environmental Protection Technology	01.05.2017 / 30.06.2017	FP1
Hagen Leipprand	G. Lamanna	Aerospace Engineering	15.05.2017 / 15.08.2017	FP1
Maha Badri	G. Lamanna	Aerospace Engineering	01.06.2017 / 30.09.2017	FP1
Alaa Bejaoui	G. Lamanna	Aerospace Engineering	01.06.2017 / 31.03.2018	FP1
Lukas Münch	R. Bernard	Aerospace Engineering	01.06.2017 / 30.09.2017	FP1
Carl Quast	R. Bernard	Aerospace Engineering	01.06.2017 / 31.12.2017	FP1
Victor Dubois	P. Foltyn	Aerospace Engineering	01.09.2017 / 31.12.2017	FP1
Sai Ranjith Reddy Lomada	B. Weigand	Aerospace Engineering	15.10.2017 / 31.12.2017	FP1
Christian Kunzi	B. Weigand	Aerospace Engineering	15.10.2017 / 31.12.2017	FP1
Matthias Ringkowski	R. Bernard	Aerospace Engineering	01.11.2017 / 31.01.2018	FP1
Maha Badri	G. Lamanna	Aerospace Engineering	01.01.2018 / 31.03.2018	FP1
Sai Ranjith Reddy Lomada	B. Weigand	Aerospace Engineering	01.01.2018 / 30.09.2018	FP1
Victor Dubois	P. Foltyn	Aerospace Engineering	01.01.2018 / 31.03.2018	FP1

Carl Quast	R. Bernard	Aerospace Engineering	01.01.2018 / 15.05.2018	FP1
Christian Kunzi	B. Weigand	Aerospace Engineering	15.01.2018 / 30.09.2018	FP1
Johannes Müller	G. Lamanna	Aerospace Engineering	01.03.2018 / 30.06.2018	FP1
Matthias Ringkowski	R. Bernard	Aerospace Engineering	01.03.2018 / 30.06.2018	FP1
Tobias Kurtz	G. Lamanna	Aerospace Engineering	17.03.2018 / 17.09.2018	FP1
Andrea Hinkel	W. Reschke	Aerospace Engineering	01.04.2018 / 31.03.2019	FP1
Calvin Ebert	R. Bernard	Aerospace Engineering	15.04.2018 / 31.07.2018	FP1
Steven Jöns	C. Müller	Aerospace Engineering	01.06.2018 / 31.07.2018	FP1
Ferdinand Restle	P. Foltyn	Aerospace Engineering	15.06.2018 / 31.08.2019	FP1
Julian Härter	G. Lamanna	Aerospace Engineering	01.07.2018 / 31.08.2018	FP1
Dominik Münch	A. Beck	Aerospace Engineering	01.07.2018 / 31.12.2018	FP1
Tihomir Varchev	P. Foltyn	Aerospace Engineering	01.07.2018 / 31.03.2019	FP1
Tobias Kurtz	G. Lamanna	Aerospace Engineering	01.11.2018 / 30.09.2019	FP1
Eli Aoun Abi	R. Bernard	Aerospace Engineering	01.01.2019 / 15.03.2019	FP1
Pia Schuler	P. Foltyn	Aerospace Engineering	01.03.2019 / 31.08.2019	FP1
Thomas Bernhard	R. Bernard	Mechatronics	01.06.2019 / 30.09.2019	FP1
Julian Schmid	V. Vaikuntanathan	Aerospace Engineering	16.06.2019 / 31.07.2019	FP1
Calvin Ebert	R. Bernard	Aerospace Engineering	01.08.2019 / 01.12.2019	FP1
Daniel Maurat	A. Beck	Aerospace Engineering	15.08.2019 / 14.12.2019	FP1
Elie Abi Aoun	W. Ren	Aerospace Engineering	14.02.2020 / 09.09.2020	Numerical computations with FS3D, DNS.
Heinz Heinrich Heinzer	A. Beck	Aerospace Engineering	01.05.2020 / 31.07.2020	Setting up regression testing and continuous integration for a FLEXI branch for multiphase flows with interface tracking for droplet-dynamic processes
Frederic-Wilhelm Rapp	A. Beck / D. Appel	Aerospace Engineering	15.06.2020 / 31.12.2020	Wall modeling in LES simulations of droplet flows (literature research, code development, definition of test cases, model comparisons, etc.).
Ferdinand Restle	P. Foltyn	Aerospace Engineering	01.07.2020 / 31.12.2020	Scientific support for data management, visualization and processing; development of evaluation routines
Tihomir Varchev	P. Foltyn	Aerospace Engineering	01.10.2020 / 20.09.2020	Numerical simulation of droplets on regular-structured surfaces (code development and evaluation)
Facen Zhou	W. Ren	Aerospace Engineering	01.10.2020 / 31.03.2021	Webhiwi for DROPIT Homepage
Hamza Dounassre	S. Schubert	Aerospace Engineering	01.10.2020 / 30.09.2021	Support in the implementation of the "defocusing" method for drop-wall film interactions
Lynn Rihm	P. Foltyn / A. K. Geppert	Aerospace Engineering	01.12.2020 / 28.02.2022	Further development of the LPSM measurement method and evaluation of experiments/results
Chris Bürger	A. K. Geppert	Aerospace Engineering	01.12.2020 / 31.05.2021	Support in the automation of the analytical calculation of crown parameters and their comparison with

				experimental data using the programming language Matlab
Elizabeth Gutierrez Rojas	A. Geppert	Aerospace Engineering	16.12.2020 / 31.07.2021	Implementation of constructive designs of new test equipment, scientific support
Frederic-Wilhelm Rapp	A. Beck / D. Appel	Aerospace Engineering	01.02.2021 / 31.03.2021	Wall modeling in LES simulations of droplet flows (literature research, code development, definition of test cases, model comparisons, etc.).
Martin Froch	A. K. Geppert	Aerospace Engineering	01.03.2021 / 30.04.2021	Preparation of droplet impact experiments on stand-alone surface structures; literature review
Christian Beck	A. Baric / C. Rohde	Mathematics	16.03.2021 / 31.03.2022	Research on diffuse interface models and conduction of simulations
Frederic Rapp	A. Beck / D. Appel	Aerospace Engineering	01.04.2021 / 30.04.2021	Wall modeling in LES simulations of droplet flows (model comparisons).
Yuchen Wan	W. Ren	Aerospace Engineering	15.04.2021 / 15.11.2021	Numerical computations with FS3D, DNS.
Nils Janßen	S. Schubert	Aerospace Engineering	15.04.2021 / 31.01.2022	Support by development of the optical setup for micro and macro high-speed imaging experiments on droplet impact
Vibhav Sawant	R. Bernard	Aerospace Engineering	15.06.2021 / 31.08.2021	Support by experiments on multiple and simultaneous droplet impact
Sicong Wu	A. K. Geppert	Aerospace Engineering	15.06.2021 / 15.11.2021	Programming post-processing routines
Lisa Bachmann	A. K. Geppert	Aerospace Engineering	01.07.2021 / 31.03.2023	Conducting and analyzing of early impact dynamics experiments for droplet impact onto thin liquid films
Patrick Wolff	P. Palmetschofer	Aerospace Engineering	15.07.2021 / 31.08.2021	Support for experimental investigations
Nathalie Kludt	S. Schubert	Aerospace Engineering	01.10.2021 / 31.10.2021	Support by development of the optical setup for microscopic particle imaging velocimetry
Yasmine Saidi	R. Helmig	Civil Engineering	01.10.2021 / 31.03.2022	Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Leonardo Burkart	P. Palmetschofer	Aerospace Engineering	15.11.2021 / 30.06.2022	Support for experimental investigations
Dantong Li	W. Ren	Aerospace Engineering	16.11.2021 / 31.03.2022	FS3D, DNS calculations, data analysis
Dorian Benedict Brajer	A. K. Geppert / G. Lamanna	Aerospace Engineering	01.03.2022 / 31.12.2025	Support for analyzing experiments for droplet impact onto stand-alone surface structures
Qianrong Yi	W. Ren	Aerospace Engineering	15.04.2022 / 31.03.2023	FS3D, DNS calculations, data analysis
Max Hohenberger	S. Schubert	Aerospace Engineering	15.05.2022 / 15.09.2022	Support for the measurement campaign by microscopic particle imaging velocimetry in the crown crater
Alexander Ring	J. Stober	Aerospace Engineering	01.07.2022 / 31.08.2022	Support postprocessing of FS3D simulation results
Tim Arcos Marz	P. Palmetschofer	Aerospace Engineering	01.08.2022 / 15.05.2023	Experimental investigations on droplet impact on structured walls
Alexander Mühleisen	P. Palmetschofer	Aerospace Engineering	15.08.2022 / 30.04.2023	Experimental investigations on droplet impact on structured walls

				Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Helena Kschidok	R. Helmig	Civil Engineering	01.09.2022 / 28.02.2023	
Joelle Schäfer	R. Tietz	Aerospace Engineering	15.09.2022 / 31.03.2023	Numerical computations with PICLas
Simon Weitling	J. Stober	Aerospace Engineering	15.10.2022 / 15.01.2023	Support in experimental work on oblique impacts on thin films
				Support for the measurement campaign by microscopic particle imaging velocimetry in the uprising lamella
Salih Baykal	G. Lamanna	Aerospace Engineering	01.12.2022 / 31.05.2023	
Johanna Hintz	D. Appel / A. Travnicek	Aerospace Engineering	01.01.2023 / 31.07.2024	Extension of the compressible sharp-interface framework to the full Navier-Stokes equations by introducing viscous interface fluxes
Luis Wüstner	P. Molina Vogelsang	Aerospace Engineering	15.03.2023 / 29.02.2024	Matlab PIV postprocessing
Marvin Tigre Larschow	A. K. Geppert	Aerospace Engineering	01.04.2023 / 30.09.2023	Support installing experimental facilities
Leonard Brandstetter	S. Schubert / P. Molina Vogelsang	Aerospace Engineering	01.04.2023 / 30.11.2023	CAD work for adaptations of experimental test stand
				Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Torben Schiz	R. Helmig	Civil Engineering	01.04.2023 / 31.03.2024	
Sarah Altenkrüger	C. Marianowski	Aerospace Engineering	15.05.2023 / 31.01.2024	Numerical computations with PICLas
				Support in the development of a post-processing code for measuring velocity in the uprising lamella with geometrical transposition
Moritz Stein	S. Schubert	Aerospace Engineering	15.05.2023 / 15.08.2023	
				Development and testing of the optical flow software for event-based velocimetry, applied to droplet impact
Nicklas Essig	G. Lamanna	Aerospace Engineering	01.06.2023 / 30.09.2024	
Sarah Altenkrüger	C. Marianowski		01.10.2023 / 31.01.2024	Numerical computations with PICLas
Lucas Kytölä	P. Palmethofer	Aerospace Engineering	01.12.2023 / 31.05.2024	Experimental investigations on droplet impact on wall structures
				Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Torben Schiz	R. Helmig	Civil Engineering	07.12.2023 / 31.03.2024	
Hannes Minzenmay	P. Molina Vogelsang	Aerospace Engineering	15.01.2024 / 31.07.2024	Postprocessing script for Davis picture processing
				Extension of the compressible sharp-interface framework to the full Navier-Stokes equations by introducing viscous interface fluxes
Johanna Hintz	A. Travnicek	Aerospace Engineering	01.02.2024 / 31.07.2024	
Nikolaus Vohwinkel	A. Travnicek	Aerospace Engineering	01.03.2024 / 31.08.2024	Implementation of a high-order finite volume shock-capturing for discontinuous Galerkin methods
Elias Ennulath	R. Helmig	Civil Engineering	01.04.2024 / 30.09.2024	Supported program development and collaborated on the verification of models developed for droplet

				formation at the interface between free and porous flow
Cemre Inal	R. Saha	Aerospace Engineering	15.04.2024 / 15.06.2024	Support in literature review on droplet impact dynamics
Mert Meric	P. Palmetshofer / G. Lamanna	Aerospace Engineering	01.07.2024 / 30.09.2025	Support on experimental work on event-based velocimetry for droplet impact
Frederik Hemminger	P. Palmetshofer	Aerospace Engineering	15.07.2024 / 28.02.2025	Experimental investigations on droplet impact on wall structures
Cemre Inal	R. Saha	Aerospace Engineering	16.07.2024 / 31.03.2025	Support in DNS of water droplet impact onto smooth surfaces
Lukas Walle	J. Stober	Aerospace Engineering	12.08.2024 / 11.10.2024	Support in experimental work on oblique impacts on thin films
Johanna Hintz	A. Travnicek	Aerospace Engineering	01.09.2024 / 31.12.2024	Extension of the compressible sharp-interface framework to the full Navier-Stokes equations by introducing viscous interface fluxes
Lukas Walle	J. Stober	Aerospace Engineering	14.10.2024 / 13.11.2024	Support in experimental work on oblique impacts on thin films
Philip Klostermeier	G. Lamanna	Aerospace Engineering	01.11.2024 / 31.05.2025	Feasibility study on the applicability of event-based velocimetry
Phil Kreul	A. Kosteletzky	Civil Engineering	01.11.2024 / 31.03.2025	Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Carlos Fengler	P Molina-Vogelsang	Aerospace Engineering	15.11.2024 / 30.09.2025	Support in experimental work of droplet impacts on impact onto thin films with PIV
Simon Grether	A. Kosteletzky	Civil Engineering	16.11.2024 / 30.09.2025	Supported program development and collaborated on the verification of models developed for droplet formation at the interface between free and porous flow
Cemre Inal	R. Saha	Aerospace Engineering	16.12.2024 / 30.09.2025	Support in understanding the spreading dynamics of ethanol and water droplet impact onto a smooth surface
Ayush Thakur	G. Lamanna	Aerospace Engineering	01.01.2025 / 31.03.2025	Droplet impact simulations with FS3D
Timon Wendt	G. Lamanna	Aerospace Engineering	15.03.2025 / 14.09.2025	Support on experimental testing: calibration and comparison PIV
Jens Kriesamer	G. Lamanna	Aerospace Engineering	15.03.2025 / 14.09.2025	Support on experimental testing and analysis of defocusing
Lyane-Antonin Mavougou	R. Saha	Aerospace Engineering	16.03.2025 / 15.06.2025	Support in understanding the contact angle dynamics on a heated surface
Nicklas Essig	G. Lamanna	Aerospace Engineering	01.04.2025 / 31.05.2025	Development and testing of the optical flow software for event-based velocimetry, applied to droplet impact
Yogesh Magare	R. Saha	Aerospace Engineering	15.04.2025 / 30.09.2025	Support in DNS of heated substrate and DNS code verification

Marie-Sophie Hepp	A. Travnicek	Aerospace Engineering	15.05.2025 / 30.09.2025	Validation of high-order finite volume shock-capturing schemes for DG methods
Mert Meric	G. Lamanna	Aerospace Engineering	15.10.2025 / 31.12.2025	Support on experimental work on event-based velocimetry for droplet impact

6.2.2 Duration of employment and fellowship contracts

Normally, the doctoral and postdoctoral researchers for DROPIT are employed for the full time, e.g. three years for doctoral researchers. However, as it can be seen in the table below a large number of shorter contracts (up to 12 months and up to 24 months) had to be issued. This had mainly three reasons:

First, for our foreign doctoral researcher, employment contracts could not be issued for longer than the duration of their residence permits. This increased the number of employment contracts under 12 months drastically. Second, the contracts had to be increased by some additional months provided by the DFG for the researchers employed during the Coronavirus Pandemic. Third, for the doctoral researcher of the second generation, the contracts could only be made up to the end of FP1. Only after the funding for FP2 was available these contracts could be extended.

Contract duration	Number of contracts for		Number of contracts for		Total number of contracts
	Female doctoral researchers	Male doctoral researchers	Female postdoctoral researchers	Male postdoctoral researchers	
Up to 12 months	12	21	0	1	34
Up to 24 months	2	11	1	1	15
Up to 36 months	4	12	0	0	16
Exceeding 36 months	0	0	1	0	1

6.2.3 Promotion of equality: Female RTG Participants by Qualification Level

The University of Stuttgart offers an elaborate infrastructure that is tailored to implement gender equality and more specifically to attract, keep and promote the career chances of talented women within the university. The university has chosen a holistic approach to increase the participation of women in science, which targets female students from high-school to post-doctoral and professional levels. The entire portfolio encompasses the following initiatives, e.g.: **Girl's Day; TryScience; StartScience; Praktikums-Info-Börse; a gender- and diversity-sensitive Mentoring Programme; Femtec.Network; Service Uni & Familie and Service Gender Consulting**. Thanks to this elaborate gender concept, the University of Stuttgart is certified as “family-friendly higher education institution”. In this context, the present IRTG could benefit from a number of already established measures and programmes:

- To promote work/family balance
- To support women's career chances in industry, science and academia
- To endorse women's understanding and access to key technologies, which is also a key-aspect for fostering gender equality.

Concerning work/family balance, the university pre-school institutions and kids-vacation programme provide a first basis for child-care solutions. Supporting the career opportunities for women in the scientific and technical sector can take place within the framework of the Femtec.Network. The Femtec.Network allows ambitious females in engineering and science to prepare for future leadership positions in industry or academia through training, internships, and networking. The Femtec initiative comprises nine national technical universities (among which the University of Stuttgart) and several leading international companies. Within a three-semester career building programme, participants establish close contacts to the partner companies and acquire leadership and management skills. Furthermore, in an innovation workshop, participants solve a problem from one of the Femtec companies and experience individual counselling on their career path.

In the framework of this IRTG, the mentoring activities have been expanded by setting up an international mentoring programme as explained before. Each doctoral researcher had been assigned an international mentor. For example, during the annual doctoral seminar there was the possibility to discuss career possibilities and career development strategies also with the Italian/German supervisors. In addition, during visits of the doctoral researchers at the partner university, this could be done. The inclusion of foreign mentors enabled to improve the researcher's intercultural competence. This encompasses the enhancement of intercultural understanding and tolerance, the capability of communicating effectively and gaining more confidence in interacting with professionals from cultural backgrounds different to their own.

In addition, researchers and PIs of DROPIT acted as mentors or role models. Here, e.g. the DROPIT postdoc Anne Geppert participated in **TryScience**, **StartScience** and had a Mentee in the framework of the CyberMentor-programme (<https://www.cybermentor.de/index.php>).

DROPIT also supported gender equality through pooling fond. The pooling fond is centrally located in the administration. With these funds, comprehensive gender equality measures have been implemented, which supported different DFG research projects and could be shared, e.g. equipment of a parent-child room on the Vaihingen campus or participation in the basic fee for flexible childcare with two providers.

In collaboration with **uniquUS**, career support measures were offered to female postdocs (e.g. A. Geppert), covering different aspects from visibility measures to the development of didactic, communication and management skills to foster their independence. Finally, a strong support was given also on promoting the visibility of female scientists and their work. This occurred in three parallel ways. First, in collaboration with uniuUS within the Pooling initiatives, the **Junior Professor Carina Bringedal** (former PI in DROPIT) and **Dr. Anne Geppert** (PostDoc) were selected for a scientific portrait in the programme "**Women in STEM make a difference**" to make their research visible to the general public and as successful role models for early careers researchers (ECR) (<https://www.youtube.com/playlist?list=PLGuevu4esbwEy-0zrvy7GeeFD0Lra6cQa>).

Second, for the general public, the research topic of Anne Geppert was also proposed to **SWR** to be broadcasted in the TV show "**Das Wissen**" (<https://www.swr.de/swrkultur/wissen/tropfenforschung-eine-wissenschaft-fuer-alle-lebenslagen-100.html>).

This visibility measure opened also the opportunity for a joint research project with the Paul Scherrer Institute (CH) on neutron imaging investigation of liquids mixing during droplet impact onto liquid films. The project had Dr. Anne Geppert as lead scientist and included several DROPIT doctoral and associated members. The result of this collaboration, partly financed by DROPIT, were presented at the ILASS-Europe 2025 conference in Lund (S), which represents one of the most successful measures in promoting the research and visibility of female, early career researchers. Third, DROPIT also actively supported the participation of female ECRs in the **Fellowship-Programme Zia – Visible Women in Science**. During the one-year fellowship program, female ECRs gain a unique opportunity to support further their professional and personal development by interacting with female role models, institutional representatives from academia, research, business, and the public sector. This enables also the creation of a wide, professional network while simultaneously improving their communication skills. In DROPIT, the doctoral researcher **Claudia Marianowski** could benefit from the ZIA Fellowship. Today she is working as mechanical and thermal engineer at TESAT.

Table A shows the current composition of the doctoral and postdoctoral researchers participating in DROPIT. It can be seen that in the first funding period we were not able to fully obtain our very ambitious goal of having 30% female doctoral researchers. However, in FP2 we were able to fulfil

this goal. In the group of postdoctoral researchers, we were able to fulfil our goal in FP1. In FP2, Anne Geppert, had the Postdoc position in DROPIT from 01.06.2019 to 30.09.2024. She left then DROPIT, in order to become a department manager at the Leibniz Institute for material-oriented technologies (IWT) in Bremen. The postdoc position was then used only by 50% for the last year of the project for the employment of Patrick Palmetshofer and Manish Kumar. See also **Section 6.2.1**.

- Female IRTG members

	First Funding Period (FP1)		Second Funding Period (FP2)	
	% Goal	% Goal	% Status Quo	% Status Quo
	<i>As per establish- ment proposal</i>	<i>As of renewal proposal submis- sion</i>	<i>As per renewal proposal</i>	<i>As of last day of funding</i>
Female Doctoral Researchers	30	25	30	36
Female Postdoctoral Researchers	30	50	30	0

As a big achievement, it can be noted that two doctoral researchers of DROPIT (**Martina Baggio** and **Karin Schlottke**) have won the very prestigious **Amelia Earhart Fellowship in 2019**. In an effort to carry out its mission that women have access to all resources and are represented in decision-making positions on an equal basis with men, **Zonta International offers the Amelia Earhart Fellowship**. The Amelia Earhart Fellowship was established in 1938 in honour of famed pilot and Zontian, Amelia Earhart. **The US\$10,000 Fellowship is awarded annually to up to 30 women worldwide pursuing Ph.D. degrees in aerospace engineering and space sciences**. See also: <https://foundation.zonta.org/Our-Programs/Educational-Programs/Amelia-Earhart-Fellowship>.

Finally, it should be stated that we believe that already in school females have to be educated in a way to interest them in engineering sciences. Therefore, DROPIT has started in FP1 to take additional measures to promote the topic of research of this IRTG to children. Thus, a lecture has been given in the “**kids university**” in Heilbronn on 15.01.2020 by K. Schulte and B. Weigand entitled: “**The wonder world of drops**”. Here roughly 250 kids participated. Furthermore, DROPIT participates in **TryScience** at the University of Stuttgart. In this workshop the children receive information

about the courses of study and research, they carry out small experiments themselves, receive mini-lectures and visit laboratories and experimental halls. The CampusTour provides an insight into the everyday life of the students.

Table B reports the participating female researchers of this project. Due to several promotions (see also the personnel development in **Section 6.1.**) the percentage of C4/C3 professors could be increased. The overall percentage of female researchers slightly decrease in FP2 because Carina Bringedal left for taking a professorship in Norway.

- Participating female researchers

	1. Funding period % Status Quo <i>As per establishment proposal</i>	2. Funding period % Status Quo <i>As of renewal proposal submission</i>	End of funding % Status Quo <i>As of last day of funding</i>
Female postdoctoral researchers*	100	50	0
Junior Professors, Independent Junior Research Group Leaders	60	100	100
Professors C3/W2	0	25	0
Professors C4/W3	0	0	15
Total	27	31	27

*Research staff with a doctoral degree but without their own working group

6.2.4 Research stays

There had been several research stays by doctoral and postdoctoral researchers from DROPIT at other institutes. The doctoral researcher **Anna Mareike Kostelecky** went for 2,5 months to the NTU Trondheim in Norway for a long research stay, because of the outstanding expertise of this group in porous media thermodynamics. This was already been explained at the beginning of Chapter 6.1 as an answer to comment 3 from the reviewers of the renewal proposal.

Name	From / to	Institute visited	Local supervisors (if applicable)	Research activities performed and skills acquired during stay
Ronan Bernard (1. Gen. PhD, US)	23.10.2017 / 27.10.2017	Centre for Combustion Science and Technology (CECOST), Lund University, Sweden	E. Berrocal	Course: Laser Imaging of Spray systems Attended lectures by experts on various aspects of laser imaging of spray systems; conducted MC simulations using an in-house tool to understand multiple scattering effects in laser imaging. Participation in laboratory sessions to better understand the concepts learnt and the imaging instrumentation discussed during the lectures.
	02.07.2018 / 06.07.2018	Institute of Fluid Mechanics and Heat Transfer (IWS), TU Graz, Graz, Austria	C. Planchette	Visit for preparation of the DROPIT summer school
Patrick Foltyn (1. Gen. PhD, US)	11.10.2017 / 13.10.2017	Center of Smart Interfaces, Thermo-Fluids & Interfaces, TU Darmstadt, Germany	Dr.-Ing. Benjamin Lambie (Organizational Chairman)	Short Course on Complex Wetting at TU, Darmstadt, Germany: Fundamental knowledge about wetting phenomena; preparation & characterization of surfaces, optical measurement techniques of wetting processes, characterization of surface & interfacial tension, contact angles, dynamic wetting of soft, liquid and soluble surfaces, dynamic wetting of textured surfaces, wetting of membranes and porous media
Jonas Zeifang (1. Gen. PhD, US)	22.08.- 23.08.2019	Hasselt, Belgium	--	Research visit
Visakh Vaikuntanathan (1. Postdoc, US)	23.10.2017 / 27.10.2017	Centre for Combustion Science and Technology (CECOST), Lund University, Sweden	--	Course: Laser Imaging of Spray systems Attended lectures by experts on various aspects of laser imaging of spray systems; conducted MC simulations using an in-house tool to understand multiple scattering effects in laser imaging. Participation in laboratory sessions to better understand the concepts learnt and the imaging instrumentation discussed during the lectures.
Robert Haide (1. Gen. PhD, BG)	16.09.2019 / 23.09.2019	Imec-Vision Lab, University of Antwerp, Belgium	J. Sijbers	Familiarization and training on ASTRA reconstruction framework

Simone Chiocchetti (1. Gen. PhD, Trento)	19.06.2018 / 01.07.2018	Aix-Marseille Univer- sité, Marseille, France	S. Gavriluk	Research stay to discuss mathematical modelling of compressible multiphase flow
Simone Chiocchetti (1. Gen. PhD, Trento; StartUp Grant in 05 / 2023)	12.07.2023 / 22.07.2023	Università di Catania, Catania, Italy	G. Russo	Research collaboration
Weibo Ren (2. Gen. PhD, US)	17.02.2020 / 20.02.2020	Technische Universität Darmstadt	--	Short Course on Atomization and Sprays: Obtained current understanding and state-of-the-art of atomization fundamentals, their realization in atomizer systems and their application in a wide variety of engineering branches, including spray drying, spray coating, spray cooling, fuel injection, etc. These aspects were first addressed theoretically in terms of hydrodynamic instabilities of liquid jets and sheets – primary atomization, followed by considerations about the break-up of single droplets – secondary atomization. Engineering solutions for realizing the different atomization mechanisms were then presented.
Stefan Schubert (2. Gen. PhD, US)	04.07.2022 / 08.07.2022	Technical University of Denmark (DTU), Den- mark	M. Rossi	Code development for the GDPT method
Anna Mareike Kosteletzky (3. Gen. PhD, US)	08.2024 / 10.2024	NTNU Trondheim & Pore Lab, Trondheim, Norway	Prof. Signe Kjelstrup	Study non-equilibrium thermodynamics
Rishav Saha (3. Gen. PhD, US)	06.05.2024 / 10.05.2024	Udine University		Drops, Jets and Films Seminar, CISM

6.2.5 Conference travel undertaken by (post)doctoral researchers

The doctoral and postdoctoral researchers had been very active in attending conferences. In nearly all cases they were presenting their work as poster and/or talks at these conferences.

Name	From / to	Name of conference	Location	Title of own presentation/title of own poster presentation/participation only
Sina Ackermann (1. Gen. PhD, US)	05.10.2016 / 07.10.2017	NUPUS Meeting 2016	Mühlhausen, Germany	Participation only
	07.05.2017 / 12.05.2017	Interpore	Netherlands	Poster: Upscaling of coupled free-flow and porous-medium-flow processes
	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Upscaling of coupled free-flow and porous-medium-flow processes
	25.06.2017 / 29.06.2017	Multiscale methods summer school	Hasselt, Belgium	Participation only
	09.10.2017 / 11.10.2017	NUPUS Meeting 2017	Mühlhausen, Germany	Presentation: Upscaling of coupled free-flow and porous-medium-flow processes
	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster: Upscaling of coupled free-flow and porous-medium-flow processes
	06.05.2019 / 10.05.2019	Interpore	Valencia, Spain	Poster: Upscaling of coupled free-flow and porous-medium-flow processes
	17.05.2019	DIPSI 2019	Bergamo, Italy	2 Posters: Upscaling of coupled free-flow and porous medium flow processes, A multiscale approach for drop detachment in a coupled free-flow/porous-medium-flow system
Martina Baggio (1. Gen. PhD, US)	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Numerical computation for drop impact on textured surfaces
	26.03.2018 / 28.03.2018	2 nd Int. Conf. on Simulation Technology	Stuttgart, Germany	Poster: Implementation of structured Surfaces in FS3D
	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster: On the implementation of Structured Surfaces to FS3D
	22.06.2018 / 26.06.2018	ICLASS 2018	Chicago, USA	Presentation: A numerical method for handling arbitrary solid boundaries within the DNS multiphase code Free Surface 3D (FS3D)
	17.05.2019	DIPSI 2019	Bergamo, Italy	Poster: On the implementation of Structured Surfaces to FS3D
	02.09.2019 / 04.09.2019	ILASS Europe 2019	Paris, France	Presentation: Direct numerical simulations for drop impact on superhydrophobic patterned surfaces using the VoF method
Ronan Bernard (1. Gen. PhD, US)	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Micro and macro drop impact dynamics with miscible liquids
	05.09.2017 / 08.09.2017	ILASS Europe 2017	Valencia, Spain	Presentation: Generalized analysis of the deposition/splashing for one- and two-component droplet impacts upon thin films

	23.10.2017 / 27.10.2017	Short course on Laser Imaging of Spray systems	Lund, Sweden	Participation only
	18.05.2018	DIPSI 2018	Bergamo, Italy	Participation only
	22.06.2018 / 26.06.2018	ICLASS 2018	Chicago, USA	Presentation: On the scaling of crown rim diameter during droplet impact on thin wall-films
	17.05.2019	DIPSI 2019	Bergamo, Italy	Poster: On the crown rim expansion kinematics during droplet impact on wall-films
	24.06.2019 / 28.06.2019	8th Conference Bubble & Drops	Sofia, Bulgaria	Presentation: Crown rim kinematics for droplet impact on wall-films
Simone Chiocchetti (1. Gen. PhD, Trento; StartUp Grant in 05 / 2023)	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster: High order schemes for a first order hyperbolic formulation of compressible flow with surface tension
	01.04.2019 / 01.05.2019	HONOM	Madrid, Spain	Presentation: high order ADER schemes for a first order hyperbolic formulation of compressible flow with surface tension
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
	20.05.2019 / 24.05.2019	SHARK-FV	Povoa de Varzim, Portugal	Presentation: Presentation: high order ADER schemes for a first order hyperbolic formulation of compressible flow with surface tension
	22.07.2019	Lava-flow modeling Workshop	Catania, Italy	Presentation: High order numerical methods for a hyperbolic formulation of viscous flows.
	09.09.2019 / 13.09.2019	MultiMath conference	Trento, Italy	Participation only
	30.09.2019 / 05.10.2019	ENUMATH conference	Egmond aan Zee, the Netherlands	Presentation: High order space-time adaptive ADER schemes for compressible multi-phase flows
	26.07.2021 / 30.07.2021	NumHyp conference	Trento, Italy	Presentation: Tips and tricks for Simple Hyperbolic Viscous Flow
	07.03.2022 / 11.03.2022	CIRM Int. workshop "Perspectives on Multi-phase Fluid Dynamics, Continuum Mechanics & Hyperbolic Balance Laws"	Marseille, France	Presentation: Hyperbolic viscous flow made simple
	04.04.2022 / 08.04.2022	HONOM	Braga, Portugal	Presentation: Hyperbolic viscous flow with hyperbolic surface tension
	23.05.2022 / 27.05.2022	SHARK-FV	Povoa de Varzim, Portugal	Participation only

	13.06.2022 / 15.06.2022	UnTRIM workshop	Trento, Italy	Participation only
	22.08.2022 / 26.08.2022	MultiMat conference	Zürich, Switzerland	Presentation: Semi-implicit schemes with efficient finite rate stiff relaxation for compressible two-phase flow and plasticity
	21.05.2023 / 26.05.2023	SHARK-FV	Portugal	Presentation: --
	24.06.2023 / 01.07.2023	Num Hyp 2023	Bordeaux, France	Presentation: --
	20.08.2023 / 25.08.2023	ICIAM 2023	Tokyo, Japan	Presentation: --
	10.09.2023 / 16.09.2023	Hirschegg Workshop	Hirschegg, Austria	Presentation: --
	30.10.2023 / 03.11.2023	FVCA 10 Congress	Straßburg, France	Presentation: --
	30.11.2023 / 01.12.2023	Godunov in memoriam Congress	Marseille, France	Presentation: --
Patrick Foltyn (1. Gen. PhD, US)	26.05.2017	DIPSI 2017	Bergamo, Italy	Participation only
	11.10.2017 / 13.10.2017	Short Course on Complex Wetting	Darmstadt, Germany	Participation only
	17.10.2017 / 18.10.2017	Stemmer Technologieforum	Munich, Germany	Participation only
	19.02.2018 / 21.02.2018	KNMF UserMeeting	Karlsruhe, Germany	Presentation: Droplet collisions with solid superhydrophilic surfaces – investigation strategies
	18.05.2018	DIPSI 2018	Bergamo, Italy	Participation only
	19.02.2019 / 20.02.2019	KNMF UserMeeting	Karlsruhe, Germany	Participation only; Specialized short course about: Introduction to Electron Microscopy (SEM, FIB, TEM)
	17.05.2019	DIPSI 2019	Bergamo, Italy	Presentation: 360° Evaluation of Apparent Contact Angles of Static Droplets on Structured Surfaces
	16.06.2019 / 19.06.2019	9th International Colloids	Barcelona, Spain	Poster: Oxygen and Hydrogen Plasma Activation Effects on Capillary Thermodynamics in Fibrous Materials
	02.09.2019 / 04.09.2019	ILASS 2019	Paris, France	Presentation: Development and calibration of the LASER Pattern Shift Method for measuring the lamella topology during drop impact on walls

	10.03.2020 / 11.03.2020	KNMF UserMeeting	Karlsruhe, Germany	Participation only; Discussion with KNMF representatives about proposal for new surface structure geometries and reconciliation with them for journal publication.
Manuel Giovanni Gritti (1. Gen. PhD, BG)	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
Robert Haide (1. Gen. PhD, BG)	26.05.2017	DIPSI 2017	Bergamo, Italy	Participation only
	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster "Characterization of Porous Media by X-Ray Micro Computed Tomography"
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
Gautham Varma Raja Kochanattu (1. Gen. PhD, BG)	26.05.2017	DIPSI 2017	Bergamo, Italy	Participation only
	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster: Modelling of heat and mass transfer from spheroidal drops with general non-uniform Dirichlet boundary conditions.
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
Christoph Müller (1. Gen. PhD, US)	27.03.2017 / 31.03.2017	HONOM 2017	Stuttgart, Germany	Participation only
	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Sharp interface methods for droplet simulation
	04.06.2017 / 10.06.2017	Oberwolfach Seminar: Compressible and Incompressible Multiphase Flows: Modelling, Analysis, Numerics	Oberwolfach, Germany	Participation only
	24.06.2017 / 29.06.2017	ICNMMF-III	Japan	Presentation: Simulation of compressible multiphase flows near the critical point using a sharp interface method
	17.09.2017 / 22.09.2017	13th Hirschegg Workshop on Conservation Laws	Hirschegg, Austria	Presentation: Ghost Fluid Methods for Multi-Phase Flow with Phase Change
	18.05.2018	DIPSI 2018	Bergamo, Italy	Participation only
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
	14.07.2019 / 20.07.2019	ICIAM	Valencia, Spain	Presentation: Sharp interface methods for compressible multiphase flow including heat conduction phenomena
Lukas Ostrowski (1. Gen. PhD, US)	06.03.2017 / 10.03.2017	GAMM	Weimar, Germany	Presentation: Compressible multicomponent porous media flow with Maxwell-Stefan diffusion
	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Compressible effects in droplet interaction with textured walls

	25.06.2017 / 29.06.2017	Multiscale methods summer school	Hasselt, Belgium	Poster: Numerical Upscaling for Droplet Impact on rough Walls
	04.06.2017 / 09.06.2017	Oberwolfach Seminar: Compressible and Incompressible Multiphase Flows: Modelling, Analysis, Numerics	Oberwolfach, Germany	Participation only
	17.09.2017 / 22.09.2017	13th Hirschegg Workshop on Conservation Laws	Hirschegg, Austria	Presentation: An Approach to Numerical Upscaling for Compressible Phase Field Flow
	26.03.2018 / 28.03.2018	SimTech	Stuttgart, Germany	Poster: Numerical Upscaling for Compressible Phase Field Flow
	23.06.2018 / 29.06.2018	HYP2018	Pennsylvania, USA	Presentation: Compressible droplet impingement on perfect and rough walls
	30.09.2018 / 05.10.2018	Fall school on hyperbolic conservation laws and mathematical fluid dynamics	Würzburg, Germany	Poster: Compressible Droplet Impingement, Fall school on hyperbolic conservation laws and mathematical fluid dynamics
	17.05.2019	DIPSI 2019	Bergamo, Italy	Poster1: Compressible Droplet Impingement Poster2: A multi-scale approach for drop detachment in a coupled free-flow/porous-medium-flow system
	24.11.2019 / 30.11.2019	Oberwolfach Seminar: Structure-preserving Methods for Nonlinear Hyperbolic Problems	Oberwolfach, Germany	Participation only
	09.02.2020 / 15.02.2020	Course on Advanced Numerical Methods for Hyperbolic Equations	Trento, Italy	Participation only
Wladimir Reschke (1. Gen. PhD, US)	26.05.2014	DIPSI 2017	Bergamo, Italy	Poster: Introduction to the DROPIT Subproject SP-A3: Gas-Kinetic Simulation of Micro-droplet-Gas Interaction.
	27.08.2017 / 30.08.2017	Direct Simulation Monte Carlo	Santa Fe, USA	Presentation: Catalytic models in the DSMC module within the plasma solver PICLas
	17.05.2018	DIPSI 2018	Bergamo, Italy	Poster: Current State of the DROPIT Subproject SP-A3: Gas-Kinetic Simulation of Micro-droplet-Gas Interaction.

	23.07.2018 / 27.07.2018	RGD31	Glasgow, GB	Presentation: Validation of DSMC and CFD Based Catalysis Modelling Using Plasma Wind Tunnel Flows
	29.09.2019 / 04.10.2019	FAR	Monopoli	Presentation: Modelling and Simulation of Heterogeneous Reactions with Statistical Particle Methods
Alexander Straub (1. Gen. PhD, US)	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Visualization of Droplet Internal Flow
	26.03.2018 / 28.03.2018	SimTech 2018	Stuttgart, Germany	Poster: Visual Analysis of Interface Deformation in Multiphase Flow
	18.05.2018	DIPSI 2018	Bergamo, Italy	Poster: Visual Analysis of Interface Deformation in Multiphase Flow
	08.10.2018 / 12.10.2018	TRR161/GCPR/VMV	Stuttgart, Germany	Participation only
	20.10.2018 / 26.10.2018	IEEE VIS	Berlin, Germany	Participation only
	17.05.2019	DIPSI 2019	Bergamo, Italy	Poster: Visualization and Visual Analysis for Multiphase Flow
	16.06.2019 / 19.06.2019	TopoInVis 2019	Sweden	Presentation: Implicit Visualization of 2D Vector Field Topology for Periodic Orbit Detection
Visakh Vaikuntanathan (1. Postdoc, US)	26.05.2017	DIPSI 2017	Bergamo, Italy	Participation only
	18.05.2018	DIPSI 2018	Bergamo, Italy	Participation only
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
Jonas Zeifang (1. Gen. PhD, US)	27.03.2017 / 31.03.2017	HONOM	Stuttgart, Germany	Participation only
	26.05.2017	DIPSI 2017	Bergamo, Italy	Poster: Mathematical and numerical modelling of droplet dynamics in weakly compressible multi-component flows
	17.09.2017 / 22.09.2017	ACOMEN	Gent, Netherlands	Presentation: Investigation of a novel splitting scheme for the weakly compressible Euler equations
	18.05.2018	DIPSI 2018	Bergamo, Italy	Participation only
	28.05.2018 / 01.06.2018	CANUM	Cap d'Agde, France	Presentation: A comparison of splitting techniques for the isentropic Euler equations
	17.05.2019	DIPSI 2019	Bergamo, Italy	Participation only
	28.08.2019 / 30.08.2019	GACM 2019	Kassel, Germany	Presentation: High order IMEX splitting methods for low Mach number flows

2. Generation of (post)doctoral students				
Daniel Appel (2. Gen. PhD, US)	02.2022	Conference SIAM Parallel Processing	Online Conference	Presentation: Narrow Band-Based Dynamic Load Balancing for Multiphase Flow Simulations
	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Narrow band-based dynamic load balancing for multiphase flow simulations
	06 / 2023	ECCOMAS Adaptive Modeling & Simulation (ADMOS)	Gothenburg, Sweden	Presentation: An Efficient hp-Adaptive Approach for Compressible Two-Phase Flows
	09 / 2023	GACM Colloquium on Computational Mechanics	Vienna, Austria	Presentation: Towards Direct Numerical Simulation of Compressible Droplet Grouping in Turbulent Flows
Anne K. Geppert (2. Postdoc, US)	17.05.2019	DIPSI 2019	Bergamo, Italy	Presentation: An analytical solution for modelling the crown spreading on wetted walls based on stagnation point flow: Effects of film thickness and viscosity.
	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Effect of edge length and surface wettability on droplet impact onto a stand-alone cubic pillar
	02.04.2022 / 07.04.2022	ICMF 2023	Kobe, Japan	Presentation: Spreading dynamics during single droplet impact onto a free-standing cubic surface element
	16.06.2023	DIPSI 2023	Bergamo, Italy	Presentation: Ground spreading dynamics during single droplet impact onto a free-standing cubic surface element
	04.09.2023 / 07.09.2023	ILASS Europe 2023	Naples, Italy	Presentation: Towards spatial and time-resolved film thickness measurements during droplet impact on thin liquid films
	17.06.2024	DIPSI 2024	Bergamo, Italy	Poster: Mechanism behind the Generation of a Rectangular Liquid Umbrella after Droplet Impact onto a Small Square Surface
	23.06.2024 / 06.06.2024	ICLASS 2024	Hybrid Conference (Shanghai, China)	Presentation: Unravelling the Generation of a Rectangular Liquid Umbrella after Droplet Impact onto a Small Square Surface
Daniel Klötzl (2. Gen. PhD, US)	20.03.2022 / 24.03.2022	Hygraphics 2022	Hirschegg, Austria	Presentation: Local Bilinear Computation of Jacobi Sets
	14.10.2022 / 21.10.2022	IEEE VIS 2022	Oklahoma City, USA	Presentation: Reduced Connectivity for Local Bilinear Jacobi Sets
	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: Relation between shear rate and structural parameter in non-Newtonian droplet oscillations via Jacobi sets and STL decomposition: a visual analysis
	16.08.2023 / 26.08.2023	Norrköping Summer School	Norrköping, Sweden	Participation only
	17.06.2024	DIPSI 2024	Bergamo, Italy	Poster: NMF - Based Analysis of Droplet Wall-Film Interactions

Pascal Mossier (2. Gen. PhD, US)	12.07.2021 / 16.07.2021	ICOSAHOM	Online Conference	Presentation: A p-adaptive discontinuous Galerkin method with hp-shock capturing
	13.09.2021 / 17.09.2021	Hirschegg Workshop on Conservation Laws	Hirschegg, Austria	Presentation: The Riemann Problem with Evaporation
	04.04.2022 / 08.04.2022	HONOM conference	Braga, Portugal	Presentation: An efficient hp-adaptive strategy for the Level-set Ghost fluid method
	16.06.2023	DIPSI 2023	Bergamo, Italy	Participation only
	24.07.2023 / 27.07.2023	USNCCM conference	Albuquerque, USA	Presentation: An Efficient hp-Adaptive Strategy for a Level-Set Ghost Fluid Method
Weibo Ren (2. Gen. PhD, US)	30.06.2022	DIPSI 2022	Bergamo, Italy	Poster: Numerical Computation for Drop Impact on Textured Surfaces.
	06.08.2022 / 08.08.2022	ILASS 2022	Virtual Conference	Presentation: Splashing During Simultaneous Two Drops Impact Onto A Micropillared Superhydrophilic Surface
	16.06.2023	DIPSI 2023	Bergamo, Italy	Participation only
Raphael Tietz (2. Gen. PhD, US)	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Symmetric simulations of droplets with a particle based Vlasov-Enskog-solver
	04.07.2022 / 08.07.2022	32 nd Int. Symposium on Rarefied Gas Dynamics (RGD)	Seoul, South Korea	Presentation: Symmetric Simulations of Droplets with a Particle based Vlasov-Enskog-Solver
	19.06.2022 / 23.06.2022	2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-entry Missions Engineering (FAR)	Heilbronn, Germany	Presentation: Shock Tube Simulations with the PIC-DSMC Code PICLAS
	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: Simulation of sub-micrometer droplet evaporation with a particle-based spherically-symmetric Enskog-Vlasov solver
	04.09.2023 / 07.09.2023	ILASS Europe 2023	Naples, Italy	Presentation: Simulation of Sub-Micrometer Droplet Evaporation with a Particle-Based Spherically-Symmetric Enskog-Vlasov Solver
	17.06.2024	DIPSI 2024	Bergamo, Italy	Presentation: Enskog-Vlasov Simulations in High Pressure Environments
Stefan Schubert (2. Gen. PhD, US)	29.08.2021 / 02.09.2021	ICLASS 2021	Virtual Online Conference	Participation only
	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Experimental 2D velocity measurement in the area of the crown crater with μ PIV technique

	03.11.2022 / 04.11.2022	3rd Int. Workshop on Near-Wall Reactive Flows 2022	Darmstadt, Germany	Poster: Experimental 2D velocity measurement in the area of the crown crater with μ PIV technique
	23.06.2024 / 27.06.2024	ICLASS 2024	Shanghai, China	Presentation: PIV measurements in the inner crown wall resulting from droplet impact on a thin liquid film
Maziar Veyskarami (2. Gen. PhD, US)	30.06.2022	DIPSI 2022	Bergamo, Italy	Poster: Formation, growth and detachment of multiple droplets at the free flow-porous medium interface
	31.05.2021 / 04.06.2021	Interpore 2021	Online Conference	Presentation: Droplet formation, growth and detachment on the interface of a coupled free flow-porous medium system.
	30.05.2022 / 02.06.2022	Interpore 2022	Online Conference	Presentation: Coupling porous medium-free flow: Formation and evaporation of multiple droplets at the interface.
	22.05.2023 / 25.05.2023	Interpore 2023	Online Conference	Presentation: Droplet evaporation on the interface of a coupled free-flow-porous medium system: Modelling and analysis.
3. Generation of doctoral students				
Rishi Dhar (3. Gen. PhD, US)	17.06.2024	DIPSI 2024	Bergamo, Italy	Poster: Investigation of Droplet-Pool Phenomena and Droplet-Film Interaction for Immiscible Fluids
Davide Ferrari (3. Gen. PhD. BG)	2024	MULTIMAT 2024	Breckenridge, USA	Presentation: Numerical solution of a unified HTC multiphase model of continuum mechanics
Yuequiao Hu (3. Gen. PhD, BG)	16.06.2023	DIPSI 2023	Bergamo, Italy	Participation only
	17.06.2024	DIPSI 2024	Bergamo, Italy	Participation only
	24.06.2024 / 07.06.2024	ICLASS 2024	Shanghai	Presentation: A numerical approach to the study of the evaporation of an oscillating droplet under unsteady conditions.
	13.06.2025	DIPSI 2025	Bergamo, Italy	Presentation: A Semi-Analytical Model for the Evaporation of an Oscillating Spheroidal Droplet.
	30.08.2025 / 04.09.2025	ILASS 2025	Lund, Sweden	Presentation: A Semi-Analytical and Numerical Model for the Evaporation of an Oscillating Droplet in a Potential Flow
Hasel Cicek Konan (3. Gen. PhD, US)	04.06.2023 / 07.06.2023	Workshop on Compressible Multiphase Flows 2023	Straßburg, France	Participation only
	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: An incompressible two-phase model with hyperbolic Navier-Stokes-Cahn-Hilliard Approach

	26.06.2023 / 30.06.2023	Num. Methods for Hyperbolic Problems 2023	Bordeaux, France	Participation only
	17.06.2024 / 19.06.2024	Workshop on Compressible Multiphase Flows 2024	Straßburg, France	Poster: Approximation of the Navier-Stokes-Cahn-Hilliard system for two-phase flows
	01.07.2024 / 05.07.2024	HYP2024: The 19th International Conference on Hyperbolic Problems	Shanghai, China	Presentation: Approximation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows
	09.09.2024 / 13.09.2024	Numdiff-17: Conference on the Numerical Solution of Differential & Differential-Algebraic Equations	Halle (Saale), Germany	Presentation: Approximation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows
	09.06.2025 / 13.06.2025	Numerical Methods for Hyperbolic Problem 2025	Darmstadt, Germany	Poster: Hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows
Anna Mareike Kostecky (3. Gen. PhD, US)	13.03.2023 / 17.03.2023	CSD Winter School	Norway	Participation only
	13.05.2024 / 16.05.2024	InterPore Conference 2024	Online Conference	Presentation: Local thermal non-equilibrium processes influenced by interfaces within porous media systems
	19.05.2025 / 22.05.2025	InterPore Conference 2025	Albuquerque, USA	Presentation: Local thermal non-equilibrium processes in porous media: Comparison of different models from the pore- to the REV-scale
Claudia Marianowski (3. Gen. PhD, US)	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: Simulating liquid-vapor interfaces using non-equilibrium particle schemes
	04.09.2023 / 07.09.2023	ILASS Europe 2023	Naples, Italy	Presentation: Non-Equilibrium Particle Schemes to Simulate Liquid-Vapor Interfaces
	22.04.2024 / 26.04.2024	Helmholtz GPU Hackathon	Görlitz, Germany	Participation only
	17.06.2024	DIPSI 2024	Bergamo, Italy	Poster: Gas-Kinetic Simulation of Microdroplet-Gas Interaction
	15.07.2024 / 19.07.2024	33 rd Int. Symposium on Rarefied Gas Dynamics (RGD33)	Göttingen, Germany	Participation only
Pablo Molina Vogel-sang (3. Gen. PhD, US)	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: An approach to investigate the 3D height evolution in the crown crater during early crown formation with the General Defocusing Particle Tracking (GDPT) method

	17.06.2024	DIPSI 2024	Bergamo, Italy	Participation only
	13.06.2025	DIPSI 2025	Bergamo, Italy	Participation only
	26.06.2025 / 28.06.2025	16th International Symposium on Particle Image Velocimetry 2025	Tokyo, Japan	Presentation: Shadowgraph defocusing μ PTV measurements of an impacting droplet onto a thin liquid film
	30.08.2025 / 04.09.2025	ILASS 2025	Lund, Sweden	Presentation: Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film
Patrick Palmethofer (3. Gen. PhD, US)	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Dynamic contact angle measurements of droplets spreading on surfaces of varying wettability
	02.04.2022 / 07.04.2022	ICMF 2023	Kobe, Japan	Presentation: Experimental and Numerical Investigation of Droplet Impacts onto Cubic Pil- lars with a Wetted Base
	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: 360° contact angles of water droplets on micrometric ramps, pyramids and stag- gered cuboid surfaces at two different wettabilities
	04.09.2023 / 07.09.2023	ILASS Europe 2023	Naples, Italy	Presentation: Experimental investigation of the height and angle of an uprising sheet oc- curring in near-simultaneous double droplet impacts onto smooth surfaces of varying wet- tability
	17.06.2024	DIPSI 2024	Bergamo, Italy	Presentation: Splashing modes in droplet impacts onto sub-millimeter microstructures
Daniel Regener Roig (3. Gen. PhD, BG)	08.09.2024 / 13.09.2024	HONOM 2024	Chania, Greece	Presentation: Entropy-stable discontinuous Galerkin solution of the multicomponent com- pressible Euler model with entropy balance enforcement
	13.06.2025	DIPSI 2025	Bergamo, Italy	Poster
Daniele Rusconi (3. Gen. PhD, BG)	23.09.2025 / 25.09.2025	IMAGING25 Conference	Varenna, Italy	Poster: Weak Supervision: A Solution for Image Segmentation in Challenging Datasets
Rishav Saha (3. Gen. PhD, US)	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: Numerical simulation of droplet impact on structured surfaces
	05.05.2024 / 11.05.2024	Drop, Jets and Films Seminar	Udine, Italy	Participation only
	09.06.2024 / 14.06.2024	EUROTHERM Conference	Bled, Slovenia	Presentation: Numerical simulation of droplet impact onto heated surfaces below the boil- ing point
	17.06.2024	DIPSI 2024	Bergamo, Italy	Poster: DNS of droplet impact onto heated surfaces
	13.06.2025	DIPSI 2025	Bergamo, Italy	Poster: DNS of ethanol droplet impact
	30.08.2025 / 04.09.2025	ILASS 2025	Lund, Sweden	Presentation: DNS of droplet impact onto smooth and micro-structured surfaces
Jonathan Stober (3. Gen. PhD, US)	01.07.2022	DIPSI 2022	Bergamo, Italy	Poster: Numerical investigation of the mechanisms of early crown formation at an oblique drop impact onto a wall film

	02.04.2022 / 07.04.2022	ICMF 2023	Kobe, Japan	Presentation: Numerical Investigation of the Mechanism of Early Crown Formation during an Oblique Droplet Impact onto a Wall Film
	16.06.2023	DIPSI 2023	Bergamo, Italy	Poster: Experimental observation of the collision of a crown with a thin wall film during early oblique droplet impact
	04.09.2023 / 07.09.2023	ILASS Europe 2023	Naples, Italy	Presentation: Experimental Investigation of an Oblique Droplet Impact onto a Quiescent Wall Film with Two-Perspective High-Speed Imaging
	17.06.2024	DIPSI 2024	Bergamo, Italy	Presentation: Investigation of the Flow Field at the Wall during an Oblique Droplet Impact onto a Thin Wall Film
Amalia Travnicek (3. Gen. PhD, US)	16.06.2023	DIPSI 2023	Bergamo, Italy	Participation only
	17.06.2024	DIPSI 2024	Bergamo, Italy	Participation only
	08.2025	European Fluid Dynamics Conference 2025	Dublin, Ireland	Presentation: Direct Numerical Simulation of a Droplet in Compressible Homogeneous Isotropic Turbulence

6.2.6 Stays by German (post-)doctoral researchers at foreign partner universities

As already mentioned at the beginning of Chapter 6.1 the second generation of the doctoral researchers could not do longer research stays at the foreign partner universities because of the Coronavirus Pandemic (11.03.2020 – end of 2022). However, most of them attended the three week of the second summer school in Bergamo from 10.06.2022 – 30.06.2022.

From the third generation of doctoral researchers Rishav Saha (B05), Jonathan Palmetshofer (B01) and Jonathan Stober (C02) went for longer research stays (some months) to Bergamo. Anna Mareike Kosteletzky (B06) went for a research stay of roughly 2,5 months to the NTNU Trondheim, because of their outstanding expertise in porous media thermodynamics (see Chapter 6.2.4). Unfortunately, one of the doctoral researchers of the third generation dropped out (Claudia Marianowski (A03)) before she could take a longer research stay. She moved to industry, because she had a very good offer. Furthermore, we could not fill one position (for sub-project C04 - Visualization) because here the possible applicants preferred to go to industry because of the huge difference in salary in Informatics. Some of the doctoral researchers of the third generation started in DROPIT relatively late (Rishi Dhar (A02) on 01.02.2024 and Kim-Sophie Ellenberger (A03) on 01.11.2024), so that they could not go for a longer stay to the partner University. The doctoral researcher Hasel Cicek Konan (B04) did her Master-thesis in Italy before joining DROPIT. Therefore, she decided to go only for shorter visits to Bergamo. However, it is planned that she goes to Bergamo for a longer research stay either at the end of 2026 or at the beginning of 2027. This will be financed by our own funds. Unfortunately, two of the doctoral students (Pablo Molina Vogelsang (C01), and Amalia Travnicek (A06)) were only able to attend for a few short stays. This was understandable due to personal/family reasons.

Name	From / to	Institute visited	Local supervisors	Research activities performed and skills acquired during stay	Participation in qualification activities
Sina Ackermann (1. Gen. PhD, US)	09.10.2016 / 10.10.2016	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Kick-off
	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	13.05.2019 / 14.05.2019	University of Bergamo, Italy	M. Santini, S. Fest-Santini	Discussion with M. Santini and S. Fest-Santini about possible collaboration	
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
Ronan Bernard (1. Gen. PhD, US)	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
	03.06.2019 / 07.06.2019	University of Bergamo, Italy	G.E. Cossali		"Laser Diagnostics in Sprays" given by Prof. Coghe (Milano); "Lecture on micro-CT" given by Prof. Santini (Bergamo) and Prof. Sijbers (Antwerp)
Martina Baggio (1. Gen. PhD, US)	09.10.2016 / 10.10.2016	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Kick-off
	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	06.09.2018 / 19.08.2018	University of Bergamo, Italy	G.E. Cossali		DROPIT Summer School 2018
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019

Patrick Foltyn (1. Gen. PhD, US)	01.04.2016 / 30.09.2016	University of Bergamo, Italy	M. Santini, S. Fest-Santini	Master Thesis: Measurement of Phase Transition Rates of Arbitrary Shaped Particles in Natural Convection with Micro-Computed Tomography; microCT, image processing	
	25.05.2017 / 27.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	05.09.2018 / 19.09.2018	University of Bergamo, Italy	S. Fest-Santini	Summer School Project: microCT, image processing,	
	03.06.2019 / 07.06.2019	University of Bergamo, Italy	G. E. Cossali	DROPIT courses, additional discussion of cooperation with SP-AS04	"Laser Diagnostics in Sprays" given by Prof. Coghe (Milano); "Lecture on micro-CT" given by Prof. Santini (Bergamo) and Prof. Sijbers (Antwerp)
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
Christoph Müller (1. Gen. PhD, US)	02.07.2018 / 03.08.2018	University of Trento, Italy	M. Dumbser	GPR model, path conservative schemes	
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	16.09.2019 / 17.10.2019	University of Trento, Italy	M. Dumbser	GPR model, multi-phase Riemann Problem with phase transition	
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
Lukas Ostrowski (1. Gen. PhD, US)	27.05.2019 / 26.07.2019	Dipartimento di Ingegneria e Scienze Applicate, University of Bergamo, Italy	F. Bassi	Research activities: Derivation and analysis of an incompressible-compressible phase field model. Development of numerical solver for the system. Acquired skills: Improved knowledge in thermodynamics, Riemann solver, and discontinuous Galerkin schemes	Lectures on X-ray micro-computed tomography

	09.10.2016 / 10.10.2016	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Kick-off
	25.05.2017 / 27.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	17.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DIPSI Workshop 2019
Alexander Straub (1. Gen. PhD, US)	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	04.09.2018 / 21.09.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Summer School 2018
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
	11 / 2019	University of Bergamo, Italy	M. Santini	Research stay – data analysis	
Visakh Vaikuntanathan (1. Postdo, US)	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Workshop / DIPSI Workshop 2018
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019
Jonas Zeifang (1. Gen. PhD, US)	24.05.2017 / 26.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2017
	16.05.2018 / 19.05.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2018
	04.09.2018 / 21.09.2018	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Summer School 2018
	14.05.2019 / 18.05.2019	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar / DIPSI Workshop 2019

2. Generation of (post)doctoral students					
Daniel Appel (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doc- toral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Summer School 2022
Andre Baric (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doc- toral Seminar 2021
Anne K. Geppert (2. Postdoc, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doc- toral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo & University of Stuttgart	G.E. Cossali, B. Weigand		DROPIT Summer School 2022
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numeri- cal Methods in Science
Daniel Klötzl (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doc- toral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo & University of Stuttgart	G.E. Cossali, B. Weigand		DROPIT Summer School 2022
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numeri- cal Methods in Science
Pascal Mossier (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doc- toral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini		DROPIT Summer School 2022
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023

Weibo Ren (2. Gen. PhD, US)	10.02.2020 / 14.02.2020	University of Trento, Italy	M. Dumbser		Lecture given by Prof. Michael Dumbser: Advanced Numerical Methods For Hyperbolic Equations
	25.05.2017 / 27.05.2017	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2017
	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	15.06.2023 / 16.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
Stefan Schubert (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	13.06.2022 / 02.07.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Research stay: exchange with Prof. Cossali and Prof. Tonini	
	10.06.2022 / 30.06.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini	DROPIT Summer School 2022	
	15.06.2023 / 16.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
Raphael Tietz (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini	DROPIT Summer School 2022	
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, S. Tonini	DROPIT meets Industry Workshop	
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
Maziar Veyskarami (2. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	10.06.2022 / 30.06.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini	DROPIT Summer School 2022	
3. Generation of doctoral students					

Rishi Dhar (3. Gen. PhD, US)	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
Hasel Cicek Konan (3. Gen. PhD, US)	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
Anna Mareike Kostecky (3. Gen. PhD, US)	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
Claudia Marianowski (3. Gen. PhD, US)	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
Pablo Molina Vogel- sang (3. Gen. PhD, US)	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
	12.06.2025	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop “Analytical and Numerical Methods in Science”
Patrick Palmetshofer (3. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	05.03.2023 / 23.03.2023	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Research Stay	
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024

	10.06.2024 / 20.06.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Research Stay	
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
Rishav Saha (3. Gen. PhD, US)	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science
	15.06.2024 / 15.08.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Research Stay	
	11.06.2025 / 12.06.2025	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop “AI meets Scientific Computing” and “Analytical and Numerical Methods in Science”
	10.11.2025 / 30.11.2025	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Research Stay	
Jonathan Stober (3. Gen. PhD, US)	08.09.2021 / 09.09.2021	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Kick-Off for 2nd Funding Period / Doctoral Seminar 2021
	04.10.2021 / 15.10.2021	University of Bergamo, Italy	M. Santini	Research Stay: rebuilding of the oblique droplet generator (linear motor assembly)	
	09.05.2022 / 20.05.2022	University of Bergamo, Italy	M. Santini	Research Stay: rebuilding of the oblique droplet generator (linear motor assembly)	
	22.06.2022 / 02.07.2022	University of Bergamo, Italy	G.E. Cossali, S. Tonini	Group Work for the DROPIT Summer School	DROPIT Summer School
	15.06.2023	University of Bergamo, Italy	G.E. Cossali, B. Weigand		DROPIT meets Industry 2023
	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science

	10.11.2024 / 22.01.2024	University of Bergamo, Italy	M. Santini	Research Stay: Work with M. Santini on a journal publication on the regime map of oblique droplet impacts onto a thin film	
Amalia Travnicek (3. Gen. PhD, US)	08.02.2024 / 09.02.2024	University of Bergamo, Italy	G.E. Cossali, S. Tonini		Doctoral Seminar 2024
	18.06.2024	University of Bergamo, Italy	G.E. Cossali, B. Weigand		Workshop „Experimental and Numerical Methods in Science

6.2.7 Research stays by (post)doctoral researchers from abroad in Germany

Name	From / to	Institute visited	Local supervisors	Research activities performed and skills acquired during stay	Participation in qualification activities
Gautham Varma Raja Kochanattu (1. Gen. PhD, BG)	08.07.2019 / 09.07.2019	University of Stuttgart, Germany	A. Beck, C.-D. Munz, J. Dürrwächter		DROPIT-Seminar on “Introduction to Discontinuous Galerkin methods”
	11.07.2019 / 12.07.2019	University of Stuttgart, Germany	G.E. Cossali, S. Tonini, G. Lamanna		DROPIT-Seminar on “Analytical methods and improved non-equilibrium droplet evaporation models for high pressure conditions”
	05.12.2017 / 07.12.2017	University of Stuttgart, Germany	C.J Fitzsimons		Course on Scientific Writing
Robert Haide (1. Gen. PhD, BG)	11.04.2018 / 12.04.2018	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2018
	06.09.2018 / 19.08.2018	University of Bergamo, Italy	G.E. Cossali		DROPIT Summer School 2018
	11.04.2018 / 12.04.2018	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2018

Simone Chiocchetti (1. Gen. PhD, Trento)	06.09.2018 / 19.08.2018	University of Bergamo, Italy / University of Stuttgart, Germany	G.E. Cossali		DROPIT Summer School 2018
	03.09.2018 / 20.10.2018	University of Stuttgart, Germany	C.-D. Munz	Research stay	
	10.10.2019 / 11.10.2019	University of Stuttgart, Germany	B. Weigand		Future Perspectives Workshop with Industry
Yueqiao Hu (2. Gen. PhD, BG)	01.03.2023 / 02.03.2023	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2023
	04.2023	University of Stuttgart, Germany	B. Weigand	Research stay: discussion about the evaporation model in the numerical code FS3D	
	28.01.2025 / 29.01.2025	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2025
	31.07.2025 / 01.08.2025	University of Stuttgart, Germany	B. Weigand, E. Cossali		DROPIT Final Colloquium
Davide Ferrari (2. Gen. PhD, T)	28.01.2025 / 29.01.2025	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2025
	31.07.2025 / 01.08.2025	University of Stuttgart, Germany	B. Weigand, E. Cossali		DROPIT Final Colloquium
Daniel Regener Roig (3. Gen. PhD, BG)	28.01.2025 / 29.01.2025	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2025
	31.07.2025 / 01.08.2025	University of Stuttgart, Germany	B. Weigand, E. Cossali		DROPIT Final Colloquium
Daniele Rusconi (3. Gen. PhD, BG)	28.01.2025 / 29.01.2025	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2025

	31.07.2025 / 01.08.2025	University of Stuttgart, Germany	B. Weigand, E. Cossali		DROPIT Final Colloquium
Sara Rinaldi (3. Gen. PhD, T)	28.01.2025 / 29.01.2025	University of Stuttgart, Germany	B. Weigand		Doctoral Seminar 2025
	31.07.2025 / 01.08.2025	University of Stuttgart, Germany	B. Weigand, E. Cossali		DROPIT Final Colloquium

6.2.8 Individual reports by doctoral and postdoctoral researchers and, where applicable, qualifying fellows

Individual reports by doctoral researchers funded by DROPIT (*chronologically according to date of joining DROPIT*)

- B1-1 Patrick Foltyn**
Droplet collisions with solid superhydrophilic surfaces
- A4-1 Simone Chiocchetti**
Numerical methods for compressible multi-phase flows with complex equations of state
- B2-1 Manuel Gritti**
Experimental investigation of droplet evaporation in cross-flow
- B5-2 Weibo Ren**
Direct Numerical Simulation of Droplet Impact onto Textured Surfaces
- B6-2 Maziar Veyskarami**
Coupled free-flow–porous media flow processes including drop formation
- A3-1 Rafael Tietz**
Modelling and Simulation of Multi-Phase Flows with a Particle-Based Enskog-Vlasov Solver
- C1-2 Stefan Schubert**
Micro and macro drop impact dynamics with miscible and immiscible liquids
- A6-2 Daniel Appel**
Multiscale Modelling of Compressible Droplet Clustering for Lagrangian Models
- A2-2 Pascal Mossier**
Multi-scale modelling of the evaporation process
- B4-2 Andrej Barič**
Diffuse-interface modelling for the interaction of droplets with textured walls
- C4-2 Daniel Klötzl**
Visualization of droplet-liquid interaction
- C2-1 Jonathan Stober**
Oblique drop impact onto a wetted wall
- B1-2 Patrick Alois Palmetshofer**
Droplet collisions with solid superhydrophilic surfaces
- B6-3 Anna Mareike Kostelecky**
Visualization of droplet-liquid interaction
- A1-2 Yueqiao Hu**
Modelling of deformed, multi-component time dependent liquid drop evaporation
- B4-3 Hasel Cicek Konan**
Diffuse-interface modelling for the interaction of droplets with textured walls

- A4-2 Davide Ferrari**
Numerical methods for compressible multi-phase flows with surface tension and complex equation of state
- A3-3 Claudia Marianowski**
Gas-kinetic simulation of microdroplet – gas interaction
- C1-3 Pablo Molina Vogelsang**
Micro and macro drop impact dynamics with miscible liquids
- B5-3 Rishav Saha**
Numerical computation for drop impact on textured surfaces
- A6-3 Amalia Travnicsek**
Multi-scale modelling of compressible droplet clustering for Lagrangian models
- AX Daniel Regener Roig**
Towards the Discontinuous Galerkin solution of multi-phase flows
- A2-3 Rishi Dhar**
Numerical Investigation of Drop-film Interaction for Immiscible Fluid.
- B3-1 Daniele Rusconi**
Towards More Accurate Porous Media Modelling Using High-Resolution X-ray Imaging and Deep Learning Segmentation.
- A4-3 Sara Rinaldi**
High order structure-preserving numerical schemes for continuum mechanics of mixtures with phase transition
- A3-4 Claudia Ellenberger**
Gas-kinetic simulation of microdroplet – gas interaction

Report of doctoral researcher (SP-B1-1)

Project number: SP-B1-1

First and last name of doctoral researcher: Patrick Foltyn

(Working) title of doctoral project: Droplet collisions with solid superhydrophilic surfaces

Name of supervisors: Prof. B. Weigand, Prof. G. E. Cossali, Dr.-Ing. N. Roth

1. General information:

Duration of International Research Training Group membership: 01 May 2017 to 31 March 2021
Funding (type and duration): IRTG position (01 May 2017 to 31 March 2021) at Institute of Aerospace Thermodynamics
Start of doctoral training and (anticipated) date of doctoral degree: Started 01.05.2017 – Ph.D. defence: 30.08.2022
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 10/2011 Aerospace Engineering, University of Stuttgart, Bachelor of Science, 22 April 2015 04/2015 Aerospace Engineering, University of Stuttgart, Master of Science, 27 March 2017
Where applicable, please note any circumstances which could have affected your scientific work: COVID-19 strongly influenced the travels in generally, especially the participation on conferences and travels to IRTG-partners
Occupation after leaving the IRTG, if applicable: Alfred Kärcher SE & Co. KG / TRUMPF SE & Co. KG

2. Description of doctoral project and research results achieved to date:

Motivation

Droplet-wall interactions are observable in a large variety of technical and natural processes. Such interactions can be collisions or static interactions, i.e. droplets resting on surfaces, respectively. In technical applications, such interactions are often used deliberately to achieve the desired physical process. In order to provide desired droplet-wall behaviour, however, it is necessary to understand the physical processes during the impact. The focus of DROPIT subproject SP-B1-1 is exclusively on dry, smooth or micro-structured surfaces. Furthermore, the influence of different wettabilities ranging from superhydrophilic to slightly hydrophobic onto the respective process is investigated.

Objectives and scientific approach

The investigations of the complex physical phenomena for different impact or interaction scenarios are very complex and divers. Therefore, four subtopics are created in order to assess the respective phenomena in a more structured manner. Firstly, the surface manufacturing, reproduction and quality assessment is carried out. Secondly, a process to modify the wetting behaviour of the polymeric surface samples is worked out. Thirdly, the static droplet-wall interactions at varying wettability and surface topography i.e. smooth or structured, are investigated. Finally, the droplet-wall collisions for the latter surface samples are examined systematically under variable impact and surface conditions.

Experimental results and achieved goals

The manufactured surface samples by the Karlsruhe Micro and Nano Facility (KNMF) are reproducibly treated by plasma activation and plasma polymerization in order to modify the free surface energy. As a consequence, the apparent contact angle θ_{app} can be modified independently to the surface structure and material. This is very advantageous as usually for such evaluations with different surface wettabilities the complete surface materials and usually also the surface structure itself are exchanged. Measurements of the polymeric free surface energies can confirm the predicted contact angles using the Lewis Acid-Base approach or the OWRK-model.

Single droplets, which are gently placed on top of such surface specimens, show strong pinning effects at the three-phase contact line along the structure. The forming non-spherical shape and the alignment to the surface pattern lead to a non-uniform apparent contact angle, which depends on the azimuthal angle α . A specially modified Optical Contour Analysis (OCA) device measures contact angle hysteresis within a range of about 42° for the used micro-metric grooves and 15° to 7° for the squared pillars. Consequently, a measurement of the apparent contact angle $\theta_{app}(\alpha)$ over an azimuthal range that takes into account the symmetry of the droplet shape and the surface roughness pattern is highly recommendable. Furthermore, as Huang and Gates [1] reported such behaviour also for isotropic rough surfaces this recommendation can be extended to rough surfaces in general.

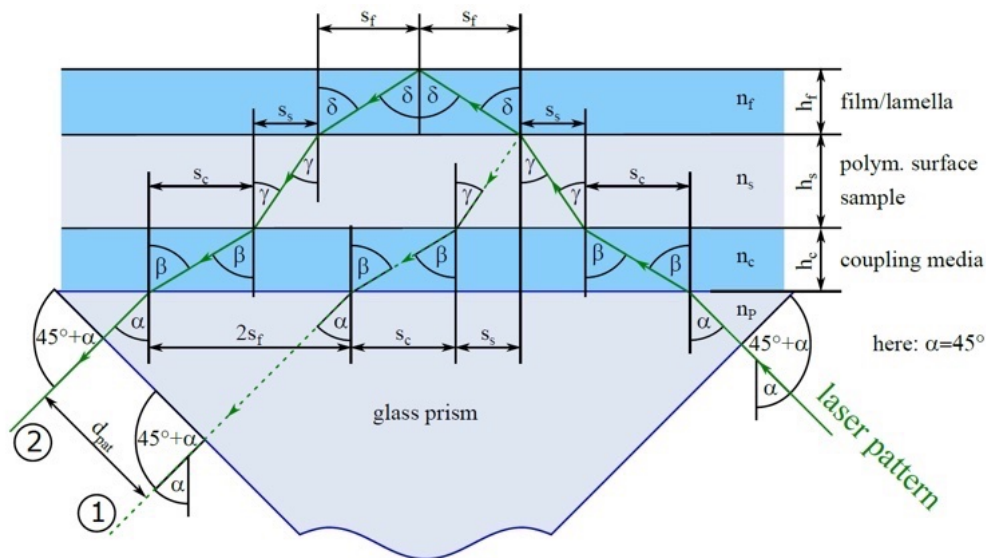


Figure 1: Measurement principle of the LASER Pattern Shift Method (LPSM) to measure the thickness of optical transparent liquid films and layers [3].

The behaviour of impacting droplets is investigated for single water and isopropanol droplets at Re numbers between 1,135 and 12,240, We numbers between 80 and 1,165 and apparent contact angles θ_{app} of 0° to 122° . With increasing impact energy an increase in spreading diameter, in spreading velocity and in case of splashing an increase in the splashing magnitude is observable independently from the surface topography. For the description of the splashing limit a modified splashing limit based on the K-parameter definition of Vander Wal et al. [2] is developed taking also the surface roughness into account. The wettability, however, shows no influence on the splashing limits, due to its inertia driven character. Wettability mainly has an influence while capillary forces are dominant. The surface structure shows its effect in directional splashing, spreading and receding movements. Furthermore, the splashing tendency is strongly increased for rougher surfaces, as expected. With the help of the extended LASER Pattern Shift Method (LPSM) the lamella thickness during the spreading phase can be determined. As expected, surfaces with higher wettability show lower lamella thicknesses during the complete spreading phase. The measurement technique has been developed within this IRTG-subproject. It is integrated into the completely new and built up test rig for this subproject. With the help of a prism and the shown principle of LASER beam tracking in Fehler! Verweisquelle konnte nicht gefunden werden., a naturally existing air bubble forming during the impact can be used for measurement. The bubble forms two shadows, a static and a moving shadow. The thinner the lamella is, the closer the moving shadow approaches the static shadow. The distance of the two shadows can be converted into a lamella thickness.

Cooperation benefits

The partnership with SP-AS04 inside the IRTG DROPIT made a deeper analysis of the surface structure possible. In cooperation with SP-B5-1 the surface export format and further research topics were realized and defined. With SP-B5-2 the formation of a bubble under an isolated pillar was experimentally and numerically investigated in detail. Moreover, the discussion with other members of DROPIT and PIs, mainly with SP-C1, SP-C5 and SP-AS01, strongly supported the build-up of the test rig, proper selection of suitable measurement techniques, the definition of the parameter space, and a significant improvement of the scientific results in this subproject.

References

- [1] Huang, X. and Gates, I. Sci. Rep., 05/2020, 10 (1),p. 8220.
- [2] Vander Wal, R. L., Berger, G. M., and Mozes, S. D., Exp. in Fluids, 01/2006, 40 (1), pp. 53–59.
- [3] Foltyn, P., PhD thesis. University of Stuttgart, 2023.

3. Comments on the qualification programme and supervision strategy:

The IRTG DROPIT offers a very comprehensive qualification program deepening the knowledge in the own field of research but also in the other topics inside the IRTG. Since the group members of the IRTG have different scientific backgrounds, there were vivid scientific discussions improving the own research output. For the course program of the IRTG, all members could suggest topics in addition to the proposed ones. They were then offered within the next half year, e.g. a workshop given by IRTG members talking about problems and solutions in their research topic, or a visiting patent attorney. The summer school, which was mainly organized by the IRTG members was very important for each researcher's scientific output. As one of the leading organizers and spokesperson for the IRTG PhD students, I was deeply involved in the organization. The stay at the foreign university during my Master's thesis and during the summer school required careful planning due to the complexity of experiments.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Analytical and numerical methods
- Analysis of droplet processes
- Elastic & inelastic light scattering
- Multiphase flow seminar

Specialised courses:

- Introduction to discontinuous Galerkin methods
- Short Course on Complex Wetting 2017 on October 11-13, 2017 in Darmstadt
- Lecture on Laser Diagnostics in Sprays given by Prof. Coghe (Milano)
- Lecture on micro-CT given by Prof. Santini (Bergamo) and Prof. Sijbers (Antwerp)

Elective and soft skill courses:

- Writer's Workshop for Scientific and Technical English
- Introductory course in Italian language and culture
- Critical Thinking for Scientists
- Polymer Microfabrication 1 & 2 (Fabrication of nickel molds and polymer replication)
- Introduction to Electron Microscopy (SEM, FIB, TEM)

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: Stay abroad at the University of Bergamo for 6 months from April to September 2016 during my M.Sc.-thesis and for 1 month again at the University of Bergamo during the summer school of the IRTG in September 2018.

Participation in the following International Research Training Group events:

- Doctoral Seminar on 25. Mai 2017 in Bergamo
- Doctoral Seminar on 11./12. April 2018 in Stuttgart
- DROPIT Workshop on 17. Mai 2018 in Bergamo
- Summer School 2018 from 4. to 21. September 2018 in Stuttgart
- Doctoral Seminar 15./16. Mai 2019 in Bergamo
- DROPIT Seminar 4./5. June 2019 in Bergamo

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI), Italy, in 2017, 2018 and 2019
- Technologieforum Bildverarbeitung 2017 in Munich, Germany
- KNMF User Meeting in Karlsruhe, Germany, in 2018 and 2019
- 9th International Colloids Conference, Sitges (Spain), 2019
- 29th ILASS-Europe, Paris (France), 2019

4. Individual publications:

I. Peer-reviewed articles:

- **P. Foltyn**, D. Ribeiro, A. Silva, G. Lamanna, B. Weigand; Influence of wetting behavior on the morphology of droplet impacts onto dry-patterned micro-structured surfaces. *Physics of Fluids* 1 December 2022; 34 (12): 123322. <https://doi.org/10.1063/5.0124692>
- Ben Ikenson; Bubble shadows add significant new dimension to film thickness measurement methodology. *Scilight* 29 October 2021; 2021 (44): 441105. <https://doi.org/10.1063/10.0007044>
- **P. Foltyn**, L. K. Rihm, D. Ribeiro, A. Silva, B. Weigand; Measurement of the lamella thickness during droplet impact onto differently wettable smooth surfaces using an extension of the LASER Pattern Shift Method with naturally occurring patterns. *Rev. Sci. Instrum.* 1 October 2021; 92 (10): 105111. <https://doi.org/10.1063/5.0060407>
- Ren, W., **Foltyn, P.**, Geppert, A. & Weigand, B. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. *Sci Rep* **11**, 18018 (2021). DOI: 10.1038/s41598-021-97376-3
- **P. Foltyn**, D. Ribeiro, A. Silva, G. Lamanna, B. Weigand; Influence of wetting behavior on the morphology of droplet impacts onto dry smooth surfaces. *Physics of Fluids* 1 June 2021; 33 (6): 063305. <https://doi.org/10.1063/5.0053539>
- **Foltyn P.**, Restle F, Wissmann M, Hengsbach S, Weigand B. The Effect of Patterned Micro-Structure on the Apparent Contact Angle and Three-Dimensional Contact Line. *Fluids*. 2021; 6(2):92. <https://doi.org/10.3390/fluids6020092>
- **Foltyn, P.**, Guttman, M., Schneider, M., Fest-Santini, S., Wildenschild, D., Weigand, B. (2020). Fabrication and Evaluation Methods of Micro-structured Surfaces for Droplet Impact Experiments. In: Lamanna, G., Tonini, S., Cossali, G., Weigand, B. (eds) *Droplet Interactions and Spray Processes. Fluid Mechanics and Its Applications*, vol 121. Springer, Cham. https://doi.org/10.1007/978-3-030-33338-6_6
- Santini, M., Fest-Santini, S., **Foltyn, P.**: On the local mass transfer rates around arbitrary shaped particles calculated by X-ray computed microtomography: Prospective for a novel experimental technique. *International Communications in Heat and Mass Transfer*, 2016, 79, 135 – 139.

II. Further publications

a) Proceedings

- **Foltyn P.**, Ribeiro D., Silva A., Lamanna G., and Weigand B.: Effect of surface wettability on the droplet impact morphologies on dry smooth polycarbonate surfaces. *ICLASS, 15th Triennial International Conference on Liquid Atomization and Spray Systems*. Edinburgh, UK, 08/2021, doi: 10.2218/iclass.2021.5907
- Ribeiro D., **Foltyn P.**, Silva A., Lamanna G., and Weigand B.: The influence of wettability on the droplet impact onto micro-structured surfaces. *ICLASS, 15th Triennial International Conference on Liquid Atomization and Spray Systems*. Edinburgh, UK, 08/2021. doi: 10.2218/iclass.2021.5908.
- Bernard, R., **Foltyn, P.**, Geppert, A., Lamanna, G. and Weigand, B.: Generalized analysis of the deposition/splashing limit for one- and two-component droplet impacts upon thin films. 28th ILASS-Europe, Valencia, Spain, 2017.
- **Foltyn, P.**, Restle F., Weigand, B.: 360° Evaluation of Projected Contact Angles of Static Droplets on Structured Surfaces. DIPSI, Bergamo, Italy, 2019.

b) Technical and scientific reports

- **Foltyn P.**, Weigand B., and Roth N.: 'Measuring apparatus and method for determining the depth of field of an optical system'. German Patent application DE10 2021 206 221.0. 06/2021.
- **Foltyn, P.**: DROPIT Midterm Review: DROPIT Subproject B1: Droplet collisions with solid superhydrophilic surfaces, May 2019.

- **Foltyn, P.**, Roth, N., Weigand, B., inventor; University of Stuttgart, assignee. Verfahren zur Messung der Schichtdicke einer optisch transparenten Schicht, insbesondere einer Flüssigkeitsschicht. European Patent application 19 192 074.3. 2019 Aug 16.

III. Presentations

a) Oral

- **Foltyn, P.**, Roth, N. & Weigand, B.: Development and calibration of the LASER Pattern Shift Method for measuring the lamella topology during drop impact on walls *29th European Conference on Liquid Atomization and Spray Systems*, 2019.

b) Posters

- **Foltyn, P.**, Schlottke, A., and Weigand, B.: Effect of plasma activation on the contact angle of smooth polymer surfaces and its long-term durability in ambient air. *10th International Colloids Conference*, 7-9 December 2020, (Online) Palma de Mallorca, Spain. Vol. 10. 12/2020.
- **Foltyn, P.**, Restle, F., Weigand, B.: 360° Evaluation of Projected Contact Angles of Static Droplets on Structured Surfaces. *Droplet Impact Phenomena & Spray Investigations (DIPSI)*, Bergamo, Italy, 2019.
- Schegk, G., **Foltyn, P.**, Li, W.-J., Terzis A., Weigand B.: Oxygen and Hydrogen Plasma Activation Effects on Capillary Thermodynamics in Fibrous Materials, *9th International Colloids Conference*, 16-19 June 20

Report of doctoral researcher (SP-A4-1)

Project number: SP-A4

First and last name of doctoral researcher: Simone Chiocchetti

Title of doctoral project: Numerical methods for compressible multi-phase flows with complex equations of state

Name of supervisors: Prof. Dr.-Ing Michael Dumbser, Prof. Dr.-Ing Claus-Dieter Munz

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT since May 2018
Funding (type and duration): PhD student at the Laboratory of Applied Mathematics, Department of Civil, Environmental, and Mechanical Engineering in Trento, Nov 2019 – June 2022
Start of doctoral training and (anticipated) date of doctoral degree: 5 Nov 2018 to 10 June 2022
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09/2011 Environmental Engineering, University of Trento, Bachelor's degree, 18 Dec. 2014 09/2014 Environmental Engineering, University of Trento, Master's degree, 29 March 2018 11/2018 Civil, Environmental, and Mechanical Engineering, University of Trento, PhD degree, 10 June 2022
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

The project aims at developing robust, accurate, and efficient numerical methods for multiphase flow with complex equations of state. A showcase practical application might be the simulation of freezing droplets, where the change of phase is taken into account by means of a specifically designed equation of state.

The chosen simulation approach is the following: First, one writes a system of first order hyperbolic partial differential equations, for example starting from models of the Baer-Nunziato type, and then includes new terms/equations in order to take into account surface tension, phase change, viscosity, heat transport. Subsequently, such equations are discretized and solved by means of a High Order Finite Volume or Discontinuous Galerkin method. Such methods can be formulated for a general balance law including conservative fluxes, as well as non-conservative differential terms and algebraic sources, so that one is allowed to operate changes to the mathematical model under development, without having to change the numerical method to accommodate the new physics.

The task of simulating such complex multiphase flows, besides the construction of the mathematical model itself, forwards a number of challenges. Namely, one has to deal with complex systems generally counting several fully coupled equations (ten, twenty, thirty, ...), with non-conservative products, strong resolution requirements, phase vacuum, dimensional scaling issues, stiff algebraic source terms.

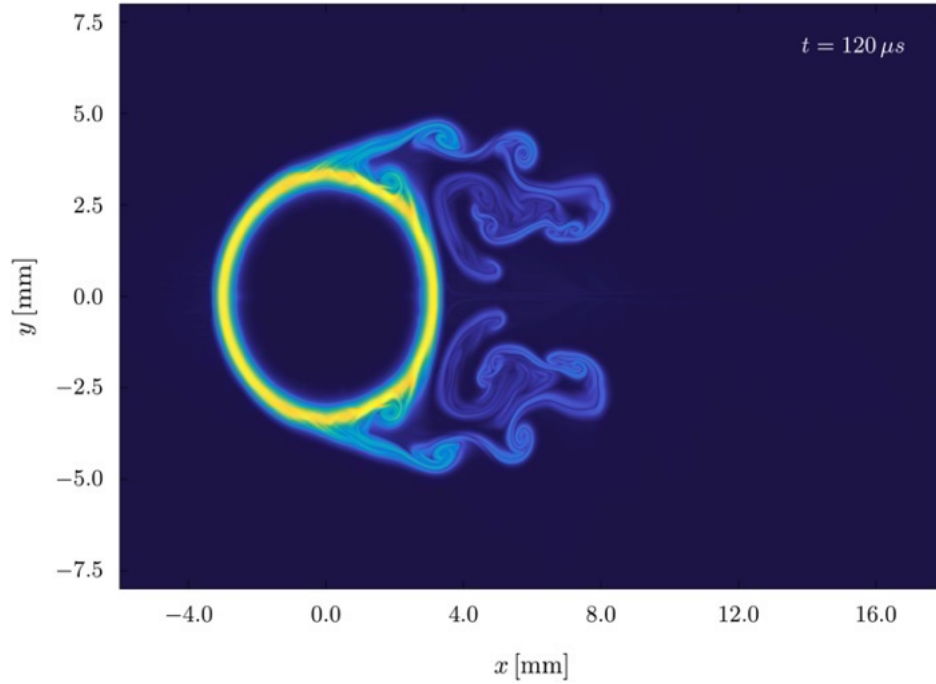


Figure 1: Visualization of the interface field in the early stages of a Shockwave-Water column interaction problem. The result is obtained by solving, with a fourth-order ADER Discontinuous Galerkin scheme on a uniform Cartesian grid, the equations of the five-equation-Baer-Nunziato model, with the addition of a set of equations governing the evolution of the interface, introduced for the purpose of modelling surface tension effects.

Thanks to the generality and modularity of the employed numerical machinery (fully discrete one-step arbitrary high order ADER methods), each challenge may be tackled somehow separately from the others. In particular, attention has been devoted to four topics:

1. The inclusion of capillarity effects in the five-equation-Baer Nunziato model, with a hyperbolic formulation of surface tension which tracks the movement of geometrical interfaces directly, rather than reconstructing them from phase-field data at each time step. The methodology yielded very promising results on fixed Cartesian grids. In the near future the model will be tested also on space-time-adaptive quadtree-type meshes, on which the transfer of data between grid of different coarseness is a nontrivial issue, due to potential ill-conditioning given from the presence of two or more phases and material interfaces.
2. The treatment of stiff sources by means of exponential-integration-type techniques: friction between the air and water phases, as well as pressure relaxation and freezing (to be accounted for in the future) require that special care be taken in adequately resolving the effects of the algebraic source terms used to model these phenomena. For the purpose, a novel splitting method based on exponential integration has been developed and carefully tested on the six and seven equation Baer-Nunziato models, as well as on a mathematical model for crack propagation in elastoplastic solids (for which the modelling approach is the same as the one we are planning to employ for freezing dynamics), again this methods gave satisfactory results.
3. The construction of efficient methods for flows where the velocity scales of the phenomena are much lower than the speed of acoustic waves (this would be quite often the case for freezing droplets): for this purpose, a parallel, semi-implicit Finite Volume code that solves the equations of Kapila (five-equation-Baer-Nunziato) with hyperbolic surface tension has been written and is currently being tested.
4. The possibility of increasing the accuracy of interface tracking by means of methods of the Arbitrary-Lagrangian-Eulerian (ALE) type has been considered and a number of innovative mesh motion strategies have been implemented in a new Discontinuous Galerkin and Finite Volume ALE code in which the grid connectivity graph may change at each time step. The code has been validated on challenging test problems that are typically used to assess the quality of Lagrangian codes, but not yet on droplet dynamics.

With more detail, the three years of the doctoral studies can be broken down as follows:

1st year. The project aims at developing robust, accurate, and efficient numerical methods for compressible multiphase flows. The models being investigated are those of the Baer--Nunziato type, as well

as a new class of models derived from the theory of Symmetric Hyperbolic Thermodynamically Compatible systems (SHTC). Both approaches feature several challenges which can be tackled with the classical engineering method of splitting a large problem in many smaller and simpler subproblems to be solved separately. Four main lines of work have been outlined:

1. SHTC models often require that solutions of the governing equations satisfy some stationary differential constraint, in addition to the evolution laws per-se. Namely often it is necessary that the curl of certain vector fields be zero at all times. Initially, it was unclear whether such constraints played a significant role in the long-term stability of simulations, but after having implemented two different solutions to the issue, it is apparent that these involution constraints must indeed be carefully accounted for. The two solutions are:

- 1.1. The construction of an enlarged PDE system for two-phase flow with surface tension that allows to propagate the errors in the constraints outside of the computational domain, with an approach which is analogous to the Generalised Lagrangian Multiplier divergence cleaning commonly employed in the solution of the equations of Magnetohydrodynamics.

- 1.2. The development of a numerical scheme in which the discrete differential operators for gradient and curl are defined in such a way that initially curl-free vector fields preserve the property to machine accuracy. The code has been tested on twodimensional flows yielding very good results.

2. Stiff sources are often employed for modelling mechanical relaxation and phase transition in multiphase flow. They are often treated with robust methods like the implicit Euler scheme, which I found to give results that are not satisfactory in many test problems. For this reason I developed a new code for solving stiff ODEs. The code is based on Exponential Integration, and features a decomposition of the source jacobian which allows to robustly deal with the extreme range of timescales. The code has also been used (in collaboration with M. Tavelli and E. Romenski) for solving fracture kinetics in a SHTC model for elasto-plastic solids. The mathematical structure of such kinetics strongly resembles the one that we are envisaging for modelling freezing droplets.

3. For low Mach number flows, explicit schemes can be inefficient. To solve the issue, the exactly-curl-free method introduced in paragraph 1.2, is implemented in a code that treats acoustic waves implicitly. The code gives good solutions in little time thanks to the parallel implementation employing efficient neighborhood-collective MPI operations. I also developed another semi-implicit code for a mixture-energy consistent six-equation Baer-Nunziato model in one dimension.

4. I collaborated with E. Gaburro and W. Boscheri on the construction of an Arbitrary Lagrangian-Eulerian code using a moving Voronoi grid and allowing changes of connectivity between consecutive timesteps. Specifically, I worked on the task of grid generation and on implementing several strategies to constrain the mesh motion so that the quality of the initial grid is maintained throughout the simulation.

2nd year. During the second year of my PhD, my research activities have been focused on the numerical solution of diffuse interface models for two-phase flow, as well as diffuse interface models for elasto-visco-plastic solids with material failure. During this period of time, I tackled several sub-topics:

1. I have completed the development of the GLM curl cleaning approach for treating differential involution constraints of the curl type in the context of solving a novel strongly hyperbolic diffuse interface model for two phase flow with surface tension, by means of high order ADER Discontinuous Galerkin and Finite Volume methods.

2. I have applied the same GLM approach in the solution of surface tension-driven flows with semi-implicit methods that are specifically designed for treating low Mach number flows. This work has made apparent that, when using methods of second order of accuracy, the surface tension model under investigation can be solved with a much higher degree of stability and with lower error if one employs particular discretisation techniques that allow to preserve curl involutions at the discrete level exactly. I have written an MPI-parallel Fortran computational code implementing such structure-preserving methods and began its extension to viscous flows with surface tension, heat conduction and phase change (freezing). This work has been carried forward in the context of my visit to the Institute of Aerodynamics and Gasdynamics of the university of Stuttgart.

3. I have continued my collaboration with M. Tavelli on the topic of diffuse interface modeling of material failure and specifically, I investigated the capabilities of the GPR model in the description of materials exhibiting weakening and failure due to fatigue.

4. I have collaborated with the Geophysics research group led by A. Gabriel at the Ludwig-Maximilians-Universität of Munich on the topic of modeling and computation of nonlinear dynamic rupture processes in diffuse fracture zones. This work has been based on the diffuse interface fracture model object of my previous collaboration with M. Tavelli, and involved its extension to materials with phase change (melting and solidification) and weakness to tensile stresses. In the context of this work, I also developed a framework for the estimation of material parameters of the Godunov-Peshkov-Romenski model with material failure, using particle swarming optimisation.

3rd year. I devoted the third year of my doctoral studies to the development of a high performance code for the simulation of viscous two-phase flows with surface tension. The code is based on the SHTC unified formulation of continuum mechanics and exactly curl-preserving discretisation discussed in the previous years. The main topics that have been tackled are:

1. The coupling between several different sub-systems (two-phase flow convection, acoustic waves, viscosity/strain dynamics surface tension) which form the large set of governing equations. In particular, it is computationally impossible to obtain exact signal speeds of the full hyperbolic system, while eigenvalues of each sub-system can be computed, although this is a non-trivial task on itself, even with the aid of computer algebra systems. Thus I have computed and tested some relatively simple but accurate approximation formulas for the eigenvalues of the full system.
2. The accurate and efficient integration of the relaxation source terms governing strain relaxation and thus viscosity. A new approach in the treatment of such highly nonlinear strain relaxation source terms, exploiting semi-analytic solutions like in my previous attempts at tackling this problem, allows now to compute accurate solutions to arbitrarily stiff relaxation sources with unparalleled robustness and efficiency. The main novelty with respect to previously developed semi-analytical techniques is that the new one does not use source splitting and preserves the asymptotic coupling between strain production by convective effects and strain relaxation operated the algebraic source. Moreover, this new technique does not require numerical evaluation of matrix exponentials or inversion of matrices of size larger than 3 by 3. Comparatively, even the very simple implicit Euler method would require the inversion of a sequence of 9 by 9 matrices instead, while not achieving the same accuracy or even failing altogether. The same technique has been applied in the context of a collaboration with W. Boscheri as a building block for an Updated Lagrangian code for fluids, elastic, and plastic solids on moving two and three dimensional unstructured simplicial grids.
3. The maximisation of the computational efficiency of the code. The computational efficiency of the code has been assessed and vastly improved with respect to earlier versions. The first step has been the streamlining the use of inter-cpu data exchange routines by so-called communication masking via non-blocking MPI routines. A second important improvement has been the reduction of the number of RAM read/write operations which involved an extensive restructuring of the Fortran implementation of the code. The code has been shown to efficiently scale up to 65000 CPU cores on the HPE Hawk supercomputer in Stuttgart and was capable of handling simulations involving several billions of degrees of freedom even in single-file-MPI-parallel disk I/O tasks.

Cooperation benefits

I collaborated with other DROPIT doctoral researchers for an extended period of time in two instances: first, during Christoph Mueller's (project SP-A2) stay in Trento (five weeks in the summer of 2018, and he going to be back again to Trento starting from mid-September 2019). We discussed at length numerical methods for the solution of non-conservative hyperbolic equations (path-conservative schemes specifically), scientific programming (Christoph shared some of his coding expertise), and advantages and drawbacks of the diffuse-interface approach (when compared to so-called sharp-interface methods). Second, in occasion of the DROPIT Summer School in September, followed by a three-week visit to the IAG in Stuttgart, I worked on a method for treating of stiff thermochemical relaxation source terms, as well as developing a preliminary semi-implicit method for the six-equation Baer-Nunziato system. Regarding this latter point, consultation with Jonas Zeifang, (project SP-A6), who is specifically working on methods for weakly compressible flows, has been very valuable.

Participation in the following courses/seminars of the Int. Research Training Group:

- Advanced numerical methods for free-surface flows
- Advanced thermodynamics
- Mathematical and Numerical methods for Partial Differential Equations

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

1. Research stay in Marseille (France) with Prof. Sergey Gavriluk, 19 June to 1 July 2018, during which we discussed mathematical modelling of compressible multiphase flow
2. Research stay in Stuttgart from 3 September to 20 October 2018, period during which:
 - I participated in the Summer School from 3 September to 21 September
 - I visited the IAG and worked with fellow DROPIT doctoral researchers Christoph Mueller and Jonas Zeifang, both students of Prof. Dr.-Ing. Claus-Dieter Munz, on shock-capturing methods for Baer-Nunziato-type systems.
 - I attended the MPI-OpenMP parallel programming course held at the HLRS.

3. Research stay in Stuttgart from 1 Sep. to 30 Nov. 2020, aimed at the realisation of a high performance parallel code for the computation of low-Mach two-phase flows with surface tension.

Participation in the following International Research Training Group events:

- Doctoral Seminar, Stuttgart, 11-12 April 2018
- Workshop on Experimental and Numerical Methods, Bergamo, 17 May 2018
- Summer School, Stuttgart, 4-21 September 2018
- Doctoral Seminar, Bergamo, 15-16 May 2019
- DROPIT Workshop, Stuttgart, Germany, 10-11 Oct 2019
- DROPIT Seminar, Bergamo, Italy, 8-9 Sep 2021

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2018
- HONOM conference, Madrid (Spain), 1-5 April 2019
- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2019
- SHARK-FV workshop, Povia de Varzim (Portugal), 20-24 May 2019
- Lava-flow modelling workshop, Catania (Italy), 22 July 2019
- MultiMat conference, Trento (Italy), 09-13 Sep 2019
- Enumath conference, Egmond aan Zee (the Netherlands), 30 Sept. to 5 October 2019
- NumHyp conference, Trento (Italy), 26-30 July 2021
- CIRM International workshop "Perspectives on Multiphase Fluid Dynamics, Continuum Mechanics and Hyperbolic Balance Laws", Marseille (France), 7-11 March 2022
- HONOM conference, Braga (Portugal), 4-8 April 2022
- SHARK-FV workshop, Povia de Varzim (Portugal), 23-27 May 2022
- UnTRIM workshop, Trento (Italy), 13-15 June 2022
- MultiMat conference, Zürich (Switzerland), 22-26 August 2022

4. Individual publications:

I. Peer-reviewed articles:

- E. Gaburro, W. Boscheri, **S. Chiocchetti**, C. Klingenberg, V. Springel, M. Dumbser. High order direct Arbitrary-Lagrangian-Eulerian schemes on moving Voronoi meshes with topology changes. *Journal of Computational Physics*, Volume 407, 109167, 2020.
- S. Busto, **S. Chiocchetti**, M. Dumbser, E. Gaburro, I. Peshkov. High order ADER schemes for continuum mechanics. *Frontiers in Physics*, Volume 8, 32, 2020.
- **S. Chiocchetti**, C. Müller. A solver for stiff finite-rate relaxation in Baer–Nunziato two-phase flow models. in G. Lamanna, S. Tonini, G.E. Cossali, B. Weigand (Eds.), in 'Dropet Interaction and Spray Processes', Springer, Heidelberg, Berlin, 31-44, 2020.
- C. Müller, T. Hitz, S. Jöns, J. Zeifang, **S. Chiocchetti**, C.D. Munz. Improvement of the Level-Set Ghost-Fluid Method for the Compressible Euler Equations. in G. Lamanna, S. Tonini, G.E. Cossali, B. Weigand (Eds.), in 'Dropet Interaction and Spray Processes', Springer, Heidelberg, Berlin, 17-29, 2020.
- M. Tavelli, **S. Chiocchetti**, E. Romenski, A.A. Gabriel, M. Dumbser. Space-time adaptive ADER discontinuous Galerkin schemes for nonlinear hyperelasticity with material failure. *Journal of Computational Physics*, Volume 422, 109758, 2020.
- M. Dumbser, **S. Chiocchetti**, I. Peshkov. On numerical methods for hyperbolic PDE with curl involutions. in G.V. Demidenko, E. Romenski, E.F. Toro, M. Dumbser (Eds.), 'Continuum Mechanics, Applied Mathematics and Scientific Computing: Godunov's Legacy', Springer International Publishing, 125-134, 2021.
- **S. Chiocchetti**, I. Peshkov, S. Gavrilyuk, M. Dumbser. High order ADER schemes and GLM curl cleaning for a first order hyperbolic formulation of compressible flow with surface tension. *Journal of Computational Physics*, Volume 426, 109898, 2021.
- A.A. Gabriel, D. Li, **S. Chiocchetti**, M. Tavelli, I. Peshkov, E. Romenski, M. Dumbser. A unified first order hyperbolic model for nonlinear dynamic rupture processes in diffuse fracture zones. *Philosophical Transactions of the Royal Society A*, Volume 379, 20200130, 2021.
- I. Peshkov, M. Dumbser, W. Boscheri, E. Romenski, **S. Chiocchetti**, M. Ioriatti. Simulation of non-Newtonian viscoplastic flows with a unified first order hyperbolic model and a structure-preserving semi-implicit scheme. *Computers and Fluids*, Volume 224, 104963, 2021.

- W. Boscheri, **S. Chiocchetti**, I. Peshkov. A cell-centered implicit-explicit Lagrangian scheme for a unified model of nonlinear continuum mechanics on unstructured meshes. *Journal of Computational Physics*, Volume 451, 110852, 2022.

II. Presentations

a) Oral

- S. Chiocchetti, M. Dumbser, S. Gavriluk, I. Peshkov: High order ADER schemes for a first order hyperbolic formulation of compressible flow with surface tension. HONOM, Madrid (Spain), 1-5 April 2019.
- S. Chiocchetti, M. Dumbser, S. Gavriluk, I. Peshkov: High order ADER schemes for a first order hyperbolic formulation of compressible flow with surface tension. SHARK-FV, Povo de Varzim (Portugal), 20-24 May 2019.
- S. Chiocchetti, M. Dumbser: High order numerical methods for a hyperbolic formulation of viscous flows. Lava-flow modeling workshop, Catania (Italy), 2019.
- S. Chiocchetti, M. Dumbser. High order space-time adaptive ADER schemes for compressible multi-phase flows, (Enumath), Egmond aan Zee (the Netherlands), 30 Sept. - 4 Oct. 2019.
- S. Chiocchetti, M. Dumbser, I. Peshkov. Tips and tricks for Simple Hyperbolic Viscous Flow. NumHyp, Trento (Italy), 26-30 July 2021.
- S. Chiocchetti, M. Dumbser, I. Peshkov. Hyperbolic viscous flow made simple. CIRM International workshop, Marseille (France), 7-11 March 2022.
- S. Chiocchetti, M. Dumbser, I. Peshkov. Hyperbolic viscous flow with hyperbolic surface tension. HONOM conference, Braga (Portugal), 4-8 April 2022.
- S. Chiocchetti, M. Dumbser, I. Peshkov. Semi-implicit schemes with efficient finite rate stiff relaxation for compressible two-phase flow and plasticity. MultiMat conference, Zürich (Switzerland), 22-26 August 2022.

b) Poster

- S. Chiocchetti, M. Dumbser: High order schemes for a first order hyperbolic formulation of compressible flow with surface tension. DIPSI, Bergamo (Italy), 2018.

Report of doctoral researcher (SP-B2-2)

Project number: SP-B2 First and last name of doctoral researcher: Manuel Giovanni Gritti (Working) title of doctoral project: Experimental investigation of droplet evaporation in cross-flow (in Italian: “Indagine sperimentale sull'evaporazione di gocce in flusso incrociato”) Name of supervisors: Prof. M. Santini, Prof. B. Weigand
--

1. General information:

Duration of International Research Training Group membership: Doctoral research of DROPIT from 01.10.2018 to 31.12.2021
Funding (type and duration): Doctoral scholarship from Università degli Studi di Bergamo for three years
Start of doctoral training and (anticipated) date of doctoral degree: start: 10/2018; doctoral degree: 03/2023
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09/2012 Mechanical Engineering, Università degli Studi di Bergamo, Bachelor Degree 03/2015 Mechanical Engineering, Università degli Studi di Bergamo, Master Degree 03/2018 Civil Engineering, Università degli Studi di Bergamo, Master Degree
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Motivation

The main objective of this sub-project is to provide experimental data for the validation of numerical models and assumptions of two main research topics: wetting transition and evaporation on porous surfaces.

Part 1 Wetting transition

A wide range of applications depends on the self-cleaning mechanism of surfaces. Wettability can be induced and controlled by surface's topography. Super-hydrophobicity is associated with the heterogeneous air trapping regime known as Cassie wetting. When the Cassie wetting state is not stable, transition to the Wenzel homogeneous one may occur spontaneously or is promoted by external stimuli. The proceeding from Cassie to Wenzel state leads to the loss of the repellent and self-cleaning surface's properties. Here, the understanding of wetting transition mechanisms is crucial for design optimizing of hydrophobic and superhydrophobic materials.

Part 2 Evaporation on porous surfaces

Porous media flow phenomena are largely conducted through tortuous paths of interconnected voids in complex networks of solid matrices. As a result, these flows are inherently complicated and are still subject to many unresolved fundamental questions. In cases where there is an adjoining parallel flow, an additional difficulty that is presented is the proper definition of boundary conditions at the interface between the porous medium and the cross-flow. As these issues are still analytically and numerically challenging, there is a pressing demand for experimental studies to serve as a basis for further under-

standing of the underlying physics involved. Therefore, an experimental setup will be developed to provide experimental data. The latter could help to resolve these outstanding concerns pertaining to cross-flow over porous media.

Objective and scientific approach

Part 1 Wetting transition

The main objective of this sub-project is to visualize the triple line for different wetting states and to improve the understanding of the impact exerted by surface topography on wetting transition. Therefore, micro-CT is used to generate 3D images of the drop-substrate system.

Part 2 Evaporation on porous surfaces

The main objective of this sub-project is to design an experimental setup to observe an evaporating drop grown from modeled porous media in natural or forced convection. This experimental setup needs to fit the requirements to be installed within the X-ray micro computed tomography unit (microCT unit) as well as needs to be adaptable for high-speed measurements.

Results and achieved goals

Part 1 Wetting transition

Structured surfaces were scanned by microCT and surface models were extracted and converted to triangulated meshes described by the unit normal and vertices of the triangles using a three-dimensional Cartesian coordinate system in SP-AS04. For minimizing X-ray interaction, reflection and refraction, with the topographic pattern, these surfaces were acquired to be inclined with respect to the horizontal. The geometrical characterization of surface structures requires a tool for measuring automatically those patterns requiring a previous roto-translation to re-align the digitalized geometry. Here fore, the geometry is translated into its center of mass. Further, a plane is defined by three non-collinear points allowing the determination of the required rotation angle.

The rotation matrix can be described with

$$\text{Mod}_{\text{rot}} = \begin{bmatrix} V_x^2 + (1 - V_x^2)\cos(\alpha) & V_x V_y (1 - \cos(\alpha)) - V_z \sin(\alpha) & V_x V_z (1 - \cos(\alpha)) + V_y \sin(\alpha) \\ V_x V_y (1 - \cos(\alpha)) - V_y \sin(\alpha) & V_y^2 + (1 - V_y^2)\cos(\alpha) & V_x V_z (1 - \cos(\alpha)) - V_x \sin(\alpha) \\ V_x V_z (1 - \cos(\alpha)) - V_y \sin(\alpha) & V_y V_z (1 - \cos(\alpha)) + V_x \sin(\alpha) & V_z^2 + (1 - V_z^2)\cos(\alpha) \end{bmatrix} p \quad (1)$$

where: $V_x, V_y, V_z \rightarrow$ Versor of the components of the rotation axis,

$p \rightarrow$ Points of pre-translation geometry

$\alpha \rightarrow$ Rotation Angle

A second rotation along the z-axis orientates the surface pattern parallel to the x- and y-axis.

$$\text{Mod}_{\text{fin}} = \begin{bmatrix} \cos(\vartheta) & -\sin(\vartheta) & 0 \\ \sin(\vartheta) & \cos(\vartheta) & 0 \\ 0 & 0 & 1 \end{bmatrix} p \quad (2)$$

Pillar profiles along the xz and yz planes can be extrapolated and their regularity evaluated. Figure 1a shows overlaid profiles of all single pillars and the averaged one.

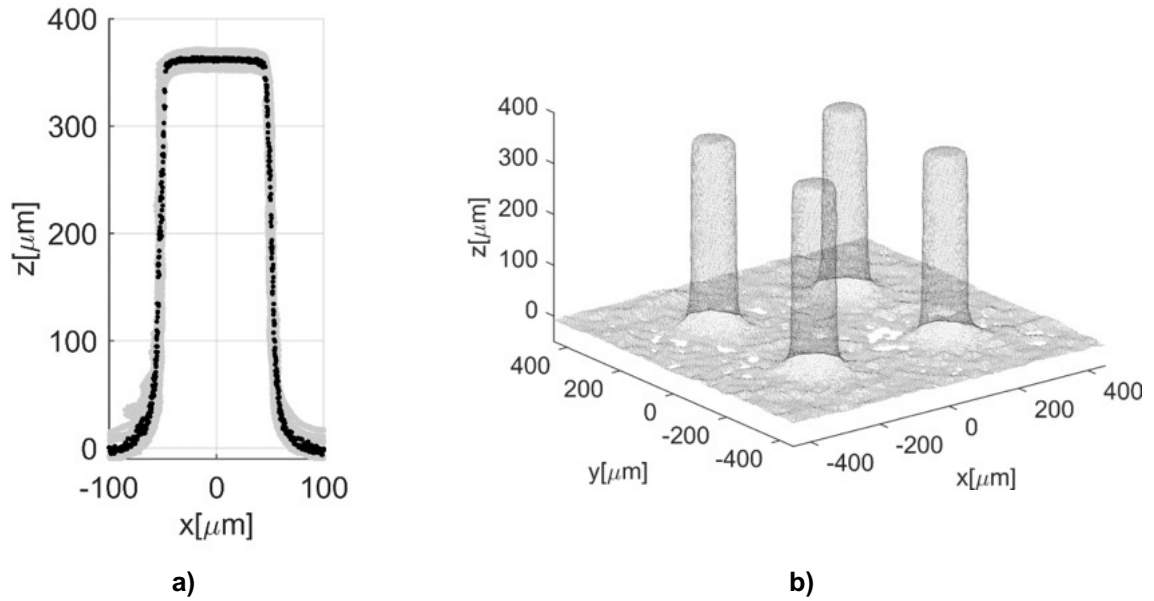


Figure 1: a) trend of pillars profile in xz plane and median profile. b) aligned central pillars.

This tool is therefore ready and usable for any sample that needs to be analyzed in the future independent from the microstructure's dimension. Thus, a tool to fulfill the first sub-project goal, the geometrical characterization of surface pattern, is available.

The main goal of this sub-project, 3D visualization of the triple line, requires a method for its detection as well as the automatic calculation of the contact angle distribution and if applicable local capillary pressures. Here fore, in-situ wettability measurements and the corresponding

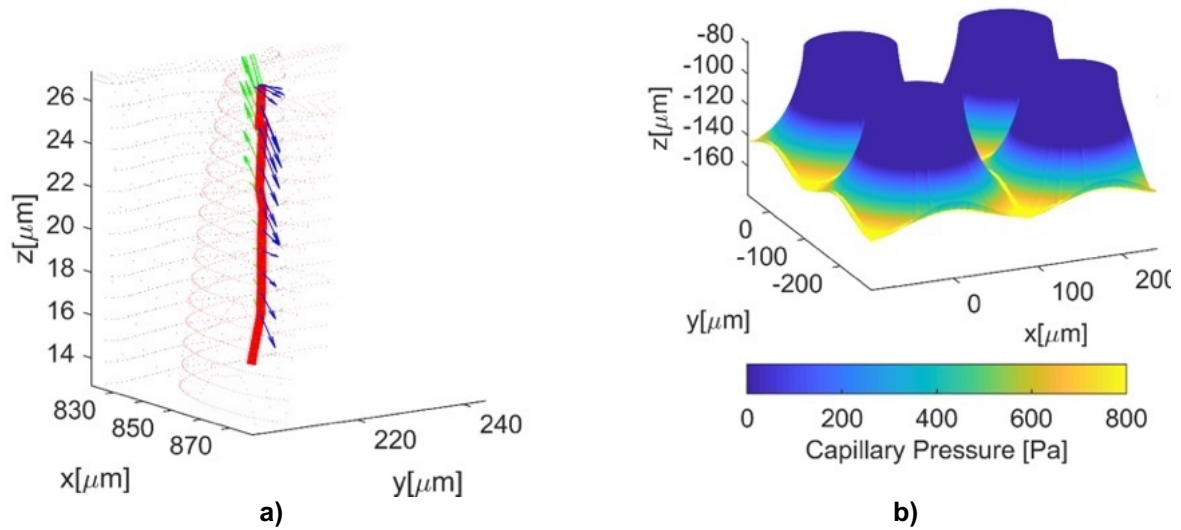


Figure 2: a) Contact line zone showing in green the normal vectors of the gas/liquid interface and in blue the normal vectors of the liquid/solid interface; b) Trend of the capillary pressure inside the hole of pillars.

extracted multi-zone surface mesh from a segmented voxelized microCT volume were provided from SP-AS04.

The extracted meshes are sets of vertexes, and the three-phase contact line is the set of vertexes, which are shared by all of them.

First, it is necessary to perform a smoothing procedure as suggested by Garcia [1].

A smooth surface method was chosen. This method allows to reduce the voxel effect due to the reconstruction, thus proceeding with the analysis. There are different smoothing methods. The one chosen is one of the best performing methods, a Gaussian 3D smoother. It will then be necessary to precisely set the characteristic parameters when using the reconstructed models concerning the experiment with structured surfaces, e.g. the number of iterations for smoothing.

Based on the smoothed meshes, the contact angle on each vertex that belongs to the contact line set is computed, whereby the procedure suggested by AlRatrou et al [2]. is followed. Each vertex is represented with two vectors normal to the gas/liquid interface and the liquid/solid interface. Then, the contact angle for each vertex on the contact line is determined through the liquid phase. Contact line set and normal vectors are shown in Figure 2a. The cosine of the contact angle is calculated from the dot product of these two normals.

$$\theta_i = \pi - \cos^{-1}(\mathbf{n}_i \cdot \mathbf{z}_2 \cdot \mathbf{n}_i \cdot \mathbf{z}_3) \quad (3)$$

where: $\mathbf{n}_{z_2}$: normal vector to the gas/liquid interface.
 $\mathbf{n}_{z_3}$: normal vector to the liquid/solid interface.

Capillary pressure is inherently a pore-scale phenomenon and could be also calculated for the heterogeneous wetting regime. It is defined at equilibrium as the difference in pressure across a curved interface separating two immiscible fluids, expressed as

$$P_c = P_{nw} - P_w \quad (4)$$

here P_{nw} is the non-wetting phase pressure at an interface and P_w is the wetting phase pressure at an interface. In a multiphase system, P_c is attributable to the normal component of the interfacial forces at the interfacial boundaries.

By considering a small segment of a curved interface, balancing pressure and interfacial forces, and assuming static equilibrium, the Young-Laplace equation can be written as

$$P_c = 2\sigma\kappa \quad (5)$$

where σ is the interfacial tension between the wetting and the non-wetting phase, and κ is the mean curvature. The calculation of the mean curvature uses the surface treated with the previously described smoothing technique. This calculation is carried out in the area where there are only two phases, liquid and gas. In Fig. 2b, instead, a trend of the capillary pressure of the 3D model is represented.

Preliminary wetting experiments depositing a sessile drop onto a structured surface were carried out (see Figure 3) Those were necessary in order to decide adequate scanning parameters, as e.g. x-ray tube voltage and amperage, compromise between resolution and sampling time according to the evaporation of the fluid phase. It has been established that the x-ray has to be pre-filtered in order to reduce the beam hardening artefacts, and the fluid phase needs to be dropped with a contrast medium.

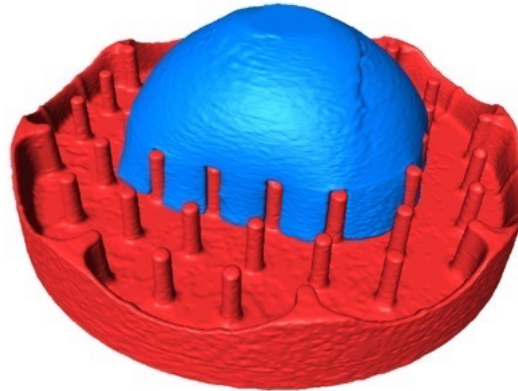


Figure 3: Micro-CT render of a sessile drop on a structured surface in the Wenzel wetting regime.

Part 2 Evaporation on porous surfaces

The designed experimental set-up mounted onto the rotating table of the microCT unit is shown in figure 4.

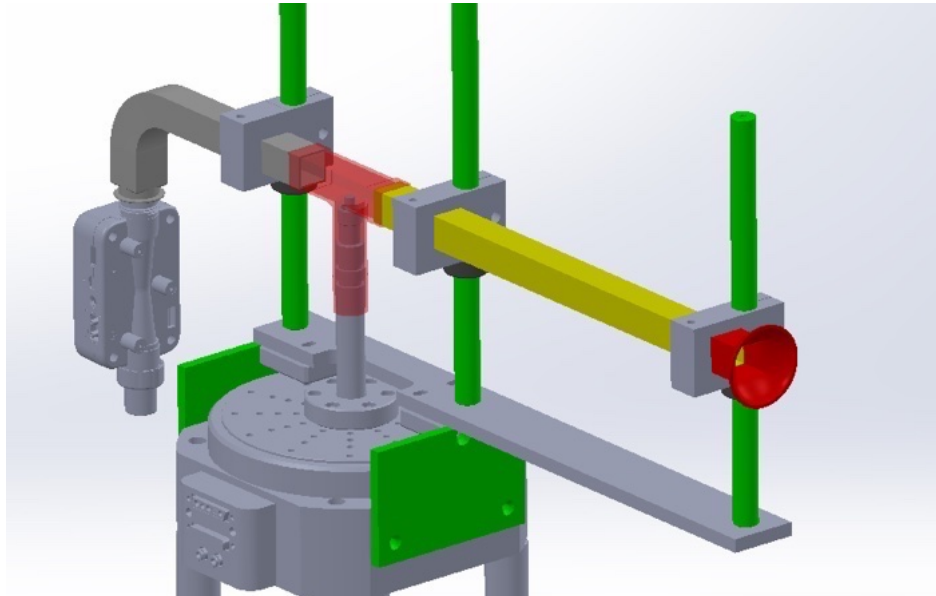


Figure 4: designed flow channel.

The flow inlet has been modelled and optimized to have a fully developed velocity profile in the region of interest, highlighted in red in figure 4. The region of interest contains the modeled porous media and the evaporating drop. Instrumentations had been foreseen those measure important data describing the flow conditions, like mean velocity, air temperature and humidity.

The numerical results of the velocity profile were verified by hot wire measurements.

First evaporation measurements had been acquired using a high-speed setup. This data had been compared to analytical models available in the literature.

Cooperation benefits

Part 1 Wetting transition

The partnership with the IRTG DROPIT, especially with the SP-B3 and SP-AS04 projects is very strong and will be more and more so in the future as the tools developed can also be used in their research project. The technique underlying the analysis is in common with all of these.

Another cooperation in the future will be with the SP-B5 project for a data exchange and for a comparison between experimental and numerical part.

It is also possible to cooperate with the SP-B1 project by making a comparison between hydrophobic and hydrophilic surfaces

Part 2 Evaporation on porous surfaces

A cooperation in the future could be established with the SP-B6 project. The experimental data could be used for numerical validation of growing and evaporating drops.

3. Comments on the qualification programme and supervision strategy:

I followed the doctorate courses of both the DROPIT project and the Bergamo engineering school. One of the most important courses that I followed and that allowed me to go into the topics was that related to the Practical seminar on Micro-CT and the lesson of the prof. Sijbers about the Reconstructing X-ray images. During this I could better understand the working of this technique that I will use during all my future experiments.

References

- [1] D. G. Garcia: Robust smoothing of gridded data in one and higher dimensions with missing values. Comput. Stat. Data Anal. 54(4) (2010), pp. 1167-1178.
- [2] A. A. AlRatrou, Q. Ali , B. Branco and J. B. Martin: Automatic measurement of contact angle in pore-space images. Adv Water Resour 109 (2017), pp. 158-169.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Heat transfer (in Italian “Trasmissione del calore”)
- Electronic systems design (in Italian “Progettazione dei sistemi elettronici”)

Specialised courses:

- Advance Numerical Methods for Engineering
- Strumenti informatici a supporto del ricercatore
- Seismic improvement techniques (in Italian “Tecniche di miglioramento sismico”)
- Multibody system dynamics
- Technique and characterization of materials (in Italian “Tecnica e caratterizzazione dei materiali”)
- Modelling of renewable energy systems
- Electronic instrumentation
- Advanced Prefabricated Systems (in Italian “Sistemi Prefabbricati avanzati”)
- Dimensional analysis for heat and mass transfer problems
- Heat and mass transfer in droplets
- Modelling of Sprays: Simple solution to complex problems
- Design Methodologies for electronic System in Wearable Applications
- Time series analysis
- Direct Numerical Simulation (DNS) of primary jet Break-up
- Practical seminar on Micro-CT
- Reconstructing X-ray images
- Selected solution methods for nonlinear partial differential equations for flow and heat transfer problems

Elective and soft skill courses:

- Scientific Writing
- English intensive course for Phd students

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --

Participation in the following International Research Training Group events:

- Doctoral Seminar on 15.-16. May 2019 in Bergamo
- Doctoral Seminar on 25.-26. June 2020 in Bergamo
- Doctoral Seminar on 08.-09. March 2021 in Bergamo
- Doctoral Seminar on 08.-09. September 2021 in Bergamo

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI) 2019
- Droplet Impact Phenomena & Spray Investigation (DIPSI) 2021

4. Individual publications:

I. Presentations

a) Oral

- **Gritti M.:** SP-B2 - Drop impact/deposition onto micro-structured hydrophobic and superhydrophobic surface. Doctoral Seminar 05/2019, Bergamo, Italy, 2019
- **Gritti M.:** Presentazione di fine primo anno sulle attività svolte. Doctoral seminar of Phd students of University of Bergamo, Dalmine, Italy, 2019.
- **Gritti M.:** SP-B2 - Drop impact/deposition onto micro-structured hydrophobic and superhydrophobic surface. Doctoral Seminar 06/2020, online, 2020

- **Gritti M.** Presentazione di fine secondo anno sulle attività svolte. Doctoral seminar of Phd students of University of Bergamo, Dalmine, Italy, 2020
- **Gritti M.:** SP-B2 - Drop impact/deposition onto micro-structured hydrophobic and superhydrophobic surface. Doctoral Seminar 03/2021, Bergamo, Italy, 2021
- **Gritti M.:** SP-B2 - Drop impact/deposition onto micro-structured hydrophobic and superhydrophobic surface. Doctoral Seminar 09/2021, Bergamo, Italy, 2021
- **Gritti M.** Presentazione di fine terzo anno sulle attività svolte. Doctoral seminar of Phd students of University of Bergamo, Dalmine, Italy, 2021

Report of doctoral researcher (SP-B5-2)

Project number: SP-B5

First and last name of doctoral researcher: Weibo Ren

(Working) title of doctoral project: Direct Numerical Simulation of Droplet Impact onto Textured Surfaces

Name of supervisors: Prof. B. Weigand, Prof. M. Dumbser

1. General information:

Duration of International Research Training Group membership:
Since 1st October 2019

Funding (type and duration): Research associate at the Institute of Aerospace Thermodynamics at the University of Stuttgart (DFG IRTG DROPIT, GRK 2160/1 and GRK 2160/2)

Start of doctoral training and (anticipated) date of doctoral degree:
1st July 2019 to Summer 2024

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2013 Aerospace Engineering, University of Stuttgart, Bachelor of Science

05/2019 Aerospace Engineering, University of Stuttgart, Master of Science

Where applicable, please note any circumstances which could have affected your scientific work: -

Occupation after leaving the IRTG, if applicable: -

2. Description of doctoral project and research results achieved to date:

Motivation

Droplet impact onto textured surfaces is widely encountered in numerous technical applications and natural scenarios. Despite extensive experimental investigations over recent years, the physical mechanisms behind droplet behaviours such as bouncing, spreading, or splashing on surfaces with various structures and hydrophobic or hydrophilic coatings remain unclear. To enhance our understanding of these impact processes, it is essential to examine the flow field within the droplet and around the surface microstructures closely. This project aims to conduct a detailed analysis of droplet impacts on textured surfaces using Direct Numerical Simulations (DNS) with our in-house multiphase flow code, Free Surface 3D (FS3D).

During the first funding period of DROPIT, FS3D has been successfully extended to handle arbitrary structured surfaces [1][2]. Comparison with experiments have demonstrated that the simulations are able to correctly capture the droplet morphological changes and the contact time reduction in the case of superhydrophobic complex solid surfaces. In the current phase, our research extends to a broader range of textured surfaces. We focus on a comprehensive analysis of various phenomena occurring during impact, such as air entrapment, splashing, sheet formation, and flow features in between the surface microstructures.

Simulation Results

Overview of investigated cases

We have conducted simulations for a variety of cases, from drop impact onto superhydrophobic surfaces textured with cubic pillars, to impact onto superhydrophilic surfaces with microstructures of different spacing and shapes. Drop impact onto surfaces textured with a single cubic pillar is validated through comparison with experimental results, contributed by SP-B1. The microstructures investigated included such as cubic pillars, hemispheres, tetrahedrons, and cones, with sizes varying from 100 μm to 500 μm . The impact velocities of the droplets were between approximately 1 and 2.5 m/s.

Additionally, we conducted simulations of two simultaneous droplet impacts onto micropillared surfaces. We compared the splashing behaviour for cases with micropillars of different spacings. Flow features were analysed in detail to understand different outcomes of the impact. Moreover, we explored other scenarios, including droplet impacts into deep pools with various underlying structures. In the following sections, we will briefly discuss a selection of the conducted cases. This discussion begins with the validation case for droplet impact onto a surface textured with a single cubic pillar, then moves on to impacts on surfaces covered with tetrahedrons and concludes with the scenario of two simultaneous droplet impacts.

Validation for drop impact onto superhydrophilic textured surfaces

The vertical impact of an isopropanol droplet onto a cubic pillar embedded on a flat surface was simulated and compared with experimental results. The diameter of the droplet is around 2 mm; the edge length of the cubic pillar is 1 mm. In the simulations, the contact angle between the solid and the liquid phase was set to 0 degree, corresponding to a full-wetting condition. The investigated impact velocity is 1.46 m/s, with the Weber and Reynolds numbers around 160, and 1100, respectively. Very good agreement concerning the droplet morphological changes has been found between the experimental and the numerical results, as is shown in Figure 1. Furthermore, during the impact a large volume of air was found to be trapped around the pillar side faces, based on the numerical results. More details of the air entrapment process as well as simulation results regarding other impact position relative to the pillar can be found in [3].

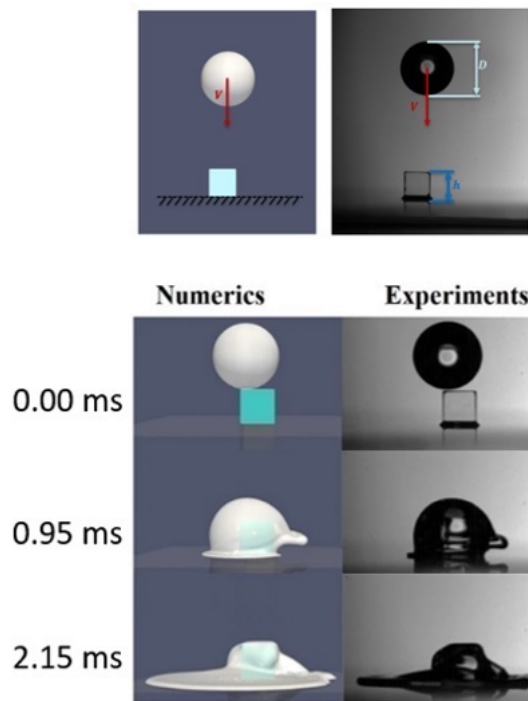


Figure 1: Comparison between FS3D simulations and experimental results from SP-B1 for drop impact at Weber number $We = (\rho RU^2)/\sigma = 158$.

Two drops impact onto micropillared superhydrophilic surfaces

We computed the impact of two neighbouring droplets onto superhydrophilic surfaces, textured with cubic pillars[4]. The distance between the impact centres is 3 mm. The length of the cubic pillars are 0.125 mm, corresponding to 1/16 of the droplet diameter, where the spacings between the pillars are 0.125 mm, 0.35 mm. In the initial impact stage, fingers are formed between the pillars. A central uprising sheet is produced during the interaction of the fingers and the rim of both droplets. In case of the densely placed pillars, the uprising sheet forms a semi-lunar shape. Fingers appear and grow bigger on the rim of the uprising sheet in a later time stage. In case of the sparsely placed pillars, fingers form due to the influence of the pillars, which eventually evolve into cylindrical jets before disintegrating into secondary

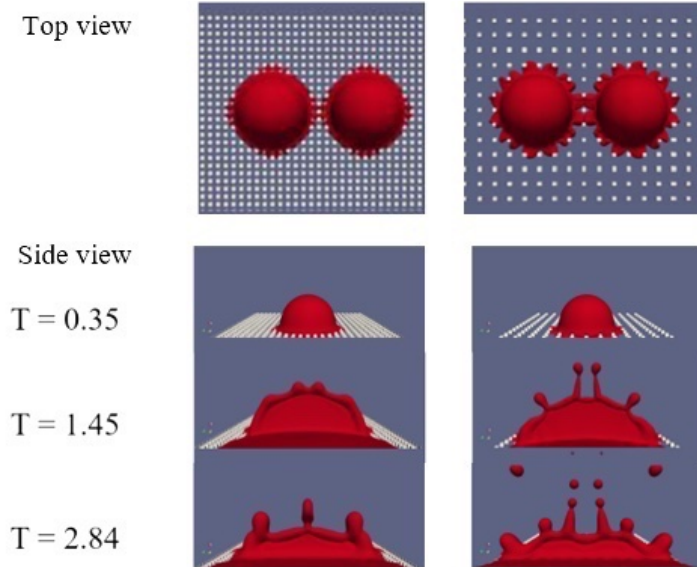


Figure 2: Simulation of a droplet impacting onto surfaces with cubic pillars, $We = 119$, $Re = 965$.

droplets. To understand the formation and the evolution of the uprising sheet, the velocity field at the sheet, both in between and above the cubic pillars were discussed in detail and can be found in [4].

Droplet impact onto surfaces textured with tetrahedrons

We simulated the impact of an isopropanol droplet at a speed of 2.5 m/s onto superhydrophilic surfaces covered with tetrahedrons. At an early stage of the impact, spreading of the droplet between the structures is dominated by the inertial forces. Fingers are then observed to form between the tetrahedrons. As these fingers collides with the tetrahedron sidewalls, they are redirected along the surface of the structures up into the air, as shown in the side view of Figure 3. As these flying fingers elongate with high velocities, the surface tension forces eventually become too weak to hold them together. This causes secondary droplets to detach from the fingers due to Rayleigh-Plateau instabilities.

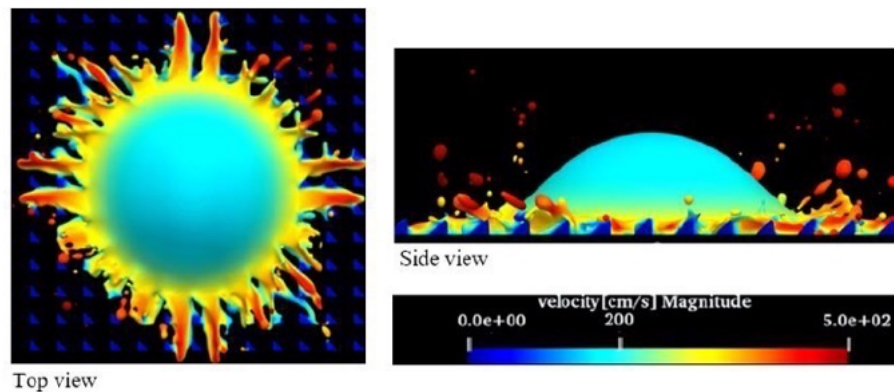


Figure 3: Droplet impact onto surfaces textured with tetrahedrons, with $We = 466$, $Re = 1916$.

Cooperation Benefits

The simulation results of droplet impacts on superhydrophilic surfaces, which are textured with a single pillar, have been validated through collaborative efforts with SP-B1. Within this collaboration, SP-B1 conducted experiments that provided morphological changes of droplet during the impact.

References

- [1] Baggio, M., & Weigand, B. (2019). Numerical simulation of a drop impact on a superhydrophobic surface with a wire. *Physics of Fluids*, 31(11), 112107.
- [2] Baggio, M. (2020). *Volume of Fluid Numerical Simulations of Drop Impact on Superhydrophobic Complex Solid Surfaces Embedded in a Staggered Cartesian Grid* (Verlag Dr. Hut).

- [3] Ren, W., Foltyn, P., Geppert, A., Weigand, B., (2021, Sep). Air entrapment and bubble formation during droplet impact onto a single cubic pillar. Scientific Reports, 11 (1), 18018
- [4] Ren, W., Hu, Y., Weigand, B., (2021, Sep). [Splashing during simultaneous two drops impact onto a micropillared superhydrophilic surface](#) ILASS–Europe 2022, 31th Conference on Liquid Atomization and Spray Systems

3. Comments on the qualification programme and supervision strategy:

This program is very well-structured, where a variety of courses and workshops were offered to expand my knowledge and understanding in droplet dynamics and relevant research methodologies. Professor Bernhard Weigand's guidance has always provided me with valuable insights as well as necessary independence for my research. During the midterm review, Professor Michael Dumbser also provided valuable feedback. In addition, my involvement in developing the DROPIT website, as one of the DROPIT members, enhanced my technical skills and provided a platform for showcasing our research as well as further facilitating our academic exchange.

Overall, the program's diverse courses, great opportunities for scientific collaboration, and the balance of guidance and freedom in research work, significantly aided my professional and research development.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Nonlinear Partial Differential Equations
- Heat and Mass Transfer in Droplets
- Modelling of Micro-explosion

Specialised courses:

- Analytical Methods and Improved Non-equilibrium Droplet Evaporation Models for High Pressure Conditions, University of Stuttgart, Stuttgart (DE), July 11-12, 2019
- Parallel Programming Workshop (MPI, OpenMP and Advanced Topics), High Performance Computing Center (HLRS), Stuttgart (DE), October 14-18, 2019
- Atomization and Sprays, Technische Universität Darmstadt, Darmstadt (DE), February 17-20, 2020
- Advanced Numerical Methods for Hyperbolic Equations, University of Trento, Trento (IT), February 10-14, 2020
- Measurement Techniques, University of Bergamo, June 14, 2023

Elective and soft skill courses:

- Writers Course, University of Stuttgart, Stuttgart (DE), Juli 22-24, 2020

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: Due to the COVID-19 pandemic and resulting travel restrictions, I was not able to pursue the research stay or internships abroad.

Participation in the following International Research Training Group events:

- DROPIT meets Industry Workshop 15.06.2023 in Bergamo
- Doctoral Seminar on 01-02/03/2023 in Stuttgart
- DROPIT Summer School 09-30/06/2022 in Stuttgart and Bergamo
- Kick-Off for the Second Funding Period and Doctoral Semina on 08-09/10/2021, remote
- Doctoral Seminar on 08-09/03/2021, remote

- Doctoral Seminar on 25-26/06/2020, remote
- Future Perspective Workshop 10-11/10/2019 in Stuttgart

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (IT), 16/06//2023
- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (IT), 31/06//2022
- 31th ILASS-Europe, Conference on Liquid Atomization and Spray Systems, Virtual Conference, 6-8/09/2022

4. Individual publications:

I. Peer-reviewed articles:

- **W. Ren**, P. Foltyn, A. Geppert, B. Weigand. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. Scientific Reports 11.1 (2021): 18018.
- **W. Ren**, J. Reuttsch, B. Weigand Direct Numerical Simulation of Water Droplets in Turbulent Flow. Fluids 2020, 5(3), 158.

II. Further publications

a) Proceedings

- P. Palmetshofer, **W. Ren**, J. Steigerwald, A. Geppert, B. Weigand Experimental and Numerical Investigation of Droplet Impacts onto Pillars with a Wetted Base. International Conference on Magnetic Fluids (ICMF), 2023.
- **W. Ren**, Y. Hu, B. Weigand Splashing During Simultaneous Two Drops Impact Onto A Micropillared Superhydrophilic Surface. 31th ILASS-Europe, Conference on Liquid Atomization and Spray Systems, 2022.

b) Technical and scientific reports

- Midterm Review of DROPIT Subproject SP-B5

III. Presentations

a) Oral

- **W. Ren**, Y. Hu, B. Weigand Splashing During Simultaneous Two Drops Impact Onto A Micropillared Superhydrophilic Surface. 31th ILASS-Europe, Conference on Liquid Atomization and Spray Systems, Virtual Conference, 2022.

b) Posters

- **W. Ren**, B. Weigand, M. Dumbser: Numerical Computation for Drop Impact on Textured Surfaces. DIPSI Summer School 2022, Bergamo, Italy, 2022.

Report of doctoral researcher (SP-B6-2)

Project number: SP-B6-2

First and last name of doctoral researcher: Maziar Veyskarami

(Working) title of doctoral project: Coupled free-flow–porous media flow processes including drop formation

Name of supervisors: Prof. Dr.-Ing. Rainer Helmig, Assoc. Prof. Dr. Carina Bringedal, Prof. Dr.-Ing. Maurizio Santini

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT from 1st October 2019 to 31st March 2023

Funding (type and duration):

Doctoral researcher full position, from 1st October 2019 to 31st March 2023

Start of doctoral training and (anticipated) date of doctoral degree:

Start of doctoral training: 01.09.2019

date of doctoral degree: 12.07.2023

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

09/2010; Petroleum Engineering; Petroleum University of Technology, Iran; B.Sc.; July, 2014

09/2014; Reservoir Engineering; Sharif University of Technology, Iran; M.Sc.; January, 2017

09/2019; Civil and environmental engineering; University of Stuttgart; PhD degree, July 2023

Where applicable, please note any circumstances which could have affected your scientific work: COVID-19-pandemic during years 2020-2022.

Occupation after leaving the IRTG, if applicable: Post-doctoral researcher in SFB1313

2. Description of doctoral project and research results achieved to date:

2.1. The initial hypothesis and aims of the project, the current project status, a description of results achieved to date

2.1.1. The initial hypothesis and aims of the project

In a coupled free-flow–porous medium system, the interface plays a crucial role. Phenomena occurring at the interface determine the interaction between the free flow and the porous medium. Formation of droplets at the interface of a coupled free-flow-porous medium system occurs in many natural processes (e.g., human perspiration, in which droplets form at the skin to control the body temperature) and technical applications (e.g., fuel cells, cooling systems, membrane emulsification and filtration, thermal insulation and air conditioning of buildings).

Droplet formation at the interface influences the interactions between the free flow and the porous medium significantly, in particular, the exchange of mass, momentum, and energy between the two domains. When a droplet covers a part of the interface, the exchange between the two domains occurs through the droplet. Thus, at the part covered with a droplet, droplet–porous medium and droplet–free-flow interactions govern the coupling between the free flow and the porous medium. Droplet formation tends to increase the complexity of the system through turning a simple interface to a complex interface, which not only handles the exchange between the free flow and the porous medium, but also stores mass and energy.

The aim of our work was to develop a model being able to describe multiple droplets at the interface of a coupled free-flow–porous medium system. In this work, we focused mainly on three aspects:

1. Developing a coupling concept for free flow and porous medium with droplets at the interface,
2. Developing a model for droplet dynamics (droplet formation, growth, and detachment),
3. Developing a model for droplet evaporation,

which I go through them in detail in what follows. We implemented all the models developed in this work in DuMu^x, an open-source simulator for flow and transport in porous media.

2.1.2. The current project status and a description of results achieved to date

Developing a coupling concept for free flow and porous medium with droplets at the interface

We extended the coupling concept for a coupled free-flow-porous medium system without droplets at the interface to include the droplet impact on the mass, momentum, and energy exchange between the two domains. The new coupling concept not only describes the interaction between a droplet formed at the interface with the free flow and porous medium, but also accounts for storing mass and energy by the droplet. The new coupling concept developed in our work to include droplet impact at the interface was implemented and used in simulations of droplet dynamics and droplet evaporation.

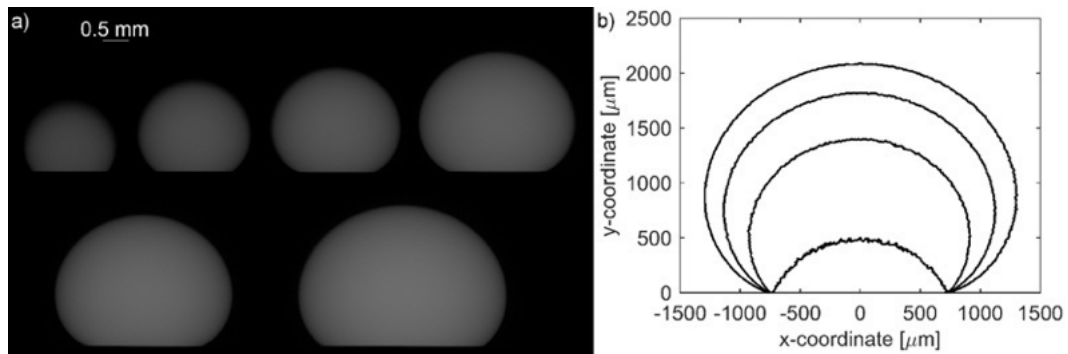


Figure 1. a) Post-processed radiographies for a growing droplet, b) extracted contours for a droplet pinned to the pore at different time steps [1].

Developing a model for droplet dynamics (droplet formation, growth, and detachment)

Having developed the proper coupling conditions to include droplet effect on the free-flow-porous medium interactions, the droplet formation and growth is predicted by summing up the total mass exchanged at the droplet-pore interface and at the droplet-free-flow interface (i.e., droplet surface). We compared the simulations results with experimental data provided by Prof. Maurizio Santini and his group for a single droplet forming and growing on a hydrophobic porous medium. This comparison showed that our model is able to accurately describe such a process. It was shown that the description of the growing mode, i.e., variation of contact angle and radius of curvature, is in a good agreement with the experiment.

We developed a concept for droplet detachment by identifying the forces involved, i.e., free-flow forces and the adhesion force acting on the triple contact line. Comparing the free-flow drag force and the adhesion force provides a criterion to predict the droplet detachment.

We used our model to predict the droplet detachment from the interface and compared the results with experimental data provided by Theodorakakos et al. [3]. We obtained a good match between the simulation results and the experimental data for prediction of droplet detachment volume, as Figure 2b shows.

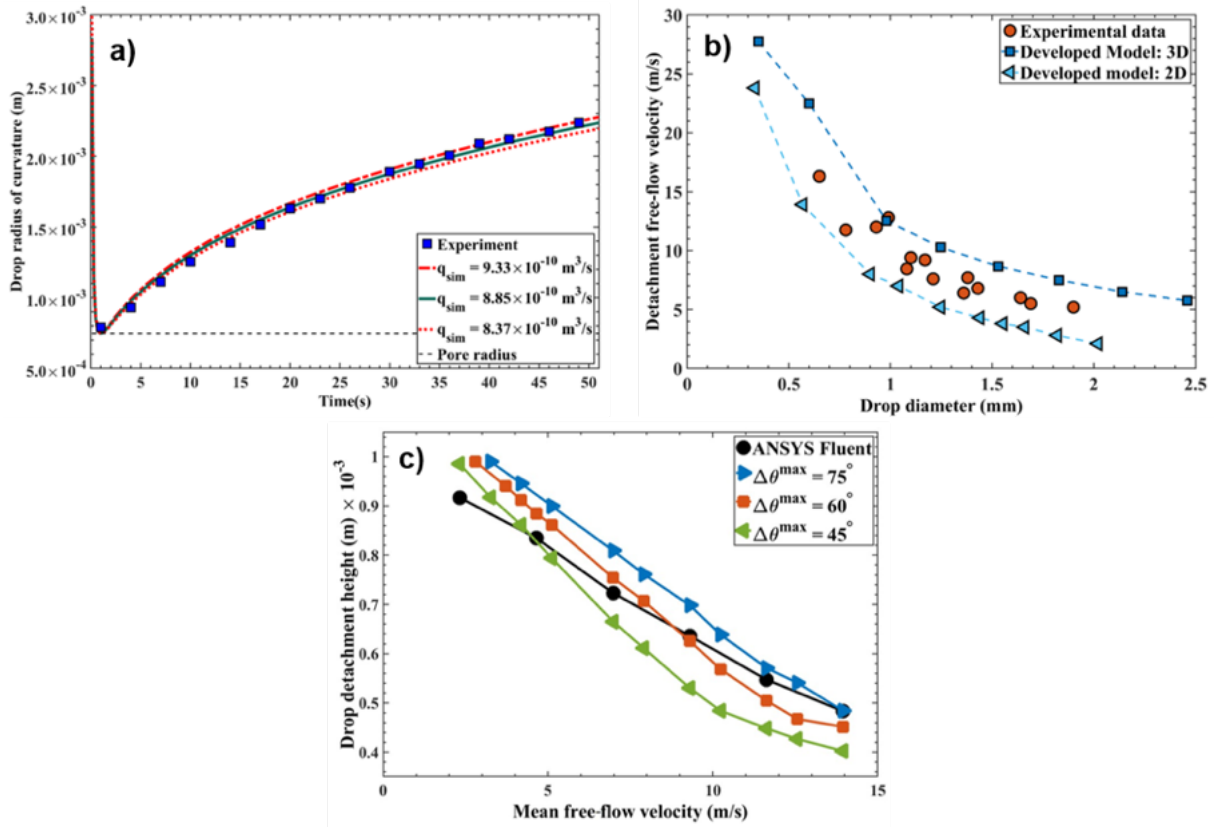


Figure 2. (a) Comparison of drop radius of curvature from experiment (squares) and simulation (lines). (b) Comparison of the experimental data provided by [3] with results of our model for the free flow velocity detaching the droplets in an air channel. (c) Impact of the maximum contact angel hysteresis on the detachment predictions by our model [4].

Comparing the simulation results using our model with ANSYS Fluent simulations for a growing droplet at the interface showed that while the predictions for droplet detachment height are in a close agreement, there are differences in predictions of droplet detachment volume (as shown in Figure 2c). How a neighbouring growing droplet affects the droplet detachment is examined. The results show that a neighbouring droplet located upstream might significantly delay the droplet detachment (see Figure 3).

Developing a model for droplet evaporation and analysis

In our work, droplet evaporation into the free flow is described as a diffusion-driven phenomenon. That means that evaporation occurs due to diffusive transport of vapour from the droplet surface to the surrounding free flow, driven by the vapour concentration gradient. To estimate the evaporation rate, we assume local thermodynamic equilibrium at the droplet surface. Using Raoult's law and Dalton's law for ideal solutions, we then compute the vapour concentration at the droplet surface as a function of pressure and temperature. We employ the Fick's law of diffusion to calculate the evaporative flux at each point of the droplet surface.

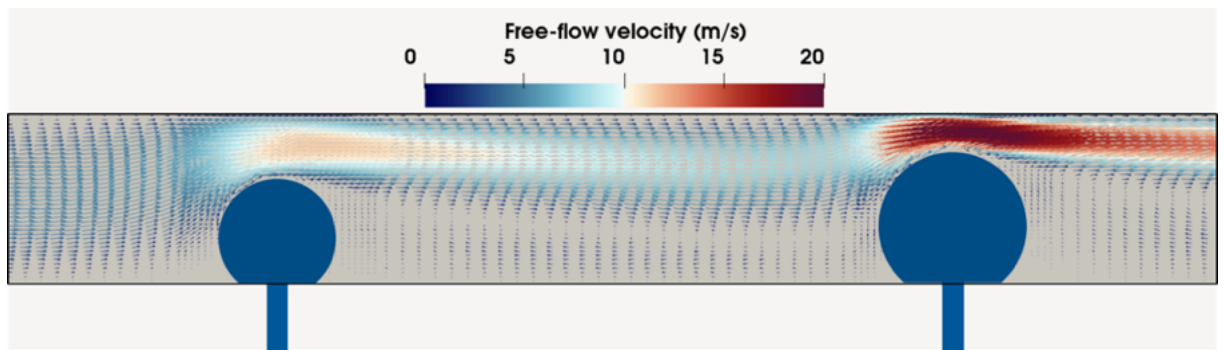


Figure 3. Impact of two droplets at the interface on the free-flow velocity field.

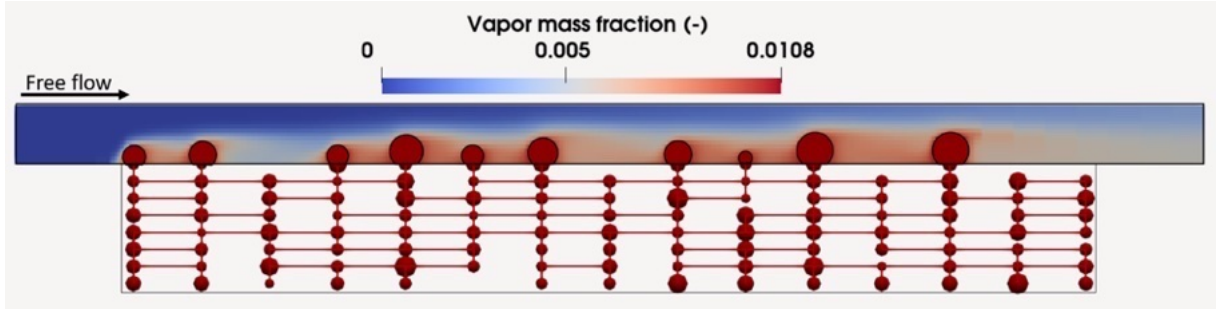


Figure 5. Evaporation of multiple droplets at the interface between a free flow and a porous medium (pore network).

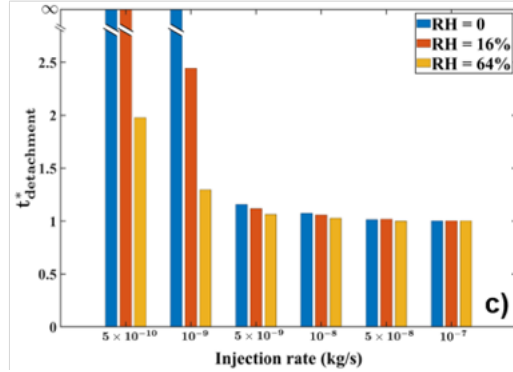


Figure 4. (a) Comparison of simulation results with experimental data given in [2] for impact of channel length on the evaporation of a single droplet confined in a channel. (b) Impact of free-flow velocity on the average evaporation rate of the droplet in three free-flow relative humidities (RH). (c) Impact of droplet evaporation on the change in the dimensionless detachment time, $t^*_{\text{detachment}}$, which is the ratio of the detachment time predicted with and without evaporation, with injection rate to the throat.

Integrating the evaporative mass flux over the droplet surface gives the total mass flux leaving the droplet due to the evaporation. Using the developed model for droplet evaporation to simulate evaporation of a confined single droplet in a channel with different lengths and comparing the results with the experimental data provided by Bansal et al. [2] showed that our model is able to provide a proper description of the evaporation process and the impact of confinement on the evaporation rate (as shown in Figure 4a).

We carried out an analysis to examine how free-flow properties (e.g., free-flow velocity, relative humidity and temperature) and porous medium properties (e.g., pore size and temperature) affect the droplet evaporation (see e.g., Figure 4b, which shows the impact of free-flow velocity on droplet evaporation). The results show that increasing free-flow velocity and decreasing relative humidity increases the evaporation rate. The connected pore size affects the evaporation rate by affecting the evaporation mode. Increasing the pore temperature boosts the droplet evaporation by increasing the vapour pressure. Through examining the impact of droplet evaporation on detachment predictions, we found that the droplet evaporation could delay the droplet detachment by decreasing the droplet growing rate depending on the rate of fluid fed to the droplet by the connected pore. Such an impact could be so significant that prevents the droplet detachment. Figure 4c shows a showcase of formation and evaporation of multiple droplets at the interface between a free flow and a porous medium (pore network). The modeling concepts introduced in our work help us to gain a better understanding about the impact of multiple droplets formation at the interface between a free flow and a porous medium on the exchange of mass, energy, and momentum between these two flow domains. They could be used as a tool to analyse interactions of a droplet at the interface with the free flow and the porous medium and examine the influence of the free-flow and porous medium properties. Being able to predict the droplet detachment and describing the droplet evaporation, the model developed here could be used in technical applications (e.g., water management in fuel cells and cooling systems) for optimization purposes. Our model could serve as a basis for further developments, e.g., describing film flow at the interface. It also could be used as an input provider for homogenization and upscaling of transport in coupled free-flow–porous medium systems, including droplets.

2.1.3. References

- [1] S. Ackermann, S. Fest-Santini, M. Veyskarami, R. Helmig & M. Santini. *Int. J. Multiphase Flow*, 160:104371, 2023.
- [2] L. Bansal, S. Chakraborty & S. Basu. *Soft Matter*, 13(5):969-977, 2017.
- [3] A. Theodorakakos, T. Ous, M. Gavaises, J. Nouri, N. Nikolopoulos & H. Yanagihara. *J. Colloid Interface Sci.*, 300(2):673-687, 2006b.
- [4] M. Veyskarami, C. Michalkowski, C. Bringedal & R. Helmig. *Transport Porous Media*, 149(2):389-419, 2023.

2.2. Networking with other RTG projects

In a close collaboration with SP-B3, we did a comparison study on formation and growth of a droplet. In this collaboration, the experimental analysis was carried out by our colleagues in SP-B3 (Prof. Santini group at University of Bergamo) and the model development and simulation were done by us in SP-B6, at University of Stuttgart. We profited from fruitful discussions with other project especially during our annual doctoral seminars and DIPSI workshops. In DROPIT summer school we worked for three weeks on a concept for droplet evaporation with external partners from Utrecht University, The Netherlands, and University of Arizona, USA. We also worked closely with project A02, SFB1313 on coupled free-flow-porous medium systems.

3. Comments on the qualification programme and supervision strategy:

Participation in the following courses/seminars of the Int. Research Training Group: <i>Core courses:</i> • Modelling of Hydrosystems: 6 ECTS credits, SS2020, University of Stuttgart. • Environmental Fluid Mechanics II: 6 ECTS credits, SS2020, University of Stuttgart. • Multiphase Modelling in Porous Media: 6 ECTS credits, WS2020/2021, University of Stuttgart <i>Specialised courses:</i> • Capillarity in porous media at different scales: given by Prof. Majid Hassanizadeh (Utrecht University), 07-11.10.2019, University of Stuttgart. • Heat and mass transfer in droplets: given by Prof. Sergei Sazhin (University of Brighton), 01-02.03.2021, Online course • Nonlinear partial differential equations by Prof. Bernhard Weigand (University of Stuttgart), Online course.
<i>Elective and soft skill courses:</i> • Professional Communications: Feb. 2020, University of Stuttgart. • Effective (Online) Presentations for PhD Students: Dec. 2020, University of Stuttgart. • Research Proposals and Grant Writing for PhD Students: Dec. 2020, University of Stuttgart. • Ethics of Science: June 2021, online round table discussion, University of Stuttgart.
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:--
Participation in the following International Research Training Group events: • Future Perspectives Workshop: 10-11.10.2019, Stuttgart • DROPIT doctoral seminar: 25-26.06.2020, Online. • DROPIT doctoral seminar: 08-09.03.2021, Online. • DROPIT doctoral seminar: 08-09.09.2021, Bergamo. • DROPIT summer school: 10.06.2022 (Stuttgart)-30.06.2022 (Bergamo) • DIPSI workshop: 30.06.2022, Bergamo • DROPIT doctoral seminar: 01-02.03.2023, Stuttgart.
Participation in conferences, congresses, etc., at home and abroad: • Interpore2021 online conference: 31.05-04.06.2021. • Interpore2022 online conference: 30.05-02.06.2022. • Interpore2023 online conference: 22-25.05.2023. • International Conference on Data-Integrated Simulation Science (SimTech2023): 04-06.10.2023, Stuttgart, Germany

4. Individual publications:

I. Peer-reviewed articles:

- S. Ackermann, S. Fest-Santini, **M. Veyskarami**, R. Helmig & M. Santini. Experimental validation of a coupling concept for drop formation and growth onto porous materials by high-resolution x-ray imaging technique. *Int. J. Multiphase Flow*, 160:104371, 2023.
- C. Michalkowski, **M. Veyskarami**, C. Bringedal, R. Helmig & V. Schleper. Two phase flow dynamics at the interface between gdl and gas distributor channel using a pore-network model. *Transport Porous Media*, 144(2):429-458, Sep 2022. DOI: 10.1007/s11242-022-01813-4.
- **M. Veyskarami**, C. Michalkowski, C. Bringedal & R. Helmig. Droplet formation, growth and detachment at the interface of a coupled free-flow-porous medium system: A new model development and comparison. *Transport Porous Media*, 149(2):389-419, 2023. DOI: 10.1007/s11242-023-01944-2.
- H. Wu, **M. Veyskarami**, M. Schneider, R. Helmig. A New Fully Implicit Two-Phase Pore-Network Model by Utilizing Regularization Strategies. *Transport Porous Media*. 17:1-26, 2023.

II. Submitted articles:

- M. Veyskarami, C. Bringedal, and R. Helmig. Droplet evaporation at the interface of a coupled free-flow-porous medium system: A new model development and analysis. Submitted to *Transport in Porous Media*.

III. Presentations

a) Oral

- Droplet formation, growth and detachment on the interface of a coupled free flow-porous medium system. Interpore2021 conference. 31.05-04.06.2021.
- Coupling porous medium-free flow: Formation and evaporation of multiple droplets at the interface. Interpore2022 conference. 30.05-02.06.2022.
- Droplet evaporation on the interface of a coupled free-flow-porous medium system: Modelling and analysis. Interpore2023 conference. 22-25.05.2023.

b) Posters

- Coupling Free Flow-Porous Media flow Including Drop Formation. SFB1313 Status Seminar. 09-11.03.2020. Gültstein, Germany.
- Formation, growth and detachment of multiple droplets at the free flow-porous medium interface. DIPSI workshop. 30.06.2022. Bergamo.
- Formation, growth and detachment of multiple droplets at the free flow-porous medium interface. SFB1313 Status Seminar. 27.29.03.2023. Gültstein, Germany
- Formation, growth and detachment of multiple droplets at the free flow-porous medium interface. International Conference on Data-Integrated Simulation Science (SimTech2023). 04-06.10.2023. Stuttgart, Germany.

Report of doctoral researcher (SP-A3-1)

Project number: SP-A3-1

First and last name of doctoral researcher: Raphael Tietz

(Working) title of doctoral project: Modeling and Simulation of Multi-Phase Flows with a Particle-Based Enskog-Vlasov Solver

Name of supervisors: Prof. Dr.-Ing. Stefanos Fasoulas, Prof. Dr.-Ing. Lorenzo Alessio Botti

1. General information:

Duration of International Research Training Group membership: 3.5 years
Funding (type and duration): PhD Student 01.01.2020 – 30.06.2023
Start of doctoral training and (anticipated) date of doctoral degree: 01.11.2019 – Mid 2024
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 01.10.2012; aerospace engineering; University of Stuttgart; 24.10.2016; B.Sc. 01.10.2016; aerospace engineering; University of Stuttgart; 30.04.2019; M.Sc.
Where applicable, please note any circumstances which could have affected your scientific work: Covid-Pandemic
Occupation after leaving the IRTG, if applicable: Continue my research at the Institute of Space Systems

2. Description of doctoral project and research results achieved to date:

The doctoral project delves into the complex dynamics of sub-micrometer droplet evaporation, a subject with significant implications in fields like atmospheric science and pathogen transmission. This project is a step towards a deeper understanding of the intricate behaviors of atoms and molecules at the liquid-vapor interface under non-equilibrium conditions.

Initial Hypothesis and Aims:

The project was predicated on the hypothesis that conventional models based on Navier-Stokes equations were inadequate for accurately capturing the behavior of atoms and molecules at the liquid-vapor interface, particularly under non-equilibrium conditions marked by high accelerations due to surface tension. The primary objective was to employ a kinetic solver, specifically a Direct Simulation Monte Carlo (DSMC)-based Enskog-Vlasov solver, to simulate and gain insights into the evaporation and condensation processes of these droplets.

Current Project Status:

The project employs a particle-based, spherically-symmetric Enskog-Vlasov solver, a kinetic equation that describes the evolution of particle properties due to collisions and force fields. This equation extends the Boltzmann equation to dense gas regimes and includes a Vlasov term to account for particle attraction.

A DSMC method (Figure 1) is used to solve the kinetic equations, utilizing a statistical discretization based on particles. The DSMC method handles the motion of particles and their collisions, applying an operator-splitting scheme for the different mathematical forms of the equation.

The simulation algorithm is a multi-step process. Initially, the simulation domain is populated with particles, each embodying multiple physical particles, and their velocities are distributed according to the Maxwell-Boltzmann distribution.

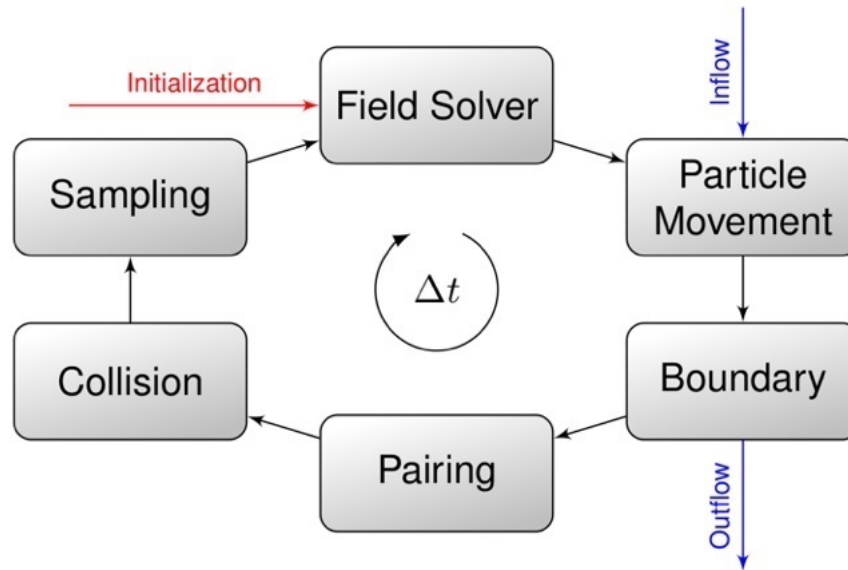


Figure 2: Schematic of the used Enskog-Vlasov solver.

The algorithm then addresses the intermolecular attraction field by solving the convolution of the density field with the Sutherland potential derivative.

Particle movements are computed by integrating their paths based on velocity and acceleration. Boundary conditions are applied as particles interact with domain edges. In the subsequent step, particles are paired based on proximity, and collision probabilities are calculated. By comparison of this probability with a random number it is decided if a collision takes place. The algorithm finalizes a time-step by sampling particle properties to compute macroscopic flow variables like density and temperature.

Results Achieved:

Comparison to existing literature: In the comparative analysis the planar solver's effectiveness was assessed by evaluating a gas without attraction, examining a liquid-gas boundary, and analyzing derived evaporation coefficients (Figure 2a). Additionally, the performance of the spherically symmetric solver was compared against the planar solver for simulations in domains distant from the origin. These comprehensive comparisons consistently demonstrated the equivalence and reliability of the different solvers, underscoring their accuracy and effectiveness in various simulation scenarios.

Saturated Vapor Density: The study shows a temperature-dependent increase in saturated vapor density inversely correlated with droplet size. This phenomenon aligns with predictions made by the Kelvin equation. However, there is the necessity for more comprehensive analysis in future work to do a quantitative analysis.

Evaporation Coefficient: A key finding is that the evaporation coefficient increases with a decrease in droplet diameter (Figure 2b). This trend starts at approximately 300 nm, with the coefficient reaching about 1.7 times higher than that of a planar interface for smaller droplets. This suggests that traditional models might underestimate the evaporation rate in these scenarios.

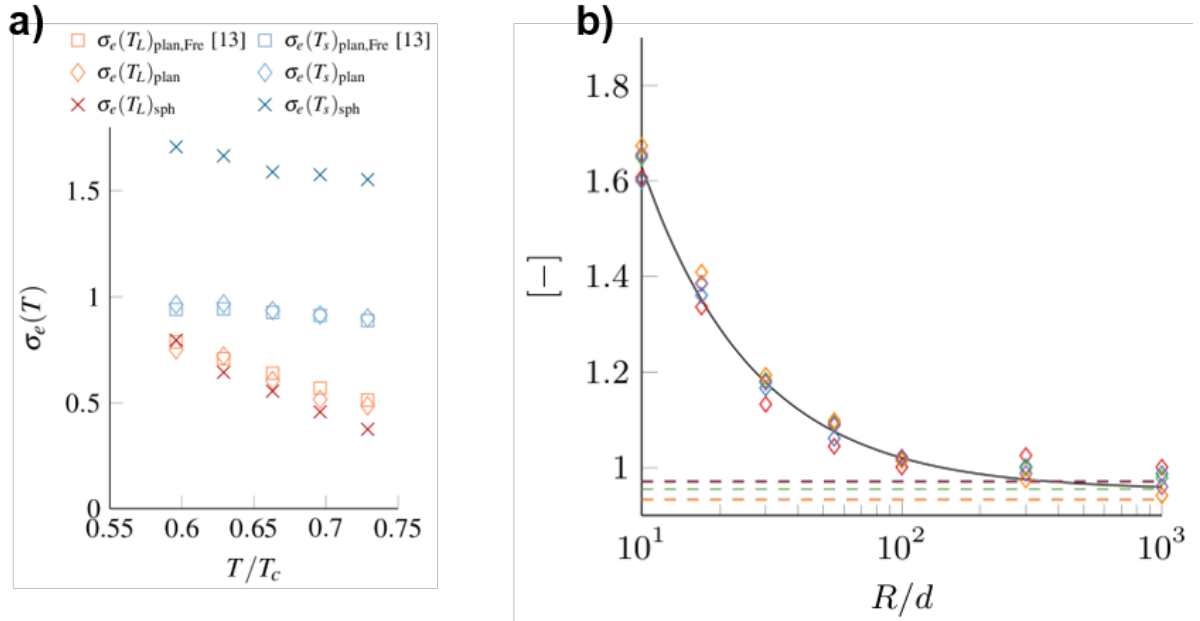


Figure 3: (a) Resulting evaporation coefficients of the flat simulations compared to the reference. (b) Resulting evaporation coefficients of droplets with different diameters in comparison with their coefficients in planar simulations.

Future Work: The study lays a foundation for future research, which could include extending these simulations to 2D axisymmetric setups to simulate droplet interactions and incorporating different species to study evaporation into other gases.

Cooperation within the IRTG: In this project, we discovered that our method is effective for determining evaporation coefficients of liquids near their critical point. These coefficients hold significant value for the research conducted by another Principal Investigator within the IRTG.

3. Comments on the qualification program and supervision strategy:

I found the qualification program to be immensely valuable in enhancing my expertise in the field of droplet dynamics. The courses provided a solid foundation, allowing me to delve into intricate aspects of the subject matter, ultimately contributing to the advancement of my work. The scope of the courses was well-balanced, covering essential topics and providing a comprehensive understanding of the field. Active involvement in the organization of the summer school was a unique and enriching experience. It offered insights into event management on a larger scale, a skill set that is often overlooked in traditional academic training. Working collaboratively within a team during the summer school allowed us to validate and improve our respective codes collectively, fostering a collaborative and supportive environment.

The doctoral seminars were instrumental in broadening my perspective. They provided a platform to explore potential connections between sub-projects and individuals, facilitating a more holistic approach to my research. This interdisciplinary engagement was particularly beneficial, encouraging cross-pollination of ideas and methodologies.

The visits to the foreign partner university were transformative. These experiences not only honed my scientific English skills but also exposed me to working within a diverse cultural setting. Interacting with individuals from different backgrounds expanded my cultural awareness and enhanced my ability to collaborate effectively in an international context. Understanding their culture was not only academically enriching but also crucial for fostering a collaborative and inclusive research environment.

The collaborative mentoring approach not only provided a comprehensive support system but also offered diverse perspectives, enriching my academic journey. The blend of expertise from both institutions created a synergistic environment, ensuring a well-rounded and globally informed doctoral experience. The dual-supervision strategy proved instrumental in fostering autonomy, while the combined wealth of knowledge from the two mentors strengthened the foundation for my academic and research pursuits.

Participation in the following courses/seminars of the Int. Research Training Group: <i>Specialised courses:</i> • Heat and Mass Transfer in droplet • Modelling of micro-explosions: simple solutions to complex problems • Analytical Methods for non-linear partial differential equations • Analytical modelling of species and energy transport from evaporating drops • Orthogonal curvilinear coordinate systems in analytical modelling of drop evaporation • Optical Methods in Spray Research • Fundamentals of X-ray experimental techniques <i>Elective and soft skill courses:</i> • Scientific Writing • Italian Course
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: -
Participation in the following International Research Training Group events: • Doctoral Seminar, online, 25.-26.06.2020 • Doctoral Seminar, online, 08.-09.03.2021 • Doctoral Seminar, online, 08.-09.09.2021 • Summer School, Stuttgart/Bergamo, 09.06.-30.06.2022 • Doctoral Seminar, Stuttgart, 01.-02.03.2023 • DROPIT meets Industry, Bergamo, 15.06.2023
Participation in conferences, congresses, etc., at home and abroad: • Short Course on Atomization and Sprays, online, 2021 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2022 • 2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-entry Missions Engineering (FAR), Heilbronn (Germany), 2022 • 32nd International Symposium on Rarefied Gas Dynamics (RGD), Seoul (South Korea), 2022 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023 • European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023

4. Individual publications:

I. Further publications

a) Proceedings

- R. Tietz, J. Beyer, M. Pfeiffer, S. Fasoulas, SHOCK TUBE SIMULATIONS WITH THE PIC-DSMC CODE PICLAS, 2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-entry Missions Engineering (FAR), Heilbronn (Germany), 2022.
- R. Tietz, S. Fasoulas, M. Pfeiffer, Symmetric Simulations of Droplets with a Particle based Vlasov-Enskog-Solver, 32nd International Symposium on Rarefied Gas Dynamics (RGD), Seoul (South Korea), 2022.
- R. Tietz, S. Fasoulas, M. Pfeiffer, Simulation of Sub-Micrometer Droplet Evaporation with a Particle-Based Spherically-Symmetric Enskog-Vlasov Solver, European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023.

b) Technical and scientific reports

- Tietz, Raphael, Prof. Dr.-Ing. Stefanos Fasoulas, and Prof. Dr.-Ing. Lorenzo Alessio Botti. DROPIT SP-A3: Gas-kinetic Simulation of Microdroplet-Gas Interaction, Midterm Review, 2021.

II. Presentations

a) Oral

- R. Tietz, S. Fasoulas, L. Botti, SP-A3: Gas-Kinetic Simulation of Microdroplet - Gas Interaction, DROPIT Doctoral Seminar, online, 2020.

- R. Tietz, S. Fasoulas, L. Botti, SP-A3: Gas-Kinetic Simulation of Microdroplet - Gas Interaction, DROPIT Doctoral Seminar, online, 2021.
- R. Tietz, S. Fasoulas, L. Botti, SP-A3: Gas-Kinetic Simulation of Microdroplet - Gas Interaction, DROPIT Doctoral Seminar (Midterm Review), online, 2021.
- R. Tietz, J. Beyer, M. Pfeiffer, S. Fasoulas, SHOCK TUBE SIMULATIONS WITH THE PIC-DSMC CODE PICLAS, 2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-entry Missions Engineering (FAR), Heilbronn (Germany), 2022.
- R. Tietz, P. Mossier, S. Tonini, J. Keim, S. Jöns, Investigation of Surface Tension Effects on Evaporating Droplets with Micro and Macroscopic Models, DROPIT Summer School, Bergamo (Italy), 2022.
- R. Tietz, S. Fasoulas, M. Pfeiffer, Symmetric Simulations of Droplets with a Particle based Vlasov-Enskog-Solver, 32nd International Symposium on Rarefied Gas Dynamics (RGD), Seoul (South Korea), 2022.
- R. Tietz, S. Fasoulas, L. Botti, SP-A3: Gas-Kinetic Simulation of Microdroplet - Gas Interaction, DROPIT Doctoral Seminar, Stuttgart, 2023.
- R. Tietz, S. Fasoulas, M. Pfeiffer, Simulation of Sub-Micrometer Droplet Evaporation with a Particle-Based Spherically-Symmetric Enskog-Vlasov Solver, European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023.

b) Posters

- R. Tietz, SP-A3: Gas-Kinetic Simulation of Microdroplet-Gas Interaction, Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2022.
- R. Tietz, SP-A3: Gas-Kinetic Simulation of Microdroplet-Gas Interaction, Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023.

Report of doctoral researcher (SP-C1-2)

Project number: SP-C1-2

First and last name of doctoral researcher: Stefan Schubert

(Working) title of doctoral project:

SP-C1: Micro and macro drop impact dynamics with miscible and immiscible liquids

Name of supervisors: PD Dr.-Ing. G. Lamanna, Prof. S. Tonini

1. General information:

Duration of International Research Training Group membership:

15.01.2020 to 14.07.2023

Funding (type and duration):

IRTG fellowship (15.01.2020 to 14.07.2023)

Start of doctoral training and (anticipated) date of doctoral degree:

Started 15.01.2020, anticipated date of doctoral degree in 2024

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2012 Mechanical Engineering, RWTH Aachen University, Bachelor of Science, 09.10.2017

10/2017 Mechanical Engineering, RWTH Aachen University, Master of Science, 29.10.2019

Where applicable, please note any circumstances which could have affected your scientific work: COVID-19 - University closure – Substantial delays in the delivery of optical equipment

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Growing interest in droplet impact on thin wall films has been driven by research into bio-recyclable fuels for diesel engines. Recent observed phenomena in the splashing regimes are reported e.g. hole formation in the crown wall (Thoroddsen et al. [1]), bottom break up in the crown neck region as well as web-like structures of the crown wall [2,3]. These ruptures strongly influence the resulting drop size distribution and consequently the mixture formation. For binary droplet impacts miscibility-gaps and surface tension differences are reputed to have an influence. Although, observed are web-like structures in the crown lamella as well as bottom break up for unary impacts on very thin films. Hence, the controlling mechanism is unclear so far. The objective of SP-C1 is to understand these collisional phenomena by studying the fluid dynamic processes at the micro-scale, which are coupled to the overall macroscopic flow features like the crown shape. Moreover, the final goal is to translate this understanding into a semi-empirical model to predict these ruptures and crown shape.

Studying thin film dynamic is aimed for in cooperation with SP-C5. The main scientific goal of SP-C5 is to develop optical measurement techniques to understand the microscopic mechanisms leading to the macroscopic outcomes observed in other sub-projects dealing with liquid droplet impacts. In fact, a strong cooperation with SP-C5 was already established in FP-1 by applying the time resolved micro-PIV technique to capture the velocity field of shadow graphic images. In FP-1, the necessity of prioritizing development for new quantitative measurement techniques to study droplet impacts was recognized. As an example the mechanisms of the bottom break up is not understood. There are at least two competing explanations among the scientific community. Stumpf et al. [4] proposed that Kolmogorov eddies resulting from the turbulent eddy cascade generated by a strong shear field at the interface of the impacting drop and the pre-existing liquid film on the wall explain a 'perfect corona detachment'. In the case that the corona wall is sufficiently thin ($\sim 1\mu\text{m}$). Villiermaux [5], instead, proposed kinematic thinning

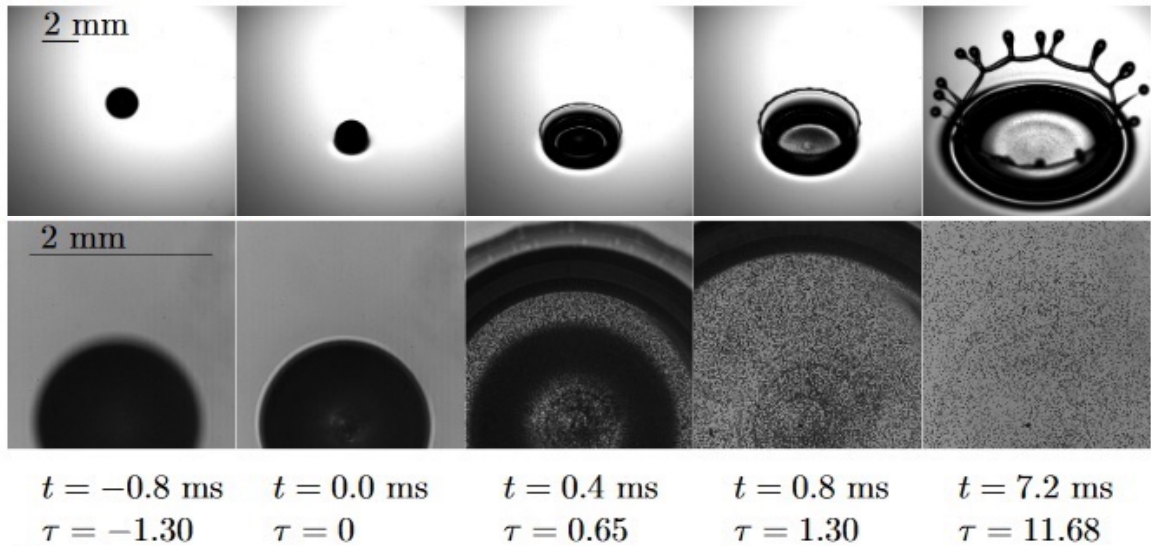


Figure 1: Evolution of unary droplet impact onto a thin wall film. The first row shows the oblique top view, whereas the second row presents the bottom view of the impact. The pool bottom is made of sapphire glass. The droplet is seeded with spherical polystyrol tracer particles ($d_p = 10\mu\text{m}$) which can be recognized as circular small black shadows. Both views are captured at 12,500 fps with two synchronized

as an alternative mechanism for controlling the rupture of thin liquid films. In this case, the film thickness is strongly dependent upon the velocity in the liquid films and its temporal derivative. An alternative explanation foresees insufficient feeding of the crown wall could be the main reason. The assumption is that the fluid velocity in the neck region is insufficient, resulting in the formation of holes or complete detachment also referred as bottom break up. However, a certain level of disturbance and instability are required. To validate the different theories, quantitative data are essential. Consequently, a shift in priority was required in project C1 due to the need of acquiring quantitative velocity data with a high spatial and temporal resolution. For this purpose, both the micro and macro investigation of the droplet impact dynamics have been extended by spatially and temporally resolved velocity measurements. Moreover, the focus was set on unary droplet film interactions (see Fig. 1), where the splashing, transition as well as the deposition regime have been studied with focus on the evolution phase. Three approaches were developed in SP-C1:

1. Micro-PIV was utilized to determine the radial velocity of liquid droplets in the crown crater region. Only the bottom view is used to determine the velocity. The measured radial velocity field exhibits point symmetry. The impact center aligns with the location where a small air bubble is encapsulated upon the impact center point. It was confirmed that velocity increases linearly with radial position for a given moment in time and decreases over time. The temporal decrease in velocity is the focus of a recently submitted paper titled 'Micro-PIV study on the influence of viscosity on droplet impact dynamics and induced velocity field'. It shows that velocity decreases more significantly at higher viscosities and that this decrease follows an exponential trend. Initial development of the method occurred in FP-1 and was further refined in FP-2. The main improvement to the test rig is the new oblique top view. The reason for this was that the impact conditions could not be determined precisely before. A new optical zoom lens, mirror mount and mirror have been fitted for this purpose. The illumination of both views were enhanced. They are now homogeneous and stronger, allowing a shorter exposure time to min-

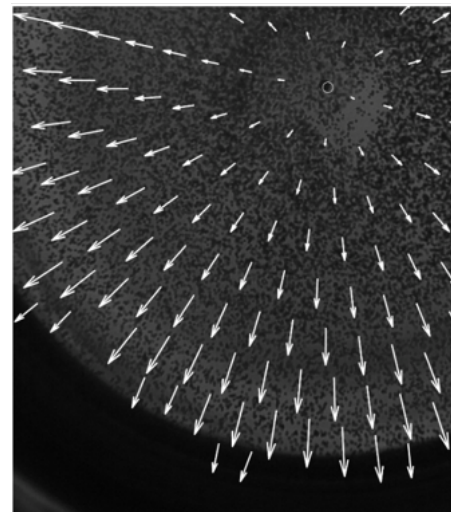


Figure 2: Two dimensional velocity field illustrated by white arrows. They are derived with the MATLAB tool PIVlab. The center of impact is marked with a white circle, which is calculated with the help of streamlines. The image is taken at $t = 0.64$ ms after impact.

imize motion blur. Both SA-X2 cameras were synchronized. The suspension of the cameras has been improved to ensure repeatable positioning. A laser barrier was installed so that the cameras trigger automatically when a drop falls. The microscope is now controlled via PC using LaVision's Davis software, allowing the microscope optics to be adjusted more quickly and accurately. In addition to the test rig improvements, the evaluation routines have been improved by rewriting the code from scratch.

2. The General Defocusing Particle Tracking (GDPT) method is applied to investigate the three-dimensional velocity distribution in the crone crater region. The same region was studied with the micro-PIV approach, observed from the bottom view. The aim is to better understand the effects of viscosity and impulse-driven forces. The number of tracer particles has been reduced compared to the micro-PIV investigations to prevent overlap, allowing for a better exploitation of the defocusing effect. Defocusing simply means that the objective is out of focus, appearing blurred and therefore larger. This additional perceptual change can be utilized to assign a height position to each particle using a single perspective. The GDPT method, originally developed by M. Rossi and R. Barnkob [6], was successfully adapted to droplet impact studies in FP-2. Spherical particles as tracer particles and a continuous light source (resp. a LED) were applied in this adaptation of the GDPT method. The intended use of fluorescent particles at a frame rate of 12,500 fps can only be achieved with a high-speed laser or a high power continuous beam laser. However, due to safety concerns for both personnel and laboratory equipment (microscope and cameras), the use of such a laser was deemed impractical, leading to the exclusion of this approach. As part of the Summer School 2022, experiments were conducted in collaboration with Prof. M. Rossi. The MATLAB code to analysis the 3D velocity field was further adjusted to suit the characteristics of droplet impacts. In this context, I undertook a one-week research stay at the Technical University of Denmark (DTU) in Copenhagen. There is an ongoing effort to pursue a joint publication of the findings. Now, it is possible to observe the particles in their vertical position (see Fig. 3). It should be noted that each individual particle still undergoes significant fluctuations in its vertical position but in a statistic perspective a clear trend of descent is measured. The radial spreading velocity aligns well with measurements previously determined using micro-PIV.

3. The third approach is to determine the velocity field of the droplet liquid in the crown wall. The geometric shape of the crown is simply assumed as a truncated cone. The impact point and the resulting crown are observed from the oblique top view. The contours of the crown are detected using a self-developed MATLAB routine. This routine determines parameters like height, base, and crown diameter at any given frame. Subsequently, a truncated cone is calculated based on these measurements. The PIV code (PIVlab from W. Thielicke [7]) previously employed for the bottom view, is applied to the 2D images. However, to accurately determine the true velocities of the fluid in the crown wall, the calculated velocities must be projected onto the surface of the actual truncated cone. The necessary transformation for this purpose was also developed in this subproject in FP-2.

The latter two approaches were developed in the second phase (FP-2). To obtain valid results, the test setup developed in FP-1 had to undergo substantial improvements. Enhancements, such as an improved spatial resolution of the oblique top view, have made it possible for the first time to observe microscopic

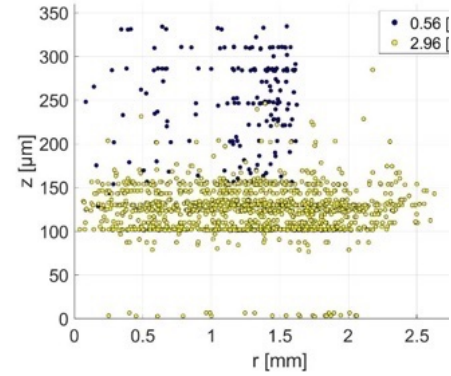


Figure 3: The MATLAB tool Defocus-tracker uses the GDPT method to find and track each particle. All detected particles with their height position z are presented over their radial position for two selected times. With time, the particles decrease in height and spread in radial position.

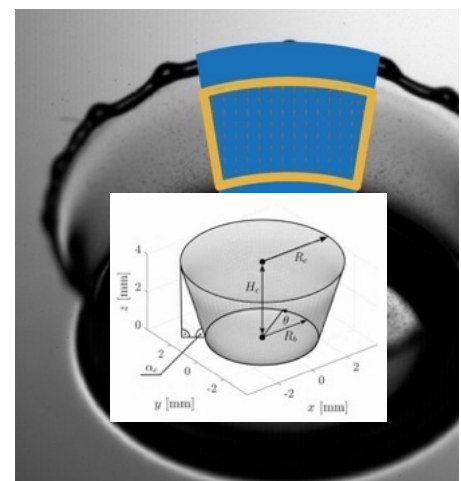


Figure 4: PIV velocity measurement of droplet liquid of the crown wall and an illustration of an derived truncated cone. Velocity vectors are printed in red surrounded by the yellow box that highlights the analyzable area.

flow features, such as the tracer particles of the impacting droplet in the crown wall. The micro-PIV measurement technique has been rarely studied in literature, primarily in cases of droplet impact on dry surfaces. The determination of the 3D velocity field within the crown crater and the velocity in the crown wall, as applied in this subproject, represents a novel contribution to the research field. There are many challenges, such as the shortness of the process (millisecond scale), the process's inherent unsteadiness and the limited repeatability under identical impact conditions. The uncertainties of the measurement methods must be taken into account. Velocity fields have been determined in master and bachelor's theses with financial support of this subproject. The estimation of error and uncertainty is subject of ongoing investigation. Consequently, fluid parameters have been restricted so far. Therefore, the focus was on unary droplet impacts. Furthermore, as stated in the project proposal, subsequent investigations should be conducted for both miscible and immiscible fluids, now that the foundational groundwork for measuring the phenomena's qualities has been established. It is important to note, however, that deviations in the crown shape from a truncated cone require an expansion of the current method. Additionally, disturbances or artifacts in the crown typically diminish the homogeneity of crown wall illumination, thereby posing additional challenges to the measurement process. As originally planned, the future focus could be directed towards comparing the experimental results with numerical simulations. For instance, the outcomes of the radial velocity micro-PIV measurements in the crown crater have already been compared with simulations (with the in-house code FS3D), revealing a very good agreement. Results from this comparison have been submitted for publication and are currently under review.

In general, the insights gained so far in SP-C1 not only contribute to SP-C1 and SP-C5, but also have implications for other sub-projects, such as SP-C2. In SP-C2 significant benefits were gained, particularly in setting up the test rig, leveraging both purchased equipment and acquired knowledge. Hence, the cooperation should be further explored in the future. Additionally, collaboration with SP-C4 focusing on crown visualization should be emphasized.

References

- [1] S. T. Thoroddsen, T. G. Etoh & K. Takehara. Crown breakup by marangoni instability. *J. Fluid Mech.*, 557:63-72, 2006.
- [2] A. Geppert, A. Terzis, G. Lamanna, M. Marengo and B. Weigand. On the formation of secondary droplets from crown bottom breakdown during drop impact on very thin films. Proc. ILASS - Europe 2016, 27th Annual Conference on Liquid Atomization and Spray Systems, Brighton, UK, 2016. (Paper ID 139)
- [3] A. K. Geppert. Experimental investigation of droplet wall-film interaction of binary systems. Ph.D. Thesis, University of Stuttgart, 2019
- [4] B. Stumpf, I. V. Roisman, A. L. Yarin and C. Tropea. Drop impact onto a substrate wetted by another liquid: corona detachment from the wall film. *Journal of Fluid Mechanics*, 956:A10, 2023
- [5] E. Villiermaux. Fragmentation versus Cohesion. *J. Fluid Mech.*, 898:P1, 2020
- [6] R. Barnkob and M. Rossi. General defocusing particle tracking: fundamentals and uncertainty assessment. *Exp Fluids* 61:110, 2020
- [7] W. Thielicke and R. Sonntag. Particle Image Velocimetry for MATLAB: Accuracy and enhanced algorithms in PIVlab. *J. Open Research Software*, 9:12, 2021

3. Comments on the qualification programme and supervision strategy:

Facilitating scientific exchange, the Summer School and research stays in Bergamo have proven particularly beneficial. These experiences allowed for collaborative team engagement, fostering mutual exchange of ideas. The option to choose special courses was highly beneficial, as it allowed me to gain in-depth knowledge in specific areas, such as image and video processing.

My involvement in designing and organizing the program was particularly noticeable in tasks such as the devolvement of the GDPT and Crown Wall Velocity Measurement. I actively participated in the shift from C1 to C5, taking on responsibilities for collaboration with Prof. M. Rossi and leading the development of the defocusing task.

The structure of the supervision was outlined by the framework of the DROPIT proposal. A more detailed plan and progression of my dissertation was documented in my dissertation agreement in consultation with my supervisor, Grazia Lamanna. From my perspective, there could be improvements in graduate projects like DROPIT by extending the doctoral candidates' funding period from three to five years. Additionally, there should be an overlap in time between generations to facilitate more effective knowledge transfer. My suggestion is that, upon completing the dissertation, the doctoral candidate should receive an additional 6-month contract to integrate the next generation. This approach could help mitigate conflicts of interest between completing one's own dissertation and the goals of onboarding a new doctoral candidate. Moreover, it will increase the number of publications since not all research

conducted during the dissertation may be published by the time the doctoral candidate leaves the university without ongoing employment prospects for disseminating that knowledge.

For IRTGs: I would rate my stays at the foreign partner university UniBG as well as the stay at DTU as a great win for the project as I gained a greater network in the research area.

Participation in the following courses/seminars of the Int. Research Training Group: <i>Core courses:</i> - Analytical and numerical methods (6 ECTS), course tutor – SS2020, WS2020, SS2021, WS2021, SS2022, WS2022 - Scientific Visualization (6 ECTS), SS2021 <i>Specialised courses:</i> - Analytical Methods for non-linear partial differential equations (18-19.02.2021) - Atomization and Sprays (22-25.02.2021) - Heat and Mass Transfer in droplet (01-02.03.2021) - Experimental techniques I: X-Ray computed Tomography (XRCT) - from pore space imaging to numerical simulations (06.07.2021) <i>Elective and soft skill courses:</i> - Public Speaking für Nachwuchs-WissenschaftlerInnen (02.12.2020) - Self-management for Ph.D.-Candidates- how to deal with research, teaching and writing the Ph.D.-Thesis, (24.06, 08.07, 12.07, 26.07.2021) - Technical and Academic Writing Skills for PhD Students (10.01.2023, 17.01.2023) - Italian language course (13.02-15.02.2023) - Scientific Writing course led by Dr. CJ Fitzsimons (25-27.04.2023)
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: - Research stay at UniBG, exchange with Prof. Cossali and Prof. Tonini (13.06-02.07.2022) - Research stay at DTU, Department of Physics, code development of the GDPT method with Prof. Rossi (04.07-08.07.2022)
Participation in the following International Research Training Group events: - DROPIT, Doctoral Seminar 2020 (25-26.06.2020) - DROPIT, Doctoral Seminar 2021 (08-09.03.2021) - DROPIT, Doctoral Seminar 2021 (08-09.09.2021) - DROPIT, Summer School 2022 (9-10.06.2022, 30.06.2022)
Participation in conferences, congresses, etc., at home and abroad: - ICLASS 2021 - A Virtual Conference (30.08-02.09.2021) 3rd International Workshop on Near-Wall Reactive Flows 2022 (03-04.11.2022) - ICLASS 2024 (23.06.-27.06.2024)

4. Individual publications:

I. Peer-reviewed articles:

- S. Schubert**, J. Steigerwald, A. Geppert, B. Weigand, G. Lamanna. Micro-PIV study on the influence of viscosity on the droplet impact dynamics and induced velocity field. *Exp. Fluids*, 65:69, DOI: 10.1007/s00348-024-03800-5, 2024.

II. Articles in preparation

- A. Geppert, **S. Schubert**, G. Lamanna. Towards spatial and time-resolved film thickness measurements during droplet impact onto thin liquid films. *Atomization & Spray Systems*, 2024

III. Further publications

a) Technical and scientific reports

- S. Schubert**, SP-C1: Micro and macro drop impact dynamics with miscible liquids. DROPIT Midterm Review, 2021

- N. Janßen. Parameterstudie des Tropfenaufpralls auf dünne Filme mittels micro-PIV Setup (experimentell). Bachelor Thesis, University of Stuttgart, 2021
- Y. Matis. Investigation of drop impact on thin wall-films Analysis of the velocity field with a micro-PTV setup and the General Defocusing Particle Tracking (GDPT) method. Bachelor Thesis, University of Stuttgart, 2022
- N. Kludt. Improvement of the combined macro and micro visualization of droplet impact phenomena. Bachelor Thesis, University of Stuttgart, 2022
- M. Stein. Investigation of the Crown Wall Velocity Field of Drop Impact Onto Thin Liquid Films With the PIV Method. Master Thesis, University of Stuttgart, 2023

IV. Presentations

a) Oral

- **S. Schubert**. Velocity measurement of an impacting droplet onto a thin liquid film. Multiphase Flow Seminar (01.12.2022, Stuttgart)
- **S. Schubert**, M. Stein, A. K. Geppert & G. Lamanna. PIV measurements in the inner crown wall resulting from droplet impact on a thin liquid film. ICLASS 2024, 16th Triennial International Conference on Liquid Atomization and Spray Systems (23.06.-27.06.2024, Shanghai, China)

b) Posters

- **S. Schubert**, G. Lamanna. Experimental 2D velocity measurement in the area of the crown crater with μ PIV technique. DIPSI 2022 (01.07.2022, Bergamo)
- **S. Schubert**, G. Lamanna. Experimental 2D velocity measurement in the area of the crown crater with μ PIV technique. 3rd International Workshop on Near-Wall Reactive Flows 2022 (03-04.11.2022, Darmstadt)
- P. Molina Vogelsang, **S. Schubert**, G. Lamanna. An approach to investigate the 3D height evolution in the crown crater during early crown formation with the General Defocusing Particle Tracking (GDPT) method. DIPSI 2023 (16.07.2023, Bergamo)

Report of doctoral researcher (SP-A6-2)

Project number: SP-A6-2

First and last name of doctoral researcher: Daniel Appel

(Working) title of doctoral project: Multiscale Modelling of Compressible Droplet Clustering for Lagrangian Models

Name of supervisors: Prof. Dr. Beck, Prof. Dr. Claus-Dieter Munz

1. General information:

Duration of International Research Training Group membership: 03/2020 - 08/2023
Funding (type and duration): Phd-Student in the Numerics Research Group of the Institute of Aerodynamics and Gas Dynamics (IAG), University of Stuttgart
Start of doctoral training and (anticipated) date of doctoral degree: 11/2019 – 2024
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 10/2011 – 09/2015 B.Sc. Energy and Process Engineering, TU Munich 04/2016 – 04/2019 M.Sc. Mechanical Engineering, TU Munich
Where applicable, please note any circumstances which could have affected your scientific work: COVID-19 Pandemic
Occupation after leaving the IRTG, if applicable: –

2. Description of doctoral project and research results achieved to date:

Motivation and Objective

The second phase of DROPIT sub-project SP-A6 seeks to gain insights into droplet clustering, particularly in compressible gas flows, by means of interface-resolved simulations.

The grouping effect itself can be attributed to the modulation of the gaseous bulk phase by each droplet, which affects the flow field encountered by neighbouring droplets. When those get entrapped in the wake of another droplet, they experience a lower drag force, causing them to convene. Despite its technical relevance, through impacting e.g. evaporation rates and combustion dynamics, the phenomenon and its influencing factors are not yet fully understood.

Highly resolved two-phase flow simulations conveniently allow to develop a deeper understanding of the fundamental grouping mechanics by treating both, the liquid and the gaseous phase, as continua, implying an Eulerian-Eulerian description. In terms of the physically relevant length scales, the sharp-interface method models the phase interface adequately as infinitely thin discontinuity, while still describing its geometry and evolution accurately.

With view to the third phase of the present project, the outlined simulations serve to incorporate the clustering effect in the equations of motion used by Lagrangian surrogate models. Compared to Eulerian-Eulerian approaches, this Eulerian-Lagrangian description reduces the computational cost significantly by treating the individual droplets as point masses, which obey Newton's second law of motion. The available literature considers e.g. the exerted forces due to drag, buoyancy and the added-mass effect, whereas the impact of the clustering as well as turbulence modulation and shape oscillations, for example, are not covered.

Work Packages and Achieved Results

In the first phase of the present project, a novel implicit-explicit flux splitting scheme was proposed to efficiently simulate droplet dynamics in weakly compressible flows. The second project phase has continued this work by enhancing the efficiency of the developed sharp-interface framework towards massively parallelized large-scale simulations of compressible droplet grouping. This contribution has been structured along the following work packages.

Consistent Discretization of the Level-set Equations

The given sharp-interface framework employs the level-set ghost-fluid method, where the fluid flow in each phase is governed by its own set of equations and material parameters. The phase coupling is achieved by solving a two-phase Riemann problem at the phase interface, while its geometry and evolution are captured by the level-set method. The latter describes the phase interface implicitly as the root of a signed-distance function. To this end, it introduces three additional PDEs: the advection equation for the level-set transport, a Hamilton-Jacobi (HJ) type re-initialization equation to preserve the signed-distance property over time, and another HJ-type PDE to extrapolate the interface velocity from the interphase into the domain.

Throughout the second phase of sub-project SP-A6, the discretization of these PDEs has been improved in several regards. First, the Discontinuous Galerkin (DG) / Finite Volume (FV) hybrid discretization of the advection PDE was enhanced by a second-order FV scheme and a consistent flux computation at mixed DG/FV element interfaces. Second, the eminent impact of the smoothing parameters in the HJ equations on the level-set method's inherent mass loss was unveiled and studied in detail. Third, consistent symmetry boundary conditions were formulated for the all mentioned PDEs in order to exploit physical symmetries of the droplet configurations in the numerical setups. Since the grouping behaviour is associated with large time scales, the aforementioned and further improvements aimed at minimizing the approximation errors accumulated over the runtime.

Narrow Band Masking and Dynamic Load Balancing

Driven by the significant computational cost of the grouping simulations, various measures were taken to improve the computational efficiency of the given sharp-interface framework. Aside from several implementational and algorithmic modifications to enhance the node-level performance, the main contribution in this regard consists in proposing a dynamic load balancing scheme for the level-set ghost-fluid method. The approach recognizes that the information on the geometry and dynamics of the phase interface, imparted by the level-set field, is only required in the vicinity of the interface. It therefore applies the costly interface-tracking algorithm solely to the grid elements in this confined region, which is commonly referred to as narrow band.

The induced element masking results in a significant performance gain on the node level, but introduces severe workload imbalances in parallelized simulations. It was thus complemented by a load balancing scheme that repartitions the computational domain at runtime, based on a static domain decomposition through space-filling curves (SFCs). The current workload distribution is determined through element-local wall time measurements, which makes use of the element masking for an efficient code instrumentation.

Investigation of Partitioning Algorithms

Since the load balancing schemes relies on a domain decomposition through SPCs, the partitioning problem conveniently reduces to one dimension. It consists in distributing the grid elements, each of which is associated with an individual computational weight, among the available processors, such that the maximum processor load, also referred to as bottleneck, is minimized. Various algorithms for this optimization problems have been proposed over the years, but comprehensive comparisons of them are scarce, especially in the context of actual simulations and with the large number of processors in today's HPC systems.

The contributions in this area, made during the second phase of sub-project SP-A6, were to implement and benchmark established partitioning algorithms in order to identify the best choice for the present sharp-interface framework as a state-of-art, massively parallelized simulation environment. The investigations unveiled that the differences among the algorithms in the quality metrics do not necessarily transfer to the actual simulation runtimes. This could be attributed to simplistic assumptions on the computational weights, which are made in the design of the algorithms but do not apply to the workload distribution of complex numerical solvers. Possible remedies to this problem were suggested, but more work is needed to further increase the effectiveness of the load balancing.

Grouping in an One-Dimensional Monodisperse Droplet Stream

In order to simulate droplet grouping scenarios, a few more extensions of the outlined sharp-interface framework were necessary. While the first phase of sub-project SP-A6 focussed only inviscid two-phase flows, described by the Euler equations, the full Navier-Stokes equations need to be solved to reproduce clustering effects. In particular, this necessitated to define an effective viscosity at the phase interface to model the interfacial viscous flux. Moreover, a simple region-based tracking algorithm was devised to follow the individual droplets during the grouping process and monitor their geometric shape.

The grouping was eventually studied by considering two equisized droplets aligned in streamwise direction. This constitutes the most fundamental configuration subject to grouping and has been investigated analytically, numerically as well as experimentally. The setup was aligned to the literature and served as an experimentally validated, incompressible baseline test case. Qualitatively, the front-back pressure asymmetry of the leading droplet, which is more pronounced than for the trailing droplet and thereby causes the grouping, grows as the Reynolds number increases, while the grouping time decreases. Both trends could be reproduced well. Quantitatively, the results at the lower Reynolds numbers were in good agreement with the reference data. However, for higher Reynolds numbers the grouping tendency was overpredicted, i.e. an earlier coalescence was observed than in the experiments. On the one hand, this difference may be attributed to minor modifications in the numerical setup compared to the experiment, owing to the high computational cost. On the other hand, the results did not yet seem to be fully grid-converged. Both issues are subject of ongoing work.

3. **Comments on the qualification programme and supervision strategy:**

The qualification program associated with DROPIT has been an integral part of my research, both from a scientific and a professional point of view. The lectures and specialized courses listed below deepened my understanding in the numerical methods employed in the given code framework and expanded my knowledge in droplet physics. The attended soft skill courses made me reflect on my time management and question working styles I had adopted over the years. Besides, it allowed me to broaden my scientific horizon by discussing ethical questions related to my work and aspects of logical reasoning, for example. Eventually, the courses permitted to connect with other PhD students of the university in general and of DROPIT in particular. The latter was crucial to a fruitful cooperation within DROPIT, especially since the personal exchange among the PhD students in the first two years was inhibited by remote work due to the lockdowns of the COVID-19 pandemic. Certainly, also the joint organization of the summerschool and the doctoral meetings, which involved inviting keynote speakers and initiating collaborations with other institutes, for instance, have fostered my personal development towards an independent researcher.

The supervision during the project consisted in regular meetings, depending on the progress but also on-demand, with the research group leader at my affiliation (Prof. Beck, IAG). In particular, the discussions with the colleagues of our research group that also work on the sharp-interface framework have been indispensable. In the beginning, I benefitted profoundly from being introduced to the topic and the project directly by my predecessor (Dr. Jonas Zeifang, first phase of SP-A6). Throughout the present project phase, the joint work with SP-A2 (Pascal Mossier) on the efficiency of the shared code framework was very valuable. However, I also enjoyed the aforementioned interdisciplinary collaboration with colleagues from other institutes because it always offered a new perspective on my own work.

In retrospective, more insights into research data management might have been beneficial. Although drawing from the experience of my predecessor in this regard turned out to be helpful, the third project phase of DROPIT could benefit from offering a systematic introduction to the topic and fostering discussions on best-practice among the PhD students.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Discontinuous Galerkin Methods (Prof. Beck, University of Stuttgart)

Specialised courses:

- Advanced Numerical Methods for Hyperbolic Equations (Prof. Dumbser, University of Trento)
- Modeling of Drop Heating and Evaporation (Prof. Sazhin, University of Stuttgart)

Elective and soft skill courses:

- Academic writing in English (Dr. Michaels, University of Stuttgart)
- Critical Thinking for Scientists (Prof. Zoglauer, University of Stuttgart)
- Zeit- und Selbstmanagement (Meyer-Koch, University of Stuttgart)
- Technikethik (Dr. Gammel, University of Stuttgart)

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

- Short course “Atomization and Sprays”, 05/2021, TU Darmstadt (online)
- Short course “MPI+X – Introduction to Hybrid Programming in HPC”, 06/2021, High-Performance Computing Center Stuttgart (online)
- Workshop “Optimization of Scaling and Node-level Performance on Hawk”, 11/2021, High-Performance Computing Center Stuttgart (online)

Participation in the following International Research Training Group events:

- Doctoral Seminar 2020, 06/2020, online
- Doctoral Seminar 2021, 03/2021, online
- Doctoral Seminar 2021, 09/2021, Bergamo
- Summerschool 2022, 06/2022, Stuttgart / Bergamo
- Doctoral Seminar 2023, 03/2023, Stuttgart

Participation in conferences, congresses, etc., at home and abroad:

- Workshop “Efficient Implementation of DG-Methods for the Solution of PDEs”, 09/2021, Waldachtal
- Conference “SIAM Parallel Processing” (SIAM-PP), 02/2022, online
- Workshop “Droplet Impact Phenomena and Spray Investigations” (DIPSI), 07/2022, Bergamo
- Workshop “Efficient Implementation of DG-Methods for the Solution of PDEs”, 10/2022, Waldachtal
- Conference “ECCOMAS Adaptive Modeling and Simulation” (ADMOS), 06/2023, Gothenburg
- Conference “GACM Colloquium on Computational Mechanics”, 09/2023, Vienna
- Workshop “Efficient Implementation of DG-Methods for the Solution of PDEs”, 09/2023, Waldachtal

4. Individual publications:

I. Peer-reviewed articles:

- M. Gao, **D. Appel**, A. Beck, C.-D. Munz. *A high-order fluid–structure interaction framework with application to shock-wave/turbulent boundary-layer interaction over an elastic panel*. *J. Fluids*, Volume 121, August 2023
- P. Mossier, **D. Appel**, A. Beck, C.-D. Munz. *An Efficient hp-Adaptive Strategy for a Level-Set Ghost-Fluid Method*. *J. Sci. Comp.*, Volume 97, October 2023

II. Further publications

a) Proceedings

- **D. Appel**, S. Jöns, J. Keim, C. Müller, J. Zeifang, C.-D. Munz. *A narrow band-based dynamic load balancing scheme for the level-set ghost-fluid method*. *High Performance Computing in Science and Engineering '21*, 2023, Springer
- **D. Appel**, P. Mossier, J. Keim, S. Jöns, A. Beck, C.-D. Munz. *A high-order framework for compressible droplet dynamics*. *High Performance Computing in Science and Engineering '23*, Springer – accepted

III. Presentations

a) Oral

- **D. Appel**. *Narrow Band-Based Dynamic Load Balancing for Multiphase Flow Simulations*. Conference SIAM Parallel Processing, 02/2022, online
- P. Mossier, **D. Appel**, A. Beck, C.-D. Munz. *An Efficient hp-Adaptive Approach for Compressible Two-Phase Flows*. Conference ECCOMAS ADMOS, 06/2023, Gothenburg
- **D. Appel**, A. Beck. *Towards Direct Numerical Simulation of Compressible Droplet Grouping in Turbulent Flows*. GACM Colloquium on Computational Mechanics, 09/2023, Vienna

b) Posters

- **D. Appel**, A. Beck. *Narrow Band-Based Dynamic Load Balancing for Multiphase Flow Simulation*. DIPSI Workshop, 2022

Report of doctoral researcher (SP-A2-2)

Project number: SP-A2-2

First and last name of doctoral researcher: Pascal Mossier

Title of doctoral project: Multi-scale modelling of the evaporation process

Name of supervisors: Prof. Dr. Claus-Dieter Munz, Prof. Dr. Michael Dumbser, Prof. Dr. Christian Rohde

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT from 01 March 2020 to 31 August 2023

Funding (type and duration):

PhD student at the Institute of Aerodynamics and Gas Dynamics of the University of Stuttgart, 01 March 2020 – 31 August 2023

Start of doctoral training and (anticipated) date of doctoral degree:

01 March 2020 to 31 February 2024

Commencement of studies; subject; university and date of degree:

10/13 Aerospace Engineering, University of Stuttgart, Bachelor's degree 12 May 2017

15/17 Aerospace Engineering, University of Stuttgart, Master's degree 31 Jan 2020

Where applicable, please note any circumstances which could have affected your scientific work: The COVID-19 pandemic

Occupation after leaving the IRTG, if applicable: –

2. Description of doctoral project and research results achieved to date:

The process of phase transition is a defining characteristic of two-phase flows and at the heart of fundamental environmental processes like the water cycle. In engineering, the understanding and reliable prediction of multiphase flows with phase transition is essential for current and future propulsion systems of airliners and orbital launch systems. Here, an efficient vaporization and mixing of carbon-based fuels and the upcoming e-fuels is vital to achieve a stable and complete combustion and thus reduce the emission of greenhouse gases. However, the development of numerical models and simulation methods to provide high-fidelity simulations for such processes is still the subject of ongoing research.

A key challenge hereby is the multi-scale character of the underlying problem: While the flow field can be described by macroscopic continuum models, the interface dynamics during phase transition are driven by microscopic interactions on a molecular level. Further, the flow field itself poses a multi-scale problem when turbulent structures appear in the bulk flow. Finally, under extreme ambient conditions, compressible effects cannot be neglected and force a tight coupling between the hydrodynamics and interface local thermodynamics as well as a careful treatment of shocks in the flow field. Given the complexity of the problem, high-fidelity simulations in multiple space dimensions incur immense computational costs and require a scalable implementation to harness the computational resources of today's massively parallel high-performance computing architectures.

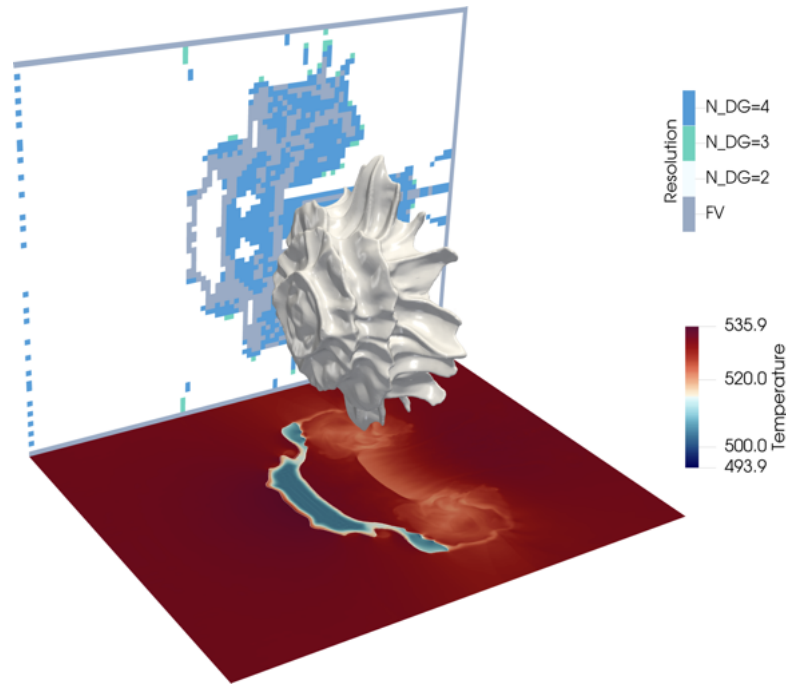


Figure 4: Visualization of the droplet geometry, temperature distribution and element local discretization of an evaporating n-dodecane droplet, interacting with an incident shock wave. The simulation is performed with a sharp interface approach and employs an hp-adaptive hybrid Discontinuous Galerkin and Finite Volume discretization. The evaporating phase boundary is modeled with a two-phase Riemann solver that relies on a local thermodynamic evaporation model.

To address these challenges, the project focuses on *three* main aspects to facilitate accurate simulations of compressible two-phase flows with phase transition:

1. The simulation of complex multiphase flows in the compressible regime introduces demanding requirements on a numerical scheme. While a high-order method is advantageous for scale-resolving simulations of the bulk flow, the presence of shocks and the phase interface necessitates the introduction of a shock and interface capturing scheme. Further, an efficient usage of the given computational resources calls for a locally adaptive discretization method.
2. To study multiphase flows on a macroscopic scale, the sharp interface method is employed where the phase boundary is modeled as a sharp interface of zero thickness. Consequently, a key challenge for sharp interface models is a thermodynamically consistent coupling of the bulk phases across the interface. A promising strategy is the solution of a two-phase Riemann problem across the phase boundary. In the scope of this project, the construction of two-phase Riemann solvers with phase transition is of particular interest. Therefore, a first-order hyperbolic continuum model is studied that allows for the consistent incorporation of interface local evaporation models.
3. The dynamically adaptive discretization method, the accurate computation of the phase interface position and geometry and the solution of two-phase Riemann problems at the interface introduce significant variations of the local computational costs. Thus, a dynamic load-balancing procedure (DLB) is vital to avoid load imbalances and idle time among the processor units in parallel computations. The implementation of an efficient and flexible DLB scheme is therefore a central requirement for the project.

In the following, an overview is provided how these tasks were tackled during the three and a half year of the doctoral project:

1st year: The first year of my doctoral project was dedicated to the development and implementation of a novel hp-adaptive discretization scheme in the high-order code framework *FLEXI* which is developed at the *Institute of Aerodynamics and Gas Dynamics*. The goal was to extend the present discretization scheme, which uses a Discontinuous Galerkin Spectral Element Method (DGSEM) with a Finite Volume (FV) shock capturing scheme towards an hp-adaptive method. This task was addressed as follows:

1. In a first step, an adaptive extension for the hybrid DG/FV operator of *FLEXI* was constructed. In general, local refinement can be achieved in two ways: through local mesh refinement called *h-refinement* or through a local increase in the order of the scheme called *p-refinement*. While *p-refinement* profits from an exponential convergence in smooth regions, it leads to oscillatory solutions at discontinuities. Here, a robust low-order scheme with *h-refinement* is the method of choice to achieve both a robust discretization and good localization of discontinuous solution features. This motivated the construction of a hybrid operator that employs a *p*-adaptive DGSEM in smooth regions and an FV scheme on an *h*-refined grid at discontinuities.
2. The proposed *hp*-adaptive hybrid DG/FV discretization was first implemented in an experimental 1D solver and studied for well-known 1D gas dynamics benchmarks like the SOD and Shu-Osher problem. Results indicated that the novel *hp*-adaptive operator provides excellent shock capturing while maintaining overall lower computational costs due to *p*-adaptation.
3. With promising results in 1D, the adaptive method was implemented in the *FLEXI* framework to study the operator for single-phase flows. Since *FLEXI* relies heavily on a static data structure, with a constant number of degrees of freedom (DOFs) per element, the implementation required major modifications throughout the entire code.
4. A key ingredient for an adaptive discretization is a suitable indicator. Therefore, an indicator, which evaluates the magnitude of the solution modes was developed. From the decay rate of the solution modes, the indicator infers an error estimate and the smoothness. Thus both *p*-adaptation and *h*-refinement can be controlled with a single indicator.
5. The proposed *hp*-adaptive method was then tested for 1D, 2D and 3D benchmarks from single-phase gas dynamics. Thereby, particular attention was dedicated to the single core performance to demonstrate that the novel scheme maintains the efficiency of the previous implementation.

2nd year: During the second year of my PhD project, my research focused on the incorporation of the novel *hp*-adaptive discretization in a sharp interface method to advance towards multiphase flows. A key building block of a sharp interface method is an interface tracking procedure. Common techniques are the Volume of Fluid method (VOF) and the Level-set method. I focused on the latter, which requires the transport of a signed distance function to obtain the interface position as the level-set zero iso-contour. To improve the accuracy of the interface representation and geometry, the idea was to utilize the novel adaptive scheme for both the fluid flow and level-set transport. Towards this goal, the following steps were taken during the second year:

1. First the *hp*-adaptive method was adapted for level-set transport equation and implemented in the multiphase code. The scheme proved well suited to deal with irregularities in the level-set due to approaching level-set contours and topological changes like merging droplets.
2. As a next step, the parallelization of the novel operator was addressed. Communication between elements of different solution representations was achieved through a masked MPI communicator. Thereby, sides of each representation are gathered in a buffer and communicated for each representation separately. However, a major challenge for the parallelization of the resulting framework are load imbalances between elements.
3. In the multiphase code, a variety of factors contribute to load imbalances among the elements, the most prominent being the *hp*-adaptive discretization and the interface tracking, performed only in a narrow band around the interface. In a joint work with my colleague Daniel Appel in SP-A6, we developed a DLB procedure based on a partitioning along space-filling curves. The method relies on wall time measurements at run time and thus adapts automatically to the current setup, without the need for any user input.
4. The resulting adaptive sharp interface method was subsequently validated against challenging shock-droplet and shock-bubble benchmarks. The scheme proved to be well suited for the most complex test cases with excellent interface capturing while reducing the wall time significantly when compared to the previous non-adaptive method.

3rd year: Research during my third year was directed towards a thermodynamically consistent coupling at the interface in the presence of phase transition. Thereby, the work of my predecessor in SP-A2 Christoph Müller, who studied the two-phase Riemann problem for the hyperbolic Peshkov Romenski (HPR) model, was continued. While the approximate Riemann solvers developed by Müller et al. provided results in reasonable agreement with molecular dynamics (MD) reference data in the bulk away from the interface, they underpredicted the evaporating and thus deviated significantly from the reference. My work on the two-phase Riemann problem was structured as follows:

1. First, the definition of the jump conditions for the HPR equation system at an evaporating interface was revisited. A key aspect thereby was the incorporation of an interfacial temperature jump that

accounts for the entropy production across the phase boundary. This interfacial entropy production is predicted via a local thermodynamic evaporation model. Using the tools of irreversible thermodynamics, phenomenological relations for the interfacial mass and heat transfer were derived and included in the jump conditions to close the equation system. The resulting non-linear system can then be solved iteratively. Following the ghost-fluid method, the intermediate Riemann states are then used as boundary conditions to couple the phases.

2. The improved two-phase Riemann solver was validated against MD data and achieved a significantly better agreement. Further, in the scope of a collaboration with Steven Jöns from SFB-TRR 75 the HPR Riemann solver was compared against the Euler-Fourier system. The study revealed that both solvers provide near equivalent results while the HPR system is advantageous in the presence of high thermal conductivities due to the lack of a parabolic time step constraint.
3. A crucial aspect when discretizing the HPR equation system is the treatment of the possibly stiff relaxation source terms. Following a fruitful discussion with Simone Chiocchetti from SP-A4, I adopted the semi-analytical source term integration developed by Chiocchetti and Müller. This allowed for efficient and robust computations in the stiff regime as encountered when simulating n-alkanes like n-dodecane.
4. Finally, the developed approximate two-phase Riemann solver and the semi-analytical source term integration were integrated in the multiphase code and applied to the simulation of fully three-dimensional shock-droplet interactions with phase transition.

3. Cooperation benefits

I cooperated continuously with other DROPIT doctoral researchers during my PhD project. In particular, the collaboration with Daniel Appel in SP-A6 resulted in lengthy discussion about the level-set interface tracking and possible DLB strategies. This interaction contributed decisively to the development of a robust hp-adaptive interface tracking algorithm and the implementation of an efficient DLB scheme as a joint work. In the course of the DROPIT summer school 2022, the cooperation with Rafael Tietz from SP-A3 enabled an interesting study of evaporating shock tubes with a macroscopic sharp interface solver and the mesoscopic Vlasov-Enskog model developed in SP-A3. Noticeably, both models produced comparable results which supports the validity of the developed two-phase Riemann solvers. Finally, the cooperation with Simone Chiocchetti from SP-A4 was particularly valuable and led to fruitful discussions about the HPR continuum model and strategies for the treatment of algebraic source terms and thus benefited my research greatly.

Participation in the following courses/seminars of the Int. Research Training Group: • <i>Advanced Numerical Methods for Hyperbolic Equations and Applications</i>
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • Doctoral Seminar, Online, 25-26 Jun 2020 • Doctoral Seminar, Online, 8-9 May 2021 • Doctoral Seminar, Bergamo, Italy, 8-9 Sep 2021 • Summer School, Stuttgart, 10-30 Jun 2022 • Doctoral Seminar, Stuttgart, Germany, 1-2 Mar 2023 • DIPSI Workshop, Bergamo, Italy 15-16 Jun 2023
Participation in conferences, congresses, etc., at home and abroad: • ICOSAHOM conference, (online), 12-16 July 2021 • Hirschegg Workshop on Conservation Laws, Hirschegg (Austria), 13-17 Sep 2021 • HONOM conference, Braga (Portugal), 4-8 April 2022 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023 • USNCCM conference, Albuquerque (USA), 24-27 Jul 2023 • 26th Results and Review Workshop of the HLRS, Stuttgart (Germany), 12-13 Oct

4. Individual publications:

I. Peer-reviewed articles:

- **P. Mossier**, A. Beck, CD. Munz. *A p-Adaptive Discontinuous Galerkin Method with hp-Shock Capturing*. *J. Sci. Comp.* **91**, 4 (2022). <https://doi.org/10.1007/s10915-022-01770-6>
- C. Müller, **P. Mossier**, CD. Munz. A sharp interface framework based on the inviscid Godunov-Peshkov-Romenski equations: Simulation of evaporating fluids. *J. Comp. Physics* 473, (2023). <https://doi.org/10.1016/j.jcp.2022.111737>
- **P. Mossier**, D. Appel, A. Beck, CD. Munz. *An Efficient hp-Adaptive Strategy for a Level-Set Ghost-Fluid Method*. *J. Sci. Comp.* **97**, 50 (2023). <https://doi.org/10.1007/s10915-023-02363-7>
- M. Gao, **P. Mossier**, CD. Munz. *Shock capturing for a high-order ALE discontinuous Galerkin method with applications to fluid flows in time-dependent domains*. *Computers & Fluids* **269**, (2023). <https://doi.org/10.1016/j.compfluid.2023.106124>

II. Presentations

a) Oral

- **P. Mossier**, A. Beck, CD. Munz: *A p-adaptive discontinuous Galerkin method with hp-shock capturing*. ICOSAHOM (online), 12-16 July 2021
- **P. Mossier**, CD. Munz: *The Riemann Problem with Evaporation*. Hirschegg Workshop, Hirschegg (Austria), 13-17 September 2021
- **P. Mossier**, A. Beck, CD. Munz: *An efficient hp-adaptive strategy for the Level-set Ghost fluid method*. HONOM, Braga (Portugal), 4-8 April 2022
- **P. Mossier**, A. Beck, CD. Munz: *An Efficient hp-Adaptive Strategy for a Level-Set Ghost Fluid Method*. USNCCM, Albuquerque (USA), 24-27 July 2023
- **P. Mossier**, D. Appel, S. Jöns, J. Keim, A. Beck, CD. Munz: *A High-Order Framework for Compressible Droplet Dynamics*. HLRS Results and Review Workshop, Stuttgart (Germany) 12-13 October

Report of doctoral researcher (SP-B4-2)

Project number: SP-B4-2

First and last name of doctoral researcher: Andrej Barič

(Working) title of doctoral project: Diffuse-interface modelling for the interaction of droplets with textured walls

Name of supervisors: Prof. C. Rohde, Prof. L. Botti

1. General information:

Duration of International Research Training Group membership:
Member of DROPIT from 01.03.2020 to 31.03.2022

Funding (type and duration): IRTG funding (01.03.2020 to 31.03.2022)

Start of doctoral training and (anticipated) date of doctoral degree: 01.03.2020

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

October 2017 – February 2020 (MSc Mathematics) University of Stuttgart

October 2017 – September 2017 (BSc Mathematics) University of Stuttgart

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Motivation

The impact of single liquid droplets onto surfaces occurs in many real-life situations and is equipped with various interesting effects that come along with widely different flow regimes. The interaction dynamics is governed by incompressible, weakly compressible or even fully compressible regimes in the liquid droplet phase and the surrounding gaseous phase. For high velocities rebound pressure waves turn into compression shock waves sensitively influencing the lateral wave jetting. An air bubble within the droplet might subject to strong compressibility effects. These phenomena have been studied for isolated flow regimes and flat perfect walls, but very little has been done for arbitrary regimes and textured (superhydrophobic) walls, when micro-level structures affect the macroscopic flow field.

Objectives and methodology

The goal of the SP-B4 project is to derive mathematical models and efficient numerical methods to describe the phenomena mentioned above. In addition to a model for the interaction of droplets with textured walls the validation with a suitable numerical simulation is provided. In order to achieve these tasks, the following approach was tried:

First, we considered a diffuse interface between the phases of the fluids. The two phases can be governed by the fourth order Cahn-Hilliard equation while the flow can be described by the Navier-Stokes equations. Combined we studied the Navier-Stokes-Cahn-Hilliard (NSCH) system. Since such a system is unstable, we examined thermodynamically consistent approximations of the (NSCH) which in the limit solves the system.

One such approximation is achieved by hyperbolizing the system. Another is by introducing a further elliptical equation which decreases the system to the second order.

Second, Direct Numerical Simulations can be done:

The extreme geometry of a realistic textured (superhydrophobic) wall (in the nm scale) does not permit a direct numerical simulation with a flow model as described above. Nevertheless, we tested the diffuse-interface model on a wall with a small number of ripples.

Current state

The (NSCH) system was again split into its two systems the Navier-Stokes (NS) system and the fourth order Cahn-Hilliard (CH) equation. In this version the (CH) equation reads

$$\partial_t \phi = \Delta (W'(\phi) - \Delta \phi).$$

For this equation we developed a relaxed system, the so-called Relaxed Cahn-Hilliard (RCH) system which is given by the following two equations

$$\partial_t \phi \alpha = \Delta (W'(\phi \alpha) - \alpha (z \alpha - \phi \alpha)), \Delta z \alpha = \alpha (z \alpha - \phi \alpha).$$

Here α is a positive big enough parameter. This (RCH) system was shown to converge weakly as α approaches infinity to the (CH) equation. This theoretical result is currently being simulated to verify it numerically.

In addition, another model was tried by adding a second time derivative of the phase field ϕ to the (CH) equation multiplied by a small parameter. This second time derivative made the system appear hyperbolic. The idea was that if this small parameter approaches 0, this new “hyperbolic” system would again converge weakly to the original (CH) equation.

3. Comments on the qualification programme and supervision strategy:

We were encouraged to exchange with international students and from different backgrounds. As a mathematician I got a different and more applied view of the researching world. In addition, planning the summer school helps developing responsibilities and brings the students closer together. It was only unfortunate that due to the Covid-pandemic we were unable to see each other more often however even if many events were online, they were still helpful.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Solution methods for nonlinear partial differential equations
- Porous media

Specialised courses:

- Special aspects in numerical analysis
- Homogenization

Elective and soft skill courses:

- Scientific Writing Workshop

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: Was not possible due to the Covid-Pandemic situation

Participation in the following International Research Training Group events:

- 2nd Funding Period Kick-Off Doctoral Seminar Stuttgart October 2021 Bergamo
- Doctoral Seminar March 2021 Online
- Doctoral Seminar June 2020 Online

Participation in conferences, congresses, etc., at home and abroad:

- DIPSI Workshop 2021

4. Individual publications:

I. Oral Presentations

- Doctoral Seminar 2021 Bergamo
- Doctoral Seminar 2021
- Doctoral Seminar 2020

Report of doctoral researcher (SP-C4)

Project number: SP-C4

First and last name of doctoral researcher: Daniel Klötzl

(Working) title of doctoral project: Visualization of droplet-liquid interaction

Name of supervisors: T. Ertl, D. Weiskopf, M. Santini

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT: 01.2021 – 12.2023, since 01.2024 associate member

Funding (type and duration):

Doctoral researcher at the Visualization Research Center of the University of Stuttgart (VISUS),

Funding by GRK 2160: 01.2021 – 12.2023, since 2024 funded by basic funds of the University of Stuttgart

Start of doctoral training and (anticipated) date of doctoral degree:

11.05.2020 to 2026

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

01.09.2013 - 19.09.2017: Mathematics, B. Sc., Eberhard Karl University of Tübingen

19.09.2017 - 03.03.2020: Mathematics, M. Sc., Eberhard Karl University of Tübingen

Where applicable, please note any circumstances which could have affected your scientific work: ---

Occupation after leaving the IRTG, if applicable: Continue as research associate at VISUS

2. Description of doctoral project and research results achieved to date:

Motivation

Droplet–liquid interaction phenomena are inherently complex, requiring advanced visual data analysis to support understanding. SP-C4 focuses on developing novel visualization methods and integrating them into interactive tools, with a particular emphasis on droplet–impact processes. In the second funding period (FP2), we built on prior work while introducing new directions. Key challenges include capturing the dynamics of droplet impacts, handling data uncertainty, supporting multi-scale and multi-physics scenarios, and understanding fluid–solid interactions, such as flow in porous media or droplet impacts on surfaces.

To address these challenges, we pursue three complementary research lines. First, we develop a reduced-connectivity method for Jacobi set extraction that improves topological clarity and robustness in multi-field scalar data, particularly for complex droplet–liquid simulations. Second, we apply seasonal-trend decomposition to extract and compare structural patterns in non-Newtonian droplet oscillations, enabling phase-based analysis of temporal dynamics. Third, we employ nonnegative matrix factorization for the analysis of droplet wall–film interactions to enable an exploratory analysis of multiple experiment recordings.

In the following, we lay out the three topics worked on during the IRTG membership in cooperation with interdisciplinary project partners.

Project 1: Reduced Connectivity for Local Bilinear Jacobi Sets for Droplet–Liquid Interaction

This work proposes an improved method to reduce the connectivity complexity of locally bilinear computed Jacobi sets [2], aiming to minimize visual clutter while preserving topological and geometric properties. Building upon the prior introduced local bilinear Jacobi sets [1], simplicial collapse, and strong collapse, the method combines their advantages by retaining original Jacobi points and collapsing higher-dimensional simplices. We show that this ensures homotopy equivalence and guarantees topological correctness.

In collaboration with SP-C2, we applied the approach to a high-resolution DNS dataset of a droplet impacting a thin liquid film, featuring velocity fields at 1024×1024 resolution over 85 time steps. In this context, the Jacobi set of two consecutive magnitudes of the velocity field was computed and analyzed, see Figure 1. Compared to the local bilinear connectivity, the reduced connectivity representation provides clearer, less cluttered Jacobi sets, especially in complex regions such as large and small air bubbles, the highly curved crown, and the thickened crown rim.

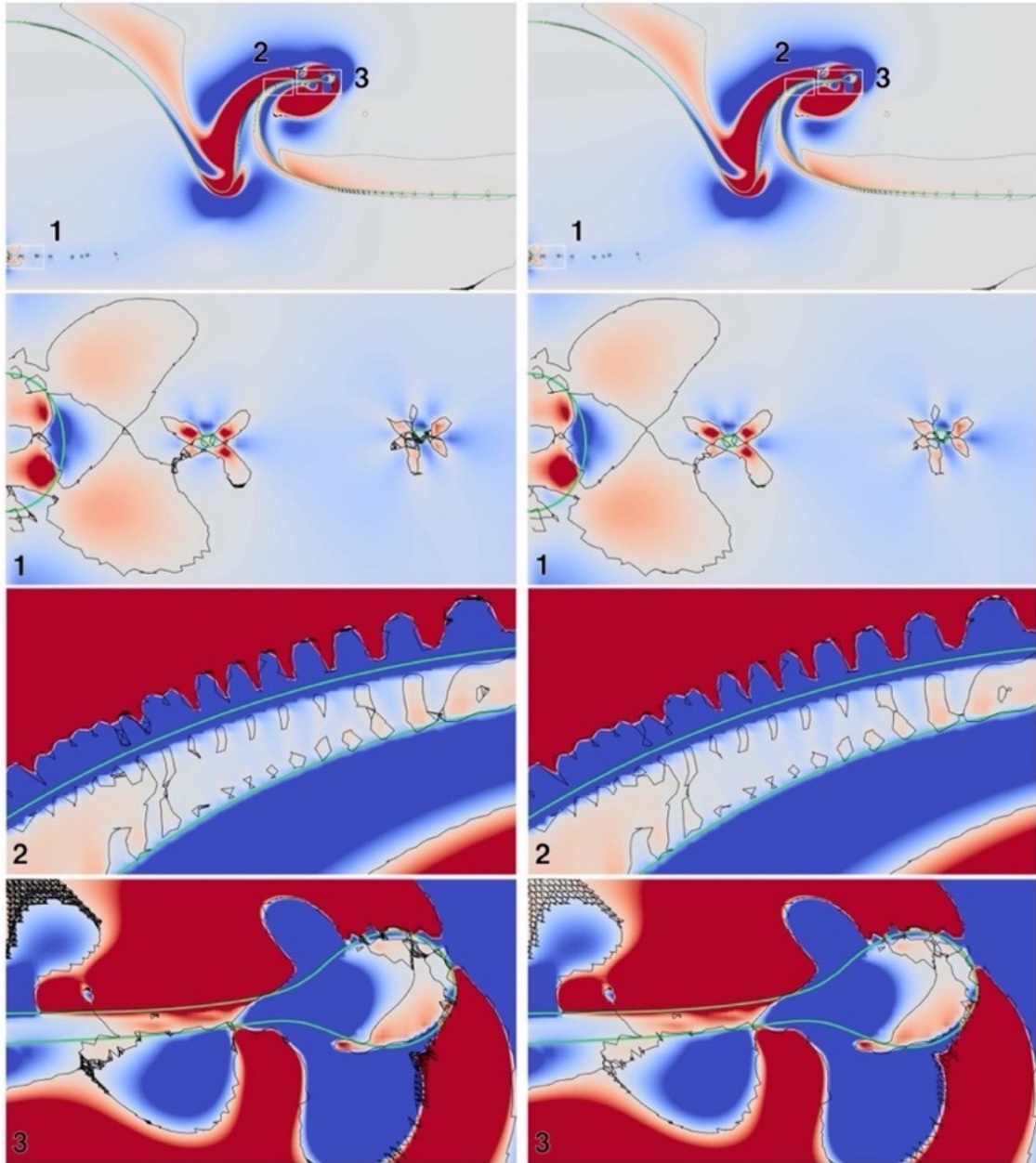


Figure 1. Comparison of the local bilinear Jacobi set construction (left column) and our reduced connectivity (right column) for the Jacobi set of the x and y components of the velocity field. The color coding shows the gradient alignment field of two consecutive time steps of the magnitudes. Source: [2, Fig. 11].

This simplification enables faster and more reliable identification of critical topological structures related to secondary phenomena like splashing and crown formation. Overall, the reduced connectivity method significantly enhances the visualization and interpretability of Jacobi sets in complex flow fields without losing topological or geometric information.

Project 2: Visual Analysis of Non-Newtonian Droplet Oscillations Using Jacobi Sets and STL

This study investigates the complex relationship between shear rate and structural parameter in oscillating non-Newtonian, thixotropic droplets via advanced visualization techniques [8]. Thixotropy describes fluids whose viscosity depends on shear rate and structural breakdown/rebuilding dynamics. It is critical in various engineering processes, but difficult to isolate experimentally. Numerical simulations with the FS3D multiphase solver generate detailed scalar fields of shear rate and structure evolution inside oscillating droplets.

Seasonal-trend decomposition based on loess (STL) is applied to the time series of the structural parameter averaged over the droplet, illustrated in Figure 2. STL separates the signal into trend, seasonal oscillations, and residuals, facilitating direct comparison across simulations with varying k_1 and Mu . Results confirm a phase shift between structural recovery maxima and droplet oscillation peaks.

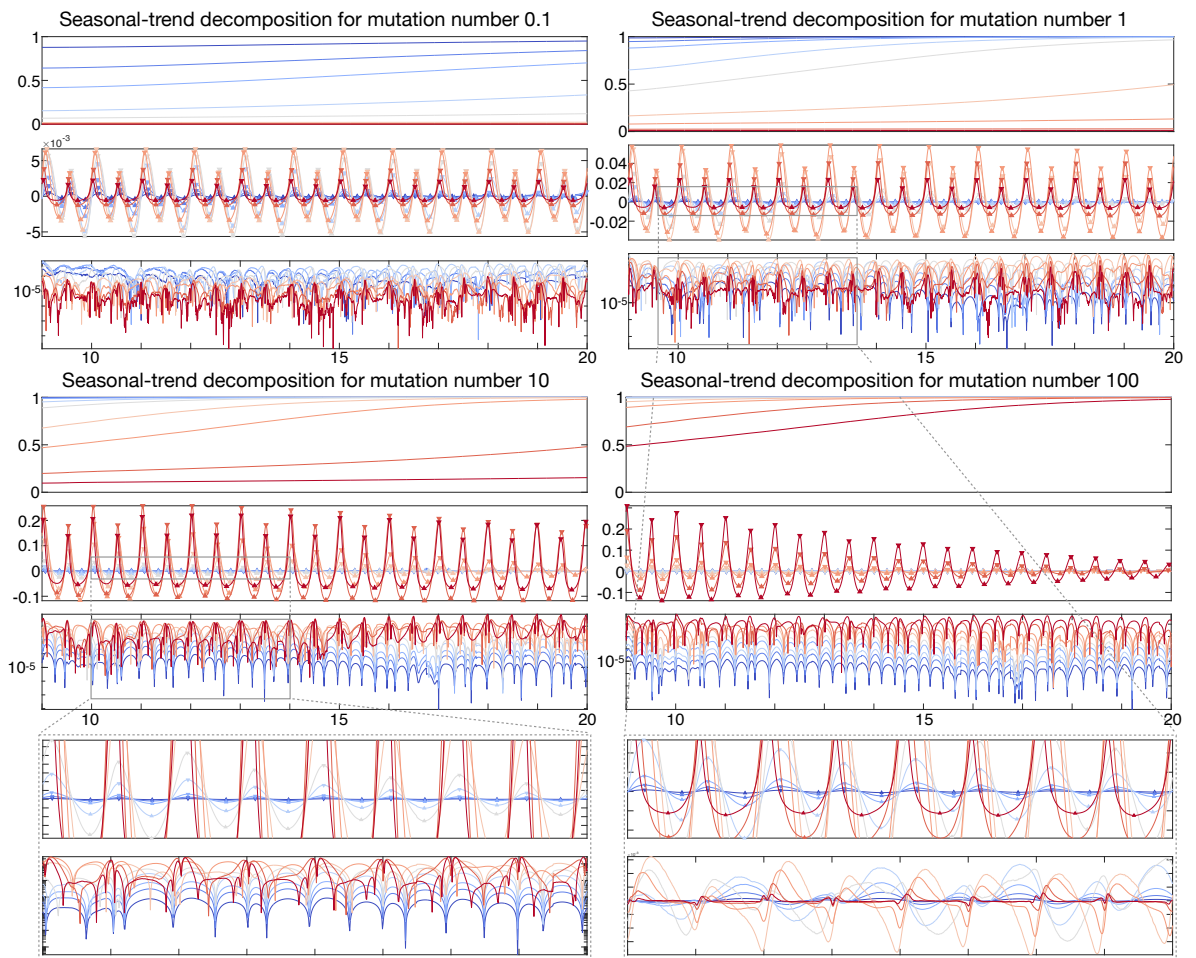


Figure 2. STL decomposition of the evolution of the structural parameter λ over the non-dimensional time $\tilde{t}$ for the mutation numbers 0.1 and 10 (left) as well as 1 and 100 (right). The different k_1 values are visually encoded in the discrete warm cold color scale. Zoomed-in areas highlight two specific regions of the resulting seasonal and residual components (upper and lower part, respectively). Source: [8].

Project 3: NMF-Based Analysis of Droplet Wall-Film Interactions

We apply nonnegative matrix factorization (NMF) to analyze ensembles of droplet impact recordings on thin liquid films [9]. NMF decomposes these videos into a set of nonnegative spatial and temporal components, enabling identification of recurring phenomena such as crown formation, jets, and bubble entrainment. Each component represents a spatial pattern and a temporal activation profile across the experiments, facilitating comparison of impact dynamics.

The dataset comprises 1300 monochrome recordings from experiments by Geppert¹, covering a range of droplet sizes (2.2–4.0 mm), velocities (0.1–5.0 m/s), film thicknesses (0.1–1.3 mm), and fluid combinations. To reduce noise, every tenth frame is used, and recordings are preprocessed by subtracting the initial frame. The frames are vectorized and jointly factorized following established NMF algorithms.

After the initial analysis of 50 hyspin-hexa droplet wall-film interactions, we focused on 12 preprocessed droplet wall-film combinations, i.e., hyspin-hexa (0.27, 0.23, 0.24, 0.25), diesel-engine oil (0.26, 0.74), hyspin-hyspin (0.26, 0.26), and hexa-hexa (0.25, 0.22, 0.51, 0.51) with different film thickness (in mm), as depicted in Figure 3. NMF successfully decomposes the data into six spatiotemporal components and isolate different impact outcomes and secondary phenomena, e.g., lamella (comp. 2) and resulting jet (comp. 5), as well as crown detachment (comp. 1, 4, and 6). Furthermore, the temporal indicator plots show the temporal occurrence of the extracted secondary phenomena and therefore enable the differentiation between impact velocities, fluids, and droplet sizes, e.g., the hexa-hexa recordings (rec. 309 and rec. 310) can be identified via their resulting jet formation.

While effective for moderate dataset sizes (~50 recordings), NMF loses clarity for larger ensembles (>200 recordings), due to component mixing. The choice of component number k critically affects results: too few components miss details; too many introduce noise. Overall, NMF provides an interpretable, unsupervised framework to extract and compare impact phenomena across varied experimental conditions.

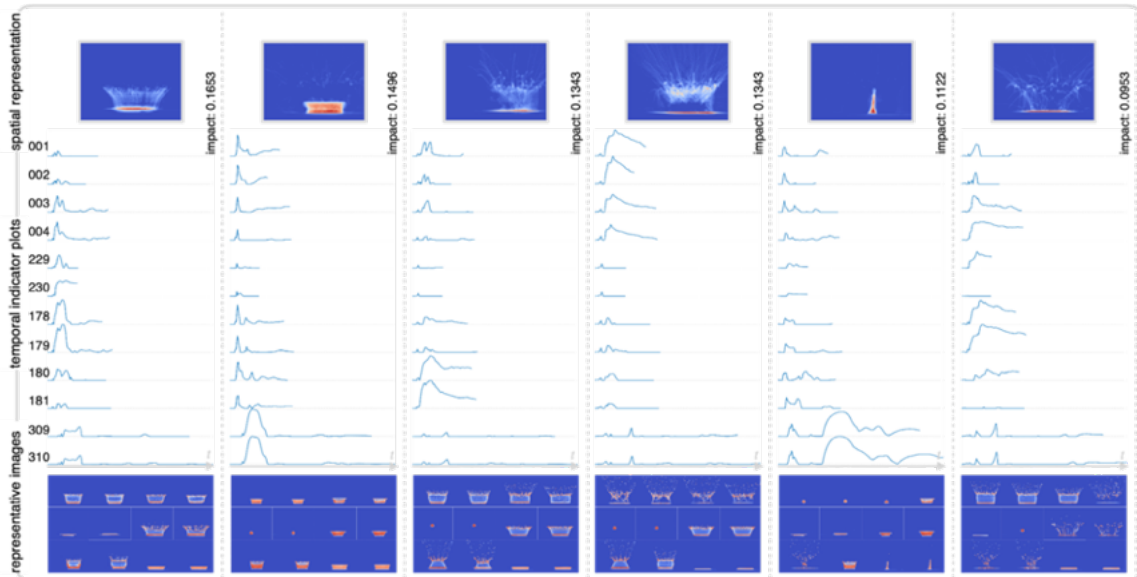


Figure 3. Application of our NMF-based visual analysis on 12 droplet wall-film interaction recordings (recording numbers 001-310 as shown) with the following different droplet wall-film combinations from top to bottom: hyspin-hexa (0.27, 0.23, 0.24, 0.25), diesel-engine oil (0.26, 0.74), hyspin-hyspin (0.26, 0.26), and hexa-hexa (0.25, 0.22, 0.51, 0.51). The numbers in brackets denote the measured film thickness (in mm). The data was decomposed into six spatiotemporal components. Source: [9].

Discussion and Future Work

Further explored topics are the segmentation of simulation data via contour trees and particle identification in high-speed experiments. However, both approaches require formalization and further methodological development.

For segmentation, the use of contour trees combined with topological persistence needs to be extended into a robust and generalizable framework. This includes formal criteria for persistence-based thresholding, stability analysis across temporal slices, and integration into interactive visual analysis workflows. In collaboration with SP-C1, particle identification via topological descriptors has shown initial success in detecting tracer particles embedded in impacting droplets. To make this approach reliable and reproducible, future work will focus on robust feature extraction under noise, particle trajectory reconstruction, and synchronization with simulation data for joint analysis.

¹ A. K. Geppert, "Experimental investigation of droplet wall-film interaction of binary systems," Dissertation, University of Stuttgart, 2019.

Overall, both strands of work should be advanced toward a cohesive pipeline for topological analysis of droplet impact phenomena, bridging simulation and experiment and supporting multi-scale, time-resolved interpretation.

3. Comments on the qualification program and supervision strategy:

Coming from a background in mathematics, the qualification program was essential for building the foundations needed to work in the interdisciplinary field of droplet dynamics and visualization. The range of courses, seminars, and workshops was appropriate in scope and content, enabling me to quickly gain the necessary theoretical and practical knowledge.

The DROPIT Summer School was particularly beneficial; it not only deepened my technical understanding but also facilitated contact with colleagues across sub-projects. These interactions led to enduring collaborations and even joint publications. The format fostered both professional exchange and personal connections. Regular doctoral seminars and structured workshops provided a platform for presenting research, receiving feedback, and identifying opportunities for collaboration. Being involved in the organization of some events offered further experience in academic communication and project coordination.

Supervision was clearly structured. I was supported by two main supervisors whose expertise complemented each other well, covering both methodological and applied aspects of my research.

A point for improvement would be the onboarding of new doctoral researchers. A concise introduction to the structure of the IRTG, key events (e.g., summer school, doctoral seminar), and reporting obligations would help clarify expectations from the outset.

In summary, the program offered a strong framework for both scientific and personal development, with a well-balanced mix of structured training, collaborative exchange, and individual guidance.

Participation in the following courses/seminars of the Int. Research Training Group:

- Mathematical and Numerical Methods for Partial Differential Equations
- Solution Methods for Non-Linear PDE
- Heat and Mass Transfer in Droplets

Elective and soft skill courses:

- Basic Italian Course

Research stays or internships at other research institutions, both at home and abroad, as well as stays at the partner university:

June 2022 (3 weeks): Summer School at the University of Bergamo

Participation in the following International Research Training Group events:

- 08.03. – 09.03.2021: Doctoral Seminar, Stuttgart
- 08.09. – 09.09.2021: Doctoral Seminar, Bergamo
- 10.06. – 30.06.2022: DROPIT Summer School, Bergamo
- 01.03. – 02.03.2023: Doctoral Seminar, Stuttgart
- 15.06.2023: DROPIT Meets Industry, Bergamo

Participation in conferences, congresses, etc., at home and abroad:

- HiGraphics Workshop, Hirschegg, Austria, 2022.
- The 39th International Conference on Computer Graphics, Geneva, Switzerland, 2022.
- IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis) in conjunction with IEEE VIS, Oklahoma, United States, 2022.
- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo, Italy, 2023
- Norrköping Summer School: Topological Data Analysis in Visualization: Theory, Applications, Software, Norrköping, Sweden, 2023.
- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo, Italy, 2024.
- Workshop on Eye Tracking and Visualization (ETVIS) in conjunction with ACM ETRA, Glasgow, United Kingdom, 2024.

4. Individual publications:

I Peer-reviewed articles:

- **D. Klötzl**, T. Krake, Y. Zhou, I. Hotz, B. Wang, and D. Weiskopf, “Local Bilinear Computation of Jacobi Sets,” *The Visual Computer*, vol. 38, pp. 3435–3448, 2022.
- **D. Klötzl**, T. Krake, Y. Zhou, J. Stober, K. Schulte, I. Hotz, B. Wang, and D. Weiskopf, “Reduced Connectivity for Local Bilinear Jacobi Sets,” in *Topological Data Analysis and Visualization (TopoIn-Vis)*, pp. 39–48, 2022.
- T. Krake, **D. Klötzl**, B. Eberhardt, and D. Weiskopf, “Constrained Dynamic Mode Decomposition,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, pp. 182–192, 2023.
- **D. Klötzl**, T. Krake, F. Heyen, M. Becher, M. Koch, D. Weiskopf, and K. Kurzhals, “NMF-Based Analysis of Mobile Eye-Tracking Data,” in *Proceedings of the 2024 Symposium on Eye Tracking Research and Applications*, pp. 1–9, 2024.
- T. Krake, **D. Klötzl**, D. Hägele, and D. Weiskopf, “Uncertainty-Aware Seasonal-Trend Decomposition Based on Loess,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 2, pp. 1496–1512, 2024.

II Presentations

c) Oral

- [6] “Local Bilinear Computation of Jacobi Sets,” HiGraphics, Hirschegg, Austria, 2022.
- [7] “Topological Visualization Methods in the Context of Droplet Liquid Interaction,” DROPIT Meets Industry, Bergamo, Italy, 2023.

d) Posters

- [8] **D. Klötzl**, M. Ibach, and D. Weiskopf, “Shear Rate and Structural Parameter in Non-Newtonian Droplet Oscillations via Jacobi Sets and STL Decomposition: A Visual Analysis,” In DIPSI Workshop 2023 on Droplet Impact Phenomena & Spray Investigation, 14th June 2023, Bergamo, Italy
- [9] **Daniel Klötzl** and Daniel Weiskopf, “NMF - Based Analysis of Droplet Wall-Film Interactions,” In DIPSI Workshop 2024 on Droplet Impact Phenomena & Spray Investigation, 17th June 2024, Bergamo, Italy

Report of doctoral researcher (SP-C2)

Project number: SP-C2

First and last name of doctoral researcher: Jonathan Lukas Stober

Title of doctoral project: Oblique droplet impact onto a wetted surface

Name of supervisors: Dr.-Ing. Kathrin Schulte, Prof. Dr.-Ing. Maurizio Santini

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT since April 2021

Funding (type and duration):

PhD researcher, Institute of Aerospace Thermodynamics (ITLR), University of Stuttgart, DROPIT (15.04.2021 – 31.03.2025), Basic Funds (01.04.2025 – 31.10.2025)

Start of doctoral training and (anticipated) date of doctoral degree:

15.04.2021 to end of 2025

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2014 - 04/2018: Aerospace Engineering, University of Stuttgart, Germany, B.Sc. (24.04.2018)

09/2019 - 01/2020: Aerospace Engineering, TU Delft, Netherlands (Erasmus exchange semester)

10/2018 - 03/2021: Aerospace Engineering, University of Stuttgart, Germany, M.Sc. (31.03.2021)

Where applicable, please note any circumstances which could have affected your scientific work: -

Occupation after leaving the IRTG, if applicable: -

2. Description of doctoral project and research results achieved to date:

a) Motivation of the Project

Droplet impacts on a thin film of the same liquid are fundamental to numerous technical and environmental processes. Some examples are spray cooling, medical inhalers, and spray coating. In these applications, droplets impact with different velocities and angles onto a wetted wall. If the impact energy is high enough, a crown forms from which secondary droplets detach, called splashing. To improve the efficiency of the abovementioned applications, it is crucial to understand impact dynamics and splashing behaviour during oblique droplet impacts onto a thin wall film.

b) Experimental Method

The custom-built droplet generator employs a mechanically accelerated needle system. It allows for a precise and independent control of impact velocity and angle. A high-speed imaging system captures the crown formation and splashing behaviour from multiple perspectives at the same time. The film thickness is measured during the experiment using a confocal-chromatic sensor from below. The impact parameters such as droplet size, velocity, and angle are extracted from the side view recordings using automated image processing. A consistent experimental procedure and a systematic uncertainty estimation ensure high repeatability and accuracy of the measured data.

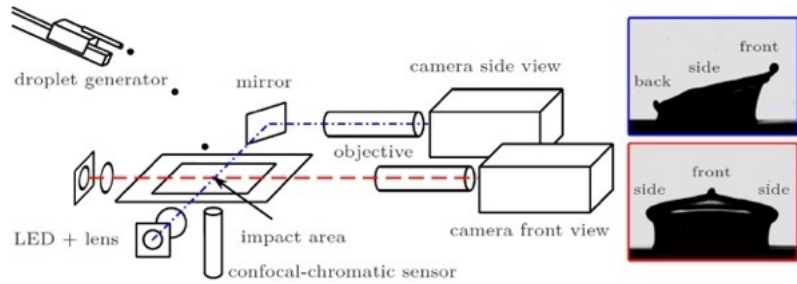


Figure 5: Sketch of the experimental setup.

c) Numerical Method

Direct Numerical Simulations (DNS) are used to complement the experimental findings. The simulations are performed using the ITLR's in-house code Free Surface 3D (FS3D), which solves the incompressible Navier-Stokes equations with a Volume-of-Fluid (VOF) method. The numerical approach was validated by a grid refinement study and comparisons with experimental recordings, focusing on both early-time crown formation and later-stage crown propagation. The results confirm that the simulations are able to predict the key physical mechanisms and provide access to flow field details that are not measurable in experiments.

d) Experimental Results

An extensive parameter study consisting of over 600 impact experiments across a wide range of impact angles and Weber numbers was conducted. With that, distinct splashing types were observed and characterized. Each splashing type is characterised by specific crown morphologies, finger positions, triggering mechanisms, and secondary droplet trajectories. These splashing types are: deposition, 90°-like splashing (90L), side-only (SO), front-only (FO), combined side and front splashing (S+F), crown-film interaction-induced splashing (I), and interaction-induced splashing combined with side splashing (I+S). The classification is summarized in a regime map, see Figure 2.

The term 90°-like splashing includes phenomena such as crown splash and Worthington splashing as it closely resembles splashing behaviour found in normal impacts. In contrast, side-only splashing is unique to oblique impacts and it is triggered by the oblique mechanism of crown formation at the back side of the impact. Fingers form at the side parts of the crown, with one or two droplets detaching per side, blue in Figure 2. Front-only splashing, on the other hand, involves droplet ejection from fingers that form at the front part of the rim, green in Figure 2. It can be either from a single-finger or from multiple fingers, which become more pronounced at higher Weber numbers. These splashing types can also superpose in a combined side and front splashing (S+F). The S+F regime exhibits droplet ejection from both side and front fingers, indicating that the triggering mechanisms of both splashing types are independent from each other and can superpose without interference.

At low impact angles and high Weber numbers, the forward crown interacts with the film in the early phase of crown propagation. This introduced disturbance to the rim that lead to fingers and atomisation, or directly causes a rupture of the front rim with the ejection of many tiny secondary droplets. This splashing type is referred to as crown-film interaction-induced splashing (I), orange in Figure 2. This splashing is independent from the growth of side fingers and, therefore, it can superpose to a combination of interaction-induced splashing and side splashing (I+S).

The results highlight the strong link between early crown dynamics, particularly the presence of the oblique mechanism on the back side, or crown-film interaction.

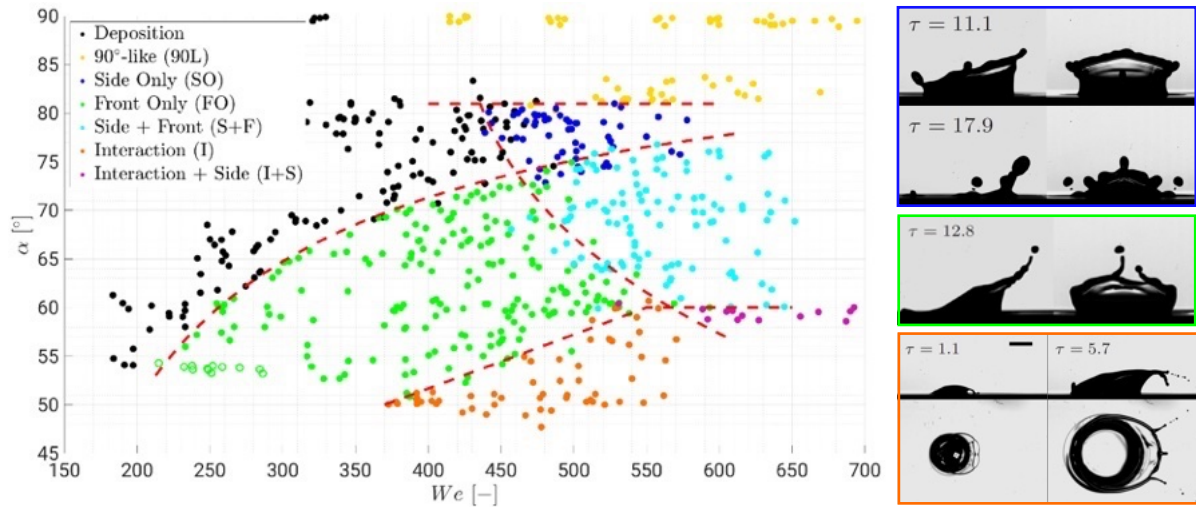


Figure 6: Overview of different observed splashing types and threshold functions, $\delta = 0.22$, $Oh = 0.013$.

Based on the classification of splashing types, regime thresholds are derived and described by physically meaningful mathematical functions. These thresholds define the conditions at which a specific splashing type starts to occur, depending on the impact angle and Weber number. The onset of side splashing is described accurately by the wall-normal Weber number only $We_n = 425$. The normal Weber number is calculated with the normal component of the impact velocity. This reveals that side splashing is governed solely by the normal component of the impact velocity and is independent of the tangential motion of the droplet. In contrast, the front splashing threshold is described by the relation $\cos(\alpha) \cdot We = 128$, or $\sqrt{We_t We} = 128$, indicating that front splashing is promoted equally by the tangential and total Weber numbers.

The transition from 90°-like splashing to side-only splashing is identified to occur at a fixed impact angle of $\alpha = 81^\circ$, which corresponds to the appearance of the oblique mechanism on the back side of the crown. For the crown-film interaction-induced splashing, the threshold is described empirically by a piecewise linear function. It occurs at low impact angles ($\alpha \leq 60^\circ$) and requires sufficiently high Weber numbers.

e) Numerical Results

The oblique mechanism of crown formation, which is observed on the back side in the experiments, is further investigated with DNS. In contrast to the front ($\varphi = 0^\circ$), where the crown develops continuously from an initial lamella, the crown on the back ($\varphi = 180^\circ$) forms differently. The simulations reveal that the initially ejected lamella collides and coalesces with the bulk of the droplet at $\tau = 0.2$, see Figure 3. This hinders the further growth of the lamella and the lamella is dragged backwards. The liquid stops feeding into the lamella anymore, causing it to recede. As a result, a crown forms with a thicker wall and an inward bend shape, $\tau = 0.45$.

The simulations further reveal that this oblique mechanism extends up to an azimuth position $\varphi \geq 140^\circ$ for the $\alpha = 78^\circ$ impact case, while the normal mechanism is present for $\varphi \leq 120^\circ$. As the impact angle increases, the oblique mechanism disappears between $\alpha = 80^\circ$ and $\alpha = 82^\circ$ confirming the experimentally observed transition from side-only splashing to 90°-like splashing. With decreasing α , the azimuthal extension of the oblique mechanism increases with a transition azimuth of $\varphi = 110^\circ$ for $\alpha = 70^\circ$ impact angle.

The transition zone aligns with the characteristic azimuth position of side finger formation and splashing found in the experiments. This proved that the presence of the oblique mechanism introduces disturbance to the rim on the sides which is the source for finger formation and subsequent side-only splashing.

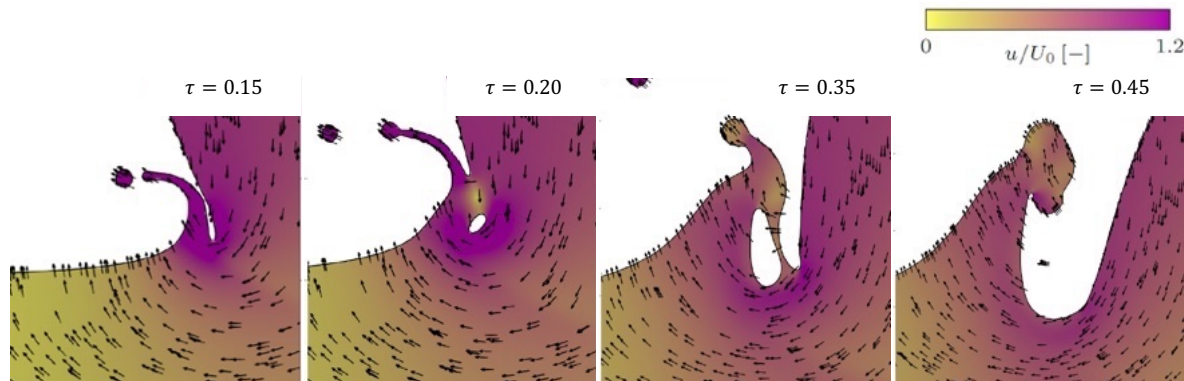


Figure 7: Oblique mechanism of crown formation and velocity field on the back side ($\varphi = 180^\circ$).

Within this project, we also investigated the flow dynamics within the inner wall film during the crown spreading phase of a $\alpha = 90^\circ$ droplet impact. The focus is to determine the governing forces during the decay of the stagnation point flow field, which are crucial for many analytical models. The results revealed that wall friction forces govern the spreading process. The temporal decay follows an exponential progression, deviating from the inviscid theory, even at high Reynolds numbers $Re = 15070$. Only a free-slip wall boundary condition is able to reproduce the inviscid theory. Furthermore, a comparison of the DNS results with an analytical boundary layer model demonstrated the negligible influence of pressure, gravity, and surface tension forces during the spreading phase, and wall friction appearing to be the sole relevant force.

3. Comments on the qualification programme and supervision strategy:

The qualification programme of the International Research Training Group has been an enriching component of my PhD journey. Through a well-balanced combination of scientific seminars, specialised workshops, and research stays abroad, the programme provided me with valuable interdisciplinary training beyond my core research. Multiple research stays and summer schools in Bergamo fostered a strong collaborative spirit and offered unique opportunities to engage with researchers from all three partner institutions. Regular doctoral seminars and workshops, including the “DROPIT meets Industry” and the final colloquium, not only encouraged scientific exchange but also strengthened the international network within our research community. Attending major conferences such as ICMF and ILASS enabled me to present my work to a broader scientific audience and receive valuable feedback. Overall, the programme provided an excellent framework for academic growth and international collaboration.

Participation in the following courses/seminars of the Int. Research Training Group:

- “Scientific Writing Skills for Doctoral Students in Aerospace Engineering” by Dr. Joseph Michaels, Language Centre of the University of Stuttgart, Winter Semester 2022/2023
- “Analytical modelling of species and energy transport from evaporating drops” by Prof. S. Tonini, Stuttgart, 28.02.2023
- “Orthogonal curvilinear coordinate systems in analytical modelling of drop evaporation” by Prof. G.E. Cossali, Stuttgart, 28.02.2023
- “Optical Methods in Spray Research” by Prof. A. Coghe, Bergamo, 14.06.2023
- “Fundamentals of X-ray experimental techniques” by Prof. M. Santini, Bergamo, 14.06.2023

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

- 2 Weeks, Research Stay, Bergamo, 04.10.2021 – 15.10.2021
- 2 Weeks, Research Stay, Bergamo, 09.05.2022 – 21.05.2022
- 2 Weeks, Summer School, Bergamo, 22.06.2022 – 02.07.2022
- 2 Weeks, Research Stay, Bergamo, 10.11.2024 – 22.11.2024

Participation in the following International Research Training Group events:

- Kick-Off Second Funding Period + Doctoral Seminar, Bergamo, 08.09.2021 – 09.09.2021
- Summer School, Stuttgart and Bergamo, 09.06.2022 – 30.06.2022
- Doctoral Seminar, Stuttgart, 01.03.2023 – 02.03.2023
- Workshop “DROFIT meets Industry”, Bergamo, 15.06.2023
- Doctoral Seminar, Bergamo, 08.02.2024 – 09.02.2024
- Workshop “Experimental and Numerical Methods in Science”, Bergamo, 18.06.2024
- Doctoral Seminar, Stuttgart, 28.01.2025 – 29.01.2025
- Final Colloquium, Stuttgart, 31.07.2025 – 01.08.2025

Participation in conferences, congresses, etc., at home and abroad:

- DIPSI 2022, Droplet Impact Phenomena & Spray Investigation, Bergamo (Italy), 01.07.2022
- ICMF 2023, 11th International Conference on Multiphase Flow, 2-7 April 2023, Kobe, Japan
- DIPSI 2023, Droplet Impact Phenomena & Spray Investigation, Bergamo (Italy), 16.06.2023
- ILASS Europe 2023, 32nd Conference on Liquid Atomization & Spray Systems, 4-7 September 2023, Napoli, Italy
- DIPSI 2024, Droplet Impact Phenomena & Spray Investigation, Bergamo (Italy), 17.06.2024

4. Individual publications:

I. Peer-reviewed articles:

- Potyka, J., **Stober, J.**, Wurst, J., Ibach, M., Steigerwald, J., Weigand, B., & Schulte, K. (2022). Towards DNS of Droplet-Jet Collisions of Immiscible Liquids with FS3D. In W. E. Nagel, D. H. Kröner, & M. M. Resch (Eds.), High Performance Computing in Science and Engineering '22 (pp. 197–212). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-46870-4_14
- **Stober, J. L.**, Santini, M., & Schulte, K. (2023). Influence of Weber Number on Crown Morphology during an Oblique Droplet Impact on a Thin Wall Film. *Fluids*, 8, Article 11. <https://doi.org/10.3390/fluids8110301>
- **Stober, J. L.**, Potyka, J., Ibach, M., Weigand, B., & Schulte, K. (2023). DNS of the Early Phase of Oblique Droplet Impact on Thin Films with FS3D. In High Performance Computing in Science and Engineering '23. Springer International Publishing. <https://doi.org/10.48550/arXiv.2311.17690>
- Geppert, A. K., **Stober, J. L.**, Steigerwald, J., Schulte, K., Tonini, S., & Lamanna, G. (2024). Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. *Physics of Fluids*, 36, Article 1. <https://doi.org/10.1063/5.0179716>
- **Stober, J. L.**, & Schulte, K. (2025). What governs droplet spreading on a wetted wall. *Physics of Fluids*, 37, Article 3. <https://doi.org/10.1063/5.0258742>
- **Stober, J. L.**, Santini, M., & Schulte, K. (2025). Characterization of splashing and regime thresholds for oblique droplet impact on thin wall films. *Experimental Thermal and Fluid Science*, 167, 111493. <https://doi.org/10.1016/j.expthermflusci.2025.111493>

II. Submitted articles:

- **Stober, J. L.**, Heinemann, M., Santini, M., Schulte, K. (2026) Quantitative Analysis of the Asymmetry of Inner Wall Film Flow in Oblique Droplet Impacts, Atomization and Sprays, <https://doi.org/10.1615/AtomizSpr.2026062531>

III. Further publications

a) Proceedings

- Klötzl, D., Krake, T., Zhou, Y., **Stober, J.**, Schulte, K., Hotz, I., Wang, B., & Weiskopf, D. (2022). Reduced Connectivity for Local Bilinear Jacobi Sets. 2022 IEEE Workshop on Topological Data Analysis and Visualization, 39–48. <https://doi.org/10.1109/TopoInVis57755.2022.00011>
- **Stober J.**, Santini, M.; Geppert, A.; Schulte, K. Experimental Investigation of an Oblique Droplet Impact onto a Quiescent Wall Film with Two-Perspective High-Speed Imaging. Conference: ILASS-Europe 2022, 31th Conference on Liquid Atomization and Spray Systems, 4-7 September 2023

IV Presentations

a) Oral

- **Stober J.L.**; Schulte, K. Numerical Investigation of the Mechanism of Early Crown Formation during an Oblique Droplet Impact onto a Wall Film. ICMF 2023: 11th International Conference on Multiphase Flow, April 2-7, 2023, Kobe, Japan, 2023
- **Stober J.L.**; Schulte K.: Investigation of the Asymmetry of the Flow Field at the Wall during an Oblique Droplet Impact onto a Thin Wall Film. DIPSI, Bergamo (Italy), 2024

b) Posters

- **Stober J.L.**; Schulte K.: Numerical Investigation of the Mechanisms of early Crown Formation at an oblique Drop Impact onto a Wall Film. DIPSI, Bergamo (Italy), 2022
- **Stober J.L.**; Santini M.; Schulte K.: Experimental Observation of the Collision of a Crown with a thin Wall Film during Early Oblique Droplet Impact. DIPSI, Bergamo (Italy), 2023

Report of doctoral researcher (SP-B1-2)

Project number: SP-B1-2

First and last name of doctoral researcher: Patrick Palmetshofer

(Working) title of doctoral project: Droplet collisions with superhydrophilic surfaces

Name of supervisors: Bernhard Weigand, Gianpietro Elvio Cossali

1. General information:

Duration of International Research Training Group membership: May 1, 2021 – June 15, 2025t
Funding (type and duration): PhD student at Institute of Aerospace Thermodynamics, University of Stuttgart May 1, 2021 – March 30, 2025
Start of doctoral training and (anticipated) date of doctoral degree: May 1, 2021 – March 4, 2025 (date of doctoral defense)
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 10/2015 – 09/2018 Aerospace Engineering, University of Stuttgart, B.Sc. 10/2018 – 03/2021 Aerospace Engineering, University of Stuttgart, M.Sc. 05/2021 – current Aerospace Engineering, University of Stuttgart, Dr.-Ing.
Where applicable, please note any circumstances which could have affected your scientific work: -
Occupation after leaving the IRTG, if applicable: -

2. Description of doctoral project and research results achieved to date:

The project aimed to improve the understanding of droplet impacts onto superhydrophilic surfaces and adjacent phenomena.

Overall, this project was characterized by a non-linear parallel workflow that explored various topics in the overall context of droplet impacts onto dry walls, especially structured walls. Later, due to the focus on superhydrophilic surfaces, which normally cause the formation of a thin liquid film after a droplet impact, “successive” droplet impacts onto structured surfaces submerged in a liquid film became a focus of the project. This broad orientation of the project lead to a variety of findings in different areas related to droplet impacts onto superhydrophilic surfaces.

In the scope of my PhD thesis, different structure size ranges were investigated, for both dry surfaces and pre-wetted surfaces.

Findings that resulted in journal publications or represent significant chapters in my PhD thesis were made in four distinct categories:

1. *Droplet impacts onto and sessile droplets on non-cuboidal microstructures*

Following up directly on the work done by my predecessor in the first phase of the IRTG, surface structures manufactured by the Karlsruhe Nano Micro Facility (KNMF) were investigated. Due to developments in their manufacturing capabilities, it was now possible to create structures which are non-cuboid in shape.

Non-cuboid surface structures influence sessile droplet behavior differently compared to cuboid structures. However, in both cases, the apparent contact angles measured at all azimuthal angles are higher than those on smooth surfaces, regardless of whether the wetting state is Cassie-Baxter or Wenzel or whether the surfaces are untreated or hydrophobized through plasma polymerization.

For dynamic droplet impacts, the simple splashing limit proposed by Foltyn et al. [1] effectively predicts splashing behavior even for non-cuboid geometries and hexagonal surface patterns under the tested impact conditions. Partial crown formations tend to appear and remain more stable along the diagonals relative to the surface structure and in the up-ramp direction on sloped surfaces. Air entrapment most often occurs behind the vertical sides of ramps and is less common in areas where crown splashing happens, for both ramp and pyramid configurations. If a receding phase is present during droplet behavior, the maximum spreading factor can be accurately predicted using the correlation developed by Lee and Lee [2]. On hydrophobic surfaces, as droplets retract, certain areas dry out sooner, leading to the breakup of the droplet into smaller, separate parts that reflect the underlying surface geometry. Interestingly, these early dewetting regions are the same locations where crown splashing initially occurred during impact.

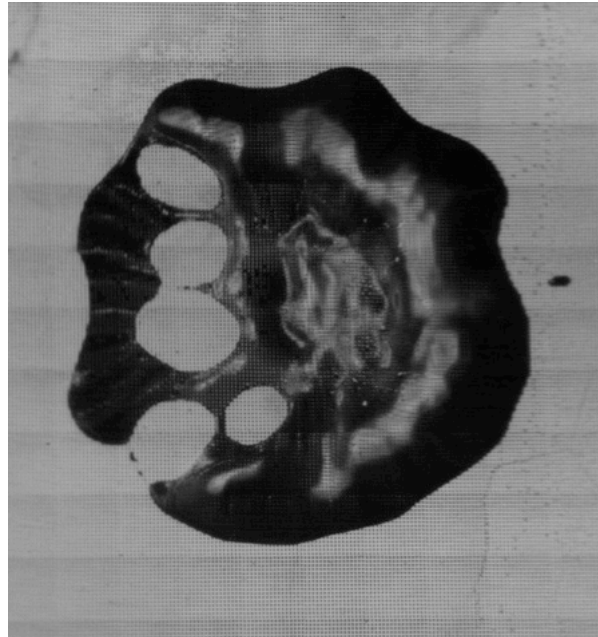


Figure 8: Top view of a receding droplet on a polymerized surface with ramp-shaped structures, illustrating asymmetric receding behavior.

2. Droplet impacts onto dry sub-millimeter structures

During droplet impacts onto regularly structured surfaces which feature structure sizes between 0.1 mm and 0.5 mm, droplet impacts show some characteristics which are distinct from those seen on smaller and on larger structures. While smaller structures mainly act on the early impact phases and similar to directional roughness, larger structures can affect the impact process at every stage. Depending on the structure size and impact parameters and time after impact, a to p-dominant, a top-driven and a groove-

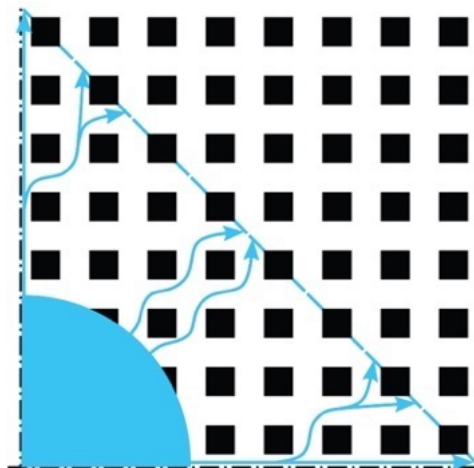


Figure 9: Schematic of formation mechanism leading to rhomboid spreading shapes.

driven regime were identified. The two former regimes are characteristic for smaller structures or low velocity impacts and have been described previously, while the groove-driven regime exhibits some unique characteristics and can lead to the formation of star-like or rhomboid spreading shapes. Utilizing numerical simulations of Associated Project AS11, the formation of such spreading shapes can be described in the detail to form due to zig-zagging flow in the gaps between structures.

Additionally, a Weber-number dependent finger splashing regime was identified on sub-millimeter structures in addition to prompt splashing. Also, additional secondary droplets can be generated by popping air bubbles which are entrapped in the wake of the pillars early in the impact process.

3. Droplet impacts onto submerged structures

In droplet dynamics research, droplet impacts onto dry surfaces and droplet impacts onto films and pools have mainly been studied separately from each other. However, some research suggested that especially for thin films, the underlying surface structure may have a significant effect on the impact dynamics. As such, I studied droplet impacts onto fully and barely submerged single pillars, finding that the crown formation is suppressed in the wake of the pillar and is less affected in other regions. When droplets strike the pillars more centrally, the maximum crown height decreases.

In the case of submerged sub-millimetric surface structures, several key observations were made. Although grooves make up approximately 75% of the total projected area during droplet impacts on these submerged structures, the development of the crown is more strongly influenced by the film height above the tops of the pillars than by the height above the groove bottoms. This dependence extends to both crown height and crown lifetime. Additionally, crown angles are more sensitive to the size of the surface structures than to the film height within the examined parameter ranges.

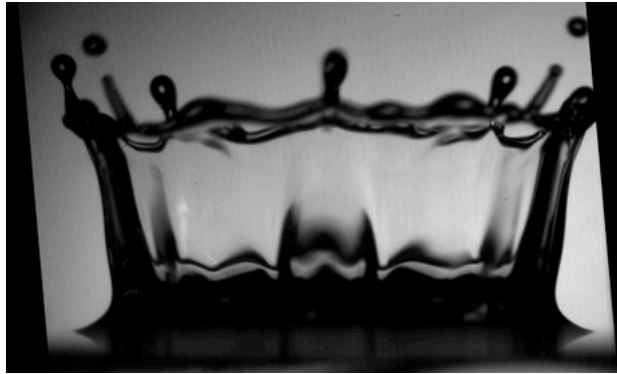


Figure 10: Crown streaks as a result of underlying surface structures after droplet impacts onto submerged sub-millimeter structures.

The splashing threshold is affected in a non-linear way by the structure size; when surface features approach the natural wavelength for crown breakup, they enhance splashing, while larger structures can lead to the formation of regular, finger-like extensions that align with the crown's pattern. These larger structures also promote a more gradual crown disintegration through the emergence of thick fingers, and they tend to produce larger secondary droplets on average. Lateral views reveal thickness variations within the crown, seen as streaks, likely caused by azimuthal differences in bulk flow velocity inside the crown crater—an effect driven by the surface geometry.

When there is no liquid film above the pillars, the liquid in the grooves becomes critical to crown formation, often resulting in highly irregular crowns with many secondary droplets. In these barely submerged conditions, symmetrical breakdown of the crown base can occur along the diagonals of the surface pattern. This phenomenon resembles diamond-shaped spreading seen on dry, structured surfaces. Similarly, the base of the crown sometimes adopts rhomboid shapes, reflecting the underlying structure. To account for these effects, a consistent empirical model was developed using an effective film height that incorporates both the film height above the pillars and the height of the pillars themselves, providing accurate predictions for maximum film height and the timing at which it is reached.

4. Effects of wettability on droplet impacts onto heated smooth surfaces

Outside the scope of the thesis, published findings were made for droplet impacts onto smooth heated surfaces. Taking advantage of the Total Internal Reflection (TIR) imaging method applied in the test stand with a sapphire glass prism, the combined effects of the surface temperature and wettability of the surface on the maximum spreading diameter could be investigated in detail.

Additionally, a phenomenon, which has been appearing frequently in images of droplet impacts onto heated surfaces, but – to my knowledge – never been identified before was studied in detail using this

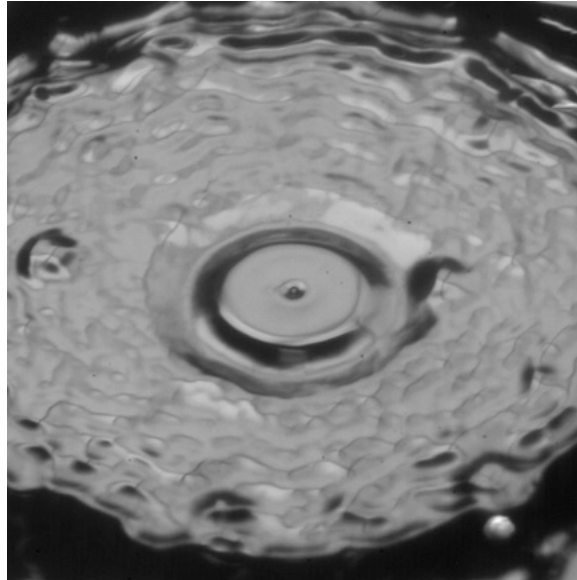


Figure 11: Bottom view of a droplet impact onto a smooth heated sapphire glass surface, showing a thermocapillary lamella recess at the center of the impact zone.

measurement method. The *thermocapillary lamella recess* happens due to the steep temperature gradients within the forming lamella, which cause a steep temperature gradient along the gas-liquid interface of the centrally enclosed bubble in the earliest phase of the droplet impact process. The LPSM method, developed in the first phase of the IRTG, which uses the double shadow of this bubble to determine the height of the lamella, was then used to characterize the thinning lamella in the center. Using numerical simulations of a stagnant thin film with an enclosed bubble at the surface heated from the bottom confirmed the explanation of this phenomenon.

Additional minor results were obtained in the following areas:

1. Simultaneous droplet impacts onto smooth surfaces

As a preliminary investigation to study simultaneous side-by-side droplet impacts onto structures surfaces as explored numerically in SP-B5, a dropper which can generate two droplets simultaneously was developed and the central uprising sheet was measured.

However, the experiments showed that the falling process of two droplets falling next to each other at a distance where they interact significantly on the surface presents a somewhat unstable state and the phenomenology of such droplet impacts are very sensitive to the temporal separation between the droplet impacts.

2. Droplet impacts at non-isothermal conditions

As surprising new results were found for the “classic case” of droplets impacting onto smooth heated surfaces, I used a test chamber previously used for near-critical droplet evaporation investigations to study droplet impacts at high pressures and high temperatures. Acetone droplets were released onto dry surfaces in a high-pressure, high-temperature nitrogen atmosphere, including near-critical conditions w.r.t. the critical point of Acetone. Under isothermal conditions, classical spreading occurs, but the maximum spreading diameter shows a complex, non-monotonic dependence on ambient pressure and temperature. In non-isothermal cases (droplet injection temperature below ambient and surface temperatures), increased evaporation reduces spreading and can trigger droplet rebound or Leidenfrost-breakups, with occasional ejection of dense fluid at the contact line during evaporation.

3. Neutron imaging of droplet impacts (Main work in SP-C4)

In joint work with the Paul Scherrer Institute (PSI) in Villigen, Switzerland, we used high-temporal resolution neutron imaging to visualize the mixing process of a water droplet impacting a heavy water film - two miscible liquids distinguishable by their neutron attenuation properties. By varying the film thickness while maintaining constant droplet velocity and size, we aimed to understand the fluid dynamics and mixing behavior within the crown crater and lamella formed during impact.

[1] Foltyn, P., Ribeiro, D., Silva, A., Lamanna, G., and Weigand, B.: Influence of wetting behavior on the morphology of droplet impacts onto dry-patterned micro-structured surfaces. *Physics of Fluids*, 2022, 34 (12), p. 123322. doi: 10.1063/5.0124692.

[2] Lee, J. B. and Lee, S. H.: Dynamic wetting and spreading characteristics of a liquid droplet impinging on hydrophobic textured surfaces. *Langmuir*, 2011, 27 (11), pp. 6565–6573. issn: 0743-7463. doi: 10.1021/la104829x.

Cooperation benefits within the IRTG

The combination of experimental projects with numerical projects tremendously assisted in the understanding of a variety of phenomena observed in my experiments. As such, joint work together with SP-B5 resulted in interpretable results on droplet impacts onto submerged single cubes and droplet impacts onto submerged sub-millimeter structures.

To study droplet impacts onto dry sub-millimeter structures, it soon became apparent that a dynamic contact angle model is necessary to capture the contact line dynamics numerically. Working together with AS-11 made a comparison of the results possible, ultimately leading to a publication in the *Journal of Colloid and Interface Science*.

Additionally, as later in the project, droplet impacts onto films became a focus of my work, the experience and knowledge of researchers in SP-C1 and SP-C2 became an important asset and allowed me to quickly ramp up experimental campaigns in this field.

3. **Comments on the qualification programme and supervision strategy:**

The courses offered in the scope of the IRTG were useful to get a more comprehensive overview of droplet and spray research outside of the boundaries of my own project and the IRTG itself. Additionally, the possibility to invite people for presentations and cooperative work, such as in the summer school, was highly beneficial.

Many of the soft skill courses were also interesting and important to hone such skills outside of the lab.

For me, the supervision strategy of my advisors was characterized by vast freedoms to choose my focus and my work even beyond the bounds of my subproject. The regular meetings such as DIPSI or doctoral seminars, which provided opportunities to get “sanity checks” on my research also benefitted me greatly. That the events alternated between Bergamo and Stuttgart gave the whole IRTG a collaborative spirit that I highly enjoyed.

At the same time, longer-term stays in Bergamo were great opportunities to spend more time on scientific discussions with both my advisors, advancing my research and ideas greatly.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- *Analytical and numerical methods*
- *Multiphase seminar*
- *Compressible Flows*
- *Numerical methods in two-phase flows*

Specialised courses:

- Analytical methods for non-linear partial differential equations
- Short course on atomization and sprays, TU Darmstadt
- Analytical modelling of species and energy transport from evaporating drops
- Orthogonal curvilinear coordinate systems in analytical modelling of drop evaporation
- PIV and laser-based measurement techniques in sprays

Elective and soft skill courses:

- Introductory Italian language course
- Intercultural communication
- Unconscious bias awareness
- Scientific writing skills
- Effective self-marketing

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

- March 6, 2023 – March 23, 2023 Stay at University of Bergamo, Italy
- June 13, 2023 – June 16, 2023 Stay at University of Bergamo, Italy
- August 4, 2024 – August 7, 2024 Measurement campaign at Paul Scherrer Institute, Villigen, Switzerland

Participation in the following International Research Training Group events:

- Doctoral Seminar March 2021, Remote
- Doctoral Seminar/Kick-Off 8.-9.9.2021, Bergamo
- Summer School 2022
- Doctoral Seminar 1.-2.3.2023
- Doctoral Seminar 8.-9.2.2024
- Doctoral Seminar January 28.-29.1.2025

Participation in conferences, congresses, etc., at home and abroad:

- DIPSI 2022, 1.7.2022, Bergamo, IT
- ICMF 2023, 2.-7.4.2023, Kobe, Japan
- DIPSI 2023, 16.6.2023, Bergamo, IT
- ILASS Europe 2023, 4.-7.9.2023, Napoli, IT
- DIPSI 2024, 17.6.2024, Bergamo, IT
- Thermodynamik-Kolloquium, 25.-27.9.2024, Stuttgart

4. Individual publications:

I Peer-reviewed articles:

- Palmetshofer, P., Wurst, J., Geppert, A. K., Schulte, K., Cossali, G. E., and Weigand, B.: Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science*, 2025, 682, pp. 413–422. issn: 0021-9797. doi: <https://doi.org/10.1016/j.jcis.2024.11.154>
- Palmetshofer, P. and Weigand, B.: Experimental study on splashing, spreading, and air entrapment dynamics in droplet impacts onto micrometric ramps, pyramids, and staggered cubes. *Physics of Fluids*, 02/2025, 37 (2), p. 022112. doi: 10.1063/5.0251394
- Palmetshofer, P., Hengsbach, S., Guttman, M., Worgull, M., and Weigand, B.: Azimuthal variation of apparent contact angles on structured surfaces with micrometric ramps, pyramids and staggered cubes at two different inherent wettabilities. *Langmuir* (in press), 2025. doi: 10.1021/acs.langmuir.4c04554

- Palmetshofer, P., Geppert, A. K., Steigerwald, J., Arcos Marz, T., and Weigand, B.: Thermocapillary central lamella recess during droplet impacts onto a heated wall. *Scientific Reports*, 2024, 14 (1102). doi: 10.1038/s41598-024-51382-3

II Submitted articles:

- Palmetshofer, P., Saha, R., Bryce, J., Geppert, A. K., Cossali, G. E., and Weigand, B.: Interfacial phenomena in droplet impacts onto submerged structured surfaces. Submitted to *Physics of Fluids*
- Palmetshofer, P., Wirth, M., Steinhausen, C., Trtik, P., Lehmann, E., Geppert, A. K.: Neutron imaging investigation of liquids mixing during droplet impact onto liquid films. Submitted to *ILASS Europe 2025*

III Further publications

a) Proceedings

- Palmetshofer, P., Ren, W., Steigerwald, J., Geppert, A., and Weigand, B.: Experimental and Numerical Investigation of Droplet Impacts onto Cubic Pillars with a Wetted Base. 11th International Conference on Multiphase Flow, ICMF 2023, Kobe, Japan, April 2–7, 2023. 04/07/2023
- Palmetshofer, P., Heckel, J., Geppert, A., and Weigand, B.: Experimental investigation of the height and angle of an uprising sheet occurring in near-simultaneous double droplet impacts onto smooth surfaces of varying wettability. *ILASS Europe 2023, 32nd European Conference on Liquid Atomization & Spray Systems*, 4-7 Sept 2023, Napoli, Italy. 09/07/2023. isbn: 978-88-88104-28-7
- Palmetshofer, P., Arcos Marz, T., and Weigand, B.: 360° contact angles of water droplets on micro-metric ramps, pyramids and staggered cuboid surfaces at varying wettability. *Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigation*, Bergamo, Italy, 16th June 2023, Bergamo, Italy. Università degli studi di Bergamo. 06/16/2023, pp. 22–24. doi: 10.13122/DIPSI2023
- Palmetshofer, P., Foltyn, P., Geppert, A., and Weigand, B.: Dynamic contact angle measurements of droplets spreading on surfaces of varying wettability. *Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations*, Bergamo, Italy, 1st July 6, 2022. Università degli studi di Bergamo. 07/01/2022, pp. 30–31. doi: 10.13122/DIPSI2022

b) Technical and scientific reports

- Palmetshofer, P.: DROPIT Midterm Review: DROPIT Subproject B1: Droplet collisions with solid superhydrophilic surfaces, July 2023

IV Presentations

a) Oral

- Palmetshofer, P., Heckel, J., Geppert, A., and Weigand, B.: Experimental investigation of the height and angle of an uprising sheet occurring in near-simultaneous double droplet impacts onto smooth surfaces of varying wettability. *ILASS Europe 2023, 32nd European Conference on Liquid Atomization & Spray Systems*, 4-7 Sept 2023, Napoli, Italy. 09/07/2023. isbn: 978-88-88104-28-7
- Palmetshofer, P., Ren, W., Steigerwald, J., Geppert, A., and Weigand, B.: Experimental and Numerical Investigation of Droplet Impacts onto Cubic Pillars with a Wetted Base. 11th International Conference on Multiphase Flow, ICMF 2023, Kobe, Japan, April 2–7, 2023. 04/07/2023

b) Posters

- Palmetshofer, P., Arcos Marz, T., and Weigand, B.: 360° contact angles of water droplets on micro-metric ramps, pyramids and staggered cuboid surfaces at varying wettability. *Proceedings of the DIPSI Workshop 2023: Droplet Impact Phenomena & Spray Investigation*, Bergamo, Italy, 16th June 2023, Bergamo, Italy. Università degli studi di Bergamo. 06/16/2023, pp. 22–24. doi: 10.13122/DIPSI2023
- Palmetshofer, P., Foltyn, P., Geppert, A., and Weigand, B.: Dynamic contact angle measurements of droplets spreading on surfaces of varying wettability. *Proceedings of the DIPSI Workshop 2022: Droplet Impact Phenomena & Spray Investigations*, Bergamo, Italy, 1st July 6, 2022. Università degli studi di Bergamo. 07/01/2022, pp. 30–31. doi: 10.13122/DIPSI2022

Report of doctoral researcher (SP-B6-3)

Project number: SP-B6-3

First and last name of doctoral researcher: Anna Mareike Kosteletzky

(Working) title of doctoral project: Coupled free flow-porous media flow processes including drop formation

Name of supervisors: Prof. Rainer Helmig, Prof. Maurizio Santini, Prof. Carina Bringedal

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT: October 2022-September 2025

Funding (type and duration):

DFG – DROPIT, October 2022 – September 2025 basic funds of University

Start of doctoral training and (anticipated) date of doctoral degree:

04.10.2022 – 31.12.2026

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

B.Sc. Simulation Technology, University of Stuttgart, July 29th, 2019

M.Sc. Simulation Technology, University of Stuttgart, August 25th, 2022

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Motivation & Aim of the project

Application fields for coupled free-flow and porous media systems including drop formation at the interface span from water management in fuel cells, thermal insulation, and membrane emulsification to evaporation from biological porous structures such as leaves and skin.

Pore-scale droplet formation at the interface can have a large impact on the surrounding macroscopic flow behaviour. Therefore, an accurate description of mass, momentum and heat transfer between different domains are crucial to model the drop dynamics, including evaporation of the droplets, correctly. In previous work ([1], [2], [3]), the droplet dynamics (drop formation, growth and detachment) were modelled on different scales. Ackermann et al. (2021) models the coupled system at the REV-scale and includes the interface as a lower dimensional domain. For this, the droplet dynamics are approximated in an upscaled manner. In addition to this multiscale approach, a pore-scale model was developed to gain further insight into the ongoing interfacial processes [3]. On the pore-scale, a pore-network model is taken to represent the pore-scale porous structure and droplet dynamics are included through force balances.

A common assumption of those models is that the phases are in local thermal equilibrium (LTE) within the porous medium as well as at the porous-medium free-flow and the drop-free-flow interface. This means that instantaneous heat transfer between different phases is assumed. However, for large differences in temperatures, thermal properties or very dynamic systems this assumption may not be valid. Furthermore, if heat transport is combined with mass transport across an interface, e.g. in case of evaporation, accounting for jumps in temperature as well in chemical potential might be necessary to capture the interfacial processes accurately. For porous media systems, a dual network model [4] can be taken to describe local thermal non-equilibrium (LTNE) processes at the pore-scale, while on the REV-scale different models (e.g. Nuske et al. [5]) aim to describe these processes in an effective way. Therefore, we aim to first evaluate the differences between the models and to find suitable relations for effective up-scaled quantities from the pore-scale. Coupling those models to the free-flow domain and considering droplet dynamics, will enable us to investigate under which conditions the assumption of LTE is valid. All in all, this will lead to a more accurate description of the interfacial processes of coupled free-flow porous-medium system including droplet formation and will enhance the understanding of those.

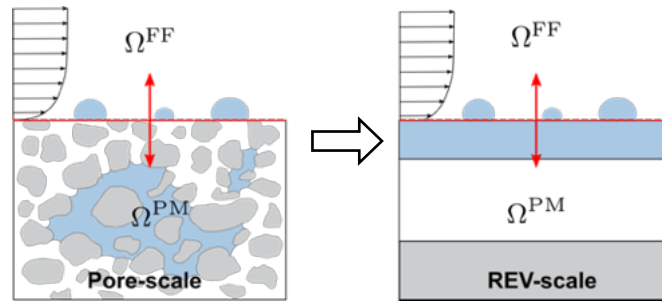


Figure 12: Schematic for Aim of Project

Current status and achieved results

First, we investigated the different models available for porous media systems, before coupled systems are looked at.

For investigating the difference between LTNE porous media models on different scales, we consider two pore-scale and one REV-scale model. The pore-resolved model is not only chosen to serve as a reference, but also to obtain effective quantities for the REV-scale model from homogenization theory. The model hereby accounts for the pore-scale geometry of the porous medium explicitly and incorporates the heat transfer of different phases through flux continuity and temperature discontinuity at the underlying sharp interface (see Fig. 2). To reduce computational costs the dual network model, accounting for the pore-scale geometry in an approximate way, is considered. The dual-network model is represented through two interconnected pore-network models. One network is representing the void space, while the other describes the solid of a porous structure. Connections between both networks allow for interfacial heat exchange. To represent the porous structure in a simplified way, the pore-network mod-

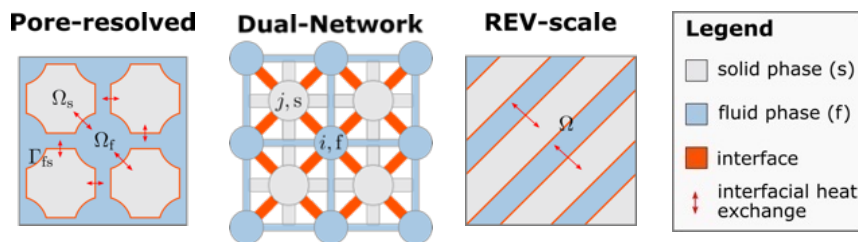


Figure 13: Schematic Sketch of Porous Media Models for LTNE processes

els consist of idealized shapes, such as spheres and tubes. On the REV-scale, the effective heat transfer within and between the phases can be obtained through empirically motivated laws (see e.g. Nuske et al. [5]) or through mathematically derived upscaled values, e.g. through homogenization.

Comparing averaged temperatures of the models for a pure heat conduction case shows that REV models with the effective parameters obtained through homogenization theory are in close agreement with the pore-resolved results. This holds for both high and low interfacial resistance values (see Fig. 3 for results in case of high interfacial resistance value). Results for the dual network model differ slightly more from the two beforementioned models. This can however be addressed to the fixed spatial resolution of the porous domain, which had to be chosen small enough such that pore-resolved calculations are still feasible. Moreover, for large interfacial resistance values non-equilibrium effects were obtained, as results from LTE and LTNE models differ (see Fig. 3). For small interfacial resistance values however, LTE is a valid assumption in case of a stagnant fluid. Evaluating different formulations of effective properties for the REV-scale model showed the deficiency of the empirical formulations, which do not take the interfacial resistance into account.

Currently, we are conducting comparisons between the same models in case of a moving fluid through the porous domain. This convective flux is then expected to increase the heat transfer between the phases. The importance of the different mechanisms is evaluated for different non-dimensional number, such as Peclet or Reynolds number as well as the conductivity ratio of the two phases.

A first step towards coupling the dual network model to the free-flow domain has been done such that regimes and setups can be investigated in terms of the relevance in including LTNE processes.

Moreover, first ideas were developed to consider multiple fluid phases in the dual-network model, as the current model is restricted to fully saturated porous systems. To overcome the difficulty of modelling heat transport at the three-phase contact points, a liquid film will be assumed to cover the solid, when the liquid saturation is larger than zero. As velocities are assumed to be very slow within the porous medium, i.e. creeping flow, we simplify multiphase Nusselt numbers to the single-phase ones.

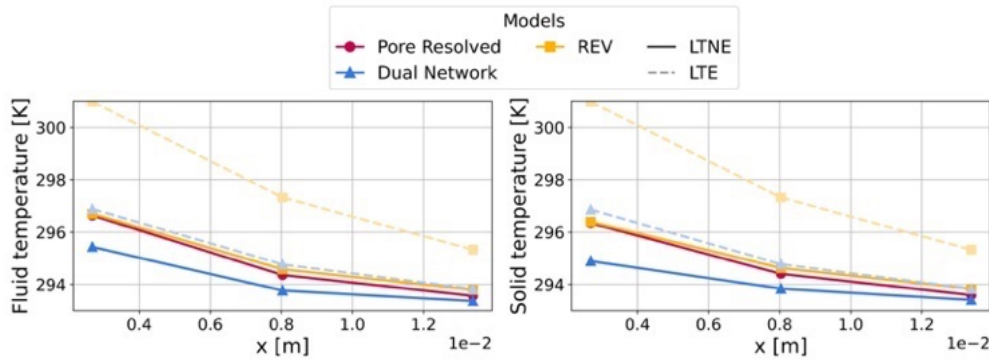


Figure 14: Averaged temperatures for different models for a high interfacial resistance value between solid and fluid phase. Comparison of LTE and LTNE models.

Outlook

As a next step, the comparison of models will be finished soon. Moreover, the implemented coupling between the free-flow domain and dual-network model will be used to analyse systems in order to find out under which conditions LTNE should be considered. Next, the coupling can be extended to incorporate the droplet dynamics described in Veyskarami et al. [3]. The extension of the dual network model towards handling multiple fluid phases within the void space for LTNE processes, will enable us to study more relevant cases, where also the porous domain can dry-out due to evaporation. All in all, coupled systems incl. droplet should be studied under LTNE on different scales with the before-mentioned models.

Stays abroad

For my stay abroad, I chose to stay at our partners at the Norwegian University of Science and Technology (NTNU) in Trondheim for three months in 2024. Here, I joined the PoreLab group, a joint collaboration between researchers at the NTNU and the University of Oslo (UiO) conducting fundamental research on the topic of porous media systems. I had the pleasure to discuss multiple times per week with Prof. Kjelstrup and Prof. Bedeaux about non-equilibrium thermodynamics to understand the underlying thermodynamic mechanisms and derivations of temperature and concentration jumps across sharp interfaces [6]. While I was first revising the basics of thermodynamic laws and properties relevant for the next steps, we quickly focused on the derivation of temperature jump conditions for solid-liquid interfaces as they e.g. appear in saturated porous media. Reformulation of those conditions enabled us to include these jump conditions in the pore-resolved model for upscaling as well as in the dual network model. While pore-scale parameters like the interfacial resistivity are crucial for describing the temperature jump between two phases, they're not always available from experiments or modelling. Values appropriate for granite-water systems can be e.g. found in Gonçalves et al. [7]. Further work has been done on the formulation of jump conditions for liquid-gaseous interfaces, e.g. for liquid drops in surrounding air, where mass transport is present in addition to the energy transport across the interface. As those two fluxes are tightly coupled, also the jump conditions for chemical potentials and temperatures are tightly coupled, leading to a linear system of equations that needs to be solved to obtain the values of the physical properties from each side of the interface. Values for the transport properties in case of water-air systems are given by Wilhelmsen et al. [8]. Furthermore, Wilhelmsen et al. [8] postulate the relevance of including the coupled jump conditions to described evaporation and condensation of water. The implementation of those coupled jump conditions for a pore-network model fully saturated with water coupled to a gaseous free-flow system turned out to be more challenging than previously expected and is therefore still an ongoing work. This is due to the fact that for our used marker and cell (MAC) scheme in the free-flow domain, degrees of freedom are not available directly at the coupled free-flow porous medium interface.

References

- [1] Baber, K., et al. Modeling drop dynamics at the interface between free and porous-medium flow using the mortar method. *Int. J. Heat Mass Transf.* 99 (2016), pp. 660-671.
- [2] Ackermann, S., et al. Multi-scale three-domain approach for coupling free flow and flow in porous media including droplet-related interface processes. *Journal of Computational Physics* (2021).
- [3] Veyskarami, M., et al. Droplet formation, growth and detachment at the interface of a coupled free-flow-porous medium system: A new model development and comparison. *Transport in porous media* (2023).
- [4] Koch, T., et al. A (Dual) Network Model for Heat Transfer in Porous Media: Toward Efficient Model Concepts for Coupled Systems from Fuel Cells to Heat Exchangers. *Transport in Porous Media* 140 (2021), 107–141.
- [5] Nuske, P. Modeling two-phase flow in a micro-model with local thermal non-equilibrium on the darcy scale. *International Journal of Heat and Mass Transfer* 88 (2015), 822–835.
- [6] Kjelstrup, S., Bedeaux, D. Non-equilibrium thermodynamics of heterogeneous systems. (2008).
- [7] Gonçalves, William, et al. "Interfacial thermal resistance between nanoconfined water and silicon: Impact of temperature and silicon phase." *Surfaces and Interfaces* 33 (2022): 102188
- [8] Wilhelmsen, Ø., et al. "Coherent description of transport across the water interface: From nanodroplets to climate models." *Physical Review E* 93.3 (2016): 032801.

3. Comments on the qualification programme and supervision strategy:

Both the technical and soft-skill courses were highly beneficial for me. The core and specialized courses deepened my understanding of topics aligned with my research area and laid a strong foundation for my research project. Courses beyond my immediate research focus broadened my perspective and helped me better understand the wider research context I am working within. Technical courses, such as programming, supported the proper implementation of my models. Additionally, soft-skill courses - particularly the scientific writing course and the poster workshop - significantly enhanced my ability to communicate my research effectively, both in writing and visually. The scope of the courses I selected was well-suited to my research.

The IRTG's qualification programme actively encouraged suggestions for new courses and visiting external researchers. Further I was involved among other PhDs in the organization of the final colloquium. With my main supervisor Prof. Helmig and co-supervisor Prof. Bringedal I had regular meetings for feedback and discussions about my research topic.

The scheduled mid-term review provided a valuable opportunity to receive feedback from all of my supervisors, helping me assess my progress and clarify the next steps toward completing my PhD. Moreover, the programme strongly encouraged a research stay abroad. My stay at the foreign partner university (NTNU) was highly valuable to me. It deepened my understanding of the underlying processes relevant to my research and allowed me to expand my professional network. Engaging in interdisciplinary collaboration enhanced my scientific communication skills, allowing me to effectively discuss my topic using diverse terminology. Getting familiar with a different method and discussing the relation of a topic to my own research project helped me.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

- Lecture on 'Fundamentals and modelling of turbulent flow' given by Xu Chu and Heng Xiao (6 ETCS)

Specialised courses:

- InterPore short course on 'Porous Media Free-Flow Coupling - from REV to Pore Scale and back' given by Rainer Helmig
- SFB1313 short course on 'Modeling and simulation of blood flow in microvascular networks' given by Tobias Köppl.
- C++ advanced course of HLRS
- C++ intermediate course of HLRS

Elective and soft skill courses:

- Scientific-Writing Course
- Poster Workshop

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

- August 2024 – October 2024: Research stay at NTNU Trondheim & PoreLab with Prof. Signe Kjølstrup to study non-equilibrium thermodynamics

Participation in the following International Research Training Group events:

- March 2023: Doctoral Seminar, Stuttgart
- February 2024: Doctoral Seminar, Bergamo
- January 2025: Doctoral Seminar, Stuttgart
- August 2025: DROPIT final colloquium

Participation in conferences, congresses, etc., at home and abroad:

- March 2023: CSD Winter School in Myrkdalen (Norway) on 'Coupled Subsurface Dynamics'
- April 2023: SFB 1313 Status Seminar - presentation and poster presentation
- September 2023: Summer School SFB1313 with a project on 'Single phase flow in a porous media with a single crack: Micro experiment and numerical simulation'
- October 2023: SimTech Conference 2023, Stuttgart, attendance
- March 2024: SFB 1313 Status Seminar - poster presentation
- May 2024: InterPore Conference 2024, online participation with oral presentation
- June 2024: Workshop on Coupled Free-Flow Porous Media Systems
- March 2025: SFB 1313 Status Seminar - poster presentation
- March 2025: Workshop on Local Thermal Non-Equilibrium Processes in Porous Media
- May 2025: InterPore Conference 2025, participation with in-person presentation and poster presentation

4. Individual publications:

I. Peer-reviewed articles:

II. Submitted articles:

- **Kosteletzky, A. M.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H. K., Helmig, R.: Local Thermal Non-Equilibrium Models in Porous Media: A Comparative Study of Conduction Effects, International Journal of Heat and Mass Transfer, 2025a, submitted 2025

- **Kosteletzky, A. M.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H. K., Helmig, R.: Local Thermal Non-Equilibrium Models in Porous Media: A Comparative Study of Convection Including Conduction. Submission planned for International Journal of Heat and Mass Transfer, 2025b, (in preparation)
- Grether, S., **Kosteletzky, A. M.**, Kiemle, S., Schneider, M., Helmig, R.: A dual-continuum approach for precipitated salt in porous media: Accounting for coupled transport processes. Submission planned for International Journal of Heat and Mass Transfer, 2025, (in preparation)

III. Further publications

a) Proceedings

- **Kosteletzky, A. M.**, Helmig, R., Stefansson, I., Berre, I.: Local thermal non-equilibrium processes influenced by interfaces within porous media systems. InterPore Conference 2024, Abstract, 2024
- **Kosteletzky, A. M.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H. K., Helmig, R.: Local thermal non-equilibrium processes in porous media: Comparison of different models from the pore- to the REV-scale. InterPore Conference 2025, Abstract, 2025

b) Technical and scientific reports

IV. Presentations

a) Oral

- **Kosteletzky, A. M.**, Helmig, R., Stefansson, I., Berre, I.: Local thermal non-equilibrium processes influenced by interfaces within porous media systems. InterPore Conference 2024, remote, 2024
- **Kosteletzky, A. M.**, Helmig, R.: Modeling and analysis of local thermal non-equilibrium processes at the interface between free- and porous media flow. Coupled free-flow porous medium workshop, Stuttgart, 2024
- **Kosteletzky, A. M.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H. K., Helmig, R.: Local thermal non-equilibrium processes in porous media: Comparison of different models from the pore- to the REV-scale. InterPore Conference 2025, Albuquerque, US, 2025

b) Posters

- **Kosteletzky, A. M.**, Schneider, M., Helmig, R., Bringedal, C.: Modelling Local Thermal Non-Equilibrium (LTNE) for coupled free-flow and porous-media systems. SFB1313 Status Seminar, Gültstein, Germany, 2023
- Schneider, M., **Kosteletzky, A. M.**, Wu, H., Müller, J., Helmig, R., Weigand, B.: Coupling free flow with porous-media flow on the REV scale. SFB1313 Status Seminar, Gültstein, Germany, 2023
- **Kosteletzky, A. M.**, Schneider, M., Ghosh, T., Helmig, R., Dahle, H. K., Bringedal, C.: Modelling Local Thermal Non-Equilibrium (LTNE) for interfacial processes: From the pore- to the REV-scale. SFB1313 Status Seminar, Gültstein, Germany, 2024
- **Kosteletzky, A. M.**, Stefansson, I., Bringedal, C., Ghosh, T., Dahle, H. K., Helmig, R.: Modelling Local Thermal Non-Equilibrium (LTNE) for interfacial processes: From the pore- to the REV-scale. SFB1313 Status Seminar, Gültstein, Germany, 2025
- Koch, T., **Kosteletzky, A.M.**: DuMux –an open-source simulator for solving flow and transport problems in porous media with a focus on model coupling. InterPore Conference 2025, Albuquerque, US, 2025
- **Kosteletzky, A. M.**, Veyskarami, M., Ackermann S., Michalkowski, C., Helmig, R.: Coupled free flow-porous media flow processes including drop formation. DROPIT final colloquium, Stuttgart, Germany, 2025

Report of doctoral researcher (SP-A1)

Project number: SP-A1

First and last name of doctoral researcher: Yueqiao, Hu

(Working) title of doctoral project: Modelling of deformed, multicomponent time-dependent liquid drop evaporation

Name of supervisors: Prof. G.E.Cossali, Prof. S.Tonini, Prof. B.Weigand

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT, October 2022-September 2025
Funding (type and duration): PhD student at the Department of Engineering and Applied Science, University of Bergamo, October 2022-September 2025
Start of doctoral training and (anticipated) date of doctoral degree: October 2022-March 2026
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09.2013-07.2017, Energy and Power Engineering, Northwestern Polytechnical University. Bachelor's degree. 04.2019-02.2022, Aerospace Engineering, University Stuttgart. Master's degree. 10.2022-09.2025, Engineering and Applied Science, University of Bergamo. Doctoral degree.
Where applicable, please note any circumstances which could have affected your scientific work:
Occupation after leaving the IRTG, if applicable:

2. Description of doctoral project and research results achieved to date:

The subproject SP-A1 aims at an evaporation model for an oscillating droplet. In the past investigation on droplet evaporation, the non-sphericity, multicomponent, non-uniform temperature distribution and many other effects have already been discussed in detail [1]. But the quasi-steady assumption was always taken. The evolution of the interface and flow fields is much slower than the mass and heat transfer through the interface. The droplet surface is assumed to be still. But for an **oscillating** droplet, the droplet has a moving interface and this assumption has to be released. An analytical/numerical model for the evaporation of an oscillating droplet is the aim of the present project. The effect of the moving interface on droplet evaporation is the focus point. The numerical code FS3D (Free Surface 3D) is shared by the partner ITLR of University Stuttgart, which is developed for the direct numerical simulation of multiphase flows [2].

A simple analytical droplet oscillation model is first introduced. Two basic assumptions hold in this model: a spheroidal droplet and potential flows. The assumed shape of spheroidal droplets provides great convenience in employing the prolate/spherical/oblate coordinate systems, which can be summarized as a generalized spheroidal coordinate system (ψ, η, φ) in a unified way [3]. Therefore, it is very suitable for a droplet oscillating in a spheroidal shape. The droplet shape and this generalized spheroidal coordinate system are completely described by a time-varying deformation parameter, which can be

numerically solved for from a nonlinear energy conservation equation [3]. In this time-varying generalized spheroidal coordinate system, the moving interface between the droplet and gas can be simply described at $\psi = 1$. The moving boundary problem is then avoided. The assumption of potential flow allows us to describe the velocity as the gradient of a potential. The velocity field can be derived from the Laplace equation of a velocity potential, instead of solving the momentum equation. Solving the Laplace equation in this generalized spheroidal coordinate system, the velocity field in both phases can be given fully analytically. The simulation results for an oscillating droplet by the in-house code FS3D are used as a comparison to better show how these assumptions in the analytical model possibly deviate from the real situation, as reported in Fig.(1). Inside the droplet, the velocity field obtained by the analytical model coincides very well with that from the simulations by FS3D. From the simulation results, in the gaseous phase, the vortex is observed where the droplet surface is almost still during the oscillation, while the analytical model assumes potential flows. This vortex is strong for a spherical droplet when the velocity is fast, and weak when the droplet approaches a maximum extended shape.

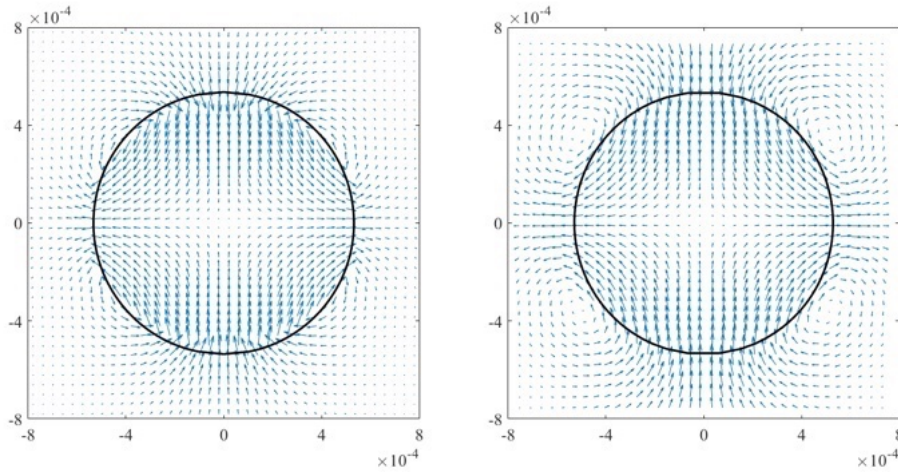


Figure 1: Comparison of the velocity field by analytical model (left) and FS3D (right) for a spherical droplet, which contracts from a prolate shape to an oblate shape.

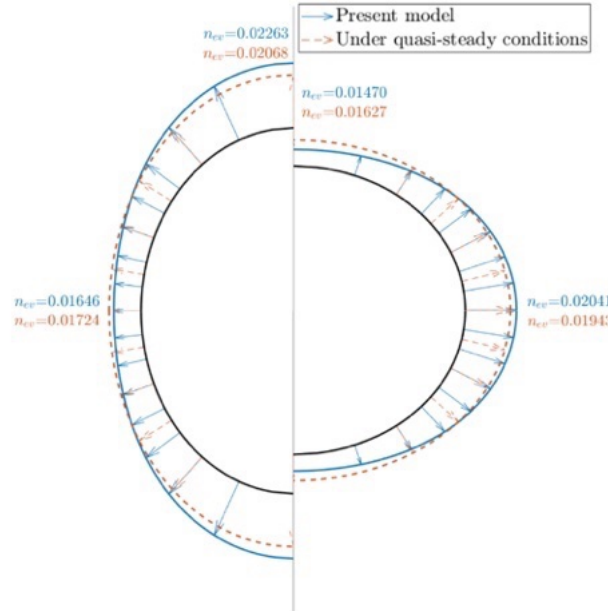


Figure 2: Comparison of the evaporation flux n_{ev} distribution along the droplet surface by the present model (blue line) and by an analytical model under quasi-steady assumptions (red line) [4].

From this oscillation model, combined with the evaporation, an evaporation model for an oscillating droplet is proposed. Besides the Laplace equation of velocity potential, the species conservation equation in the gaseous phase is solved. The generalized spheroidal coordinate system is still employed. The main focus in the present work is to investigate the effect of a moving interface on the evaporation

characteristics. Thus, additional simplifications have been made temporarily. The temperature distribution over the surface is assumed to be fixed and uniform, and the temperature dependence of the physical properties is therefore ignored. Gas density is assumed to be constant at each location, even though the vapour mass fraction varies spatially. The normal velocities within the two phases are discontinuous due to the evaporation. The evaporation flux is calculated in a diffusion form through the jump conditions. Compared with pure mass diffusion, the evaporation flux is enhanced by a fraction $\frac{1}{1-\chi_s}$ due to the convection (χ_s is the saturated vapour mass fraction at the interface). The instantaneous evaporation rate is evaluated by integration of the evaporation flux over the surface. The mass loss within one period can be obtained by integrating the evaporation rate over one period. Since the species conservation equation and the Laplace equation are coupled, they are numerically solved by a computational loop, using the General Form of PDE in COMSOL Multiphysics. Due to the moving interface, the evaporation characteristics of an oscillating droplet are different from those of the quasi-steady model [4]. The evaporation flux shows a similar tendency, but quantitatively different, as shown in Fig.(2). This difference leads to a phase shift between the evaporation rate and droplet oscillation. The peak of the evaporation rate is always ahead of the maximum extended droplet shape, and the minimum is always ahead of the spherical droplet shape. The mass loss within one period for an oscillating droplet is close to that of the quasi-steady model for a small initial droplet deformation, but larger than that for an increasing initial droplet deformation.

References

- [1]. S. S. Sazhin, Droplets and sprays: Simple models of complex processes, *Springer*, 2022.
- [2]. K. Eisenschmidt, M. Ertl, H. Gomma, C. Kieffer-Roth, C. Meister, P. Rauschenberger, M. Reitzle, K. Schlottke, B. Weigand, Direct numerical simulations for multiphase flows: An overview of the multiphase code FS3D, *Appl. Math. Comput.* 272 (2016) 508–517.
- [3]. S. Tonini, G. Cossali, Analytical model of small-and large-amplitude drop oscillation dynamics, *Physics of Fluids* 36 (11) (2024).
- [4]. S. Tonini, G. Cossali, An exact solution of the mass transport equations for spheroidal evaporating drops, *International Journal of Heat and Mass Transfer* 60 (2013) 236–240.

3. Comments on the qualification programme and supervision strategy:

The courses/seminars organised by the IRTG help a lot. The courses/speeches given by the invited professors/researchers introduce various methods/tools, which can be applied and bring inspiration to our research scope. I'm very thankful to my supervisors for their professional skills/knowledge, full patience and encouragement. The research program is well organised. The research content was well planned, and we met regularly to discuss the questions and procedures. Much useful literature is shared, so I can know more details rather than a simple conclusion. I can get strong and patient support when I have any problems, not only skills/answers/solutions, but also the methods/ideas on how to think and solve the problems. On the other hand, the working atmosphere is very pleasant and free. I can completely take my time to deal with the problems and am encouraged to try all the possibilities without any pressure, when I was involved in any questions/difficulties. Any chances to improve my skills and broaden my horizons are recommended, but not forced. I have also visited the partner University Stuttgart to discuss some problems in the numerical code. It is a very helpful experience. During this period, I realised the importance of communication and cooperation with other researchers, which I should learn and improve in future work. Totally, this is the most pleasant education period of mine. I sincerely thank my supervisors for the opportunity of this project of appropriate scope, their professional skills/abilities and kindness in the last three years.

Participation in the following courses/seminars of the Int. Research Training Group: <i>Core courses:</i> • Heat and mass transfer in droplets. • Simple models of complex processes: heating and evaporation of droplets on a superhydrophobic surface and other. <i>Specialised courses:</i> • Analytical modelling of species and energy transport from evaporating drops. • Orthogonal curvilinear coordinate systems in analytical modelling of drop evaporation. • Selected solution methods for non-linear partial differential equations for flow and heat transfer problems. • Short course on dimensional analysis for engineering applications. <i>Elective and soft skill courses:</i> • Information literacy skills, research strategies and sharing your findings. • Scientific Writing. • Use of IT tools to support the research.
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: 04.2023. Research stay in University Stuttgart, discussion about the evaporation model in the numerical code.
Participation in the following International Research Training Group events: • DROPIT Doctoral Seminar. Stuttgart. 01.03-02.03.2023 • DROPIT meets industry. Bergamo. 15.06.2023 • DROPIT Hybrid Doctoral Seminar. Bergamo. 08.02-09.02.2024 • DROPIT Workshop "Experimental and Numerical Methods in Science". Bergamo. 18.06.2024 • DROPIT Doctoral Seminar. Stuttgart. 28.01-29.01.2025 • DROPIT Workshop "Analytical and Numerical Methods in Science". Bergamo. 12.06.2025 • DROPIT Final Colloquium. Stuttgart. 31.07-01.08.2025
Participation in conferences, congresses, etc., at home and abroad: • Droplet Impact Phenomena & Spray Investigations. Bergamo. 16.06.2023 • Droplet Impact Phenomena & Spray Investigations. Bergamo. 17.06.2024 • ICLASS Conference. Shanghai. 24.06-27.06.2024 • Droplet Impact Phenomena & Spray Investigations. Bergamo. 13.06.2025 • ILASS Conference. Lund. 30.08-04.09.2025

4. Individual publications:

I Peer-reviewed articles:

- Y.Hu, S.Tonini, J.Reutzsch, M.Frey, G.E.Cossali, B.Weigand, "Direct Numerical Simulation of an Oscillating Droplet under Evaporating Conditions." *International Journal of Heat and Mass Transfer* 232 (2024): 125891.

II Submitted articles:

- Y.Hu, S.Tonini, G.E.Cossali, "An Analytical/Numerical Evaporation Model for an Oscillating Spheroidal Droplet in a Potential Flow." *International Journal of Heat and Mass Transfer*.

III Further publications

a) Proceedings

- W.Ren, Y.Hu, B.Weigand, "Splashing during simultaneous two drops impact onto a micropillared superhydrophilic surface." *31st Conference on Liquid Atomization and Spray Systems*. 2022.
- Y.Hu, S.Tonini, B.Weigand, G.E.Cossali, "A numerical approach to the study of the evaporation of an oscillating droplet under unsteady conditions." *Proceedings of ICLASS 2024, 16th Triennial International Conference on Liquid Atomization and Spray Systems, Shanghai, China*. ICLASS, 2024.

- Y.Hu, S.Tonini, G.E.Cossali, “A Semi-Analytical and Numerical Model for the Evaporation of an Oscillating Droplet in a Potential Flow.” *33rd Conference on Liquid Atomization and Spray Systems*. 2025.

IV Presentations

a) Oral

- Y.Hu, S.Tonini, G.E.Cossali: Modelling of deformed, multicomponent, time-dependent liquid drop evaporation. Bergamo (Italy), 08.02-09.02.2024.
- Y.Hu, S.Tonini, B.Weigand, G.E.Cossali: A numerical approach to the study of the evaporation of an oscillating droplet under unsteady conditions. Shanghai (China), 24.06-27.06.2024.
- Y.Hu, S.Tonini, G.E.Cossali: Modelling of deformed time-dependent liquid drop evaporation. Stuttgart (Germany), 28.01-29.01.2025.
- Y.Hu, S.Tonini, G.E.Cossali: A Semi-Analytical Model for the Evaporation of an Oscillating Spheroidal Droplet. Bergamo (Italy), 13.06.2025.
- G.E.Cossali, Y.Hu, S.Tonini: Modelling of deformed, multicomponent, time-dependent liquid drop evaporation. Stuttgart (Germany), 31.08-01.09.2025.

b) Posters

- Yueqiao Hu, Gianpietro Elvio Cossali, Simona Tonini, Bernhard Weigand, Jonathan Reutzs, Markus Frey. DIPSI. Bergamo. 2023
- Yueqiao Hu, Simona Tonini, Gianpietro Elvio Cossali, Bernhard Weigand. Analytical and numerical modelling of deformed, time-dependent liquid drop evaporation. DROPIT Final Colloquium. Stuttgart. 2025

Report of doctoral researcher (SP-B4-3)

Project number: SP-B4-3

First and last name of doctoral researcher: Hasel-Cicek Konan

Title of doctoral project: Hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for two-phase incompressible flows

Name of supervisors: Prof. Dr. Christian Rohde, Prof. Dr. Lorenzo Botti

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT since December 2022
Funding (type and duration): PhD student at Applied Mathematics of IANS, University of Stuttgart
Start of doctoral training and (anticipated) date of doctoral degree: 8 December 2022 to 1 June 2026
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09/2011 Environmental Engineering, Middle East Technical University, Bachelor's degree, 20 June 2016 09/2016 Mechanical Engineering, Izmir Institute of Technology, Master's degree, 13 August 2018 04/2022 Applied Mathematics, University of Siena, Master's degree, 3 Mai 2022
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

The project aims to develop a hyperbolic approximate system for the diffuse-interface approach based on the Navier-Stokes-Cahn-Hilliard (NSCH) system. Here we focus on the Euler-part of the NSCH system to construct a hyperbolic approximation.

The NSCH system is employed to simulate the interface dynamics in two-phase flows because it is capable of modelling the topological changes such as droplet breakup and coalescence. Moreover, the system is thermodynamically consistent. As a starting point, we consider a simplified version of the NSCH system, known as Model-H, for two incompressible and immiscible fluids at a constant temperature. Due to the incompressibility constraint in the Navier-Stokes equations, we have to solve a Poisson equation at each time step to compute the pressure. Moreover, the non-linear terms, namely the convection term in the Navier-Stokes equations and the chemical potential term in the Cahn-Hilliard (CH) equation, can cause stability issues for the numerical methods. The fourth-order derivatives lead to severe time-steps restrictions for explicit time discretization. All of these challenges result in high computational cost. Therefore, the goal of this project is to improve the computational efficiency of solving the NSCH system by developing a suitable approximate system.

We apply three relaxation approaches to eliminate the computational challenges of the NSCH system. The first is the artificial compressibility method to remove the non-locality arising from the incompressibility constraint. Next, we approximate the CH equation using a friction-type approximation, inspired by the Euler-Korteweg with friction. This yields a third-order system of equations. The last relaxation approach involves applying the elliptic approximation to reduce the third-order derivatives to the first-order ones. As a result, we obtain a first-order approximate system for the Euler-part of the NSCH system. The approximate system for the NSCH system is governed by

$$p_t^\varepsilon + \frac{1}{\alpha} \nabla \cdot u^\varepsilon = 0, (1)$$

$$u_t^\varepsilon + (u^\varepsilon \cdot \nabla) u^\varepsilon + \frac{1}{2} (\nabla \cdot u^\varepsilon) u^\varepsilon + \nabla p^\varepsilon + c^\varepsilon \left(\nabla \left(W'(c^\varepsilon) + \frac{1}{\beta} c^\varepsilon \right) \right) = \frac{1}{\beta} c^\varepsilon \nabla \omega^\varepsilon + \gamma \Delta u^\varepsilon, (2)$$

$$c_t^\varepsilon + \nabla \cdot (c^\varepsilon u^\varepsilon) + \nabla \cdot v^\varepsilon = 0, (3) \in \Omega_T,$$

$$v_t^\varepsilon + \frac{1}{\delta} \left(\nabla \left(W'(c^\varepsilon) + \frac{1}{\beta} c^\varepsilon \right) \right) = \frac{-1}{\delta} v^\varepsilon + \frac{1}{\delta \beta} \nabla \omega^\varepsilon, (4)$$

$$-\gamma \Delta \omega^\varepsilon + \frac{1}{\beta} \omega^\varepsilon = \frac{1}{\beta} c^\varepsilon, (5)$$

where $\Omega_T = \Omega \times (0, T)$. $\Omega \subseteq \mathbb{R}^d$ is an open and bounded domain, $d = \{2, 3\}$. $(0, T)$ represents a time interval with end time $T > 0$. The unknowns are $(p^\varepsilon, u^\varepsilon, c^\varepsilon, v^\varepsilon, \omega^\varepsilon)$, which are the pressure, the velocity, the phase-field variable, the artificial flux and the relaxation variable, respectively. ε is a set of the relaxation parameters, $\varepsilon = \{\alpha, \delta, \beta\}$, in which α, δ, β represent the artificial compressibility parameter, the friction parameter and the elliptic parameter, respectively. A characteristic analysis in one spatial dimension shows that the first-order approximation possesses four distinct and real eigenvalues. Therefore, it is a strictly hyperbolic approximation. Additionally, the approximate system remains thermodynamic consistency. On the other hand, we increase the number of equations by introducing auxiliary variables for the hyperbolic approximation. Although the number of unknowns increases, this system offers notable advantages: classical numerical methods developed for systems of balance laws can be applied, and implicit–explicit (IMEX) time-stepping schemes can be used to overcome time-step restrictions. In our primary results, we obtain the wave structure and verify thermodynamic consistency at the discrete level in one spatial dimension. In the left plot of Fig.1 the wave structure of c^ε is displayed for fixed δ and varying α . We can observe that the outer waves correspond to rarefaction waves and inner waves represent shock waves. In addition, the size of the plateau region changes depending on the value of α . In the right plot of Fig 1 indicates two consistency results for the approximate system and the CH equation. The evolution of both total energy of the CH equation, E , and the total energy of the approximate system, E^ε , are consistent over time. A slight inconsistency between them are observed only for $\beta = 10^{-1}$. On the other hand, two results match each other for $\beta = 10^{-2}, 10^{-3}$ well. Moreover, E^ε decreases over time, confirming thermodynamic consistency of the approximate system.

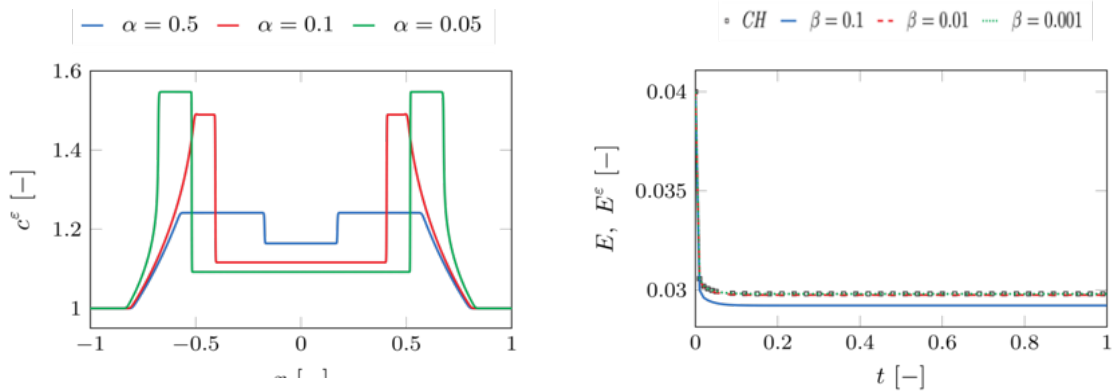


Figure 15. Left: The wave structure of c^ε as $\alpha \rightarrow 0$ for $\delta = 0.0625$. Right: Comparison between total energy of the original CH equation with the approximate system over time as $\beta \rightarrow 0$ for $\alpha = 10^{-4}, \delta = 10^{-8}, \gamma = 10^{-4}$.

3. Comments on the qualification programme and supervision strategy:

During my time at IRTG, I had the opportunity to discuss my research topic with colleagues. These discussions deepened my understanding of the subject and helped me plan future work. Moreover, the multidisciplinary structure of IRTG allowed me to gain insight into engineers' perspectives on droplets, as well as experimental studies on droplets and their interactions. Therefore, IRTG offers a broad and stimulating research scope on droplet-related topics.

With more detail, I can tell my work during my doctoral studies in each year as follows:

1st year: During my master's study, I studied the incompressible Navier-Stokes equations. Therefore, in the first year, my search focused on understanding two-phase flows, the Cahn-Hilliard equation and the NSCH system. While doing research, I learnt different mathematical challenges related to solve the NSCH system. Moreover, I also explored different approaches to deal with these challenges. Notably, I also recognized that our approach for overcoming the drawbacks of the NSCH system is different from the approaches in the literature. Our first attempt involved coupling the incompressible Navier-Stokes equations with the artificial compressibility method, called the Navier-Stokes system, and a hyperbolic version of the Cahn-Hilliard equation. We derived an energy inequality of the new system to ensure the thermodynamic consistency and to obtain a-priori estimate of the solutions. However, upon deriving the energy inequality, we observed that the total energy does not decrease over time due to the existence of an auxiliary variable. Hence, we decided to abandon the coupling of the incompressible Navier-Stokes equations with the hyperbolic version of the Cahn-Hilliard equation. Furthermore, I participated in *Advanced Numerical Methods for Hyperbolic PDE* to study finite volume and discontinuous Galerkin methods for the numerical solutions of the hyperbolic systems.

2nd year: We could not achieve our desired results during our first attempt in the first year. Therefore, we began to consider a new approximate system. The Cahn-Hilliard equation can be formulated as a high-friction limit of the Euler-Korteweg system with friction. It gives a third-order system of equations. In this way, the fourth-order Cahn-Hilliard equation can be approximated by a third-order system. Then, we coupled the Navier-Stokes system and the Euler-Korteweg with friction. Moreover, based on prior studies of the Navier-Stokes-Korteweg system, it is known that an elliptic relaxation approach can reduce the third-order derivatives to first-order ones. This technique provides a first-order approximate system for the Euler-part of the NSCH system. After applying the relaxation steps, we derive the energy inequality, which decreases over time, ensuring both the well-posedness and physical admissibility of the solutions. On the other hand, the third-order approximate system for the Cahn-Hilliard equation was changed. We simplified the Euler-Korteweg system with friction. Its simplified version is called a friction-type approximate system. In the end, we obtained a system of five equations involving three relaxation parameters. The purpose of our approach was to derive a hyperbolic-parabolic system because this structure facilitates the computation of numerical solutions using FLEXI. To implement the approximate system in FLEXI, we have collaborated with Jens Keim from Institute of Aerodynamics and Gas Dynamics at University of Stuttgart. Furthermore, we started the characteristic analysis of the approximate system and reduced the approximate system in one spatial dimension for the four equations (we ignored the elliptic approximation) to compute the eigenvalues. However, it was not easy calculation process. Therefore, firstly, we were able to obtain real and distinct eigenvalues numerically. We solve the approximate system with a classical finite difference method to illustrate numerically the convergence of the approximate model towards NSCH system in one spatial dimension. Moreover, we discuss the sensitivity of the results on the choice of the approximate parameters. The primary results shown the accuracy between original system and the approximate one. Moreover, it also shows the thermodynamic consistency of the system in the discrete level. On the other hand, to study the numerical analysis of the hyperbolic systems, I attended to *Mathematical Methods in Fluid Mechanics*. In addition, in *Introduction to Asymptotic Analysis and Homogenization Theory* I studied the asymptotic analysis. Altogether, these courses were beneficial in helping me understand the structure of the approximate system and in guiding the development new numerical schemes.

3rd year: We have continued to work on modifying the FLEXI code to implement the approximate system. Additionally, the formulae of the approximate eigenvalues were obtained using perturbation theory for the approximation in one spatial dimension because classical algebraic methods were not able to compute them. These approximate eigenvalues will enable the development of a Riemann solver, which is planned as a future extension of this project. Concurrently, we are investigating a possible relaxation between three relaxation steps, which simplifies the numerical implementation. After the investigation of the current approximate system, we aim to extend this approximation to model more complex two-phase flow problems.

Participation in the following courses/seminars of the Int. Research Training Group: <i>Specialised courses:</i> • <i>Advanced Numerical Methods for Hyperbolic PDE</i> • Drops, Jets and Films <i>Elective and soft skill courses:</i> • Scientific Writing • Italian Language Course for DROPIT students
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • Doctoral Seminar, Stuttgart, 2-3 March 2023 • DROPIT meets Industry Workshop, Bergamo, 15 June 2023 • Doctoral Seminar, Bergamo, 8-9 February 2024 • Doctoral Seminar, Stuttgart, 28-29 January 2025
Participation in conferences, congresses, etc., at home and abroad: • Numhyp2023, Numerical Methods for Hyperbolic Problems Bordeaux (France), 26-30 June 2023 • Fifth Workshop on Compressible Multiphase flows, Strasbourg (France), 5-7 June 2023 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 16 June 2023 • Sixth Workshop on Compressible Multiphase flows, Strasbourg (France), 17-19 June 2024 • HYP2024: The 19th International Conference on Hyperbolic Problems: Theory, Numerics & Applications, Shanghai (China), 1-5 July 2024 • Numdiff-17: Conference on the Numerical Solution of Differential & Differential-Algebraic Equations, Halle (Saale) (Germany), 9-13 September 2024 • Numhyp25: Numerical Methods for Hyperbolic Problems, Darmstadt (Germany), 9-13 June 2025 • SPP2410 Summer School: Mathematical Fluid Dynamics – Hyperbolic Balance Laws across the Scales, Hirschegg (Austria), 7-14 September 2025 • The 17th Hirschegg Workshop on Conservation Laws, Hirschegg (Austria), 14-20 September 2025

4. Individual publications:

I. Submitted articles:

- Keim, J., **Konan, H. C.**, & Rohde, C. (2024). A Note on Hyperbolic Relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flow. *arXiv preprint (accepted by ESAIM Proceedings)*.

II. Technical and scientific reports

- **Hasel-Cicek Konan.** (2025). Hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. Milestone report. 27 January 2025.

III. Presentations

a) Oral

- **Hasel-Cicek Konan**, Christian Rohde: Approximation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. HYP2024, Shanghai (China), 1-5 July 2024.
- **Hasel-Cicek Konan**, Christian Rohde: Approximation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. Numdiff-17, Halle (Saale) (Germany), 9-13 September 2024.

b) Posters

- **Hasel-Cicek Konan**, Christian Rohde: Incompressible two-phase model. DIPSI, Bergamo, (Italy), 17 June 2024.
- **Hasel-Cicek Konan**, Christian Rohde: Approximation of the Navier-Stokes-Cahn-Hilliard system for two-phase flows, Strasbourg (France), 17-19 June 2024.
- **Hasel-Cicek Konan**, Christian Rohde: Hyperbolic relaxation of the Navier-Stokes-Cahn-Hilliard system for incompressible two-phase flows. Numhyp25, Darmstadt (Germany), 9-13 June 2025.

Report of doctoral researcher (SP-A4-2)

Project number: SP-A4-2

First and last name of doctoral researcher: Davide Ferrari

(Working) title of doctoral project: Compressible multiphase flows, from model formulation to numerical solution

Name of supervisors: Prof. Dr.-Ing Michael Dumbser

1. General information:

Duration of International Research Training Group membership:

January 2023 – May 2024

Associate member since June 2024

Funding (type and duration):

PhD student at the Laboratory of Applied Mathematics, Department of Civil, Environmental, and Mechanical Engineering in Trento, Nov 2020 – May 2024

Start of doctoral training and (anticipated) date of doctoral degree:

Start of Doc. Train. in DROPIT: Jan. 23

Doctoral degree date: 17th May 2024

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

01/11/2020 – 01/11/2023 Ph.D in Civil, Environmental and Mechanical Engineering

Defense of the PhD thesis May 17, 2024, University of Trento, Italy

Doctor Europaeus label

01/11/2020 – 01/11/2023 Honours Programme Modelling and Simulation

An elite (numerus clausus) post-graduate honours program by DICAM, University of Trento, Italy

01/11/2020 – 01/11/2023 Master's degree

Environmental and Land Engineering (LM-35), University of Trento, Italy

01/11/2020 – 01/11/2023 Bachelor's degree

Environmental and Land Engineering (L-7), University of Trento, Italy

Where applicable, please note any circumstances which could have affected your scientific work:

Occupation after leaving the IRTG, if applicable:

01/01/2024 – current, Postdoctoral fellow within Prof. Michael Dumbser's group

Laboratory of Applied Mathematics, DICAM, University of Trento, Italy

Research topic: Structure preserving methods for compressible multiphase flows

2. Description of doctoral project and research results achieved to date:

The doctoral project aims to study both the theoretical aspects of modelling compressible and incompressible multiphase flows and the design of suitable numerical algorithms for solving such models.

Currently, there is no universally accepted mathematical model to describe two-phase flows, consequently the extension to multi-phase flows (with more than two phases) is even less clear. A popular approach to describe two-phase flows is to use so-called homogenised mixture models,

which are based on the diffuse interface approach. One of the most established ones is the Baer-Nunziato (BN) model. Due to its relatively simple mathematical structure, it is possible to derive reduced models, such as a model for incompressible two-phase flows, with straightforward assumptions. However, there are modelling and numerical reasons for the need to develop a mathematical model formulated in a well-defined mixture theory. Therefore, the **derivation of a multi-phase and multi-material model within the framework of Symmetric Hyperbolic Thermodynamically Compatible (SHTC) theory and its numerical solution** are of relevant interest.

The three-year doctoral programme and the topics covered can be broken down as follows:

1. I begin by addressing the problem of a reduced BN-type model for incompressible two-phase flows. In incompressible models, the pressure acts as a Lagrange multiplier which ensures the divergence free condition of the velocity field.

A natural numerical approach is to treat pressure-related terms implicitly, thus in this context two efficient mass and momentum conservative semi-implicit FV schemes for different applications are developed, e.g. for complex non-hydrostatic free surface flows, or for flows interacting with moving solid obstacles. These new numerical algorithms are implemented in a distributed memory MPI-parallel Fortran code. In addition, the semi-implicit discretisation adopted for the pressure subsystem, rather than a fully implicit one for the complete Partial Differential Equation (PDE) system, leads to a small-size, symmetric positive definite discrete linear system that can be solved efficiently with a preconditioned, matrix-free conjugate gradient method. This directly results in a much higher parallel scalability compared to fully implicit schemes.

2. A semi-implicit scheme boils down to the solution of a discrete linear system that considers the contributions of implicitly treated terms involved in the governing equations. As a result, understanding the numerical problems associated with the discretization of each term is more complex. Some of the numerical problems usually encountered in the numerical solution of compressible multi-phase model are related to the presence of complex interfaces described by the volume fraction of each phase. Therefore, to become more familiar with and better address these challenges, explicit schemes are considered. In particular, the numerical scheme initially adopted is an explicit second order FV method combined with the path-conservative technique of Castro and Pares for the treatment of the non-conservative products appearing in BN-type models.

With the insights from this numerical scheme, the complete seven-equation BN model for compressible two phase flows is discretized.

However, despite the variety of literature addressing this model, it still presents some problems, such as the fact that the model is not closed (in the sense that it requires the specification of an interface pressure and an interface velocity, the choice of which is not unique) and that it is not clear how to generalise the BN formulation to mixtures with an arbitrary number of phases.

3. To overcome these problems, I have collaborated with I. Peshkov and E. Romensky on the derivation of a model based on a well-defined mixture theory and from first principles as causality and the laws of thermodynamics. This model can be written in a conservative form and, as formulated, is already generalised to an arbitrary number of constituents. It takes the form of a monolithic system of first order hyperbolic PDEs that include a unified multi-phase description of fluids and solids.

Moreover, to compare the new mixture theory with the original BN model, the new model can be rewritten in terms of the equations of conservation of mass, phase momenta and phase energies. The resulting BN-type form of the new HTC mixture theory is given by a large non-linear PDEs system that includes highly non-linear stiff algebraic source terms and many non-conservative products.

4. The different challenges presented by the model are tackled numerically by adopting an operator splitting approach. The homogeneous part of the PDE system is discretized with a MUSCL-Hancock finite volume scheme using a primitive variable reconstruction and positivity preserving limiting, combined with a path-conservative technique to deal with the non-conservative products. Furthermore, the scheme employs semi-analytical time-integration methods for the stiff source terms governing the various relaxation processes. In addition, special attention was paid to the parallel implementation of the numerical scheme by employing efficient neighbourhood-collective MPI operations.

I provide results for complex flows resulting from the interaction of three different phases including gases, liquids and solids. Results are shown for multiphase flows in the limit behaviour of the Newtonian inviscid and viscous fluid, as well as in the limit of nonlinear hyperelasticity for phases behaving as elastic and elasto-plastic solids. In both cases the numerical results are comparable with results obtained from established standard models, i.e. the Euler or Navier-Stokes equations for fluids, or the classical hypo-elastic model with plasticity, but, notably, everything within a unified multi-phase model of continuum mechanics.

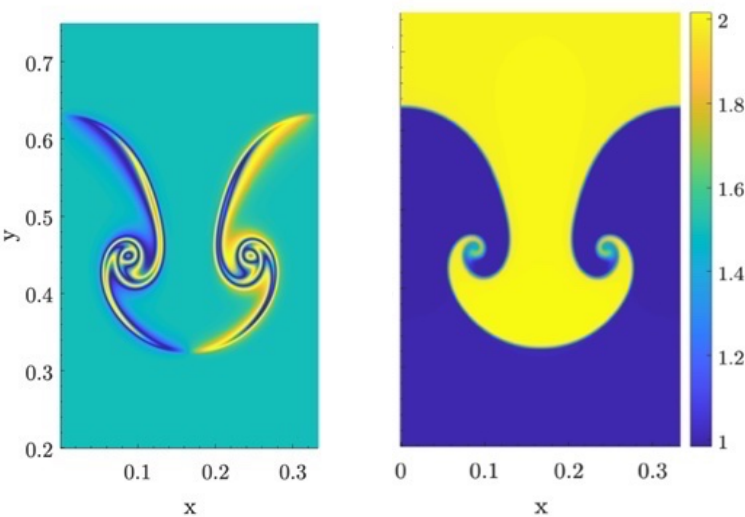


Figure 1: Visualisation of a component of the distortion field (left) and the volume fraction of the first phase (right)for the multiphase Rayleigh-Taylor instability problem. The result is obtained by solving the BN-type form of the new mixture theory of continuum mechanics.

3. Comments on the qualification programme and supervision strategy:

The qualification programme played a central role in my scientific training, but above all, it was a concrete opportunity to start building a network of international contacts by enthusiastically participating in numerous International Research Training Group events. Discussions with other DROPIT doctoral researchers were also crucial in achieving the goals of my doctoral project, in particular with Simone Chiocchetti (SP-A4). In occasion of the different DROPIT events. We have discussed at length numerical methods to correctly address the discretization of distortion field equations that present particular non-linear algebraic constraints and stiff and non-linear sources.

Participation in the following courses/seminars of the Int. Research Training Group: • Advanced numerical methods for free-surface flows • Mathematical and Numerical methods for Partial Differential Equations • MPI-OpenMP parallel programming course held at the HLRS
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: 1. Research stay at Inria Strasbourg (France), guest of Philippe Helluy's applied mathematics group, from 21 August to 26 November 2024, during which we discussed the new mixture theory and its asymptotically reduced versions.
Participation in the following International Research Training Group events: • DROPIT- Hybrid Doctoral Seminar, Stuttgart, Germany, March 1-2, 2023 • DROPIT- Hybrid Doctoral Seminar, Stuttgart, Germany, February 8-9, 2024 • DROPIT- Hybrid Doctoral Seminar, Stuttgart, Germany, January 28-29, 2025

Participation in conferences, congresses, etc., at home and abroad:

- HONOM 2022, Braga (Portugal), April 4-8, 2022
- 19th UnTRIM Workshop, Trento (Italy), June 13-15, 2022
- Shark-FV Conference, Minho (Portugal), May 22-26, 2023
- 20th UnTRIM Workshop, Trento (Italy), June 12-14, 2023
- DIPSI Workshop 2023, Bergamo (Italy), June 16, 2023
- NumHyp23, Bordeaux (France), June 26-30, 2023
- Seminaire EDP, Université de Strasbourg (France), November 14, 2023
- ProHyp 2024, Trento (Italy), April 22-26, 2024
- MULTIMAT 2024, Breckenridge (US), August 26-30, 2024
- HONOM 2024, Chania (Greece), September 8-13, 2024

4. Individual publications:

I Peer-reviewed articles:

- **D. Ferrari**, M. Dumbser, A mass and momentum-conservative semi-implicit finite volume scheme for complex non-hydrostatic free surface flows, *Int. J. Numer. Meth. Fluids.* (2021). DOI: [10.1002/fld.5017](https://doi.org/10.1002/fld.5017)
- **D. Ferrari**, M. Dumbser, A semi-implicit finite volume scheme for incompressible two-phase flows, *Commun. Appl. Math. and Comput.* (2024). DOI: [10.1007/s42967-024-00367-0](https://doi.org/10.1007/s42967-024-00367-0)
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser, A unified SHTC multiphase model of continuum mechanics, *J. Comput. Phys.* (2024). DOI: [10.1016/j.jcp.2024.113553](https://doi.org/10.1016/j.jcp.2024.113553)

II Presentations

a) Oral

- **D. Ferrari**, M. Dumbser: A mass and momentum-conservative semi-implicit finite volume scheme for incompressible two-phase flows. HONOM, Braga (Portugal), 2022
- **D. Ferrari**, M. Dumbser: A semi-implicit finite volume scheme for incompressible two-phase flows. 19th UnTRIM Workshop, Trento (Italy), 2022
- **D. Ferrari**, M. Dumbser: Semi-implicit and explicit finite volume schemes for incompressible and compressible multiphase flows. Shark-FV, Minho (Portugal), 2023
- **D. Ferrari**, M. Dumbser: An explicit finite volume scheme for compressible multiphase flows. 20th UnTRIM Workshop, Trento (Italy), 2023
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser: Towards a unified description of multiphase flows. DIPSI Workshop 2023, Bergamo (Italy), 2023
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser: An extension and numerical solution of a multiphase hyperbolic model of continuum mechanics. Seminaire EDP, Université de Strasbourg (France), 2023
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser: Formulation of a unified HTC multiphase model of continuum mechanics. ProHyp 2024, Trento (Italy), 2024
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser: Numerical solution of a unified HTC multiphase model of continuum mechanics. MULTIMAT 2024, Breckenridge (US), 2024
- **D. Ferrari**, I. Peshkov, E. Romenski and M. Dumbser: A unified HTC multiphase model of continuum mechanics. HONOM 2024, Chania (Greece), 2024

b) Posters

- **D. Ferrari**, M. Dumbser: An explicit finite volume scheme for a unified hyperbolic model for multiphase continuum mechanics. NumHyp23, Bordeaux (France), 2023.

Report of doctoral researcher (SP-A3-3)

Project number: SP-A3-3

First and last name of doctoral researcher: Claudia Marianowski

(Working) title of doctoral project: Gas-kinetische Simulation von Gas-Tropfen-Interaktionen

Name of supervisors: Prof. Dr.-Ing. Stefanos Fasoulas, Prof. Dr.-Ing. Lorenzo Alessio Botti

1. General information:

Duration of International Research Training Group membership: 1 year and 9 months
Funding (type and duration): PhD Researcher at IRS from 01.01.2023 – 30.09.2024
Start of doctoral training and (anticipated) date of doctoral degree: 01.01.2023 – aborted (30.09.2024)
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 01.10.2015; aerospace engineering; University of Stuttgart; 11.10.2019; B.Sc. 01.10.2019; aerospace engineering; University of Stuttgart; 04.10.2022; M.Sc.
Where applicable, please note any circumstances which could have affected your scientific work:
Occupation after leaving the IRTG, if applicable: Tesat Spacecom GmbH, mechanical & thermal engineer

2. Description of doctoral project and research results achieved to date:

The doctoral project dealt with the gas-kinetic simulation of multiphase flows. It investigated approximate methods for the simulation of the Enskog-Vlasov equation, which describes the temporal evolution of the velocity distribution of atoms or molecules. With the spatially resolved velocity information, macroscopic quantities such as densities and temperatures can be derived. These approximate methods should make the previous PhD student's simulations much more efficient with minimal loss of accuracy, allowing simulations of larger systems.

Initial Hypothesis and Aims:

The project was based on the hypothesis that conventional models based on the Navier-Stokes equations are inadequate to accurately capture the behavior of atoms and molecules at the liquid-vapor interface, especially under non-equilibrium conditions characterized by high accelerations due to surface tension. To close this gap, kinetic solvers, specifically a Direct Simulation Monte Carlo (DSMC)-based Enskog-Vlasov (ESMC) solver, have been developed to simulate and gain insight into the evaporation and condensation processes of these droplets. Due to the non-local behavior of these solvers, they are rather slow and challenging to parallelize. Therefore, a Fokker-Plank solver that solves a simplified version of the Enskog-Vlasov equation has been developed to increase the efficiency of such simulations. The goal of this work was to extend the capabilities of state-of-the-art solvers.

Current Project Status:

The state-of-the-art Dense Fokker-Plank (DFP) solver developed by Sadr, Gorji, and Pfeiffer was used and tested in new setups. It is a particle-based approach where a simulation particle represents a defined number of physical particles in the simulation domain. This is the same in the ESMC approach. However, the interaction between the particles is not modeled directly, but via moments of the particle distribution function such as density, flow velocity or temperature. This interaction results in acceleration and position changes of the particles.

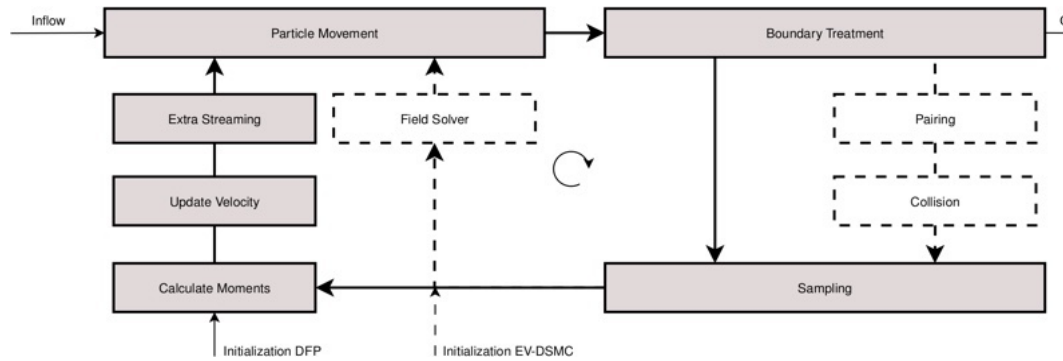


Figure 1: Comparison of a PICLas timestep for DFP (solid) and modified DSMC (dashed).

A graphical representation of the steps of the solvers is shown in Figure 1. The necessary Pairing and Collision steps of the ESMC are omitted and replaced by the calculation of moments in the Fokker-Plank solver. Next, the new velocities of all simulation particles are calculated and the additional flow is determined to obtain the correct virials. These are needed to determine the new position of the particles. The steps Particle Movement, Boundary Treatment and Sampling remain mostly the same. All steps together are a time step and are repeated until a stationary result with low noise is obtained.

Results Achieved:

Pressurized Box: To test if the DFP can reproduce the correct densities of a dense gas, a box filled with a certain number of simulation particles was simulated. The pressure was sampled for different numbers of particles inside the box and compared to results from the Enskog DSMC solver (ESMC) and analytical expressions. The results are shown in Figure 2 and demonstrate the ability of the solver to reproduce the correct pressure.

Fourier Flow: The solver was then tested in a 1D test case. This case is a confined gas between two tempered plates. When a dense gas is in contact with a solid surface, damped oscillations in the number density develop. The solver was tested against ESMC and Molecular Dynamics (MD) simulations and the results are shown in Figure 3. In the results of the DFP solver, the typical oscillations are missing. We suspect that this is due to the Free Streaming step, which is intended to replicate the pressure virials by shifting the particle positions. This can cause particles to leave the simulation domain. By applying boundary conditions to these particles, the virials can no longer be maintained and therefore the dense gas effects are not simulated correctly. This leads to the vanishing of the oscillations that originate at the wall. A solution to this problem has not been found.

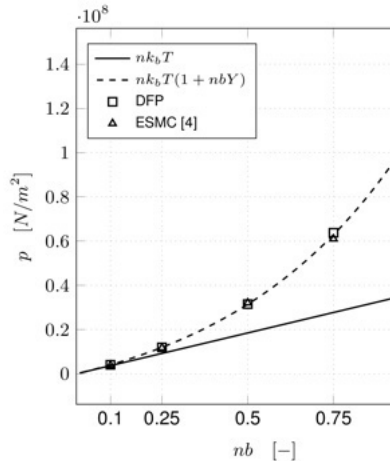


Figure 3: Resulting pressures of the solver at different densities (DFP). These are compared to the ideal gas law (solid), dense

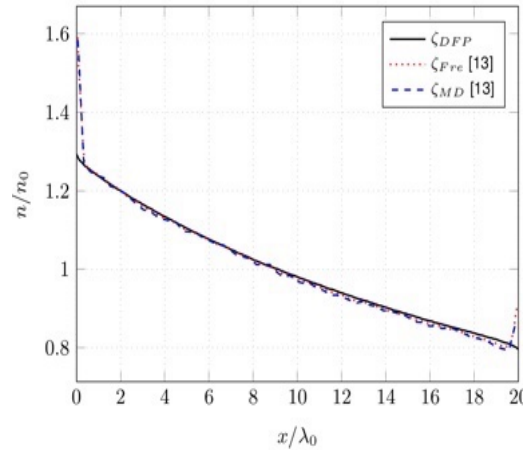


Figure 2: Resulting density profiles of a dense gas between different tempered plates. The DFP solution shows a smooth undisturbed profile between the plates. The references with an ESMC solver (Fre) and Molecular Dynamics (MD) simulations in contrast show the expected oscillations

Future Work: Extend the theory of the solver to solve the problem of solid walls.

Cooperation within the IRTG: In this project, we anticipated to provide detailed reference results to validate macroscopic CFD simulations and help to develop models with simulations results. Due to the problems of the solver, this goal could not be achieved.

3. Comments on the qualification programme and supervision strategy:

Being part of the international research group as a scientific assistant and PhD student was a very formative experience. The program offered an environment for academic growth through diverse activities such as doctoral seminars, language lessons, international conferences, and collaborative publications. These opportunities allowed me to develop both professionally and personally, enhancing my research skills and intercultural competence. One of the most defining aspects of my time in the program was the mentorship I received from Dr. Anne Geppert. Her insightful guidance, dedication to the program, and integral role in shaping the structure of the program were pivotal to my positive experience. Dr. Geppert's support helped me navigate challenges and inspired me to experience all the different aspects and opportunities within this academic position. I am very grateful for this opportunity and for having had such an inspiring mentor.

Participation in the following courses/seminars of the Int. Research Training Group:

- Online Short Course on Atomization and Sprays
- Scientific Writing
- Italian Course

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: -
Participation in the following International Research Training Group events: • Doctoral Seminar, Stuttgart, 01.-02.03.2023 • DROPIT meets Industry, Bergamo, 15.06.2023
Participation in conferences, congresses, etc., at home and abroad: • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023 • European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2024

4. Individual publications:

I. Peer-reviewed articles:

II. Submitted articles:

III. Further publications

a) Proceedings

- C. Marianowski, S. Fasoulas, M. Pfeiffer, Non-Equilibrium Particle Schemes to Simulate Liquid-Vapor Interfaces, European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023

IV. Presentations

a) Oral

- C. Marianowski, S. Fasoulas, M. Pfeiffer, Non-Equilibrium Particle Schemes to Simulate Liquid-Vapor Interfaces, European Meeting on Liquid Atomization and Spray Systems (ILASS), Naples (Italy), 2023

b) Posters

- C. Marianowski, SP-A3: Gas-Kinetic Simulation of Microdroplet-Gas Interaction, Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023
- C. Marianowski, SP-A3: Gas-Kinetic Simulation of Microdroplet-Gas Interaction, Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2024

Report of doctoral researcher (SP-C1-3)

Project number: SP-C1-3

First and last name of doctoral researcher: Pablo Molina Vogelsang

(Working) title of doctoral project: 3D analysis and modelling of droplet impact dynamics on wetted walls based on SD μ PTV

Name of supervisors: Dr.-Ing. Grazia Lamanna, Prof. Dr.-Ing. Simona Tonini

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT since January 2023
Funding (type and duration): PhD researcher, ITLR, Funding: DROPIT (15.01.2023 - 31.01.2026)
Start of doctoral training and (anticipated) date of doctoral degree: 15.01.2023 - 30.07.2026
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 08/2010 State-certified design and manufacturing engineering assistant, Metal technology, Hans-Böckler-Berufskolleg, 07/ 2013 09/2013 Aerospace Engineering, Aachen University of Applied Sciences, bachelor's degree, 10/2018 10/2018 Aerospace Engineering, University of Stuttgart, master's degree, 09/2021 01/2022 Chair of Thermodynamics of Mobile Energy Conversion Systems, RWTH Aachen University, Research Associate, 08/2022 01/2023 Aerospace Engineering, Universität Stuttgart, PhD degree
Where applicable, please note any circumstances which could have affected your scientific work: Parental leave, long-distance commuter
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

This project aims to experimentally investigate the three-dimensional (3D) flow dynamics of a droplet impacting a thin liquid film. Droplet impact plays a central role in numerous technical and natural processes, including spray coating, printing technologies, and pesticide application [1]. The behaviour of the droplet and the resulting liquid lamella after impact depends strongly on the initial conditions, the properties of the liquid, and the film thickness [2]. Previous 2D studies have shown that the radial velocity of the lamella increases linearly with radius, while decreasing exponentially over time [3]. However, the interplay between viscous losses and the decaying strength of the pressure force at the stagnation point is still not fully understood and remain an active field of research.

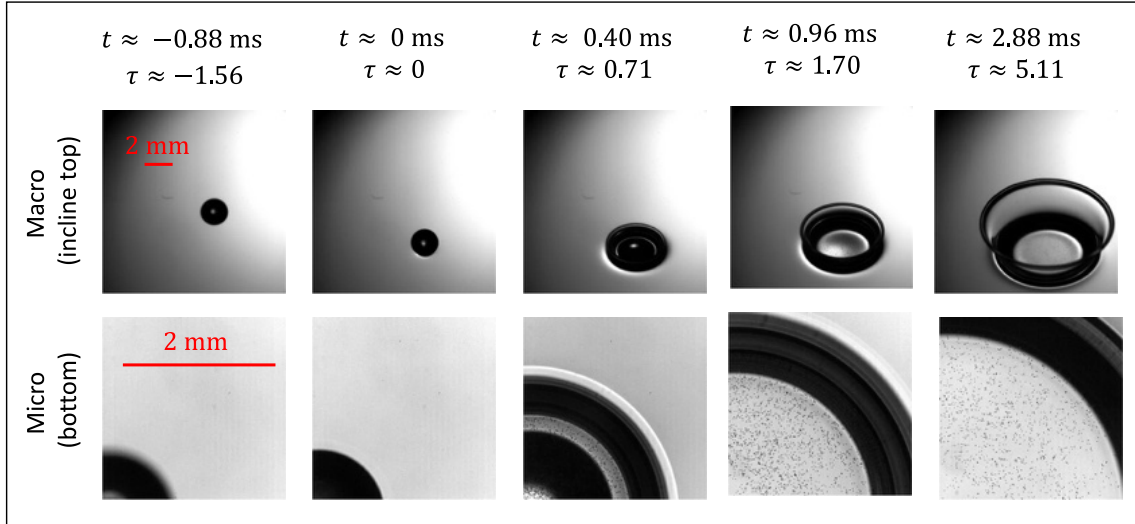


Figure 1: Temporal sequence of the droplet impact on a thin film. The macroscopic images show the process from an incline top view, while the microscopic [4]. The small black dots in the shadow images represent the tracer particles used for tracking the droplet's motion.

Experimental Setup and Imaging Technique

Microliter-sized droplets were released vertically onto the film, and the impact sequence was recorded simultaneously from a microscopic bottom view and an inclined top view. This setup provided crucial data for analyzing the dynamics of the droplet impact. Figure 1 from [4] shows selected frames of the recordings, just before and shortly after impact, illustrating the detailed motion of the droplet as it interacts with the thin liquid film.

Methodological Background

This work builds on the feasibility study by Stefan Schubert, who applied the DefocusTracker open-source toolbox [7], based on Generalized Defocusing Particle Tracking (GDPT), to explore the defocusing principle for droplet research. While the toolbox demonstrated feasibility, it offered only limited accuracy in depth reconstruction.

Development of SDμPTV

Building on this, the Shadow imaging based Defocusing Micro Particle Tracking Velocimetry (SDμPTV) was developed and applied in collaboration with Thomas Fuchs from the University of the Bundeswehr Munich. In these experiments, micro-particles are added to the droplet, under the assumption that they faithfully follow the fluid motion and thus represent the flow. To justify this assumption, the particle–fluid coupling is quantified using the Stokes number $St = t_p/t_f$, here defined conservatively as the ratio between the particle response time and a characteristic flow time scale. Using $t_p = \rho_p d_p^2 / (18\mu_f)$ and $t_f = L/U$. A small Stokes number indicates that particles rapidly adapt to fluid accelerations. According to [8], particles with $St \ll 1$ behave essentially inertia-free and follow the flow without measurable lag, while [9] recommend $St < 0.1$ as a robust general criterion.

The SDμPTV method exploits the variation in particle image diameter on the camera sensor as a function of the distance from the focal plane, whereby the particle image is smallest and exhibits the strongest intensity drop in the focal plane. Since particles appear darker than the background in shadowgraphy, these intensity valleys can be reliably detected and directly used to determine the particle depth from the particle image diameter. Before diameter determination, a simple two-step preprocessing is applied. First, a thresholding based on the median image intensity preserves only the dark particle images (and a few noise pixels). The particle edge is then determined sub-pixel accurately using a Thin-Plate Spline interpolation of the local intensity field. The edge is defined by crossing a specified relative intensity threshold, from which the particle image diameter is calculated. Through a prior calibration, the diameter is mapped unambiguously to the z -position. Combined with conventional μPTV analysis of the x/y positions, this yields the full three-dimensional SDμPTV method. The measured particle positions are transformed into cylindrical coordinates, exploiting the rotational symmetry of the droplet impact, and are further processed into velocity fields using a time-resolved nearest-neighbor tracking algorithm with trajectory prediction.

Current status and achieved results

The current status of the doctoral project demonstrates that the Shadowgraph Defocusing Micro-Particle-Tracking Velocimetry (SD μ PTV) can successfully capture the three-dimensional crown crater droplet flow after a droplet impacting on a thin liquid film. Using SD μ PTV, detailed radial velocity profiles of the droplet as functions of radius, height, and time could be obtained. For the first time, the flow in 3D within the droplet in the crown crater region was recently resolved and published, with velocity profiles recorded as a function of height z at spatially and temporally discrete points (Figure 2 from [4]). The significance of this achievement was underscored by the fact that the study was honored with the Best Paper Award at ILASS Lund in 2025 in the category Atomization & Drop Impacts.

Measurements of the film flow have the potential to further extend these profiles and provide a more comprehensive understanding of the overall flow. Preliminary investigations of the film flow have already been conducted. These studies indicated that the particles follow the flow accurately in viscous media, thereby providing a reliable representation of the fluid motion. In less viscous media, however, the particles were only able to partially represent the flow. This behavior can be attributed to the Stokes number, which increases when the fluid viscosity decreases or when the particle size is comparatively large. As a result, the tracer particles no longer respond quickly enough to local fluid accelerations and therefore fail to fully satisfy the common criteria. To address these limitations, the method is currently being further developed for the use of smaller tracer particles.

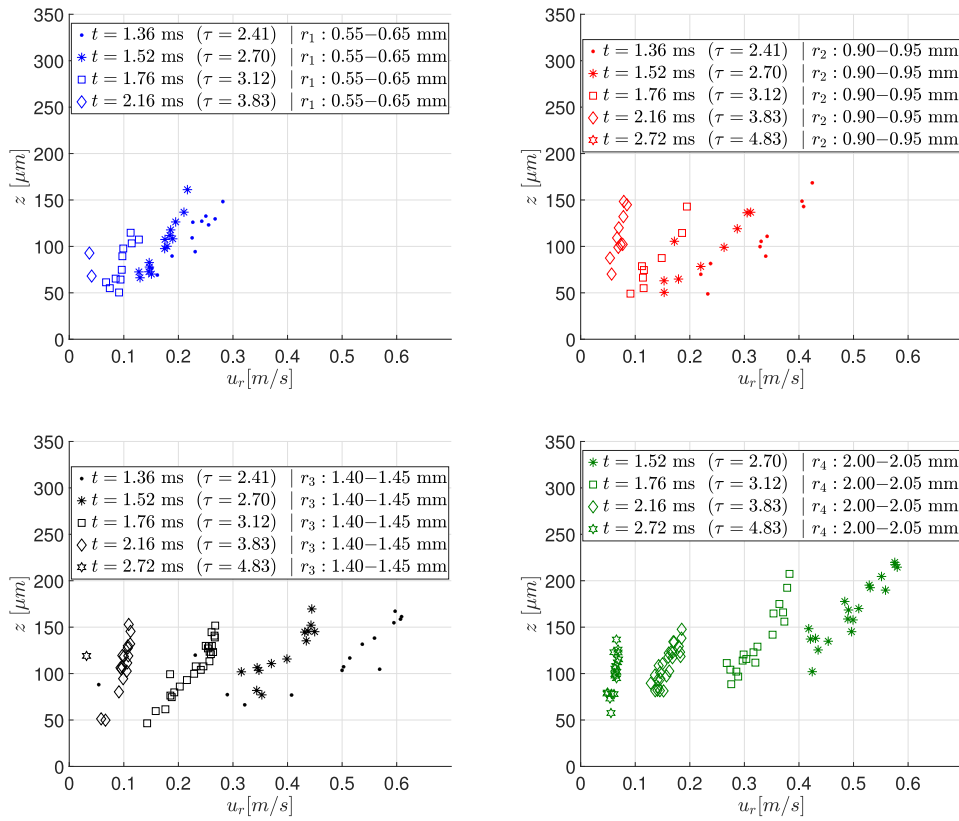


Figure 2: Velocity profiles of the droplet crown crater region as a function of height z , at spatially and temporally discrete points. [4]

Future goals

To improve the accuracy of the film flow measurements, the SD μ PTV method will be further developed to allow the use of significantly smaller tracer particles. This enables a more precise mapping of the local flow fields, particularly in low-viscosity fluids.

For future work, the focus will be on a quantitative analysis of the relative importance of inertial forces, the decreasing pressure at the stagnation point, and viscous dissipation mechanisms throughout the spreading process. By capturing both the droplet and the film lamella using SD μ PTV, physical insights can be gained that will contribute to the development of analytical models.

References

[1] A. L. Yarin, "Drop impact dynamics: splashing, spreading, receding, bouncing," *Annual Review Fluid Mechanics* 38, 159–192 (2006).

- [2] A. L. Yarin and D. A. Weiss, "Impact of drops on solid surfaces: self-similar capillary waves, and splashing as a new type of kinematic discontinuity," *Journal of Fluid Mechanics* 283, 141–173 (1995).
- [3] S. Schubert, J. Steigerwald, A. K. Geppert, B. Weigand, and G. Lamanna, "Micro-piv study on the influence of viscosity on the dynamics of droplet impact onto a thin film," *Experiments in Fluids* 65, 1–23 (2024).
- [4] P. Molina Vogelsang, T. Fuchs, G. Lamanna, "Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film", ILASS Europe 2025, Lund, Schweden, 31.08- 04.09.2025.
- [5] T. Fuchs, C. Burkhardt, and C. Kähler, "Defocusing PTV with Increased Measurement Depth Using Shadowgraphy," in 20th International Symposium on the Application of Laser and Imaging Techniques to Fluid Mechanics, Lisbon, Portugal, 11-14 July. Lisbon, (2022).
- [6] N. Erkan, "Full-field spreading velocity measurement inside droplets impinging on a dry solidheated surface." *Experiments in Fluids* 60, 88 (2019).
- [7] R. Barnkob, and M. Rossi. DefocusTracker: A Modular Toolbox for Defocusing-based, Single-Camera, 3D Particle Tracking. *Journal of Open Research Software*. 9(1). (2021). doi: <http://doi.org/10.5334/jors.351>
- [8] C. Tropea, A. Yarin, and J.F. Foss, Eds., *Springer Handbook of Experimental Fluid Mechanics*, Springer Handbooks, Springer Berlin Heidelberg, Berlin, Heidelberg, 2007, 22.
- [9] Raffel, M., Willert, C.E., Scarano, F., Kähler, C.J., Wereley, S.T., and Kompenhans, J., *Particle Image Velocimetry: A Practical Guide*, 3rd Edition, Springer International Publishing, Cham, 2018, description based on publisher supplied metadata and other sources. 22.

3. **Comments on the qualification programme and supervision strategy:**

During my doctoral studies, I expanded both my technical and academic skills through participation in a wide range of scientific events and workshops. I presented my research at DIPSI 2023 and 2025 in Bergamo, the DFG-supported Workshop 2024 "Future Perspectives on Defocus Particle Tracking" in Lisbon, and the DROPIT Final Colloquium 2025, where I also served as a presenter. At the University of Bergamo, I attended the Doctoral Seminar 2024, the workshop "Experimental and Numerical Methods in Science," and DROPIT meets Industry 2023. In Germany, I participated in the Doctoral Seminars 2023 and 2025 at the University of Stuttgart, where I contributed as a member of the organizing team for the 2025 seminar and the DROPIT Final Colloquium, assisting in the planning and execution of these events.

I also attended interdisciplinary workshops such as "AI Meets Scientific Computing in Bergamo 2025" and "Analytical and Numerical Methods in Science 2025," which broadened my methodological perspective and strengthened the integration of experimental and data-driven approaches. Complementing these technical activities, I completed soft skills training, including Professional Presentation Skills for the Academic Context, Time and Self-Management at the Graduate Academy GRADUS, and Scientific Writing offered by CEO • Leadership Sculptor, enhancing my transferable competencies.

My doctoral supervision was provided by Dr.-Ing. Grazia Lamanna (ITLR, Stuttgart, Germany) and Dr.-Ing. Simona Tonini (Department of Engineering, Bergamo, Italy). Additionally, a collaboration with Dr.-Ing. Thomas Fuchs (University of the Bundeswehr Munich and ITLR, Stuttgart) was established during the DFG Workshop 2024 in Lisbon, which has been instrumental in advancing the project and achieving key results. The collaboration also proved highly successful at the • ILASS Europe 2025 conference, where our work was recognized with the Best Paper Award in the category Atomization & Drop Impacts.

Overall, this structured qualification programme and supervision strategy have effectively supported the progress of my doctoral research, while also fostering my professional development and international research experience.

Participation in the following courses/seminars of the Int. Research Training Group: • DFG-supported Workshop 2024, Lisbon, Portugal – Presentation: “Investigation of the 3D micro-fluid dynamics within a crown crater formed by the impact of a single droplet on a thin wall film using the GDPT method” • DROPIT meets Industry Workshop 2023, University of Bergamo, Italy • “Optical Methods in Spray Research”, A. Coghe, Bergamo, Italy • “Fundamentals of X-ray Experimental Techniques”, M. Santini, Bergamo, Italy • “Analytical and Numerical Methods in Science 2025”, Bergamo, Italy • “AI Meets Scientific Computing in Bergamo 2025”, Bergamo, Italy • Professional Presentation Skills for the Academic Context • Time and Self-Management • Scientific Writing
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • Doctoral Seminar on 01.-02.03.2023 in Stuttgart, Germany • Doctoral Seminar on 08.-09.02.2024 in Bergamo, Italy • Doctoral Seminar on 28.-29.01.2025 in Stuttgart, Germany • DROPIT Final Colloquium, 31.07-01.08.2025 in Stuttgart, Germany
Participation in conferences, congresses, etc., at home and abroad: • DIPSI Workshop 2023, June 16, 2023, Bergamo, Italy • DIPSI Workshop 2024, June 17, 2024, Bergamo, Italy • DIPSI Workshop 2025, June 13, 2025, Bergamo, Italy • ILASS Europe 2025, 31st Aug. - 4th Sept, Lund, Sweden

4. Individual publications:

I. Peer-reviewed articles:

- P. Molina Vogelsang, T. Fuchs, G. Lamanna. Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film. ILASS Europe 2025, 33rd Conference on Liquid Atomization & Spray Systems, 31st Aug. - 4th Sept. 2025, Lund, Sweden
-

II. Further publications

a) Proceedings

- Molina Vogelsang, P., Fuchs, T., Lamanna, G. (2025). Droplet splashing measurements using shadowgraph defocusing. In Experiments in Fluids: Collection (Still in progress)
- Invitation to submit an Atomization and Sprays paper - Gold Open Access offered (due to best Paper award at ILASS 2025 (First Draft is written))
- P. Molina Vogelsang, T. Fuchs, G. Lamanna. Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film. ILASS Europe 2025, 33rd Conference on Liquid Atomization & Spray Systems, 31st Aug. - 4th Sept. 2025, Lund, Sweden
- P. Molina Vogelsang, T. Fuchs, G. Lamanna. Shadowgraph defocusing μ PTV measurements of an impacting droplet onto a thin liquid film. ISPIV 2025, 16th International Symposium on Particle Image Velocimetry, June 26-28. 2025, Tokyo, Japan
- Molina Vogelsang, P., Geppert, A., Lamanna, G. (2024). SP-C1/C5: Development of a novel optical techniques for investigating 3D-micro-fluid dynamics within crown crater induced by droplet impact onto thin wall-films. ICLASS 2024 (p. 3-5)
- Molina Vogelsang, P., Schubert, S., Lamanna, G. (2023). An approach to investigate the 3D height evolution in the crown crater during early crown formation with the General Defocusing Particle Tracking (GDPT) method. DIPSI WORKSHOP 2023 (p. 18-21). DOI: 10.13122/DIPSI2023

III. Presentations

a) Oral

- ILASS Europe 2025 31st Aug. - 4th Sept. 2025, Lund, Sweden | Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film
- DROPIT Final Colloquium 2025 | Stuttgart, Germany. Comparison of micro-PIV and defocused μ PTV measurements to analyze the influence of viscosity on the dynamics of droplet impact on a thin film
- Workshop 2024 | Lisbon, Portugal. Investigation of the 3D microfluid dynamics within a crown crater formed by the impact of a single droplet on a thin wall film using the GDPT method
- Workshop „Experimental and Numerical Methods in Science“ 2024 | University of Bergamo, Italy
- Molina Vogelsang, P., Geppert, A., Lamanna, G. (2024). SP-C1/C5: Development of a novel optical techniques for investigating 3D-micro-fluid dynamics within crown crater induced by droplet impact onto thin wall-films. ICLASS 2024 (p. 3-5)

b) Posters

- Molina Vogelsang, P., Schubert, S., Lamanna, G.: An approach to investigate the 3D height evolution in the crown crater during early crown formation with the General Defocusing Particle Tracking (GDPT) method. DIPSI Workshop 2023, Bergamo, Italy, 2025
- Molina Vogelsang, P., Fuchs, T., Lamanna, G.: Defocused shadowgraphy μ PTV (SD μ PTV) methodology for analysing the droplet spreading mechanism on wetted walls. DROPIT Final Colloquium 2025, Stuttgart, Germany, 2025
- Molina Vogelsang, P., Fuchs, T., Lamanna, G.: Defocused shadowgraphy μ PTV methodology for analyzing the spreading mechanism of an impacting droplet onto a thin liquid film ILASS Europe 2025 31st Aug. - 4th Sept. 2025, Lund, Sweden

Report of doctoral researcher (SP-B5-3)

Project number: SP-B5-3

First and last name of doctoral researcher: Rishav Saha

Title of doctoral project: Direct Numerical Simulation of droplet impact onto heated surfaces

Name of supervisors: Prof. Dr.-Ing. Bernhard Weigand, Prof. Dr.-Ing. Michael Dumbser

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT from 15.01.2023 – 31.12.2025

Funding (type and duration):

PhD researcher, ITLR, Funding: DROPIT (15.01.2023 - 31.12.2025), since 01.01.2026 – funds of the University of Stuttgart (ITLR)

Start of doctoral training and (anticipated) date of doctoral degree:

15.01.2023 and 31.12.2026

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

06/2014 Mechanical Engineering, Maulana Abul Kalam Azad University of Technology, Bachelor's degree, Dec 2018

10/2019 Chemical and Energy Engineering, OVGU Magdeburg, Master's degree, Nov 2022

Where applicable, please note any circumstances which could have affected your scientific work: -

Occupation after leaving the IRTG, if applicable:

2. Description of doctoral project and research results achieved to date:

This project aims to conduct a numerical study in drop impact dynamics and wetting behaviour on textured heated surfaces. The process encompasses various physical phenomena, such as droplet spreading, retraction, phase change, and heat transfer. This can be influenced by numerous factors, i.e. drop parameters, surrounding gas parameters and surface characteristics [1]. Understanding this process needs a clear idea about hydrodynamics and heat-mass transfer phenomena, and the coupling between them [2]. This research primarily analyses the heat transfer mechanism during the droplet wetting process through simulation results. Recent studies showed that the wetting area can be enhanced by introducing structured surfaces, which can contribute to the overall heat transfer rate from the solid surface to the liquid. Therefore, it is worth investigating the heat transfer mechanisms during a droplet impact onto structured heated surfaces. Therefore, it is very important to simplify the problem and gradually introduce complex physics into it.

This study employs a computational fluid dynamics (CFD) framework to simulate the droplet impact process. The Direct Numerical Simulations (DNS) are conducted using the in-house multiphase flow solver Free Surface 3D (FS3D) [3] running on a supercomputer of the High-Performance Computing Center Stuttgart (HLRS), HPE Apollo Hawk platform. The incompressible Navier-Stokes equations are solved using the Finite-Volume method, where the interface is captured using the Volume-of-Fluid (VOF) method [4], the reconstruction of their surface is also accomplished by the Piecewise Linear Interface Calculation (PLIC) scheme [5] in a Cartesian grid.

1st year

In this project, we first studied the hydrodynamics of the process and then introduced heat transfer effects into it. When a droplet impacts a surface with a high impact velocity, the droplet mostly splashes and forms secondary droplets, and some portion of the liquid sticks on the surface. Whereas a droplet with a relatively lower impact velocity spreads on the surface without splashing. Thus, with more contact

area the heat removal process becomes more efficient for spreading than splashing. Therefore, the spreading process is studied for liquids such as water [6], ethanol [7], isopropanol [8], refrigerant perfluorohexane (FC-72) and 3M™ Novec™ 7000 Engineered Fluid. A detailed hydrodynamic study is conducted with focus on the spreading regime, and the simulations are performed till the droplet becomes stable after the impact. Figures 1 and 2 show simulation results of an isopropanol droplet impacting onto various micro-structured surfaces, highlighting droplet spreading and splashing behavior [8].

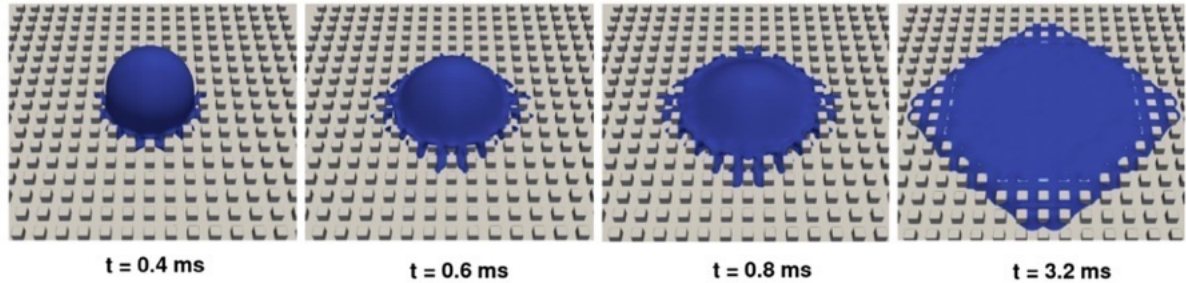


Figure 1: Spreading dynamics: A 2mm diameter isopropanol droplet impacts onto a super-hydrophilic micro-structured surface at 1.7m/s. The surface comprises a uniformly spaced array of cubic pillars (200µm size, 200µm spacing). DNS results are shown at different time instants t , with the droplet visualized in blue.

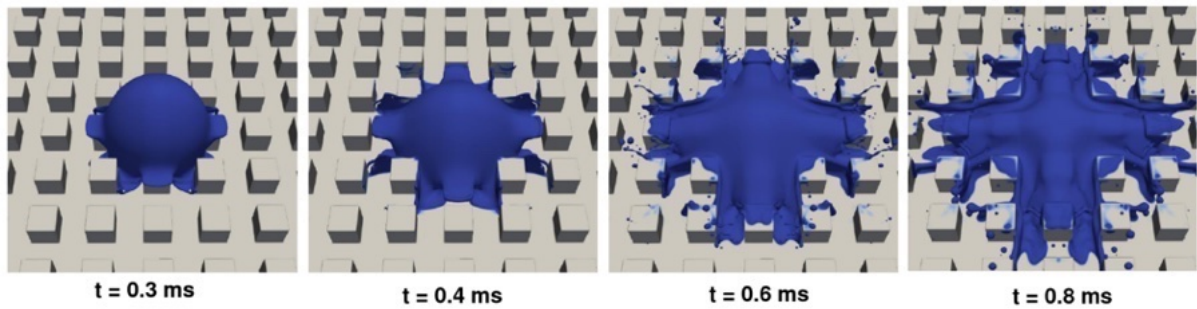


Figure 2: Splashing dynamics: A 2mm diameter isopropanol droplet impacts onto a super-hydrophilic micro-structured surface at 3.2m/s. The surface consists of a uniformly spaced array of cubic pillars (500µm size, 500µm spacing). DNS results are shown at different time instants t , with the droplet visualized in blue.

2nd year

A detailed investigation was conducted on the spreading phenomena of droplets with different Weber and Reynolds numbers impacting smooth surfaces. This study investigated water droplet impact onto smooth surfaces [6] with static contact angles of $CA = 0^\circ$, 90° , and 180° , validating the spreading factor against empirical correlations. Droplet spreading behaviour was analysed for Weber numbers ranging from 20 to 120, beyond which splashing might occur [9]. Dynamic wetting and the evolving contact area [10], were also examined. It was also concluded that contact angle dynamics becomes a critical factor for predicting the spreading factor, for droplets impacting at lower Weber numbers. Entrapped air bubbles, consistent with experimental observations, were present during the impact, but their dynamics cannot be simulated alongside the droplet due to the large difference in length scales.

Subsequently, the surface was replaced with a heated surface, where the solid material was assumed to have very high thermal conductivity (i.e., copper). The CFD model was extended with the energy equation, using a Dirichlet boundary condition (isothermal surface) for a smooth, highly conducting surface maintained at a constant temperature of 363K, which is below the boiling point of water. The results indicate a maximum temperature rise of 5K in the liquid during spreading, which is insufficient to significantly alter the liquid's material properties, such as viscosity, density, specific heat, or thermal conductivity [10,11]. Furthermore, the characteristic timescales for heat conduction and evaporation are much longer than the droplet spreading timescale [10,11]. This highlights that droplet spreading, heat conduction, and evaporation can occur on distinctly different temporal scales. Therefore, to study the coupled heat transfer and hydrodynamic mechanisms, the relevant process timescales must overlap to influence one another. A more suitable liquid with low specific heat can be used to promote this timescale overlap and have both hydrodynamic and heat transfer effect together.

During my two months visit to the partner university of Bergamo, I collaborated with Prof. Cossali and Prof. Tonini to investigate this problem. We simplified the process into two separate problems, refer to figure 3. Firstly, focussing on the heat transfer between the solid and the liquid interface (case I). And secondly, focussing on the liquid gas interface during spreading, where both heat and mass transfer effects are seen (case II). However, these two problems require separate liquids to study in detail the mechanisms involved. Then a detailed liquid selection study was conducted. Studies showed that FC-72 droplets are suitable for studying the heat transfer effects during spreading [12,13]. Therefore, for the first case, FC-72 droplets were selected, while for the second case, a highly volatile liquid, 3M™ Novec™ 7000 Engineered Fluid, was chosen.

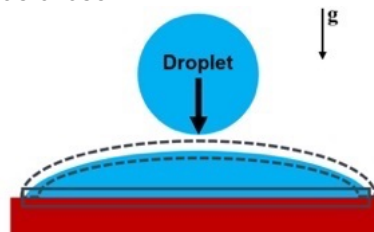


Figure 3: Solid line indicates the focus area for Case I, where heat transfer is the dominant mechanism. Dashed line indicates the focus area for Case II, where both heat and mass transfer play a significant role.

3rd year

The impact of an FC-72 (refrigerant perfluorohexane) droplet onto a smooth heated surface was investigated using calcium fluoride glass as the solid substrate, to study the solid-liquid heat transfer. The liquid and the surrounding gas were initially at ambient temperature, while the solid was maintained at 13K above the saturation temperature of liquid FC-72 ($T_{\text{sat, FC-72}} = 329\text{K}$). The simulations were performed for a Weber number of 14 and a Reynolds number of 955 on a superhydrophobic surface ($\text{CA} = 180^\circ$). A mesh resolution of 4.1 million cells, corresponding to approximately 60 cells per droplet diameter, was employed. In Figure 4, DNS results show the temperature profiles at 2ms and 4ms after impact. Cooling of the solid surface was observed as heat was transferred to the spreading droplet, giving the opportunity to understand the cooling mechanism.

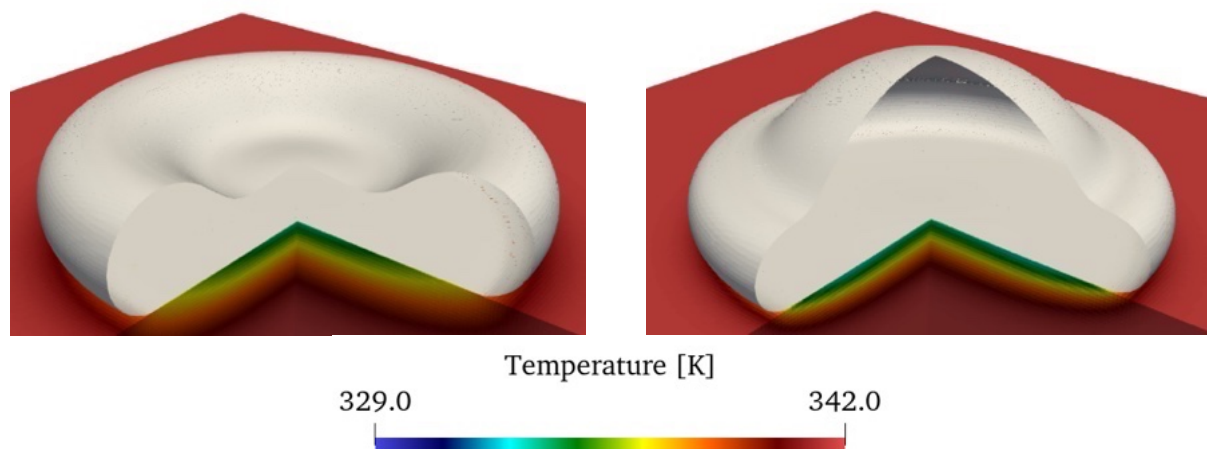


Figure 4: Temperature distribution during the impact of an FC-72 droplet onto a smooth heated calcium fluoride surface at 2ms (left) and 4ms (right) after contact. The substrate is maintained 13K above the liquid saturation temperature ($T_{\text{sat, FC-72}} = 329\text{K}$), with $We = 14$, $Re = 955$, and a superhydrophobic surface condition ($\text{CA}=180^\circ$).

We then introduced a heated micro-pillared surface to modify the wetted area, thereby enhancing the heat transfer rate. The smooth surface was replaced with an array of uniformly spaced (spacing of $200\mu\text{m}$) cubic micro-pillars, each with a characteristic size of $200\mu\text{m}$. Figure 5 presents preliminary DNS results showing the substrate temperature distribution during impact at different time instants t .

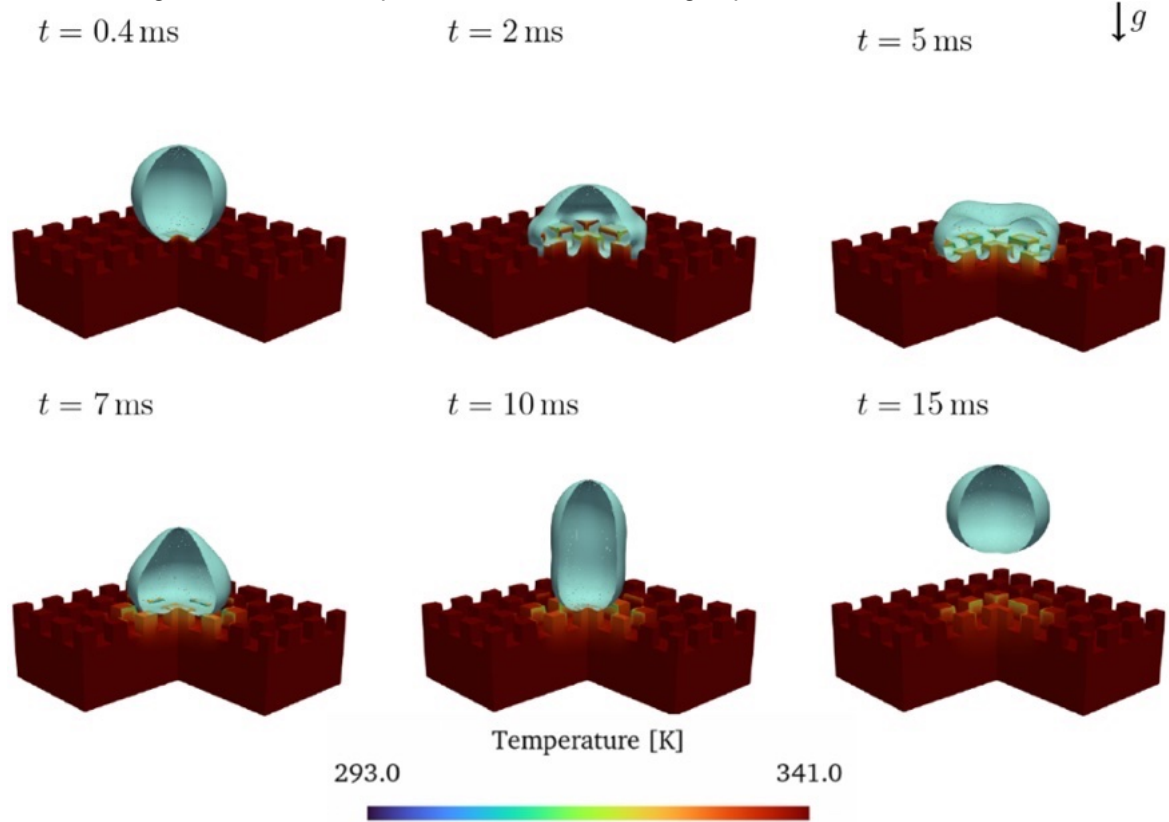


Figure 5: Temperature distribution of the solid substrate from the DNS simulation of an FC-72 droplet impact ($We = 14$) onto a heated calcium fluoride (CaF_2) structured surface ($T_{\text{solid}} = 341\text{K}$), shown at different time instants t . The surface features $200\mu\text{m}$ cubic pillars and the gas-liquid interface is shown in cyan.

The simulations show that the micro-pillars significantly alter both the droplet hydrodynamics and the heat transfer process. The resulting change in contact area has an influence on heat transfer, leading to a difference in substrate cooling. These combined effects highlight the importance of surface structuring in controlling both fluid dynamics and heat transfer during the droplet impact process.

For the second case, we will investigate the coupled heat and mass transfer phenomena from a wetting perspective, with a focus on how variations in the wetting behavior influence the overall heat transfer performance. Particular attention will be given to the evaporation-driven effects and thermal Marangoni flow, which are expected to strongly influence the spreading dynamics. In this context, the droplet (3M™ Novec™ 7000 Engineered Fluid) and substrate will be initialized at the same temperature to neglect the heat transfer effects between the liquid and the substrate. Then we can separately focus on the liquid-gas interface during the spreading and recoiling phase.

References

- [1] C. Josserand and S. T. Thoroddsen. "Drop impact on a solid surface". *Annual Review of Fluid Mechanics* 48.1 (2016), pp. 365–391.
- [2] G. Liang and I. Mudawar. "Review of drop impact on heated walls". *International Journal of Heat and Mass Transfer* 106 (2017), pp. 103–126.
- [3] K. Eisenschmidt, M. Ertl, H. Gomma, C. Kieffer-Roth, C. Meister, P. Rauschenberger, M. Reitzle, K. Schlottke and B. Weigand. "Direct numerical simulations for multiphase flows: An overview of the multiphase code FS3D". *Applied Mathematics and Computation* 272 (2016), pp. 508–517.
- [4] C. W. Hirt and B. D. Nichols. "Volume of fluid (VOF) method for the dynamics of free boundaries". *Journal of Computational Physics* 39.1 (1981), pp. 201–225.
- [5] W. J. Rider and D. B. Kothe. "Reconstructing volume tracking". *Journal of Computational Physics* 141.2 (1998), pp. 112–152.

- [6] R. Saha, C. Inal, M. Kumar, and B. Weigand. "DNS of droplet impact onto smooth and micro-structured surfaces". *ILASS Europe, 33rd Conference on Liquid Atomization and Spray Systems*, 31 August–4 September 2025, Lund, Sweden.
- [7] R. Saha, C. Inal, and B. Weigand. "DNS of ethanol droplet impact onto a smooth surface". Extended abstract. *Droplet Impact Phenomena & Spray Investigation (DIPSI) Workshop*, 13 June 2025, Bergamo, Italy.
- [8] R. Saha and B. Weigand. "Numerical simulation of droplet impact onto structured surfaces". Extended abstract. *Droplet Impact Phenomena & Spray Investigation (DIPSI) Workshop*, 16 June 2023, Bergamo, Italy.
- [9] P. Foltyn, D. Ribeiro, A. Silva, G. Lamanna, and B. Weigand. "Influence of wetting behavior on the morphology of droplet impacts onto dry-patterned micro-structured surfaces". *Physics of Fluids* 34.12 (2022).
- [10] M. Kumar, R. Saha, J. Potyka, K. Schulte, and B. Weigand. "Direct Numerical Simulations of Droplet Impact onto Heated Surfaces using the Program Free Surface 3D (FS3D)". *HLRS Annual Report 2024*, Stuttgart, Germany.
- [11] R. Saha and B. Weigand. "Numerical simulation of droplet impact onto heated surfaces below the boiling point". *Journal of Physics: Conference Series* 2766.1 (2024), 012045.
- [12] S. Batzdorf. "Heat transfer and evaporation during single drop impingement onto a superheated wall". PhD thesis, Technische Universität Darmstadt, 2015.
- [13] S. [Henrik](#). "Numerical investigation of drop impingement onto superheated walls". PhD thesis, Technische Universität Darmstadt, 2025.

3. Comments on the qualification programme and supervision strategy:

During my doctoral studies, students were encouraged to participate in the planning of specialized courses by inviting visiting scientists, so that the events met our research needs. I attended the Drops, Jets and Films seminar organised by CISM in Udine, Italy, May 2024, which along with my participation in DROPIT events and other courses, significantly enhanced my technical expertise. Soft skills courses also contributed to my development and were highly valuable. I am especially grateful for the opportunity to spend an extended period at the partner university in Bergamo, where I gained numerous scientific benefits. Although the two-month stay was not sufficient to complete all planned work, ongoing collaboration through mutual visits has established a strong foundation for international cooperation. I was involved in the organization of the DROPIT Final colloquium 2025 in Stuttgart.

Participation in the following courses/seminars of the Int. Research Training Group:

- Fortran for Scientific Computing (2023/FTN1), HLRS Stuttgart, February 13 - 17, 2023, Stuttgart, Germany
- Course of Prof. Coghe and Prof. Santini about measurement techniques, 14 June 2023, Bergamo, Italy
- DROPIT Meets Industry Workshop 2023, 15 June 2023, Bergamo, Italy
- Scientific Writing Skills for Doctoral Students in Aerospace Engineering, 4 March 2024, Stuttgart, Germany
- Drops, Jets and Films Seminar, CISM, May 6-10, 2024, Udine, Italy
- Experimental and Numerical Methods in Science, 18 June 2024, Bergamo, Italy
- AI meets Scientific Computing, 11 June 2025, Bergamo, Italy
- Analytical and Numerical Methods in Science, 12 June 2025, Bergamo, Italy

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: Research stay at the University of Bergamo from 15.06.2024 to 15.08.2024 (2 months) and from 10.11.2025 to 30.11.2025 (20 days).

Participation in the following International Research Training Group events:

- Doctoral Seminar on 01.-02.03.2023 in Stuttgart, Germany
- Doctoral Seminar on 08.-09.02.2024 in Bergamo, Italy
- Doctoral Seminar on 28.-29.01.2025 in Stuttgart, Germany
- DROPIT Final Colloquium, 31.07-01.08.2025 in Stuttgart, Germany

Participation in conferences, congresses, etc., at home and abroad:

- DIPSI Workshop 2023, June 16, 2023, Bergamo, Italy
- European Thermal Sciences Conference, June 10-13, 2024, Lake Bled, Slovenia
- DIPSI Workshop 2024, June 17, 2024, Bergamo, Italy
- DIPSI Workshop 2025, June 13, 2025, Bergamo, Italy
- ILASS Europe 2025, 31st Aug. - 4th Sept, Lund, Sweden

4. Individual publications:

I Peer-reviewed articles:

- P. Palmetshofer, **R. Saha**, J. Bryce, A. K. Geppert, G. E. Cossali, and B. Weigand. "Droplet impact onto sub-millimeter microstructures fully and barely submerged in a liquid film". *Physics of Fluids* 37.8 (2025).

II Further publications

a) Proceedings

- **R. Saha** and B. Weigand. "Numerical simulation of droplet impact onto heated surfaces below the boiling point". *Journal of Physics: Conference Series* 2766.1 (2024), 012045.
- **R. Saha**, C. Inal, M. Kumar, and B. Weigand. "DNS of droplet impact onto smooth and micro-structured surfaces". *ILASS Europe, 33rd Conference on Liquid Atomization and Spray Systems*, 31 August–4 September 2025, Lund, Sweden.

b) Technical and scientific reports

- **R. Saha** and B. Weigand. "Numerical simulation of droplet impact onto structured surfaces". Extended abstract. *Droplet Impact Phenomena & Spray Investigation (DIPSI) Workshop*, 16 June 2023, Bergamo, Italy.
- M. Kumar, **R. Saha**, J. Potyka, K. Schulte, and B. Weigand. "Direct Numerical Simulations of Droplet Impact onto Heated Surfaces using the Program Free Surface 3D (FS3D)". *HLRS Annual Report 2024*, Stuttgart, Germany.
- **R. Saha** and B. Weigand. "DNS of droplet impact onto heated surfaces". Extended abstract. *Droplet Impact Phenomena & Spray Investigation (DIPSI) Workshop*, 17 June 2024, Bergamo, Italy.
- **R. Saha**, C. Inal, and B. Weigand. "DNS of ethanol droplet impact onto a smooth surface". Extended abstract. *Droplet Impact Phenomena & Spray Investigation (DIPSI) Workshop*, 13 June 2025, Bergamo, Italy.
- **R. Saha**, M. Dumbser, and B. Weigand. "Numerical Simulation of Droplet Impact onto Textured Heated Surface". Midterm Review Report, 2 October 2025, Institute of Aerospace Thermodynamics (ITLR), University of Stuttgart, Germany.

III Presentations

a) Oral

- "Numerical simulation of droplet impact onto heated surfaces below the boiling point". At European Thermal Sciences Conference (Eurotherm), 10–13 June 2024, Lake Bled, Slovenia.
- "Numerical Simulation of Droplet–Wall Interaction Using the Volume-of-Fluid Method". At Analytical and Numerical Methods in Science Workshop, 12 June 2025, Bergamo, Italy.
- "Numerical simulation for drop impact on macro-scaled structures". At DROPIT Final Colloquium 2025, Stuttgart, Germany (combined talk with Martina Baggio).
- "DNS of droplet impact onto smooth and micro-structured surfaces". At ILASS Europe, 33rd Conference on Liquid Atomization and Spray Systems, 31 August–4 September 2025, Lund, Sweden.
- "Numerical Simulation of Droplet Impact onto Textured Heated Surface". At Multiphase Seminar, 23 October 2025, Institute of Aerospace Thermodynamics (ITLR), University of Stuttgart, Germany.

b) Posters

- **R. Saha** and B. Weigand. "Numerical simulation of droplet impact on structured surfaces". At DIPSI Workshop 2023, Bergamo, Italy.
- **R. Saha** and B. Weigand. "DNS of droplet impact onto heated surfaces". At DIPSI Workshop 2024, Bergamo, Italy.

- **R. Saha**, C. Inal and B. Weigand. “DNS of ethanol droplet impact”. At DIPSI Workshop 2025, Bergamo, Italy.
- **R. Saha**, M. Baggio, M. Dumbser and B. Weigand. “Numerical computation of drop impact on textured surface”. At DROPIT Final Colloquium 2025, Stuttgart, Germany.

Report of doctoral researcher (SP-A6-3)

Project number: SP-A6-3

First and last name of doctoral researcher: Amalia Travnicek

(Working) title of doctoral project:

Multi-scale modeling of compressible droplet clustering for Lagrangian models

Name of supervisors: Prof. Dr.-Ing. Andrea Beck, Prof. Dr.-Ing. Stefanos Fasoulas, Prof. Dr. Gianpietro Elvio Cossali

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT since April 2023

Funding (type and duration):

PhD researcher at the Institute of Aerodynamics and Gasdynamics, University of Stuttgart, April 2023 - December 2025, GRK 2150/2,
January 2026 - March 2026 basic funds, University of Stuttgart

Start of doctoral training and (anticipated) date of doctoral degree:

1 April 2023 - September 2027

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2016 Aerospace Engineering, University of Stuttgart, Bachelor's degree, 07 May 2020
04/2020 Aerospace Engineering, University of Stuttgart, Master's degree, 29 November 2022
04/2023 Aerospace Engineering, University of Stuttgart, PhD degree

Where applicable, please note any circumstances which could have affected your scientific work:

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

A unifying theme across all phases of the project was the progression from incompressible to compressible droplet dynamics, encompassing numerical schemes, efficiency and validation, as well as analysis and modeling. The first phase focused on simulating droplet dynamics in weakly compressible flows, introducing a novel implicit–explicit flux-splitting scheme. The second phase built upon this foundation in terms of validation and computational efficiency, laying the groundwork for massively parallel large-scale simulations of droplet grouping in compressible flows. Finally, the last phase concentrates on analysis and modeling, applying the complete framework to investigations of droplets in compressible turbulence. Here, two complementary research directions are pursued: first, the investigation of droplet–turbulence interactions in decaying compressible turbulence; and second, the development of a forcing algorithm to sustain compressible multiphase turbulence, thereby enabling systematic studies of compressibility effects in multiphase turbulence with statistically stationary data. These findings then represent a first step in the development of Lagrangian models for droplets in compressible turbulence, which can eventually pave the way for large scale simulations of sprays. The work of the last phase presented here is divided into the following packages.

Pre- and Postprocessing pipeline for compressible multiphase turbulence

The current framework had to be extended to handle compressible multiphase turbulence. First, a turbulence generator was integrated by developing a tool that constructs a velocity field from a prescribed kinetic energy spectrum and sets the pressure to obtain a target turbulent Mach number. In this config-

uration, the density is set constant, and pressure fluctuations are computed from the velocity fluctuations. This produces an initially incompressible velocity field with non-physical higher-order velocity moments, which progressively develop compressible characteristics according to the turbulent Mach number as the simulation evolves. Next, an additional tool was implemented to enable droplet insertion into arbitrary velocity fields. This capability allows droplets to be introduced into the compressible turbulence only after the energy cascade is fully developed, thereby minimizing disturbances to droplet dynamics arising from non-physical higher-order moments and incomplete cascade formation of the gaseous carrier flow. Combined with the interface viscosity approach introduced in Phase 2, the framework is thus capable of handling compressible multiphase turbulence.

For post-processing, a spherical harmonics (SH) framework was incorporated into the solver. Spherical harmonics form a complete set of orthonormal functions and serve as the spherical analogue of Fourier series, enabling the representation of functions on spherical surfaces in frequency space. Here, spherical harmonics are employed for two purposes:

1. Spectral decomposition of the droplet radius, describing deformation and oscillation
2. Analysis of interface force distributions, quantifying the forces exchanged between the droplet and the carrier flow.

These analyses enable a quantitative description of the mutual influence between the liquid and gaseous phases, accounting for droplet parameters and the turbulent Mach number. Combined with Fourier transforms of the whole flow field for a global spectral analysis, the reciprocal influence of the liquid and gaseous phases can be studied extensively in the pre-break-up regime. Compressibility effects are assessed by extracting the dilatational velocity component.

Direct numerical simulation of a droplet in decaying homogeneous isotropic turbulence

With the pre- and postprocessing pipeline in place, first investigations of singular droplets in decaying compressible homogeneous isotropic turbulence (HIT) were conducted using direct numerical simulations (DNS). The liquid phase is assigned thermodynamic properties distinct from those of the gas using appropriate equations of state, resulting in a droplet which is closer to realistic applications. By varying droplet density, viscosity, and surface tension, the influence of the Weber and Ohnesorge numbers on the droplet behavior was examined. Analyses included the droplet's kinetic energy, dissipation rate and surface area, and the carrier flow kinetic energy and dissipation rate. Figure 1 shows a droplet in decaying weakly compressible HIT at an initial turbulent Mach number $M_t=0.2$ and $Re_\lambda=50$, with $M_t=\sqrt{3}u_{rms}/a$ being the ratio of the turbulent velocity fluctuations to the speed of sound, and $Re_\lambda=\rho u_{rms}\lambda/\mu$ being the Taylor Reynolds number.

The temporal evolution of the carrier flow properties was found to be largely unaffected by variations in droplet parameters. In contrast, droplet surface area, kinetic energy, and dissipation rate increased with increasing Weber number and decreasing Ohnesorge number, consistent with the physical interpretation of these dimensionless groups. Analogously to the incompressible, purely solenoidal case, the dilatational kinetic energy spectrum exhibits larger kinetic energy at the smaller scales for the droplet-laden simulations compared to the droplet-free reference case.

Investigation of interfacial pressure forces in relation to the droplet radius revealed that larger deviations in radius correlated with higher interfacial pressure forces, again in agreement with the physical implications of the Weber and Ohnesorge numbers. These results represent an first step toward studying compressibility effects on droplet dynamics in compressible carrier flows, defining a relevant parameter space for studies of droplets in sustained compressible turbulence.

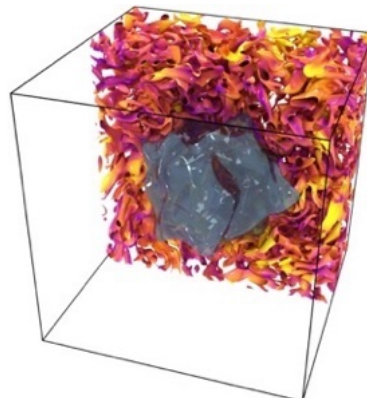


Figure 1: DNS of a droplet in decaying compressible HIT at $M_t = 0.2$ and $Re_\lambda = 50$.

Forcing of compressible multiphase turbulence

Maintaining a statistically stationary state of the compressible carrier flow is essential to thoroughly quantify the effects of compressible turbulence on droplet behavior. This can be achieved by employing a forcing algorithm to sustain the turbulence. Current efforts are focused on developing an algorithm based on spectral forcing for compressible multiphase turbulence, enabling rigorous studies of compressibility effects without the limitations introduced by the temporal decay of turbulence.

The developed forcing algorithm combines different approaches known from forcing of compressible and incompressible turbulence in physical space. The forcing term is computed in spectral space to account for both solenoidal and dilatational velocity components, and is then transformed back to physical space before being added to the momentum and energy equations. This procedure ensures the consideration of the non-zero dilatational motions of compressible turbulence to maintain a statistically stationary state, while also being applicable in non-spectral codes, and in this case, multiphase flows, due to the option of applying the forcing separately to each phase. By applying a relaxation approach to the temporal evolution equations of the dilatational and solenoidal kinetic energy, the respective velocity components can be controlled and a statistically stationary state can be sustained. Initial tests for a compressible single-phase case have shown promising results, with the temporal evolution of the solenoidal and dilatational kinetic energy shown in Figure 2. Ongoing work aims to develop an efficient parallel implementation and extend the approach to multiphase flows by applying the forcing term exclusively to the gaseous carrier phase, consistent with studies of multiphase turbulence with large density ratios between the phases.

Building on this, the next step is to investigate the behavior of a single droplet in sustained compressible homogeneous isotropic turbulence, focusing on its oscillation, deformation, and potential disintegration under turbulent conditions. These studies will be conducted through direct numerical simulations, in alignment with the objectives of the previous work package. This will guide the selection of relevant droplet properties and help characterize clustering behavior. The use of sustained, statistically stationary turbulence makes the resulting data especially well-suited for the construction of appropriate models applicable to Lagrangian frameworks. Additionally, the insights gained from the SH framework will be incorporated. Here, models for the exerted forces due to, e.g., drag, buoyancy, and the added-mass effect are readily available in the literature, whereas the impact of the local particle concentration and shape oscillations are still an open question. The proposed highly-resolved sharp interface simulations thus fill a gap in knowledge necessary to gain insights into compressible droplet grouping. This novel knowledge can progress Lagrangian surrogate models of droplet clustering and potentially provide an important building stone in enabling simulations of whole sprays in the near future.

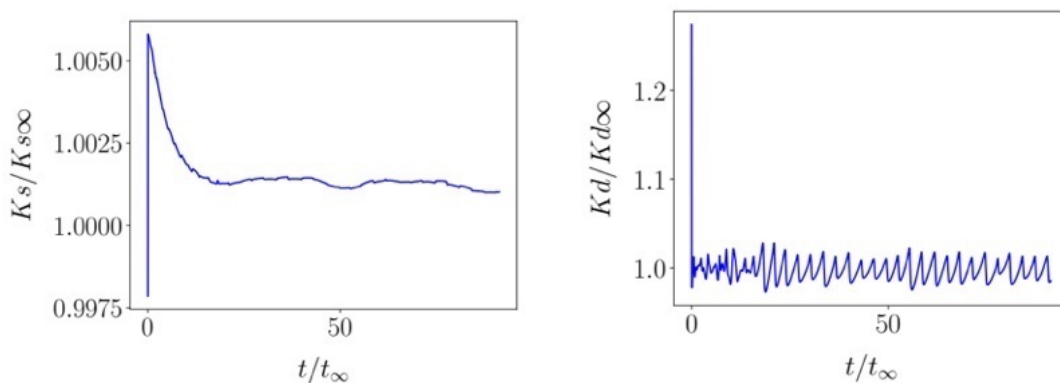


Figure 2: Temporal evolution of the solenoidal (left) and dilatational (right) kinetic energy, normalized with their respective target value, during the application of the developed forcing algorithm for single-phase compressible turbulence

3. Comments on the qualification programme and supervision strategy:

Through the IRTG, the qualification program shaped me as a researcher, both scientifically and professionally. I appreciated the interdisciplinary nature of DROPIT, which ignited fruitful discussions of droplet dynamics from different perspectives. The courses and regular meetings at the doctoral seminars and workshops provided opportunities to connect with fellow PhD researchers, each bringing a unique background, which helped me view my own research in a new light. Overall, the IRTG offered the invaluable opportunity to exchange knowledge about droplet phenomena in a broader

scope. In the beginning, I benefitted profoundly from being introduced to the topic and the project directly by my predecessor Daniel Appel and our colleague from the sub-project SP-A2 Pascal Mossier.

Furthermore, I was able to strengthen my soft skills, particularly in organization and time management. Coordinating schedules, structuring the program, and ensuring smooth execution of the Final Colloquium required both attention to detail and adaptability.

Participation in the following courses/seminars of the Int. Research Training Group: • Discontinuous Galerkin Methods (Prof. Beck), University of Stuttgart • Analytical and numerical methods (Prof. Weigand and Prof. Beck), University of Stuttgart • English for Space Engineering (Dr. Micheals), University of Stuttgart • Arbeitsmethoden, Zeitmanagement und Selbstorganisation (Schüpp-Niewa), University of Stuttgart
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: • EuroCC2 N-Ways to GPU Programming Bootcamp, 2024, High-Performance Computing Center Stuttgart, online • Hackathon: Porting and Optimization for Hunter, 2025, High-Performance Computing Center Stuttgart, Stuttgart, Germany
Participation in the following International Research Training Group events: • Doctoral Seminar, Bergamo, 8-9 February 2024 • Workshop on Experimental and Numerical Methods, Bergamo, 18 June 2024 • Final Colloquium, Stuttgart, 31 July – 1 August 2025
Participation in conferences, congresses, etc., at home and abroad: • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2023 • Workshop “Efficient Implementation of DG-Methods for the Solution of PDEs”, 2023, Waldachtal (Germany) • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (Italy), 2024 • European Fluid Dynamics Conference, Dublin (Ireland), 2025

4. Individual publications:

I. Peer-reviewed articles:

II. Submitted articles:

III. Further publications

a) Proceedings

- D. Appel, J. Keim, P. Mossier, P. Oestlinger, **A. Travnicsek**, A. Beck. *Recent Contributions to the High-Order Numerical Simulation of Multiphase and Multispecies Flows*. High Performance Computing in Science and Engineering '25, Springer - submitted

IV. Presentations

a) Oral

- **A. Travnicsek**. Numerical methods for compressible multi-phase simulations. DROPIT Workshop of Experimental and Numerical Methods, 07/2024, Bergamo

- **A. Travnicek**, P. Mossier, D. Appel, J. Zeifang, C. Müller. Multi-Phase, Multi-Scale, Multi-Physics: A Sharp-Interface Perspective on Compressible DROplet InteracTion Dynamics, DROPIT Final Colloquium, 07/2025, Stuttgart
- **A. Travnicek**, D. Appel, S. Jöns, A. Beck. Direct Numerical Simulation of a Droplet in Compressible Homogeneous Isotropic Turbulence. EFDC2, 08/2025, Dublin

b) Posters

- J. Zeifang, D. Appel, **A. Travnicek**, A. Beck. Multi-scale modeling of compressible droplet clustering for Lagrangian models. DROPIT Final Colloquium, 07/2025, Stuttgart

Report of doctoral researcher (SP-AX)

Project number: SP-AX

First and last name of doctoral researcher: Daniel Regener Roig

(Working) title of doctoral project: Development of an efficient high-fidelity simulation tool for multiphase flow simulations and application to the study of drop-gas interaction

Name of supervisors: Alessandro Colombo, Andrea Beck

1. General information:

Duration of International Research Training Group membership: November 2023 to December 2025
Funding (type and duration): Scholarship funded by the university, 3 years (November 2023 to October 2026)
Start of doctoral training and (anticipated) date of doctoral degree: Start at 1 st of November of 2023 and anticipated end at 31 st of October of 2026
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): Sep. 2015; Bachelor's degree in aerospace vehicle engineering; Universitat Politècnica de Catalunya; ended Feb. 2021. Feb. 2021; Master's degree in aeronautical engineering; Universitat Politècnica de Catalunya; ended Jun. 2023.
Where applicable, please note any circumstances which could have affected your scientific work:
Occupation after leaving the IRTG, if applicable:

2. Description of doctoral project and research results achieved to date:

The project investigates multiphase flows, specifically drop-gas interactions. Simulating compressible multiphase flows is crucial for engineering and scientific applications involving shock waves and material interfaces. To resolve resulting small-scale structures, we aim to develop a high-fidelity and efficient simulation tool utilizing CPU-GPU architectures. We also take account of machine learning techniques to mitigate possible spurious oscillations. We adopt a high-order interface-capturing method to preserve conservation and resolve complex interfaces efficiently.

We study the symmetric multi-component Euler (MCE) model [1]. While assuming equilibrium, the MCE can present numerical artefacts that can be tackled imposing Entropy Stability (ES) properties to the numerical scheme. In the first year, we addressed the high computational cost of ES in modal discontinuous Galerkin (DG) discretizations [2] by applying DEEB [3], [4] and entropy projection [5] strategies (extending [6]). Our approach is approximately 75% faster than other state-of-the-art strategies (Fig. 1) and its robustness is shown in the shock-helium bubble benchmark (Fig. 2).

For the second year, we adopted the stiffened gas (SG) EOS and added phase transition terms [7], without imposing the ES properties. This multiphase modelling can present even more complex dynamics like split rarefaction fans and steep gradients near liquid-gas interfaces, posing significant numerical challenges. To address negative apparent densities near these interfaces, we implemented an order-reducing limiter after finding pressure projection [8] insufficient. Fig. 3 shows a CO₂ pipe depressurization [9], stabilized using the limiter (κ -PPS) and a viscous shock capturing. Finally, Fig. 2 shows a multiphase explosion without separated species, simulated on a curved-edge mesh (Fig. 3) tailored to the initial discontinuity.

Future work involves accelerating the κ -PPS strategy via machine learning and enhancing time integration robustness. We will also replace equilibrium constraints with the Baer-Nunziato model, utilizing the entropy framework [10] and stabilization strategies developed during this project

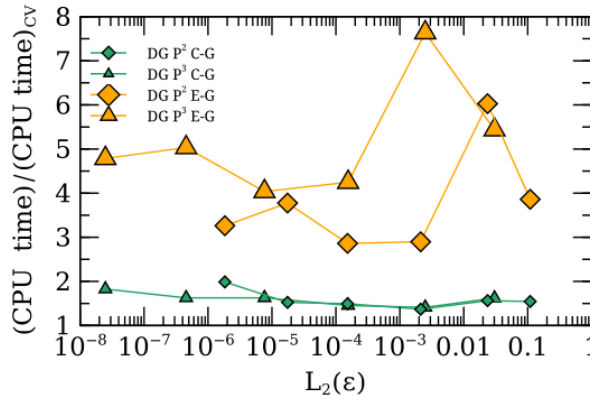


Fig. 1: Advection of a density wave [1] computational time vs. error comparison. The results of the P-G and E-G schemes are plotted against the fastest (but less robust) C-G strategy.

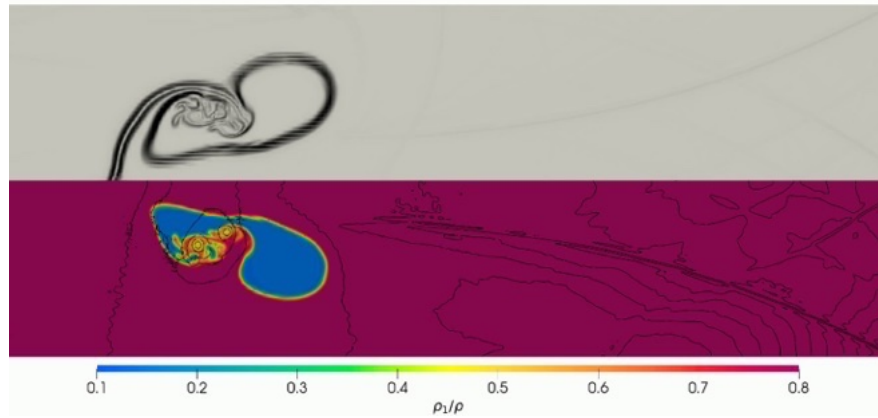


Fig. 2: Results of the shock-bubble interaction [1] for $t = 983 \mu s$. The spatial discretization is performed with a mesh consisting of 579×45 elements using a polynomial approximation $\mathbb{P}^3$. On top-half, pseudo-Shlieren contours. On bottom half, mass fraction of the first compo-

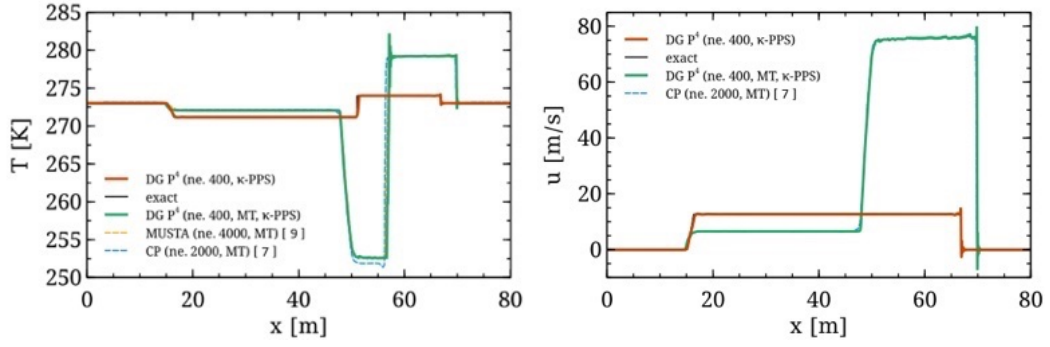
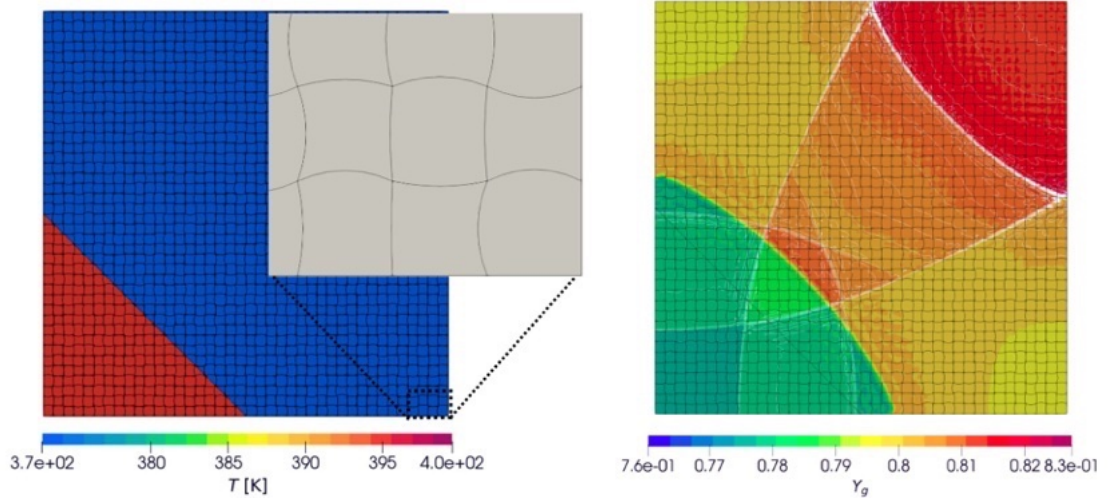


Fig. 3: Results of a depressurization of a CO_2 pipe without and with mass transfer. The lines “DG” are the ones obtained in the present thesis, where “MT” indicates that mass transfer was taken into



(a) Temperature at the initial condition of an explosion problem, simulated in a mesh composed of 1 620 elements with randomly curved parabolic edges. At the initial

(b) Results in the mass fraction of gas for the explosion problem. The explosion has provoked a large condensation zone in the bottom-

Fig. 4: Multi-phase explosion with phase transition, as a two-dimensional extension of the liquid-vapor shock tube problem of Ref. [7].

References

- [1] F. Renac, "Entropy stable DGSEM for nonlinear hyperbolic systems in nonconservative form with application to two-phase flows," *Journal of Computational Physics*, vol. 382, pp. 1–26, 2019.
- [2] F. Bassi, L. Botti, A. Colombo, D. Di Pietro, y P. Tesini, "On the flexibility of agglomeration based physical space discontinuous Galerkin discretizations," *Journal of Computational Physics*, vol. 231, no. 1, pp. 45–65, 2012.
- [3] R. Abgrall, "A general framework to construct schemes satisfying additional conservation relations. application to entropy conservative and entropy dissipative schemes," *Journal of Computational Physics*, vol. 372, pp. 640–666, 2018.
- [4] R. Abgrall, P. Öffner, H. Ranocha, "Reinterpretation and extension of entropy correction terms for residual distribution and discontinuous Galerkin schemes: Application to structure preserving discretization," *Journal of Computational Physics*, vol. 453, p. 110955, 2022.
- [5] J. Chan, H. Ranocha, A. M. Rueda-Ramírez, G. Gassner, y T. Warburton, "On the entropy projection and the robustness of high order entropy stable discontinuous Galerkin schemes for under-resolved flows," *Frontiers in Physics*, vol. 10, 2022.

- [6] L. Alberti, E. Carnevali, A. Colombo, A. Crivellini, "An entropy conserving/stable discontinuous Galerkin solver in entropy variables based on the direct enforcement of entropy balance," *Journal of Computational Physics*, vol. 508, p. 113007, 2024.
- [7] P. Bacigaluppi, J. Carlier, M. Pelanti, P. M. Congedo, y R. Abgrall, "Assessment of a non-conservative four-equation multiphase system with phase transition," *J. Sci. Comput.*, vol. 90, no. 1, 2022.
- [8] N. Franchina, M. Savini, y F. Bassi, "Multicomponent gas flow computations by a discontinuous Galerkin scheme using L2-projection of perfect gas EOS," *Journal of Computational Physics*, vol. 315, pp. 302–322, 2016.
- [9] H. Lund y P. Aursand, "Two-phase flow of CO2 with phase transfer," *Energy Procedia*, pp. 1–10, 2011.
- [10] D. Regener Roig, A. Crivellini, y A. Colombo, "Efficient entropy-conserving/stable Discontinuous Galerkin solution of the multicomponent compressible Euler equations," *J. Comput. Phys.*, 2025. (under revision).

3. Comments on the qualification programme and supervision strategy:

The courses proved extremely helpful to complement the knowledge acquired in my day-to-day work. Since I am mainly focused on numerical methods, different courses organized my DROPT allowed me to learn about the physics of the multiphase flows from the main experts of the field, enhancing the quality my overall research. The courses and seminars were well-planned, structured and announced with good anticipation. For my previous knowledge, the scope of the lectures and seminars was excellent since I had the opportunity to learn about new topics keeping always in the flow of the lecture. I had the opportunity to suggest topics for the "AI meets scientific computing" workshop, and the principal investigators did an effort to invite speakers that best matched the interests of the PhD students. I also helped in the in-situ organization of the event to ensure that it went smoothly, and the event was a success. For me, the largest value that DROPT added to my thesis comes from the opportunity of presenting my work to the principal investigators and other PhD students and get their perspectives and advice. This periodic meetings and exchanges during the DROPT seminars were also used to monitor the progress of the work. The presence of pre-scheduled doctoral seminars was also exploited to set clear milestones for the project development. Unfortunately, since my collaboration in DROPT lasted only two years, I had not the opportunity to do a stay at University of Stuttgart.

Participation in the following courses/seminars of the Int. Research Training Group:

Core courses:

-

Specialised courses:

- Heat and mass transfer in droplets I (Prof. Sazhin)
- Heat and mass transfer in droplets II (Prof. Sazhin)
- Simple models of complex processes: recent developments (Prof. Sazhin)
- An introduction to discrete Rham methods (Prof. di Pietro)
- Lecture by Firas Dhaouadi from the University of Trento (Dott. Dhaoudi)
- A talk by Prof. Bernhard Weigand: Wonderworld of Droplets (Prof. Weigand)
- Transport equations for modelling high-speed droplet impact and brain neural activity (Dr. Chizhov)
- Short course on homogenization (Prof. Melnyk)

Elective and soft skill courses:

-

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: -

Participation in the following International Research Training Group events:

- DROPT doctoral seminar 2024
- DROPT workshop in numerical and experimental methods 2024
- DROPT doctoral seminar 2025
- DROPT workshop in numerical and analytical methods 2025
- AI meets scientific computing 2025
- DROPT final colloquium

Participation in conferences, congresses, etc., at home and abroad:

- HONOM 2024 (Chania, Greece)

4. Individual publications:

I. Peer-reviewed articles:

II. Submitted articles:

- D. Regener Roig, A. Crivellini, y A. Colombo, "Efficient entropy-conserving/stable Discontinuous Galerkin solution of the multicomponent compressible Euler equations," *J. Comput. Phys.*, 2025. (under revision).

III. Presentations

a) Oral

- DROPIT doctoral seminar (5th Feb. 2024, Bergamo)
- DROPIT workshop in numerical and experimental methods (18th Jun. 2024, Bergamo)
- DROPIT doctoral seminar 2025 (28th-29th Jan. 2024, Stuttgart)
- DROPIT workshop in numerical and analytical methods 2025 (12th Jun. 2025, Bergamo)
- DROPIT final colloquium (1st of Aug. 2025, Stuttgart)

b) Posters

- DIPSI 2025 (19th Jun. 2025, Bergamo)
- DROPIT final colloquium (31st Jul. and 1st of Aug. 2025, Stuttgart)
- PhD Day of Università degli Studi di Bergamo (26th Sep. 2025, Bergamo)

Report of doctoral researcher (SP-A2-3)

Project number: SP-A2-3

First and last name of doctoral researcher: Rishi Dhar

(Working) title of doctoral project: Numerical Investigation of Drop-film Interaction for Immiscible Fluids

Name of supervisors: Dr.-Ing. Kathrin Schulte, Prof. Dr.-Ing. Simona Tonini

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT since Feb 2024 until Dec 2025.
Funding (type and duration): PhD researcher, ITLR, Funding: DROPIT (01.02.2024 - 31.12.2025)
Start of doctoral degree: 01.02.2024 (anticipated) date of doctoral degree: 31.08.2027
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 06/13 Mechanical Engineering, Maulana Azad National Institute of Technology, Bachelor's Degree, April 2017 10/21 Computer Aided Conception in Mechanical Engineering, RWTH Aachen, Master's Degree, Jan 2024 02/24 Aerospace Engineering, Universität Stuttgart, PhD degree
Where applicable, please note any circumstances which could have affected your scientific work:
Occupation after leaving the IRTG, if applicable:

2. Description of doctoral project and research results achieved to date:

Droplet collision and droplet–film interactions play a critical role in a range of applications, including spray-cooling methods [1], inkjet printing [2], and additive manufacturing [3]. The fluid properties are one of the key parameters that influence the outcomes of these interactions. Depending on whether the fluid is Newtonian [4–6] or non-Newtonian [3,12,13], the behaviour can vary significantly. Among Newtonian systems, several applications—such as oil–water remediation [6] and drug delivery [7]—involve immiscible fluids. Although these applications often involve spray breakup, the fundamental physics can be better understood by studying individual droplet–film interactions as an elementary process, particularly the collision of secondary droplets on wet surfaces. These interactions are inherently multiscale, and both experimental [8,9] and numerical studies [10,11] have revealed the complexity introduced by parameters such as the viscosity ratio, surface tension ratio, and film thickness.

To systematically analyse the process, it can be divided into four stages: (1) initial contact between the droplet and the film, (2) spreading of the droplet over the film surface, (3) crown formation and propagation, and (4) rim breakup leading to secondary droplet formation [20]. While these stages have been extensively characterized for miscible fluids [20], immiscible systems are more difficult to study experimentally due to overlapping and indistinct transitions between phases. As a result, this work adopts a numerical approach to investigate the dynamics introduced by variations in viscosity ratio. The numerical simulations are conducted using Free Surface 3D (FS3D) [18], a program package developed at ITLR which solves incompressible Navier–Stokes equations for multiphase flow situations, including three phases [19].

Literature classifies droplet–liquid interaction scenarios based on the non-dimensional film thickness (δ), defined as the ratio of the film height (h) to the droplet diameter (D_0). These interactions are grouped into four categories as per Geppert [20]:

- Droplet–pool interaction: $\delta > 4$
- Droplet–thick film or shallow pool interaction: $4 > \delta > 0.6$
- Droplet–thin film interaction: $0.6 > \delta > 0.08$
- Droplet–very thin film interaction: $\delta < 0.08$

This project focuses on investigating of droplet–thin film interactions. A numerical setup in FS3D for a drop–film interaction as shown in Fig. 1 is created using a quarter droplet of diameter D_0 impacting a film of height h with a Weber number $We = \frac{\rho u^2 D_0}{\sigma}$, where U_{drop} represents the velocity of drop before impact. A symmetry condition is applied at $x=0$ and $y=1$ with a resolution of D_0/x being around 1/3, where x is the number of cells along each side.

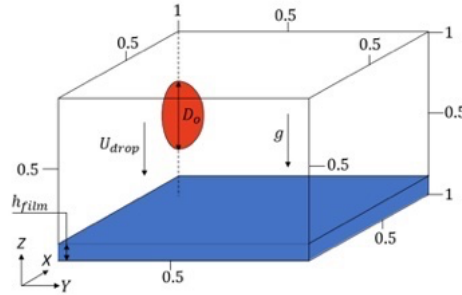


Figure 1: Typical numerical setup of a drop–film interaction.

To reduce complexity during the initial stages of investigation, a simplified droplet–pool scenario was considered. This allows observation of droplet–liquid interaction without interference from the drop–wall interaction and film thickness effects.

The droplet–pool case was set up based on the experimental work by Castillo-Orozco et al. [14], in which a silicon B5 oil droplet with a diameter of 1.8 mm was released vertically onto a pool of the same liquid. The impact occurred with $We = 230$.

A grid independence study was first conducted to determine the appropriate resolution for the numerical simulations. A timestep-wise comparison between numerical and experimental results showed very good agreement (Fig.2). The simulations accurately captured key phenomena, including the initial crater formation, the emergence of the Worthington jet, and the detachment of secondary droplets, validating the reliability of the numerical method for this regime.

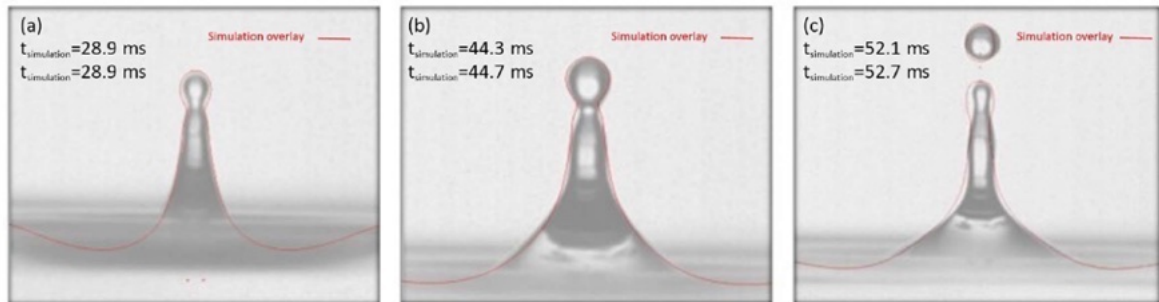


Figure 2: Evolution of the drop–pool interaction for silicon oil S5 drop of diameter (D_0) of 1.8 mm impacting a S5 silicon oil pool with velocity $u = 2.1$ m/s and $We_{drop} = 230$.

Given the promising results of the previous validations using the same fluids, the next step involves evaluating the solver's predictions for different fluid combinations. Hence as a next step, the film height was decreased to a range of ($0.6 > \delta > 0.08$) in order to replicate the experimental setup by Che and Matar [9]. Among the multiple cases investigated by Che and Matar [9], two cases where the drop and film liquids were interchanged for drop–film interaction with similar initial and boundary conditions were numerically investigated. The impact conditions for setup shown in Fig.1 included a film height of 0.7 mm and a U_{drop} of 2.4 m/s for both cases. A silicon oil S50 and water + 40% glycerol mixture was used interchangeably as drop and film liquid. Simulations could imitate the physics correctly for both cases as shown in Fig. 3. In the first cases as shown in Fig. 3(a), when a silicon oil S50 as drop liquid impacted

water + 40% glycerol mixture, a crown with the oil engulfing the water formed in the experiments as well as in the numerical simulations. On interchanging the liquids, a compound crown formation was experimentally observed where the water drop liquid formed a smaller crown being attached to define a larger crown formed by the silicon oil film liquid. The spreading factor ($S = \sigma_{\text{water-air}} - (\sigma_{\text{oil-air}} + \sigma_{\text{water-oil}})$) where $\sigma_{\text{water-oil}}$ is the interfacial tension and $\sigma_{\text{water-air}}$, $\sigma_{\text{oil-air}}$ are the surface tension) evaluated for first case is positive which explains the engulfing of the water film liquid by oil drop liquid during crown formation. The spreading factor for the latter case is negative, as the surface tension of the water is much higher than silicon oil, hence the spreading height for water over oil is lower. The case described in Fig. 3(b) provides distinguishable geometric parameters which can be used to identify the variations between experimental and numerical results.

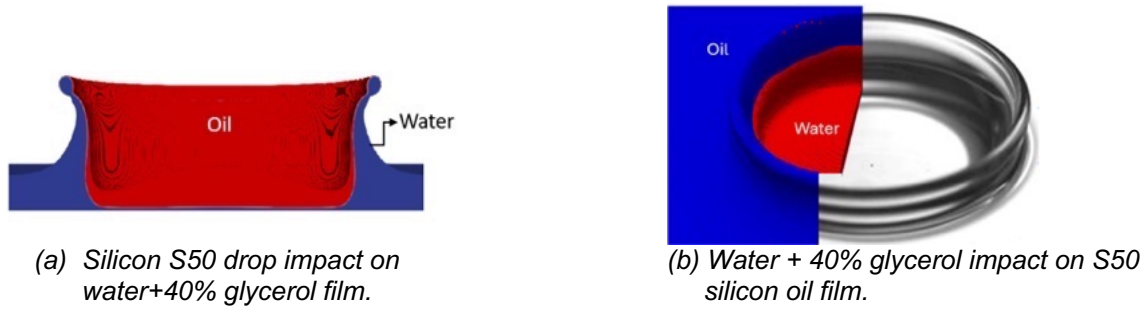


Figure 3: Immiscible drop film interaction with drop diameter $D_0 = 3$ mm with $We = 174$ on film of thickness $h = 0.7$ mm.

In order to further investigate the case shown in Fig. 3(b), the impact conditions of an experiment done by Che and Matar [9], where a case of pure water is impacted on a silicon oil S10 liquid film, has been setup. The interfacial tension of 36 mN/m was used between water and oil where a droplet of diameter 3 mm was impacted with $We = 174$ on the film of $h = 0.7$ mm. A grid independence study was done comparing the crown height as represented in Fig. 4(b) for four cases of grid resolution as per Tab. 1. Fig. 4(b) showed that 1024^3 had a resolution post which the crown height remains the same. Hence the same resolution has been used for further investigations. The setup used for grid independence was run till $\tau = 6$, where $\tau = \frac{tU_{\text{drop}}}{D}$ represents the non-dimensional time. The crown height evaluated numerically as shown in Fig. 4(a) was compared with experimental results. It showed a good agreement with the experimental results as shown in Fig. 4(c).

Grid	I	II	III	IV
Resolution	512^3	768^3	1024^3	2048^3

Table 1: Grid resolutions used for grid independence.

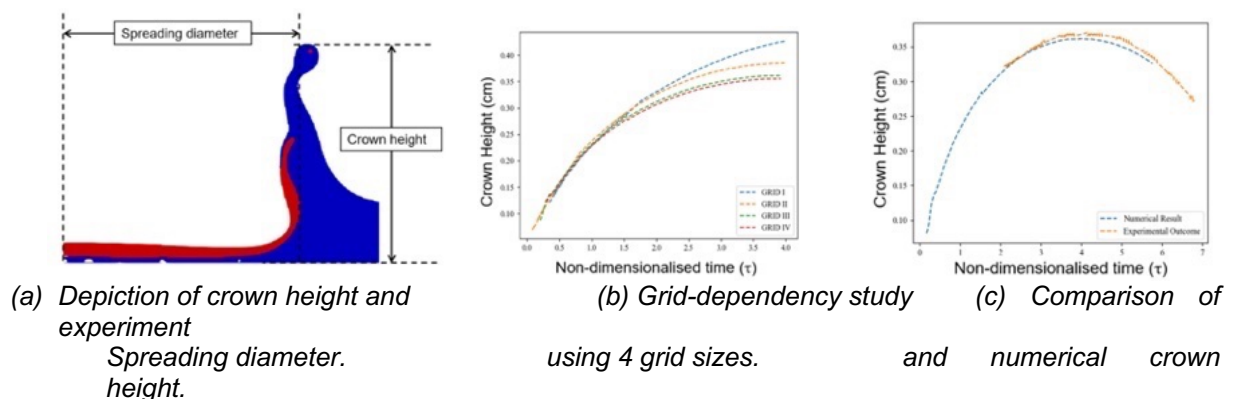


Figure 4: Immiscible drop film interaction with drop diameter $D_0 = 3$ mm with $We = 174$ on film of thickness $h = 0.7$ mm.

Accurate specification of interfacial tension is essential for the numerical evaluation of the drop-film interface between immiscible fluids. Huge discrepancies are found in literature for the interfacial tension values [10,14] observed for same set of liquids, which can be attributed to impurities within the liquids. Since the validation case provides the accurate interfacial tension when comparing numerical results with the experimental result of Che and Matar [9], a comparison study can reveal the effects of changing interfacial tension on the outcomes.

Subsequently, the same setup for the validation case was extended to 5 cases, where the interfacial tension was varied to 0, 10, 20, 30 and 40. The compound crown diameter and spreading diameter were compared for all cases. The variation in the spreading diameter and compound crown diameter were negligible, but the internal crown structure was very different for cases where the interfacial tension was lower than the reference case. This showed that while the macro parameters of the interaction remain largely unaffected by the interfacial tension, the internal crown structure varied significantly. Interfacial tension, similar to surface tension, quantifies the surface energy between two immiscible fluids. A higher interfacial tension leads to greater conversion of a droplet's kinetic energy into surface energy upon impact. This can be observed as a reduction in crown height with increasing interfacial tension during droplet–film interactions as shown in Fig. 5.

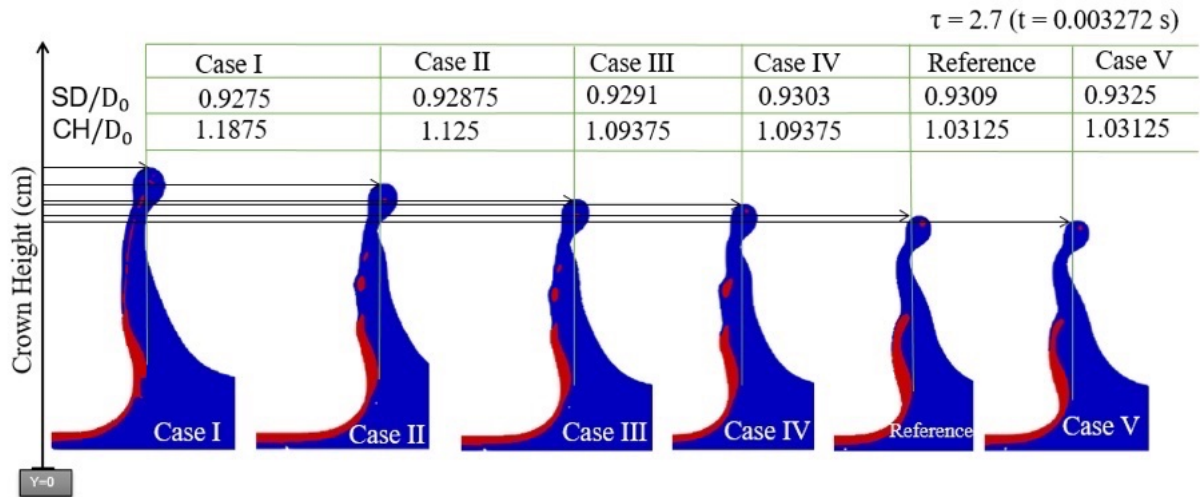


Figure 5: Immiscible drop film interaction with drop diameter $D_0 = 3$ mm with $We = 174$ on film of thickness $h = 0.7$ mm. with different cases of interfacial tension. Here SD denotes the spreading diameter and CH denotes the crown height.

Other variations in the fluid's properties also have a huge impact on collision dynamics. One example includes non-Newtonian fluids. Non-Newtonian fluids, particularly thixotropic fluids exhibit shear-thinning behaviour with a time-dependent recovery of viscosity once shear is removed. Thixotropic fluids are commonly encountered in applications such as foam handling [16] and inkjet printing [17]. However, there is limited fundamental research on how thixotropy affects droplet–droplet collisions. To address this gap, a binary droplet collision study is conducted using thixotropic fluid models. The Weber number is varied in the range of 15 to 50 to examine regimes of coalescence, transitional separation, and head-on separation between droplets. A grid independence study has been performed to ensure sufficient resolution and numerical accuracy. The results of the ongoing analysis will provide further insight into the role of thixotropy on droplet collision outcomes.

References:

1. Kim, J. (2007). Spray cooling heat transfer: The state of the art. *International Journal of Heat and Fluid Flow*, 28, 753.
2. van Dam, D. B. and Le Clerc, C. (2004). Experimental study of the impact of an ink-jet printed droplet on a solid substrate. *Physics of Fluids*, 16(9), 3403–3414.
3. Sheng, Y. X., et al. (2024). Rheological properties of gel foam co-stabilized with nanoparticles, xanthan gum, and multiple surfactants. *Polymers*, 9(7), 534. (Foams show Cross-model shear-thinning.)
4. Pasandideh-Fard, M., et al. (2001). Cooling effectiveness of a water drop impinging on a hot surface. *International Journal of Heat and Fluid Flow*, 22(2), 830–836.
5. Harkins, W. D. and Feldman, A. (1922). Films: The spreading of liquids and the spreading coefficient. *Journal of the American Chemical Society*, 44(12), 2665–2685.
6. Davanlou, A. and Kumar, R. (2015). Thermally induced collision of droplets in an immiscible outer fluid. *Scientific Reports*, 5, 9531.
7. Bhushan, B. (2019). Bioinspired oil–water separation approaches for oil spill clean-up and water purification. *Philosophical Transactions of the Royal Society A*, 377, 20190120.
8. Damiati, S., Kompella, U. B., Damiati, S. A., and Kodzius, R. (2018). Microfluidic devices for drug delivery systems and drug screening. *Genes (Basel)*, 9(2), 103.
9. Che, Z. and Matar, O. (2018). Impact of droplets on immiscible liquid films. *Soft Matter*, 14(9).

10. Kittel, H., Roisman, I., and Tropea, C. (2018). Splash of a drop impacting onto a solid substrate wetted by a thin film of another liquid. *Physical Review Fluids*, 3, 073601.
11. Li, J. and Renardy, Y. (2001). Numerical study of flows of two immiscible liquids at low Reynolds number. *SIAM Journal on Applied Mathematics*, 42(3), 417–439.
12. Qin, Z., Wu, J., Cai, Y., Lin, L., and You, H. (2023). Effects of an immiscible liquid film on an impacting droplet. *International Journal of Multiphase Flow*, 163, 104422.
13. Gardiner, B. S., Dlugogorski, B. Z., Jameson, G. J., and Chhabra, R. P. (2003). Yield stress of firefighting foams. NIST Report.
14. Sen, U., Datt, C., Segers, T., Wijshoff, H., Snoeijer, J. H., Versluis, M., and Lohse, D. (2021). The retraction of jetted slender viscoelastic liquid filaments. *Journal of Fluid Mechanics*.
15. Castillo-Orozco, E., Davanlou, A., Choudhury, P., and Kumar, R. (2015). Droplet impact on deep liquid pools: Rayleigh jet to formation of secondary droplets. *Physical Review E*, 92, 053022. <https://doi.org/10.1103/PhysRevE.92.053022>
16. Shaikh, S., Toyofuku, G., Hoang, R., and Marston, J. O. (2017). Immiscible impact dynamics of droplets onto millimetric films. *Experiments in Fluids*, 59(1). <https://doi.org/10.1007/s00348-017-2461-4>
17. Wang, Y., et al. (2022). Suppression and utilization of satellite droplets for inkjet printing. *Processes*, 10, 932.
18. Eisenschmidt, K., Ertl, M., Goma, H., Kieffer-Roth, C., Meister, C., Rauschenberger, P., Reitzle, M., Schlotke, K., and Weigand, B. (2016). Direct numerical simulations for multiphase flows: An overview of the multiphase code FS3D. *Applied Mathematics and Computation*, 272(Part 2), 508–517.
19. Potyka, J., Schulte, K., and Planchette, C. (2023). Liquid distribution after head-on separation of two colliding immiscible liquid droplets. *Physics of Fluids*, 35(10).
20. Geppert, A. K. (2019). Experimental investigation of droplet wall-film interaction of binary systems. <https://doi.org/10.18419/OPUS-10577>

3. Comments on the qualification programme and supervision strategy:

The qualification programme offered by DROPIT has provided a well-rounded combination of core scientific training and interdisciplinary soft skills, essential for building a successful research career. Annual meetings such as DIPSI and the Doctoral Seminars have created valuable opportunities for collaboration, idea exchange, and a deeper structural understanding of research challenges. The supervision strategy has offered a clear and detailed roadmap for problem analysis and the development of effective solution strategies. The individual support and guidance from my supervisors, along with the collaborative spirit of peers and senior researchers, have fostered a holistic and motivating environment conducive to achieving the project's goals.

Participation in the following courses/seminars of the Int. Research Training Group: Core courses:

Specialised courses:

- N-Ways to GPU Programming Bootcamp from April 3-5, 2024
- AMD Instinct GPU Training from April 22-25, 2024
- Advanced Parallel Programming with MPI and OpenMP from Dec 2-5, 2024

Elective and soft skill courses:

- Scientific Writing Skills for Doctoral Students in Aerospace Engineering, Feb 20, 2025

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

Participation in the following International Research Training Group events:

- Doctoral Seminar, Bergamo, 8-9 Feb, 2024
- “Experimental and Numerical Methods in Science” on 18.06.2024 in Bergamo, Italy
- Doctoral Seminar, Stuttgart, 28-29 Jan 2025

Participation in conferences, congresses, etc., at home and abroad:

- DIPSI 2024, Bergamo, 16-17 June 2024

4. Individual publications:

I. Peer-reviewed articles:

II. Submitted articles:

III. Further publications

a) Proceedings

- **R. Dhar** and K. Schulte, “Investigation of Droplet-Pool Phenomena and Droplet-Film Interaction for Immiscible Fluids” In DIPSi Workshop 2024 on Droplet Impact Phenomena & Spray Investigation, 17th June 2024, Bergamo, Italy

b) Technical and scientific reports

- **R. Dhar**, D. Gösele, P. Saumet, B. Weigand, K. Schulte. “The Role of Interfacial Tension in Direct Numerical Simulations of Drop-Film Interaction for Immiscible Fluids”. *HLRS Annual Report 2025*, Stuttgart, Germany

IV. Presentations

a) Oral

b) Posters

- DIPSi 2024, ‘Investigation of Droplet-Pool Phenomena and Droplet-Film Interaction for Immiscible Fluids’ by Rishi Dhar, Kathrin Schulte
- Dropit Final Colloquium 2025 ‘Numerical Investigation of Drop-film Interaction for Immiscible Fluids’ by Rishi Dhar, Kathrin Schulte

Report of doctoral researcher (SP-B3)

Project number: SP-B3

First and last name of doctoral researcher: Daniele Rusconi

(Working) title of doctoral project:

Towards More Accurate Porous Media Modelling Using High-Resolution X-ray Imaging and Deep Learning Segmentation.

Name of supervisors: Prof. M. Santini, Prof. S. Fest-Santini, Prof. D. Weiskopf

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT since 1.11.2024

Funding (type and duration):

PhD student at the Laboratory of X-ray microtomography, Department of Engineering and Applied Sciences in Bergamo (November 2024 – December 2027)

Start of doctoral training and (anticipated) date of doctoral degree:

November 2024 to January 2028

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2013; Industrial Engineering (Bachelor); Politecnico di Milano; 07/2017

10/2020; Mechanical Engineering (Master); Università degli studi di Bergamo; 09/2024

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Motivation:

The project aims to improve the modelling and simulation of porous media by integrating advanced X-ray imaging techniques, Deep Learning image segmentation methods, and multi-phase multi-component modelling approaches. The objective is to enhance the geometric realism and predictive capability of pore-scale models, by enabling a more reliable description of transport phenomena in porous materials.

The central hypothesis of the project is that limitations in existing porous media models largely arise from oversimplified geometric representations and insufficient characterisation of pore connectivity and heterogeneity. These simplifications are often a consequence of limited imaging resolution and inadequate segmentation methodologies. By leveraging high-resolution X-ray tomography in combination with modern deep learning-based segmentation techniques, it is expected that pore-scale structures can be reconstructed with greater fidelity. This detailed structural information will form the basis for the development of representative pore network models and, in later stages, for pore-scale simulations of coupled thermal and flow processes that more accurately reflect the physical behaviour of real porous systems.

Project structure and results:

The project is structured into several interconnected steps. The first step focuses on X-ray imaging techniques for the reconstruction of porous media. This includes the acquisition, reconstruction, and pre-processing of volumetric X-ray tomography data, with particular emphasis on noise reduction, artefact mitigation, and the assessment of resolution-dependent effects on pore-scale features. A key aspect of this work is the evaluation of how different imaging modalities influence the visibility and interpretability of complex pore geometries.

In the past year, I actively contributed to the design and installation of a new X-ray source in Bergamo. This activity involved participation in the technical planning and setup of the system and represents a significant step towards improving the overall quality and flexibility of X-ray experiments within the research environment. The new installation is expected to substantially enhance imaging performance and to enable the acquisition of higher-quality datasets of porous media. In particular, it provides access to a range of advanced imaging techniques, including phase-contrast imaging, edge illumination methods, photon-counting detection, and X-ray microscopy. In general, it will allow for higher spatial resolution, improved contrast, and enhanced sensitivity to fine structural features, which are essential for the accurate reconstruction of complex porous materials.

The second step addresses the segmentation of X-ray tomographies using neural networks and constitutes the main focus of the research conducted to date. Initial investigations were carried out using a U-Net-based architecture as a baseline approach for pore–solid phase segmentation. This model served as a reference point for evaluating segmentation accuracy and robustness in the presence of noise and heterogeneous pore structures. Building on this baseline, attention mechanisms inspired by Vision Transformer architectures were explored to improve the representation of long-range spatial dependencies and complex morphological patterns that are difficult to capture with purely convolutional approaches.

More recently, the work has progressed towards the investigation of weakly supervised learning strategies. This direction is motivated by the limited availability of fully annotated ground truth data for high-resolution X-ray tomography of porous media, particularly for large and complex datasets. Weak supervision offers the potential to reduce annotation effort while maintaining acceptable segmentation performance and generalisability. Current efforts focus on assessing the suitability of such approaches for various applications, identifying their limitations in comparison to fully supervised methods.

The subsequent development of pore network models that realistically represent the topology, connectivity, and transport-relevant properties of the porous media constitutes the next phase of the project and is planned as future work. These models will be derived from the segmented volumetric data and are intended to preserve key structural characteristics while remaining computationally efficient. Based on these networks, multi-phase modelling will be performed in a later stage of the project to simulate transport processes at the pore scale. These modelling activities have not yet started and will be addressed in the future year.

1st year overview:

At the current stage, the main results achieved include the implementation and evaluation of multiple neural network-based segmentation strategies, the successful contribution to the installation of an advanced X-ray imaging facility, and the design of X-ray imaging workflows. Preliminary results indicate that attention-based architectures and weakly supervised approaches can provide advantages over classical segmentation methods when dealing with complex pore geometries and limited annotations. Networking and qualification measures form an integral part of the doctoral training within the research training group. In this context, the doctoral candidate participated in the DuMux course in September 2025, which provided structured training in the DuMux simulation framework. The knowledge acquired through this course is expected to support the planned pore network modelling and contributes to the long-term integration of imaging, segmentation, and simulation activities within the project.

3. Comments on the qualification programme and supervision strategy:

I was part of DROPIT for one year before the end of it. During this period, the qualification programme provided a well-structured framework that encouraged consistent engagement with research activities and supported sustained effort in the advancement of my doctoral project.

The seminars offered within the programme contributed to broadening my scientific perspective and allowed me to deepen my knowledge beyond the immediate focus of my project. They also facilitated interdisciplinary discussion and exchange, offering insight into different subprojects and different methodological approaches. The interactions and shared moments with colleagues during seminars and joint activities were particularly motivating and helped establish a constructive and collaborative working environment.

Among other closely related training activities, the DuMux course was closely related to the methodological needs of my research. Participation in this course provided practical insights into numerical modelling of coupled transport processes in porous media, which will be valuable for future stages of my work.

Overall, despite the limited duration of my participation, the qualification programme was well structured and effective in fostering both scientific development and personal motivation.

Participation in the following courses/seminars of the Int. Research Training Group: • DROPIT doctoral seminar, Stuttgart February 2025 • DuMux course, Stuttgart, September 2025
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • DROPIT Doctoral Seminar, Stuttgart, February 2025 • DROPIT Final colloquium, Stuttgart, July 2025
Participation in conferences, congresses, etc., at home and abroad: • Ultrafast X-ray Summer School, Villigen (CH), June 2025 • IMAGING25 conference, Varenna (IT), September 2025

4. Individual publications:

I. Presentations:

a) Oral

- D. Rusconi: "Neural networks for the segmentation of porous media." DROPIT Doctoral seminar, Stuttgart (DE), February 2025.

b) Posters

- D. Rusconi, S. Fest-Santini: "From U-Net to UNETR: Advancing 3D Image Segmentation with Transformers." DROPIT Final colloquium, Stuttgart (DE), July 2025.
- D. Rusconi: "Weak Supervision: A Solution for Image Segmentation in Challenging Datasets." IMAGING25 Conference, Varenna (IT), September 2025.

Report of doctoral researcher (SP-A4-3)

Project number: SP-A4-3
First and last name of doctoral researcher: Sara Rinaldi
Title of doctoral project: High order structure-preserving numerical schemes for continuum mechanics of mixtures with phase transition
Name of supervisors: Prof. Dr.-Ing Michael Dumbser, Dr. Olindo Zanotti

1. General information:

Duration of International Research Training Group membership: Doctoral researcher in DROPIT January 2025-current
Funding (type and duration): PhD student at the Laboratory of Applied Mathematics, Department of Civil, Environmental, and Mechanical Engineering in Trento, Nov 2024 – current
Start of doctoral training and (anticipated) date of doctoral degree: 4 Nov 2024 to Nov 2027
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09/2019-07/2022 Mathematics, University of Rome La Sapienza, Bachelor's degree 09/2022-07/2024 Applied Mathematics, University of Rome La Sapienza, Master's degree
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

This project deals with the development of new high order accurate structure-preserving finite volume and discontinuous Galerkin finite element schemes for the description of nonlinear continuum mechanics of mixtures (liquid-solid and liquid-gas mixtures) with phase transition. The governing equations are based on the unified thermodynamically compatible model of continuum mechanics of Godunov, Peshkov and Romenski that describes at the same time the dynamics of elasto-plastic solids at large deformations, as well as fluids.

We aim at modelling compressible and nearly incompressible two-phase flows with phase transition (liquid-gas and liquid-solid mixtures), also obtaining simplified depth-averaged versions of the governing equations for future environmental and geophysical applications. The considered first order hyperbolic PDE systems with relaxation will be discretized with new innovative numerical schemes that preserve essential features of the underlying continuous equations exactly also at the discrete level. Particular care will be taken in the low Mach number limit of the equations. Potential applications are industrial liquid-vapour flows, additive manufacturing and environmental and geophysical applications such as debris flows and lava flows.

Current Research

I started my Ph.D by analyzing the Zel'dovich-von Neumann-Doering profile of a detonation wave. In this case the leading PDE system is defined by the conservation of mass, momentum and total energy together with a reaction process equation that describes the rate at which the material is combusted/burned.

In the 1D case the PDE system can be reduced into a single ODE. Thus the detonation wave can be obtained by solving the ODE and a system of algebraic equations. Currently I am working with the compressible Navier–Stokes equations with chemical reactions and viscosity.

With the viscosity terms the equations become stiff, making the computation of the solution more difficult. In the near future I am planning to develop a numerical scheme that compute its solution accurately.

Participation in the following courses/seminars of the Int. Research Training Group: • Advanced numerical methods for free-surface hydrodynamics • Advanced numerical methods for Hyperbolic equations
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:
Participation in the following International Research Training Group events: • DROPIT Doctoral Seminar, Stuttgart, 28-29 January 2025
Participation in conferences, congresses, etc., at home and abroad:

4. Individual publications:

-

Report of doctoral researcher (SP-A3-4)

Project number: SP-A3-4

First and last name of doctoral researcher: Kim Sophie Ellenberger

(Working) title of doctoral project: A Multi-Species Enskog-Vlasov Solver for Multi-Phase Flows

Name of supervisors: Prof. Dr.-Ing. Stefanos Fasoulas, Prof. Dr.-Ing. Lorenzo Alessio Botti

1. General information:

Duration of International Research Training Group membership: 11 months
Funding (type and duration): PhD Researcher at IRS from 01.11.2024 – 30.09.2025
Start of doctoral training and (anticipated) date of doctoral degree: 01.11.2024 – not known
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 01.03.2018; Technophysics; Technical University of Kaiserslautern; 01.10.2021; B.Sc. 01.10.2021; Technophysics; RPTU Kaiserslautern; 01.03.2024; M.Sc.
Where applicable, please note any circumstances which could have affected your scientific work:
Occupation after leaving the IRTG, if applicable: Remain at the Institute of Space Systems

2. Description of doctoral project and research results achieved to date:

The doctoral project dealt with the gas-kinetic simulation of multi-phase and multi species (multi-component) flows. The existing solver of a former PhD researcher was extended from single-species to multi-species simulations. Therefore, the different particle diameters have to be considered in the simulation framework. This solver was used to predict the composition of argon-neon mixtures at different temperatures and pressures. These compositions were compared to the results of the SAFT-VRQ model. Furthermore, the evaporation of this mixtures were determined.

Initial Hypothesis and Aims:

The project was based on the hypothesis that conventional models based on the Navier Stokes equations are inadequate to accurately capture the behavior of atoms and molecules at the liquid-vapor interface, especially under non-equilibrium conditions characterized by high accelerations due to surface tension. To close this gap, kinetic solvers, specifically a Direct Simulation Monte Carlo (DSMC)-based Enskog-Vlasov (EV-DSMC) solver, have been further developed to simulate and gain insight into the evaporation and condensation processes of these droplets under pressurized conditions. The derived evaporation and condensation coefficients are needed for the development of models for the behavior of liquid fuel droplets in combustion engines. These models can increase their efficiency in new generations of engines.

Current Project Status:

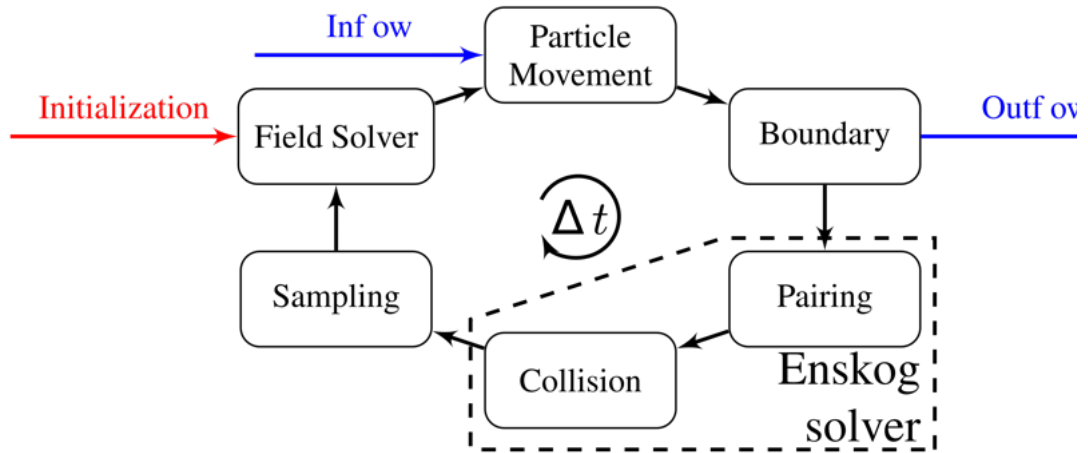


Figure 4: Flowchart of the EV-DSMC time step

The single-species EV-DSMC solver was extended to handle multiple species with separate diameters, masses and attraction strengths. A flowchart of the different steps of the solver can be seen in Figure 1. The *Field Solver* was adjusted to simulate the resulting force of different species onto the simulation particles of different species. In the boundary stage, new boundary interactions are included as a pressure boundary which maintain a constant pressure in the computational domain. The *Pairing* step was adjusted to handle different displacements of the particles depending on the involved species. In the *Collision* the calculation of the post collision velocities is changed to remain momentum and energy conserving if the masses of the colliding particles differs. Also a new pair correlation function to cover multi-species cases is used. For the *Sampling* a new multi-species pressure sampling was developed to yield correct pressures.

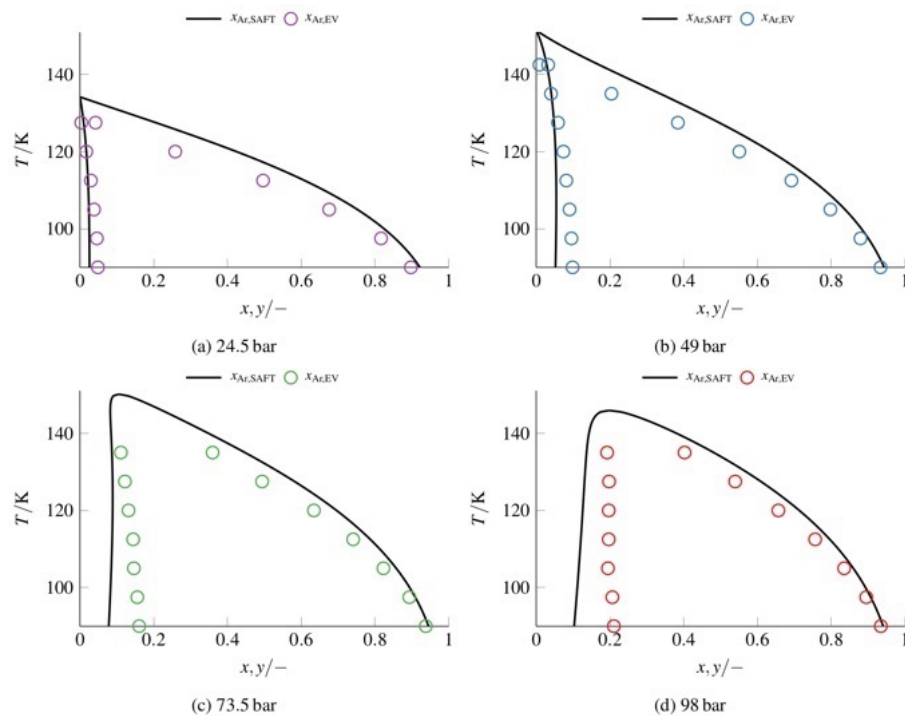


Figure 5: Composition compared to SAFT-VRQ Mie. The solid black line are the results of the SAFT-VRQ Mie EoS, whereas the circles are those from the Vlasov-Enskog solver. The left branch is the liquid state and the right one is the liquid composition, respectively.

Results Achieved:

Composition of neon-argon mixtures: To test if the extended EV-DSMC can reproduce the correct phase compositions it was tested against a SAFT-VRQ method. This solver uses a Mie potential instead of the Sutherland potential of the EV-DSMC. Therefore, discrepancies are expected. The results are shown in Figure 2, where the results of the SAFT method are indicated with the black lines and the EV-DSMC solutions are the colored circles. The deviation between both solvers are for most cases below 10%.

Evaporation coefficients of neon-argon mixtures: Of the previous simulations the evaporation and condensation coefficients were sampled. These are shown in Figure 3. It can be seen, that the evaporation coefficient decreases exponentially with increasing pressures for argon and decreases almost linearly with increasing temperature. For neon the evaporation coefficient decreases linearly with the pressure and is almost constant in respect to the temperature.

Future Work: Extending the framework to other potentials.

Cooperation within the IRTG: In this project, we cooperated with Grazia Lamanna. She uses the derived evaporation coefficients in their model to predict the droplet evaporation and heating in hot and pressurized environments as they occur combustion chambers of engines.

3.

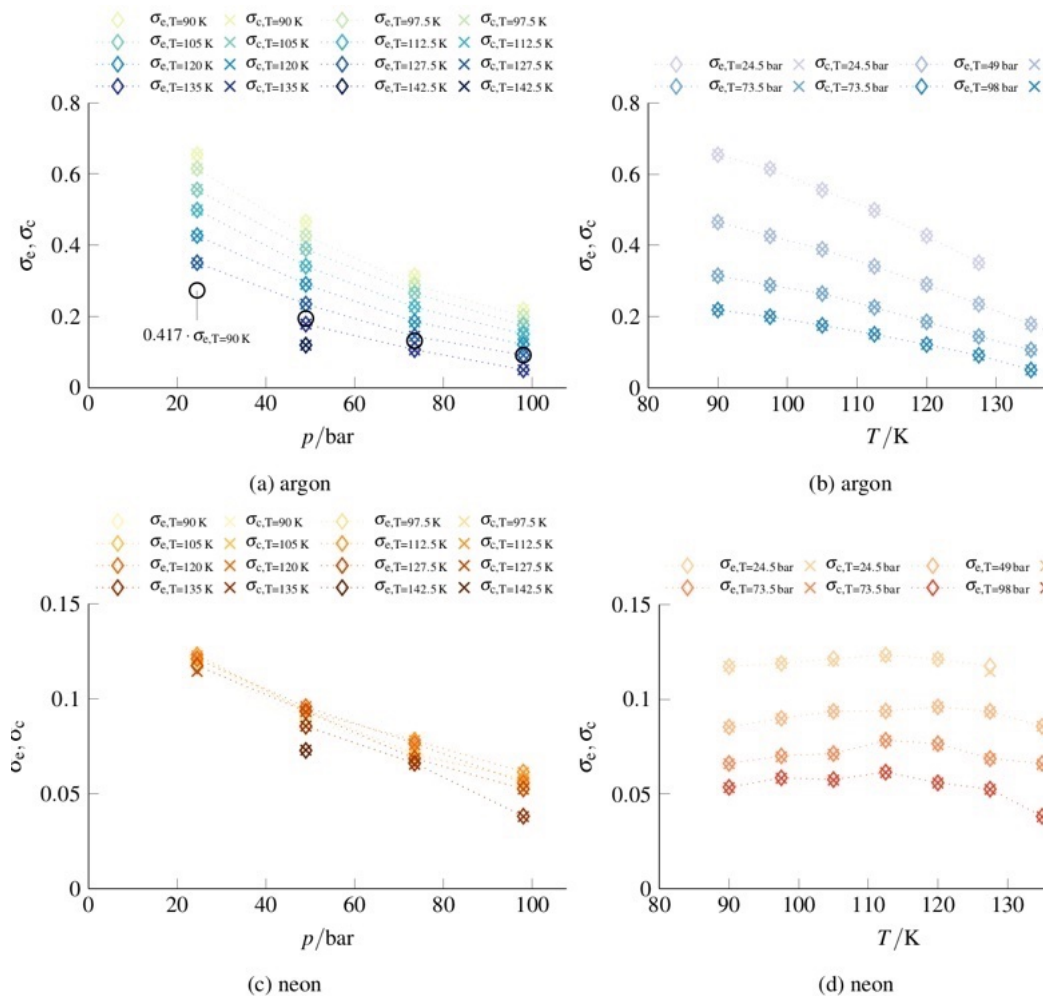


Figure 6: Evaporation coefficients of argon and neon at different temperatures and pressures. The dotted lines are eye-guides to connect data with same pressures or temperatures.

Comments on the qualification program and supervision strategy:

Due to the short time in the Dropit project, no comments about the qualification program and supervision strategy can be made.

Participation in the following courses/seminars of the Int. Research Training Group: <i>Core courses:</i> • <i>Specialised courses:</i> • <i>Elective and soft skill courses:</i> •
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: -
Participation in the following International Research Training Group events: • Doctoral Seminar, Stuttgart, 28.-29.01.2025
Participation in conferences, congresses, etc., at home and abroad: • 3rd International Conference on Flight Vehicles, Aerothermodynamics and Re-Entry (FAR), Arcachon (France), 2025

4. Individual publications:

I. Peer-reviewed articles:

-

II. Submitted articles:

- R. Tietz, R. Stierle, K.-S. Ellenberger, S. Fasoulas, M. Pfeiffer, A Multi-Species Enskog-Vlasov Solver to Determine Evaporation Coefficients of Fluids in High Pressure Environments (<https://arxiv.org/abs/2506.22162>)

-

III. Further publications

a) Proceedings

- K.-S. Ellenberger, A. Schlitzer, M. Pfeiffer, S. Fasoulas, Numerical Study on the Setup of an Atomic Oxygen Ground Testing Facility, 3rd International Conference on Flight Vehicles, Aerothermodynamics and Re-Entry (FAR), Arcachon (France), 2025

b) Technical and scientific reports

-

IV. Presentations

a) Oral

- K.-S. Ellenberger, A. Schlitzer, M. Pfeiffer, S. Fasoulas, Numerical Study on the Setup of an Atomic Oxygen Ground Testing Facility, 3rd International Conference on Flight Vehicles, Aerothermodynamics and Re-Entry (FAR), Arcachon (France), 2025

b) Posters

Individual reports by associated doctoral researchers

AS04 Yanchao Liu

Droplet interaction at high velocities

AS05 Melanie Lipp

Modelling and analysis of coupled porous-medium and free-flow with application to evaporation processes

AS06 Dennis Hoch

Micro-scale modelling and simulation of two phase flow processes in oil mist filtration

AS07 Luca Verzeroli

DG methods for linear elasticity: implementation, validation and perspectives

AS09 Verena Kunberger

Interactions of Supercooled Water Droplets

AS10 Florian Wendt

Data-Integrated Two-Scale Modelling of Compressible Multiphase Flow

AS11 Jonathan Wurst

Direct Numerical Simulation of Complex Drop-Wall Interactions with FS3D

Report of doctoral researcher (SP-AS04)

Project number: Associated Project 04

First and last name of doctoral researcher: Yanchao Liu

(Working) title of doctoral project: Numerical investigation on high velocity liquid dynamics

Name of supervisors: Prof. B. Weigand

1. General information:

Duration of International Research Training Group membership: Since 1 st February 2019
Funding (type and duration): Chinese Scholarship Council (CSC) Research associate at the Institute of Aerospace Thermodynamics at the University of Stuttgart
Start of doctoral training and (anticipated) date of doctoral degree: 15 th September 2018 to 28 th February 2024
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 09/2008 Flight Vehicle Propulsion Engineering, Northwestern Polytechnical University, Bachelor of Science, 06/2012 10/2013 Aerospace Engineering, University of Stuttgart, Master of Science, 06/2017
Where applicable, please note any circumstances which could have affected your scientific work: -
Occupation after leaving the IRTG, if applicable: -

2. Description of doctoral project and research results achieved to date:

Motivation

Medical inhalers, particularly jet inhalers, are designed for converting liquid medication into an aerosol spray, aiming to efficiently deliver drugs to the lungs. These devices have traditionally been developed through empirical methods, with a limited understanding of the intricate fluid dynamics involved, including the atomization process and droplet formation. The complexity of these dynamics, characterized by the interaction of a high-velocity gas jet with the liquid medication and the resultant droplet behavior, has left the detailed mechanisms at the nozzle's vicinity largely unexplored. The current project seeks to investigate these processes through a detailed numerical analysis, aiming to enhance inhaler design and drug delivery efficiency by understanding the mechanics of liquid breakup, gas-liquid interactions, and the impact dynamics of the resulting droplets.

Numerical Methods

The simulations were performed based on the solver called “*compressibleInterFoam*”, which is an OpenFOAM solver using the Volume of Fluid (VOF) approach. In the VoF method, the liquid is indicated by its volume fraction $\alpha \in [0, 1]$ in each computational cell. The interface can then be defined by the iso-surface of $\alpha = 0.5$. A detailed description of the schemes in OpenFOAM can be found in Jasak (1996) [1], Liu et al. (2023) [3] and Liu et al. (2024) [4]. The interactions between the high-velocity gas jet and the liquid are simulated by Large-Eddy Simulations (LES), while in the simulation of high velocity droplet impact, no turbulence model has been applied, and the grid resolution is fine enough to capture the concerned flow features. Thus, here a DNS has been done.

Results

Firstly, the project focused on understanding the atomization process occurring at the nozzle exit, where

the liquid undergoes atomization due to the interaction with a high-velocity gas jet. This process is crucial for the effective delivery of medications via inhalers. The simulation reveals that the liquid jet oscillates radially near the nozzle exit, experiencing constriction and expansion which lead to the breakup of the jet into ligaments. This behaviour is influenced by the surface tension of the liquid, the geometry of the nozzle, and the pressure dynamics of the gas jet. Further quantitative analysis can be found in Liu et al. (2021) [2].

Additionally, the instability of the liquid sheet under various conditions has been investigated, employing a simplified geometry to analyse the effects of inlet pressure and gas jet swirl velocities. The findings indicated that higher gas jet inlet pressures and the presence of swirl can significantly influence the instability and breakup patterns of the liquid sheet. The use of spatial Dynamic Mode Decomposition (DMD) provided insights into the growth rates of instabilities. As shown in Fig.1, for the high-pressure cases (4-6), the growth rates are significantly higher than those in the low-velocity cases due to the inertial effect. In the axial direction, the flow shows an increasing instability up to a maximum at around $x/h \approx 1.8$. When the swirl velocity is increased, the leading growth rates do not decrease as they do in the low-velocity cases. Rather, a slight increase in the growth rate is observed in case 6. Given the reduction in momentum of the gas jet caused by the swirl, it appears that the instability is further enhanced by the addition of a swirl. In other words, the energy transfer from the gas stream to the liquid sheet is more efficient with the presence of swirling in the gas stream. For case 6, this additional instability overcomes the stabilization effect of the reduced gas momentum. Furthermore, transfer entropy analysis has been used to quantify the interaction between the gas jet pressure and the liquid's radial motion, highlighting the complex interplay between these elements. Figure 2 summarizes this relationship with blue lines representing low-pressure cases and red lines representing high-pressure cases. The transfer entropy from pressure to momentum $\tilde{T}_{P \rightarrow M}$ is generally higher than vice versa, i.e. $\tilde{T}_{M \rightarrow P}$ indicating the pressure as a cause and the momentum as an effect. Furthermore, the swirling effect can increase the causal relationship $\tilde{T}_{P \rightarrow M}$ in the low-pressure condition but decrease the causality in the high-pressure condition. This suggests that the swirling effect is highly dependent on the pressure level in the chamber. Further details and results can be found in Liu et al. (2023) [3].

The study also investigated the dynamics of droplet impacts on cylindrical pillars, proposing new correlations for predicting maximum impact pressure and droplet behaviour, especially concerning cavitation. It was observed that shock waves generated during impact cause lateral jetting and cavitation, with the pillar diameter playing a crucial role in the interaction of low-pressure areas within the droplet.

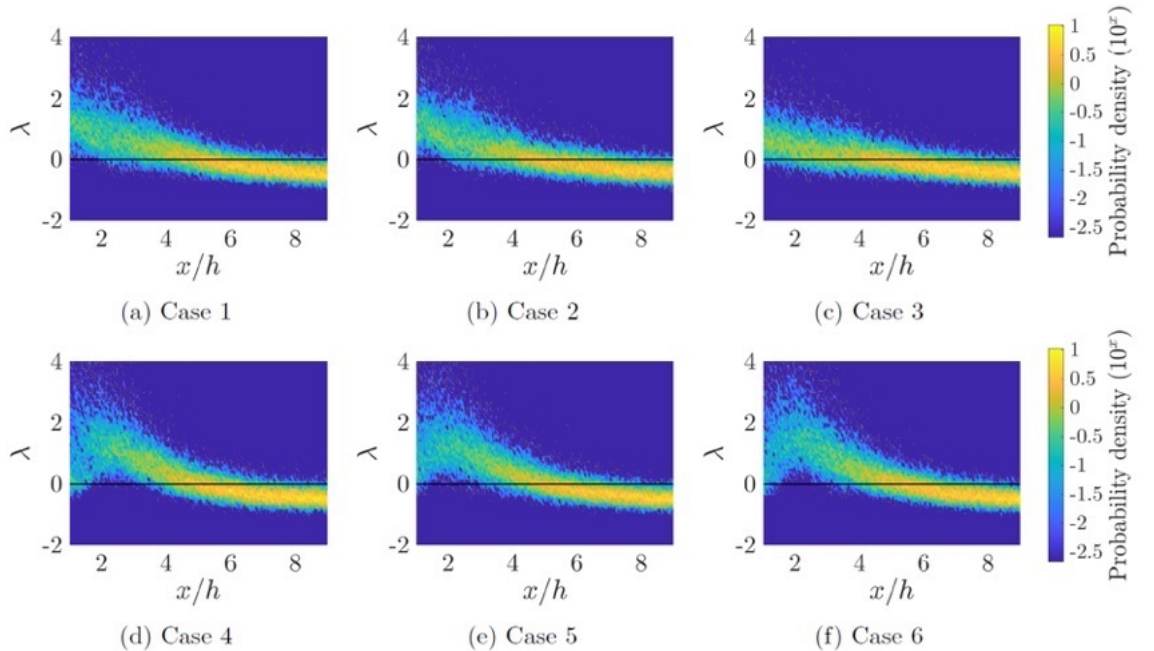


Figure 1: Probability density of maximal spatial growth rate λ versus x/h (axial position) and R (radius of the liquid mass center) for the 6 investigated cases. Case 1-3 have an inlet pressure of 2 bar and case 4-6 have an inlet pressure of 2.6 bar. The swirl velocity of the gas jet in case 4&6 is 0 and 100 rad/s for case 2&5 and 500 rad/s for case 3&6. [3]

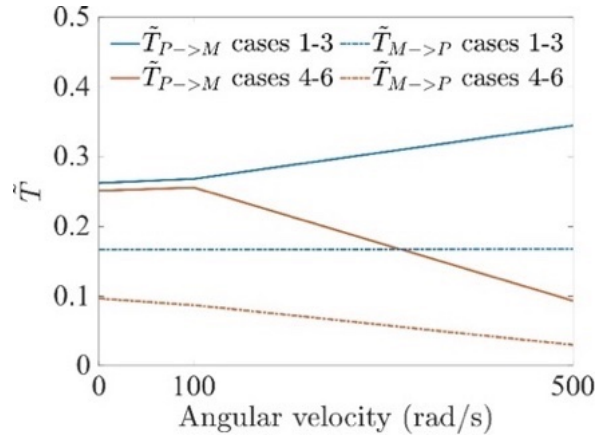


Figure 2: Transfer entropy of first modes for all cases. The solid lines indicate transfer entropy from the pressure field to the momentum field. The dash lines indicate transfer entropy from the momentum field to the velocity field. The blue lines represent cases 1-3 and the red lines represent cases 4-6. [4]

A new model for predicting the radius of a droplet's turning line during impact on a rigid wall was introduced. As shown in Fig. 3, this model, which is different to traditional approaches, showed enhanced accuracy across various impact velocities and droplet sizes by incorporating energy equilibrium equations and calibrated data. Furthermore, the pressure evolution within a droplet when it impacts at high velocity onto a cylindrical pillar has been analysed in Liu et al. (2024) [4].

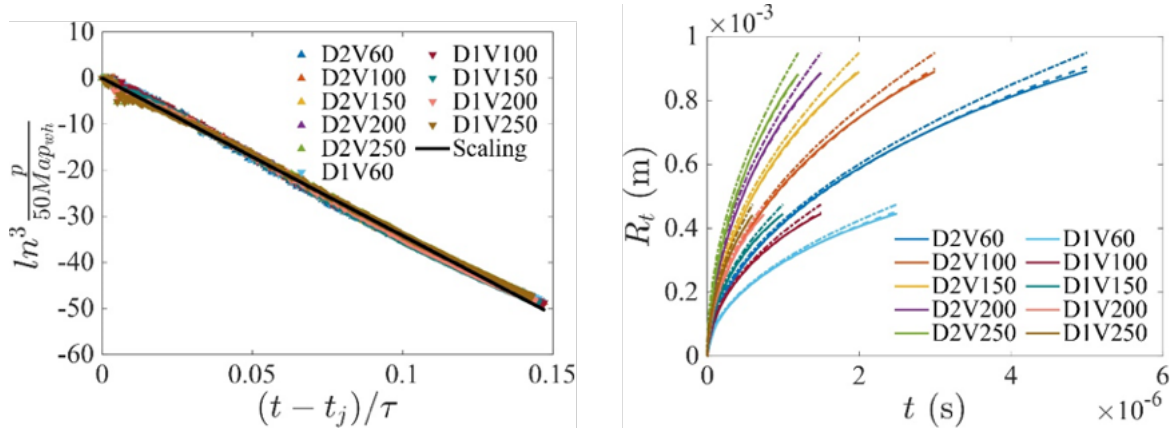


Figure 3: New model proposed in the present project: (a) Linear scaling of the maximum pressure in the contact region over time; (b) Depiction of the radius of the turning line's evolution over time for all cases. Solid lines denote simulation data. Dashed lines represent the current model. Dash-dotted lines correspond to the incompressible model [4].

Summary

The project investigates high-velocity liquid dynamics, particularly in medical inhalers, using simulations due to the challenges of experimental studies at high velocities. Large-Eddy Simulations (LES) reveal a strong correlation between the oscillatory behaviour of an annular liquid jet and the supersonic gas jet's velocity and pressure. This finding is crucial for understanding atomization in inhalers. The study also examines the impact of a swirling supersonic gas stream on the liquid sheet, revealing significant instabilities, particularly in the nozzle's near-field. Additionally, a new approach to model the radius of a droplet's turning line during high-velocity impacts was developed, showing improved accuracy for various velocities and droplet sizes. This comprehensive study enhances the understanding of fluid dynamics in medical and industrial applications.

References

- [1] Jasak, H. (1996). Error analysis and estimation for the finite volume method with applications to fluid flows. Direct, M.
- [2] Liu, Y., Geppert, A., Chu, X., Heine, B., and Weigand, B. (2021). Simulation of an annular liquid jet with a coaxial supersonic gas jet in a medical inhaler. *Atomization and Sprays*, 31(9):95–116.

[3] Liu, Y., Chu, X., Wang, W., and Weigand, B. (2023). Large-eddy simulation, convective instability, and modal causality of coaxial supersonic air–water jets considering a swirl effect. *Physics of Fluids*, 35(6):063316.

[4] Liu, Y., Chu, X., Yang, G., and Weigand, B. (2024). Simulation and analytical modeling of high-speed droplet impact onto a surface. *Physics of Fluids*, 36(1):012137.

3. **Comments on the qualification programme and supervision strategy:**

The core courses in my master's program provided me with the essential knowledge required to understand the complexities of fluid dynamics and various literatures. The specialized courses offered during the IRTG were tailored to address specific problems encountered in my doctoral project. Additionally, the regularly held multi-phase flow seminar afforded me opportunities to engage with international researchers in this field, thereby enriching my research project with new insights. I met regularly with my supervisor, Professor Bernhard Weigand, whose feedback was instrumental in guiding my efforts towards the successful completion of the project. At the DROPIT seminars and the institute colloquium, I had the chance to present the progress of my work and receive feedback from other group members, professors, and colleagues. During the summer school, collaboration with other group members led to results that were subsequently published. The partnership with Pari GmbH proved to be beneficial, offering insights into how multiphase flow is applied in the industry. Throughout the entire program, these experiences enabled me to successfully complete my doctoral project.

Participation in the following courses/seminars of the Int. Research Training Group:

- Analytical and numerical methods
- Dimensional analysis
- Compressible flows I & II
- Numerical fluid mechanics
- Turbulence
- Introduction to Discontinuous Galerkin Methods, Institute of Aerodynamics and Gas Dynamics, Stuttgart, 8-9 July 2019
- Short Course on Atomization and Sprays, Technische Universität Darmstadt, Darmstadt, 17-20 February 2020

Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university:

Stays abroad have not been possible because of the Corona pandemic. However, cooperation with colleagues from other institutes, such as SimTech, the Max Planck Institute in Stuttgart, and Shanghai Jiao Tong University, has been fruitful.

Participation in the following International Research Training Group events:

- Doctoral Seminar 15-16/05/2019 in Bergamo
- Doctoral Seminar 25-26/06/2020 in Stuttgart
- Doctoral Seminar 8-9/09/2021 in Stuttgart
- DROPIT Summer School Kick-Off Meeting on 10/06/2022 in Stuttgart
- DROPIT Summer School 2022 30/06/2022 in Bergamo

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (IT), 17/05/2019
- Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo (IT), 01/07/2022

4. **Individual publications:**

I. Peer-reviewed articles:

- Liu, Y., Geppert, A., Chu, X., Heine, B., and Weigand, B. (2021). Simulation of an annular liquid jet with a coaxial supersonic gas jet in a medical inhaler. *Atomization and Sprays*, 31(9):95–116.
- Liu, Y., Chu, X., Wang, W., and Weigand, B. (2023). Large-eddy simulation, convective instability, and modal causality of coaxial supersonic air–water jets considering a swirl effect. *Physics of Fluids*, 35(6):063316.
- Liu, Y., Chu, X., Yang, G., and Weigand, B. (2024). Simulation and analytical modeling of high-speed droplet impact onto a surface. *Physics of Fluids*, 36(1):012137.

II. Presentations

a) Oral

- Y. Liu. Droplet interaction at high velocities. Doctoral Seminar, Bergamo, Italy, 15.05.2019
- Y. Liu. Droplet interaction at high velocities. Doctoral Seminar, Stuttgart, Germany, 25.06.2020
- Y. Liu. Droplet interaction at high velocities. Doctoral Seminar, Bergamo, Italy, 09.09.2021
- P. Palmetshofer, W. Ren, Y. Liu. Experimental and Numerical Investigation of Droplet Impacts onto Pillars with a Wetted Base. Summer School, Bergamo, Italy, 30.06.2022

Report of associated doctoral researcher (SP-AS05)

Project number: Associated Project 05

First and last name of doctoral researcher: Melanie Lipp

(Working) title of doctoral project: Capturing local details in fluid-flow simulations: options, challenges and applications using marker-and-cell schemes

Name of supervisors: Prof. Rainer Helmig (Universität Stuttgart, Germany), Prof. Claus-Dieter Munz (Universität Stuttgart, Germany), Prof. Ivan Yotov (University of Pittsburgh, USA)

1. General information:

Duration of International Research Training Group membership:
Associated member of DROPIT since March 2019

Funding (type and duration):

- September 2017: Employee at the Institute for Modelling Hydraulic and Environmental Systems
- October 2017-December 2020: Scholarship of the Landesgraduiertenförderung Baden-Württemberg + 10.12 h/week employee at the Institute for Modelling Hydraulic and Environmental Systems
- January 2021 - April 2022: Employee at the Institute for Modelling Hydraulic and Environmental Systems
- May 2022 – August 2022: Employee of the University of Stuttgart
- September 2022-August 2024: Employee of the Collaborative Research Center 1313

Start of doctoral training and (anticipated) date of doctoral degree:
28.06.2017 – 28.06.2024

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2011 Physics, University of Stuttgart, B.Sc., 01 August 2014
08/2014 Physics, University of Missouri, St. Louis, (MO, USA), MS., 19 December 2015
10/2014 Physics, University of Stuttgart, M.Sc., 06 April 2017

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Aims of the project

Complex local flow structures appear in a wide range of free-flow systems, e.g. vortices build after obstacles. For understanding and predicting numerous processes, it is important to capture local details in free fluid flow, which is the focus of this work. Particularly, we are interested in local flow structures in free-flow coupled to porous-medium flow. A better resolution of local structures in free flow can be achieved by refining computational grids, which is studied in this project. Also, in the case of stair-shaped boundaries of drop-shaped obstacles, refining computational grids helps to better capture the boundary shape.

Particularly, we focus on finite-volume and finite-difference methods for the two-dimensional Navier-Stokes equations with constant density and constant viscosity, using the marker-and-cell method (pressures in cell centers, velocities on cell faces) and rectangular control volumes.

There exists a variety of methods, with a range of characteristics, which can be used to refine computational grids.

The first objective of this work is to develop for many different available approaches (the just-mentioned ones and related methods) one common way of description of the class of methods within our focus and to display their similarities and differences. The second objective is to gain insight and in-detail understanding of the local-refinement-methods' behavior by examining one chosen method before numerical solution, i.e. by examining the local truncation errors. The third objective is to gain further understanding of the local-refinement-methods' behavior as well as display examples, in which the chosen method is beneficial when neglecting computational-efficiency issues, by examining our chosen method after numerical solution, i.e. by examining the actual numerical solutions.

Current project status and results

Systematization and analysis of a class of free-flow refinement methods

We start by presenting different equations that the methods consider, options to design and handle quadtree grids, possibilities for the arrangement of the unknown pressures and unknown velocities -- we call them computational variables -- and options to define control volumes. We present the way, the equations are discretized: by first integrating over the control volumes and then discretizing similar to how one discretizes on a regular grid but using stencils, which may contain variables at points at which no computational variables are located - so-called virtual variables. Various choices of the positions of the (possibly-virtual) variables are possible and options are presented. Virtual variables are approximated from computational variables. Similarly, some methods within the scope of the literature review require additional approximations. Many options for those approximations, different choices of different authors and some consequences of the choices are presented. Apart from the choices when approximating virtual variables from computational variables, also the choices regarding the positioning of possibly-virtual variables or the choices regarding the discretization of the fluxes can influence if mass and momentum are conserved locally. Sometimes, further approaches are followed to design a conservative discretization, which are presented and compared.

Our chosen spatial discretization method

We choose a method, treating the current locally-refined grid as one grid, disregarding the grid's predecessors before refinement, using two velocity computational variables where one coarse cell touches two fine ones, defining a single pressure computational variable in any grid-cell center, using as scalar-associated control volumes the grid cells and using as vector-component-associated control volumes two rectangular control volumes where one coarse cell touches two fine ones.

Examining our chosen spatial discretization method before numerical solution

For our chosen method, local refinement contributes to local truncation errors - the differences between the continuous terms from the continuous partial differential equation and the discretized versions of those terms evaluated for the expected solutions - due to the following contributions, which are of the same order of magnitude in our example: due to not placing certain stencil variables symmetrically with respect to certain axes, due to interpolations of virtual variables and due to enforcement of conservation. The theoretical and numerical grid-convergence analysis of local truncation errors of individual control volumes for our chosen method gives: In the inner of a non-refined regular grids, the rate is 2 (Stokes) or 1 (Navier-Stokes). In the inner of non-regular rectilinear grids, without local refinement, the rate is in general 0, when treating the lateral diffusion term with a common approach. With local refinement, rates of individual terms can decrease depending on local-refinement method choices. Boundary effects can lead to decreased rates in any case. The grid convergence analysis of sums of local truncation errors over all control volumes for our chosen method and one numerical example gives: The rates are equal to the worst of the involved individual-control-volume rates plus 1, but maximally the ones of the inner of regular grids.

When introducing local refinement by dividing some cells of a non-refined regular grid, local truncation errors tend to decrease in the refinement region as cell sizes decrease (grid convergence) and tend to increase due to the above-mentioned contributions (non-symmetric stencil variable placement, interpolation and enforcement of conservation) at the interface between the grid parts with different degree of refinement (refinement interface). Depending mostly on the question if the refinement interfaces are positioned in regions of (practically-) constant primary variables, the increase can significantly exceed the decrease despite of the larger number of computational variables. We also saw examples, in which the decrease exceeds the increase - sometimes enough to let a quadtree grid have better local truncation errors than a Cartesian reference grid with more computational variables. In quadtree grid convergence studies, in cases without first-order upwind convective terms and with zeroth-order local truncation errors, the increase at refinement interfaces becomes more and more important compared to the decrease as all grid cells are successively divided.

Examining our chosen spatial discretization method after numerical solution

The solution errors -- the differences between the solutions of the discrete equations and the expected primary variables -- are discussed with respect to, on the one hand, the solution-error rates and, on the other hand, the solution-error changes when dividing some cells of a non-refined regular grid. The solution-error rates are sometimes larger than the local-truncation error rates -- a phenomenon called supraconvergence. As to the changes when dividing some cells of a non-refined regular grid, the increase can again exceed the decrease despite of the larger number of computational variables. We again saw examples, in which the decrease exceeds the increase. For an example with refinement interfaces positioned in regions of practically-constant primary variables, the decrease exceeded the increase enough to let a quadtree grid overall have better solution errors than a Cartesian reference grid with more computational variables. From the behavior of local truncation errors, guesses can be made about the behavior of solution errors. However, the behavior is not always the same.

Apart from an example resulting in better solution errors, we show an example in which an existing vortex flow in cavities can be observed by a quadtree grid which can not be observed by a Cartesian grid with more computational variables.

We, furthermore, show that a quantity of interest -- a mean deviation of the pore pressures from reference values -- may be approximated better with a quadtree grid than with a Cartesian grid with more computational variables. The context is a free flow coupled to a flow through a pore network.

Conclusion

Overall, finite-volume and finite-difference methods for the Navier-Stokes equations with constant density and constant viscosity on refined grids with pressures in cell centers, velocities on cell faces and rectangular control volumes can be used to resolve local complex flow structures. However, it has to be kept in mind, that, when refinement interfaces are not put into regions of (practically) constant primary variables, refinement-interface-specific contributions (not only from interpolations but also from stencil asymmetry or enforcing conservation) can easily dominate local truncation errors, especially without first-order upwind convective terms as grids get finer everywhere.

Anticipated work schedule

A journal article as well as a dissertation will be handed in by mid March 2024, the thesis defense is planned for June 2024.

Networking with other IRTG projects

Some insights of this project regarding refinement using non-regular rectilinear, which have not been locally refined, could be applied to the work of Maziar Veyskarami (SP-B6). Similarities and differences of the respective local refinement methods have been discussed with Pascal Mossier (SP-A2).

3. Comments on the qualification programme and supervision strategy:

The core courses were a great help for me, as a Physicist, to dive faster into the main field of my doctoral thesis -- Numerics. Additionally, I had the chance to gain knowledge about the Physics of the system that I am studying. The specialised courses helped me to dig deeper into the aforementioned topics, as well as to expand my methodological skills of C++ programming. Many softskill courses provided valuable tools, reaching beyond my research work into everyday life. Most recently, the course "Karrierekompass Wirtschaft" helped me develop further career goals as well as taught me ways towards reaching them.

The concept of having core courses throughout the semester, while specialized courses are treated in short full-day courses allowed me both to digest the new basic topics and to attend a specialised course, although situated in another city. Choosing specialised and soft skill courses myself increased my motivation.

My second examiner, Prof. Munz, from aerospace engineering and my third examiner, Prof. Yotov, from Mathematics, build a valuable addition to the supervision by my main advisor, Prof. Helmig, adding the perspective of their respective fields. I enjoyed discussing with Prof. Munz in the framework of Dropit and with Prof. Yotov during his several months of stay in Stuttgart.

DROPIT provides a great opportunity for me to discuss numerical methods for free-flow systems, complementing my mostly porous-medium-flow work environment.

Participation in the following courses/seminars of the Int. Research Training Group: • 10.04.-21.07.2017 Fluid Mechanics I • 10.04.-21.07.2017 Modeling of Hydrosystems • 16.10.2017-09.02.2018 Fluid Mechanics II • 09.04.-20.07.2018 Environmental Fluid Mechanics II • 18./19.09.2017: Multiscale Methods for Multiphase Flow in Fractured Porous Media • 26.2.-2.3.2017: DUNE workshop • 14.-17.5.2018: Advanced C++ with Focus on Software Engineering • 09.04.-20.07.2018 Stochastic and Statistical Topics in Modeling and Simulation • 05.-06.02.2019 Solvers for nonlinear and/or coupled problems in porous media • 03.-04.06.2019 Fundamentals of multiphase flow in porous media • 08.-09.07.2019 Introduction to Discontinuous Galerkin (DG) methods • 26.09.2018 Basics of persuasion • 04.04.-19.12.2018 Leadership in academia • 22.6.-25.6.2020 Scientific writing • 30.03.-25.07.2023: Karrierekompass Wirtschaft
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • 01.-02.04.2019 Workshop on evaporation across the free-flow porous medium interface and salt precipitation, DROPIT and SFB1313, Stuttgart, Germany • 2019, 2020, Spring 2021, Autumn 2021, 2023 DROPIT Doctoral Seminar
Participation in conferences, congresses, etc., at home and abroad: • 26.-28.03.2018 Conference on Simulation Technology, Stuttgart, Germany • 6.-10.05.2019 InterPore, 11th Annual Meeting of the international society of porous media, Valencia, Spain • 15.-19.07.2019 ICIAM, 9th International Congress on Industrial and Applied Mathematics, Valencia, Spain • 11.01.-15.01.2021 WCCM, joint 14th World Congress in Computational Mechanics and ECCOMAS Congress
• 31.05.-04.06.2021 InterPore 13th International Conference on Porous Media • 21.-24.06.2021 SIAM, Conference on Mathematical & Computational Issues in the Geosciences

4. Individual publications:

I. Submitted articles:

- **Lipp M.**, Helmig R (2019): "A locally-refined quadtree finite-volume staggered-grid scheme". Book chapter in a Springer book of DROPIT.
- Coltman, E., Vescovini, A., **Lipp, M.**, Fetzer, T. Helmig, R. (2019): "Turbulence, Interfacial Forms, and Obstacles: Numerical Analysis of Soil-Water Evaporation". Submission planned for WRR.
- Fraundorf, P., Hundley, T., and **Lipp, M** (2019). "Synthesis of unlayered graphene from carbon droplets: In stars and in the lab.", *ApJ Letters*.
- Koch, T., Gläser, D., Weishaupt, K., Ackermann, S., Beck, M., Becker, B., Burbulla, S., Class, H., Coltman, E., Emmert, S., Fetzer, T., Grüninger, C., Heck, K., Hommel, J., Kurz, T., **Lipp, M.**, Mohammadi, F., Scherrer, S., Schneider, M., ... Flemisch, B. (2021). DuMux 3 – an open-source simulator for solving flow and transport problems in porous media with a focus on model coupling. *Computers & Mathematics with Applications*.

II. Further publications

a) Proceedings

- Fraundorf, P., **Lipp, M.**, and Savage, T. "Analogues for Unlayered-Graphene Droplet-Formation in Stellar Atmospheres." *Microscopy and Microanalysis* 22.S3 (2016): 1816-1817.

- **Lipp, M.**, Savage, T., Osborn, D., and Fraundorf, P. "Laboratory evidence of slow-cooling for carbon droplets in red-giant atmospheres." *Microscopy and Microanalysis* 23.S1 (2017): 2192-2193.

b) Technical and scientific reports

- Fraundorf, P., Hundley, T., and **Lipp, M.** "Synthesis of unlayered graphene from carbon droplets: In stars and in the lab.", open e-print archive HAL, hal-02238804 (2019).

III. Presentations

a) Oral

- 6.-10.05.2019 InterPore, 11th Annual Meeting of the international society of porous media, Valencia, Spain, Talk "Coupling free flow and porous-medium flow: Comparison of solution strategies for solving saddle-point problems"
- 11.01.-15.01.2021 WCCM, joint 14th World Congress in Computational Mechanics and ECCOMAS Congress
- 21.-24.06.2021 SIAM, Conference on Mathematical & Computational Issues in the Geosciences

b) Posters

- 19.-21.6.2018: Summer school on Domain Decomposition Methods, Laboratoire Dieudonné, Nice, France, "Adaptive Staggered 2D Grids for DuMuX - Plans/Ideas"
- 26.-28.03.2018: 2nd International Conference on Simulation Technology, Stuttgart, Germany, "Adaptive Staggered 2D Grids for DuMuX - Plans/Ideas"
- 15.-19.07.2019 ICIAM, 9th International Congress on Industrial and Applied Mathematics, Valencia, Spain, "A locally-refined quadtree finite-volume staggered-grid scheme"
- 31.05.-04.06.2021 InterPore 13th International Conference on Porous Media

Report of doctoral researcher (SP-AS06)

Project number: Associated Project 06

First and last name of doctoral researcher: Dennis Hoch

(Working) title of doctoral project: A multi-scale approach for modeling and simulation of aerosol filtration

Name of supervisors: Prof. Rainer Helmig, Prof. Bernard Weigand, Prof. Jennifer Niessner

1. General information:

Duration of International Research Training Group membership:

Doctoral researcher in DROPIT: October 2019-December 2023

Funding (type and duration):

(various publicly funded projects at Heilbronn University of Applied Sciences, MWK BW / EFRE)

Start of doctoral training and (anticipated) date of doctoral degree:

October 2019 – (dropped out Dec. 2023)

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

B.Sc. Mechanical Engineering, Heilbronn University of Applied Sciences, 2017

M.Sc. Mechanical Engineering, Heilbronn University of Applied Sciences, 2019

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: *industry*

2. Description of doctoral project and research results achieved to date:

Motivation & Aim of the project

Liquid aerosol particles are generated in numerous industrial processes—such as machining operations, pneumatic brake systems of trucks and buses, and other mechanical applications—where they can pose significant risks to human health, the environment, and safety. Similar to fine dust, these mist aerosols can penetrate deep into the pulmonary system, potentially causing cancer, respiratory diseases, and allergies. Consequently, filtration of such aerosols is essential, and current mist filtration systems, while effective, are often energy-intensive and still struggle to efficiently remove harmful droplets of all sizes.

Research over past decades has predominantly focused on solid aerosols, where capture mechanisms are well understood. In contrast, liquid aerosol filtration is substantially more complex: droplets colliding with fibers form films that coalesce into a continuous liquid phase, eventually creating multiphase flow conditions governed by capillary and permeability effects [1-3], see Fig. 1. These processes occur across several length scales—from droplet-fiber interactions at the micro-scale to fluid redistribution throughout an entire filter element at the macro-scale. The complexity and computational cost of resolving all scales have historically limited liquid aerosol filtration research largely to experimental studies.

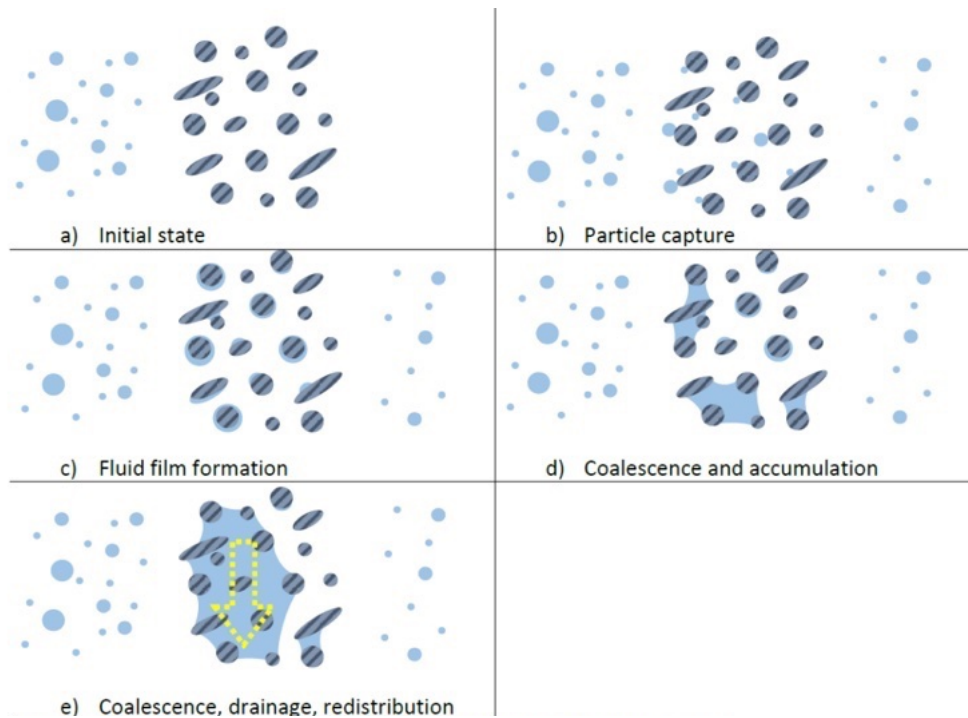


Fig. 1: Processes occurring during oil mist filtration – from particle capture over fluid film formation, coalescence to drainage and redistribution

Recent advances in imaging and computational power now enable a more physics-based and multiscale approach to this problem [4,5]. However, no existing model combines droplet transport, capture, coalescence, and resulting pressure loss in realistic fibrous structures without relying on empirical or phenomenological assumptions.

The overall aim of the project is therefore to develop a physics-based, multiscale simulation framework for liquid aerosol filtration that:

- Uses μ CT-derived geometries of real filter media as a basis
- Determines macro-scale parameters and constitutive relationships directly from micro-scale structures and simulations
- Models continuous-phase flow of the gas phase and dispersed liquid droplets from fundamental conservation laws
- Simulates coalescence and liquid-film formation
- Couples micro-scale predictions of filtration efficiency and liquid distribution to macro-scale simulations and
- Enables optimization of industrial filter designs by predicting pressure drop and efficiency evolution, especially the pressure drop at steady state.

In summary, the project seeks to produce a comprehensive numerical model capable of simulating the essential processes of liquid aerosol filtration in realistic 3D filter structures.

Current status and achieved results

Micro-scale modeling has been successfully implemented using a two-phase Eulerian–Lagrangian approach, where gas (air) acts as the incompressible continuous phase and droplets are tracked using an Lagrangian approach. Validation against the empirical single-fiber efficiency theory demonstrated excellent agreement, with both structured and unstructured meshes reproducing the expected behavior. Differences in the location of the “most penetrating particle size” (MPPS) were traced to geometric representation details [6].

High-resolution μ CT scans of fibrous media have been processed into voxel-based and both surface- and volume meshes suitable for numerical simulations, see Fig. 2. The workflow—from CT image acquisition, threshold segmentation, voxel stacking, surface extraction, and smoothing—was demonstrated and validated.

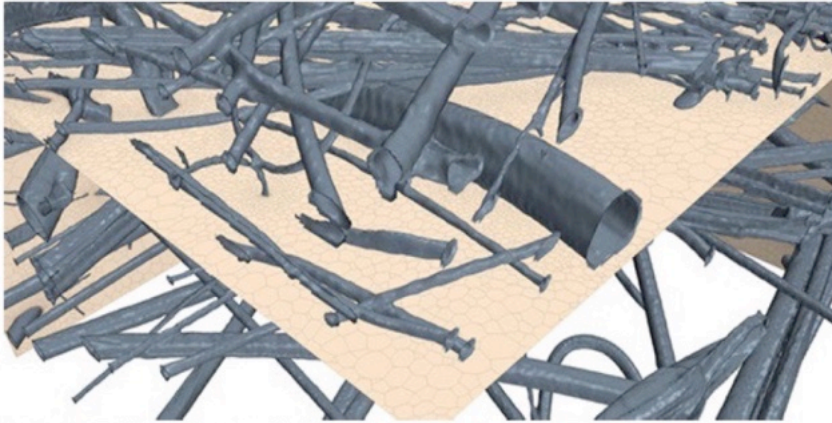


Fig. 2: Exemplary volume mesh of a fibrous filter medium, scanned by μ CT [6].

Micro-scale simulations of a real face-mask filter medium showed strong agreement with experimental measurements. Specifically, pressure drop and fractional filtration efficiency—after electrostatic discharge via IPA treatment—matched experimental data closely, demonstrating that the physically based simulation approach is accurate for real porous materials [7].

Filtration efficiency matrices were generated to connect micro-scale results with macro-scale simulations. These matrices express efficiency as a function of saturation, droplet size, and flow velocity. A 5×5 parameter sweep for each droplet size enabled smooth interpolation during macro-scale simulations.

Coalescence modeling was carried out using a volume-of-fluid (VOF) method and a novel distance-map technique (in cooperation with Andreas Weber, Math2Market). This approach accurately handled droplet capture, surface film formation, and coalescence, and was validated using analytical and numerical benchmarks such as capillary rise and Surface Evolver geometry comparisons [8].

A macro-scale two-phase flow model [9] was implemented in OpenFOAM using the modified porousMultiphaseFoam toolbox. A passive scalar was introduced to represent aerosol concentration, along with physically motivated source terms describing droplet capture and mass transfer to the liquid phase. The whole workflow is illustrated in Fig. 3. Flow and transport processes in a full-scale filter cartridge section was simulated, achieving a filtration efficiency within 0.02% of the analytical prediction, demonstrating excellent numerical performance. Contour and line-profile analyses showed physically consistent distributions of aerosol concentration, liquid saturation, and source-term magnitudes throughout the filter domain.

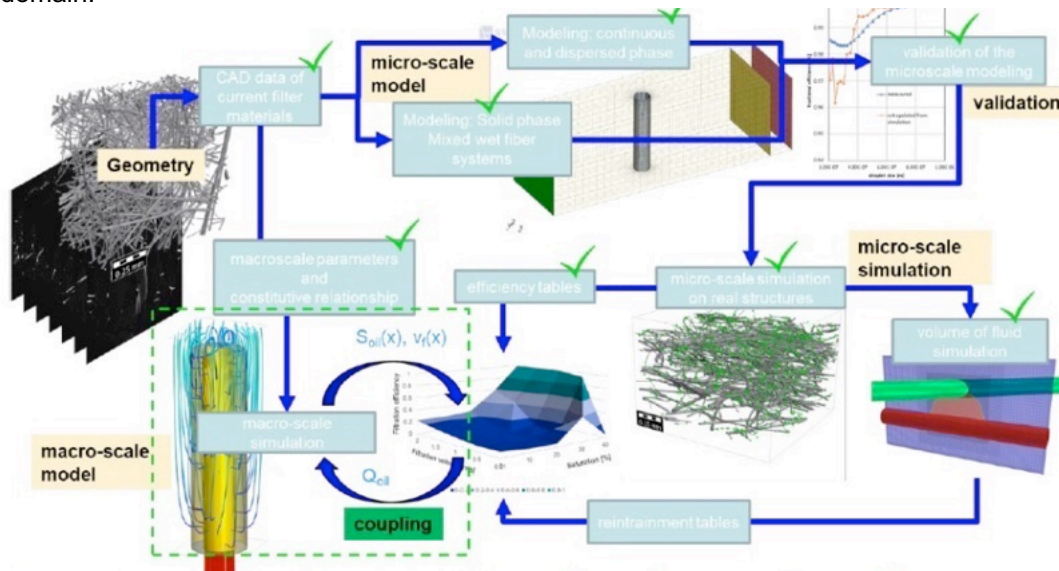


Fig. 3: Workflow from μ CT scans over micro-scale modelling and simulation to macro-scale modelling and simulation of oil mist filtration

Outlook

The next steps focus on enhancing physical fidelity and expanding the multiscale capability of the model:

1. A physically based IMPESC formulation will replace the preliminary passive-scalar implementation. This will yield fully coupled mass-balance interactions between aerosol

- concentration, liquid deposition, and air–liquid flow fields. Accurate computation of the incoming scalar value is essential for predictive accuracy.
2. The model will be extended to multiple droplet sizes, enabling simultaneous representation of a full aerosol spectrum. Each droplet class will have its own transport equation and capture source term, allowing summation of mass contributions to the liquid phase.
 3. Although re-entrainment is generally insignificant in the intended operating regimes, the framework allows for future inclusion if specific applications require it.
 4. Macro-scale model validation will proceed using experiments where μ CT geometries are available, allowing direct comparison of time-dependent saturation, total deposited mass, and steady-state efficiency.
 5. Once validated, the simulation framework will be applied to the optimization of full-scale filter elements. This includes exploring fiber-layer arrangements, modifying material properties, and minimizing pressure drop while increasing filtration efficiency.
 6. Together, these steps will refine the predictive capability of the multiscale model and expand its applicability to real-world industrial filtration challenges.

Stays abroad

- None -

References

- [9] Charvet, A.; Gonthier, Y.; Bernis, A.; Gonze, E. (2008): Filtration of liquid aerosols with a horizontal fibrous filter. In Chemical Engineering Research and Design 86 (6), pp. 569–576. DOI: 10.1016/j.cherd.2007.11.008.
- [10] Frising, T.; Thomas, D.; Bémer, D.; Contal, P. (2005): Clogging of fibrous filters by liquid aerosol particles: Experimental and phenomenological modelling study. In Separation and Purification 60, pp. 2751–2762.
- [11] Hinds, W. (1999): Aerosol technology - Properties, behaviour and measurement of airborne particles: John Wiley & Sons: New York.
- [12] King, A.J.C.; Mullins, B. J.; Ladenburger, G. (2010): Hybrid volume-of-fluid and discrete particle solver for oil-mist filter simulations. In : 17th Australasian Fluid Mechanics Conference.
- [13] Mead-Hunter, R.; King, A.J.C.; Kasper, G.; Mullins, B. J. (2013): Computational fluid dynamics (CFD) simulation of liquid aerosol coalescing filters. In Journal of Aerosol Science 61, pp. 36–49.
- [14] Hoch, Dennis; Azimian, Mehdi; Baumann, André; Behringer, Jens; Niessner, Jennifer (2020): Comparison of Voxel-Based and Mesh-Based CFD Models for Aerosol Deposition on Complex Fibrous Filters. In Chem. Eng. Technol. 43 (12), pp. 2538–2547. DOI: 10.1002/ceat.202000318.
- [15] Berger, Simon; Azimian, Mehdi; Eichheimer, Philipp; Hoch, Dennis; Cheng, Liping; Niessner, Jennifer (2023): CFD based optimization of face mask filter media based on parametric studies using μ CT-scans. In Chem. Eng. Technol., Article ceat.202200460. DOI: 10.1002/ceat.202200460.
- [16] Hoch, D., Weber, A., and J. Niessner, Modeling and simulation of coalescence in the context of oil mist filtration, Transport in Porous Media, 2023, doi: 10.1007/s11242-023-01908-6.
- [17] Helmig, R. (1997): Multiphase Flow and Transport Processes in the Subsurface: A Contribution to the Modeling of Hydrosystems: Springer.

3. Comments on the qualification programme and supervision strategy:

With the supervisor of my home institution, Prof. Niessner, I had bi-weekly meetings for feedback and discussion on my research. With my IRTG supervisors, Prof. Helmig and Prof. Weigand, I extensively exchanged during the IRTG doctoral seminars. The interaction with the IRTG members deepened my understanding of the underlying processes relevant to my research and allowed me to expand my professional network. Engaging in interdisciplinary collaboration enhanced my scientific communication skills, allowing me to effectively discuss my topic using diverse terminology. Getting familiar with a different method and discussing the relation of a topic to my own research project helped me.

The scheduled mid-term review provided a valuable opportunity to receive feedback from all of my supervisors, helping me assess my progress and clarify the next steps toward completing my PhD. Due to personal reasons and with a heavy heart, at the end of 2023, I decided to abandon my plan to pursue a PhD and took up an employment in industry.

Participation in the following courses/seminars of the Int. Research Training Group: • <i>none</i>
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: • <i>none</i>
Participation in the following International Research Training Group events: • May 2018: DROPIT Seminar and DIPSY workshop, Bergamo • September 2021: DROPIT doctoral seminar, Bergamo • March 2023: Doctoral Seminar, Stuttgart
Participation in conferences, congresses, etc., at home and abroad: • February 2023: FILTECH Conference • March 2022: FILTECH Conference • October 2021: GeoDict user meeting • September 2020: GeoDict user meeting • October 2019: FILTECH Conference • September 2019: GeoDict user meeting • May 2019: InterPore Conference 2019

4. Individual publications:

I. Peer-reviewed articles:

- Baumann, A., Hoch, D. and Niessner, J. (2023), Fates of Emitted Particles Depending on Mask Wearing Using an Approach Validated Across Spatial Scales. Global Challenges 2300008. <https://doi.org/10.1002/gch2.202300008>
- Hoch, D., Weber, A., and J. Niessner, Modeling and simulation of coalescence in the context of oil mist filtration, Transport in Porous Media, 2023, doi: 10.1007/s11242-023-01908-6.
- Berger, S., Azimian, M., Eichheimer, P., Hoch, D., Cheng, L. and Niessner, J. (2023), Computational Fluid Dynamics-Based Optimization of Face Mask Filter Media Using Micro-Computed Tomography Scans. Chem. Eng. Technol., 46: 1039-1046. <https://doi.org/10.1002/ceat.202200460>
- Hoch, D., Azimian, M., Baumann, A., Behringer, J. und J. Niessner: Comparison of Voxel-Based and Mesh-Based CFD Models for Aerosol Deposition on Complex Fibrous Filters. Chem. Eng. Technol., 43: 2538-2547, 2020. <https://doi.org/10.1002/ceat.202000318>
- Baumann, A., Hoch, D., Behringer, J. und J. Niessner: Macro-scale modeling and simulation of two-phase flow in fibrous liquid aerosol filters. Engineering Applications of Computational Fluid Mechanics, 2020, <https://doi.org/10.1080/19942060.2020.1828174>

II. Further publications

a) Proceedings

- Bittel, L., Jüngling, N., Hoch, D. and J. Niessner: Generation of datasets using POD for artificial intelligence to predict and control an energy-efficient HVAC system. Proceedings of the FILTECH Conference 2023, Cologne, Germany.
- Baumann, A., Hoch, D. and J. Niessner: Investigation of the near-field and far-field protection effect of face masks using micro-scale and macro-scale CFD simulations. Proceedings of the FILTECH Conference 2022, Cologne, Germany.
- Hoch, D., Weber, A. and J. Niessner: Modeling and simulation of coalescence in the context of oil mist filtration using a distance map. Proceedings of the FILTECH Conference 2022, Cologne, Germany.
- Hoch, D., Azimian, M., Baumann, A., Behringer, J. and J. Niessner: Comparing mesh-based and voxel-based CFD models for aerosol deposition in complex fibrous filters. Proceedings of the FILTECH Conference 2022, Cologne, Germany.
- Hoch, D., Baumann, A. und J. Niessner: Modeling and simulation of liquid aerosol deposition in textile filters, Proceedings of the Sixth bwHPC Symposium, September 30th 2019, KIT, Karlsruhe, Germany.

- Hoch, D., Baumann, A. und J. Niessner: Micro-Scale Simulation of Fibrous Liquid Aerosol Filters, Proceedings of the FilTech Conference 2019, October 22nd-24th 2019, Cologne, Germany.
- Baumann, A., Hoch, D. und J. Niessner: Macro-Scale Simulation of Fibrous Liquid Aerosol Filters, Proceedings of the FilTech Conference 2019, October 22nd-24th 2019, Cologne, Germany.

III. Presentations

a) Oral

- Modeling and simulation of coalescence in the context of oil mist filtration using a distance map. FILTECH conference, Cologne (16.2.2022).
- Comparing mesh-based and voxel-based CFD models for aerosol deposition in complex fibrous filters. FILTECH conference, Cologne (10.3.2022).
- Modeling and simulation of coalescence in the context of oil mist filtration using a distance map. GeoDict user meeting, Kaiserslautern (06.10.2021).
- Multiscale modelling and simulation of liquid aerosol filtration. GeoDict user meeting, Kaiserslautern (21-23.09.2020).
- Micro-scale simulation of fibrous liquid aerosol filters. FILTECH conference, Cologne (23.10.2019).
- From μ CT scans to optimized mist filters. GeoDict user meeting, Kaiserslautern (24.09.2019).

b) Posters

- Micro-scale simulation of fibrous liquid aerosol filters. Pitch & poster presentation at the InterPore Conference, Valencia (08.05.2019).

Report of associated doctoral researcher (SP-AS07)

Project number: Associated Project 07
First and last name of doctoral researcher: Luca Verzeroli
(Working) title of doctoral project: Nonconforming Finite Elements Methods for Nonlinear Elasticity Problems Featuring Finite Deformations And Frictionless Contact Constraints
Name of supervisors: Prof. F. Bassi, Prof. L. A. Botti

1. General information:

Duration of International Research Training Group membership: Associated member of DROPIT since 01.10.2018
Funding (type and duration): Project founded by Serioplast, duration three years
Start of doctoral training and (anticipated) date of doctoral degree: 01.10.2018 to 31.09.2021
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): Mechanical Engineering, University of Bergamo, Bachelor's diploma, 30.09.2015 Mechanical Engineering, University of Bergamo, Master's diploma , 21.12.2017
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Motivation

Extrusion Blow Moulding (EBM) and Stretch Blow Moulding (SBM) are two of the main manufacturing processes involved in the production of rigid plastic packages for the home care, personal hygiene, food and beverage distribution and automotive markets. EBM and SBM consist in deforming and blowing a hot plastic preform of standardized shape until contact with a mould is reached. After contact, the plastic cools down and solidifies providing the final desired shape of the plastic container. EBM and SBM differ with regard to the type and the shape of the polymers introduced in the blowing machine. In EBM, the polymeric raw material is mixed with additives and pigments, melted and extruded through a die in the shape of a parison, that is cylindrical tube-like piece of plastic. On the other hand, SBM normally employs PET preforms, intermediate injection-moulded products with a fully formed bottle/jar neck and a thick tube attached. Preforms are heated in an oven and then stretched with a rod while the pressurized air is blown.

Before starting the large scale production of new plastic containers, extensive prototyping is required and several process parameters need to be adjusted e.g. polymer temperature, parison thickness, blowing time, cooling time, mould shape.

The quality of products has to be verified by means of mechanical tests designed to evaluate, for example, the tightness ("leak test"), the resistance to axial load ("top load") or the ability to withstand impacts ("drop test"). The mechanical behaviour of plastic packages is not the sole aspect to consider: while thick walls are desirable from the resistance viewpoint they result in heavy packaging that lack cost effectiveness. Mathematical modelling and numerical simulation might be employed for investigating EBM and SBM with the goal to improve efficiency and sustainability of the overall process.

Objectives and scientific approach: computational modelling tools for blow moulding simulations

The blowing process is modelled based on a continuum mechanics approach, in particular we consider the classical problem of seeking the static equilibrium of an elastic body undergoing finite deformations caused by external solicitations. In order to encompass contact with the mould we admit unilateral frictionless contact of the elastic solid with a rigid obstacle. The equation of motion is treated in Lagrangian formulation introducing the first Piola-Kirchhoff stress tensor, a stress measure that describes the response of the body to the external solicitations. To derive discrete versions of the problem that are numerically solvable using a computer we rely on nonconforming finite elements formulations. Nonconforming finite elements methods are gaining popularity in the field of Computational Solid Mechanics (CSM) due to the following attractive features: robustness with respect to mesh distortion, ability to deal with arbitrarily unstructured polytopal elements meshes, possibility to locally increase the accuracy by raising the polynomial degree in those regions where the solution is expected to be smooth, availability of locking-free formulations in the incompressible and nearly-incompressible limits.

Numerical solution strategy

We devise, implement and validate a discontinuous Galerkin (dG) and a Hybridizable Discontinuous Galerkin (HDG) framework for nonlinear elasticity problems featuring large deformations and frictionless contact constraints.

The dG framework [1] is based on a Bassi-Rebay (BR2) formulation [3], encompasses compressible and incompressible hyperelastic materials and is capable of dealing with large deformations. In order to achieve stability, we combine higher-order lifting operators for the BR2 stabilization term with an adaptive stabilization strategy which relies on the BR2 Laplace operator stabilization and a penalty parameter based on the spectrum of the fourth-order elasticity tensor. Dirichlet boundary conditions for the displacement can be imposed by means of Lagrange multipliers and Nitsche's method. Efficiency of the solution strategy is achieved by means of state-of-the-art agglomeration based h-multigrid preconditioners [2] and the code implementation supports distributed memory execution on modern parallel architectures. Several benchmark test cases are proposed in order to investigate some relevant computational aspects, namely the performance of the h-multigrid iterative solver varying the stabilization parameters and the influence of Dirichlet boundary conditions on Newton's method globalisation strategy.

The HDG framework [4] extends the HDG formulation devised by Soon [5] and later applied to solve nonlinear elasticity problems featuring large deformations [6], introducing the possibility to enforce non-penetration frictionless contact constraints. Accordingly, the region of space occupied by a rigid fixed obstacle can be precluded to the deforming body. Signorini contact conditions are handled by means of the Lagrange multipliers method and we consider obstacles admitting an analytical description. The convergence of Newton's method towards the equilibrium configuration is globalized by means of an incremental load method coupled with an active set strategy aimed at identifying and tracking the evolution of the contact interface. In addition to frictionless contact constraints, also Dirichlet boundary conditions can be enforced weakly relying on the Lagrange Multipliers method. Interestingly, as compared with the DG formulation, imposition of boundary conditions by means of Lagrange multipliers does not require to introduce additional stabilization terms.

Results

The accuracy of nonconforming finite elements implementations has been validated by assessing the spatial convergence rates based on smooth manufactured solutions. In order to demonstrate the effectiveness of the computational contact mechanics HDG formulation, we tackle challenging 2D and 3D computations featuring contact with rigid obstacles: we consider impact of a bar with unit radius semicircular cross section and infinite extension against a planar and a circular obstacle and the interaction of thin membrane with a sphere. All computations are carried out relying on second, third and fourth accurate HDG formulations demonstrating the ability to reach high-order convergence rates for the displacement, the displacement gradient and the gap function.

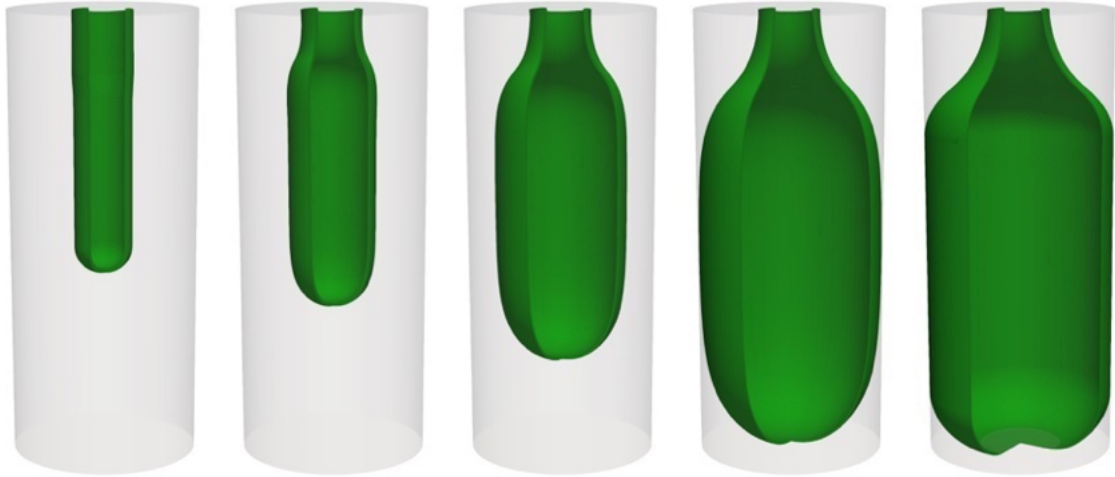


Figure 1: Simulation of preform inflation in a cylindrical cavity, sequence of frames captured along the loading path counting of 300 incremental steps. Results are obtained with a third order accurate HDG formulation over a 16k tetrahedral elements mesh of a realistic preform. The neck of the preform is constrained by imposing a homogeneous Dirichlet boundary condition, non-penetration contact constraints are imposed on the outer surface of the preform (whose outer normal points towards the mould) and normal stresses are prescribed on the internal surface (due the presence of pressurized air blown through the neck).

A simplified blow molding computation is carried out by inflating a hyperelastic preform while enforcing non-penetration contact constraints with respect to a fixed cylindrical obstacle surrounding the preform. Results depicted in Figure 1 and 2 are obtained show that inflation is carried out until contact with the mold is reached. Although the polymer is treated as an hyperelastic material, thereby disregarding its viscoplastic behavior, all the complexity related to large deformation of a thin walled structure in contact with the mould is retained.

Future development

We envisage that the geometrical flexibility of non-conforming formulations can be exploited to further improve robustness and accuracy of contact detection algorithms. In particular, adaptive mesh refinement strategies for the boundary mesh, whose mesh faces are the support of Lagrange multipliers polynomial functions, might be employed to improve the resolution near the boundaries of the contact surface. Thanks to the ability of dealing with polytopal elements meshes, mesh faces that are partially in contact with the body can be subdivided into subfaces without modifying the main computational grid of the elastic body. Specifically, mesh elements whose faces have been refined might be treated as general polytopes with increasingly high number of faces.

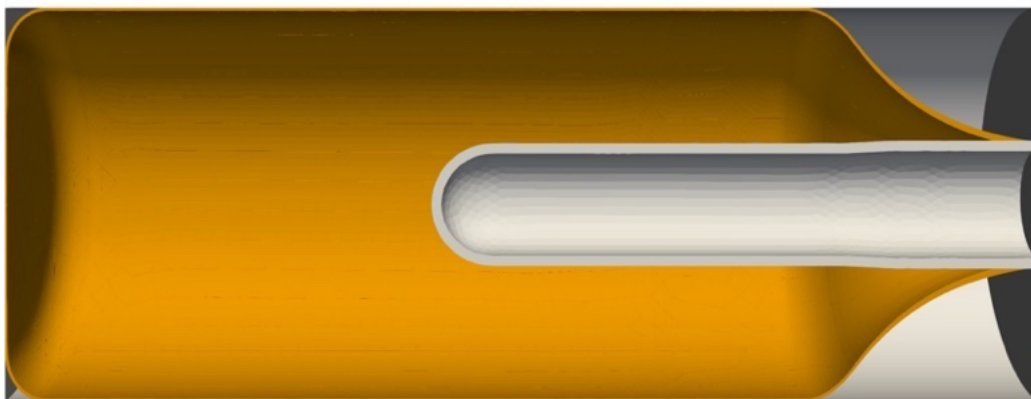


Figure 2: Simulation of preform inflation in a cylindrical cavity, undeformed configuration (white) and final deformed configuration (orange). Results are obtained with a third order accurate HDG formulation over a 16k tetrahedral elements mesh. In the final deformed configuration 4k mesh elements have reached contact with the mould. Note that, since the gap function is non differentiable near the intersection between the bottom surface and the lateral surface of the cylinder, the simulation is halted before the material approaches those sharp corners.

As a simplifying assumption, we tackle obstacles admitting an analytical description, such that all the geometrical quantities involving the gap function and its derivatives are readily available. Future works will consider contact with generic obstacles of complex shape, whose boundary is discretized with a grid. Specific strategies for the computation of distance functions and its derivatives will be required. Several improvements to the blow moulding computational modelling approach are required in order to achieve predictive capabilities regarding wall thickness and material distribution of the plastic container. Since heated up polymers behave as viscoplastic materials more complex constitutive laws need to be introduced for modelling the rate-dependent inelastic contributions. Moreover, thermal effects need to be taken into account.

Cooperation benefits

Projects dealing the development of numerical methods and schemes may offer useful suggestion for this project. In particular, **SP-A4** which investigates high order numerical schemes, adaptive mesh refinement techniques and Lagrangian methods on moving meshes and **SP-C3** that focuses on high order DG methods and time integration schemes.

References

- [1] L. Botti & L. Verzeroli, BR2 discontinuous Galerkin methods for finite hyperelastic deformations, *Journal of Computational Physics*, 463, 2022.
- [2] L. Botti, A. Colombo & F. Bassi: h-multigrid agglomeration based solution strategies for discontinuous Galerkin discretizations of incompressible flow problems. *Journal of Computational Physics*, 347, 2017.
- [3] F. Bassi, and S. Rebay: A high-order accurate discontinuous finite element method for the numerical solution of the compressible Navier-Stokes equations. *Journal of Computational Physics*, 131(2), 1997.
- [4] L. Verzeroli, F. C. Massa & L. Botti, A HDG formulation for nonlinear elasticity problems featuring finite deformations and frictionless contact constraints, *Finite Elements in Analysis and Design*, 215, 2023.
- [5] S.-C. Soon, B. Cockburn & H. Stolarski. A hybridizable discontinuous Galerkin method for linear elasticity. *International Journal for Numerical Methods in Engineering*, 80(8), 2009.
- [6] H. Kabaria, A. J. Lew & B. Cockburn, A hybridizable discontinuous Galerkin formulation for non-linear elasticity, *Computer Methods in Applied Mechanics and Engineering*, 283, 2015.

3. Comments on the qualification programme and supervision strategy:

The seminars were not strictly related to my research topic but they introduced some general concepts and tools useful in the fields of numerical methods and engineering. Professors have been able to explain clearly the main topics of their lectures and to clarify doubts. In the future, I would like to attend courses concerning applied mathematics (i.e. advanced methods to solve PDEs) and numerical analysis (i.e. the theoretical foundations of Discontinuous Galerkin Method).

Participation in the following courses/seminars of the Int. Research Training Group: <i>Core courses:</i> • Direct Numerical Simulation (DNS) of Primary Jet Break-up • Dimensional analysis for heat and mass transfer problems • Selected Solution Methods for Nonlinear Partial Differential Equations for Flow and Heat Transfer Problems
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • Doctoral Seminar, 15. & 16. May 2019 in Bergamo • Doctoral Seminar, 9 November 2021 in Bergamo • Doctoral Seminar, 8-9 March 2021 in Bergamo
Participation in conferences, congresses, etc., at home and abroad: • WCCM-ECCOMAS 2020, 11-15 January, 2021, Paris.

4. Individual publications:

I. Peer-reviewed articles

- L. Botti, and **L. Verzeroli**, BR2 discontinuous Galerkin methods for finite hyperelastic deformations, *Journal of Computational Physics*, 463, 2022.
- **L. Verzeroli**, F. C. Massa, and L. Botti, A HDG formulation for nonlinear elasticity problems featuring finite deformations and frictionless contact constraints, *Finite Elements in Analysis and Design*, 215, 2023.

II. Presentations

a) Oral

- **Verzeroli, L.:** DG methods for linear elasticity: implementation, validation and perspectives. DROPIT - Doctoral seminar 2019, Università degli Studi di Bergamo, Italy.
- **Verzeroli, L.,** Botti, L. Bassi F.: DG Methods for Finite Strain Elasticity Problems, WCCM-ECCOMAS, 11-15 January 2021, Paris.
- **Verzeroli, L.:** Discontinuous Galerkin methods for nonlinear elasticity problems, DROPIT seminar 2021, Università degli Studi di Bergamo, 8-9 March 2021, Bergamo.
- **Verzeroli, L.:** High Order methods for finite elastic deformations, DROPIT seminar 2021, Università degli Studi di Bergamo, 9 November 2021, Bergamo.
- **Verzeroli, L.:** Nonconforming Finite Elements Methods for Nonlinear Elasticity Problems Featuring Finite Deformations and Frictionless Contact Constraints. Ph.D. dissertation defense at Università degli Studi di Bergamo – Ph.D. program in Engineering and Applied Sciences, XXXIV cycle - April 5th, 2022, Bergamo.

Report of doctoral researcher (SP-AS09)

Project number: Associated Project 09

First and last name of doctoral researcher: Verena Kunberger

(Working) title of doctoral project: Interactions of Supercooled Water Droplets

Name of supervisors: Bernhard Weigand, Christian Rohde

1. General information:

Duration of International Research Training Group membership: Associated member of DROPIT from 02/2023 till 07/2024
Funding (type and duration): SFB-TRR 75 (2 years)
Start of doctoral training and (anticipated) date of doctoral degree: Start 02/2019, doctoral degree 2025
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): Aerospace Engineering, 02/2019 M.Sc.
Where applicable, please note any circumstances which could have affected your scientific work:--
Occupation after leaving the IRTG, if applicable: TESAT Spacecom

2. Description of doctoral project and research results achieved to date:

Interactions of supercooled water droplets, such as collisions, occur in clouds and significantly influence the formation of precipitation, such as sleet, graupel and hail. These micro-processes in the cloud environment are still hardly investigated experimentally, as the scale and environmental conditions are difficult to reproduce in the laboratory. Therefore, experiments are conducted at the ITLR to investigate these processes in detail and the influencing parameters on them. Subsequently, the knowledge of the small-scale processes can be applied to models in climate research and weather forecasting. Furthermore, understanding these processes is very important for all processes in which supercooled droplets and/or ice particles occur, such as aircraft icing, icing of wind turbines, high-voltage insulators, etc.

Binary Droplet Collisions

For the binary droplet collisions, a droplet is optically levitated in a laser beam to avoid external disturbances. The levitation takes place in a chamber. The chamber can be cooled down to -40°C by a cryostat. The atmosphere in the chamber can be controlled in terms of humidity via a regulated nitrogen supply and the addition of water vapor. The droplets are generated as individual droplets by a droplet generator from microdrop Technologies with constant but variable parameters. As a result, the droplet size and speed do not change during the experiment and their influence can be measured independently of other parameters. The droplet collision is observed by backlight-illumination imaging with an LED and 12MP CCD camera. Additionally, the scattered light of the laser is led to a Wollaston prism that splits it into two beams for each polarization direction. Two PSD sensors capture those beams, and with those measurements, the setup is triggered and the state of the droplet (liquid or frozen) can be determined.

While a droplet is levitated in liquid or frozen state, another droplet is released from the droplet generator to collide with the levitated one. Since evaporation and sublimation occur during levitation of the droplet, it is smaller than the falling droplet, as can be seen in the first results in Figure 1. The influence of the ambient conditions on the collision outcome can be determined by testing at different temperatures and humidity levels. Figure 1 shows shadow images of a collision at -40°C , the larger droplet is $45\text{ }\mu\text{m}$ in size, both droplets are frozen which can be recognized by the spike visible in the left image on the larger

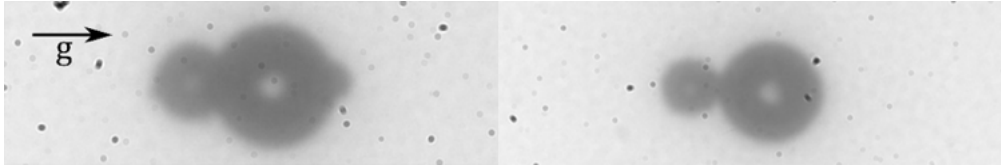


Figure 16: Droplet collision at -40°C.

droplet. On the right, it can be seen that the droplet freezes almost immediately on impact without much deformation, a small bridge is formed to connect both droplets.

In Figure 2, results of collisions at -23°C are shown and they looked quite different than collisions at higher supercooling. The two droplets do not freeze on impact but coalesce and freeze subsequently. The freezing can be seen by the bulge in the left image and the spike in the center image. The right image shows an almost spherical droplet and it has to be determined by other means, like the PSD sensors, if it is frozen.

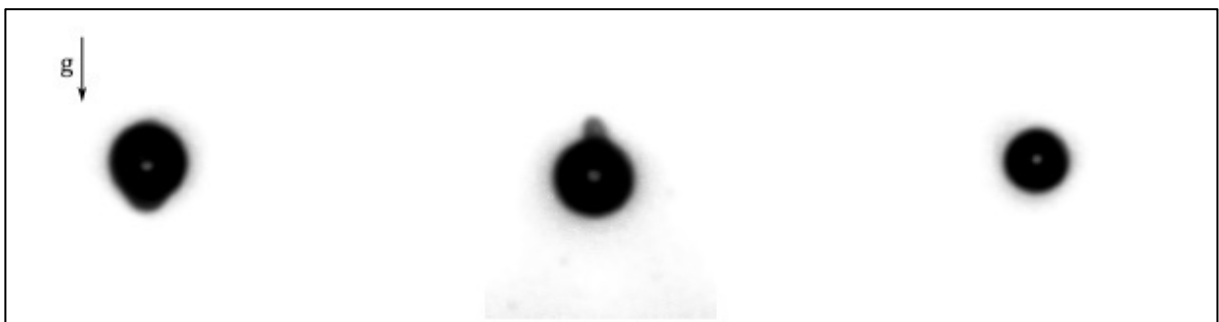


Figure 2: Droplet collision at -23°C.

Rime Formation

For the investigation of rime formation through interactions of supercooled water droplets on a surface, there is a second similar experimental setup. There, also a cryostat cools the chamber down to -40°C and the inflow of nitrogen/water vapor is controlled to allow a set humidity within the chamber. Sensors are inserted to measure the temperature and humidity inside the chamber before the experiment begins.

In addition, two thermocouples are located directly below the target surface on which the droplets impact to ensure that the surface and the ambience have the same temperature. The droplet generator is positioned vertically above the surface and can be moved electrically with piezo-electric stages. These stages are programmed to move randomly before another droplet falls to simulate the random droplet distribution in a cloud. The structures on the surface are observed by a shadowgraphy setup. Figure 3 shows results of droplet collisions at an ambient temperature of -20°C.

At this temperature, straight turrets form and the individual droplets are barely distinguishable. The structures seem dense and more like clear ice. Sublimation and evaporation on these structures is also visible, e.g. in the middle image on the right side.

At -40°C ambient temperature, as shown in Figure 4, the structures look different. They are much broader and more branched out and the individual droplets are clearly visible. The falling droplets seem

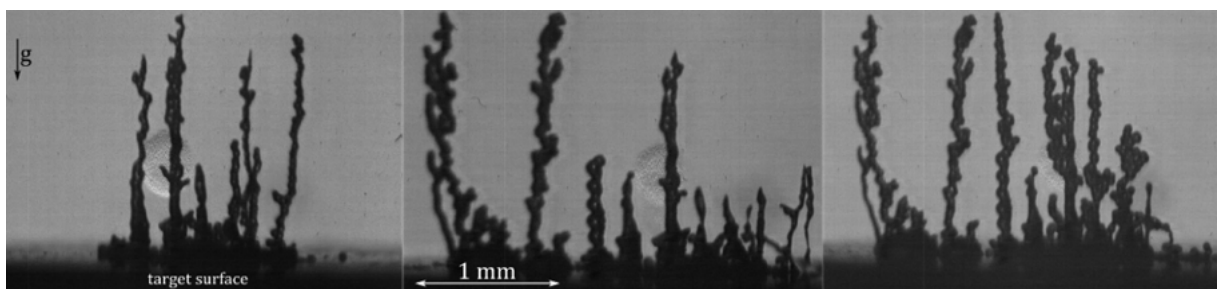


Figure 3: Turrets forming on the target surface at -20°C

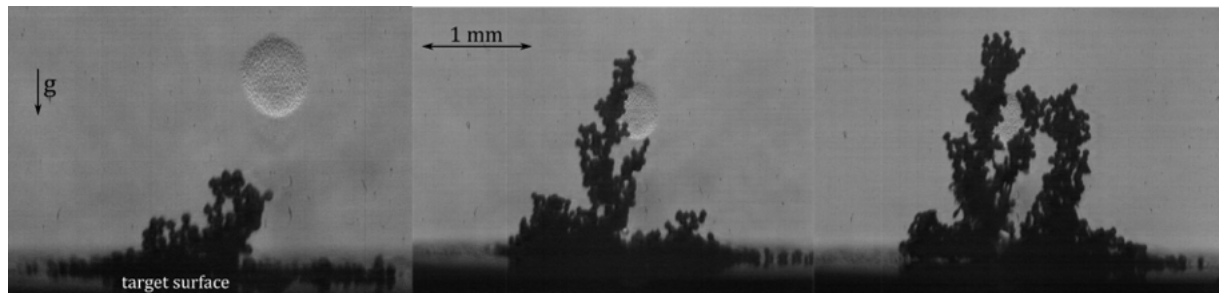


Figure 4: Structures on the target surface at -40°C .

to freeze immediately on impact without much deformation just like in the case of binary collisions at -40°C . A reason for the further spread and branching of the structure could be a possible freezing of the droplets during the fall. In that case, the spike of the frozen drop would change its aerodynamics leading to changing impact paths.

Figure 5 shows different freezing timescales within the structures. In the left image, hard rime can be seen and individually frozen droplets are visible which means that they froze very quickly. The structures in the middle image show larger droplets and deformations, especially in the red circle, which are likely due to the drops still being partly liquid while the next droplet impacts. In that case, it is important to analyze the time until the droplet is completely frozen and compare it to the time until the next droplet impacts. The image on the right shows mostly deformed droplets forming glaze ice at the bottom, which is very dense. In that case, a high relative humidity in the chamber was measured and the target surface was warmer than the ambient conditions, which led to the integration of the thermocouples underneath the target surface.



Figure 5: Different freezing timescales of the structures on a surface

3. Comments on the qualification programme and supervision strategy:

Participation in the following courses/seminars of the Int. Research Training Group: Italian Course
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: Doctoral Seminars, DROPIT Meets Industry 2023
Participation in conferences, congresses, etc., at home and abroad: ILASS 2023, DIPSI 2022 & 2023

4. Individual publications:

I. Peer-reviewed articles:

- V. Kunberger & B. Weigand. "Binary Collisions of Supercooled Water Droplets", Atomization and Sprays, Vol.34, pp. 19-30, 2024.

II. Further publications

a) Proceedings

- V. Kunberger & B. Weigand. (2022). "Interactions of Supercooled Water Droplets on a Surface". 31th Conference on Liquid Atomization and Spray Systems, ILASS-Europe 2022, Tel-Aviv (Virtual).

- **V. Kunberger & B. Weigand.** (2023). "Binary Droplet Collisions of Supercooled Water Droplets". 32nd Conference on Liquid Atomization & Spray Systems, ILASS Europe 2023, 4-7 Sept 2023, Napoli, Italy.

III. Presentations

a) Oral

- **V. Kunberger & B. Weigand.** (2022). "Phase Change and Interaction Behaviour of Supercooled Water Droplets". DIPSI 2022, Bergamo, Italy.
- **V. Kunberger & B. Weigand.** (2023). "Phase Change and Interaction Behaviour of Supercooled Water Droplets". DIPSI 2023, Bergamo, Italy.
- **V. Kunberger.** (2021). "Tropfeninteraktion von unterkühlten Wassertropfen", Multiphase Flow Seminar 2021.
- **V. Kunberger & B. Weigand.** (2024). "The influence of heat transfer on the outcome of droplet collisions of supercooled water droplets". DIPSI 2024, Bergamo, Italy.

Report of doctoral researcher (SP-AS10)

Project number: Associated Project 10

First and last name of doctoral researcher: Florian Wendt

(Working) title of doctoral project: Data-Integrated Two-Scale Modelling of Compressible Multiphase Flow

Name of supervisors: Prof. Dr. Christian Rohde

1. General information:

Duration of International Research Training Group membership:

Associated member of DROPIT since January 2024

Funding (type and duration):

PhD student at the Institute of Applied Mathematics and Numerical Simulation at the University of Stuttgart, Jan 2024 – Dec 2025.

Start of doctoral training and (anticipated) date of doctoral degree:

1 Jan 2024 – 31 Dec 2026

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

04/2018 Mathematics, University of Tübingen, Bachelor's degree, 09 March 2021

04/2021 Mathematics, University of Tübingen, Master's degree, 04 October 2023

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

In this project, we consider a compressible viscous fluid that can occur in a liquid and a vapor phase. For such a fluid, a physically well-accepted diffuse interface model is given by the Navier–Stokes–Korteweg (NSK) equations. However, if the number of phase transitions is very high (e.g. in a spray), this model becomes computationally unfeasible. On the other hand, there is the class of Baer–Nunziato (BN) type models that describe the two-phase flow on a larger scale. These models are of first order and thus, computationally cheaper than the detailed NSK equations. However, these models suffer from closure problems as their derivation follows an ad hoc averaging procedure from the detailed scale model which results in an inaccuracy of the model. It is the goal of this project to develop a two-scale model that describes the two-phase fluid on the larger scale with a BN type model that finds its closure in the underlying detailed scale model, that is given by the NSK model, in a mathematical rigorous manner. Concerning the compressible Navier–Stokes equations, such a two-scale model was developed in [1], where the procedure from the detailed scale to the larger scale was carried out mathematically rigorous, but without phase transition effects on the detailed scale. This result serves as starting point for this project.

The challenges and results faced during the (almost) two years of doctoral studies can be broken down as follows:

1st year. This project aims at developing a two-scale averaged model for a compressible two-phase flow that is described by the compressible NSK equations on the detailed scale in a mathematical rigorous manner. Thus, we first have to precisely describe the NSK equations that we use on the detailed scale and explain how this model is capable to model phase transition effects. As a first start, we choose the simplest framework and thus worked in the isothermal framework. Let $T > 0$, $\Omega \subseteq \mathbb{R}^3$ be a bounded domain and denote $\Omega_T := (0, T) \times \Omega$. Then, the NSK model describes the dynamics of the fluid via the density $\rho: \Omega_T \rightarrow \mathbb{R}_+$ and the velocity $\mathbf{u}: \Omega_T \rightarrow \mathbb{R}^3$ that obey in Ω_T the system of equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (1)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla P(\rho) = \mu \Delta \mathbf{u} + \kappa \rho \nabla \Delta \rho, \quad (2)$$

subject to appropriate boundary conditions and initial conditions $\rho(0) = \rho^0$, $\mathbf{u}(0) = \mathbf{u}^0$ in Ω for given functions $\rho^0, \mathbf{u}^0$. Here, the positive constants $\mu > 0$ and $\kappa > 0$ are called the viscosity coefficient and capillarity coefficient, respectively, and $P: [0, \infty) \rightarrow [0, \infty)$ denotes the pressure function. In order to model phase transition effects, we assume the pressure function P to be of Van-der-Waals type, more specifically, we assume that there exist constants $0 < b_1 < b_2$ such that P is monotonically increasing on $[0, b_1] \cup [b_2, \infty)$ and monotonically decreasing on (b_1, b_2) . Accordingly, we then say that the fluid's state is liquid, spinodal and vapor, if the density's value lies in $[0, b_1]$, (b_1, b_2) and $[b_2, \infty)$, respectively (cf. Figure 1).

In order to derive now an averaged scale model starting from this NSK model on the detailed scale, we take as a starting point the methodology in [1]. This methodology, relies on the investigation of the propagation of initial density oscillations. More specifically, we take a sequence of initial data $(\rho_n^0, \mathbf{u}_n^0)_{n \in \mathbb{N}}$, where the sequence of initial densities is expected to oscillate widely in space mimicking the large number of phase transition that are initially present. Then, we construct a sequence of solutions $(\rho_n, \mathbf{u}_n)_{n \in \mathbb{N}}$ to this sequence of initial data in an appropriate solution framework that can be viewed as the solution on the detailed scale and we anticipate that the model on the averaged scale can be found in the limit $n \rightarrow \infty$. Due to the hyperbolic character of the continuity equation, we expect that the initial density oscillations propagate in time, and thus, the averaged system is expected to contain some probabilistic parts. Therefore, a further reduction as done in [1] is needed in order to enter the BN framework.

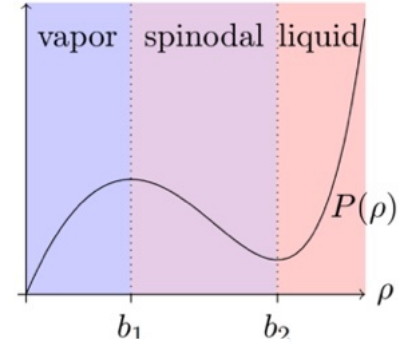


Figure 1: Example of a pressure function of Van-der-Waals type.

In order to make this methodology rigorous for the NSK equations, we have to answer the following three questions:

1. Which solution framework is appropriate for the homogenization procedure and allows for a rigorous existence result?
2. Can we pass to the limit $n \rightarrow \infty$ by deriving uniform bounds that do not exclude strong oscillations in the sequence of initial densities?
3. Can we reduce the limit system to a system of BN type in a mathematical rigorous manner?

To answer these difficult questions, we have first considered the simplest framework, that is, the one-dimensional, periodic and isothermal case. However, even in this case, we were not able to answer these questions for the NSK equations. The reason is the third order differential operator acting on the density that seems to render the system incompatible with the homogenization methodology. To overcome this remedy, we have chosen a non-local version of the NSK model that approximates the full NSK model by an additional unknown c that is given by a linear elliptic equation. More precisely, in this approximative model, the two-phase fluid is described by the fluid's density $\rho: [0, T) \times \mathbb{T} \rightarrow \mathbb{R}_+$, the fluid's velocity $u: [0, T) \times \mathbb{T} \rightarrow \mathbb{R}$ and the order parameter $c: [0, T) \times \mathbb{T} \rightarrow \mathbb{R}$ that satisfy in $(0, T) \times \mathbb{T}$ the system of equations

$$\partial_t \rho + \partial_x (\rho u) = 0, \quad (3)$$

$$\partial_t (\rho u) + \partial_x (\rho u^2) + \partial_x P(\rho) = \mu \partial_{xx} u + \lambda \rho \partial_x (c - \rho), \quad (4)$$

$$-\kappa \partial_{xx} c + \lambda (c - \rho) = 0, \quad (5)$$

subject to initial conditions $\rho(0) = \rho^0$, $u(0) = u^0$ in $\mathbb{T}$ for given functions ρ^0 , u^0 . Here, μ , κ and P are as before and the positive constant $\lambda > 0$ is called coupling coefficient. Formally, (3)-(5) approaches the NSK model for $\lambda \rightarrow \infty$. Moreover, we can rewrite the momentum equation as

$$\partial_t(\rho u) + \partial_x(\rho u^2) + \partial_x P_\lambda(\rho) = \mu \partial_{xx} u + \lambda \rho \partial_x c, \quad (6)$$

with an artificial pressure function $P_\lambda(\rho) := P(\rho) + \frac{1}{2}\rho^2$ for which we can assume monotonicity as we are interested in the regime $\lambda \rightarrow \infty$ and P is assumed to be of Van-der-Waals type. This property was crucial in order to obtain a homogenization result for (3)-(5). We were able to answer the questions 1.) – 3.) in the following sense:

1. We identified the solution concept of strong solutions as an appropriate solution framework. We were able to provide a rigorous global-in-time well-posedness result of strong solutions to (3)-(5) for initial data $\rho^0, u^0 \in H^1(\mathbb{T})$ away from vacuum for pressure functions that grow at infinity like ρ^γ with $\gamma \geq 2$. In particular, this global-in-time result applies for pressure functions of Van-der-Waals type.
2. We were able to pass to the limit $n \rightarrow \infty$ and to identify a closed limit system by assuming the sequence of initial data $(\rho_n^0, u_n^0)_{n \in \mathbb{N}}$ is uniformly bounded in $L^\infty(\mathbb{T}) \times H^1(\mathbb{T})$ and the sequence of initial densities is uniformly bounded away from vacuum, in particular, without excluding strong oscillations in the sequence of initial densities. The limit system can be seen as an averaged scale model and contains a probabilistic part in terms of a Young measure ν .
3. We were able to reduce the averaged scale model to a one-velocity BN type model by just assuming that the Young measure ν is initially given by a convex combination of two Dirac-measures. The resulting BN type model has not been considered in the literature so far.

The resulting averaged scale model is a closed model for the unknowns $(\rho_+, \rho_-, \alpha_+, u, c)$, where ρ_+ and ρ_- denotes the phase density of the fluid's liquid phase and gas phase, respectively, and α_+ and α_- denotes the volume fraction of the fluid's liquid phase and gas phase, respectively. The system of equations reads

$$\partial_t \alpha_+ + u \partial_x \alpha_+ = \mu^{-1} \alpha_+ \alpha_- (P_\lambda(\rho_+) - P_\lambda(\rho_-)), \quad (7)$$

$$\partial_t \rho_\pm + \partial_x(\rho_\pm u) = \mu^{-1} \rho_\pm \alpha_\mp (P_\lambda(\rho_-) - P_\lambda(\rho_+)), \quad (8)$$

$$\partial_t(\rho u) + \partial_x(\rho u^2) + \partial_x \bar{P} = \mu \partial_{xx} u - \lambda \rho \partial_x c, \quad (9)$$

$$-\kappa \partial_{xx} c + \lambda(c - \rho) = 0, \quad (10)$$

$$\alpha_+ + \alpha_- = 1, \quad (11)$$

with the closure $\rho = \alpha_+ \rho_+ + \alpha_- \rho_-$ and $\bar{P} = \alpha_+ P_\lambda(\rho_+) + \alpha_- P_\lambda(\rho_-)$.

These results were bundled into a preprint and submitted for publication [2].

2nd year. With the results obtained in the first year of the doctoral studies, a natural next step is to generalize the results to a more physically relevant framework. More specifically, the goal was to generalize this result to a three-dimensional setting. Motivated by the problems encountered in the one-dimensional setting, we choose again a lower-order approximative system of the NSK equations. More specifically, we describe the two-phase fluid on the detailed scale with the following set of equations for the unknowns $(\rho, \mathbf{u}, c)$ in the domain $\Omega_T = (0, T) \times \Omega$, where $\Omega \subseteq \mathbb{R}^3$ is a bounded domain:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (12)$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla P = \mu \Delta \mathbf{u} + \lambda \rho \nabla(c - \rho), \quad (13)$$

$$\beta \partial_t c - \kappa \Delta c + \lambda(c - \rho) = 0, \quad (14)$$

subject to boundary conditions $\mathbf{u}_{\partial\Omega} = 0$, $\nabla c \cdot \mathbf{n}_{\partial\Omega} = 0$ and initial conditions $\rho(0) = \rho^0$, $\mathbf{u}(0) = \mathbf{u}^0$, $c(0) = c^0$ for given functions ρ^0 , $\mathbf{u}^0$, c^0 . Here, μ , κ , λ , P are as before and we have an additional constant coupling coefficient $\beta > 0$. By formal arguments, we infer that (12)-(14) approaches the NSK model for $\lambda \rightarrow \infty$ and $\beta \rightarrow 0$. Also, we can rewrite the momentum equations as

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla P_\lambda(\rho) = \mu \Delta \mathbf{u} + \lambda \rho \nabla c, \quad (15)$$

with the artificial pressure function P_λ defined as before. Again, we can assume that P_λ is monotonically increasing, as P is of Van-der-Waals type and as we are interested in the regime $\lambda \rightarrow \infty$. The answer of the questions 1.) – 3.) is more delicate in the three-dimensional setting. For instance, a strong solution framework that is appropriate for the homogenization procedure seems out of reach for (12)–(14). However, we were able to partially answer these questions resulting into an averaged scale model, but not into an averaged scale model of BN type:

1. We were able to identify the concept of finite energy weak solutions as an appropriate solution framework. For pressure functions that grow at infinity like ρ^γ with $\gamma \geq 2$, a global-in-time existence result for initial data in the class $\rho^0 \in L^\gamma(\Omega)$, $\rho \geq 0$, $\mathbf{u} \in L^{\frac{2\gamma}{\gamma-1}}(\Omega)$, $c \in H^1(\Omega)$ seems possible by adapting methods available for the compressible Navier–Stokes equations. We identify the elaboration of such an existence result as a future direction for the doctoral studies. In particular, such an existence result applies for pressure functions of Van-der-Waals type.
2. We were able to pass to the limit $n \rightarrow \infty$ and identify a closed limit system by assuming that the sequence of initial data $(\rho_n^0, \mathbf{u}_n^0, c_n^0)_{n \in \mathbb{N}}$ is uniformly bounded in $L^\gamma(\Omega) \times L^{\frac{2\gamma}{\gamma-1}}(\Omega) \times H^1(\Omega)$, in particular, without excluding the initial density sequence to be highly oscillating. The limit system can be seen as an averaged scale model and contains a kinetic equation for a Young measure ν .
3. We were not able to reduce the model into a BN type model due to the lack of regularity for the velocity $\mathbf{u}$. We identify this issue as a future direction for the doctoral studies.

The resulting averaged scale model describes the two-phase fluid via three unknowns that consist of a deterministic part in the form of the velocity $\mathbf{u}$ and the order parameter c that satisfy in the domain Ω_T the system of equations

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla \bar{P} = \mu \Delta \mathbf{u} + \lambda \rho \nabla c, \quad (16)$$

$$\beta \partial_t c - \kappa \Delta c + \lambda(c - \rho) = 0, \quad (17)$$

and a probabilistic part in the form of a Young measure $\nu: [0, T) \times \Omega \rightarrow \mathbb{P}(\mathbb{R})$, where $\mathbb{P}(\mathbb{R})$ denotes the set of probability measures on $\mathbb{R}$, that satisfies in the domain $\Omega_T \times \mathbb{R}$ the kinetic equation

$$\partial_t \nu + \nabla \cdot (\nu \mathbf{u}) + \partial_\xi ((\mathcal{H} - \nabla \cdot \mathbf{u}) \xi \nu) + (\mathcal{H} - \nabla \cdot \mathbf{u}) \nu = 0. \quad (18)$$

The closure of this set of equations is given by

$$\rho(t, x) = \int_0^\infty \xi \, d\nu_{(t,x)}(\xi), \quad \bar{P}(t, x) = \int_0^\infty P_\lambda(\xi) \, d\nu_{(t,x)}(\xi) \quad (19)$$

for almost all $(t, x) \in \Omega_T$ and

$$\mathcal{H}(t, x, \xi) = \frac{\bar{P}(t, x) - P_\lambda(\xi)}{\mu} \quad (20)$$

for almost all $(t, x, \xi) \in \Omega_T \times \mathbb{R}$.

To make this effective model accessible for numerical methods for kinetic equations, we changed the viewpoint from the Young measure ν to its cumulative distribution function f that is defined for almost all $(t, x) \in \Omega_T$ through the relation

$$f(t, x, \xi) := \nu_{(t,x)}((-\infty, \xi]) \quad (21)$$

for any $\xi \in \mathbb{R}$. In terms of f , the kinetic equation on ν can be rewritten as

$$\partial_t f + \nabla \cdot (f \mathbf{u}) - \partial_\xi (\xi f \nabla \cdot \mathbf{u} - \xi \mathcal{R}[f]) = 0 \quad (22)$$

with $\mathcal{R}[f](t, x, \xi) := \int_0^\xi \mathcal{H}(t, x, \eta) \, df(t, x, \eta)$ for almost all $(t, x, \xi) \in \Omega_T \times \mathbb{R}$, where the integral with respect to $df(t, x, \cdot)$ denotes the Lebesgue–Stieltjes integral with respect to f . The closure relations can be rewritten as

$$\rho(t, x) = \int_0^\infty (1 - f(t, x, \xi)) \, d\xi, \quad (23)$$

$$\bar{P}(t, x) = \int_0^\infty (1 - f(t, x, \xi)) P'_\lambda(\xi) d\xi \quad (24)$$

for almost all $(t, x) \in \Omega_T$.

It would be of independent interest to investigate this averaged scale model without a reduction into a BN type model by numerical methods for kinetic equations. We identify this issue as a future direction for the doctoral studies.

So far, we have obtained results only in the simplified isothermal case. It is of particular interest to extend the homogenization results obtained so far to a more physically relevant framework. Thus, we identify the extension of the results to a non-isothermal framework as future direction for the doctoral studies.

References

- [1] D. Bresch and M. Hillaire. A compressible multifluid system with new physical relaxation terms. *Ann. Sci. Éc. Norm. Supér.* (4), 52(2):255–295, 2019.
- [2] C. Rohde and F. Wendt. Mathematical justification of a Baer–Nunziato model for a compressible viscous fluid with phase transition. *Preprint: arXiv:2504.10161*, 2025.

3. Comments on the qualification programme and supervision strategy:

In the course “Mathematische Methoden in der Strömungsmechanik” I have learned a lot on the mathematical theory of hyperbolic conservation laws and their numerical treatment in the form of finite volume methods. This course was very helpful for my doctoral studies as it taught me the interaction between the numerical methods and the mathematical analysis.

In the course “Foundations and Frontiers of Data-Integrated Simulations” I was introduced to the combination of data in applied simulations. This widened my view on the topic and was in particular helpful for my doctoral studies as a data-driven approach will be relevant in the future of my work.

In the course “Introduction to Asymptotic Analysis and Homogenization Theory” I was taught the fundamental principles of asymptotic analysis among several applications in homogenization theory. As my doctoral studies are concerned with a particular homogenization problem, this course was very helpful. At this point I also want to mention the annual doctoral seminar of the IRTG. In these seminars I gained a broader view on my topic and I had the chance to connect to other doctoral researchers.

The supervision was structured by regular meetings with my supervisor. There I could discuss results and also issues regarding my doctoral studies. I gained some new perspectives from these meetings that helped me improving my doctoral studies continuously. Also, I received well-suited recommendations for upcoming conferences and workshops for my doctoral studies. This supervision strategy matched perfectly to my working pattern and it strongly improved my results.

Participation in the following courses/seminars of the Int. Research Training Group: • Mathematische Methoden in der Strömungsmechanik • Foundations and Frontiers of Data-Integrated Simulations • Introduction to Asymptotic Analysis and Homogenization Theory
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: ---
Participation in the following International Research Training Group events: • Doctoral Seminar, Bergamo (online participation), 08-09 February 2024 • Doctoral Seminar, Stuttgart, 28-29 January 2025 • Final Colloquium, Stuttgart, 31 July – 01 August 2025
Participation in conferences, congresses, etc., at home and abroad: • Analysis of Dissipation in Inviscid and Compressible Fluid Dynamics, Konstanz (Germany), 05-07 June 2024 • Sixth Workshop on Compressible Multiphase Flow, Strasbourg (France), 17-19 June 2024 • First Young-Scientists Retreat, Kleinwalsertal (Austria), 16-20 September 2024 • Numerik im Ländle, Stuttgart (Germany), 05-06 May 2025 • Seventh Workshop on Compressible Multiphase Flow, Strasbourg (France), 26-28 May 2025 • Mathematics of Compressible Fluids, Clausthal-Zellerfeld (Germany), 23-27 June 2025

4. Individual publications:

I Peer-reviewed articles:

II Submitted articles:

- C. Rohde and **F. Wendt**. Mathematical justification of a Baer-Nunziato model for a compressible viscous fluid with phase transition. Preprint: arXiv:2504.10161, 2025

III Further publications

IV Presentations

a) Oral

- **F. Wendt**. Mathematical justification of a Baer–Nunziato model for a compressible viscous fluid with phase transition. Mathematics of Compressible Fluids Workshop, Clausthal-Zellerfeld (Germany), 23-27 June 2025.

b) Posters

- **F. Wendt**. Derivation of a Baer–Nunziato mixture model for fluids with internal capillarity in 1-D. Analysis of Dissipation in Inviscid and Compressible Fluid Dynamics. Konstanz (Germany), 05-07 June 2024.
- **F. Wendt**. Derivation of a Baer–Nunziato mixture model for fluids with internal capillarity in 1-D. Sixth Workshop on Compressible Multiphase Flow. Strasbourg (France), 17-19 June 2024.
- **F. Wendt**. Effective equations for a compressible liquid-vapor flow with highly oscillating initial density. Seventh Workshop on Compressible Multiphase Flow. Strasbourg (France), 26-28 May 2025.

Report of doctoral researcher (SP-AS11)

Project number: Associated Project 11

First and last name of doctoral researcher: Jonathan Wurst

(Working) title of doctoral project: Direct Numerical Simulation of Complex Drop-Wall Interactions with FS3D

Name of supervisors: Dr.-Ing. Kathrin Schulte

1. General information:

Duration of International Research Training Group membership: January 2024 – December 2025
Funding (type and duration): Associated member (funding from EXC 2075 - 39074001, Friedrich and Elisabeth Boysen Foundation - BOY-160) January 2024 – December 2025
Start of doctoral training and (anticipated) date of doctoral degree: March 2021 – June 2026
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 10/2013, Process Engineering, University of Stuttgart, Bachelor's degree, 03/2017 04/2017, Process Engineering, University of Stuttgart, Master's degree, 09/2020 03/2021, Aerospace Thermodynamics, University of Stuttgart, PhD candidate
Where applicable, please note any circumstances which could have affected your scientific work: --
Occupation after leaving the IRTG, if applicable: --

2. Description of doctoral project and research results achieved to date:

Moving from the basic scenario of a droplet impact on a smooth wall closer to real-world applications, this process easily gets very involved when considering e.g. non-newtonian effects, multiple fluids, structured walls or effects like contact line pinning. The general goal of this project is the development of a robust simulation framework within the in-house solver Free Surface 3D (FS3D) [1] to simulate such complex drop-wall interactions. In particular, the focus is on the two applications of a droplet impact onto a micro-structured surface and a ternary system in contact with a smooth wall. FS3D is a Direct Numerical Simulation tool for multiphase flows and developed for over 25 years already. The Finite Volume method and a staggered, Cartesian grid discretization are employed. To track the phase distribution the Volume-of-Fluid method (VoF) is utilized and the piecewise linear interface calculation (PLIC) ensures a sharp interface.

Three aspects have to be considered for the simulation of multiphase systems in contact with a wall. The reconstruction of the free surface, capturing also the dynamic contact angle with a model and incorporating correctly the wettability itself in the simulation.

For the reconstruction, the implementation is based on the work by Potyka and Schulte [2]. The orientation of the PLIC plane in an interface cell is determined by the minimization of an error function. Extending the PLIC plane in neighboring interface cells results in a predicted VoF-distribution. The deviation of the predicted VoF values from the actual distribution gives an error that is minimized. As this method does not depend on a fixed stencil, it is especially advantageous in regions like the lower pillar edges of a structured surface or three phase situations in general.

The angle at which a liquid-gas interface meets the solid surface is called the contact angle. The region where these three phases meet is the contact line region and this is in general considered simply as a line. At the static contact line a force balance of the three interfacial tensions (between each pair of phases) results in the well known Young equation and can in principle predict the static contact angle. If the contact line moves, the result of capillary, inertial and viscous forces is the dynamic contact angle. On the nanoscale, the Young equation still holds. However, this scale is too small for macroscale simulations and cannot be resolved. A dynamic contact angle model can thus be understood as a subscale model for the macroscale simulation including these effects. Besides many empirical models, Cox' Hydrodynamic Theory [3] is derived from fundamental equations and is also scale dependent, which can be utilized to reduce the grid dependence; an issue commonly observed when applying the no-slip condition [4].

To incorporate the wettability in the simulation, the applied surface tension force at the contact line is adjusted to be consistent with the contact angle. The approach proposed by Sussman [5] was chosen because it is independent of the applied surface tension model and can in principle be applied to arbitrary geometries. The idea is to provide missing information at the wall boundary by artificially transporting liquid into the ghost-cells. In a first step, an artificial velocity field is constructed that points into the wall according to the contact angle and tangential to the interface. Subsequently, the advection scheme used also in the time loop is applied to transport the liquid in the ghost-cells. This approach was then also extended for a second liquid phase by treating both liquid phases separately and superimposing the result in the ghost-cells. A local weighting of the contact angle by the different phase fractions was used to distinguish between the three contact angles.

The framework was first successfully validated with several static test cases with available analytic solutions. This included sessile drops with and without gravity for varying contact angles. Without gravity, a sessile drop describes a perfect spherical cap and thus can be compared easily with the correct solution. Under gravity, the local curvature can be described as a function of the hydrostatic pressure and the Young-Laplace pressure and has to be solved numerically. Moreover, the meniscus in a square capillary also can be described as a section of a sphere and was compared for three different contact angles.

In a next step, the framework was validated for smooth surfaces in a collaboration with the Institute of Fluid Mechanics (ISTM) at the Karlsruhe Institute of Technology [6]. They conducted experiments of droplet impacts and evaluated the dynamic contact angle. Empirical correlations usually express the dynamic contact angle as a function of the dimensionless contact line speed, motivated by the strong dependency on it. A polynomial fit of the experimental data thus served as an own dynamic contact angle model. Implementing this relationship in the simulation, a comparison could be done without the uncertainty of the dynamic contact angle. In this way, the correct implementation of the contact line dynamics was validated. In Figure 1, comparisons are shown for three experiments (cases 1 to 3) by the ISTM and an additional experiment from the literature by Yokoi et al. [7] (case 4), where the dynamic contact angle evolution was also reported.

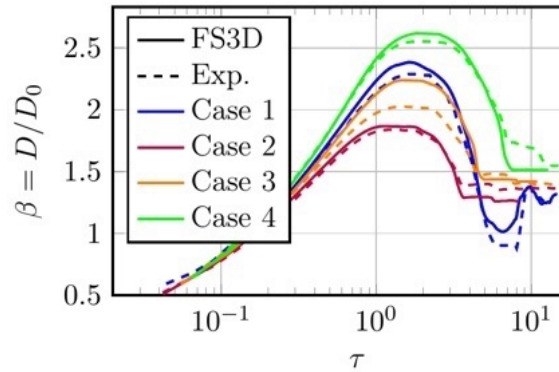


Figure 1: Comparison of the spreading factor as a function of the dimensionless time for four different cases between FS3D and experiments.

In a collaboration with SP-B2, the behavior of droplets impacting onto pillars in the sub-millimeter range was investigated. Especially the conditions and mechanism for the appearance of a so called diamond shape were revealed. For high impact conditions, the spreading occurs mainly in the grooves between pillars, as the dynamic pressure is very high. For very low conditions, the spreading takes place symmetrically just above the pillars. In addition, smaller structures promote the top spreading while larger ones the groove spreading. If the spreading process is neither top nor groove driven and the surface is hydrophilic, the diamond shape can be observed, since then at every groove intersection the flow splits up in 90 degree turns. The path the fluid takes hence resembles the taxicab distance and the resulting shape is a diamond. A snapshot of a simulation with visualized path lines can be seen in Figure 2. The collaboration resulted in a joined publication in a highly ranked international journal [8].

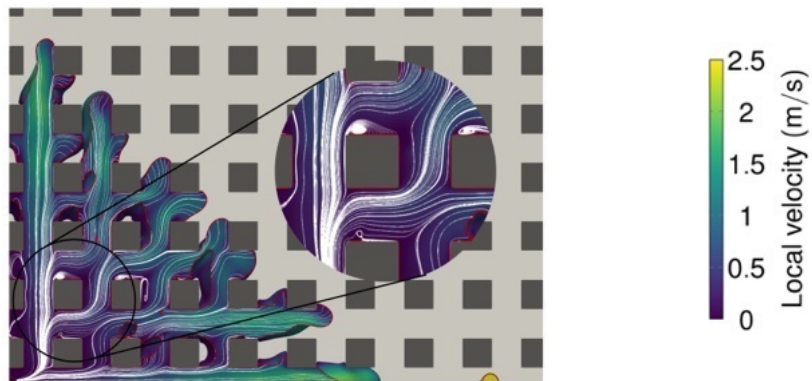


Figure 2: Simulation snapshot of a droplet impact on a micro-structured surface exhibiting the diamond shape with visualized pathlines in white [8].

Finally, an extension of the approach to a ternary system was developed. Instead of one contact angle between the liquid-gas interface, three different contact angles can appear at the same time. So far, this is only tested with static test cases. Such a sessile compound drop can in principle form four different configurations. In the Janus configuration, the two immiscible liquids form drops in contact with the wall and additionally have a common interface. The lens configuration has only one liquid in contact with the wall, while the other forms a lens on top of the base drop. The collar configuration looks similar to the lens configuration from the outside but exhibits also a hidden wall contact of the second phase. One phase totally covered by the second is called the encapsulation configuration. Simulation snapshots of all configurations are displayed in Figure 3 and found to be captured by FS3D.

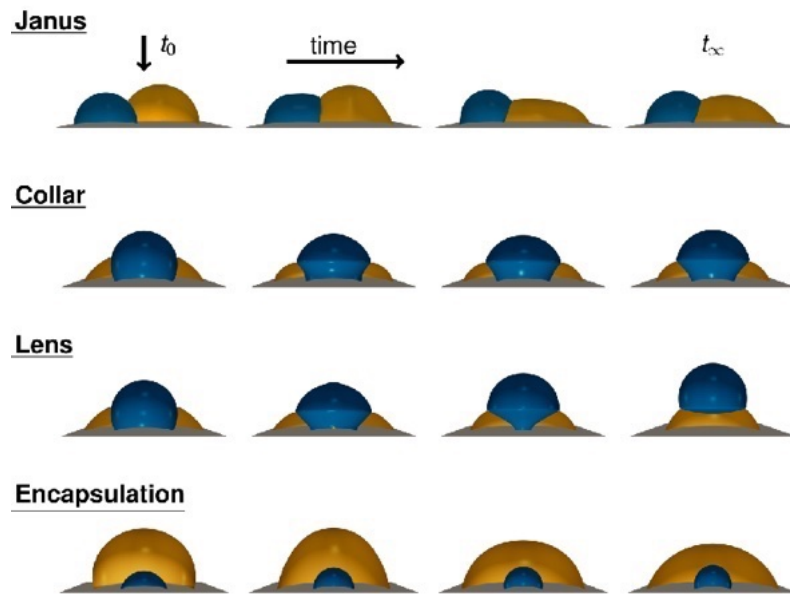


Figure 3: Evolution of different configurations towards their equilibrium configuration.

References

- [1] K. Eisenschmidt, M. Ertl, H. Goma, C. Kieffer-Roth, C. Meister, P. Rauschenberger, M. Reitzle, K. Schlottke, and B. Weigand. Direct numerical simulations for multiphase flows: An overview of the multiphase code fs3d. *Applied Mathematics and Computation*, 272:508-517, 2016.
- [2] J. Potyka, and K. Schulte. A volume of fluid method for three dimensional direct numerical simulations of immiscible droplet collisions. *International Journal of Multiphase Flow*, 170:104654, 2024.
- [3] R. G. Cox. The dynamics of the spreading of liquids on a solid surface. Part 1. Viscous flow. *Journal of Fluid Mechanics*, 168:169-194, 1986.
- [4] S. Afkhami, S. Zaleski, and M. Bussmann. A mesh-dependent model for applying dynamic contact angles to VOF simulations. *Journal of Computational Physics*, 228:5370-5389, 2009.
- [5] M. Sussman. An adaptive mesh algorithm for free surface flows in general geometries. *Chapman and Hall/CRC Boca Raton, FL, USA*, 2001.
- [6] M. Dreisbach, S. Blessing, F. Michaux, A. Brunn, A. Stroh, and J. Kriegseis. Three-dimensional encoding of a gas-liquid interface by means of color-coded glare points. *Experiments in Fluids*, 64:53, 2023.
- [7] K. Yokoi, D. Vadillo, J. Hinch, and I. Hutchings. Numerical studies of the influence of the dynamic contact angle on a droplet impacting on a dry surface. *Physics of Fluids*, 21:072102, 2009.
- [8] P. Palmethofer, J. Wurst, A. K. Geppert, K. Schulte, G. E. Cossali, and B. Weigand. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science*, 682:413-422, 2025.

3. Comments on the qualification programme and supervision strategy:

Participating in the IRTG was a good way of connecting and exchanging ideas with researchers working on similar topics. It allowed networking in the early stage of the scientific career. Moreover, the possibility of presenting the research status on a regular basis helped structuring work packages. The offered courses covered a broad field to choose from and gave a good general insight into the research topic.

Participation in the following courses/seminars of the Int. Research Training Group: • Numerical methods in two-phase flows • Seminar for two-phase flows • Modelling of drop heating and evaporation • Academic writing in English
Research stays or internships at other research institutions both at home and abroad, as well as stays at the partner university: --
Participation in the following International Research Training Group events: • Doctoral Seminar, Bergamo, 8 February 2024 • Experimental and Numerical Methods in Science workshop, Bergamo, 18 June 2024 • Doctoral Seminar, Stuttgart, 28 January 2025 • Analytic and Numerical Methods in Science workshop, Bergamo, 12 June 2025
Participation in conferences, congresses, etc., at home and abroad: • Coupled Free Flow - Porous Medium Systems Workshop, Stuttgart, 12-14 June 2024 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo, 17 June 2024 • ICNMMF-5 conference, Reykjavik, 26-28 June 2024 • Droplet Impact Phenomena & Spray Investigation (DIPSI), Bergamo, 13 June 2025 • Final Colloquium of DROPIT, 31 June - 1 August 2025, Stuttgart

4. Individual publications:

I. Peer-reviewed articles:

- Palmetshofer, P., **Wurst, J.**, Geppert, A. K., Schulte, K., Cossali, G. E., Weigand, B. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science*, 682, 413–422, 2025.
- Potyka, J., Stober, J., **Wurst, J.**, Ibach, M., Steigerwald, J., Weigand, B., Schulte, K. Towards DNS of Droplet-Jet Collisions of Immiscible Liquids with FS3D. *High Performance Computing in Science and Engineering '22* (pp. 197–212). Springer Nature Switzerland, 2024.
- Palmetshofer, P., **Wurst, J.** Replication Data for: Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces, DaRUS, 2024.

II. Submitted articles:

III. Further publications

a) Proceedings

- **Wurst, J.**, Schulte, K. Simulation framework for drop impacts on structured surfaces using PLIC-based contact line modeling. In *Proceedings of the DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations*. Università degli studi di Bergamo, 2024

IV. Presentations

a) Oral

- **Wurst, J.**, Dreisbach, M., Stroh, A., Kriegseis, J., & Schulte, K. (2024). Simulation framework for drop impact on micro-grooves using PLIC-based contact line modelling. 5th International Conference on Numerical Methods in Multiphase Flows (ICNMMF), Reykjavik, Iceland.

b) Posters

- **Wurst, J.**, & Schulte, K. Simulation framework for capillary driven two-phase flow using PLIC-based contact line modelling. Coupled Free Flow - Porous Medium Systems Workshop, Stuttgart, Germany, 2024.
- **Wurst, J.**, & Schulte, K. Simulation framework for drop impacts on structured surfaces using PLIC-based contact line modeling. DIPSI Workshop 2024: Droplet Impact Phenomena & Spray Investigations, Università degli studi di Bergamo, 2024.

- **Wurst, J., & Schulte, K.** Towards Direct Numerical Simulation of ternary systems in contact with a wall. DIPSI Workshop 2025: Droplet Impact Phenomena & Spray Investigations, Università degli studi di Bergamo, 2025.

Individual reports by postdoctoral researchers funded by DROPIT

C5-2 Dr. Anne K. Geppert

Development of novel optical techniques and analytical models for micro-fluid dynamic problems

C5-3 Dr. Manish Kumar

Direct numerical simulations of droplet evaporation

C5-4 Dr. Patrick Palmetshofer

Droplet impact on structured surfaces

Report of postdoctoral researcher (SP-C5-2)

Project number: SP-C5-2

First and last name of postdoctoral researcher: Dr. Anne K. Geppert

(Working) title of postdoctoral project: Development of novel optical techniques and analytical models for micro-fluid dynamic problems

1. General information:

Duration of International Research Training Group membership:

Associated member: 01.10.2016 – 31.05.2019

Postdoctoral researcher: 01.06.2019 – 30.09.2024

Funding (type and duration):

Type: DFG funding for Post-doctoral researcher in IRTG DROPIT

Duration: 01.06.2019 – 30.09.2024

Subject; institution; start date of doctoral programme; degree date:

Aerospace Engineering; University of Stuttgart; 01/2012; 17.04.2019

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

10/2005; Mechanical Engineering; University of Siegen; Intermediate diploma; 12.10.2007

04/2008; Aerospace Engineering; University of Stuttgart; Diploma (Dipl.-Ing.); 29.11.2011

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: Leibniz-Institut für Werkstofforientierte Technologien, Head of Department "Melt Atomization and Spray Forming"

2. Description of individual research project and research results achieved to date:

2.1 Motivation

The impact of droplets onto wetted walls is of relevance to natural phenomena as well as to many technical applications, such as internal combustion engines, icing on plane wings, medical inhalers, spraying of plant protective agents and any type of spray coating application, e.g. in food processing, robotic car painting or aerosol coating technologies. The optimisation and efficient design of these technologies requires a detailed knowledge on the spreading behaviour of the impacting droplets, i.e. on the maximum spreading diameter and the residual height of the liquid film on the solid substrate. From a macroscopic, experimental point of view, the impact dynamics of droplets onto wetted walls have been investigated during the past decades [1,2] and a basic knowledge on the spreading behaviour was established [3,4]. Nevertheless, the underlying physical mechanisms are not yet completely understood.

2.2 Objectives and scientific approach

The objective of this postdoctoral project is to deepen the understanding of the micro-scale fluid dynamics, which govern the macroscopic flow characteristics of droplets that impact onto wetted walls. Therefore, a two-fold approach is chosen: From an experimental point of view, the micro-scale flow is analysed with micro Particle Image Velocimetry (micro-PIV) and General Defocusing Particle Tracking (GDPT). A detailed description of the respective experimental setups, the limitations of the techniques as well as recent results are presented in the reports of V. Vaikuntanathan (first postdoctoral researcher in SP-C5), S. Schubert (SP-C1-2) and P. Molina Vogelsang (SP-C1-3).

In addition, the temporal evolution of the liquid layer thickness beneath the impacting droplet is investigated using a confocal chromatic imaging (CCI) technique. Furthermore, high-speed visualization with up to 200,000 frames per second is employed to study the early impact dynamics ($t < 0.5$), focusing on the motion of the contact line between droplet and wall-film liquid.

2.3 Analytical modelling approach and its validation

The second approach comprises analysing the splashing dynamics from an analytical point of view, i.e. modelling the crown evolution over time. Within the framework of DROPIT extensive experimental studies on droplet-film interactions with the same [5] or dissimilar liquids [4,6] for droplet and wall film have been conducted. These studies investigated the relevance of momentum losses arising from viscous effects during crown propagation leading to the development of an unsteady analytical solution for the modelling of the crown spreading rate, which is described in detail in Lamanna et al. [7].

Our modelling strategy encompasses an extension of the potential flow solution for crown propagation of Yarin & Weiss [8] by explicitly incorporating a boundary layer correction describing the momentum reduction due to viscous losses. For the early impact phase, the original model [8] describes the potential flow as a stagnation point flow of constant strength. In the spreading phase, however, the potential flow evolves into an unsteady stagnation point flow with decaying strength, since the peak impulse provided by the impacting droplet also decays as shown by Mitchell et al. [9]. Therefore, the boundary layer profile was explicitly incorporated in the evolution of the radial velocity component for both the flow field within the impact zone, and the propagation of the crown radius R_{KD} . The corresponding model reads

$$\bar{R}_{KD}(\tau) = \frac{R_{KD}}{D_0} = \alpha \lambda_1 \sqrt{\lambda} \sqrt{\tau - \tau_{ini}} + \bar{R}_{KD,ini} \quad (1)$$

Here α models the extent of the impact zone, λ_1 represents the momentum loss due to viscous forces, λ incorporates impact losses and $[\tau_{ini}, \bar{R}_{KD,ini}]$ represent the starting values of a fully formed crown at the end of the impact phase. The proposed modelling approach yields an excellent agreement with experiments over the entire spreading phase as shown in Figure 1 for two different liquids and two film thicknesses, respectively. Our analysis shows that, during the spreading phase, viscous losses are negligible only in the early stage of crown propagation [7]. This causes a systematic deviation from the square root time dependence, derived from the inviscid approach. In addition, viscous losses affect the temporal evolution of the crown [7] in two ways: directly by causing momentum losses during the spreading of the crown lamella, and indirectly through the momentum transfer from the droplet to the wall-film, which is inhibited in highly viscous fluids. This results in a complex interaction between impact and viscous losses, where higher impact losses lead to lower viscous losses in the spreading phase and vice versa. This balancing of viscous and impact losses explains why the crown evolution depends only weakly upon fluid viscosity.

The validity of the new model is further verified by comparison to experimental data for the crown base radius provided by sub-project SP-C1-2.

2.4 Analysing the early impact dynamics of single droplets onto wetted surfaces

In order to gain a deeper understanding of the complex physical processes that influence the propagation of the contact line (CL) and the formation of the uprising lamella during the impact process, we performed a complementary experimental (SP-C5-2) and numerical (SP-C2) study. Specifically, we target the first two phases of the impact process ($\tau < 0.5$), which lead to the formation of the kinematic discontinuity (or crown). The results of this study are described in detail in [10].

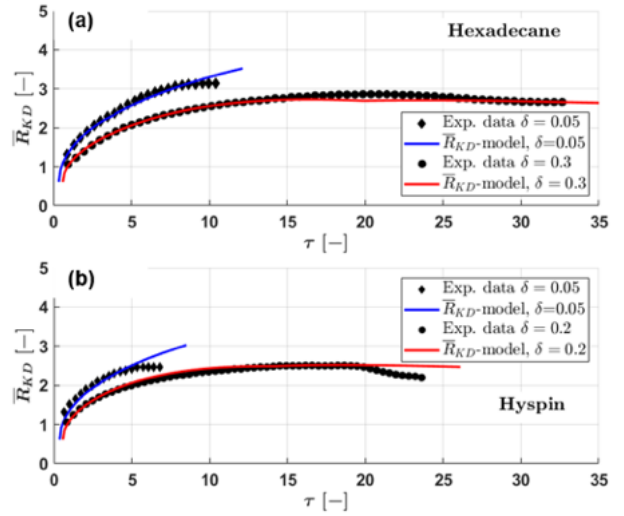


Figure 1: Temporal evolution of the crown radius $\bar{R}_{KD}(\tau)$: model predictions vs. data of experiments for (a) Hexadecane, $We \approx 1300, Re \approx 2450$ and (b) Hypsin, $We \approx 1400, Re \approx 600$.

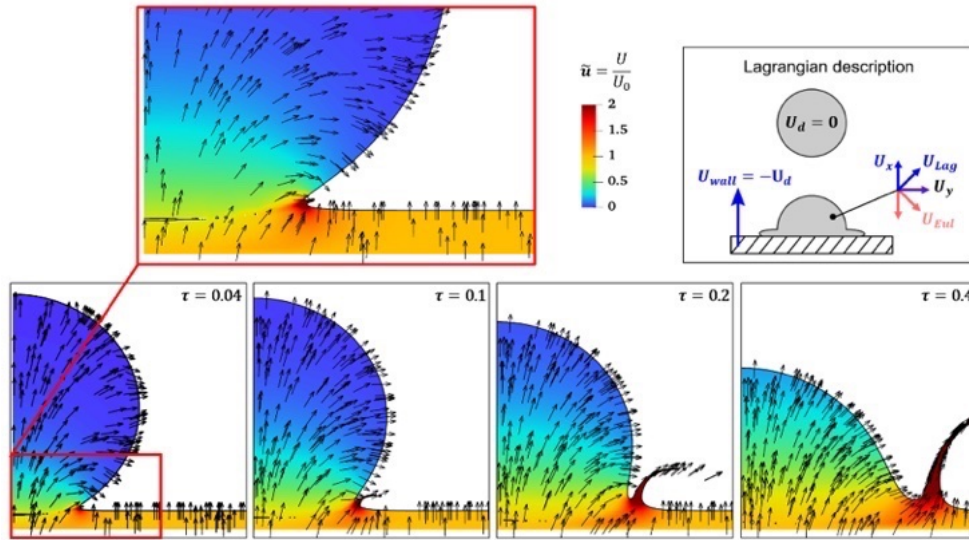


Figure 2: Lagrangian transformed flow field within the impacting droplet for different non-dimensional times for the thinnest film of $\delta = 0.07$ and silicone oil B5 (Fig. 8 of [10]).

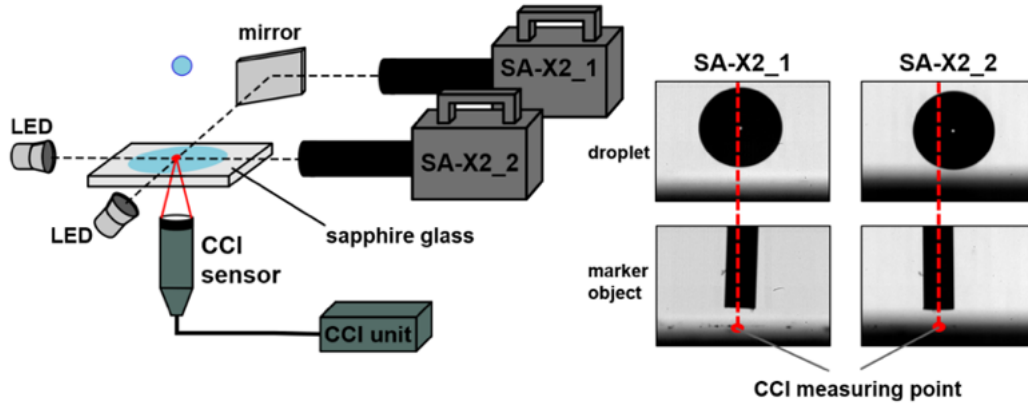


Figure 3: Scheme of experimental setup and alignment procedure between CCI-measuring point and droplet impact point.

Our analysis shows that the interaction of the droplet with the initial wall-film is the result of the interplay between three concurrent processes: droplet penetration into the wall-film, the kinematic mechanism of the rolling contact line motion and the wall-film inertia. The initial wall-film thickness δ largely controls the importance of these three processes. For impacts onto thin films ($\delta < 0.37$), the solid wall always influences the impact dynamics, because the droplet cannot fully penetrate into the wall-film and therefore must deform, which leads to a redirection of momentum. This deformation is not instantaneous and thus the contact line motion is driven by a kinematic rolling mechanism as long as the droplet maintains the shape of a truncated sphere above the liquid surface independently of initial wall-film thickness and viscosity. As soon as the deformation takes place, the lamella is accelerated radially and the effects of wall-film inertia become dominant. This rolling motion of the contact line is shown in Figure 2. It also represents a stepping-stone towards linking the impact dynamics on wetted and dry surfaces since it was observed for droplet impact onto dry surfaces [11] as well.

In a next step, we use these insights to verify the closure assumptions of our theoretical model for the crown propagation (eq. 1). Specifically, the constant of proportionality for the CL advancement α needs to be estimated accurately to achieve model closure. We derived it by fitting our early impact data to the functional relationship $\bar{R}_{CL}(\tau) = \alpha \sqrt{\tau}$. Our analysis shows that up to $\delta = 0.2$ the α -values remain unaffected by the wall-film thickness and close to the value $\alpha_{ph} = \sqrt{3/2}$. A smooth crossover from the kinematic to the inertia dominated regime occurs at $\delta \approx 0.3$. Here, an increase in film thickness leads to a decrease of the α -value, i.e. $\alpha_{YW} = [2/(3\delta)]^{1/4}$ is applicable.

2.5 Time-resolved film thickness measurements during droplet impact

Understanding the microscopic fluid dynamics of the liquid flow within the crown crater created by droplet impact, in particular the temporal evolution of the liquid layer thickness and its residual value, are crucial for the resource-efficiency of many technical applications. In order to investigate these parameters with a high spatial and temporal resolution, our group currently develops a planar, time-resolved measurement method based on UV-absorption [12]. This new technique will enable us to measure the droplet and wall-film liquid layer height separately. The validation of the measurement results to be achieved will be done in two ways, by comparison to literature data and to measurement data obtained with an already established confocal-chromatic point measuring technique (CCI-Sensor). In the experiment, the CCI-sensor measures the combined liquid layer thickness (droplet and wall-film) through a transparent pool bottom. The CCI-technique cannot differentiate between droplet and wall-film liquid because for one-component interactions no detectable interface is formed between the liquid parts. The experimental setup is shown in Figure 3 and further details can be found in Geppert et al. 2023 [13]. Here, we present CCI-measurements of the combined liquid layer thickness within the crown crater for intermediate normalized film thickness $0.09 < \delta < 0.34$ and intermediate Reynolds and Weber numbers of $Re = 925$ and $We = 925$, respectively. Our measurements show, that the liquid layer height during droplet impact first strongly increases due to the droplet entering the measurement volume. Afterward, the height starts to decrease until finally a minimum residual thickness is reached, as shown in Figure 4. Both, the temporal evolution, as well as the residual film thickness are only weakly depending on the initial height of the wall-film. These findings agree with literature data according to which the viscosity and thus the Reynolds number have the greatest influence. For impacts on thin liquid films and early times, the evolution of the combined liquid layer thickness follows the viscous solution of [14] for droplet impacts onto dry surfaces. In contrast, for thicker films and later times, the deviations become larger (see Fig. 4).

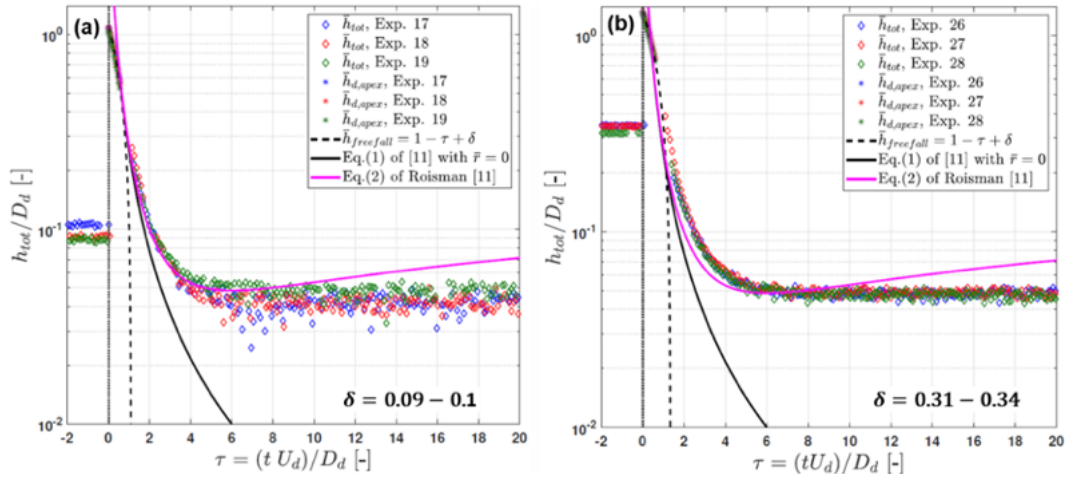


Figure 4: Evolution of liquid layer thickness during droplet impact ($Re = 925$ and $We = 925$).

2.6 Cooperation benefits

During my time as postdoctoral fellow in DROPIT, there was a strong collaboration with the projects SP-C1 and SP-C2 on the experimental investigation of droplet-film interactions. A highlight here was the Summer School project together with Jonathan Stober (SP-C2) on the early impact dynamics, which resulted in a journal publication. In addition, the collaboration with SP-B1 and SP-B5 on droplet impact dynamics onto structured surfaces was intensified, which led to a broadening of my field of research. DROPITs connections to industrial partners led to some fruitful cooperation. In general, the discussions of my scientific results within the community of DROPIT experts as well as with international scientist continued to improve my work and widened my field of research.

References

- [1] G. Liang & I. Mudawar: Review of mass and momentum interactions during drop impact on a liquid film. *Int. J. Heat Mass Transfer*, 101, 2016.
- [2] A.L. Yarin: Drop impact dynamics: Splashing, spreading, receding, bouncing... *Annu. Rev. Fluid Mech.*, 38:159-192, 2006.
- [3] A.K. Geppert: Experimental Investigation of Droplet Wall-Film Interaction of Binary Systems. PhD thesis. University of Stuttgart, 2019.

- [4] A. Geppert, A. Terzis, G. Lamanna, M. Marengo & B. Weigand: A benchmark study for the crown-type splashing dynamics of one- and two-component droplet wall-film interactions. *Exp. Fluids*, 58(12):1-27, 2017.
- [5] A. Geppert, D. Chatzianagnostou, C. Meister, H. Gomma, G. Lamanna & B. Weigand: Classification of impact morphology and splashing/deposition limit for n-hexadecane, *Atomz. Sprays* 26(10), 2016.
- [6] R. Bernard, V. Vaikuntanathan, B. Weigand & G. Lamanna: On the crown rim expansion kinematics during droplet impact on wall-films. *Exp. Therm. Fluid Sci.* 118,110168, 2020.
- [7] G. Lamanna, A.K. Geppert, R. Bernard & B. Weigand: Drop impact onto wetted walls: an unsteady analytical solution for modelling crown spreading. *J. Fluid. Mech.*, 938, A34, 2022.
- [8] A.L. Yarin & D.A. Weiss: Impact of drops on solid substrates: self-similar capillary waves, and splashing as a new type of kinematic discontinuity. *J. Fluid Mech.*, 283:141-173, 1995.
- [9] B.R. Mitchell, J.C. Klewicki, Y.P. Korkolis & B.L. Kinsey: The transient force profile of low-speed droplet impact: measurements and model. *J. Fluid Mech.* 867, 300-322. 2019.
- [10] A.K. Geppert, J.L. Stober, J. Steigerwald, K. Schulte, S. Tonini & G. Lamanna: Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. *Phys. Fluids* 36, 012005, 2024.
- [11] J. Philippi, P.-Y. Lagr  e & A. Antkowiak: Drop impact on a solid surface: short-time self-similarity. *J. Fluid Mech.* 795, 96-135, 2016.
- [12] M. Diehl: Development of a UV absorption-based measurement method for stationary and variable film thicknesses. M.Sc. thesis. University of Stuttgart, 2023.
- [13] A.K. Geppert & G. Lamanna: Towards spatial and time-resolved film thickness measurements during droplet impact onto thin liquid films. *ILASS Europe 2023*, 32nd Conference on Liquid Atomization & Spray Systems, 4-7 Sept 2023, Napoli, Italy, 2023.
- [14] I.V. Roisman: Inertia dominated drop collisions. II. An analytical solution of the Navier-Stokes equations for a spreading viscous film. *Phys. Fluids* 21, 052104, 2009

3. Description of individual IRTG-specific tasks:

My tasks in DROPIT can be divided into three main groups: supervision and training of doctoral students, organization of various meetings, coordination of the experimental work within DROPIT and my own research efforts. One of my favourite tasks it to support and counsel the PhD-students not only with regard to their scientific work, but also to their personal development. The postdoctoral position in DROPIT allowed me to do this for the two generations of PhD-students and taught me how to deal with different characters. The organization of doctoral seminars and workshops with partners from industry and academia improved my skills associated with planning, budgeting and logistic aspects of such a meetings. Especially, the collaboration with our guests helped to expand my network. The coordination of the experimental work within DROPIT broaden my expertise on various measurement techniques itself but also on the manage and maintain large equipment pools.

4. Statement on career development measures:

The supervision of PhD-students and the coordination of all experimental methods used within DROPIT, contributed to develop my supervisory skills, especially with regard to the formation and management of my own research group in the future. Therefore, the preselection of second generation PhD-candidates, which was my task during the application process, was very helpful.

I am also involved in administrative work within DROPIT, especially in the preparation of the final report of the first funding period, and the follow-up proposal for the second funding period. This includes regular meetings with all PI's, interaction with the DFG and knowledge of their funding rules, coordination of PhD-student's contributions and the attendance of a coordination talk with the university administration. The latter provided first insights into university development strategies. The interaction with academic peers, PI's and mentors within DROPIT helped me to build scientific contacts researchers all over the world and to integrate myself in the related scientific communities.

Participation in or organisation of IRTG-related events:

- DROPIT Doctoral Seminar, 15-16 May 2019, Bergamo, Italy.
- Future Perspectives Workshop, 10-11 October 2019, Stuttgart, Germany.
- DROPIT Doctoral Seminar 2020, 25-26 June 2020, virtual meeting.
- DROPIT Doctoral Seminar 2021, 08-09 March 2021, virtual meeting.
- Kick-Off meeting FP2 & Doctoral Seminar, 08-09 September 2021, Bergamo, Italy.
- DROPIT Summer School 2022, Kick-Off 10 June 2022 (S) & Closure 30 June 2022 (B).
- DROPIT Doctoral Seminar 2023, 01-02 March 2023 Stuttgart, Germany.
- “DROPIT meets Industry” Workshop, 15 June 2026, Bergamo, Italy.
- DROPIT Doctoral Seminar 2024, 08-09 February 2024, hybrid seminar, Bergamo, Italy.
- “Exp. And Num. Methods in Science” Workshop, 18 June 2024, Bergamo, Italy.

Research stays at other research institutions at home and abroad, for IRTGs also research stays at the partner university:

- Research Stay at UniBG, Summer School Project 2022, 13 June – 01 July 2022

Participation in conferences, congresses, etc., at home and abroad:

- Droplet Impact Phenomena & Spray Investigation (DIPSI) in 2019, 2022, 2023 and 2024, Bergamo, Italy.
- 30th ILASS-Europe 2020, virtual conference.
- 31th ILASS-Europe 2022, virtual conference.
- 11th International Conference on Multiphase Flow (ICMF) 2023, Kobe, Japan.
- 32nd ILASS-Europe 2023, Naples, Italy.
- 16th ICLASS 2024, hybrid conference, Shanghai, China.

5. Individual publications:**I. Peer-reviewed articles:**

- Steigerwald, J., Ibach, M., **Geppert, A. K.** & Weigand, B. Numerical Investigation of Drop Film Interactions with a Thixotropic Liquid. *J. Non-Newt. Fluid Mech.* 105259, (2024). DOI: 10.1016/j.jnnfm.2024.105259.
- Schubert, S., Steigerwald, J., **Geppert, A. K.**, Weigand, B. & Lamanna, G. Micro-PIV study on the influence of viscosity on the droplet impact dynamics and induced velocity field. *Exp. Fluids* 65, 69. (2024). DOI: 10.1007/s00348-024-03800-5.
- Palmetshofer, P., **Geppert, A. K.**, Steigerwald, J., Arcos Marz, T., & Weigand, B. Thermo-capillary central lamella recess during droplet impacts onto a heated wall. *Sci. Rep.* 14, 1102 (2024). DOI: 10.1038/s41598-024-51382-3.
- **Geppert, A. K.**, Stober, J. L., Steigerwald, J., Schulte, K., Tonini, S. & Lamanna, G. Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. *Phys. Fluids* 36 (1): 012005 (2024). DOI: 10.1063/5.0179716.
- Lamanna, G., **Geppert, A. K.**, Bernard, R., & Weigand, B. Drop impact onto wetted walls: an unsteady analytical solution for modelling crown spreading. *J. Fluid Mech.*, 938, A34 (2022). DOI: 10.1017/jfm.2022.69
- Liu, Y., **Geppert, A. K.**, Chu, X., & Weigand, B. Simulation of an annular liquid jet with a coaxial supersonic gas jet in a medical inhaler. *Atomization Sprays*, 31(9) (2021). DOI: 10.1615/AtomizSpr.2021037223.
- Ren, W., Foltyn, P., **Geppert, A. K.** & Weigand, B. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. *Sci. Rep.*, 11(1) (2021). DOI: 10.1038/s41598-021-97376-3.
- **Geppert, A.**, Bernard, R., Weigand, B., & Lamanna, G. (2020). Analytical Model for Crown Spreading During Drop Impact onto Wetted Walls: Effect of Liquids Viscosity on Momentum Transfer. In G. Lamanna, S. Tonini, G. E. Cossali, & B. Weigand (Hrsg.), *Droplet Interactions and Spray Processes. Fluid Mechanics and Its Applications*, vol. 121 (Bd. 121, S. 177-190). *Springer, Cham*. DOI: 10.1007/978-3-030-33338-6_14.
- Terzis, A., Kirsch, M., Vaikuntanathan, V., **Geppert, A.**, Lamanna, G., and Weigand, B.: Splashing characteristics of diesel exhaust fluid (AdBlue) droplets impacting on urea-water solution films. *Exp. Therm. Fluid Sci.*, 102 (2019), pp. 152-162.

II. Submitted articles:

III. Further publications

a) Proceedings

- **Geppert, A. K.**, Brajer, D. B. & Marengo, M. (2024). Unravelling the Generation of a Rectangular Liquid Umbrella after Droplet Impact onto a Small Square Surface. 16th Internat. Conference on Liquid Atomization & Spray Systems, ICLASS 2024, Shanghai, China.
- **Geppert, A. K.**, Tonini, S., Santini, M., Fest-Santini, S., Lamanna, G., Cossali, G. E. & Weigand, B. (2024). "Droplet Interaction Technologies" (DROPIT): Recent results of the International Research Training Group GRK2160/2. 16th International Conference on Liquid Atomization & Spray Systems, ICLASS 2024, Shanghai, China.
- Schubert, S., Stein, M., **Geppert, A. K.** & Lamanna, G. (2024). PIV measurements in the inner crown wall resulting from droplet impact on a thin liquid film. 16th International Conference on Liquid Atomization & Spray Systems, ICLASS 2024, Shanghai, China.
- **Geppert, A. K.**, & Lamanna, G. (2023). Towards spatial and time-resolved film thickness measurements during droplet impact onto thin liquid films. 32nd Conference on Liquid Atomization & Spray Systems, ILASS Europe 2023, 4-7 Sept 2023, Napoli, Italy.
- Palmetshofer, P., Heckel, J., **Geppert, A.** & Weigand, B. (2023). Experimental investigation of the height and angle of an uprising sheet occurring in near-simultaneous double droplet impacts onto smooth surfaces of varying wettability. 32nd Conference on Liquid Atomization & Spray Systems, ILASS Europe 2023, 4-7 Sept 2023, Napoli, Italy.
- Stober, J. L., Santini, M., **Geppert, A.** & Schulte, K. (2023). Experimental Investigation of an Oblique Droplet Impact onto a Quiescent Wall Film with Two-Perspective High-Speed Imaging. 32nd Conference on Liquid Atomization & Spray Systems, ILASS Europe 2023, 4-7 Sept 2023, Napoli, Italy.
- **Geppert, A. K.**, Foltyn, P., & Weigand, B. (2022). Effect of edge length and wettability on droplet impact onto a stand-alone cubic pillar. 31th Conference on Liquid Atomization and Spray Systems, ILASS-Europe 2022, Tel-Aviv (Virtual).
- **Geppert, A. K.**, Santini, M., Lamanna, G., Cossali, G. E., & Weigand, B. (2022). Scientific Highlights of the International Research Training Group (GRK 2160/2) „Droplet Interaction Technologies“ (DROPIT). 31th Conference on Liquid Atomization and Spray Systems, ILASS-Europe 2022, Tel-Aviv (Virtual).
- Steigerwald, J., **Geppert, A. K.**, & Weigand, B. (2021). Numerical study of drop shape effects in binary drop film interactions for different density ratios. 15th Triennial International Conference on Liquid Atomization and Spray Systems, ICLASS 2021, Edinburgh, UK.
- Lamanna, G., **Geppert, A.** and Weigand, B.: On the effect of a thin liquid film on the crown propagation in drop impact studies. 29th ILASS-Europe, Paris, France, 2019.

b) Presentations

c) Oral

- **Geppert, A. K.** (2023). Selected Results of the International Research Training Group "Droplet Interaction Technologies". 1st Sino-German Workshop "Impact of Siliceous Debris on Aviation Safety". Munich, Germany.
- **Geppert, A. K.** (2023). Tropfenaufrall auf strukturierte Oberflächen – Zwischen Ästhetik und Anwendung. Deutscher Luft- und Raumfahrtkongress 2023, Stuttgart, Germany.
- **Geppert, A. K.**, & Weigand, B. (2023). Ground spreading dynamics during single droplet impact onto a free-standing cubic surface element. DIPSI 2023, Bergamo, Italy.
- **Geppert, A. K.**, & Weigand, B. (2023). Spreading dynamics during single droplet impact onto a free-standing cubic surface element. 11th International Conference on Multiphase Flow, ICMF 2023, April 2-7, 2023, Kobe, Japan.
- Palmetshofer, P., Ren, W., Steigerwald, J., **Geppert, A.** & Weigand, B. (2023). Experimental and Numerical Investigation of Droplet Impacts onto Cubic Pillars with a Wetted Base. 11th International Conference on Multiphase Flow, ICMF 2023, April 2-7, 2023, Kobe, Japan.
- **Geppert, A.**, Bernard, R., Weigand, B. and Lamanna, G. (2019). An analytical solution for modelling the crown spreading on wetted walls based on stagnation point flow: Effect of film thickness and viscosity. DIPSI 2019, Bergamo, Italy.
- **Geppert, A.**, Bernard, R., Weigand, B. and Lamanna, G. (2019). Drop impact on wetted walls: An analytical solution for modelling the crown spreading based on stagnation-point flow. Fundamental Fluid Dynamics – Challenges in Inkjet Printing Workshop, Leiden, the Netherlands.

d) Posters

- **Geppert, A. K.**, Brajer, D. B. & Marengo, M. (2024). Mechanism behind the Generation of a Rectangular Liquid Umbrella after Droplet Impact onto a Small Square Surface. DIPSI 2024, Bergamo, Italy.
- **Geppert, A. K.**, Foltyn, P. & Weigand, B. (2022). Effect of edge length and surface wettability on droplet impact onto a stand-alone cubic pillar. DIPSI 2022, Bergamo, Italy.
- **Geppert, A.**, Bernard, R., Weigand, B. and Lamanna, G. (2019). Drop impact on wetted walls: An analytical solution for modelling the crown spreading based on stagnation-point flow. Fundamental Fluid Dynamics – Challenges in Inkjet Printing Workshop, Leiden, the Netherlands.

Report of postdoctoral researcher (SP-C5-3)

Project number: SP-C5-3

First and last name of doctoral researcher: Manish Kumar

(Working) title of postdoctoral project: Direct numerical simulations of droplet evaporation

Name of supervisors: Bernhard Weigand

1. General information:

Duration of International Research Training Group membership: Postdoctoral researcher (part time, 50%): 01.10.2024 – 31.12.2025
Funding (type and duration): 01.10.2024 – 31.12.2025, DROPIT Postdoc (50%), Other funding: 50%
Subject; institution; start date of doctoral programme; degree date: Mechanical Engineering; IIT Bombay, India; 01.01.2015 – 27.02.2021
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 07/2007; Mechanical Engineering; GBTU Lucknow, India; Bachelor of technology; 06.2011 07/2011; Thermal Engineering (CFD & HT); NIT Hamirpur, India; Master of Technology; 08.2013
Where applicable, please note any circumstances which could have affected your scientific work: -
Occupation after leaving the IRTG, if applicable: -

2. Description of post-doctoral project and research results achieved to date:

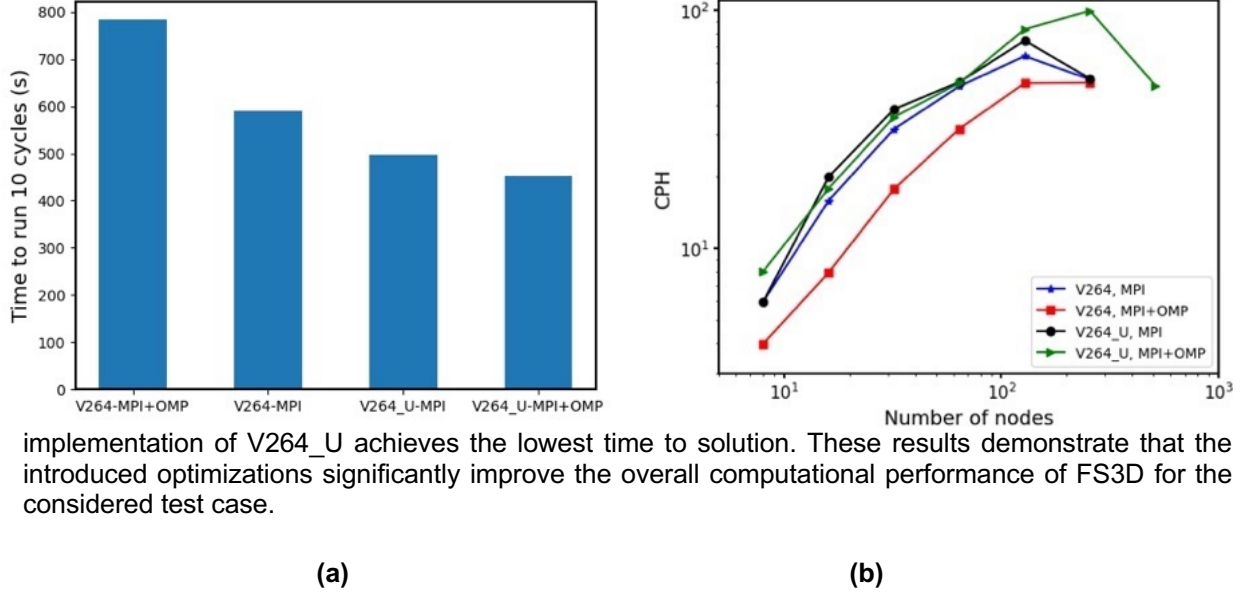
2.1 Motivation

Free Surface 3D (FS3D) is an in-house Direct Numerical Simulation (DNS) code developed at the Institute of Aerospace Thermodynamics (ITLR). It is designed to simulate multiphase flows involving liquid–gas interfaces, while resolving the hydrodynamics of single- and multi-component systems with heat and mass transport. Due to the complexity of the code, continuous support and training are required for new PhD researchers. In addition, ongoing performance enhancements are necessary to keep pace with advances in supercomputing hardware and the increasing size and resolution of simulation domains. Previous performance optimization efforts in FS3D have primarily targeted the most time-consuming routines of the hydrodynamics solver, including the multigrid (MG) solver, viscosity solver, and momentum advection solvers [1, 2]. However, for studies focusing on heat and mass transfer [3], optimizing the phase-transfer routines is essential to improve overall performance. In particular, the evaporation routines play a critical role in accurately modeling phase-change phenomena, especially when evaporation time scales become comparable to other physical processes, such as droplet wetting.

2.2 Computational performance enhancement of evaporation routines in FS3D

A test case consisting of a droplet suspended in an ambient airflow is considered as a model problem to evaluate the performance of the evaporation routines in FS3D. The performance enhancements were implemented in FS3D version V264; the optimized version is referred to as V264_U in this report for clarity. Both versions were compiled using the GNU compiler with MPI and OpenMP (MPI+OMP), as

well as in an MPI-only configuration, to enable a direct performance comparison. The performance analysis tool MAQAO was used to assess the different FS3D versions on the HPE Apollo (Hawk) system. The computational grid for the analysed test case consisted of $256 \times 256 \times 256$ cells. For the MPI-only configuration, 128 MPI processes were employed, while for the hybrid MPI+OMP configuration, 32 MPI processes with 4 OpenMP threads each were used. All simulations were executed on a single node of HPE Apollo (Hawk), ensuring that both versions were evaluated using identical computational resources. Figure 17(a) presents the total wall-clock time required to compute ten time cycles for the different FS3D versions. As shown in the bar chart, V264_U consistently requires less computational time than V264 in both the MPI-only and MPI+OMP configurations. Moreover, the hybrid MPI+OMP



implementation of V264_U achieves the lowest time to solution. These results demonstrate that the introduced optimizations significantly improve the overall computational performance of FS3D for the considered test case.

Figure 17: (a) Performance of different versions of FS3D tested for running 10 time cycles, (b) Strong scaling performance: Estimated cycles per hour (CPH) in case of utilizing MPI and MPI with OpenMP for both versions V264 and V264_U.

The effect of increasing the number of cores on performance was evaluated through a strong-scaling study, in which the grid size was kept constant while the degree of parallelization was increased. A fixed spatial resolution of $1024 \times 1024 \times 1024$ grid cells was used for this analysis. Figure 1(b) presents the measured strong-scaling performance on a logarithmic scale in terms of cycles per hour (CPH), which represents the number of time steps simulated per hour. The CPH values were obtained by running each test case for 30 minutes and extrapolating the results. The strong-scaling tests were performed on the HPE Apollo (Hawk) system, where each node provides 128 CPU cores. For the MPI-only configuration, simulations were executed with one MPI process per core, fully occupying each node. In the hybrid MPI+OMP configuration, 32 MPI processes per node were used, each employing four OpenMP threads, again utilizing all available cores. Each MPI process was pinned with a stride of four to accommodate the OpenMP threads. This setup ensures a fair comparison across all cases by using identical computational resources, problem sizes, and spatial resolutions while varying only the parallelization strategy and FS3D version. The number of nodes was systematically increased to evaluate the strong-scaling behavior until a decline in CPH was observed for each combination of FS3D version and parallelization approach. The maximum number of nodes considered in this study was 512, corresponding to 65536 CPU cores. As shown in Figure 17(b), the CPH increases monotonically for all configurations up to 128 cores. Among the tested cases, V264 with MPI+OMP exhibits the lowest overall CPH. V264_U using MPI-only achieves the best performance for 16 and 32 nodes, whereas V264_U with MPI+OMP delivers the highest CPH for all larger node counts. Except for V264_U with MPI+OMP, all configurations show a performance degradation when scaling beyond 256 nodes. In contrast, the hybrid MPI+OMP implementation of V264_U continues to scale up to 256 nodes and only shows a decline at 512 nodes. This demonstrates that the hybrid parallelization not only improves performance at commonly used node counts but also extends the strong-scaling limit by a factor of two. Overall, the maximum CPH achieved with V264_U is approximately 35% higher than that obtained with V264.

In addition to the performance improvements, several modifications/improvements have also been done for the evaporation modelling in FS3D, and several DNS have been done for evaporating droplets.

2.3 Cooperation benefits

The position acted as the central contact for all DROPIT members for numerical projects, especially for all people using FS3D. For example, there was strong collaboration and cooperation with the projects SP-A1 and SP-B5 throughout my time as a DROPIT postdoc. These projects involved two PhD researchers (during my time as a DROPIT postdoc): Hu Yueqiao and Rishav Saha, respectively. As they are users of FS3D, conducting their research work by solving the physics of droplet evaporation and wetting dynamics of falling droplets, I was able to help them resolve some issues encountered during their work with FS3D. In addition, I also benefited in terms of broadening my research experience and research domain in the field of DNS simulations. Reducing the computational time of FS3D's evaporation routines helped save the precious research time of PhD students. Overall, working and interacting with researchers in the DROPIT group and other experts helped me to improve my knowledge and gave me the experience of mentoring young researchers.

References:

- [1] J. Potyka, J. Stober, J. Wurst, M. Ibach, J. Steigerwald, B. Weigand und K. Schulte, „Towards DNS of droplet-jet collisions of immiscible liquids with FS3D,“ in *International Conference on High Performance Computing in Science and Engineering*, 2022.
- [2] J. Steigerwald, M. Ibach, J. Reutzsch und B. Weigand, „Towards the Numerical Determination of the Splashing Threshold of Two-component Drop Film Interactions,“ in *High Performance Computing in Science and Engineering '20*, W. E. Nagel, D. H. Kröner und R. M. M., Hrsg., Springer, 2022.
- [3] J. Reutzsch, C. Kieffer-Roth und B. Weigand, „A consistent method for direct numerical simulation of droplet evaporation,“ *Journal of Computational Physics*, Bd. 413, p. 109455, 2020.

3. Description of individual IRTG-specific tasks

As a DROPIT postdoc, I was responsible for several tasks. One of the tasks was training and mentoring new PhD researchers and helping them to resolve problems with FS3D direct numerical simulations on the previous supercomputer HLRS-Hawk and the new supercomputer HLRS-Hunter. I also guided them on how to implement Open-MP in their part of the FS3D code to enhance the computational performance for their cases. In addition to supporting PhD students in DROPIT projects, I participated in organizing various meetings, Multi-phase seminars, and colloquiums. Many external scientific guests were invited to give lectures in Multi-phase seminars, which helped DROPIT researchers and myself broaden our knowledge and research aptitude. Many Master's students also participated, along with the DROPIT researchers, resulting in the dissemination of knowledge within the University of Stuttgart's larger community. Lastly, the organization of doctoral seminars and workshops with partners from industry and academia strengthened my competencies in planning and logistical coordination, while fostering a structured approach to academic project management. Close collaboration with invited speakers and external partners further supported interdisciplinary exchange and the expansion of my professional network. Additionally, coordinating the numerical work within the DROPIT project enhanced my expertise in various CFD techniques and deepened my experience in the management, maintenance, and efficient use of large CFD codes.

4. Statement on career development measures:

Being a DROPIT postdoc helped me gain experience of supervising PhD students and also broaden my research expertise in numerical or computational fluid dynamics (CFD), including the aspect of the computational performance of CFD codes and how to enhance it.

Participation in or organisation of IRTG-related events:

- DROPIT Doctoral Seminar, 28-29 Jan. 2025, Stuttgart.
- AI Meets Scientific Computing workshop, 11 Jun. 2025, Bergamo, Italy.
- Analytical and Numerical Methods in Science workshop, 12 Jun. 2025, Bergamo, Italy.
- Droplet Impact Phenomena & Spray Investigation (DIPSI)-2025, 13 Jun. 2025, Bergamo, Italy.
- DROPIT final colloquium, 31 Jul. – 01 Aug. 2025, Stuttgart.

Participation in conferences, congresses, etc., at home and abroad:

- 33rd ILASS-Europe 2025 conference, 31 Aug. – 04 Sep. 2025, Lund, Sweden.

5. Individual Publications:

- **M. Kumar**, R. Saha, J. Potyka, K. Schulte and B. Weigand, “Direct Numerical Simulations of Droplet Impact onto Heated Surfaces using the Program Free Surface 3D (FS3D)”, (High Performance Computing in Science and Engineering '24, Accepted for publication by HLRS in Springer Nature).

Report of postdoctoral researcher (SP-C5-4)

Project number: SP-C5-4

First and last name of doctoral researcher: Patrick Palmetshofer

(Working) title of doctoral project: Droplet impact on structured surfaces

Name of supervisors: Bernhard Weigand

1. General information:

Duration of International Research Training Group membership: May 1, 2021 – June 15, 2025
Funding (type and duration): April 1, 2025 – June 15, 2025, DROPIT Postdoc (50%), University of Stuttgart Basic Funds (50%)
Start of doctoral training and (anticipated) date of doctoral degree: May 1, 2021 – March 4, 2025
Commencement of studies; subject; university and date of degree (diploma, MA, etc.): 10/2015 – 09/2018 Aerospace Engineering, University of Stuttgart, B.Sc. 10/2018 – 03/2021 Aerospace Engineering, University of Stuttgart, M.Sc. 05/2021 – current Aerospace Engineering, University of Stuttgart, Dr.-Ing.
Where applicable, please note any circumstances which could have affected your scientific work: -
Occupation after leaving the IRTG, if applicable: -

2. Description of post-doctoral project and research results achieved to date:

In the two months after the successful defense of my PhD thesis, I worked on getting our last two papers and my dissertation published, with a focus on droplet-film interaction. As such, my work has been focused on additional analysis and write-up of the experimental data obtained during my PhD work.

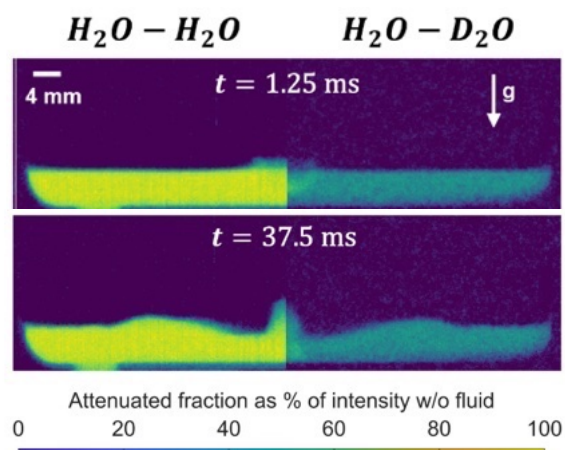


Figure 18: Neutron imaging of water droplet impacts onto water (left) and heavy water (right).

Mainly, my work has been focused on the results made regarding droplet impacts onto submerged sub-millimeter structures and the Neutron Imaging investigations at Paul Scherrer Institute.

Most importantly, a comparison between numerical results obtained in SP-B5 and my experimental data was now possible, resulting in the submission of our paper titled “Droplet impacts onto submerged structured surfaces”. This allows a proper explanation of the formation mechanisms of the crown streaks, which appear to be caused in a similarly to the rhomboid spreading shapes on dry structured surfaces. With possible higher resolutions in the future, the breakup mechanisms of the crowns could be investigated further.

For the neutron imaging investigations, additional averaging steps were implemented and several beam hardening corrections were applied. This way, qualitative results for the distribution of the droplet liquid in the film over time are obtained. However, quantitative results remain difficult to obtain due to the low signal to noise ratios for a hardened beam.

Additionally, I worked on other activities in DROPIT during my Postdoc time: I managed the test stands for droplet impact and contact angle measurements. To ensure a smooth transition for the experimental researchers in DROPIT who are still working on their PhDs, I helped with their measurement campaigns and organized measurement equipment, most importantly the four-perspective test stand. I was also involved in the organization of the final colloquium, which will take place between July 31st and August 1st.

3. Comments on the qualification programme and supervision strategy:

For overall comments on the qualification programme and supervision strategies, please refer to my PhD report.

4. Individual publications:

I. Submitted articles (just in my postdoc time):

- Palmetshofer, P., Saha, R., Bryce, J., Geppert, A. K., Cossali, G. E., and Weigand, B.: Droplet impacts onto submerged structured surfaces. Submitted to Physics of Fluids
- Palmetshofer, P., Wirth, M., Steinhausen, C., Trtik, P., Lehmann, E., Geppert, A. K.: Neutron imaging investigation of liquids mixing during droplet impact onto liquid films. Submitted to ILASS Europe 2025

Individual reports of associated postdoctoral researchers

AS02 Dr. Francesco Carlo Massa

High-order numerical methods for multi-component incompressible flows

Start-up grant Dr. Simone Chiocchetti

High-order numerical methods on moving Voronoi meshes with Voronoi-Scutoid space-time elements

Report of associated postdoctoral researcher (SP-AS02)

Project number: Associated Project 02

First and last name of postdoctoral researcher: Francesco Carlo Massa

(Working) title of postdoctoral project:

High-order numerical methods for multi-component incompressible flows

1. General information:

Duration of International Research Training Group membership:

Associated member of DROPIT since 10 October 2016

Funding (type and duration):

Tenure-Track position at the Department of Engineering and Applied Sciences (DISA), University of Bergamo.

Start: 11-Jul-2025. End: 10-Jul-2031.

Post-doc position at the Department of Engineering and Applied Sciences (DISA), University of Bergamo.

Start: 06-Jul-2020. End: 05-Jul-2025

Subject; institution; start date of doctoral programme; degree date:

Implicit Discontinuous Galerkin Methods with efficient time integration for incompressible variable density and compressible turbulent flows.

Department of Engineering and Applied Sciences (DISA), University of Bergamo.

Start: 1-Jan-2014. Degree date: 10-May-2017

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

Bachelor degree in Mechanical Engineering at University of Brescia, 10-Nov-2010.

Master degree in Mechanical Engineering at University of Brescia, 9-Jul-2013.

Where applicable, please note any circumstances which could have affected your scientific work: --

Occupation after leaving the IRTG, if applicable: --

2. Description of individual research project and research results achieved to date:

Motivation

Multicomponent flows arise in a wide range of industrial and technological applications such as hydrocarbon extraction and transport, pollutant treatment, liquid-jet processes, fuel sprays, droplet dispersion, coating technologies and porous media flows. Their physical modelling is challenging because macroscopic phenomena, e.g., surface tension, droplet coalescence and breakup, are strongly influenced by microscopic interactions occurring at the interfaces between the fluid components. A reliable numerical tool must therefore capture these complex interactions while preserving robustness and accuracy across multiple spatial and temporal scales.

During the first phase of the DROPIT project, a high-order fully implicit Discontinuous Galerkin (DG) discretization of the Variable Density Incompressible (VDI) model was developed. This approach treats density as an advected property that enables a diffuse-interface representation of multicomponent flows. The DG formulation proved particularly suitable for sharply resolving interface layers while maintaining high accuracy. The encouraging results obtained during phase 1 motivated the extension of the methodology toward a more general diffuse-interface model, capable of describing multicomponent flows featuring both incompressible phases and incorporating surface tension effects directly at the modeling level.

State of the art

Interface modelling strategies for multicomponent flows can be broadly classified into sharp-interface methods (e.g., Volume of Fluid, Level Set, ALE, front tracking) and diffuse-interface methods. Sharp-interface methods describe the material interface as a surface of zero thickness across which thermodynamic quantities are discontinuous. Diffuse-interface approaches, instead, represent the interface as a thin but finite mixing region characterized by a rapid yet continuous variation of a suitable field variable (e.g., mass/volume fraction or density). The latter do not require geometric interface reconstruction and can naturally describe topological changes such as droplet coalescence and breakup in two and three dimensions.

Given their high spatial accuracy and flexibility on unstructured meshes, DG methods represent an excellent choice for diffuse-interface formulations, particularly when coupled with high-order implicit time integration schemes. In the context of the present project, the quasi-incompressible Navier–Stokes–Cahn–Hilliard (NSCH) model, originally proposed by Lowengrub and Truskinovsky, provides two key advantages compared to the VDI formulation: (i) the interface thickness is an explicit model parameter; (ii) surface tension arises directly from the thermodynamic principles.

Project development & status

Building upon the phase-1 DG solver, the second phase of the project focused on the development of a DG discretization of the quasi-incompressible NSCH equations. The transition from VDI to NSCH required substantial methodological advancements.

From 2020 to 2021, in collaboration with subproject SP-B4, an artificial Equation-of-State Riemann solver for incompressible single-component flows was formulated and published in *Journal of Computational Physics* (Massa, Ostrowski, Bassi, Rohde 2022). Although originally developed for a not multi component setting, this solver provided the conceptual foundation for the flux formulation later adopted in the NSCH convective terms.

During the DROPIT Summer School 2022, a collaboration with SP-C2 investigated droplet–film interaction at high density ratios using the VDI code developed in the DROPIT phase 1. The comparison with experimental results and FS3D simulations offered valuable insights into the behaviour of diffuse-interface models for air–water systems and led to the publication in *Applied Sciences* (Bassi, Botti, Colombo, Massa 2022). These studies highlighted both the strengths and the limitations of the VDI model, reinforcing the need for the more general NSCH formulation.

Subsequent work led to the extension of the artificial Equation-of-State based Riemann solver to multi-component incompressible flows for the treatment of the NSCH convective numerical fluxes. Preliminary results were presented at ICLASS 2022, demonstrating the capability of the DG solver to advect immiscible droplets in a physically consistent manner at increasing polynomial degrees and high-order implicit time integration.

In 2023–2024, the DG formulation was further refined. A baseline implementation of the full NSCH system was developed during the Outgoing Visiting Fellowship at ITLR in early 2023, followed by the application of high-order implicit Rosenbrock and ESDIRK schemes for efficient time integration.

The most recent developments concern the BR2 method based implementation and validation of the mass diffusion term of the NSCH model, a fundamental component governing the interface thickness between components. This research work was conducted during the Outgoing Visiting Fellowship at ITLR in early 2025.

Results

The numerical developments carried out during the second phase of the project have progressively extended and generalised the DG solver originally designed for the VDI model, and have been accompanied by a continuous validation effort. The effectiveness of the artificial Equation-of-State Riemann solver for single-component incompressible flows was demonstrated on several benchmark cases, including the periodic-hill turbulent flow at $Re=10595$, see Figure 1, where the solver showed excellent stability and produced results in good agreement with reference numerical and experimental

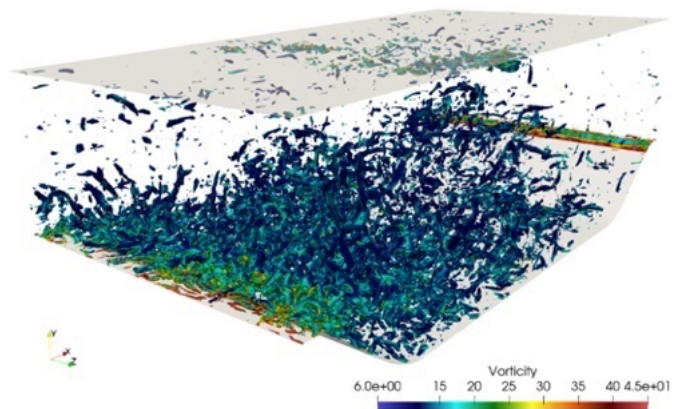


Figure 1: Q-criterion of the turbulent flow over periodic hills

data. This test case shows that the artificial-EoS approach is a robust and suitable choice for the numerical convective fluxes treatment in the context of turbulent single-component incompressible flows. In parallel, the VDI formulation developed during the first phase was assessed in detail with respect to accuracy, mass conservation and robustness at high density ratios (Bassi, Botti, Colombo, Massa 2022). The solver showed optimal spatial and temporal convergence on analytical test cases. The model was then applied to droplet-film interaction in air-water conditions, see Figure 2. The numerical results highlighted the ability to capture interface topology changes without geometric reconstruction and some inherent limitations of the VDI model, mainly related to interface spreading. These findings further motivated the transition to the quasi-incompressible Navier-Stokes-Cahn-Hilliard (NSCH) model.

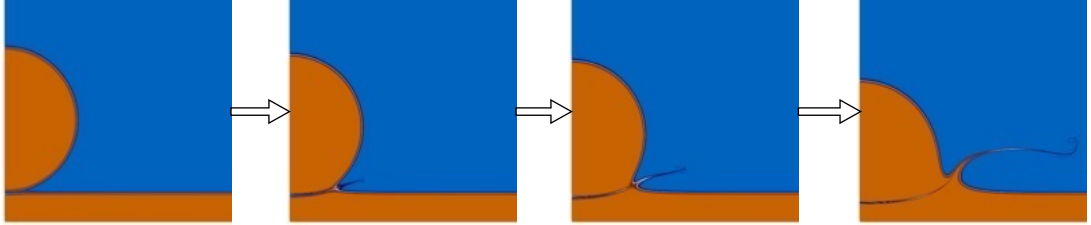


Figure 2: Density field time evolution of a water droplet impingement over the liquid film within air.

The extension of the artificial Equation-of-State Riemann solver to multicomponent incompressible flows was assessed through the pure advection of two immiscible droplets within an air-oil-water three-component system. The DG formulation of the inviscid NSCH model demonstrated that the approach can robustly advect the droplets while achieving progressively higher accuracy with increasing polynomial degree, see Figure 3. This test cases also highlighted the effectiveness and robustness of the implementation of the Rosenbrock and ESDIRK high-order implicit time integration schemes.

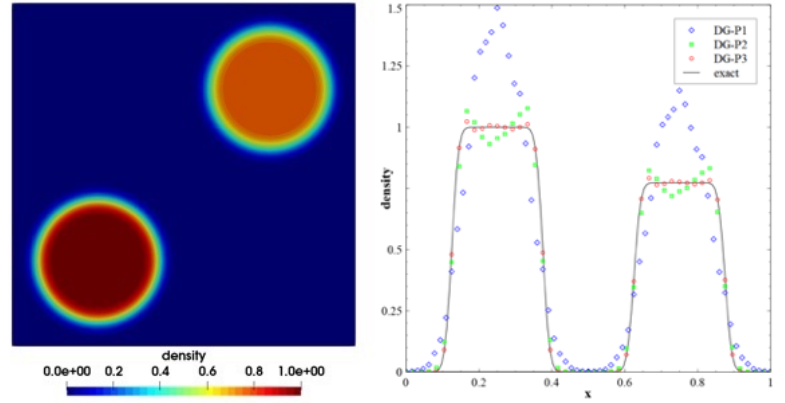


Figure 3: Raining droplet test case (water and oil in air). Initial solution (left) and dimensionless density profiles obtained from first to third degree DG polynomial approximation.

During the most recent research stay at ITLR (2025), the mass diffusion term implementation, based on the BR2 scheme, was completed and validated firstly on the raining droplet test case (water-oil configuration), see Figure 4. The test case allowed to show the capability of the computational approach to effectively control the interface thickness by simply modify a user-defined parameter, the Hilliard number.

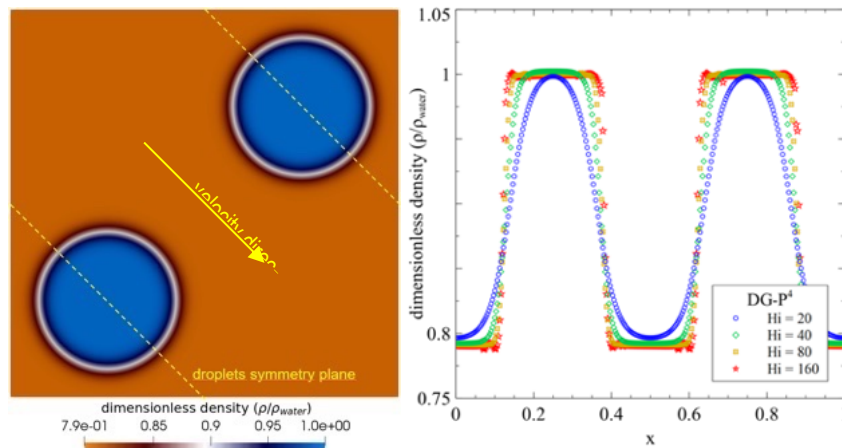


Figure 4: Raining droplet test case (water in oil). Initial solution (left) and dimensionless density profiles obtained with a fourth degree DG polynomial approximation after one period of advection (right) at different Hilliard (H_i) numbers.

The results obtained during this project demonstrate a continuous progression toward the formulation of an high-order implicit DG framework for the NSCH model. The most recent investigations have confirmed the potential of the approach, while also revealing critical aspects that will guide the next phase of research.

Networking with other projects

Throughout the duration of the DROPIT project, the research work has benefitted from close and continuous interactions with several subprojects, in particular SP-B4 and SP-C2. These collaborations have been very important for the development and assessment of the numerical approach proposed in the SP-AS05. Although the DROPIT project formally concludes, the collaborations established during its course have laid the foundation for continued research interactions beyond the project deadline, opening new opportunities for joint investigations and long-term scientific cooperation.

3. Description of individual IRTG-specific tasks:

The current associated project focused on the development and validation of high-order numerical methods for multi-component incompressible flows. During this period, several tasks were successfully completed. In collaboration with SP-B4, a new artificial Equation-of-State Riemann solver was assessed for single-component incompressible flows. The interaction with SP-C2, particularly during the DROPIT Summer School, enabled a comparative assessment of the diffuse-interface VDI model against experimental data and FS3D simulations, providing essential information about the robustness of the diffuse interface approach when dealing with high density ratio configurations. Additional activities focused on the development of a high-order implicit DG framework for the quasi-incompressible NSCH system: the extension of the artificial EoS Riemann solver to multicomponent configurations, the implementation of Rosenbrock and ESDIRK implicit time integration schemes, and the BR2-based treatment of the mass diffusion contribution at material interfaces constituted the main tasks addressed. Research stays at the ITLR played a central role in addressing key modelling aspects such as interface thickness control and mass diffusion.

Participation in or organisation of IRTG-related events:

- DROPIT Doctoral Seminar, 25-26 June 2020, online
- DROPIT Doctoral Seminar, 08-09 March 2021, online
- DROPIT Kick-Off-Meeting (II phase), 08-09 September 2021, Bergamo, Italy
- DROPIT Summer School, 9-30 June 2022, Stuttgart-Bergamo, Germany-Italy
- DROPIT Doctoral Seminar, 01-02 March 2023, Stuttgart, Germany
- DROPIT meets Industry, 15 June 2023, Bergamo, Italy
- DROPIT Doctoral Seminar, 08-09 February 2024, Bergamo, Italy
- DROPIT Workshop “Experimental and Numerical Methods in Science”, 18 June 2024, Bergamo, Italy
- DROPIT Doctoral Seminar, 28-29 January 2025, Stuttgart, Germany
- DROPIT Workshop “AI meets Scientific Computing: bridging theory and practice”, 11 June 2025, Bergamo, Italy
- DROPIT Final colloquium, 31 July – 1 August 2025, Stuttgart, Germany

Research stays at other research institutions at home and abroad, for IRTGs also research stays at the partner university:

- From 09 January 2023 to 17 February 2023, research stay at the Institut für Aerodynamik und Gasdynamik of the University of Stuttgart founded by the Outgoing Visiting Fellowship of the University of Bergamo.
- From 17 February 2025 to 07 March 2025, research stay at the Institut für Aerodynamik und Gasdynamik of the University of Stuttgart founded by the Outgoing Visiting Fellowship of the University of Bergamo.

Participation in conferences, congresses, etc., at home and abroad:

- USNCCM16, Chicago (online), 2021
- HONOM 2022, Braga, 2022
- ECCOMAS 2022, Oslo, 2022
- DLES13, Udine, 2022
- ETMM14, Barcelona, 2023
- DLES14, Erlangen, 2024
- HONOM 2024, Chania, 2024
- M2P, Valencia, 2024
- ETMM15, Dubrovnik, 2025

4. Individual publications:

I. Peer-reviewed articles:

- Bassi F., Botti L., Colombo A., Ghidoni A., Massa F.: Linearly implicit Rosenbrock-type Runge Kutta schemes applied to the Discontinuous Galerkin solution of compressible and incompressible unsteady flows. *Computers & Fluids*, 118 (2015), pp. 305-320.
- Bassi F., Botti L., Colombo A., Crivellini A., Ghidoni A., Massa F.: On the development of an implicit high-order Discontinuous Galerkin method for DNS and implicit LES of turbulent flows. *European Journal of Mechanics – B/Fluids*, 55 (2) (2016), pp. 367-379.
- Noventa G., Massa F., Bassi F., Colombo A., Franchina N., Ghidoni A.: A high-order Discontinuous Galerkin solver for unsteady incompressible turbulent flows. *Computers & Fluids*, 139 (2016), pp. 248-260.
- Massa F., Noventa G., Lorini M., Bassi F., Ghidoni A.: High-order linearly implicit two-step peer schemes for the discontinuous Galerkin solution of the incompressible Navier-Stokes equations. *Computers & Fluids*, 162 (2018), pp. 55-71.
- Bassi F., Massa F., Botti L., Colombo A.: Artificial compressibility Godunov fluxes for variable density incompressible flows. *Computers & Fluids*, 169 (2018), pp. 186-200.
- Noventa G., Massa F., Rebay S., Bassi F., Ghidoni A.: Robustness and Efficiency of an Implicit Time-Adaptive Discontinuous Galerkin Solver for Unsteady Flows. *Computers & Fluids*, 204 (2020), pp. 1-20.
- Botti L., Massa F.: HHO Methods for the Incompressible Navier-Stokes and the Incompressible Euler Equations. *Journal of Scientific Computing*, 92 (2022), pp. 1-38.
- Ghidoni A., Massa F., Noventa G., Rebay S.: Assessment of an adaptive time integration strategy for a high-order discretization of the unsteady RANS equations. *International Journal for Numerical Methods in Fluids*, (2022), pp. 1-41.
- Bassi F., Botti L., Colombo A., Massa F.: Assessment of an Implicit Discontinuous Galerkin Solver for Incompressible Flow Problems with Variable Density. *Applied Sciences*, 12 (2022), pp. 1-24.
- Massa F., Ostrowski L., Bassi F., Rohde C.: An artificial Equation of State based Riemann solver for a discontinuous Galerkin discretization of the incompressible Navier–Stokes equations. *Journal of Computational Physics*, 448 (2022), pp. 1-25.
- Verzeroli L., Massa F., Botti L.: A HDG formulation for nonlinear elasticity problems featuring finite deformations and frictionless contact constraints. *Finite Elements in Analysis and Design*, 215 (2023), pp. 1-17.
- Colombo A., Crivellini A., Ghidoni A., Massa F. C., Noventa G.: p-adaptive discontinuous Galerkin solution of transonic viscous flows with variable time step-size. *Computers & Fluids*, 282 (2024), pp. 1-15.
- Ghidoni A., Massa F. C., Noventa G.: Equal-loudness-adaptive time integration for the high-order solution of direct aeroacoustics simulations. *Computers & Fluids*, 301 (2025), pp. 106776.
- Botti L., Botti M., Di Pietro D. A., Massa F.: Stability, convergence, and pressure-robustness of numerical schemes for incompressible flows with hybrid velocity and pressure. *Mathematics of Computation*, 95 (2026), pp. 1-28.

- Botti L., Di Pietro D. A., Massa F.: Hybrid High-order formulations with turbulence modelling capabilities for incompressible flow problems. *Computers & Fluids*, 305 (2026), pp. 106915.

II. Further publications

a) Proceedings

- Massa F., Noventa G., Bassi F., Colombo A., Ghidoni A., Lorini M.: High-Order linearly implicit two-step Peer methods for the Discontinuous Galerkin solution of the incompressible RANS equations.
- Bassi F., Colombo A., Massa F., Noventa G., Ghidoni A.: Hybrid RANS-LES simulations with the discontinuous Galerkin method. In: *Proceedings of the 6th European Conference on Computational Mechanics: Solids, Structures and Coupled Problems, ECCM 2018 and 7th European Conference on Computational Fluid Dynamics, ECFD 2018*, pp. 4183-4193.
- Noventa G., Massa F., Rebay S., Colombo A., Bassi F., Ghidoni A.: Investigation of adaptive time-step strategies for high-order accurate incompressible simulations. In: *Proceedings of the 6th European Conference on Computational Mechanics: Solids, Structures and Coupled Problems, ECCM 2018 and 7th European Conference on Computational Fluid Dynamics, ECFD 2018*, pp. 3583-3594.
- Ostrowski L., Massa F.: An incompressible compressible approach for droplet impact. In: *Proceedings of the DIPSI Workshop 2019: Droplet Impact Phenomena & Spray Investigations*, pp. 18-21.

b) Technical and scientific reports

- Bassi F., Botti L., Colombo A., Ghidoni A., Massa F., Noventa G.: On the development of an implicit high-order discontinuous Galerkin solver for a hybrid RANS-LES model. In: Salvetti M., Armenio V., Fröhlich J., Geurts B., Kuerten H. (eds) *Direct and Large-Eddy Simulation XI. ERCOFTAC Series*, vol 25. Springer, Cham (2019).

III. Presentations

a) Oral

- Massa F., Bassi F., Botti L., Colombo A.: High-Order implicit time integration schemes in a DG solver for compressible and incompressible unsteady flow simulation. *HONOM 2015*, Trento, Italy, 2015.
- Massa F., Bassi F., Botti L., Colombo A.: An implicit high order discontinuous Galerkin method for incompressible variable density flows. *ECCM 6 – ECFD 7*, Glasgow, United Kingdom, 2018.
- Massa F., Bassi F., Botti L., Colombo A.: Density positivity and mass conservation for an implicit High-Order discontinuous Galerkin method applied to Variable Density Incompressible flows. *HONOM 2019*, Madrid, Spain, 2019.
- Massa F., Bassi F., Botti L., Colombo A.: An implicit high-order Discontinuous Galerkin method applied to variable-density incompressible flows. *TiciNum 2019*, Pavia, Italy, 2019.
- Botti L., Massa F.: DG Discretizations of the INS Equations, which features matter the most? *USNCCM 2021*, Chicago, USA (held online), 2021.
- Botti L., Massa F.: HHO methods for the incompressible Navier–Stokes and the incompressible Euler equations. *HONOM 2022*, Braga, Portugal, 2022.
- Bassi F., Colombo A., Crivellini A., Massa F.: Implicit discontinuous Galerkin method for the efficient scale resolving simulation of compressible turbulent flows. *ECCOMAS 2022*, Oslo, Norway, 2022.
- Bassi F., Colombo A., Massa F.: Scale-resolving simulation of compressible turbulent flows with a Discontinuous Galerkin method. *DLES13*, Udine, Italy, 2022.
- Bassi F., Colombo A., Massa F.: Scale-resolving simulation of compressible turbulent flows with a Discontinuous Galerkin method. *DLES13*, Udine, Italy, 2022.
- Botti L., Massa F.: Hybrid High-Order Methods with Turbulence Modelling Capabilities. *DLES14*, Erlangen, Germany, 2024.
- Botti L., Massa F.: Hybrid High-Order methods with hybrid pressure and improved turbulence modelling capabilities. *HONOM 2024*, Chania, Greece, 2024.

b) Posters

- Massa F., Bassi F., Botti L., Colombo A.: High-order numerical methods for multi-component incompressible flows. *DROPIT final colloquium*, 2025

Report of postdoctoral researcher – Start-up grant

Project number: Start-up grant

First and last name of doctoral researcher: Simone Chiocchetti

(Working) title of postdoctoral project: High order numerical methods on moving Voronoi meshes with Voronoi-Scutoid space-time elements

Name of supervisors: Prof. Dr.-Ing. Andrea Beck

1. General information:

Duration of International Research Training Group membership:

Postdoctoral researcher: 15.05.2023 – 31.12.2023

Funding (type and duration):

15.05.2023 – 31.12.2023, DROPIT Start-up Grant

Subject; institution; start date of doctoral programme; degree date:

Civil, Environmental, and Mechanical Engineering; University of Trento; begin 5 Nov 2018; PhD degree 10 June 2022

Commencement of studies; subject; university and date of degree (diploma, MA, etc.):

09/2011 Environmental Engineering, University of Trento, Bachelor's degree, 18 Dec 2014.
09/2014 Environmental Engineering, University of Trento, Master's degree, 29 March 2018.
11/2018 Civil, Environmental, and Mechanical Engineering, University of Trento, PhD, 10 June 2022.

Where applicable, please note any circumstances which could have affected your scientific work: -

Occupation after leaving the IRTG, if applicable: - Marie Skłodowska-Curie Postdoctoral Fellow, University of Cologne (from January 2024)

2. Description of post-doctoral project and research results achieved to date:

2.1 Motivation

Moving (Lagrangian) meshes offer significant advantages in the simulation of flows involving interfaces, sharp gradients, and complex geometries. The mesh adapts to the flow, allowing for a more natural and accurate representation of the solution. However, when the mesh elements are polygonal or polyhedral, as in the Voronoi tessellations that are of interest here, mesh motion leads to connectivity changes: elements may gain or lose neighbours as the mesh deforms. In the space-time framework underlying the ADER method, each connectivity change introduces a space-time hole between elements at consecutive time levels. These holes, which in the two-dimensional case take the shape of tetrahedra, must be catalogued and integrated over, complicating both the implementation and the analysis of the method. The central idea of this project was to replace centroid-dual polygonal grids (as used in the author's earlier work with Gaburro and Dumbser) with genuine Voronoi grids, on the basis that Voronoi grids transition smoothly through degenerate configurations involving zero-length edges, rather than undergoing abrupt connectivity flips. This eliminates the space-time holes entirely, and the resulting space-time elements take the shape of so-called scutoids, a family of prismatoid solids named after their resemblance to three-dimensional packings of

epithelial cells in biology. The project aimed at building a proof-of-concept implementation of such Voronoi-Scutoid elements and the associated subgrid generation infrastructure, in order to evaluate the viability and identify potential obstacles.

2.2 Construction of Voronoi-Scutoid elements.

A two-dimensional implementation of the Voronoi-Scutoid space-time element construction was developed. Given two consecutive Voronoi tessellations, the code identifies the connectivity changes (edge births, deaths, and transitions through zero-length degenerate configurations) and constructs the corresponding space-time control volumes. The technique is based on writing down the analytical evolution equation of the geometry of each edge, either forwards in time for edges present at the past time level or backwards in time for edges present at the future time level. In this way, one essentially only needs to track the disappearance of edges, while the appearance of new edges is simply a byproduct of looking at the mesh evolution backwards in time. The geometrically non-trivial step is the determination of the exact time at which an edge disappears, which requires solving a quartic polynomial equation. Standard root-finding (both analytical via Ferrari's formula and iterative via Newton's method) proved insufficient due to near-degenerate configurations, and the issue was resolved by employing double-double (quad-precision emulated) arithmetic.

2.3 Subgrid generation. The grant proposal envisioned the use of nonconforming quadrilateral subgrids (split-quads) for the integration and reconstruction steps on general polygonal elements. A simple prototype of this approach was implemented and tested. However, the data structures required to store and manipulate the quadrilateral subgrids must accommodate arbitrarily short edges (arising from near-degenerate Voronoi configurations), and therefore have essentially the same complexity as a general Voronoi data structure. Since no simplification is gained, the conclusion was that a nested Voronoi-based subgrid strategy is preferable, as it naturally handles all degenerate cases and unifies the main grid and subgrid representations.

2.4 Lagrangian mesh motion. In parallel, a fully implicit Lagrangian solver for the multimaterial Euler equations was developed, building on the foundational work of Munz (1994) on implicit finite-volume Lagrangian methods. This solver introduces nonconservative corrections to suppress the well-known pressure oscillations at material interfaces. The development of this solver took place during the funded period; the paper was submitted and published subsequently, and acknowledges the DROPIT start-up grant. The Lagrangian solver also served as a testbed for understanding mesh imprinting effects and the sensitivity of high-order methods to mesh quality and motion, which informed the broader Voronoi-Scutoid project.

2.5 Collaborative work on the GPR model and a quaternion reformulation of the model. The unified model of continuum mechanics (also known as the Godunov-Peshkov-Romenski, or GPR, model) describes viscous and heat-conducting flows as a first-order hyperbolic system, avoiding second-order spatial derivatives entirely. A central variable in the model is the 3 by 3 distortion field A , which simultaneously encodes the rotation, shear, and volumetric deformation of material elements. Standard numerical schemes treat the nine components of A as independent scalar fields and reconstruct them component-wise, which systematically destroys the rotation content: interpolating between two rotation matrices component-wise does not produce a rotation, but introduces spurious stretching and deformation. This well-known problem (Shoemake, 1985, in the computer graphics literature) makes the standard GPR formulation essentially unusable for turbulent flows, where A undergoes large rotations from cell to cell.

A collaboration with Pascal Mossier at IAG concerned the heat conduction aspects of the GPR model. I contributed the semi-analytical relaxation source term integrator for the thermal impulse equation, which Mossier then validated within the high-order Discontinuous Galerkin framework FLEXI, including comparisons against direct molecular dynamics simulations for multiphase flows.

Beyond this published result, the stay in Stuttgart proved decisive for the subsequent direction of my research. During the DG workshop informally known as "Zollernblick", organized by Jens Keim, Anna Schwarz, and Andrea Beck, discussions with Pascal Mossier about the applicability and competitiveness of the GPR model in real-world flows led me to begin experimenting with quaternion-based bookkeeping of the rotational part of the distortion field. What eventually became a full PDE-based reformulation of the GPR model (decomposing $A = R S$ into rotation and stretch, representing the rotation R as a unit quaternion) began as a purely computational-level experiment: since the evolution equations for the quaternion

and stretch variables were not yet algebraically derived, I used discrete time-finite differences to numerically construct and solve the new PDE system, without even knowing the analytical form of the equations. This early prototype confirmed that the idea was viable, and the quaternion reformulation ultimately became the central scientific contribution of my subsequent MSCA fellowship at the University of Cologne, reducing the computational cost of turbulence simulations with the GPR model by several orders of magnitude.

2.6 Cooperation benefits

The position at IAG during the start-up grant provided direct access to the expertise in high-performance computing on CPUs and GPUs, Discontinuous Galerkin methods, and turbulence modelling that characterizes the Stuttgart group. The collaboration with Pascal Mossier on the GPR model was the most impactful scientific outcome, resulting in a joint publication and in the exchange of ideas between the Finite Volume and Discontinuous Galerkin communities within the same model framework. The “Zollernblick workshop”, in particular, created the informal scientific environment in which the quaternion idea could take shape, specifically from Pascal Mossier’s inquiries about the applicability and competitiveness of the GPR model in real-world flows.

3. Statement on career development measures:

The start-up grant fulfilled its primary career development objective: during the funded period, I successfully secured a Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF, Grant Agreement 101109532, project “MoMeNTUM”) at the University of Cologne, which began in January 2024 and provided two years of independent research funding. The scientific groundwork laid during the start-up, in particular the Voronoi-Scutoid feasibility study and the early quaternion ideas born at the Zollernblick workshop, directly informed the MSCA work carried out in Cologne.

Participation in or organisation of IRTG-related events:

- DIPSI Workshop 2023 / DROPIT doctoral seminar, Bergamo, Italy, 16 June 2023.
- DROPIT final colloquium, 31 Jul. – 01 Aug. 2025, Stuttgart.

Participation in conferences, congresses, etc., at home and abroad:

- SHARK-FV 2023, Povia de Varzim, Portugal, 22--26 May 2023.
- ICIAM 2023, Tokyo, Japan, 20--25 August 2023.
- 16th Hirschegg Workshop on Conservation Laws, Hirschegg, Austria, 10--16 September 2023.
- DG Workshop “Zollernblick”, organized by J. Keim, A. Schwarz, and A. D. Beck (IAG Stuttgart).

4. Individual publications:

S. Chiocchetti, G. Russo. "An efficient implicit scheme for the multimaterial Euler equations in Lagrangian coordinates." *Journal of Computational Physics*, 537 (2025) 114086. DOI:10.1016/j.jcp.2025.114086, arXiv:https://arxiv.org/abs/2406.1564.

P. Mossier, S. Joens, S. Chiocchetti, A. D. Beck, C.-D. Munz. "Numerical simulation of phase transition with the hyperbolic Godunov-Peshkov-Romenski model." *Journal of Computational Physics*, 520 (2025) 113514. DOI: 10.1016/j.jcp.2024.113514, arXiv: https://arxiv.org/abs/2403.01847.

A. Schwarz, J. Keim, S. Chiocchetti, A. D. Beck. "A reinforcement learning based slope limiter for second-order finite volume schemes." *PAMM*, 23(1), e202200207, 2023. DOI: 10.1002/pamm.202200207.

APPENDIX

Appendix 1: List of Workshops / Summer Schools within DROPIT

Table A1.1 summarises the workshops held in DROPIT during the nine years of the project. There have been some workshops which have been organized together with other research initiatives like the SFB-TRR75 or the SFB1313. These workshops have been highlighted in **green** color.

Table A1.1. Summary of workshops within DROPIT

Title	from / to	Organisation/Location
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2017	26.5.2017	University of Bergamo together with DROPIT/Bergamo
Workshop on Experimental and Numerical Methods	17.5.2018	DROPIT/Bergamo
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2018	18.5.2018	University of Bergamo together with DROPIT/Bergamo
Workshop on Evaporation across the Free-Flow Porous Medium Interface and Salt Precipitation	01.4.2019 - 02.4.2019	DROPIT together with SFB 1313/Stuttgart
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2019	17.5.2019	University of Bergamo together with DROPIT/Bergamo
Workshop on Mathematical and Numerical Modelling of Multiphase Flow	07.11.2019	DROPIT together with SFB-TRR 75/Stuttgart
Future Perspectives Workshop	10.11.2019 - 11.11.2019	DROPIT/Stuttgart
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2022	01.07.2022	University of Bergamo together with DROPIT/Bergamo
DROPIT meets Industry 2023	15.06.2023	DROPIT/Bergamo
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2023	16.06.2023	University of Bergamo together with DROPIT/Bergamo
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2024	17.06.2024	University of Bergamo together with DROPIT/Bergamo

Experimental and Numerical Methods in Science & Ethics in Science	18.06.2024	DROPIT/Bergamo
Homogenization Short Course	10.02.2025-12.02.2025	University of Stuttgart together with SFB1313
Workshop on AI meets Scientific Computing	11.06.2025	DROPIT/Bergamo
Workshop on Analytical and Numerical Methods in Science	12.06.2025	DROPIT/Bergamo
International Workshop on Drop Impact Phenomena and Spray Investigation (DIPSI), 2025	13.06.2025	University of Bergamo together with DROPIT/Bergamo
DROPIT Final Colloquium	31.07.2025 / 01.08.2025	DROPIT/Stuttgart

Some of these workshops, e.g., the workshop “**Future Perspectives**”, which took place from 10.10.2019-11.10.2019 in Stuttgart and the follow up workshop “**DROPIT meets Industry 2023**”, which took place in Bergamo, have been set-up, based on the initiative of the doctoral researchers.

Doctoral Seminars

The yearly **doctoral seminars** have been held in Bergamo and in Stuttgart:

- DROPIT Doctoral Seminar 2017, Bergamo (Italy), 24.05.2017
- DROPIT Doctoral Seminar 2018, Stuttgart (Germany), 11.04.-12.04.2018
- DROPIT Doctoral Seminar 2019, Bergamo (Italy), 15.05.-16.05.2019
- DROPIT Doctoral Seminar 2020, Stuttgart (Germany), 02.04.-03.04.2020
 - ➔ Because of the spreading of the Corona virus this Doctoral Seminar had to be postponed to 25.06.-26.06.2020. Finally, it was held in an online format via Webex.
- DROPIT Virtual Doctoral Seminar 2021, 08.03.-09.03.2021
- DROPIT Doctoral Seminar 2021, 08.09.-09.09.2021 (Bergamo) in presence and virtual
 - ➔ This was the first Workshop which was allowed (under very strict limitations) in Bergamo
- DROPIT Hybrid Doctoral Seminar 2023, Stuttgart (Germany), 01.03.-02.03.2023
- DROPIT Hybrid Doctoral Seminar 2024, Bergamo (Italy), 08.02.-09.02.2024
- DROPIT Doctoral Seminar 2025, Stuttgart (Germany), 28.01.-29.01.2025

During the doctoral seminars, the doctoral researchers presented their yearly progress in their research project.

Multiphase Flow Seminar

In the “**Multiphase Flow Seminar**” the doctoral researchers presented their work during the semester. In this seminar normally also students participated and, thus, a good discussion of the work of the doctoral researchers could be achieved. Furthermore, the students could learn about subprojects in DROPIT and got interested in the project.

International Workshop on Droplet Dynamics (DIPSI)

The International Workshop builds upon the DIPSI Workshop on “**Droplet Impact Phenomena and Spray Investigation**”. This yearly workshop has a very long tradition and provides an opportunity to share recent knowledge on droplets and sprays in a variety of research fields and industrial applications. The workshop is always held in Bergamo within the second half of May and the first half of June, every year. Here the doctoral researchers have actively been involved, e.g. by special poster presentations during the DIPSI workshop. This gave the doctoral researchers the chance to actively discuss their research results with international experts and to start scientific collaborations.



Figure A.1. DIPSI workshop 2023 in Bergamo

The poster session was always introduced by a short talk about DROPIT, so that everyone could get familiar with the concept of an IRTG. The doctoral researchers had to attend minimum two DIPSI workshops.

In the week of the DIPSI workshop normally also additional workshops, lectures and/or the doctoral seminar were held. Thus, in 2018 a day before DIPSI the “**Workshop on Experimental and Numerical Methods**” took place in Bergamo (Italy) on 17.05.2018. In 2022, the closing event of the Summer School 2022 took place the day before DIPSI 2022. In 2024, the “**Workshop on Experimental and Numerical Methods in Science & Ethics in Science**” was held the day after the DIPSI international workshop 2024. It included also a session on “Ethics in Science”.

1st International summer school (04.09.2018 - 21.08.2018)

The summer schools have been fully organised by the doctoral and postdoctoral researchers. Thus, they have been responsible for the planning, budgeting, and logistic aspects of the summer schools. The summer schools have been multinational and interdisciplinary in character. It was structured as an intensive academic programme. The main objective of the summer school was to endorse doctoral researchers to achieve scientific independence at an early point in their research career. Furthermore, it offered a natural platform to support teamwork and the ability to work in a multi-disciplinary and multi-national team. In the following a short report of the first summer school is given:

The first summer school of the IRTG was conducted in 2018. The aim of this summer school was to bring multi-disciplinary scientists together to work on research topics related to droplet interaction processes using a combination of experimental, numerical, and analytical tools. As a first step to achieve this, short summer school project proposals had been submitted by different teams, and were reviewed and approved. This resulted in five approved summer school projects. In addition, there were invited lectures and keynote speeches by experts in the field of droplet dynamics and related topics during the period of the summer school. The summer school began with an official kick-off meeting on 04.09.2018 in Stuttgart where the approved summer school project proposals were presented by the DROPIT doctoral researchers. Subsequently, the project groups worked independently within themselves, collaborating with the invited experts of each group over a period of two and a half weeks at various locations specified in the proposals. After this period of intensive work, all the participants met again for the final meeting on 21.09.2018 in Stuttgart, where they presented the work accomplished and prospects of collaborative work. This was followed by a farewell evening, taking place on the TV-tower in Stuttgart. During the summer school project, the teams worked in Stuttgart and in Bergamo. The selected five summer school projects have been:

A. Diffuse interface models of Baer-Nunziato type

Participants: C. Müller¹ (SP-A2), W. Reschke¹ (SP-A3), S. Chiocchetti² (SP-A4), J. Zeifang¹ (SP-A6), F. C. Massa³ (SP-C3/AS05), F. Bassi³, A. D. Beck¹, M. Dumbser², S. Fasoulas¹, C.-D. Munz¹, F. Föll¹, T. Hitz¹, M. Ioriatti², M. Pelanti⁴

¹University of Stuttgart, ²University of Trento, ³University of Bergamo, ⁴ENSTA ParisTech

B. Experimental and numerical investigation of wetting phenomena on selected structured surfaces and porous media

Participants: P. Foltyn¹ (SP-B1), R. Haide² (SP-B3), M. Baggio¹ (SP-B5), A. Straub¹ (SP-C4), S. Fest-Santini² (SP-AS04), T. Ertl¹, N. Roth¹, M. Santini², J. Sijbers³, D. Wildenschild⁴

¹University of Stuttgart, ²University of Bergamo, ³University of Antwerp, ⁴Oregon State University

C. Coupling porous media and free flow with help of an up-scaled phase field model

Participants: S. Ackermann¹ (SP-B6), L. Ostrowski¹ (SP-B4), R. Helmig¹, C. Rohde¹, J. Magiera¹, S. Pop², L. von Wolff¹

¹University of Stuttgart, ²Hasselt University

D. Investigation on the mixing behaviour for droplet impact onto miscible and immiscible liquids

Participants: R. Bernard¹ (SP-C1), G. Lamanna¹, C. Planchette²

¹University of Stuttgart, ²TU Graz

E. Oscillation of an evaporating spheroidal drop: effect of frequency and amplitude

Participants: G. Varma Raja Kochanattu¹ (SP-A1), K. Schlottke² (SP-AS02), G. E. Cossali¹, S. Tonini¹, B. Weigand², G. Brenn³, M. Ibach², J. Reuttsch²

¹University of Bergamo, ²University of Stuttgart, ³TU Graz

The following invited lectures/talks have been held during this summer school:

- **“Shape oscillations of drops of complex fluids”** by Prof. Günter Brenn from the Graz University of Technology (TU Graz), Austria.
- **“Compressible three-phase liquid-vapour-gas flow with heat and mass transfer”** by Prof. Marica Pelanti from ENSTA ParisTech, France.
- **“Granular armors at liquid interfaces”** by Dr. Carole Planchette from Graz University of Technology (TU Graz), Austria.
- **“Visualisation of droplet dynamics”** by Prof. Filip Sadlo from the Heidelberg University, Germany.
- **“Drop impact onto super-hydrophobic surfaces with macro-structures”** by Prof. Shamit Bakshi from the Indian Institute of Technology Madras, India.
- **“Exploring wettability states and alterations in porous media multiphase flow systems using 3D imaging”** by Prof. Dorte Wildenschild from the Oregon State University, USA.

From the projects started in the summer school in 2018 a lot of very fruitful collaboration evolved. Thus, the work was continued after the end of the summer school. Out of these collaborations, several common publications between different groups could be produced.

The second summer school was originally planned for 2020. However, it had to be moved to June 2022, because of the Coronavirus Pandemic.

2nd International summer school (10.06.2022 - 30.06.2022)

The second summer school in 2022 was structured in the same way as the first one in 2018. The short summer school project proposals were submitted for review by different teams, and were reviewed and approved.



Figure A.2. Participants of the DROPI Summer School 2022 at the final meeting on 30.06.2022 in Bergamo, Italy.

This resulted this time in seven approved summer school projects. In addition, there were invited lectures and keynote speeches by experts in the field of droplet dynamics and related topics during the period of the summer school. The summer school began with an official kick-off meeting on 10.06.2022 in Stuttgart where the approved summer school project proposals were presented by the DROPI doctoral researchers. Subsequently, the project groups worked independently within themselves, collaborating with the invited experts of each group over a period of three weeks at various locations specified in the proposals. After this period of intensive work, all the participants met again for the final meeting on 30.06.2022 in Bergamo, where they presented the work accomplished and prospects of collaborative work. During the summer school project, the teams worked in Stuttgart and in Bergamo. The selected five summer school projects have been:

A. Num. & Exp. Investigation of Grouping Behavior in Monodisperse Droplet Streams

Participants: D. Appel¹ (SP-A6), M. Ibach¹, V. Vaikuntanathan²

¹University of Stuttgart, ² Shiv Nadar University

B. Evaporation in a Coupled Free Flow–Porous Medium System Including Droplets on the Interface

Participants: M. Veyskarami¹ (SP-B6), R. Helmig¹, C. Bringedal¹, M. Santini², S. Fest-Santini², A. Raoof³, E. de Vries³, S. Chen⁴

¹University of Stuttgart, ²University of Bergamo, ³Utrecht University, ⁴University of Arizona

C. Application of the General Defocusing Particle Tracking Method to Analyze the Three-Dimensional Flow Field during Droplet Impact

Participants: S. Schubert¹ (SP-C1), A. Geppert¹ (SP-C5), M. Rossi²

¹University of Stuttgart, ²University of Denmark

D. Investigation of Surface Tension Effects on Evaporating Droplets with Micro and Macroscopic Models

Participants: R. Tietz¹ (SP-A3), P. Mossier¹ (SP-A2), S. Tonini², J. Keim¹, S. Frank¹

¹University of Stuttgart, ²University of Bergamo

E. Topological Visualization Methods to Analyze Drop Impacts

Participants: D. Klötzl¹ (SP-C4), P. Palmetshofer¹ (SP-B1), W. Ren¹ (SP-B5), S. Schubert¹ (SP-C1)

¹University of Stuttgart

F. Exp. & Num. Investigation of Droplet Impacts onto Pillars with a Wetted Base

Participants: P. Palmetshofer¹ (SP-B1), W. Ren¹ (SP-B5), Y. Liu¹, J. Steigerwald¹, S. Bakshi

¹University of Stuttgart, ²IIT Madras

G. Investigation of Early Phase Contact Line Movement During Droplet Impact to Improve the Analytical Crown Base Model (Experiments, Numerical Simulation, Analytics)

Participants: A. Geppert¹ (SP-C5), J. Stober¹ (SP-C2), F. Massa², S. Schubert¹ (SP-C1)

¹University of Stuttgart, ²University of Bergamo

The following invited lectures/talks have been held during this summer school:

- **“An overview of some experimental investigations on droplet interaction with solid, liquid, and air”** by Dr. Visakh Vaikuntanathan from the Shiv Nadar University, Dehli NRC, India.
- **“Evaporation induced flow around a pendant droplet evaporating in atmospheric condition”** by Prof. Shamit Bakshi from the Indian Institute of Technology (IIT) Madras, India.
- **“Grouping Principles and Applications”** by Prof. David Katoshevski from the Ben-Gurion University of the Negev (BGU), Be’er Scheva, Israel.

From the projects started in the summer school in 2022 a lot of very fruitful collaboration evolved. Thus, the work was continued after the end of the summer school. Out of these collaborations, several common projects between different groups could be initiated and several publications have been published.

Beside the summer school, the Doctoral Seminars and a large number of specialised courses, DROPIT organised several workshops. In **Table A1.1** all workshops, which have been organised in cooperation with DROPIT are listed. The table also contains the location, where the workshops have been held.

Final Colloquium (31.07.2025 – 01.08.2025)

The Final Colloquium of DROPIT took place on 31.07.2025 – 01.08.2025. The photograph below shows the participants of the Final Colloquium.



Fig. A.3. Participants of the Final Colloquium in Stuttgart in 2025.

The colloquium was structured in such a way to give all the participants an overview of the structure of the project and the progress, which had been achieved in the programme and in the individual subprojects.

In addition, key note talks have been given by the two rectors of the involved Universities (Peter Middendorf for the University of Stuttgart and Sergio Cavalieri for the University of Bergamo) addressing the importance of this cooperation for both Universities.

We were really happy that also a large number of doctoral researchers of the first and second generation joined the Final Colloquium and gave talks about the subprojects together with the third generation of the doctoral researchers.

Appendix 2: Visiting Scientist Program

The visiting scientist program represented an important cornerstone of our qualification measures for the doctoral researchers. Its goal was to promote scholarly exchange on numerical, theoretical and experimental methods for two-phase flows and, if feasible, to establish fruitful collaborations with international leading scientists. The visiting scientists had been integrated in DROPIT in several ways. First, they presented an invited talk during their stay. Second, an intensive exchange between the visitor and interested doctoral researchers took place. Thus, the doctoral researchers could benefit, already at an early stage, from an external and qualified feedback on the status of their own research. Furthermore, comments and suggestions from experts helped in their own revision and conceptualization of their research work, thus providing a unique learning experience. In order to reach this goal, the doctoral researchers had a very active role in outlining the visiting researcher programme. They have been encouraged and motivated to propose names of visiting researchers. Thanks to the close geographical vicinity of Stuttgart and Bergamo some visitors could be hosted at both partner universities. **Table A2.1** summarizes the visitors together with the time span of their visit, the purpose of the visit and the institution they are coming from.

Table A2.1. List of visiting scientists

Visitor	from / to	Institution	Description / Purpose of visit
Jochen Schütz	10.01.2017 / 12.01.2017	Faculty of Sciences, Hasselt University, Belgium	Presentation/Research cooperation with SP-A6
Klaus Kaiser	10.01.2017 / 12.01.2017	IGPM, RWTH Aachen University	Presentation/Research cooperation with SP-A6
Kata Kurgyis	09.01.2017 / 24.02.2017	Montan University, Leoben	Research cooperation with SP-B6
Veronika Schleper	25.05.2017 / 26.05.2017	Robert Bosch GmbH	Presentation at DIPSI Workshop 2017 at University of Bergamo
Bo Gao	04.07.2017 / 31.07.2017	Colorado School of Mines, USA	Cooperation with SP-B6
Sergei Sazhin	10.12.2017 / 16.12.2017	School of Computing, Engineering & Maths, Advanced Engineering Centre, University of Brighton, United Kingdom	Lecture on "Heat and mass transfer in droplets" at University of Bergamo and University of Stuttgart
Shamit Bakshi	25.04.2018 / 25.07.2018	Department of Mechanical Engineering, Indian Institute of Technology Madras, India	Alexander von Humboldt visitor, Cooperation with SP-B5
Giancarlo Maccarini	10.04.2018 / 12.04.2018	University of Bergamo	Advisory Board Meeting and Doctoral Seminar
Jan-Philipp Schmidt	11.04.2018 / 12.04.2018	Springer Nature, Springer Heidelberg	Advisory Board Member
Günter Brenn	17.05.2018 / 18.05.2018	Institute of Fluid Mechanics and Heat Transfer, TU Graz, Austria	Presentation at DIPSI Workshop 2018 at University of Bergamo
Jan Sijbers	24.07.2018 / 25.07.2018	imec - Vision Lab, University of Antwerp, The Netherlands	Participation in Summer School Project 2018

Carole Planchette	03.09.2018 / 05.09.2018	Institute of Fluid Mechanics and Heat Transfer, TU Graz, Austria	Presentation / Participation in Summer School Project of SP-C1
Sorin Pop	04.09.2018 / 07.09.2018	Faculty of Sciences, Hasselt University, Belgium	Participation in Summer School Project of SP-B4 and SP-B6
Matteo Ioriatti	01.09.2018 / 30.09.2018	Department of Mathematics, University of Trento, Italy	Participation in Summer School Project of SP-A2, SP-A3, SP-A4 and SP-A6
Günter Brenn	06.09.2018 / 09.09.2018	Institute of Fluid Mechanics and Heat Transfer, TU Graz, Austria	Seminar during Summer School 2018
Marica Palanti	11.09.2018 / 15.09.2018	Department of Mechanical Engineering, ENSTA Paris-Tech, France	Seminar during Summer School 2018
Dorthe Wildenschild	13.09.2018 / 24.09.2018	College of Engineering: Chemical, Biological, and Environmental Engineering, Oregon State University, USA	Presentation at Summer School 2018 at University of Bergamo Cooperation with SP-B1 and SP-B6
Carole Planchette	19.09.2018 / 22.09.2018	Institute of Fluid Mechanics and Heat Transfer, TU Graz, Austria	Participation in Summer School Project of SP-C1
Michael Dumbser	27.09.2018 / 28.09.2018	Department of Civil, Environmental and Mechanical Engineering, University of Trento, Italy	Milestone Presentation of M. Baggio
André Silva	07.02.2019 / 09.02.2019	Aeronautics and Astronautics Research Center (AEROG), Portugal	Presentation / Research / cooperation with SP-B1
Andro Mikelić	01.04.2019 / 02.04.2019	Université Claude Bernard Lyon 1, Institut Camille Jordan	Lecture at Workshop on evaporation across the free-flow porous medium interface and salt precipitation at University of Stuttgart
Peter Lehmann	01.04.2019 / 02.04.2019	ETH Zürich, Institute of Biogeochemistry and Pollutant Dynamics	Lecture at Workshop on evaporation across the free-flow porous medium interface and salt precipitation at University of Stuttgart
Uri Nachshon	01.04.2019 / 02.04.2019	ARO, Volcani Research Center Israel, Institute for Soil, Water and Environmental Sciences	Lecture at Workshop on evaporation across the free-flow porous medium interface and salt precipitation at University of Stuttgart
J. A. Sander Huisman	01.04.2019 / 02.04.2019	Forschungszentrum Jülich	Lecture at Workshop on evaporation across the free-flow porous medium interface and salt precipitation at University of Stuttgart
Jan-Philipp Schmidt	14.05.2019 / 17.05.2019	Springer Nature, Springer Heidelberg	Advisory Board Member - Participation at DROPIT Doctoral Seminar and DIPSI Workshop 2019 at University of Bergamo
Majid Hassanizadeh	02.06.2019 / 04.06.2019	Faculty of Geosciences, Utrecht University, The Netherlands	Seminar on "Fundamentals of multiphase flow in porous media" at University of Stuttgart

Jan Sijbers	03.06.2019 / 06.06.2019	imec - Vision Lab, University of Antwerp, The Netherlands	Seminar on micro-CT at University of Bergamo
Aldo Coghe	04.06.2019 / 05.06.2019	Politecnico Milano, Department of Energy	Seminar on "Laser Diagnostics in Sprays" at University of Bergamo
Miloslav Feistauer	06.06.2019	Charles University in Prague, Faculty of Mathematics and Physics, Department of Numerical Mathematics	Course on "Numerical solution of two-phase flow by the finite element discontinuous Galerkin and level set method" at University of Stuttgart
Jochen Schütz	07.07.2019 / 09.07.2019	Faculty of Sciences, Hasselt University, Belgium	Seminar on "Introduction to Discontinuous Galerkin Methods" at University of Stuttgart
Daniela Ribeiro	01.10.2019 / 13.12.2019	Aeronautics and Astronautics Research Center (AEROG), Portugal	Research visit and cooperation with SP-B1
Majid Hassanizadeh	07.10.2019 / 11.10.2019	Faculty of Geosciences, Utrecht University, The Netherlands	Seminar on "Course on capillarity in porous media at different scales" at University of Stuttgart
Michael Wolf	10.10.2019 / 11.10.2019	WOLF & WOLF Patent und Rechtsanwaltsgesellschaft mbH	Presentation at "Future Perspectives" Workshop
Sebastian Spring	10.10.2019 / 11.10.2019	Tplus Engineering GmbH	Presentation at "Future Perspectives" Workshop
Nicolas Schwenck	10.10.2019 / 11.10.2019	PARI GmbH	Presentation at "Future Perspectives" Workshop
Michael Fischer	10.10.2019 / 11.10.2019	Friedrich Boysen GmbH & Co. KG	Presentation at "Future Perspectives" Workshop
Daniel Knaf	10.10.2019 / 11.10.2019	Friedrich Boysen GmbH & Co. KG	Presentation at "Future Perspectives" Workshop
Felix Jäggle	10.10.2019 / 11.10.2019	Robert Bosch GmbH	Presentation at "Future Perspectives" Workshop
Maren Hantke	07.11.2019	Martin-Luther-Universität Halle-Wittenberg, Institut für Mathematik	Presentation at "Mathematical and Numerical Modelling of Multiphase Flow" Workshop
Ferdinand Thein	07.11.2019	Otto-von-Guericke University Magdeburg	Presentation at "Mathematical and Numerical Modelling of Multiphase Flow" Workshop
Josef Winter	07.11.2019	TU Munich	Presentation at "Mathematical and Numerical Modelling of Multiphase Flow" Workshop
Gerald Warnecke	07.11.2019	Otto-von-Guericke University Magdeburg	Presentation at "Mathematical and Numerical Modelling of Multiphase Flow" Workshop
Sergei Sazhin	16.12.2019 / 20.12.2019	School of Computing, Eng. & Maths, Advanced Engineering Centre, University of Brighton, UK	Lecture on "Heat and mass transfer in droplets" at University of Bergamo and University of Stuttgart
Massimiliano Rossi	23.05.2022 / 25.05.2022	University of Denmark, Copenhagen, Denmark	Visiting Scientist for Summer School project
Woradej Manosroi	20.07.2022 / 22.07.2022	Chiang Mai University, Thailand	Lecture on "Design and Construction of Dust Collector with Evaporative Condensation Growth Particle Scrubber"

Philipp Ligrani	19.06.2022 / 02.07.2022	The University of Alabama, Huntsville, USA	Visiting Scientist for Summer School project
Somnath Bhattacharyya	24.06.2022 / 10.07.2022	IIT Madras	Visiting Scientist for Summer School project
Rafael Tadmore	27.06.2022 / 02.07.2022	Ben Gurion University of the Negev, Israel	Visiting Scientist for Summer School project
Jan-Philipp Schmidt	29.06.2022 / 01.07.2022	Springer Nature, Springer Heidelberg	Advisory Board Member - Participation at DROPIT Doctoral Seminar and DIPSI Workshop 2019 at University of Bergamo
David Katoshevski	29.06.2022 / 02.07.2022	Ben Gurion University of the Negev, Israel	Visiting Scientist for Summer School and DIPSI Workshop
Cameron Tropea	30.06.2022 / 03.07.2022	University of Darmstadt, Germany	Participation in DIPSI Workshop
Bassam Younis	17.10.2022 / 25.10.2022	University of California, Davis, USA	Research visit
Sergei Sazhin	17.12.2022 / 21.12.2022	School of Computing, Eng. & Maths, Advanced Engineering Centre, University of Brighton, UK	Lecture on "Heat and mass transfer in droplets" at University of Bergamo and University of Stuttgart
Rahul Barthwal	02.10.2022 / 05.10.2022	Department of Mathematics, Indian Institute of Technology Kharagpur, India	Research visit, Lecture on "Nonlinear wave interactions in certain hyperbolic system of conservation laws"
T. Raja Sekar	02.10.2022 / 05.10.2022	Department of Mathematics, Indian Institute of Technology Kharagpur, India	Research visit, Lecture on "Nonlinear wave interactions in certain hyperbolic system of conservation laws"
Leila Hashemi	16.03.1022 / 15.04.2022	TU Delft, The Netherlands	Research visit
Giancarlo Maccarini	28.02.2023 / 02.03.2023	University of Bergamo	Advisory Board Meeting and Doctoral Seminar
Fabien Evrard	19.04.2023	Otto-von-Guericke University Magdeburg	Online lecture on "On closing the volume-filtered equations governing particle-laden flow"
Jan Schlottke	14.06.2023 / 16.06.2023	Mahle	Participation in Doctoral Seminar and "DROPIT meets Industry" Workshop
Visakh Vaikuntanathan	24.06.2023 / 01.07.2023	Shiv Nadar University, Delhi-NCR, India	Research visit
Volfango Bertola	13.12.2023 / 15.12.2023	University of Liverpool, UK	Research visit, Lecture on "Non-Newtonian drops"
Ferdinand Thein	21.06.2023 / 22.06.2023	RWTH Aachen, Germany	Research visit; Lecture on "Explicit-Implicit Domain Splitting for Two Phase Flows with Phase Transition"
Carina Bringedal	10.07.2023 / 13.07.2023	Western Norway University of Applied Sciences, Bergen, Norway	Research visit
Fabian Denner	29.11.2023	Polytechnique Montréal, Canada	Online lecture on "Eliminating the capillary time-step constraint of interfacial flow simulations"

Daniele di Pietro	14.02.2024	University of Montpellier	Seminar on "An Introduction to Discrete de Rham (DDR) methods"
Martin Sommerfeld	21.02.2024 / 23.02.2024	Otto-von-Guericke University Magdeburg	Lecture on "Characterisation of particle transport in dry powder inhalers"
Firas Dhaouadi	03.03.2024 / 09.03.2024	University of Trento	Lecture on "Hyperbolic approximation and numerical methods for the Navier-Stokes-Korteweg equations"
Signe Kjelstrup	12.06.2024 / 14.06.2024	Western Norway University of Applied Sciences, Bergen, Norway	Research visit
Koji Hasegawa	16.09.2024 / 19.06.2024	Department of Mechanical Engineering, Kogakuin University, Tokyo	Research visit and DIPSI-Workshop Bergamo
Thomas Fuchs	11.11.2024	Universität der Bundeswehr München	Lecture on "Application of defocusing particle imaging in confined domains"
Anton Chizhov	01.06.2025 / 02.06.2025	University of Bristol	Lecture on "Transport equations for modeling high-speed droplet impact and brain neural activity"
Bernant Font	10.06.2025 / 12.06.2025	Delft University of Technology, Netherlands	Lecture on "AI meets Scientific Computing"
Xu Chu	10.06.2025 / 12.06.2025	University of Exeter, UK	Lecture on "AI meets Scientific Computing"
Phil Ligrani	16.07.2025	University of Alabama Huntsville	Research Visit together with SynTrac
Anne Geppert	30.07.2025 / 01.08.2025	Leibniz-Institut für Werkstofforientierte Technologien - IWT	Talk at the DROPIT Final Colloquium
Remi Abgrall	31.07.2025 / 01.08.2025	University of Zurich	Talk at the DROPIT Final Colloquium
Giancarlo Maccarini	30.07.2025 / 02.08.2025	University of Bergamo	Talk at the DROPIT Final Colloquium
Sergei Sazhin	30.07.2025 / 02.08.2025	University of Brighton	Talk at the DROPIT Final Colloquium
Lucio Araneo	13.10.2025 / 14.10.2025	Politecnico Milano, Department of Energy	Lecture on Phase-Doppler-Anemometrie, and measurements
Lucio Araneo	03.11.2025 / 05.11.2025	Politecnico Milano, Department of Energy	Lecture on Phase-Doppler-Anemometrie, and measurements

Appendix 3: List of courses

3.1: Specialised Courses

List of the specific (specialized) courses for this IRTG:

Table 1 provides a list of all given specialized courses, including information on course duration, frequency and location, where the course took place. These specialized courses are at a more advanced level. Most courses were given by DROPIT principle investigators, and some by visiting scientists. Attendance was not limited to DROPIT doctoral researchers, but open to all doctoral researchers of the doctoral schools **GRADUS**, **ISA** and **DICAM** of the universities of Stuttgart, Bergamo and Trento, respectively.

Table A3.1.: List of specialized courses organized by **DROPIT**

Course title	Instructor(s)	Date & Location
On the efficient implementation of discontinuous Galerkin methods for the solution of partial differential equations	C.-D. Munz (US)	16.-21.10.2017, Freudenstadt, Germany
Heat and mass transfer in droplets	S. Sazhin (University of Brighton)	11.-12.12.2017, Stuttgart 14.-15.12.2017, Bergamo 16.-17.12.2019, Stuttgart 19.-20.12.2019, Bergamo 19.-20.12.2022, Stuttgart 18.-19.12.2023, Bergamo
Understanding Dimensional Analysis	B. Weigand (US)	14.-15.03.2018, Bergamo
Analysis of Droplet Dynamic Processes	N. Roth (US)	15.-16.05.2018, Bergamo Italy, 2 days
Orthogonal functions and generalized Fourier series	G.E. Cossali (UniBG)	11.09.2018, Bergamo
Workshop: Flow and transport across the free-flow porous-medium interface	R. Helmig (US) Course together with SFB1313	01.04.2019, Stuttgart
Workshop on Evaporation and Salt Precipitation	R. Helmig (US), M. Santini (BG) Course together with SFB1313	01.-02.04.2019, Stuttgart
Fundamentals of multiphase flow in porous media	M. Hassanizadeh (University of Utrecht), R. Helmig (US)	03.-04.06.2019, Stuttgart,

	Course together with SFB1313	
Lectures on X-ray micro-computed tomography	M. Santini (UniBG), J. Sijbers (imec, University of Antwerp)	03.06. & 06.06.2019, Bergamo 14.06.2023, Bergamo
Laser Diagnostics in Sprays	A. Coghe (Politecnico di Milano), G.E. Cossali (UniBG)	04.-05.06.2019, Bergamo 14.06.2023, Bergamo
Numerical solution of two-phase flow by the finite element discontinuous Galerkin and level set method	M. Feistauer (Charles University of Prague)	06.06.2019, Stuttgart
Introduction to Discontinuous Galerkin methods	A. Beck , C.-D. Munz , J. Dürrwächter (US), J. Schütz (University of Hasselt)	08.-09.07.2019, Stuttgart
Analytical methods and improved non-equilibrium droplet evaporation models for high pressure conditions	G.E. Cossali (UniBg), S. Tonini (UniBG), G. Lamanna (US)	11.-12.07.2019, Stuttgart
Selected Solution Methods for Nonlinear Partial Differential Equations for Flow and Heat Transfer Problems	B. Weigand (US)	13.09.2019, Bergamo
Course on capillarity in porous media at different scales	M. Hassanizadeh (University of Utrecht) Course together with SFB 1313	7.-11.10.2019, Stuttgart
Mathematical and Numerical Modeling of Multiphase Flow	C.-D. Munz (US), C. Rohde (US)	07.11.2019, Stuttgart
Advanced numerical methods for hyperbolic equations	M. Dumbser , E. Gaburro (UniTN)	10.2.-14.2.2020, Trento
Porous Media Free Flow Coupling	R. Helmig (US)	2021, Self-study course, online materials, QA sessions 14.11.-12.12.2022, 15:00-18:00, Stuttgart
Analytical Methods for non-linear partial differential equations	B. Weigand (US)	2021, Self-study course, online materials, QA sessions
Defocus particle tracking: Latest improvements and application to evaporating droplet	M. Rossi (University of Denmark)	25.03.2021, online lecture

Mono- and multi-component droplet heating and evaporation	S. Sazhin (University of Brighton)	16.-17.12.2021, online lectures
Puffing and micro-explosion in composite droplets	S. Sazhin (University of Brighton)	17.12.2021, online lectures
Pore network modeling workshop	R. Helmig (US)	26.-28.08.2022, Stuttgart
Short Course "Turbulence inside porous media: physics of fluids and modeling"	X. Chu (US)	10.-13.10.2022, Stuttgart
Analytical modelling of species and energy transport from evaporating drops	S. Tonini (BG)	28.02.2023, Stuttgart
Orthogonal curvilinear coordinate systems in analytical modelling of drop evaporation	G.E. Cossali (BG)	28.02.2023, Stuttgart
Simple models of complex processes: recent developments	S. Sazhin (University of Brighton)	19.12.2023, Bergamo
AI Meets Scientific Computing	R. Vinuesa P. Cinnella X. Chu B. Font O. Lehmkuhl	11.06.2025, Bergamo

These specialised courses were complemented by several international courses. These courses were found to be very good to provide to the doctoral researchers a detailed overview about the topics of droplets, sprays and interfaces. The content of these courses is given below:

- **International Advanced course on Liquid Interfaces, Drops and Sprays (LIDESP)**

The course is aimed to provide the participants with a detailed, up-to-date knowledge on the physics of drops and sprays based on recent research results and the most updated methods for the prediction of dynamic outcomes, heat transfer, wettability effects, and its applications to technological and industrial areas. The course duration is 4 days.

- **CISM Course on Drops, Jets and Films**

The course provides the participants with knowledge on drops, jets and films, their generation, stability, mutual interactions, interaction with their surroundings as well as associated transport phenomena. It includes historical retrospection, theoretical foundations, experimental and numerical methodologies, but also demonstration of advantages and disadvantages of different approaches through example research problems and solutions. Examples offered by the lecturers are to be complemented by discussion of individual challenges from participants. The course duration is 4 days.

- **Short Course on Advanced Numerical Methods for Hyperbolic Equations**

The short course on advanced numerical methods consists of a structured intensive one-week programme of 40 hours of theoretical lectures and computer laboratory exercises on advanced numerical methods for hyperbolic partial differential equations with applications in engineering and science.

- **Short Course on Atomization and Sprays**

The course presents the current understanding and state-of-the-art of atomization fundamentals, their realization in atomizer systems and application in a wide variety of engineering branches. The course was offered by the TU Darmstadt in cooperation with the SFB-TRR 75. The course duration is 3-4 days.

- **Short Course on Complex Wetting**

The course presents the fundamentals and industry applications of wetting phenomena. Fundamentals of interfacial physics and wetting phenomena are addressed together with theoretical, experimental and numerical methods. Techniques for the preparation of surfaces and the characterisation of interfaces are described and industrial applications are given. The course duration was 3 days and the course was given at the TU Darmstadt.

3.2: Core Courses

List of core courses (complementary courses from the universities' general programmes):

The core courses are part of the local universities' general educational programmes. In addition, some courses are given at the Supercomputing Center in Stuttgart (HLRS). For the German side, doctoral researchers had to achieve 9 to 12 ETCS during their affiliation to DROPIT.

Table A3.2: List of core courses, that are part of the regular lecture programme of the Universities of Stuttgart (US) and Bergamo (UniBG)

Course name	Instructor(s)	Duration / Frequency / ECTS
Location: University of Stuttgart (US)		
Analytical and numerical methods	B. Weigand, C.-D. Munz	3 h / WS / 6
Dimensional analysis	B. Weigand	1.5 h / WS / 3
Analysis of droplet dynamic processes	N. Roth	1.5 h / WS / 3
Elastic & inelastic light scattering	N. Roth	1.5 h / WS / 3
Compressible Flows I & II	U. Gaisbauer, G. Lamanna	1.5 h / WS & SS / 6
Selected problems of convective heat transfer	G. Lamanna	1.5 h / WS & SS / 3

Kinetic Gas theory	J. v. Wolfersdorf, S. Fasoulas	1.5 h / WS / 3
Multiphase Flow Seminar	C.-D. Munz, C. Rohde, B. Weigand	1.5 h / WS & SS / 3
Experimental methods in aerospace engineering	S. Fasoulas, M. Leitner, B. Peters, R. Poser, N. Roth, J. von Wolfersdorf	1.5 h / WS / 3
Numerical methods in two-phase flows	J. Schlottke	1.5 h / WS / 3
Mathematical methods in fluid mechanics	C.-D. Munz, C. Rohde	3 h / SS / 6
Mathematical and numerical modelling of multi-phase flow in porous media	R. Helmig	1.5 h / WS / 3
Environmental Fluid Mechanics II	R. Helmig,	3 h / SS / 6
Modeling of Hydrosystems	R. Helmig, B. Flemisch, M. Schneider	3 h / SS / 6
Discontinuous Galerkin methods	A. Beck	1.5h / SS / 3
Scaling and Node-level Performance	HLRS, Stuttgart	SS / 3
Fortran for Scientific Computing	HLRS, Stuttgart	SS / 3
Advanced C++ with Focus on Software Engineering	HLRS, Stuttgart	SS / 3
Location: University of Bergamo (UniBG)		
Advanced Mathematical Methods for Engineers	L. Brandolini, G. Gigante, M. Pedroni	3 h / WS / 6
Advanced numerical methods for engineering	C. de Falco (Politecnico di Milano)	3 h / WS / 6
Thermo-fluid Dynamics	F. Bassi, M. Savini, G.E. Cossali	3 h / WS / 6
Methods for material characterisation	F. Fontana	1.5 h / WS / 3
Instrumentation and measurements	G. Barigozzi, V. Re	3 h / WS / 6

3.3: Training Courses on Key Skills

List of training courses held on key skills

As stated already in the initial proposal of DROPIT, the training courses are not compulsory. Most of them are organised together with or by the respective Graduate Academy. Some training courses, like Scientific Writing, which are tailored especially to the need of DROPIT, have been organised by DROPIT alone.

Training courses at University of Stuttgart (US)
Writer's workshop for scientific and technical English
Scientific writing (2020, 2021 & 2023)
Introductory course in Italian language and culture
Critical thinking for scientists
Research data management
Convincing presence – communicating effectively
The art of communication
Goals: With mind and heart to success
Basics of persuasion
Leadership in academia
Philosophy of science and ethics of technology
Designing effective academic posters
Berufs- und Karriereplanung für Doktoranden und Doktorandinnen
Komplexität im Arbeitsalltag mit Struktur bewältigen
Supervision and notation of student theses
Roadmap towards your PhD: Techniques, tools and troubleshooting
Criteria for sound scientific conduct
Presentation techniques: competent talk and confident appearance
Time- and self-management
Competence analysis
Konfliktkompetenzen aufbauen und vertiefen

Training courses at University of Bergamo (US)
Lecture series, provided by the University of Bergamo and the University of Naples Federico II for Technology, Innovation and Management (TIM) PhD students
Italian course A2, B1, B2
How to write a successful proposal at European level – Horizon 2020
Uso degli strumenti informatici a supporto della ricerca

Training in the rules of good scientific practice

Within the training courses also courses are offered, which are concerned with the rules of good scientific practice, e.g.:

- Philosophy of science and ethics of technology
- Critical thinking for scientists
- Ethics of Science
- Ethics of Technology

Appendix 4: Supervision Agreement of Doctoral Studies (“Promotionsvereinbarung”)



Universität Stuttgart

Promotionsvereinbarung (Fassung ab 1. 10. 2021)

Präambel

Die Doktorandin bzw. der Doktorand und ihre/seine Betreuer/in in ihrer/seiner Funktion als Mitglied der Universität schließen eine Promotionsvereinbarung ab, um das Betreuungsverhältnis inhaltlich und zeitlich transparent zu gestalten. Die Promotionsvereinbarung soll die kontinuierliche Förderung und Beratung der Doktorandin oder des Doktoranden bei seinem oder ihrem Promotionsvorhaben sicherstellen und die Anforderungen an Betreuerin bzw. Betreuer und Betreute bzw. Betreuten im gegenseitigen Einverständnis formulieren. Betreuerin bzw. Betreuer und Promovierende bzw. Promovierender erkennen die Inhalte der Vereinbarung als das Fundament des Promotionsverhältnisses an und bemühen sich, die Vorgaben bestmöglich umzusetzen. Planung und Durchführung des Promotionsvorhabens sollen von den beteiligten Personen so gestaltet werden, dass das Vorhaben mit hoher Qualität innerhalb eines angemessenen Zeitraums abgeschlossen werden kann. Die individuelle Lebenssituation der Doktorandin oder des Doktoranden ist zu berücksichtigen. Die Vereinbarung richtet sich nach dem derzeit möglichen Planungshorizont und kann hinsichtlich der wissenschaftlichen Fragestellung des Dissertationsprojektes, der einzelnen Qualifizierungselemente und der fortzuschreibenden Zeitpläne für regelmäßige Betreuungsgespräche und Sachstandsberichte im gegenseitigen Einverständnis schriftlich geändert und fortgeschrieben werden. Die Promotionsvereinbarung ersetzt nicht die Annahme als Doktorand.

1. Beteiligte

Frau/Herr _____ als Doktorand(in)
(Vorname) (Nachname)

geboren am _____ in _____.

Adresse _____;
(Straße) (Hausnr.) (PLZ) (Wohnort)

Telefon _____; Handy _____;

E-Mail _____

und die/der betreuende/n Wissenschaftler/in(nen):

1. Frau/Herr _____ (Betreuer/in)

Institut _____

E-Mail _____

2. Frau/Herr _____ (ggf. Zweitbetreuer (optional))

Institut _____

E-Mail _____

2. Thema der Dissertation

2.1 Der Arbeitstitel für die Dissertation lautet:



3. Arbeits- und Zeitplan

Die Doktorandin bzw. der Doktorand und die Betreuerin bzw. der Betreuer vereinbaren einen fortzuschreibenden Arbeits- und Zeitplan, der dem Dissertationsprojekt und der Lebenssituation der Doktorandin bzw. des Doktoranden angepasst ist. Der Arbeits- und Zeitplan sieht regelmäßige Betreuungsgespräche und Sachstandsberichte vor. Er ist Anlage zu dieser Vereinbarung. Der Arbeits- und Zeitplan ist mindestens alle zwei Jahre durch die Beteiligten zu prüfen und gegebenenfalls zu modifizieren.

4. Individuelles Qualifizierungsprogramm

Die Doktorandin bzw. der Doktorand und die Betreuerin bzw. der Betreuer vereinbaren, welche begleitenden Qualifizierungsmaßnahmen im Umfang von mindestens 9 Leistungspunkten (z.B. fachspezifische Veranstaltungen, überfachliche Qualifizierungsveranstaltungen, Auslandsaufenthalte, Teilnahme an Tagungen, Vorträge etc.) wahrgenommen werden sollen, die ebenfalls Anlage zu dieser Vereinbarung sind.

5. Integration in Arbeitsgruppe, Forschungsverbund, Promotionsprogramm (optional)

Frau/Herr _____ bearbeitet ihr/sein Thema
im Rahmen des Forschungsvorhabens/Drittmittelprojektes:

_____ als eigenständiges Einzelforschungsvorhaben ggf. mit Anbindung an folgende Forschungsgruppe:

☐ im Rahmen des strukturierten Promotionsprogrammes: _____

6. Publikationen

Die Universität hat die Verpflichtung, das Wissen und die gemeinsame Leistung ihrer Wissenschaftlerinnen und Wissenschaftler der Gesellschaft zur Verfügung zu stellen. Das Publizieren geeigneter Forschungsergebnisse ist daher von großer Bedeutung.

- ☐ Die Doktorandin bzw. der Doktorand wird auf die Publikationsrichtlinie der Universität Stuttgart hingewiesen.
- ☐ Die Doktorandin bzw. der Doktorand wird über die formelle Möglichkeit informiert, eine klassische oder eine publikationsbasierte, kumulative Dissertation anzufertigen, sofern letztere durch Richtlinie des zuständigen Promotionsausschusses nach § 2 Abs. 6 der Promotionsordnung gestattet ist.

7. Verpflichtung auf die Einhaltung der Regeln guter wissenschaftlicher Praxis

Die Doktorandin bzw. der Doktorand und die Betreuerin bzw. der Betreuer verpflichten sich, die Grundsätze guter wissenschaftlicher Praxis, wie sie in den Empfehlungen der Deutschen Forschungsgemeinschaft und in der Satzung der Universität Stuttgart zur Sicherung der Integrität wissenschaftlicher Praxis und zum Umgang mit Fehlverhalten in der Wissenschaft aufgestellt sind, zu beachten und einzuhalten.



8. Festlegung der Begutachtungszeiten nach Abgabe der Dissertation

Gemäß § 10 Abs. 1 Satz 2 der Promotionsordnung soll die Begutachtung der Dissertation in der Regel innerhalb von 3 Monaten abgeschlossen sein.

9. Regelung für Konfliktfälle

Bei Meinungsverschiedenheiten oder Konflikten zwischen der Doktorandin bzw. dem Doktoranden und der Betreuerin bzw. dem Betreuer bemühen sich alle Beteiligten um eine einvernehmliche Lösung. Bei Bedarf kann sich jede bzw. jeder Beteiligte an die Ombudsperson wenden, die nach der Promotionsordnung zu diesem Zweck zu bestellen ist.

10. Beendigung der Promotionsvereinbarung

- 10.1 Die Promotionsvereinbarung kann im gegenseitigen Einvernehmen der Beteiligten jederzeit aufgelöst werden.
- 10.2 Eine einseitige Kündigung der Promotionsvereinbarung bedarf der Schriftform. Die Doktorandin bzw. der Doktorand kann ihr bzw. sein Promotionsvorhaben jederzeit ohne Angabe von Gründen aufkündigen. Die Betreuerin oder der Betreuer hat die Kündigung der Promotionsvereinbarung schriftlich zu begründen. Wird die Beendigung einseitig durch die Betreuerin oder den Betreuer angestrebt, kann die Doktorand oder der Doktorand den Promotionsausschuss zur Vermittlung anrufen. Dieser muss auf Antrag der Doktorandin oder des Doktoranden das Vorliegen triftiger Gründe bestätigen.
- 10.3 Im Falle einer Kündigung der Promotionsvereinbarung bleibt die Annahme als Doktorand hiervon unberührt, es sei denn die durch die Doktorandin oder den Doktoranden erklärte Kündigung erstreckt sich auch hierauf. Der Promotionsausschuss prüft im Falle einer Kündigung der Promotionsvereinbarung, ob die Annahme als Doktorand nach den Bestimmungen der Promotionsordnung zu widerrufen ist.
- 10.4 Die Promotionsvereinbarung ist aufgelöst, wenn der Antrag auf Annahme als Doktorandin oder Doktorand vom zuständigen Promotionsausschuss abgelehnt oder widerrufen wird.
- 10.5 Mit Abschluss des Promotionsvorhabens endet das Betreuungsverhältnis automatisch.

11. Ausfertigung und Annahme als Doktorand/in

- 11.1 Die Promotionsvereinbarung wird in **mindestens dreifacher Ausfertigung** unterzeichnet. Je eine Ausfertigung verbleibt bei der Doktorandin oder dem Doktoranden, der Betreuerin oder dem Betreuer und beim Promotionsausschuss.
- 11.2 Der **Antrag auf Annahme als Doktorandin oder Doktorand** soll in der Regel innerhalb von zwei Monaten nach Abschluss der Promotionsvereinbarung beim zuständigen Promotionsausschuss gestellt werden.

Stuttgart, den _____

Stuttgart, den _____

Doktorand*in

Betreuer*in